

fong-spivak-seven-sketches: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_15 F00001	15	■ 1.000	a	a	Connections are symmetric, so if a is connected to b , then b is connected to a . Connec-
fong-spivak-seven-sketches_15 F00002	15	■ 1.000	b	b	Connections are symmetric, so if a is connected to b , then b is connected to a . Connec-
fong-spivak-seven-sketches_52 F00003	52	■ 0.998	$\mathcal{V}$	$\mathcal{V}$	second goal is to discuss something called $\mathcal{V}$ -categories for various monoidal preorders
fong-spivak-seven-sketches_114 F00004	114	■ 1.000	Σ	Σ	nition 3.68 always has two adjoints of its own: a left adjoint which we denote Σ and a
fong-spivak-seven-sketches_114 F00005	114	■ 1.000	Π	Π	right adjoint which we denote Π .
fong-spivak-seven-sketches_176 F00006	176	■ 0.679	$\mathbf{Mat}(R)$	$\mathbf{Mat}(R)$	Definition 5.50. Let R be a rig. We define the <i>prop of R-matrices</i> , denoted $\mathbf{Mat}(R)$,
fong-spivak-seven-sketches_219 F00007	219	■ 0.926	Ω	Ω	To be clear, the Ω are just labels; the above set is isomorphic to $\{\text{light, switch, battery}\} \sqcup$
fong-spivak-seven-sketches_264 F00008	264	■ 0.930	$\mathbb{I}\mathbb{R}$	$\mathbb{I}\mathbb{R}$	The interval domain $\mathbb{I}\mathbb{R}$ is a specific topological space, which we will use to model
fong-spivak-seven-sketches_13 F00009	13	■ 0.999	$f: X \rightarrow Y$	$f: X \rightarrow Y$	maps. A map $f: X \rightarrow Y$ is a kind of observation of object X via a specified relationship
fong-spivak-seven-sketches_13 F00010	13	■ 0.999	X	X	maps. A map $f: X \rightarrow Y$ is a kind of observation of object X via a specified relationship
fong-spivak-seven-sketches_13 F00011	13	■ 1.000	Y	Y	it has with another object, Y . For example, think of X as the subject of an experiment
fong-spivak-seven-sketches_14 F00012	14	■ 1.000	f	f	Asking which aspects of X one wants to preserve under the observation f becomes
fong-spivak-seven-sketches_14 F00013	14	■ 1.000	$\mathbb{R}$	$\mathbb{R}$	functions f from $\mathbb{R}$ to $\mathbb{R}$, and we can think of them as observations: rather than view
fong-spivak-seven-sketches_14 F00014	14	■ 1.000	x	x	x “directly”, we only observe $f(x)$. Out of all the functions $f: \mathbb{R} \rightarrow \mathbb{R}$, only some of
fong-spivak-seven-sketches_14 F00015	14	■ 1.000	$f(x)$	$f(x)$	x “directly”, we only observe $f(x)$. Out of all the functions $f: \mathbb{R} \rightarrow \mathbb{R}$, only some of
fong-spivak-seven-sketches_14 F00016	14	■ 1.000	$f: \mathbb{R} \rightarrow \mathbb{R}$	$f: \mathbb{R} \rightarrow \mathbb{R}$	x “directly”, we only observe $f(x)$. Out of all the functions $f: \mathbb{R} \rightarrow \mathbb{R}$, only some of
fong-spivak-seven-sketches_14 F00017	14	■ 0.714	$x \leq y$	$x \leq y$	(a) <i>order-preserving</i> if $x \leq y$ implies $f(x) \leq f(y)$, for all $x, y \in \mathbb{R}$; ¹
fong-spivak-seven-sketches_14 F00018	14	■ 0.714	$f(x) \leq f(y)$	$f(x) \leq f(y)$	(a) <i>order-preserving</i> if $x \leq y$ implies $f(x) \leq f(y)$, for all $x, y \in \mathbb{R}$; ¹
fong-spivak-seven-sketches_14 F00019	14	■ 0.714	$x, y \in \mathbb{R}$	$x, y \in \mathbb{R}$	(a) <i>order-preserving</i> if $x \leq y$ implies $f(x) \leq f(y)$, for all $x, y \in \mathbb{R}$; ¹
fong-spivak-seven-sketches_14 F00020	14	■ 1.000	$ x - y = f(x) - f(y) $	$ x - y = f(x) - f(y) $	(b) <i>metric-preserving</i> if $ x - y = f(x) - f(y) $;
fong-spivak-seven-sketches_14 F00021	14	■ 1.000	$f(x + y) = f(x) + f(y)$	$f(x + y) = f(x) + f(y)$	(c) <i>addition-preserving</i> if $f(x + y) = f(x) + f(y)$.
fong-spivak-seven-sketches_14 F00023	14	■ 0.766	x, y	x, y	¹ We are often taught to view functions $f: \mathbb{R} \rightarrow \mathbb{R}$ as plots on an (x, y) -axis, where x is the domain
fong-spivak-seven-sketches_14 F00024	14	■ 1.000	y	y	(independent) variable and y is the codomain (dependent) variable. In this book, we do not adhere to

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fong-spivak-seven-sketches_14 F00025	14	■ 1.000	$f(x)=x^{\wedge}\{2\}$	$f(x) = x^2$	example consider the function $f(x) = x^2$. Then f being order-preserving would say that for any $x, y \in \mathbb{R}$,
fong-spivak-seven-sketches_14 F00026	14	■ 1.000	$x^{\wedge}\{2\} \leq y^{\wedge}\{2\}$	$x^2 \leq y^2$	if $x \leq y$ then $x^2 \leq y^2$; is that true?
fong-spivak-seven-sketches_15 F00027	15	■ 1.000	c	c	tions are also transitive, meaning that if a is connected to b , and b is connected to c , then
fong-spivak-seven-sketches_15 F00028	15	■ 1.000	a, b	a, b	a is connected to c ; that is, all a, b , and c are connected. Friendship is not transitive—my
fong-spivak-seven-sketches_15 F00029	15	■ 1.000	Φ	Φ	is observing this system. Her observation of interest, which we call Φ , extracts a single
fong-spivak-seven-sketches_16 F00030	16	■ 1.000	A	A	Joining our simple systems. Joining two systems A and B is performed simply by
fong-spivak-seven-sketches_16 F00031	16	■ 1.000	B	B	Joining our simple systems. Joining two systems A and B is performed simply by
fong-spivak-seven-sketches_16 F00032	16	■ 1.000	$A \vee B$	$A \vee B$	it $A \vee B$, has a connection between points x and y if there are some points $z_1, \dots, z_n$
fong-spivak-seven-sketches_16 F00033	16	■ 1.000	$z_{\{1\}}, \ldots, z_{\{n\}}$	$z_1, \dots, z_n$	it $A \vee B$, has a connection between points x and y if there are some points $z_1, \dots, z_n$
fong-spivak-seven-sketches_16 F00034	16	■ 0.562	$z_{\{1\}}$	z_1	such that each of the following are true in at least one of A or B : x is connected to z_1 ,
fong-spivak-seven-sketches_16 F00035	16	■ 1.000	$z_{\{i\}}$	z_i	z_i is connected to z_{i+1} , and z_n is connected to y . In a three-point system, the above
fong-spivak-seven-sketches_16 F00036	16	■ 1.000	$z_{\{i+1\}}$	z_{i+1}	z_i is connected to z_{i+1} , and z_n is connected to y . In a three-point system, the above
fong-spivak-seven-sketches_16 F00037	16	■ 1.000	$z_{\{n\}}$	z_n	z_i is connected to z_{i+1} , and z_n is connected to y . In a three-point system, the above
fong-spivak-seven-sketches_16 F00038	16	■ 0.506	$\Phi\left(\odot^{\odot}\right)=\Phi\left(\{ \}^{\odot}\right)=\mathscr{C}\right)=$	$\Phi\left(\odot^{\odot}\right)=\Phi\left(\odot\mathscr{C}\right)=$	We see above in Eq. (1.2) that $\Phi(\mathfrak{V}) = \Phi(\mathfrak{V}) = \text{false}$: in both cases $\bullet$ is not connected
fong-spivak-seven-sketches_16 F00039	16	■ 0.322	$\Phi\left(\odot^{\odot}\vee\right)=\Phi\left(\odot\sigma\right)=$	$\Phi\left(\odot^{\odot}\vee\right)=\Phi\left(\cdot\sigma\right)=$	$\Phi(\mathfrak{V} \vee \mathfrak{V}) = \Phi(\mathfrak{V}) = \text{true}$: in the joined system, $\bullet$ is connected to $*$. The question that
fong-spivak-seven-sketches_16 F00040	16	■ 0.322	$\bullet$	$\bullet$	$\Phi(\mathfrak{V} \vee \mathfrak{V}) = \Phi(\mathfrak{V}) = \text{true}$: in the joined system, $\bullet$ is connected to $*$. The question that
fong-spivak-seven-sketches_17 F00041	17	■ 1.000	$A \leq B$	$A \leq B$	systems A and B , we say that $A \leq B$ if, whenever x is connected to y in A , then x is
fong-spivak-seven-sketches_17 F00042	17	■ 1.000	$\leq$	$\leq$	This notion of $\leq$ leads to the following diagram:
fong-spivak-seven-sketches_17 F00043	17	■ 1.000	$A \leq(A \vee B)$	$A \leq(A \vee B)$	smallest system that is bigger than both A and B . That is, $A \leq(A \vee B)$ and $B \leq(A \vee B)$,
fong-spivak-seven-sketches_17 F00044	17	■ 1.000	$B \leq(A \vee B)$	$B \leq(A \vee B)$	smallest system that is bigger than both A and B . That is, $A \leq(A \vee B)$ and $B \leq(A \vee B)$,
fong-spivak-seven-sketches_17 F00045	17	■ 0.999	C	C	and for any C , if $A \leq C$ and $B \leq C$ then $(A \vee B) \leq C$. Let’s walk through this with an
fong-spivak-seven-sketches_17 F00046	17	■ 0.999	$A \leq C$	$A \leq C$	and for any C , if $A \leq C$ and $B \leq C$ then $(A \vee B) \leq C$. Let’s walk through this with an
fong-spivak-seven-sketches_17 F00047	17	■ 0.999	$B \leq C$	$B \leq C$	and for any C , if $A \leq C$ and $B \leq C$ then $(A \vee B) \leq C$. Let’s walk through this with an
fong-spivak-seven-sketches_17 F00048	17	■ 0.999	$(A \vee B) \leq C$	$(A \vee B) \leq C$	and for any C , if $A \leq C$ and $B \leq C$ then $(A \vee B) \leq C$. Let’s walk through this with an
fong-spivak-seven-sketches_17 F00049	17	■ 0.993	$\{\bullet, *\}$	$\{\bullet, *\}$	1. Write down all the partitions of a two element set $\{\bullet, *\}$, order them as above,
fong-spivak-seven-sketches_18 F00050	18	■ 1.000	$\mathbb{B}=\{$	$\mathbb{B} = \{$	The set $\mathbb{B} = \{\text{true}, \text{false}\}$ of booleans also has an order, $\text{false} \leq \text{true}$:
fong-spivak-seven-sketches_18 F00051	18	■ 1.000	$\}$	$\}$	The set $\mathbb{B} = \{\text{true}, \text{false}\}$ of booleans also has an order, $\text{false} \leq \text{true}$:
fong-spivak-seven-sketches_18 F00052	18	■ 1.000	$\not \leq$	$\not \leq$	Thus $\text{false} \leq \text{false}$, $\text{false} \leq \text{true}$, and $\text{true} \leq \text{true}$, but $\text{true} \not \leq \text{false}$. In other

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fong-spivak-seven-sketches_18 F00053	18	0.637	$B.^{2}$	$B.^2$	words, $A \leq B$ if A implies B . ²
fong-spivak-seven-sketches_18 F00054	18	1.000	A, B	A, B	For any A, B in $\mathbb{B}$, we can again write $A \vee B$ to mean the least element that is greater
fong-spivak-seven-sketches_18 F00055	18	1.000	$\mathbb{B}$	$\mathbb{B}$	For any A, B in $\mathbb{B}$, we can again write $A \vee B$ to mean the least element that is greater
fong-spivak-seven-sketches_18 F00056	18	0.760	$\vee$	$\vee$	2. false $\vee$ true ?
fong-spivak-seven-sketches_18 F00057	18	1.000	$\Phi(A)=\Phi(B)=$	$\Phi(A) = \Phi(B) =$	with $\Phi(A) = \Phi(B) = \text{false}$, so $\Phi(A) \vee \Phi(B) = \text{false}$, but where $\Phi(A \vee B) = \text{true}$. In
fong-spivak-seven-sketches_18 F00058	18	1.000	$\Phi(A) \vee \Phi(B)=$	$\Phi(A) \vee \Phi(B) =$	with $\Phi(A) = \Phi(B) = \text{false}$, so $\Phi(A) \vee \Phi(B) = \text{false}$, but where $\Phi(A \vee B) = \text{true}$. In
fong-spivak-seven-sketches_18 F00059	18	1.000	$\Phi(A \vee B)=$	$\Phi(A \vee B) =$	with $\Phi(A) = \Phi(B) = \text{false}$, so $\Phi(A) \vee \Phi(B) = \text{false}$, but where $\Phi(A \vee B) = \text{true}$. In
fong-spivak-seven-sketches_18 F00060	18	1.000	P	P	² In mathematical logic, false implies true but true does not imply false. That is “ P implies Q ”
fong-spivak-seven-sketches_18 F00061	18	1.000	Q	Q	² In mathematical logic, false implies true but true does not imply false. That is “ P implies Q ”
fong-spivak-seven-sketches_19 F00062	19	1.000	a, b, c	a, b, c	certain tree, or the names of your favorite fruits, or simply some symbols a, b, c . For
fong-spivak-seven-sketches_19 F00063	19	1.000	$A=\{h, 1\}$	$A = \{h, 1\}$	example, we write $A = \{h, 1\}$ to denote the set, called A , that contains exactly two
fong-spivak-seven-sketches_19 F00064	19	0.911	h	h	elements, one called h and one called 1. The set $\{h, h, 1, h, 1\}$ is exactly the same as A
fong-spivak-seven-sketches_19 F00065	19	0.911	$\{h, h, 1, h, 1\}$	$\{h, h, 1, h, 1\}$	elements, one called h and one called 1. The set $\{h, h, 1, h, 1\}$ is exactly the same as A
fong-spivak-seven-sketches_19 F00066	19	1.000	$x \in X$	$x \in X$	than once in the notation doesn’t change the set. ³ For an arbitrary set X , we write $x \in X$
fong-spivak-seven-sketches_19 F00067	19	1.000	$h \in A$	$h \in A$	if x is element of X ; so we have $h \in A$ and $1 \in A$, but $0 \notin A$.
fong-spivak-seven-sketches_19 F00068	19	1.000	$1 \in A$	$1 \in A$	if x is element of X ; so we have $h \in A$ and $1 \in A$, but $0 \notin A$.
fong-spivak-seven-sketches_19 F00069	19	1.000	$0 \notin A$	$0 \notin A$	if x is element of X ; so we have $h \in A$ and $1 \in A$, but $0 \notin A$.
fong-spivak-seven-sketches_19 F00070	19	1.000	$\varnothing$	$\emptyset$	• $\emptyset$ denotes the empty set; it has no elements.
fong-spivak-seven-sketches_19 F00071	19	0.999	$\mathbb{N}$	$\mathbb{N}$	• $\mathbb{N}$ denotes the set of <i>natural numbers</i> ; it has elements $0, 1, 2, 3, \dots, 90^{717}, \dots$
fong-spivak-seven-sketches_19 F00072	19	0.999	$0, 1, 2, 3, \ldots, 90^{717}, \ldots$	$0, 1, 2, 3, \dots, 90^{717}, \dots$	• $\mathbb{N}$ denotes the set of <i>natural numbers</i> ; it has elements $0, 1, 2, 3, \dots, 90^{717}, \dots$.
fong-spivak-seven-sketches_19 F00073	19	0.992	$\underline{n}$	$\underline{n}$	• $\underline{n}$, for any $n \in \mathbb{N}$, denotes the n^{th} ordinal; it has n elements $1, 2, \dots, n$. For example,
fong-spivak-seven-sketches_19 F00074	19	0.992	$n \in \mathbb{N}$	$n \in \mathbb{N}$	• $\underline{n}$, for any $n \in \mathbb{N}$, denotes the n^{th} ordinal; it has n elements $1, 2, \dots, n$. For example,
fong-spivak-seven-sketches_19 F00075	19	0.992	n^{th}	n^{th}	• $\underline{n}$, for any $n \in \mathbb{N}$, denotes the n^{th} ordinal; it has n elements $1, 2, \dots, n$. For example,
fong-spivak-seven-sketches_19 F00076	19	0.992	n	n	• $\underline{n}$, for any $n \in \mathbb{N}$, denotes the n^{th} ordinal; it has n elements $1, 2, \dots, n$. For example,
fong-spivak-seven-sketches_19 F00077	19	0.992	$1, 2, \ldots, n$	$1, 2, \dots, n$	• $\underline{n}$, for any $n \in \mathbb{N}$, denotes the n^{th} ordinal; it has n elements $1, 2, \dots, n$. For example,
fong-spivak-seven-sketches_19 F00078	19	1.000	$\underline{0}=\varnothing, \underline{1}=\{1\}$	$\underline{0} = \emptyset, \underline{1} = \{1\}$	$\underline{0} = \emptyset, \underline{1} = \{1\}$, and $\underline{5} = \{1, 2, 3, 4, 5\}$.
fong-spivak-seven-sketches_19 F00079	19	1.000	$\underline{5}=\{1, 2, 3, 4, 5\}$	$\underline{5} = \{1, 2, 3, 4, 5\}$	$\underline{0} = \emptyset, \underline{1} = \{1\}$, and $\underline{5} = \{1, 2, 3, 4, 5\}$.

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fong-spivak-seven-sketches_F00080	19	0.152	$\mathbb{Z}$	$\mathbb{Z}$	• $\mathbb{Z}$, the set of <i>integers</i> ; it has elements $\dots, -2, -1, 0, 1, 2, \dots, 90^{717}, \dots$
fong-spivak-seven-sketches_F00081	19	0.152	$90^{717}, \dots$	$90^{717}, \dots$	• $\mathbb{Z}$, the set of <i>integers</i> ; it has elements $\dots, -2, -1, 0, 1, 2, \dots, 90^{717}, \dots$
fong-spivak-seven-sketches_F00082	19	1.000	$\pi, 3.14, 5 * \sqrt{2}, e, e^2, -1457, 90^{717}$	$\pi, 3.14, 5 * \sqrt{2}, e, e^2, -1457, 90^{717}$	• $\mathbb{R}$, the set of <i>real numbers</i> ; it has elements like $\pi, 3.14, 5 * \sqrt{2}, e, e^2, -1457, 90^{717}$, etc.
fong-spivak-seven-sketches_F00083	19	1.000	$X \subseteq Y$	$X \subseteq Y$	Given sets X and Y , we say that X is a <i>subset</i> of Y , and write $X \subseteq Y$, if every element
fong-spivak-seven-sketches_F00084	19	1.000	$\{h\} \subseteq A$	$\{h\} \subseteq A$	in X is also in Y . For example $\{h\} \subseteq A$. Note that the empty set $\emptyset := \{\}$ is a subset
fong-spivak-seven-sketches_F00085	19	1.000	$\varnothing := \{\}$	$\emptyset := \{\}$	in X is also in Y . For example $\{h\} \subseteq A$. Note that the empty set $\emptyset := \{\}$ is a subset
fong-spivak-seven-sketches_F00086	19	0.808	$\{y \in Y \mid P(y)\}$	$\{y \in Y \mid P(y)\}$	element of Y , we write $\{y \in Y \mid P(y)\}$ to mean the subset of those y 's that satisfy P .
fong-spivak-seven-sketches_F00087	19	0.808	y'	y'	element of Y , we write $\{y \in Y \mid P(y)\}$ to mean the subset of those y 's that satisfy P .
fong-spivak-seven-sketches_F00088	19	1.000	$\mathbb{N} = \{n \in \mathbb{Z} \mid n \geq 0\}$	$\mathbb{N} = \{n \in \mathbb{Z} \mid n \geq 0\}$	1. Is it true that $\mathbb{N} = \{n \in \mathbb{Z} \mid n \geq 0\}$?
fong-spivak-seven-sketches_F00089	19	1.000	$\mathbb{N} = \{n \in \mathbb{Z} \mid n \geq 1\}$	$\mathbb{N} = \{n \in \mathbb{Z} \mid n \geq 1\}$	2. Is it true that $\mathbb{N} = \{n \in \mathbb{Z} \mid n \geq 1\}$?
fong-spivak-seven-sketches_F00090	19	1.000	$\varnothing = \{n \in \mathbb{Z} \mid 1 < n < 2\}$	$\emptyset = \{n \in \mathbb{Z} \mid 1 < n < 2\}$	3. Is it true that $\emptyset = \{n \in \mathbb{Z} \mid 1 < n < 2\}$?
fong-spivak-seven-sketches_F00091	19	1.000	X_1	X_1	If both X_1 and X_2 are subsets of Y , their <i>union</i> , denoted $X_1 \cup X_2$, is also a subset of
fong-spivak-seven-sketches_F00092	19	1.000	X_2	X_2	If both X_1 and X_2 are subsets of Y , their <i>union</i> , denoted $X_1 \cup X_2$, is also a subset of
fong-spivak-seven-sketches_F00093	19	1.000	$X_1 \cup X_2$	$X_1 \cup X_2$	If both X_1 and X_2 are subsets of Y , their <i>union</i> , denoted $X_1 \cup X_2$, is also a subset of
fong-spivak-seven-sketches_F00094	19	0.988	$h, 1$	$h, 1$	³ If you want a notion where “ $h, 1$ ” is different than “ $h, h, 1, h, 1$ ”, you can use something called <i>bags</i> ,
fong-spivak-seven-sketches_F00095	19	0.988	$h, h, 1, h, 1$	$h, h, 1, h, 1$	³ If you want a notion where “ $h, 1$ ” is different than “ $\overline{h, h, 1, h, 1}$ ”, you can use something called <i>bags</i> ,
fong-spivak-seven-sketches_F00096	19	0.829	$Z :=$	$Z :=$	⁴ When we write $Z := \text{foo}$, it means “assign the meaning <i>foo</i> to variable Z ”, whereas $Z = \text{foo}$ means
fong-spivak-seven-sketches_F00097	19	1.000	Z	Z	simply that Z is equal to <i>foo</i> , perhaps as discovered via some calculation. In particular, $Z := \text{foo}$ implies
fong-spivak-seven-sketches_F00098	19	1.000	$Z =$	$Z =$	$Z = \text{foo}$ but not vice versa; indeed it <i>would not</i> be proper to write $3 + 2 := 5$ or $\{\} := \emptyset$.
fong-spivak-seven-sketches_F00099	19	1.000	$3 + 2 := 5$	$3 + 2 := 5$	$Z = \text{foo}$ but not vice versa; indeed it <i>would not</i> be proper to write $3 + 2 := 5$ or $\{\} := \emptyset$.
fong-spivak-seven-sketches_F00100	19	1.000	$\} := \varnothing$	$\} := \emptyset$	$Z = \text{foo}$ but not vice versa; indeed it <i>would not</i> be proper to write $3 + 2 := 5$ or $\{\} := \emptyset$.
fong-spivak-seven-sketches_F00101	20	1.000	$Y = \{1, 2, 3, 4\}$	$Y = \{1, 2, 3, 4\}$	For example if $Y = \{1, 2, 3, 4\}$ and $X_1 = \{1, 2\}$ and $X_2 = \{2, 4\}$, then $X_1 \cup X_2 = \{1, 2, 4\}$.
fong-spivak-seven-sketches_F00102	20	1.000	$X_1 = \{1, 2\}$	$X_1 = \{1, 2\}$	For example if $Y = \{1, 2, 3, 4\}$ and $X_1 = \{1, 2\}$ and $X_2 = \{2, 4\}$, then $X_1 \cup X_2 = \{1, 2, 4\}$.
fong-spivak-seven-sketches_F00103	20	1.000	$X_2 = \{2, 4\}$	$X_2 = \{2, 4\}$	For example if $Y = \{1, 2, 3, 4\}$ and $X_1 = \{1, 2\}$ and $X_2 = \{2, 4\}$, then $X_1 \cup X_2 = \{1, 2, 4\}$.
fong-spivak-seven-sketches_F00104	20	1.000	$X_1 \cup X_2 = \{1, 2, 4\}$	$X_1 \cup X_2 = \{1, 2, 4\}$	For example if $Y = \{1, 2, 3, 4\}$ and $X_1 = \{1, 2\}$ and $X_2 = \{2, 4\}$, then $X_1 \cup X_2 = \{1, 2, 4\}$.
fong-spivak-seven-sketches_F00105	20	1.000	$\varnothing \cup X = X$	$\emptyset \cup X = X$	Note that $\emptyset \cup X = X$ for any $X \subseteq Y$.
fong-spivak-seven-sketches_F00106	20	0.997	$X_1 \cap X_2$	$X_1 \cap X_2$	Similarly, if both X_1 and X_2 are subsets of Y , then their <i>intersection</i> , denoted $X_1 \cap X_2$,
fong-spivak-seven-sketches_F00107	20	1.000	$\{1, 2, 3\} \cap \{2, 5\} = \{2\}$	$\{1, 2, 3\} \cap \{2, 5\} = \{2\}$	X_1 and in X_2 , and no others. So $\{1, 2, 3\} \cap \{2, 5\} = \{2\}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_20 F00108	20	■ 1.000	$X_{\{0\}}=\varnothing, X_{\{1\}}=\{1\}, X_{\{2\}}=\{1,2\}$	$X_0 = \varnothing, X_1 = \{1\}, X_2 = \{1, 2\}$	sets $X_0 = \varnothing, X_1 = \{1\}, X_2 = \{1, 2\}$, etc. as subsets of $\mathbb{N}$, and we want to know what the
fong-spivak-seven-sketches_20 F00109	20	■ 0.984	$\bigcup_{n \in \mathbb{N}} X_n$	$\bigcup_{n \in \mathbb{N}} X_n$	union of all of them is. This union is written $\bigcup_{n \in \mathbb{N}} X_n$, and it is the subset of $\mathbb{N}$ that
fong-spivak-seven-sketches_20 F00110	20	■ 1.000	$X_{\{n\}}$	X_n	contains every element of every X_n , but no others. Namely, $\bigcup_{n \in \mathbb{N}} X_n = \{n \in \mathbb{N} \mid n \geq 1\}$.
fong-spivak-seven-sketches_20 F00111	20	■ 1.000	$\bigcup_{n \in \mathbb{N}} X_n = \{n \in \mathbb{N} \mid n \geq 1\}$	$\bigcup_{n \in \mathbb{N}} X_n = \{n \in \mathbb{N} \mid n \geq 1\}$	contains every element of every X_n , but no others. Namely, $\bigcup_{n \in \mathbb{N}} X_n = \{n \in \mathbb{N} \mid n \geq 1\}$.
fong-spivak-seven-sketches_20 F00112	20	■ 1.000	$\bigcap_{n \in \mathbb{N}} X_n$	$\bigcap_{n \in \mathbb{N}} X_n$	Similarly one can write $\bigcap_{n \in \mathbb{N}} X_n$ for the intersection of all of them, which will be empty
fong-spivak-seven-sketches_20 F00113	20	■ 0.715	$X \times Y$	$X \times Y$	Given two sets X and Y , the <i>product</i> $X \times Y$ of X and Y is the set of pairs (x, y) , where
fong-spivak-seven-sketches_20 F00114	20	■ 1.000	$y \in Y$	$y \in Y$	$x \in X$ and $y \in Y$.
fong-spivak-seven-sketches_20 F00115	20	■ 1.000	$X \sqcup Y$	$X \sqcup Y$	in common. Given two sets X and Y , their <i>disjoint union</i> $X \sqcup Y$ is the set of pairs of the
fong-spivak-seven-sketches_20 F00116	20	■ 0.994	$(x, 1)$	$(x, 1)$	form $(x, 1)$ or $(y, 2)$, where $x \in X$ and $y \in Y$.
fong-spivak-seven-sketches_20 F00117	20	■ 0.994	$(y, 2)$	$(y, 2)$	form $(x, 1)$ or $(y, 2)$, where $x \in X$ and $y \in Y$.
fong-spivak-seven-sketches_20 F00118	20	■ 1.000	$A := \{h, 1\}$	$A := \{h, 1\}$	<i>Exercise 1.11.</i> Let $A := \{h, 1\}$ and $B := \{1, 2, 3\}$.
fong-spivak-seven-sketches_20 F00119	20	■ 1.000	$B := \{1, 2, 3\}$	$B := \{1, 2, 3\}$	<i>Exercise 1.11.</i> Let $A := \{h, 1\}$ and $B := \{1, 2, 3\}$.
fong-spivak-seven-sketches_20 F00120	20	■ 1.000	$A \times B$	$A \times B$	3. There are six elements in $A \times B$; write them out.
fong-spivak-seven-sketches_20 F00121	20	■ 1.000	$A \sqcup B$	$A \sqcup B$	4. There are five elements of $A \sqcup B$; write them out.
fong-spivak-seven-sketches_20 F00122	20	■ 1.000	$A \cup B$	$A \cup B$	$A \cup B$; write them out.
fong-spivak-seven-sketches_20 F00123	20	■ 0.999	$R \subseteq X \times Y$	$R \subseteq X \times Y$	Definition 1.12. Let X and Y be sets. A <i>relation between X and Y</i> is a subset $R \subseteq X \times Y$.
fong-spivak-seven-sketches_20 F00124	20	■ 0.973	$R \subseteq X \times X$	$R \subseteq X \times X$	A <i>binary relation on X</i> is a relation between X and X , i.e. a subset $R \subseteq X \times X$.
fong-spivak-seven-sketches_20 F00125	20	■ 1.000	$R \subseteq A \times A$	$R \subseteq A \times A$	It is convenient to use something called <i>infix notation</i> for binary relations $R \subseteq A \times A$.
fong-spivak-seven-sketches_20 F00126	20	■ 0.651	$a \star b$	$a \star b$	This means one picks a symbol, say $\star$, and writes $a \star b$ to mean $(a, b) \in R$.
fong-spivak-seven-sketches_20 F00127	20	■ 0.651	$(a, b) \in R$	$(a, b) \in R$	This means one picks a symbol, say $\star$, and writes $a \star b$ to mean $(a, b) \in R$.
fong-spivak-seven-sketches_20 F00128	20	■ 1.000	$(5, 6) \in R$	$(5, 6) \in R$	$(5, 6) \in R$, we write $5 \leq 6$.
fong-spivak-seven-sketches_20 F00129	20	■ 1.000	$5 \leq 6$	$5 \leq 6$	$(5, 6) \in R$, we write $5 \leq 6$.
fong-spivak-seven-sketches_20 F00130	20	■ 0.975	$=, \approx, <, >$	$=, \approx, <, >$	Other examples of infix notation for relations are $=, \approx, <, >$. In number theory, they
fong-spivak-seven-sketches_21 F00131	21	■ 1.000	$p \in P$	$p \in P$	Definition 1.14. If A is a set, a <i>partition</i> of A consists of a set P and, for each $p \in P$, a
fong-spivak-seven-sketches_21 F00132	21	■ 1.000	$A_{\{p\}} \subseteq A$	$A_p \subseteq A$	nonempty subset $A_p \subseteq A$, such that
fong-spivak-seven-sketches_21 F00133	21	■ 1.000	$\{A_{\{p\}}\}_{p \in P}$	$\{A_p\}_{p \in P}$	We may denote the partition by $\{A_p\}_{p \in P}$. We refer to P as the set of <i>part labels</i> and if
fong-spivak-seven-sketches_21 F00134	21	■ 1.000	$A_{\{p\}}$	A_p	$p \in P$ is a part label, we refer to A_p as the p^{th} <i>part</i> . The condition (1.15) says that each

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_1 F00135	21	■ 1.000	p^{th}	p^{th}	$p \in P$ is a part label, we refer to A_p as the p^{th} part. The condition (1.15) says that each
fong-spivak-seven-sketches_1 F00136	21	■ 1.000	$a \in A$	$a \in A$	element $a \in A$ is in exactly one part.
fong-spivak-seven-sketches_1 F00137	21	■ 1.000	$\left\{A_{p^{\prime}}^{\prime}\right\}_{p^{\prime} \in P^{\prime}}$	$\left\{A_{p^{\prime}}^{\prime}\right\}_{p^{\prime} \in P^{\prime}}$	We consider two different partitions $\left\{A_p\right\}_{p \in P}$ and $\left\{A_{p^{\prime}}^{\prime}\right\}_{p^{\prime} \in P^{\prime}}$ of A to be the same if for
fong-spivak-seven-sketches_1 F00138	21	■ 1.000	$p^{\prime} \in P^{\prime}$	$p^{\prime} \in P^{\prime}$	each $p \in P$ there exists a $p^{\prime} \in P^{\prime}$ with $A_p=A_{p^{\prime}}^{\prime}$. In other words, if two ways to divide A
fong-spivak-seven-sketches_1 F00139	21	■ 1.000	$A_p=A_{p^{\prime}}^{\prime}$	$A_p=A_{p^{\prime}}^{\prime}$	each $p \in P$ there exists a $p^{\prime} \in P^{\prime}$ with $A_p=A_{p^{\prime}}^{\prime}$. In other words, if two ways to divide A
fong-spivak-seven-sketches_1 F00140	21	■ 1.000	$a, b \in\{11,12,13,21,22,23\}$	$a, b \in\{11,12,13,21,22,23\}$	For any two elements $a, b \in\{11,12,13,21,22,23\}$, let's allow ourselves to write a
fong-spivak-seven-sketches_1 F00141	21	■ 1.000	$a \sim b$	$a \sim b$	twiddle symbol $a \sim b$ between them if a and b are both in the same part. Write down
fong-spivak-seven-sketches_1 F00142	21	■ 0.560	(a, b)	(a, b)	every pair of elements (a, b) that are in the same part. There should be 10. ⁵
fong-spivak-seven-sketches_1 F00143	21	■ 0.992	$\sim$	$\sim$	give it infix notation $\sim$, satisfying the following three properties:
fong-spivak-seven-sketches_1 F00144	21	■ 1.000	$a \sim a$	$a \sim a$	(a) $a \sim a$, for all $a \in A$,
fong-spivak-seven-sketches_1 F00145	21	■ 0.960	$\{ \}^a b \sim a$	$^a b \sim a$	(b) $a \sim b$ iff ^a $b \sim a$, for all $a, b \in A$, and
fong-spivak-seven-sketches_1 F00146	21	■ 0.960	$a, b \in A$	$a, b \in A$	(b) $a \sim b$ iff ^a $b \sim a$, for all $a, b \in A$, and
fong-spivak-seven-sketches_1 F00147	21	■ 1.000	$b \sim c$	$b \sim c$	(c) if $a \sim b$ and $b \sim c$ then $a \sim c$, for all $a, b, c \in A$.
fong-spivak-seven-sketches_1 F00148	21	■ 1.000	$a \sim c$	$a \sim c$	(c) if $a \sim b$ and $b \sim c$ then $a \sim c$, for all $a, b, c \in A$.
fong-spivak-seven-sketches_1 F00149	21	■ 1.000	$a, b, c \in A$	$a, b, c \in A$	(c) if $a \sim b$ and $b \sim c$ then $a \sim c$, for all $a, b, c \in A$.
fong-spivak-seven-sketches_1 F00150	21	■ 1.000	$\{ \}^a$	a	^a “Iff” is short for “if and only if”.
fong-spivak-seven-sketches_1 F00151	22	■ 1.000	$a \in A_p$	$a \in A_p$	some $p \in P$ such that $a \in A_p$ and $b \in A_p$. It is obvious that a is in the same part as
fong-spivak-seven-sketches_1 F00152	22	■ 1.000	$b \in A_p$	$b \in A_p$	some $p \in P$ such that $a \in A_p$ and $b \in A_p$. It is obvious that a is in the same part as
fong-spivak-seven-sketches_1 F00153	22	■ 0.976	$X \subseteq A$	$X \subseteq A$	what the parts are. Say that a subset $X \subseteq A$ is $(\sim)$ -closed if, for every $x \in X$ and
fong-spivak-seven-sketches_1 F00154	22	■ 0.884	$x^{\prime} \sim x$	$x^{\prime} \sim x$	$x^{\prime} \sim x$, we have $x^{\prime} \in X$. Say that a subset $X \subseteq A$ is $(\sim)$ -connected if it is nonempty and
fong-spivak-seven-sketches_1 F00155	22	■ 0.884	$x^{\prime} \in X$	$x^{\prime} \in X$	$x^{\prime} \sim x$, we have $x^{\prime} \in X$. Say that a subset $X \subseteq A$ is $(\sim)$ -connected if it is nonempty and
fong-spivak-seven-sketches_1 F00156	22	■ 0.884	$(\sim)$	$(\sim)$	$x^{\prime} \sim x$, we have $x^{\prime} \in X$. Say that a subset $X \subseteq A$ is $(\sim)$ -connected if it is nonempty and
fong-spivak-seven-sketches_1 F00157	22	■ 0.899	$x \sim y$	$x \sim y$	$x \sim y$ for every $x, y \in X$. Then the parts corresponding to $\sim$ are exactly the $(\sim)$ -closed,
fong-spivak-seven-sketches_1 F00158	22	■ 0.899	$x, y \in X$	$x, y \in X$	$x \sim y$ for every $x, y \in X$. Then the parts corresponding to $\sim$ are exactly the $(\sim)$ -closed,
fong-spivak-seven-sketches_1 F00159	22	■ 1.000	$p \neq q$	$p \neq q$	2. Show that if $p \neq q$, i.e. if A_p and A_q are not exactly the same set, then $A_p \cap A_q=\emptyset$.
fong-spivak-seven-sketches_1 F00160	22	■ 1.000	A_q	A_q	2. Show that if $p \neq q$, i.e. if A_p and A_q are not exactly the same set, then $A_p \cap A_q=\emptyset$.
fong-spivak-seven-sketches_1 F00161	22	■ 1.000	$A_p \cap A_q=\varnothing$	$A_p \cap A_q=\emptyset$	2. Show that if $p \neq q$, i.e. if A_p and A_q are not exactly the same set, then $A_p \cap A_q=\emptyset$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_1 F00162	22	0.562	$A=\bigcup_{p\in P}A_p$	$A=\bigcup_{p\in P}A_p$	3. Show that $A=\bigcup_{p\in P}A_p$.
fong-spivak-seven-sketches_1 F00163	22	1.000	$A/\sim$	$A/\sim$	<i>quotient</i> $A/\sim$ of A under $\sim$ is the set of parts of the corresponding partition.
fong-spivak-seven-sketches_1 F00164	22	1.000	S	S	Definition 1.22. Let S and T be sets. A <i>function from S to T</i> is a subset $F\subseteq S\times T$ such
fong-spivak-seven-sketches_1 F00165	22	1.000	T	T	Definition 1.22. Let S and T be sets. A <i>function from S to T</i> is a subset $F\subseteq S\times T$ such
fong-spivak-seven-sketches_1 F00166	22	1.000	$F\subseteq S\times T$	$F\subseteq S\times T$	Definition 1.22. Let S and T be sets. A <i>function from S to T</i> is a subset $F\subseteq S\times T$ such
fong-spivak-seven-sketches_1 F00167	22	1.000	$s\in S$	$s\in S$	that for all $s\in S$ there exists a unique $t\in T$ with $(s,t)\in F$.
fong-spivak-seven-sketches_1 F00168	22	1.000	$t\in T$	$t\in T$	that for all $s\in S$ there exists a unique $t\in T$ with $(s,t)\in F$.
fong-spivak-seven-sketches_1 F00169	22	1.000	$(s,t)\in F$	$(s,t)\in F$	that for all $s\in S$ there exists a unique $t\in T$ with $(s,t)\in F$.
fong-spivak-seven-sketches_1 F00170	22	1.000	F	F	The function F is often denoted $F:S\rightarrow T$. From now on, we write $F(s)=t$, or
fong-spivak-seven-sketches_1 F00171	22	1.000	$F:S\rightarrow T$	$F:S\rightarrow T$	The function F is often denoted $F:S\rightarrow T$. From now on, we write $F(s)=t$, or
fong-spivak-seven-sketches_1 F00172	22	1.000	$F(s)=t$	$F(s)=t$	The function F is often denoted $F:S\rightarrow T$. From now on, we write $F(s)=t$, or
fong-spivak-seven-sketches_1 F00173	22	1.000	$s\mapsto t$	$s\mapsto t$	sometimes $s\mapsto t$, to mean $(s,t)\in F$. For any $t\in T$, the <i>preimage of t along F</i> is the
fong-spivak-seven-sketches_1 F00174	22	1.000	t	t	sometimes $s\mapsto t$, to mean $(s,t)\in F$. For any $t\in T$, the <i>preimage of t along F</i> is the
fong-spivak-seven-sketches_1 F00175	22	1.000	$\{s\in S\mid F(s)=t\}$	$\{s\in S\mid F(s)=t\}$	subset $\{s\in S\mid F(s)=t\}$.
fong-spivak-seven-sketches_1 F00176	22	1.000	$s_1,s_2\in S$	$s_1,s_2\in S$	$F(s)=t$. A function is called <i>injective</i> , or an <i>injection</i> , if for all $t\in T$ and $s_1,s_2\in S$
fong-spivak-seven-sketches_1 F00177	22	1.000	$F(s_1)=t$	$F(s_1)=t$	with $F(s_1)=t$ and $F(s_2)=t$, we have $s_1=s_2$. A function is called <i>bijective</i> if it is both
fong-spivak-seven-sketches_1 F00178	22	1.000	$F(s_2)=t$	$F(s_2)=t$	with $F(s_1)=t$ and $F(s_2)=t$, we have $s_1=s_2$. A function is called <i>bijective</i> if it is both
fong-spivak-seven-sketches_1 F00179	22	1.000	$s_1=s_2$	$s_1=s_2$	with $F(s_1)=t$ and $F(s_2)=t$, we have $s_1=s_2$. A function is called <i>bijective</i> if it is both
fong-spivak-seven-sketches_1 F00180	23	0.413	$\rightarrow,\rightarrow>,\stackrel{\cong}{\rightarrow}$	$\rightarrow,\rightarrow>,\stackrel{\cong}{\rightarrow}$	We use various decorations on arrows, $\rightarrow,\rightarrow>,\rightarrow\!\!\!\rightarrow,\stackrel{\cong}{\rightarrow}$ to denote these special sorts of
fong-spivak-seven-sketches_1 F00181	23	0.832	id_X	id_X	a set X , denoted id_X . It is the bijective function $\mathrm{id}_X(x)=x$.
fong-spivak-seven-sketches_1 F00182	23	0.832	$\operatorname{id}_X(x)=x$	$\mathrm{id}_X(x)=x$	a set X , denoted id_X . It is the bijective function $\mathrm{id}_X(x)=x$.
fong-spivak-seven-sketches_1 F00183	23	1.000	$f:A\rightarrow B$	$f:A\rightarrow B$	1. Find two sets A and B and a function $f:A\rightarrow B$ that is injective but not surjective.
fong-spivak-seven-sketches_1 F00184	23	1.000	$f:A\rightarrow\varnothing$	$f:A\rightarrow\varnothing$	<i>Exercise 1.25.</i> Suppose that A is a set and $f:A\rightarrow\varnothing$ is a function to the empty set.
fong-spivak-seven-sketches_1 F00185	23	1.000	$f:A\rightarrow P$	$f:A\rightarrow P$	functions out of A . Given a surjective function $f:A\rightarrow P$, where P is any other set, the
fong-spivak-seven-sketches_1 F00186	23	1.000	$f^{-1}(p)\subseteq A$	$f^{-1}(p)\subseteq A$	preimages $f^{-1}(p)\subseteq A$, one for each element $p\in P$, form a partition of A . Here is an
fong-spivak-seven-sketches_1 F00187	24	1.000	$S:=\{11,12,13,21,22,23\}$	$S:=\{11,12,13,21,22,23\}$	Consider the partition of $S:=\{11,12,13,21,22,23\}$ shown below:
fong-spivak-seven-sketches_1 F00188	24	1.000	$P=\{a,b,c,d\}$	$P=\{a,b,c,d\}$	It has been partitioned into four parts, so let $P=\{a,b,c,d\}$ and let $f:S\rightarrow P$ be given

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches- F00189	24	■ 1.000	f: S \rightarrow P	$f: S \rightarrow P$	It has been partitioned into four parts, so let $P = \{a, b, c, d\}$ and let $f: S \twoheadrightarrow P$ be given
fong-spivak-seven-sketches- F00190	24	■ 1.000	F: X \rightarrow Y	$F: X \rightarrow Y$	Definition 1.28. If $F: X \rightarrow Y$ is a function and $G: Y \rightarrow Z$ is a function, their <i>composite</i>
fong-spivak-seven-sketches- F00191	24	■ 1.000	G: Y \rightarrow Z	$G: Y \rightarrow Z$	Definition 1.28. If $F: X \rightarrow Y$ is a function and $G: Y \rightarrow Z$ is a function, their <i>composite</i>
fong-spivak-seven-sketches- F00192	24	■ 1.000	X \rightarrow Z	$X \rightarrow Z$	is the function $X \rightarrow Z$ defined to be $G(F(x))$ for any $x \in X$. It is often denoted $G \circ F$, but
fong-spivak-seven-sketches- F00193	24	■ 1.000	G(F(x))	$G(F(x))$	is the function $X \rightarrow Z$ defined to be $G(F(x))$ for any $x \in X$. It is often denoted $G \circ F$, but
fong-spivak-seven-sketches- F00194	24	■ 1.000	G \circ F	$G \circ F$	is the function $X \rightarrow Z$ defined to be $G(F(x))$ for any $x \in X$. It is often denoted $G \circ F$, but
fong-spivak-seven-sketches- F00195	24	■ 1.000	F ; G	$F; G$	we prefer to denote it $F \circ G$. It takes any element $x \in X$, evaluates F to get an element
fong-spivak-seven-sketches- F00196	24	■ 1.000	F(x) \in Y	$F(x) \in Y$	$F(x) \in Y$ and then evaluates G to get an element $G(F(x))$.
fong-spivak-seven-sketches- F00197	24	■ 1.000	G	G	$F(x) \in Y$ and then evaluates G to get an element $G(F(x))$.
fong-spivak-seven-sketches- F00198	24	■ 1.000	\{1\} \rightarrow X	$\{1\} \rightarrow X$	$\{1\} \rightarrow X$, namely the function sending 1 to x . For example, the three functions
fong-spivak-seven-sketches- F00199	24	■ 0.974	\{1\} \rightarrow \{1,2,3\}	$\{1\} \rightarrow \{1,2,3\}$	$\{1\} \rightarrow \{1,2,3\}$ shown below correspond to the three elements of $\{1,2,3\}$:
fong-spivak-seven-sketches- F00200	24	■ 0.974	\{1,2,3\}	$\{1,2,3\}$	$\{1\} \rightarrow \{1,2,3\}$ shown below correspond to the three elements of $\{1,2,3\}$:
fong-spivak-seven-sketches- F00201	24	■ 1.000	x:\{1\} \rightarrow X	$x: \{1\} \rightarrow X$	$x: \{1\} \rightarrow X$. Then evaluating F at x is given by the composite, $F(x) = x \circ F$.
fong-spivak-seven-sketches- F00202	24	■ 1.000	F(x)=x ; F	$F(x) = x; F$	$x: \{1\} \rightarrow X$. Then evaluating F at x is given by the composite, $F(x) = x \circ F$.
fong-spivak-seven-sketches- F00203	24	■ 1.000	x \leq x	$x \leq x$	(a) $x \leq x$; and
fong-spivak-seven-sketches- F00204	24	■ 1.000	y \leq z	$y \leq z$	(b) if $x \leq y$ and $y \leq z$, then $x \leq z$.
fong-spivak-seven-sketches- F00205	24	■ 1.000	x \leq z	$x \leq z$	(b) if $x \leq y$ and $y \leq z$, then $x \leq z$.
fong-spivak-seven-sketches- F00206	25	■ 0.608	y \leq x	$y \leq x$	$y \leq x$, we write $x \cong y$ and say x and y are <i>equivalent</i> . We call a pair $(X, \leq)$ consisting
fong-spivak-seven-sketches- F00207	25	■ 0.608	x \cong y	$x \cong y$	$y \leq x$, we write $x \cong y$ and say x and y are <i>equivalent</i> . We call a pair $(X, \leq)$ consisting
fong-spivak-seven-sketches- F00208	25	■ 0.608	(X, \leq)	$(X, \leq)$	$y \leq x$, we write $x \cong y$ and say x and y are <i>equivalent</i> . We call a pair $(X, \leq)$ consisting
fong-spivak-seven-sketches- F00209	25	■ 0.999	(X,=)	$(X, =)$	$(X, =)$. This means that the only order relations on X are of the form $x \leq x$; if $x \neq y$
fong-spivak-seven-sketches- F00210	25	■ 0.999	x \neq y	$x \neq y$	$(X, =)$. This means that the only order relations on X are of the form $x \leq x$; if $x \neq y$
fong-spivak-seven-sketches- F00211	25	■ 0.924	X, \leq	$X, \leq$	preorder $(X, \leq)$ by equipping it with the total binary relation $X \times X \subseteq X \times X$. This is a
fong-spivak-seven-sketches- F00212	25	■ 0.924	X \times X \subseteq X \times X	$X \times X \subseteq X \times X$	preorder $(X, \leq)$ by equipping it with the total binary relation $X \times X \subseteq X \times X$. This is a
fong-spivak-seven-sketches- F00213	25	■ 1.000	x=y	$x = y$	(c) $x \cong y$ implies $x = y$.
fong-spivak-seven-sketches- F00214	26	■ 1.000	G=(V, A, s, t)	$G = (V, A, s, t)$	Definition 1.36. A graph $G = (V, A, s, t)$ consists of a set V whose elements are called
fong-spivak-seven-sketches- F00215	26	■ 1.000	V	V	Definition 1.36. A graph $G = (V, A, s, t)$ consists of a set V whose elements are called

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_26 F00216	26	■ 0.999	$s, t: A \rightarrow V$	$s, t: A \rightarrow V$	<i>vertices</i> , a set A whose elements are called <i>arrows</i> , and two functions $s, t: A \rightarrow V$ known
fong-spivak-seven-sketches_26 F00217	26	■ 1.000	$s(a)=v$	$s(a) = v$	as the <i>source</i> and <i>target</i> functions respectively. Given $a \in A$ with $s(a) = v$ and $t(a) = w$,
fong-spivak-seven-sketches_26 F00218	26	■ 1.000	$t(a)=w$	$t(a) = w$	as the <i>source</i> and <i>target</i> functions respectively. Given $a \in A$ with $s(a) = v$ and $t(a) = w$,
fong-spivak-seven-sketches_26 F00219	26	■ 1.000	v	v	we say that a is an arrow from v to w .
fong-spivak-seven-sketches_26 F00220	26	■ 1.000	w	w	we say that a is an arrow from v to w .
fong-spivak-seven-sketches_26 F00221	26	■ 1.000	$V=\{1,2,3,4\}$	$V = \{1, 2, 3, 4\}$	It has $V = \{1, 2, 3, 4\}$ and $A = \{a, b, c, d, e\}$. The source and target functions, $s, t: A \rightarrow$
fong-spivak-seven-sketches_26 F00222	26	■ 1.000	$A=\{a, b, c, d, e\}$	$A = \{a, b, c, d, e\}$	It has $V = \{1, 2, 3, 4\}$ and $A = \{a, b, c, d, e\}$. The source and target functions, $s, t: A \rightarrow$
fong-spivak-seven-sketches_26 F00223	26	■ 1.000	$s, t: A \rightarrow$	$s, t: A \rightarrow$	It has $V = \{1, 2, 3, 4\}$ and $A = \{a, b, c, d, e\}$. The source and target functions, $s, t: A \rightarrow$
fong-spivak-seven-sketches_26 F00226	26	■ 1.000	d	d	are infinitely many paths $1 \rightarrow 2$ because one can loop and loop and loop through d as
fong-spivak-seven-sketches_26 F00227	26	■ 0.862	e	e	There is one path from 2 to 3, namely the arrow e is a path of length 1. There are no
fong-spivak-seven-sketches_26 F00228	26	■ 1.000	$1 \rightarrow 2$	$1 \rightarrow 2$	are infinitely many paths $1 \rightarrow 2$ because one can loop and loop and loop through d as
fong-spivak-seven-sketches_26 F00229	26	■ 0.833	$(P, \leq)$	$(P, \leq)$	graph $G = (V, A, s, t)$ that gives a <i>presentation</i> of a preorder $(P, \leq)$. The elements of P
fong-spivak-seven-sketches_26 F00230	26	■ 0.842	$v \leq w$	$v \leq w$	are the vertices V in G , and the order $\leq$ is given by $v \leq w$ iff ⁶ there is a path $v \rightarrow w$.
fong-spivak-seven-sketches_26 F00231	26	■ 0.842	$\wedge\{6\}$	$\wedge\{6\}$	are the vertices V in G , and the order $\leq$ is given by $v \leq w$ iff ⁶ there is a path $v \rightarrow w$.
fong-spivak-seven-sketches_26 F00232	26	■ 0.842	$v \rightarrow w$	$v \rightarrow w$	are the vertices V in G , and the order $\leq$ is given by $v \leq w$ iff ⁶ there is a path $v \rightarrow w$.
fong-spivak-seven-sketches_26 F00233	26	■ 1.000	$v \rightarrow v$	$v \rightarrow v$	For any vertex v , there is always a path $v \rightarrow v$, and this translates into the reflexivity
fong-spivak-seven-sketches_26 F00234	26	■ 1.000	$u \rightarrow v$	$u \rightarrow v$	law from Definition 1.30. The fact that paths $u \rightarrow v$ and $v \rightarrow w$ can be concatenated to
fong-spivak-seven-sketches_26 F00235	26	■ 1.000	$u \rightarrow w$	$u \rightarrow w$	a path  translates into the transitivity law.
fong-spivak-seven-sketches_27 F00236	27	■ 0.959	$\left(p_{\{1\}}, p_{\{2\}}\right)$	(p_1, p_2)	That is, what are the elements of P and write down every pair (p_1, p_2) for which $p_1 \leq p_2$.
fong-spivak-seven-sketches_27 F00237	27	■ 0.959	$p_{\{1\}} \leq p_{\{2\}}$	$p_1 \leq p_2$	That is, what are the elements of P and write down every pair (p_1, p_2) for which $p_1 \leq p_2$.
fong-spivak-seven-sketches_27 F00238	27	■ 0.999	$\{\bullet, \circ, *\}$	$\{\bullet, \circ, *\}$	<i>Exercise 1.42.</i> Let X be the set of partitions of $\{\bullet, \circ, *\}$; it has five elements and an
fong-spivak-seven-sketches_27 F00239	27	■ 0.929	(x, y)	(x, y)	(x, y) of elements in X such that $x \leq y$. There should be 12.
fong-spivak-seven-sketches_27 F00240	27	■ 1.000	$\mathbb{N}:=\{0,1,2,3, \ldots\}$	$\mathbb{N} := \{0, 1, 2, 3, \dots\}$	<i>Example 1.45</i> (Natural numbers). The natural numbers $\mathbb{N} := \{0, 1, 2, 3, \dots\}$ are a pre-
fong-spivak-seven-sketches_27 F00241	27	■ 1.000	$0 \leq 1$	$0 \leq 1$	order with the order given by the usual size ordering, e.g. $0 \leq 1$ and $5 \leq 100$. This is
fong-spivak-seven-sketches_27 F00242	27	■ 1.000	$5 \leq 100$	$5 \leq 100$	order with the order given by the usual size ordering, e.g. $0 \leq 1$ and $5 \leq 100$. This is
fong-spivak-seven-sketches_27 F00243	27	■ 1.000	$m \leq n$	$m \leq n$	a total order: either $m \leq n$ or $n \leq m$ for all m, n . One can see that its Hasse diagram
fong-spivak-seven-sketches_27 F00244	27	■ 1.000	$n \leq m$	$n \leq m$	a total order: either $m \leq n$ or $n \leq m$ for all m, n . One can see that its Hasse diagram

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F00245	27	■ 1.000	m, n	m, n	a total order: either $m \leq n$ or $n \leq m$ for all m, n . One can see that its Hasse diagram
fong-spivak-seven-sketches-F00246	27	■ 1.000	$a \leq b$	$a \leq b$	and b —namely all those in the middle row—which satisfy neither $a \leq b$ nor $b \leq a$.
fong-spivak-seven-sketches-F00247	27	■ 1.000	$b \leq a$	$b \leq a$	and b —namely all those in the middle row—which satisfy neither $a \leq b$ nor $b \leq a$.
fong-spivak-seven-sketches-F00248	27	■ 1.000	$n=m$	$n = m$	$n = m$. Another ordering is the reverse ordering, like $5 \leq 3$ and $3 \leq 2$, like how golf is
fong-spivak-seven-sketches-F00249	27	■ 1.000	$5 \leq 3$	$5 \leq 3$	$n = m$. Another ordering is the reverse ordering, like $5 \leq 3$ and $3 \leq 2$, like how golf is
fong-spivak-seven-sketches-F00250	27	■ 1.000	$3 \leq 2$	$3 \leq 2$	$n = m$. Another ordering is the reverse ordering, like $5 \leq 3$ and $3 \leq 2$, like how golf is
fong-spivak-seven-sketches-F00251	27	■ 1.000	m	m	m without remainder. In this ordering $2 \leq 4$, for example, but $2 \not\leq 3$, since there is a
fong-spivak-seven-sketches-F00252	27	■ 1.000	$2 \leq 4$	$2 \leq 4$	m without remainder. In this ordering $2 \leq 4$, for example, but $2 \not\leq 3$, since there is a
fong-spivak-seven-sketches-F00253	27	■ 1.000	$2 \not\leq 3$	$2 \not\leq 3$	m without remainder. In this ordering $2 \leq 4$, for example, but $2 \not\leq 3$, since there is a
fong-spivak-seven-sketches-F00254	27	■ 0.999	$1, 2, \ldots, 10$	$1, 2, \dots, 10$	<i>Exercise 1.46.</i> Write down the numbers 1, 2, ..., 10 and draw an arrow $a \rightarrow b$ if a
fong-spivak-seven-sketches-F00255	27	■ 0.999	$a \rightarrow b$	$a \rightarrow b$	<i>Exercise 1.46.</i> Write down the numbers $1, 2, \dots, 10$ and draw an arrow $a \rightarrow b$ if a
fong-spivak-seven-sketches-F00256	28	■ 0.967	$-500 \leq -499 \leq 0 \leq \sqrt{2} \leq 100 / 3$	$-500 \leq -499 \leq 0 \leq \sqrt{2} \leq 100/3$	ordering”, e.g. $-500 \leq -499 \leq 0 \leq \sqrt{2} \leq 100/3$.
fong-spivak-seven-sketches-F00257	28	■ 1.000	$\mathrm{P}(X)$	$P(X)$	set of X ; we denote it $P(X)$. The power set can naturally be given an order by inclusion
fong-spivak-seven-sketches-F00258	28	■ 1.000	$X=\{0, 1, 2\}$	$X = \{0, 1, 2\}$	For example, taking $X = \{0, 1, 2\}$, we depict $P(X)$ as
fong-spivak-seven-sketches-F00259	28	■ 0.975	$\mathrm{P}\{1, 2, \ldots, n\}, \{ \}^{\{a\}}$	$P\{1, 2, \dots, n\},^a$	See the cube? The Hasse diagram for the power set of a finite set, say $P\{1, 2, \dots, n\},^a$
fong-spivak-seven-sketches-F00260	28	■ 1.000	$\mathrm{P} X$	PX	^a Note that we omit the parentheses here, writing PX instead of $P(X)$; throughout this book we will
fong-spivak-seven-sketches-F00261	28	■ 0.963	$\mathrm{P}(\varnothing), \mathrm{P}\{1\}$	$P(\emptyset), P\{1\}$	<i>Exercise 1.51.</i> Draw the Hasse diagrams for $P(\emptyset)$, $P\{1\}$, and $P\{1, 2\}$.
fong-spivak-seven-sketches-F00262	28	■ 0.963	$\mathrm{P}\{1, 2\}$	$P\{1, 2\}$	<i>Exercise 1.51.</i> Draw the Hasse diagrams for $P(\emptyset)$, $P\{1\}$, and $P\{1, 2\}$.
fong-spivak-seven-sketches-F00263	29	■ 1.000	$\operatorname{Prt}(A)$	$\operatorname{Prt}(A)$	think about how we might order the set $\operatorname{Prt}(A)$ of all partitions of A , for some set A . In
fong-spivak-seven-sketches-F00264	29	■ 1.000	$q \in Q$	$q \in Q$	a partition P is finer than a partition Q if, for every part $p \in P$ there is a part $q \in Q$ such
fong-spivak-seven-sketches-F00265	29	■ 1.000	$A_{\{p\}} \subseteq A_{\{q\}}$	$A_p \subseteq A_q$	that $A_p \subseteq A_q$. We could also say that Q is coarser than P .
fong-spivak-seven-sketches-F00266	29	■ 1.000	$g: A \rightarrow Q$	$g: A \rightarrow Q$	tions out of A . Then $f: A \twoheadrightarrow P$ is finer than $g: A \twoheadrightarrow Q$ if there is a function $h: P \rightarrow Q$
fong-spivak-seven-sketches-F00267	29	■ 1.000	$h: P \rightarrow Q$	$h: P \rightarrow Q$	tions out of A . Then $f: A \twoheadrightarrow P$ is finer than $g: A \twoheadrightarrow Q$ if there is a function $h: P \rightarrow Q$
fong-spivak-seven-sketches-F00268	29	■ 1.000	$f; h=g$	$f; h = g$	such that $f \circ h = g$.
fong-spivak-seven-sketches-F00269	29	■ 0.795	$P, \leq$	$P, \leq$	<i>Example 1.54</i> (Upper sets). Given a preorder $(P, \leq)$, an upper set in P is a subset U of P
fong-spivak-seven-sketches-F00270	29	■ 0.795	U	U	<i>Example 1.54</i> (Upper sets). Given a preorder $(P, \leq)$, an upper set in P is a subset U of P
fong-spivak-seven-sketches-F00271	29	■ 1.000	$p \in U$	$p \in U$	satisfying the condition that if $p \in U$ and $p \leq q$, then $q \in U$. “If p is an element then so

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F00272	29	■ 1.000	$p \leq q$	$p \leq q$	satisfying the condition that if $p \in U$ and $p \leq q$, then $q \in U$. “If p is an element then so
fong-spivak-seven-sketches-F00273	29	■ 1.000	$q \in U$	$q \in U$	satisfying the condition that if $p \in U$ and $p \leq q$, then $q \in U$. “If p is an element then so
fong-spivak-seven-sketches-F00274	29	■ 1.000	p	p	satisfying the condition that if $p \in U$ and $p \leq q$, then $q \in U$. “If p is an element then so
fong-spivak-seven-sketches-F00275	29	■ 0.912	$\mathrm{U}(P)$	$U(P)$	is anything bigger.” Write $U(P)$ for the set of upper sets in P . We can give the set U an
fong-spivak-seven-sketches-F00276	29	■ 1.000	$U \leq V$	$U \leq V$	order by letting $U \leq V$ if U is contained in V .
fong-spivak-seven-sketches-F00277	29	■ 0.998	$(\mathbb{B}, \leq)$	$(\mathbb{B}, \leq)$	For example, if $(\mathbb{B}, \leq)$ is the booleans (Example 1.34), then its preorder of uppersets
fong-spivak-seven-sketches-F00278	29	■ 0.457	$\mathrm{U}(\mathbb{B})$	$U(\mathbb{B})$	$U(\mathbb{B})$ is
fong-spivak-seven-sketches-F00279	29	■ 0.749	$\subseteq \mathbb{B}$	$\subseteq \mathbb{B}$	The subset $\{\text{false}\} \subseteq \mathbb{B}$ is not an upper set, because $\text{false} \leq \text{true}$ and $\text{true} \notin \{\text{false}\}$.
fong-spivak-seven-sketches-F00280	29	■ 0.749	$\notin \{$	$\notin \{$	The subset $\{\text{false}\} \subseteq \mathbb{B}$ is not an upper set, because $\text{false} \leq \text{true}$ and $\text{true} \notin \{\text{false}\}$.
fong-spivak-seven-sketches-F00281	29	■ 0.998	$(Q, \leq)$	$(Q, \leq)$	Example 1.56 (Product preorder). Given preorders $(P, \leq)$ and $(Q, \leq)$, we may define a
fong-spivak-seven-sketches-F00282	29	■ 1.000	$P \times Q$	$P \times Q$	preorder structure on the product set $P \times Q$ by setting $(p, q) \leq (p', q')$ if and only if
fong-spivak-seven-sketches-F00283	29	■ 1.000	$(p, q) \leq (p', q')$	$(p, q) \leq (p', q')$	preorder structure on the product set $P \times Q$ by setting $p, q \leq p', q'$ if and only if
fong-spivak-seven-sketches-F00284	29	■ 1.000	$p \leq p'$	$p \leq p'$	$p \leq p'$ and $q \leq q'$. We call this the <i>product preorder</i> . This is a basic example of a more
fong-spivak-seven-sketches-F00285	29	■ 1.000	$q \leq q'$	$q \leq q'$	$p \leq p'$ and $q \leq q'$. We call this the <i>product preorder</i> . This is a basic example of a more
fong-spivak-seven-sketches-F00286	30	■ 0.889	$(P, \leq^{\text{op}})$	$(P, \leq^{\text{op}})$	preorder $(P, \leq^{\text{op}})$ to have the same set of elements, but with $p \leq^{\text{op}} q$ if and only if $q \leq p$.
fong-spivak-seven-sketches-F00287	30	■ 0.889	$p \leq^{\text{op}} q$	$p \leq^{\text{op}} q$	preorder $(P, \leq^{\text{op}})$ to have the same set of elements, but with $p \leq^{\text{op}} q$ if and only if $q \leq p$.
fong-spivak-seven-sketches-F00288	30	■ 0.889	$q \leq p$	$q \leq p$	preorder $(P, \leq^{\text{op}})$ to have the same set of elements, but with $p \leq^{\text{op}} q$ if and only if $q \leq p$.
fong-spivak-seven-sketches-F00289	30	■ 0.988	$A, \leq_A$	$A, \leq_A$	Definition 1.59. A monotone map between preorders $(A, \leq_A)$ and $(B, \leq_B)$ is a function
fong-spivak-seven-sketches-F00290	30	■ 0.988	$B, \leq_B$	$B, \leq_B$	Definition 1.59. A monotone map between preorders $(A, \leq_A)$ and $(B, \leq_B)$ is a function
fong-spivak-seven-sketches-F00291	30	■ 0.872	$x, y \in A$	$x, y \in A$	$f: A \rightarrow B$ such that, for all elements $x, y \in A$, if $x \leq_A y$ then $f(x) \leq_B f(y)$.
fong-spivak-seven-sketches-F00292	30	■ 0.872	$x \leq_A y$	$x \leq_A y$	$f: A \rightarrow B$ such that, for all elements $x, y \in A$, if $x \leq_A y$ then $f(x) \leq_B f(y)$.
fong-spivak-seven-sketches-F00293	30	■ 0.872	$f(x) \leq_B f(y)$	$f(x) \leq_B f(y)$	$f: A \rightarrow B$ such that, for all elements $x, y \in A$, if $x \leq_A y$ then $f(x) \leq_B f(y)$.
fong-spivak-seven-sketches-F00294	30	■ 1.000	$A \rightarrow B$	$A \rightarrow B$	A monotone map $A \rightarrow B$ between two preorders associates to each element of
fong-spivak-seven-sketches-F00295	30	■ 1.000	$x \in A$	$x \in A$	from each element $x \in A$ to its image $f(x) \in B$. Note that the order must be preserved
fong-spivak-seven-sketches-F00296	30	■ 1.000	$f(x) \in B$	$f(x) \in B$	from each element $x \in A$ to its image $f(x) \in B$. Note that the order must be preserved
fong-spivak-seven-sketches-F00297	30	■ 1.000	$f(y)$	$f(y)$	lefthand preorder A , then the image $f(x)$ will be above the image $f(y)$ in the righthand
fong-spivak-seven-sketches-F00298	31	■ 1.000	$\mathbb{B} \rightarrow \mathbb{N}$	$\mathbb{B} \rightarrow \mathbb{N}$	preorder of natural numbers from Example 1.45. The map $\mathbb{B} \rightarrow \mathbb{N}$ sending false to 17

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_31 F00299	31	■ 1.000	$x \rightarrow y$	$x \rightarrow y$	because it preserves order: whenever there is a path $x \rightarrow y$ upstairs, there is a path
fong-spivak-seven-sketches_31 F00300	31	■ 1.000	$F(x) \rightarrow F(y)$	$F(x) \rightarrow F(y)$	$F(x) \rightarrow F(y)$ downstairs.
fong-spivak-seven-sketches_31 F00301	31	■ 0.988	$ \cdot : \mathrm{P}(X) \rightarrow \mathbb{N}$	$ \cdot : \mathrm{P}(X) \rightarrow \mathbb{N}$	from Example 1.50. The map $ \cdot : \mathrm{P}(X) \rightarrow \mathbb{N}$ sending each subset S to its number of
fong-spivak-seven-sketches_31 F00302	31	■ 1.000	$ S $	$ S $	elements $ S $, also called its <i>cardinality</i> , is a monotone map.
fong-spivak-seven-sketches_31 F00303	31	■ 1.000	$0 \leq 1 \leq 2 \leq 3$	$0 \leq 1 \leq 2 \leq 3$	2. Draw the Hasse diagram for the preorder $0 \leq 1 \leq 2 \leq 3$.
fong-spivak-seven-sketches_32 F00304	32	■ 0.997	$ \cdot $	$ \cdot $	3. Draw the cardinality map $ \cdot $ from Example 1.62 as dashed lines between them.
fong-spivak-seven-sketches_32 F00305	32	■ 0.995	$\mathrm{U}(P) \rightarrow \mathrm{P}(P)$	$\mathrm{U}(P) \rightarrow \mathrm{P}(P)$	$(P, \leq)$, the map $\mathrm{U}(P) \rightarrow \mathrm{P}(P)$ sending each upper set of $(P, \leq)$ to itself—considered as
fong-spivak-seven-sketches_32 F00306	32	■ 0.939	$\mathrm{P}(\mathbb{B})$	$\mathrm{P}(\mathbb{B})$	Example 1.54, and you drew the Hasse diagram for $\mathrm{P}(\mathbb{B})$ in Exercise 1.51. Now draw
fong-spivak-seven-sketches_32 F00307	32	■ 1.000	$\uparrow p := \{p' \in P \mid p \leq p'\}$	$\uparrow p := \{p' \in P \mid p \leq p'\}$	1. Show that the set $\uparrow p := \{p' \in P \mid p \leq p'\}$ is an upper set, for any $p \in P$.
fong-spivak-seven-sketches_32 F00308	32	■ 0.671	$\uparrow: P^{\mathrm{op}} \rightarrow \mathrm{U}(P)$	$\uparrow: P^{\mathrm{op}} \rightarrow \mathrm{U}(P)$	2. Show that this construction defines a monotone map $\uparrow: P^{\mathrm{op}} \rightarrow \mathrm{U}(P)$.
fong-spivak-seven-sketches_32 F00309	32	■ 1.000	$\uparrow(p') \subseteq \uparrow(p)$	$\uparrow(p') \subseteq \uparrow(p)$	3. Show that if $p \leq p'$ in P if and only if $\uparrow(p') \subseteq \uparrow(p)$.
fong-spivak-seven-sketches_32 F00310	32	■ 1.000	$b \geq a \leq c$	$b \geq a \leq c$	4. Draw a picture of the map $\uparrow$ in the case where P is the preorder ($b \geq a \leq c$) from
fong-spivak-seven-sketches_32 F00311	32	■ 1.000	$f: (P, \leq_P) \rightarrow (Q, \leq_Q)$	$f: (P, \leq_P) \rightarrow (Q, \leq_Q)$	Exercise 1.67. As you yourself well know, a monotone map $f: (P, \leq_P) \rightarrow (Q, \leq_Q)$
fong-spivak-seven-sketches_32 F00312	32	■ 1.000	$f: P \rightarrow Q$	$f: P \rightarrow Q$	consists of a function $f: P \rightarrow Q$ that satisfies a “monotonicity” property. Show that
fong-spivak-seven-sketches_32 F00313	32	■ 1.000	$(P, \leq_P)$	$(P, \leq_P)$	when $(P, \leq_P)$ is a discrete preorder, then <i>every</i> function $P \rightarrow Q$ satisfies the monotonicity
fong-spivak-seven-sketches_32 F00314	32	■ 1.000	$P \rightarrow Q$	$P \rightarrow Q$	when $(P, \leq_P)$ is a discrete preorder, then <i>every</i> function $P \rightarrow Q$ satisfies the monotonicity
fong-spivak-seven-sketches_32 F00315	32	■ 1.000	$\leq_Q$	$\leq_Q$	property, regardless of the order $\leq_Q$.
fong-spivak-seven-sketches_32 F00316	32	■ 1.000	$\mathrm{Prt}(X)$	$\mathrm{Prt}(X)$	Example 1.68. Recall from Example 1.52 that given a set X we define $\mathrm{Prt}(X)$ to be the
fong-spivak-seven-sketches_32 F00317	32	■ 1.000	$s: X \rightarrow P$	$s: X \rightarrow P$	$s: X \twoheadrightarrow P$ for some set P .
fong-spivak-seven-sketches_32 F00318	32	■ 1.000	$f^*: \mathrm{Prt}(Y) \rightarrow \mathrm{Prt}(X)$	$f^*: \mathrm{Prt}(Y) \rightarrow \mathrm{Prt}(X)$	Any surjective function $f: X \twoheadrightarrow Y$ induces a monotone map $f^*: \mathrm{Prt}(Y) \rightarrow \mathrm{Prt}(X)$,
fong-spivak-seven-sketches_32 F00319	32	■ 0.994	$s: Y \rightarrow P$	$s: Y \rightarrow P$	going “backwards.” It is defined by sending a partition $s: Y \twoheadrightarrow P$ to the composite
fong-spivak-seven-sketches_32 F00320	32	■ 0.905	s	s	$f \circ s: X \twoheadrightarrow P$. ⁷
fong-spivak-seven-sketches_32 F00321	32	■ 0.905	$X \rightarrow P$	$X \rightarrow P$. ⁷	$f \circ s: X \twoheadrightarrow P$
fong-spivak-seven-sketches_32 F00322	32	■ 1.000	$f^*(P)$	$f^*(P)$	partitions P and Q of Y , and then find $f^*(P)$ and $f^*(Q)$.
fong-spivak-seven-sketches_32 F00323	32	■ 1.000	$f^*(Q)$	$f^*(Q)$	partitions P and Q of Y , and then find $f^*(P)$ and $f^*(Q)$.
fong-spivak-seven-sketches_33 F00324	33	■ 0.979	$(Q, \leq_Q)$	$(Q, \leq_Q)$	If $(Q, \leq_Q)$ and $(R, \leq_R)$ are preorders and $f: P \rightarrow Q$ and $g: Q \rightarrow R$ are monotone,
fong-spivak-seven-sketches_33 F00325	33	■ 0.979	$(R, \leq_R)$	$(R, \leq_R)$	If $(Q, \leq_Q)$ and $(R, \leq_R)$ are preorders and $f: P \rightarrow Q$ and $g: Q \rightarrow R$ are monotone,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_1 F00326	33	■ 0.979	$g: Q \rightarrow R$	$g: Q \rightarrow R$	If $(Q, \leq_Q)$ and $(R, \leq_R)$ are preorders and $f: P \rightarrow Q$ and $g: Q \rightarrow R$ are monotone,
fong-spivak-seven-sketches_1 F00327	33	■ 0.960	$(f; g): P \rightarrow R$	$(f; g): P \rightarrow R$	then $(f \circ g): P \rightarrow R$ is also monotone.
fong-spivak-seven-sketches_1 F00328	33	■ 0.629	$\operatorname{id}_P: P \rightarrow P$	$\operatorname{id}_P: P \rightarrow P$	identity function $\operatorname{id}_P: P \rightarrow P$ is a monotone map $(P, \leq) \rightarrow (P, \leq^{\text{op}})$ if and only if for
fong-spivak-seven-sketches_1 F00329	33	■ 0.629	$(P, \leq) \rightarrow (P, \leq^{\text{op}})$	$(P, \leq) \rightarrow (P, \leq^{\text{op}})$	identity function $\operatorname{id}_P: P \rightarrow P$ is a monotone map $(P, \leq) \rightarrow (P, \leq^{\text{op}})$ if and only if for
fong-spivak-seven-sketches_1 F00330	33	■ 1.000	$p, q \in P$	$p, q \in P$	all $p, q \in P$ we have $q \leq p$ whenever $p \leq q$. For historical reasons connected to linear
fong-spivak-seven-sketches_1 F00331	33	■ 0.997	$p_1 = p_2$	$p_1 = p_2$	made into a discrete preorder ($p_1 \leq p_2$ iff $p_1 = p_2$), and the round-trips return us where
fong-spivak-seven-sketches_1 F00332	33	■ 1.000	$g: Q \rightarrow P$	$g: Q \rightarrow P$	is called an <i>isomorphism</i> if there exists a monotone function $g: Q \rightarrow P$ such that
fong-spivak-seven-sketches_1 F00333	33	■ 0.555	$f; g = \operatorname{id}_P$	$f; g = \operatorname{id}_P$	$f \circ g = \operatorname{id}_P$ and $g \circ f = \operatorname{id}_Q$. This means that for any $p \in P$ and $q \in Q$, we have
fong-spivak-seven-sketches_1 F00334	33	■ 0.555	$g \circ f = \operatorname{id}_Q$	$g \circ f = \operatorname{id}_Q$	$f \circ g = \operatorname{id}_P$ and $g \circ f = \operatorname{id}_Q$. This means that for any $p \in P$ and $q \in Q$, we have
fong-spivak-seven-sketches_1 F00335	33	■ 1.000	g	g	We refer to g as the <i>inverse</i> of f , and vice versa: f is the inverse of g .
fong-spivak-seven-sketches_1 F00336	34	■ 1.000	P, Q	P, Q	Example 1.76. Here are the Hasse diagrams for three preorders P, Q , and R , all of which
fong-spivak-seven-sketches_1 F00337	34	■ 1.000	R	R	Example 1.76. Here are the Hasse diagrams for three preorders P, Q , and R , all of which
fong-spivak-seven-sketches_1 F00338	34	■ 1.000	$f(a)=v, f(b)=w, f(c)=x, f(d)=y$	$f(a) = v, f(b) = w, f(c) = x, f(d) = y$	The map $f: P \rightarrow Q$ given by $f(a) = v, f(b) = w, f(c) = x, f(d) = y$, and $f(e) = z$ has
fong-spivak-seven-sketches_1 F00339	34	■ 1.000	$f(e)=z$	$f(e) = z$	The map $f: P \rightarrow Q$ given by $f(a) = v, f(b) = w, f(c) = x, f(d) = y$, and $f(e) = z$ has
fong-spivak-seven-sketches_1 F00340	34	■ 1.000	$x \rightarrow z$	$x \rightarrow z$	is an arrow $x \rightarrow z$ in the Hasse diagram for R and not one in Q , but in fact this arrow
fong-spivak-seven-sketches_1 F00341	34	■ 1.000	$v \rightarrow y$	$v \rightarrow y$	drawn an arrow $v \rightarrow y$ in either Q or R and it would not have changed the preorder.
fong-spivak-seven-sketches_1 F00342	34	■ 0.546	$\operatorname{Prt}(\{*, \bullet, \circ\}) \rightarrow \mathbb{B}$	$\operatorname{Prt}(\{*, \bullet, \circ\}) \rightarrow \mathbb{B}$	‘Is $\bullet$ connected to $*$?’ is a monotone map $\operatorname{Prt}(\{*, \bullet, \circ\}) \rightarrow \mathbb{B}$; see also Eq. (1.5).
fong-spivak-seven-sketches_1 F00343	34	■ 1.000	$P \rightarrow \mathbb{B}$	$P \rightarrow \mathbb{B}$	Proposition 1.78. Let P be a preorder. Monotone maps $P \rightarrow \mathbb{B}$ are in one-to-one
fong-spivak-seven-sketches_1 F00344	34	■ 0.656	$f: P \rightarrow \mathbb{B}$	$f: P \rightarrow \mathbb{B}$	<i>Proof.</i> Let $f: P \rightarrow \mathbb{B}$ be a monotone map. We will show that the subset $f^{-1}(\text{true}) \subseteq P$
fong-spivak-seven-sketches_1 F00345	34	■ 0.656	$f^{-1}(\text{true})$	$f^{-1}(\text{true})$	<i>Proof.</i> Let $f: P \rightarrow \mathbb{B}$ be a monotone map. We will show that the subset $f^{-1}(\text{true}) \subseteq P$
fong-spivak-seven-sketches_1 F00346	34	■ 0.656	$f^{-1}(\text{true}) \subseteq P$	$f^{-1}(\text{true}) \subseteq P$	<i>Proof.</i> Let $f: P \rightarrow \mathbb{B}$ be a monotone map. We will show that the subset $f^{-1}(\text{true}) \subseteq P$
fong-spivak-seven-sketches_1 F00347	34	■ 1.000	$p \in f^{-1}(\text{true})$	$p \in f^{-1}(\text{true})$	is an upper set. Suppose $p \in f^{-1}(\text{true})$ and $p \leq q$; then $\text{true} = f(p) \leq f(q)$. But in
fong-spivak-seven-sketches_1 F00348	34	■ 1.000	$f(p) \leq f(q)$	$f(p) \leq f(q)$	is an upper set. Suppose $p \in f^{-1}(\text{true})$ and $p \leq q$; then $\text{true} = f(p) \leq f(q)$. But in
fong-spivak-seven-sketches_1 F00349	34	■ 0.842	$f(p) \leq f(q)$	$f(p) \leq f(q)$	$\mathbb{B}$, if $\text{true} \leq f(q)$ then $\text{true} = f(q)$. This implies $q \in f^{-1}(\text{true})$ and thus shows that
fong-spivak-seven-sketches_1 F00350	34	■ 0.842	$f(p) \leq f(q)$	$f(p) \leq f(q)$	$\mathbb{B}$, if $\text{true} \leq f(q)$ then $\text{true} = f(q)$. This implies $q \in f^{-1}(\text{true})$ and thus shows that
fong-spivak-seven-sketches_1 F00351	34	■ 0.842	$q \in f^{-1}(\text{true})$	$q \in f^{-1}(\text{true})$	$\mathbb{B}$, if $\text{true} \leq f(q)$ then $\text{true} = f(q)$. This implies $q \in f^{-1}(\text{true})$ and thus shows that
fong-spivak-seven-sketches_1 F00352	34	■ 1.000	$f^{-1}(\text{true})$	$f^{-1}(\text{true})$	$f^{-1}(\text{true})$ is an upper set.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F00353	34	■ 1.000	$f_{\{U\}}: P \rightarrow \mathbb{B}$	$f_U: P \rightarrow \mathbb{B}$	Conversely, if U is an upper set in P , define $f_U: P \rightarrow \mathbb{B}$ such that $f_U(p) = \text{true}$ when
fong-spivak-seven-sketches F00354	34	■ 1.000	$f_{\{U\}}(p) =$	$f_U(p) =$	Conversely, if U is an upper set in P , define $f_U: P \rightarrow \mathbb{B}$ such that $f_U(p) = \text{true}$ when
fong-spivak-seven-sketches F00355	34	■ 1.000	$p \notin U$	$p \notin U$	$p \in U$, and $f_U(p) = \text{false}$ when $p \notin U$. This is a monotone map, because if $p \leq q$, then
fong-spivak-seven-sketches F00356	34	■ 1.000	$f(p) =$	$f(p) =$	either $p \in U$, so $q \in U$ and $f(p) = \text{true} = f(q)$, or $p \notin U$, so $f(p) = \text{false} \leq f(q)$.
fong-spivak-seven-sketches F00357	34	■ 0.996	$f^*: \mathrm{U}(Q) \rightarrow \mathrm{U}(P)$	$f^*: \mathrm{U}(Q) \rightarrow \mathrm{U}(P)$	map. Then we can define a monotone map $f^*: \mathrm{U}(Q) \rightarrow \mathrm{U}(P)$ sending an upper set
fong-spivak-seven-sketches F00358	34	■ 1.000	$U \subseteq Q$	$U \subseteq Q$	$U \subseteq Q$ to the upper set $f^{-1}(U) \subseteq P$. We call this the <i>pullback along f</i> .
fong-spivak-seven-sketches F00359	34	■ 1.000	$f^{-1}(U) \subseteq P$	$f^{-1}(U) \subseteq P$	$U \subseteq Q$ to the upper set $f^{-1}(U) \subseteq P$. We call this the <i>pullback along f</i> .
fong-spivak-seven-sketches F00360	34	■ 1.000	f^*	f^*	can be understood in terms of composition. Indeed, show that the f^* is defined by
fong-spivak-seven-sketches F00361	34	■ 1.000	$u: Q \rightarrow \mathbb{B}$	$u: Q \rightarrow \mathbb{B}$	taking $u: Q \rightarrow \mathbb{B}$ to $(f \circ u): P \rightarrow \mathbb{B}$.
fong-spivak-seven-sketches F00362	34	■ 1.000	$(f \circ u): P \rightarrow \mathbb{B}$	$(f \circ u): P \rightarrow \mathbb{B}$	taking $u: Q \rightarrow \mathbb{B}$ to $(f \circ u): P \rightarrow \mathbb{B}$.
fong-spivak-seven-sketches F00363	35	■ 0.637	$(\mathbb{R}, \leq)$	$(\mathbb{R}, \leq)$	Consider the preorder $(\mathbb{R}, \leq)$ of real numbers ordered in the usual way. The subset
fong-spivak-seven-sketches F00364	35	■ 1.000	$\mathbb{N} \subseteq \mathbb{R}$	$\mathbb{N} \subseteq \mathbb{R}$	$\mathbb{N} \subseteq \mathbb{R}$ has many lower bounds, namely -1.5 is a lower bound: every element of $\mathbb{N}$ is
fong-spivak-seven-sketches F00365	35	■ 1.000	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} \subseteq \mathbb{R}$	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} \subseteq \mathbb{R}$	1. Why is 0 a lower bound for $\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} \subseteq \mathbb{R}$?
fong-spivak-seven-sketches F00366	35	■ 0.719	$A \subseteq P$	$A \subseteq P$	Definition 1.81. Let $(P, \leq)$ be a preorder, and let $A \subseteq P$ be a subset. We say that an
fong-spivak-seven-sketches F00367	35	■ 1.000	$p \leq a$	$p \leq a$	(a) for all $a \in A$, we have $p \leq a$, and
fong-spivak-seven-sketches F00368	35	■ 1.000	q	q	(b) for all q such that $q \leq a$ for all $a \in A$, we have that $q \leq p$.
fong-spivak-seven-sketches F00369	35	■ 1.000	$q \leq a$	$q \leq a$	(b) for all q such that $q \leq a$ for all $a \in A$, we have that $q \leq p$.
fong-spivak-seven-sketches F00370	35	■ 1.000	$p = \bigwedge A, p = \bigwedge_{a \in A} a$	$p = \bigwedge A, p = \bigwedge_{a \in A} a$	We write $p = \bigwedge A, p = \bigwedge_{a \in A} a$, or, if the dummy variable a is clear from context, just
fong-spivak-seven-sketches F00371	35	■ 0.999	$p = \bigwedge_{a \in A} a$	$p = \bigwedge_A a$	$p = \bigwedge_A a$. If A just consists of two elements, say $A = \{a, b\}$, we can denote $\bigwedge A$ simply
fong-spivak-seven-sketches F00372	35	■ 0.999	$A = \{a, b\}$	$A = \{a, b\}$	$p = \bigwedge_A a$. If A just consists of two elements, say $A = \{a, b\}$, we can denote $\bigwedge A$ simply
fong-spivak-seven-sketches F00373	35	■ 0.999	$\bigwedge A$	$\bigwedge A$	$p = \bigwedge_A a$. If A just consists of two elements, say $A = \{a, b\}$, we can denote $\bigwedge A$ simply
fong-spivak-seven-sketches F00374	35	■ 1.000	$a \wedge b$	$a \wedge b$	by $a \wedge b$.
fong-spivak-seven-sketches F00375	35	■ 1.000	$a \leq p$	$a \leq p$	(a) for all $a \in A$ we have $a \leq p$, and
fong-spivak-seven-sketches F00376	35	■ 1.000	$a \leq q$	$a \leq q$	(b) for all q such that $a \leq q$ for all $a \in A$, we have that $p \leq q$.
fong-spivak-seven-sketches F00377	35	■ 0.990	$p = \bigvee A$	$p = \bigvee A$	We write $p = \bigvee A$ or $p = \bigvee_{a \in A} a$, or when $A = \{a, b\}$ we may simply write $p = a \vee b$.
fong-spivak-seven-sketches F00378	35	■ 0.990	$p = \bigvee_{a \in A} a$	$p = \bigvee_{a \in A} a$	We write $p = \bigvee A$ or $p = \bigvee_{a \in A} a$, or when $A = \{a, b\}$ we may simply write $p = a \vee b$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_35 F00379	35	■ 0.990	p=a \vee b	$p = a \vee b$	We write $p = \bigvee A$ or $p = \bigvee_{a \in A} a$, or when $A = \{a, b\}$ we may simply write $p = a \vee b$.
fong-spivak-seven-sketches_36 F00380	36	■ 0.980	p=\wedge A	$p = \wedge A$	$p = \wedge A$ and $q = \wedge A$ with $p \neq q$, which does not make sense if $p \neq q$!
fong-spivak-seven-sketches_36 F00381	36	■ 0.980	q=\wedge A	$q = \wedge A$	$p = \wedge A$ and $q = \wedge A$ with $p \neq q$, which does not make sense if $p \neq q$!
fong-spivak-seven-sketches_36 F00382	36	■ 0.980	p \neq q!	$p \neq q!$	$p = \wedge A$ and $q = \wedge A$ with $p \neq q$, which does not make sense if $p \neq q$!
fong-spivak-seven-sketches_36 F00383	36	■ 1.000	p, q	p, q	meets p, q are automatically isomorphic: the very definition of meet forces both $p \leq q$
fong-spivak-seven-sketches_36 F00384	36	■ 1.000	p \cong q	$p \cong q$	and $q \leq p$, and thus we have $p \cong q$. So for any $x \in P$, we have $p \leq x$ iff $q \leq x$ and
fong-spivak-seven-sketches_36 F00385	36	■ 1.000	x \in P	$x \in P$	and $q \leq p$, and thus we have $p \cong q$. So for any $x \in P$, we have $p \leq x$ iff $q \leq x$ and
fong-spivak-seven-sketches_36 F00386	36	■ 1.000	p \leq x	$p \leq x$	and $q \leq p$, and thus we have $p \cong q$. So for any $x \in P$, we have $p \leq x$ iff $q \leq x$ and
fong-spivak-seven-sketches_36 F00387	36	■ 1.000	q \leq x	$q \leq x$	and $q \leq p$, and thus we have $p \cong q$. So for any $x \in P$, we have $p \leq x$ iff $q \leq x$ and
fong-spivak-seven-sketches_36 F00388	36	■ 1.000	x \leq p	$x \leq p$	$x \leq p$ iff $x \leq q$. Thus as long as we are only interested in elements of P based on their
fong-spivak-seven-sketches_36 F00389	36	■ 1.000	x \leq q	$x \leq q$	$x \leq p$ iff $x \leq q$. Thus as long as we are only interested in elements of P based on their
fong-spivak-seven-sketches_36 F00390	36	■ 1.000	P=\{p, q, r\}	$P = \{p, q, r\}$	a subset A need not have a meet or a join. Consider the three element set $P = \{p, q, r\}$
fong-spivak-seven-sketches_36 F00391	36	■ 1.000	A=\{p, q\}	$A = \{p, q\}$	with the discrete ordering. The set $A = \{p, q\}$ does not have a join in P because if x
fong-spivak-seven-sketches_36 F00392	36	■ 0.965	\{a, b\}	$\{a, b\}$	Let A be the subset $\{a, b\}$ in the preorder specified by this Hasse diagram. Then both
fong-spivak-seven-sketches_36 F00393	36	■ 0.998	c \leq d	$c \leq d$	less than d . Note that, as in Remark 1.82, $c \leq d$ and $d \leq c$, so $c \cong d$. Such will always
fong-spivak-seven-sketches_36 F00394	36	■ 0.998	d \leq c	$d \leq c$	less than d . Note that, as in Remark 1.82, $c \leq d$ and $d \leq c$, so $c \cong d$. Such will always
fong-spivak-seven-sketches_36 F00395	36	■ 0.998	c \cong d	$c \cong d$	less than d . Note that, as in Remark 1.82, $c \leq d$ and $d \leq c$, so $c \cong d$. Such will always
fong-spivak-seven-sketches_36 F00396	36	■ 0.519	A=\{p\}	$A = \{p\}$	<i>Exercise 1.85.</i> Let $(P, \leq)$ be a preorder and $p \in P$ an element. Consider the set $A = \{p\}$
fong-spivak-seven-sketches_36 F00397	36	■ 1.000	\wedge A \cong p	$\wedge A \cong p$	1. Show that $\wedge A \cong p$.
fong-spivak-seven-sketches_36 F00398	36	■ 1.000	\wedge A = p	$\wedge A = p$	2. Show that if P is in fact a partial order, then $\wedge A = p$.
fong-spivak-seven-sketches_36 F00399	36	■ 0.999	\wedge	$\wedge$	3. Are the analogous facts true when $\wedge$ is replaced by $\vee$?
fong-spivak-seven-sketches_37 F00400	37	■ 1.000	p \vee p = p \wedge p = p	$p \vee p = p \wedge p = p$	<i>Example 1.86.</i> In any partial order P , we have $p \vee p = p \wedge p = p$. The reason is that our
fong-spivak-seven-sketches_37 F00401	37	■ 0.796	p \vee p	$p \vee p$	notation says $p \vee p$ means $\bigvee \{p, p\}$. But $\{p, p\} = \{p\}$ (see Section 1.2.1), so $p \vee p = p$ by
fong-spivak-seven-sketches_37 F00402	37	■ 0.796	\vee\{p, p\}	$\bigvee \{p, p\}$	notation says $p \vee p$ means $\bigvee \{p, p\}$. But $\{p, p\} = \{p\}$ (see Section 1.2.1), so $p \vee p = p$ by
fong-spivak-seven-sketches_37 F00403	37	■ 0.796	\{p, p\}=\{p\}	$\{p, p\} = \{p\}$	notation says $p \vee p$ means $\bigvee \{p, p\}$. But $\{p, p\} = \{p\}$ (see Section 1.2.1), so $p \vee p = p$ by
fong-spivak-seven-sketches_37 F00404	37	■ 0.796	p \vee p = p	$p \vee p = p$	notation says $p \vee p$ means $\bigvee \{p, p\}$. But $\{p, p\} = \{p\}$ (see Section 1.2.1), so $p \vee p = p$ by
fong-spivak-seven-sketches_37 F00405	37	■ 0.999	A, B \subseteq X	$A, B \subseteq X$	<i>Example 1.87.</i> In a power set $P(X)$, the meet of a collection of subsets, say $A, B \subseteq X$ is

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F00406	37	1.000	$A \wedge B = A \cap B$	$A \wedge B = A \cap B$	their intersection $A \wedge B = A \cap B$, while the join is their union, $A \vee B = A \cup B$.
fong-spivak-seven-sketches F00407	37	1.000	$A \vee B = A \cup B$	$A \vee B = A \cup B$	their intersection $A \wedge B = A \cap B$, while the join is their union, $A \vee B = A \cup B$.
fong-spivak-seven-sketches F00408	37	1.000	$n \mid m$	$n \mid m$	Exercise 1.90. Recall the division ordering on $\mathbb{N}$ from Example 1.45: we write $n \mid m$
fong-spivak-seven-sketches F00409	37	0.706	$A \subseteq B \subseteq P$	$A \subseteq B \subseteq P$	Proposition 1.91. Suppose $(P, \leq)$ is a preorder and $A \subseteq B \subseteq P$ are subsets that have
fong-spivak-seven-sketches F00410	37	1.000	$\wedge B \leq \wedge A$	$\wedge B \leq \wedge A$	meets. Then $\wedge B \leq \wedge A$.
fong-spivak-seven-sketches F00411	37	0.998	$\bigvee A \leq \bigvee B$	$\bigvee A \leq \bigvee B$	Similarly, if A and B have joins, then $\bigvee A \leq \bigvee B$.
fong-spivak-seven-sketches F00412	37	1.000	$m = \wedge A$	$m = \wedge A$	Proof. Let $m = \wedge A$ and $n = \wedge B$. Then for any $a \in A$ we also have $a \in B$, so $n \leq a$
fong-spivak-seven-sketches F00413	37	1.000	$n = \wedge B$	$n = \wedge B$	Proof. Let $m = \wedge A$ and $n = \wedge B$. Then for any $a \in A$ we also have $a \in B$, so $n \leq a$
fong-spivak-seven-sketches F00414	37	1.000	$a \in B$	$a \in B$	Proof. Let $m = \wedge A$ and $n = \wedge B$. Then for any $a \in A$ we also have $a \in B$, so $n \leq a$
fong-spivak-seven-sketches F00415	37	1.000	$n \leq a$	$n \leq a$	Proof. Let $m = \wedge A$ and $n = \wedge B$. Then for any $a \in A$ we also have $a \in B$, so $n \leq a$
fong-spivak-seven-sketches F00416	37	0.843	$A^{\{\prime\}}$	A'	because m is A 's greatest lower bound. The second claim is proved similarly.
fong-spivak-seven-sketches F00417	38	1.000	$\Phi: P \rightarrow Q$	$\Phi: P \rightarrow Q$	map $\Phi: P \rightarrow Q$ is a phenomenon (we might say “feature”) of P as observed by Q . He
fong-spivak-seven-sketches F00418	38	1.000	$f(a \wedge b) \cong$	$f(a \wedge b) \cong$	Definition 1.92. We say that a monotone map $f: P \rightarrow Q$ preserves meets if $f(a \wedge b) \cong$
fong-spivak-seven-sketches F00419	38	0.997	$f(a) \wedge f(b)$	$f(a) \wedge f(b)$	$f(a) \wedge f(b)$ for all $a, b \in P$. We similarly say f preserves joins if $f(a \vee b) \cong f(a) \vee f(b)$
fong-spivak-seven-sketches F00420	38	0.997	$a, b \in P$	$a, b \in P$	$f(a) \wedge f(b)$ for all $a, b \in P$. We similarly say f preserves joins if $f(a \vee b) \cong f(a) \vee f(b)$
fong-spivak-seven-sketches F00421	38	0.997	$f(a \vee b) \cong f(a) \vee f(b)$	$f(a \vee b) \cong f(a) \vee f(b)$	$f(a) \wedge f(b)$ for all $a, b \in P$. We similarly say f preserves joins if $f(a \vee b) \cong f(a) \vee f(b)$
fong-spivak-seven-sketches F00422	38	1.000	$f(a) \vee f(b)$	$f(a) \vee f(b)$	a and b , then the left hand side $f(a) \vee f(b)$ may be interpreted as the combination of
fong-spivak-seven-sketches F00423	38	1.000	$f(a \vee b)$	$f(a \vee b)$	$f(a \vee b)$ is the observation of the combined system $a \vee b$. The inequality implies that
fong-spivak-seven-sketches F00424	38	1.000	$a \vee b$	$a \vee b$	$f(a \vee b)$ is the observation of the combined system $a \vee b$. The inequality implies that
fong-spivak-seven-sketches F00425	38	1.000	$f(a \vee b) \neq f(a) \vee f(b)$	$f(a \vee b) \neq f(a) \vee f(b)$	$f(a \vee b) \neq f(a) \vee f(b)$, but in the subsequent text we seemed to imply there would be
fong-spivak-seven-sketches F00426	38	1.000	$f(a), f(b) \in Q$	$f(a), f(b) \in Q$	Prove that for any monotone map $f: P \rightarrow Q$, if $a, b \in P$ have a join and $f(a), f(b) \in Q$
fong-spivak-seven-sketches F00427	38	1.000	$f(a) \vee f(b) \leq f(a \vee b)$	$f(a) \vee f(b) \leq f(a \vee b)$	have a join, then indeed $f(a) \vee f(b) \leq f(a \vee b)$.
fong-spivak-seven-sketches F00428	39	0.889	$(3 \times -): \mathbb{Z} \rightarrow \mathbb{R}$	$(3 \times -): \mathbb{Z} \rightarrow \mathbb{R}$	Example 1.97. Consider the map $(3 \times -): \mathbb{Z} \rightarrow \mathbb{R}$ which sends $x \in \mathbb{Z}$ to $3x$, which we can
fong-spivak-seven-sketches F00429	39	0.889	$x \in \mathbb{Z}$	$x \in \mathbb{Z}$	Example 1.97. Consider the map $(3 \times -): \mathbb{Z} \rightarrow \mathbb{R}$ which sends $x \in \mathbb{Z}$ to $3x$, which we can
fong-spivak-seven-sketches F00430	39	0.889	$3x$	$3x$	Example 1.97. Consider the map $(3 \times -): \mathbb{Z} \rightarrow \mathbb{R}$ which sends $x \in \mathbb{Z}$ to $3x$, which we can
fong-spivak-seven-sketches F00431	39	0.711	$3x \in \mathbb{Z} \subseteq \mathbb{R}$	$3x \in \mathbb{Z} \subseteq \mathbb{R}$	consider as a real number $3x \in \mathbb{Z} \subseteq \mathbb{R}$. Let's find a left adjoint for the map $(3 \times -)$.
fong-spivak-seven-sketches F00432	39	0.711	$3 \times -$	$3 \times -$	consider as a real number $3x \in \mathbb{Z} \subseteq \mathbb{R}$. Let's find a left adjoint for the map $(3 \times -)$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F00433	39	1.000	$\lceil z \rceil$	$\lceil z \rceil$	Write $\lceil z \rceil$ for the smallest natural number above $z \in \mathbb{R}$, and write $\lfloor z \rfloor$ for the largest
fong-spivak-seven-sketches-F00434	39	1.000	$z \in \mathbb{R}$	$z \in \mathbb{R}$	Write $\lceil z \rceil$ for the smallest natural number above $z \in \mathbb{R}$, and write $\lfloor z \rfloor$ for the largest
fong-spivak-seven-sketches-F00435	39	1.000	$\lfloor z \rfloor$	$\lfloor z \rfloor$	Write $\lceil z \rceil$ for the smallest natural number above $z \in \mathbb{R}$, and write $\lfloor z \rfloor$ for the largest
fong-spivak-seven-sketches-F00436	39	0.582	$\lceil 3.14 \rceil = 4$	$\lceil 3.14 \rceil = 4$	integer below $z \in \mathbb{R}$, e.g. $\lceil 3.14 \rceil = 4$ and $\lfloor 3.14 \rfloor = 3$. ^a As the left adjoint $\mathbb{R} \rightarrow \mathbb{Z}$, let's see
fong-spivak-seven-sketches-F00437	39	0.582	$\lfloor 3.14 \rfloor = 3$	$\lfloor 3.14 \rfloor = 3$. ^a	integer below $z \in \mathbb{R}$, e.g. $\lceil 3.14 \rceil = 4$ and $\lfloor 3.14 \rfloor = 3$. ^a As the left adjoint $\mathbb{R} \rightarrow \mathbb{Z}$, let's see
fong-spivak-seven-sketches-F00438	39	0.582	$\mathbb{R} \rightarrow \mathbb{Z}$	$\mathbb{R} \rightarrow \mathbb{Z}$	integer below $z \in \mathbb{R}$, e.g. $\lceil 3.14 \rceil = 4$ and $\lfloor 3.14 \rfloor = 3$. ^a As the left adjoint $\mathbb{R} \rightarrow \mathbb{Z}$, let's see
fong-spivak-seven-sketches-F00439	39	1.000	$\lceil -/3 \rceil$	$\lceil -/3 \rceil$	if $\lceil -/3 \rceil$ works.
fong-spivak-seven-sketches-F00440	39	0.788	$(3 \times -)$	$(3 \times -)$	Success! Thus we have a Galois connection between $\lceil -/3 \rceil$ and $(3 \times -)$.
fong-spivak-seven-sketches-F00441	39	1.000	$\lfloor 3 \rfloor = 3 = \lceil 3 \rceil$	$\lfloor 3 \rfloor = 3 = \lceil 3 \rceil$	monotone. Anyway, $\lfloor 3 \rfloor = 3 = \lceil 3 \rceil$.
fong-spivak-seven-sketches-F00442	39	0.989	$3 \times$	$3 \times$	Exercise 1.98. In Example 1.97 we found a left adjoint for the monotone map $(3 \times$
fong-spivak-seven-sketches-F00443	39	0.945	$-): \mathbb{Z} \rightarrow \mathbb{R}$	$-): \mathbb{Z} \rightarrow \mathbb{R}$	$-): \mathbb{Z} \rightarrow \mathbb{R}$. Now find a right adjoint for the same map, and show it is correct.
fong-spivak-seven-sketches-F00444	39	1.000	$P = Q = \underline{3}$	$P = Q = \underline{3}$	Exercise 1.99. Consider the preorder $P = Q = \underline{3}$.
fong-spivak-seven-sketches-F00445	39	1.000	f, g	f, g	1. Let f, g be the monotone maps shown below:
fong-spivak-seven-sketches-F00446	39	0.999	$1 \leq p, q \leq 3$	$1 \leq p, q \leq 3$	Is it the case that f is left adjoint to g ? Check that for each $1 \leq p, q \leq 3$, one has
fong-spivak-seven-sketches-F00447	39	1.000	$f(p) \leq q$	$f(p) \leq q$	$f(p) \leq q$ iff $p \leq g(q)$.
fong-spivak-seven-sketches-F00448	39	1.000	$p \leq g(q)$	$p \leq g(q)$	$f(p) \leq q$ iff $p \leq g(q)$.
fong-spivak-seven-sketches-F00449	40	1.000	$L: \mathbb{Z} \rightarrow \mathbb{R}$	$L: \mathbb{Z} \rightarrow \mathbb{R}$	1. Does $\lceil -/3 \rceil$ have a left adjoint $L: \mathbb{Z} \rightarrow \mathbb{R}$?
fong-spivak-seven-sketches-F00450	40	1.000	$\text{Prt}(S)$	$\text{Prt}(S)$	Recall from Example 1.52 that we can understand the set $\text{Prt}(S)$ of partitions on a set S
fong-spivak-seven-sketches-F00451	40	1.000	$g: S \rightarrow T$	$g: S \rightarrow T$	Suppose we are given any function $g: S \rightarrow T$. We will show that this function g
fong-spivak-seven-sketches-F00452	40	0.933	$g_!: \text{Prt}(S) \rightarrow \text{Prt}(T): g^*$	$g_!: \text{Prt}(S) \rightarrow \text{Prt}(T): g^*$	induces a Galois connection $g_!: \text{Prt}(S) \rightarrow \text{Prt}(T): g^*$, between preorder of S -partitions
fong-spivak-seven-sketches-F00453	40	0.991	$g_!: \text{Prt}(S) \rightarrow \text{Prt}(T)$	$g_!: \text{Prt}(S) \rightarrow \text{Prt}(T)$	Start with $g: S \rightarrow T$; we first want to understand $g_!: \text{Prt}(S) \rightarrow \text{Prt}(T)$. So start with
fong-spivak-seven-sketches-F00454	40	1.000	$\sim_S$	$\sim_S$	a partition $\sim_S$ of S . To begin the process of obtaining a partition $\sim_T$ on T , say that two
fong-spivak-seven-sketches-F00455	40	1.000	$\sim_T$	$\sim_T$	a partition $\sim_S$ of S . To begin the process of obtaining a partition $\sim_T$ on T , say that two
fong-spivak-seven-sketches-F00456	40	0.997	$t_1, t_2 \in T$	$t_1, t_2 \in T$	elements $t_1, t_2 \in T$ are in the same part, $t_1 \sim_T t_2$, if there exist $s_1, s_2 \in S$ with such
fong-spivak-seven-sketches-F00457	40	0.997	$t_1 \sim_T t_2$	$t_1 \sim_T t_2$	elements $t_1, t_2 \in T$ are in the same part, $t_1 \sim_T t_2$, if there exist $s_1, s_2 \in S$ with such
fong-spivak-seven-sketches-F00458	40	0.990	$s_1 \sim_S s_2$	$s_1 \sim_S s_2$	that $s_1 \sim_S s_2$ and $g(s_1) = t_1$ and $g(s_2) = t_2$. However, the result of doing so will not
fong-spivak-seven-sketches-F00459	40	0.990	$g(s_1) = t_1$	$g(s_1) = t_1$	that $s_1 \sim_S s_2$ and $g(s_1) = t_1$ and $g(s_2) = t_2$. However, the result of doing so will not

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches- F00460	40	■ 0.990	$g\left(s_{\{2\}}\right)=t_{\{2\}}$	$g(s_2) = t_2$	that $s_1 \sim_S s_2$ and $g(s_1) = t_1$ and $g(s_2) = t_2$. However, the result of doing so will not
fong-spivak-seven-sketches- F00461	41	■ 1.000	$t_{\{2\}} \sim_{\{T\}} t_{\{3\}}$	$t_2 \sim_T t_3$	necessarily be transitive—you may get $t_1 \sim_T t_2$ and $t_2 \sim_T t_3$ without $t_1 \sim_T^? t_3$ —and
fong-spivak-seven-sketches- F00462	41	■ 1.000	$t_{\{1\}} \sim_{\{T\}^{\{?\}} t_{\{3\}}$	$t_1 \sim_T^? t_3$	necessarily be transitive—you may get $t_1 \sim_T t_2$ and $t_2 \sim_T t_3$ without $t_1 \sim_T^? t_3$ —and
fong-spivak-seven-sketches- F00463	41	■ 0.522	$g_{\{!\}}(\sim S) := \sim T$	$g_!(\sim S) := \sim T$	pieces (take the transitive closure). The result is $g_!(\sim S) := \sim T$.
fong-spivak-seven-sketches- F00464	41	■ 0.813	$g^{\{*\}}: \operatorname{Prt}(T) \rightarrow \operatorname{Prt}(S)$	$g^*: \operatorname{Prt}(T) \rightarrow \operatorname{Prt}(S)$	Again starting with g , we want to get the right adjoint $g^*: \operatorname{Prt}(T) \rightarrow \operatorname{Prt}(S)$. So start
fong-spivak-seven-sketches- F00465	41	■ 1.000	$g\left(s_{\{1\}}\right) \sim_{\{T\}} g\left(s_{\{2\}}\right)$	$g(s_1) \sim_T g(s_2)$	with a partition $\sim_T$ of T . Get a partition $\sim_S$ on S by saying that $s_1 \sim_S s_2$ iff $g(s_1) \sim_T g(s_2)$.
fong-spivak-seven-sketches- F00466	41	■ 1.000	$g^{\{*\}}\left(\sim_{\{T\}}\right) := \sim_{\{S\}}$	$g^*(\sim_T) := \sim_S$	The result is $g^*(\sim_T) := \sim_S$.
fong-spivak-seven-sketches- F00467	41	■ 1.000	$S=\{1,2,3,4\}, \quad T=\{12,3,4\}$	$S = \{1, 2, 3, 4\}, T = \{12, 3, 4\}$	Example 1.102. Let $S = \{1, 2, 3, 4\}, T = \{12, 3, 4\}$, and $g: S \rightarrow T$ by $g(1) := g(2) := 12$,
fong-spivak-seven-sketches- F00468	41	■ 1.000	$g(1):=g(2):=12$	$g(1) := g(2) := 12$	Example 1.102. Let $S = \{1, 2, 3, 4\}, T = \{12, 3, 4\}$, and $g: S \rightarrow T$ by $g(1) := g(2) := 12$,
fong-spivak-seven-sketches- F00469	41	■ 0.813	$g(3):=3$	$g(3) := 3$	$g(3) := 3$, and $g(4) := 4$. The partition shown left below is translated by $g_!$ to the
fong-spivak-seven-sketches- F00470	41	■ 0.813	$g(4):=4$	$g(4) := 4$	$g(3) := 3$, and $g(4) := 4$. The partition shown left below is translated by $g_!$ to the
fong-spivak-seven-sketches- F00471	41	■ 0.813	$g_{\{!\}}$	$g_!$	$g(3) := 3$, and $g(4) := 4$. The partition shown left below is translated by $g_!$ to the
fong-spivak-seven-sketches- F00472	41	■ 0.999	$c: S \rightarrow P$	$c: S \rightarrow P$	different ones and for each one, call it $c: S \twoheadrightarrow P$, find $g_!(c)$, where S, T , and $g: S \rightarrow T$
fong-spivak-seven-sketches- F00473	41	■ 0.999	$g_{\{!\}}(c)$	$g_!(c)$	different ones and for each one, call it $c: S \twoheadrightarrow P$, find $g_!(c)$, where S, T , and $g: S \rightarrow T$
fong-spivak-seven-sketches- F00474	41	■ 0.999	$S, \quad T$	S, T	different ones and for each one, call it $c: S \twoheadrightarrow P$, find $g_!(c)$, where S, T , and $g: S \rightarrow T$
fong-spivak-seven-sketches- F00475	41	■ 1.000	$g^{\{*\}}$	g^*	Here is a picture of how g^* takes a partition on T and “pulls it back” to a partition on S :
fong-spivak-seven-sketches- F00476	41	■ 1.000	$T=\{12,3,4\}$	$T = \{12, 3, 4\}$	$T = \{12, 3, 4\}$. Using the same S and $g: S \rightarrow T$ as in Example 1.102, determine the
fong-spivak-seven-sketches- F00477	41	■ 1.000	$g^{\{*\}}(c)$	$g^*(c)$	partition $g^*(c)$ on S for each of the five partitions $c: T \twoheadrightarrow P$.
fong-spivak-seven-sketches- F00478	41	■ 1.000	$c: T \rightarrow P$	$c: T \rightarrow P$	partition $g^*(c)$ on S for each of the five partitions $c: T \twoheadrightarrow P$.
fong-spivak-seven-sketches- F00479	42	■ 1.000	$d: T \rightarrow P^{\prime}$	$d: T \rightarrow P'$	2. Choose any coarser partition $d: T \twoheadrightarrow P'$, i.e. where $g_!(c) \leq d$.
fong-spivak-seven-sketches- F00480	42	■ 1.000	$g_{\{!\}}(c) \leq d$	$g_!(c) \leq d$	2. Choose any coarser partition $d: T \twoheadrightarrow P'$, i.e. where $g_!(c) \leq d$.
fong-spivak-seven-sketches- F00481	42	■ 1.000	$e: T \rightarrow Q$	$e: T \rightarrow Q$	3. Choose any non-coarser partition $e: T \twoheadrightarrow Q$, i.e. where $g_!(c) \not\leq e$. (If you can’t do
fong-spivak-seven-sketches- F00482	42	■ 1.000	$g_{\{!\}}(c) \not\leq e$	$g_!(c) \not\leq e$	3. Choose any non-coarser partition $e: T \twoheadrightarrow Q$, i.e. where $g_!(c) \not\leq e$. (If you can’t do
fong-spivak-seven-sketches- F00483	42	■ 1.000	$g^{\{*\}}(d)$	$g^*(d)$	4. Find $g^*(d)$ and $g^*(e)$.
fong-spivak-seven-sketches- F00484	42	■ 1.000	$g^{\{*\}}(e)$	$g^*(e)$	4. Find $g^*(d)$ and $g^*(e)$.
fong-spivak-seven-sketches- F00485	42	■ 0.999	$c \leq g^{\{*\}}(d)$	$c \leq g^*(d)$	$g_!(c) \not\leq e$, we should have $c \leq g^*(d)$ and $c \not\leq g^*(e)$. Show that this is true.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-1 F00486	42	■ 0.999	$c \not\leq g^*(e)$	$c \not\leq g^*(e)$	$g_!(c) \not\leq e$, we should have $c \leq g^*(d)$ and $c \not\leq g^*(e)$. Show that this is true.
fong-spivak-seven-sketches-1 F00487	42	■ 1.000	$q:=f(p)$	$q := f(p)$	<i>Proof.</i> Suppose f is left adjoint to g . Take any $p \in P$, and let $q := f(p)$. By reflexivity,
fong-spivak-seven-sketches-1 F00488	42	■ 1.000	$p \leq g(f(p))$	$p \leq g(f(p))$	this means $p \leq g(f(p))$. The proof that $f(g(q)) \leq q$ is similar.
fong-spivak-seven-sketches-1 F00489	42	■ 1.000	$f(g(q)) \leq q$	$f(g(q)) \leq q$	this means $p \leq g(f(p))$. The proof that $f(g(q)) \leq q$ is similar.
fong-spivak-seven-sketches-1 F00490	42	■ 1.000	$g(f(p)) \leq g(q)$	$g(f(p)) \leq g(q)$	$f(p) \leq q$ iff $p \leq g(q)$. Suppose $f(p) \leq q$; then since g is monotonic, $g(f(p)) \leq g(q)$, but
fong-spivak-seven-sketches-1 F00491	42	■ 1.000	$g^{\{\prime\}}$	g'	That means, for any other right adjoint g' , we have $g(q) \cong g'(q)$ for all $q \in Q$.
fong-spivak-seven-sketches-1 F00492	42	■ 1.000	$g(q) \cong g^{\{\prime\}}(q)$	$g(q) \cong g'(q)$	That means, for any other right adjoint g' , we have $g(q) \cong g'(q)$ for all $q \in Q$.
fong-spivak-seven-sketches-1 F00493	43	■ 1.000	$A \subseteq Q$	$A \subseteq Q$	$g: Q \rightarrow P$. Suppose $A \subseteq Q$ any subset, and let $g(A) := \{g(a) \mid a \in A\}$ be its image.
fong-spivak-seven-sketches-1 F00494	43	■ 1.000	$g(A):=\{g(a) \mid a \in A\}$	$g(A) := \{g(a) \mid a \in A\}$	$g: Q \rightarrow P$. Suppose $A \subseteq Q$ any subset, and let $g(A) := \{g(a) \mid a \in A\}$ be its image.
fong-spivak-seven-sketches-1 F00495	43	■ 1.000	$\bigwedge A \in Q$	$\bigwedge A \in Q$	Then if A has a meet $\bigwedge A \in Q$ then $g(A)$ has a meet $\bigwedge g(A)$ in P , and we have
fong-spivak-seven-sketches-1 F00496	43	■ 1.000	$g(A)$	$g(A)$	Then if A has a meet $\bigwedge A \in Q$ then $g(A)$ has a meet $\bigwedge g(A)$ in P , and we have
fong-spivak-seven-sketches-1 F00497	43	■ 1.000	$\bigwedge g(A)$	$\bigwedge g(A)$	Then if A has a meet $\bigwedge A \in Q$ then $g(A)$ has a meet $\bigwedge g(A)$ in P , and we have
fong-spivak-seven-sketches-1 F00498	43	■ 1.000	$\bigvee A \in P$	$\bigvee A \in P$	any subset that has a join $\bigvee A \in P$, then $f(A)$ has a join $\bigvee f(A)$ in Q , and we have
fong-spivak-seven-sketches-1 F00499	43	■ 1.000	$f(A)$	$f(A)$	any subset that has a join $\bigvee A \in P$, then $f(A)$ has a join $\bigvee f(A)$ in Q , and we have
fong-spivak-seven-sketches-1 F00500	43	■ 1.000	$\bigvee f(A)$	$\bigvee f(A)$	any subset that has a join $\bigvee A \in P$, then $f(A)$ has a join $\bigvee f(A)$ in Q , and we have
fong-spivak-seven-sketches-1 F00501	43	■ 1.000	$m:=\bigwedge A$	$m := \bigwedge A$	f . Let $A \subseteq Q$ be any subset and let $m := \bigwedge A$ be its meet. Then since g is monotone
fong-spivak-seven-sketches-1 F00502	43	■ 1.000	$g(m) \leq g(a)$	$g(m) \leq g(a)$	$g(m) \leq g(a)$ for all $a \in A$, so $g(m)$ is a lower bound for the set $g(A)$. We will be done if
fong-spivak-seven-sketches-1 F00503	43	■ 1.000	$g(m)$	$g(m)$	$g(m) \leq g(a)$ for all $a \in A$, so $g(m)$ is a lower bound for the set $g(A)$. We will be done if
fong-spivak-seven-sketches-1 F00504	43	■ 1.000	$b \leq g(a)$	$b \leq g(a)$	$b \leq g(a)$ and we want to show $b \leq g(m)$. Then by definition of g being a right adjoint
fong-spivak-seven-sketches-1 F00505	43	■ 1.000	$b \leq g(m)$	$b \leq g(m)$	$b \leq g(a)$ and we want to show $b \leq g(m)$. Then by definition of g being a right adjoint
fong-spivak-seven-sketches-1 F00506	43	■ 1.000	$f(b) \leq a$	$f(b) \leq a$	(Definition 1.95), we also have $f(b) \leq a$. This means that $f(b)$ is a lower bound for A in
fong-spivak-seven-sketches-1 F00507	43	■ 1.000	$f(b)$	$f(b)$	(Definition 1.95), we also have $f(b) \leq a$. This means that $f(b)$ is a lower bound for A in
fong-spivak-seven-sketches-1 F00508	43	■ 1.000	$f(b) \leq m$	$f(b) \leq m$	Q . Since the meet m is the greatest lower bound, we have $f(b) \leq m$. Once again using
fong-spivak-seven-sketches-1 F00509	43	■ 1.000	$f(3.9):=4$	$f(3.9) := 4$	as possible but with $f(3.9) := 4$. Both are f and g monotonic, and one can check that g
fong-spivak-seven-sketches-1 F00510	44	■ 1.000	$1 \vee 2=4$	$1 \vee 2 = 4$	is right adjoint to f (see Exercise 1.114). But g does not preserve joins because $1 \vee 2 \neq 4$
fong-spivak-seven-sketches-1 F00511	44	■ 1.000	$g(1) \vee g(2)=1 \vee 2=3.9 \neq 4=g(4)$	$g(1) \vee g(2) = 1 \vee 2 = 3.9 \neq 4 = g(4)$	holds in Q , whereas $g(1) \vee g(2) = 1 \vee 2 = 3.9 \neq 4 = g(4)$ in P .
fong-spivak-seven-sketches-1 F00512	44	■ 1.000	$\left\{q^{\{\prime\}} \in Q \mid p^{\{\prime\}} \leq g(q^{\{\prime\}})\right\} \subseteq \left\{q \in Q \mid p \leq g(q)\right\}$	$\{q' \in Q \mid p' \leq g(q')\} \subseteq \{q \in Q \mid p \leq g(q)\}$	need to show it is monotone. So suppose that $p \leq p'$. Then $\{q' \in Q \mid p' \leq g(q')\} \subseteq \{q \in Q \mid p \leq g(q)\}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-1 F00513	44	■ 1.000	$Q \mid p \leq g(q) \}$	$Q \mid p \leq g(q) \}$	$Q \mid p \leq g(q) \}$. By Proposition 1.91, this implies $f(p) \leq f(p')$. Thus f is monotone.
fong-spivak-seven-sketches-1 F00514	44	■ 1.000	$f(p) \leq f(\text{p}^{\{\text{prime}\}})$	$f(p) \leq f(p')$	$Q \mid p \leq g(q) \}$. By Proposition 1.91, this implies $f(p) \leq f(p')$. Thus f is monotone.
fong-spivak-seven-sketches-1 F00515	44	■ 1.000	$p_{\{0\}} \leq g(\text{f}(\text{f}(\text{p}_{\{0\}})))$	$p_0 \leq g(f(p_0))$	By Proposition 1.111, it suffices to show that $p_0 \leq g(f(p_0))$ and that $f(g(q_0)) \leq q_0$
fong-spivak-seven-sketches-1 F00516	44	■ 1.000	$\text{f}(\text{g}(\text{f}(\text{q}_{\{0\}}))) \leq \text{q}_{\{0\}}$	$f(g(q_0)) \leq q_0$	By Proposition 1.111, it suffices to show that $p_0 \leq g(f(p_0))$ and that $f(g(q_0)) \leq q_0$
fong-spivak-seven-sketches-1 F00517	44	■ 1.000	$p_{\{0\}} \in P$	$p_0 \in P$	for all $p_0 \in P$ and $q_0 \in Q$. For the first, we have
fong-spivak-seven-sketches-1 F00518	44	■ 1.000	$q_{\{0\}} \in Q$	$q_0 \in Q$	for all $p_0 \in P$ and $q_0 \in Q$. For the first, we have
fong-spivak-seven-sketches-1 F00519	44	■ 1.000	$p_{\{0\}}$	p_0	where the first inequality follows from the fact that if p_0 is below every element of a
fong-spivak-seven-sketches-1 F00520	44	■ 1.000	$\text{left}\{\text{q}_{\{0\}}\}\ \subseteq\ \text{left}\{\text{q} \in Q \mid \text{g}(\text{q}_{\{0\}}) \leq \text{q}\}$	$\{q_0\} \subseteq \{q \in Q \mid g(q_0) \leq$	where the first inequality follows from Proposition 1.91 since $\{q_0\} \subseteq \{q \in Q \mid q(q_0) \leq$
fong-spivak-seven-sketches-1 F00521	44	■ 0.999	$g(q) \}$	$g(q) \}$	$g(q) \}$, and the fact that $\bigwedge \{q_0\} = q_0$.
fong-spivak-seven-sketches-1 F00522	44	■ 0.999	$\text{wedge}\{\text{q}_{\{0\}}\} = \text{q}_{\{0\}}$	$\bigwedge \{q_0\} = q_0$	$g(q) \}$, and the fact that $\bigwedge \{q_0\} = q_0$.
fong-spivak-seven-sketches-1 F00523	44	■ 1.000	$\text{f}^{\{*\}}: \text{P}(Y) \rightarrow \text{P}(X)$	$f^*: P(Y) \rightarrow P(X)$	Then we have the monotone map $f^*: P(Y) \rightarrow P(X)$ that category theorists call
fong-spivak-seven-sketches-1 F00524	44	■ 1.000	$B^{\{\text{prime}\}} \subseteq B$	$B' \subseteq B$	“pullback along f .” This map takes a subset $B' \subseteq B$ to its preimage $f^{-1}(B') \subseteq A$: that
fong-spivak-seven-sketches-1 F00525	44	■ 1.000	$\text{f}^{-1}(B^{\{\text{prime}\}}) \subseteq A$	$f^{-1}(B') \subseteq A$	“pullback along f .” This map takes a subset $B' \subseteq B$ to its preimage $f^{-1}(B') \subseteq A$: that
fong-spivak-seven-sketches-1 F00526	44	■ 1.000	$B^{\{\text{prime}\}}$	B'	is, it takes a collection B' of buckets, and tells you all the apples that they contain in
fong-spivak-seven-sketches-1 F00527	45	■ 1.000	$\text{f}_{!}(A)$	$f_!(A)$	The left adjoint $f_!(A)$ is given by the direct image: it maps a subset $A' \subseteq A$ to
fong-spivak-seven-sketches-1 F00528	45	■ 1.000	$A^{\{\text{prime}\}} \subseteq A$	$A' \subseteq A$	The left adjoint $f_!(A)$ is given by the direct image: it maps a subset $A' \subseteq A$ to
fong-spivak-seven-sketches-1 F00529	45	■ 1.000	f_{*}	f_*	The right adjoint f_* maps a subset $A' \subseteq A$ to
fong-spivak-seven-sketches-1 F00530	45	■ 0.971	$\text{f}^{\{*\}}, \text{f}_{!}$	$f^*, f_!$	only apples are from A' . But we did not invent these mappings $f^*, f_!$, and f_* : they were
fong-spivak-seven-sketches-1 F00531	45	■ 1.000	$B_{\{1\}}, B_{\{2\}} \subseteq Y$	$B_1, B_2 \subseteq Y$	1. Choose two different subsets $B_1, B_2 \subseteq Y$ and find $f^*(B_1)$ and $f^*(B_2)$.
fong-spivak-seven-sketches-1 F00532	45	■ 1.000	$\text{f}^{\{*\}}(\text{B}_{\{1\}})$	$f^*(B_1)$	1. Choose two different subsets $B_1, B_2 \subseteq Y$ and find $f^*(B_1)$ and $f^*(B_2)$.
fong-spivak-seven-sketches-1 F00533	45	■ 1.000	$\text{f}^{\{*\}}(\text{B}_{\{2\}})$	$f^*(B_2)$	1. Choose two different subsets $B_1, B_2 \subseteq Y$ and find $f^*(B_1)$ and $f^*(B_2)$.
fong-spivak-seven-sketches-1 F00534	45	■ 1.000	$A_{\{1\}}, A_{\{2\}} \subseteq X$	$A_1, A_2 \subseteq X$	2. Choose two different subsets $A_1, A_2 \subseteq X$ and find $f_!(A_1)$ and $f_!(A_2)$.
fong-spivak-seven-sketches-1 F00535	45	■ 1.000	$\text{f}_{!}(\text{A}_{\{1\}})$	$f_!(A_1)$	2. Choose two different subsets $A_1, A_2 \subseteq X$ and find $f_!(A_1)$ and $f_!(A_2)$.
fong-spivak-seven-sketches-1 F00536	45	■ 1.000	$\text{f}_{!}(\text{A}_{\{2\}})$	$f_!(A_2)$	2. Choose two different subsets $A_1, A_2 \subseteq X$ and find $f_!(A_1)$ and $f_!(A_2)$.
fong-spivak-seven-sketches-1 F00537	45	■ 1.000	$\text{f}_{*}(\text{A}_{\{1\}})$	$f_*(A_1)$	3. With the same $A_1, A_2 \subseteq X$, find $f_*(A_1)$ and $f_*(A_2)$.
fong-spivak-seven-sketches-1 F00538	45	■ 1.000	$\text{f}_{*}(\text{A}_{\{2\}})$	$f_*(A_2)$	3. With the same $A_1, A_2 \subseteq X$, find $f_*(A_1)$ and $f_*(A_2)$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F00539	45	■ 1.000	$f ; g: P \rightarrowtail P$	$f;g : P \rightarrow P$	f and g to arrive at a monotone map $f \circ g: P \rightarrow P$ from preorder P to itself. This
fong-spivak-seven-sketches-F00540	45	■ 0.988	$p \leq (f ; g)(p)$	$p \leq (f;g)(p)$	monotone map has the property that $p \leq (f \circ g)(p)$, and that $(f \circ g \circ f \circ g)(p) \cong (f \circ g)(p)$
fong-spivak-seven-sketches-F00541	45	■ 0.988	$(f ; g ; f ; g)(p) \cong (f ; g)(p)$	$(f;g;f;g)(p) \cong (f;g)(p)$	monotone map has the property that $p \leq (f \circ g)(p)$, and that $(f \circ g \circ f \circ g)(p) \cong (f \circ g)(p)$
fong-spivak-seven-sketches-F00542	45	■ 1.000	$\geq$	$\geq$	$\leq$ and $\geq$.
fong-spivak-seven-sketches-F00543	45	■ 1.000	$j: P \rightarrowtail P$	$j : P \rightarrow P$	Definition 1.120. A closure operator $j: P \rightarrow P$ on a preorder P is a monotone map such
fong-spivak-seven-sketches-F00544	45	■ 0.966	$g \circ f$	$g \circ f$	⁸ The other composite $g \circ f$ satisfies the dual properties: $(g \circ f)(q) \leq q$ and $(g \circ f \circ g \circ f)(q) \cong (g \circ f)(q)$
fong-spivak-seven-sketches-F00545	45	■ 0.966	$(g \circ f)(q) \leq q$	$(g \circ f)(q) \leq q$	⁸ The other composite $g \circ f$ satisfies the dual properties: $(g \circ f)(q) \leq q$ and $(g \circ f \circ g \circ f)(q) \cong (g \circ f)(q)$
fong-spivak-seven-sketches-F00546	45	■ 0.966	$(g \circ f \circ g \circ f)(q) \cong (g \circ f)(q)$	$(g \circ f \circ g \circ f)(q) \cong (g \circ f)(q)$	⁸ The other composite $g \circ f$ satisfies the dual properties: $(g \circ f)(q) \leq q$ and $(g \circ f \circ g \circ f)(q) \cong (g \circ f)(q)$
fong-spivak-seven-sketches-F00547	46	■ 0.999	$p \leq j(p)$	$p \leq j(p)$	(a) $p \leq j(p)$;
fong-spivak-seven-sketches-F00548	46	■ 1.000	$j(j(p)) \cong j(p)$	$j(j(p)) \cong j(p)$	(b) $j(j(p)) \cong j(p)$.
fong-spivak-seven-sketches-F00549	46	■ 1.000	$7+2+3$	$7 + 2 + 3$	expressions. For example, a computer can rewrite the expression $7+2+3$ as the expres-
fong-spivak-seven-sketches-F00550	46	■ 0.986	$j: \text{exp} \rightarrowtail \text{exp}$	$j : \text{exp} \rightarrow \text{exp}$	and reducing it to another expression. So it is a map $j: \text{exp} \rightarrow \text{exp}$. It furthermore is
fong-spivak-seven-sketches-F00551	46	■ 1.000	$j(x)$	$j(x)$	means that if an expression x can be rewritten into expression y , then the reduction $j(x)$
fong-spivak-seven-sketches-F00552	46	■ 1.000	$j(y)$	$j(y)$	can be rewritten into $j(y)$. Moreover, the requirement $x \leq j(x)$ implies that j can only
fong-spivak-seven-sketches-F00553	46	■ 1.000	$x \leq j(x)$	$x \leq j(x)$	can be rewritten into $j(y)$. Moreover, the requirement $x \leq j(x)$ implies that j can only
fong-spivak-seven-sketches-F00554	46	■ 1.000	j	j	can be rewritten into $j(y)$. Moreover, the requirement $x \leq j(x)$ implies that j can only
fong-spivak-seven-sketches-F00555	46	■ 1.000	$j(j(x))=j(x)$	$j(j(x)) = j(x)$	$j(j(x)) = j(x)$ implies if you try to reduce an expression that has already been reduced,
fong-spivak-seven-sketches-F00556	46	■ 0.771	$\{ \}_{j}$	j	fix_j to have elements the fixed points of j ; that is,
fong-spivak-seven-sketches-F00557	46	■ 1.000	$j(p)$	$j(p)$	Note that $j(p)$ is a fixed point for all $p \in P$, since $j(j(p)) \cong j(p)$.
fong-spivak-seven-sketches-F00558	46	■ 0.502	$j: P \rightarrowtail$	$j : P \rightarrow$	We define an adjunction with left adjoint $j: P \rightarrow \text{fix}_j$ sending p to $j(p)$, and right
fong-spivak-seven-sketches-F00559	46	■ 0.502	$\{ \}_{j}$	j	We define an adjunction with left adjoint $j: P \rightarrow \text{fix}_j$ sending p to $j(p)$, and right
fong-spivak-seven-sketches-F00560	46	■ 0.910	$\{ \}_{j} \rightarrowtail P$	$j \rightarrowtail P$	adjoint $g: \text{fix}_j \rightarrow P$ simply the inclusion of the sub-preorder. To see it's really an
fong-spivak-seven-sketches-F00561	46	■ 0.990	$q \in \text{fix}_j$	$q \in \text{fix}_j$	adjunction, we need to see that for any $p \in P$ and $q \in \text{fix}_j$, we have $j(p) \leq q$ if and
fong-spivak-seven-sketches-F00562	46	■ 0.990	$j(p) \leq q$	$j(p) \leq q$	adjunction, we need to see that for any $p \in P$ and $q \in \text{fix}_j$, we have $j(p) \leq q$ if and
fong-spivak-seven-sketches-F00563	46	■ 1.000	$j(p) \leq j(q) \cong q$	$j(p) \leq j(q) \cong q$	transitivity. Conversely, since q is a fixed point, $p \leq q$ implies $j(p) \leq j(q) \cong q$.
fong-spivak-seven-sketches-F00564	47	■ 0.999	$p \Rightarrow q$	$p \Rightarrow q$	$p \Rightarrow q$. Since $p \Rightarrow p$ and since whenever $p \Rightarrow q$ and $q \Rightarrow r$, we also have $p \Rightarrow r$, this is
fong-spivak-seven-sketches-F00565	47	■ 0.999	$p \Rightarrow p$	$p \Rightarrow p$	$p \Rightarrow q$. Since $p \Rightarrow p$ and since whenever $p \Rightarrow q$ and $q \Rightarrow r$, we also have $p \Rightarrow r$, this is

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F00566	47	■ 0.999	$q \rightarrow r$	$q \Rightarrow r$	$p \Rightarrow q$. Since $p \Rightarrow p$ and since whenever $p \Rightarrow q$ and $q \Rightarrow r$, we also have $p \Rightarrow r$, this is
fong-spivak-seven-sketches-F00567	47	■ 0.999	$p \rightarrow r$	$p \Rightarrow r$	$p \Rightarrow q$. Since $p \Rightarrow p$ and since whenever $p \Rightarrow q$ and $q \Rightarrow r$, we also have $p \Rightarrow r$, this is
fong-spivak-seven-sketches-F00568	47	■ 1.000	$p \rightarrow j(p)$	$p \Rightarrow j(p)$	propositions to propositions, for which $p \Rightarrow j(p)$ and $j(j(p)) = j(p)$. An example of a
fong-spivak-seven-sketches-F00569	47	■ 1.000	$j(j(p))=j(p)$	$j(j(p)) = j(p)$	propositions to propositions, for which $p \Rightarrow j(p)$ and $j(j(p)) = j(p)$. An example of a
fong-spivak-seven-sketches-F00570	47	■ 0.966	$p^{\prime}$	p''	Bob is in San Diego, p'' means $B \Rightarrow p$. Let's see why $B \Rightarrow -$ is a closure operator.
fong-spivak-seven-sketches-F00571	47	■ 0.966	$B \rightarrow p$	$B \Rightarrow p$	Bob is in San Diego, p'' means $B \Rightarrow p$. Let's see why $B \Rightarrow -$ is a closure operator.
fong-spivak-seven-sketches-F00572	47	■ 0.966	$B \rightarrow$	$B \Rightarrow -$	Bob is in San Diego, p'' means $B \Rightarrow p$. Let's see why $B \Rightarrow -$ is a closure operator.
fong-spivak-seven-sketches-F00573	47	■ 1.000	$p^{\prime}$	p'	“assuming Bob is in San Diego it is the case that, assuming Bob is in San Diego, p' is
fong-spivak-seven-sketches-F00574	47	■ 0.622	$\mathbf{Rel}(S)$	$\mathbf{Rel}(S)$	Given any set S , there is a set $\mathbf{Rel}(S)$ of binary relations on S . An element $R \in \mathbf{Rel}(S)$
fong-spivak-seven-sketches-F00575	47	■ 0.622	$R \in \mathbf{Rel}(S)$	$R \in \mathbf{Rel}(S)$	Given any set S , there is a set $\mathbf{Rel}(S)$ of binary relations on S . An element $R \in \mathbf{Rel}(S)$
fong-spivak-seven-sketches-F00576	47	■ 1.000	$R \subseteq S \times S$	$R \subseteq S \times S$	is formally a subset $R \subseteq S \times S$. The set $\mathbf{Rel}(S)$ can be given an order via the subset
fong-spivak-seven-sketches-F00577	47	■ 1.000	$R \subseteq R^{\prime}$	$R \subseteq R'$	relation, $R \subseteq R'$, i.e. if whenever $R(s_1, s_2)$ holds then so does $R'(s_1, s_2)$.
fong-spivak-seven-sketches-F00578	47	■ 1.000	$R(s_1, s_2)$	$R(s_1, s_2)$	relation, $R \subseteq R'$, i.e. if whenever $R(s_1, s_2)$ holds then so does $R'(s_1, s_2)$.
fong-spivak-seven-sketches-F00579	47	■ 1.000	$R^{\prime}(s_1, s_2)$	$R'(s_1, s_2)$	relation, $R \subseteq R'$, i.e. if whenever $R(s_1, s_2)$ holds then so does $R'(s_1, s_2)$.
fong-spivak-seven-sketches-F00580	47	■ 0.999	$\mathbf{Rel}(\{1\})$	$\mathbf{Rel}(\{1\})$	For example, the Hasse diagram for $\mathbf{Rel}(\{1\})$ is:
fong-spivak-seven-sketches-F00581	47	■ 0.998	$\mathbf{Rel}(\{1, 2\})$	$\mathbf{Rel}(\{1, 2\})$	<i>Exercise 1.124.</i> Draw the Hasse diagram for the preorder $\mathbf{Rel}(\{1, 2\})$ of all binary
fong-spivak-seven-sketches-F00582	47	■ 0.755	$\{1, 2\}$	$\{1, 2\}$	relations on the set $\{1, 2\}$.
fong-spivak-seven-sketches-F00583	47	■ 1.000	$\mathbf{Pos}(S)$	$\mathbf{Pos}(S)$	For any set S , there is also a set $\mathbf{Pos}(S)$, consisting of all the preorder relations on S .
fong-spivak-seven-sketches-F00584	47	■ 0.961	$\sqsubseteq$	$\sqsubseteq$	In fact there is a preorder structure $\sqsubseteq$ on $\mathbf{Pos}(S)$, again given by inclusion: $\leq$ is below
fong-spivak-seven-sketches-F00585	47	■ 0.982	$\leq^{\prime}$	$\leq'$	$\leq'$ (we'll write $\leq \leq'$) if $a \leq b$ implies $a \leq' b$ for every $a, b \in S$. A preorder of preorder
fong-spivak-seven-sketches-F00586	47	■ 0.982	$\leq \leq \leq^{\prime}$	$\leq \leq \leq'$	$\leq'$ (we'll write $\leq \leq \leq'$) if $a \leq b$ implies $a \leq' b$ for every $a, b \in S$. A preorder of preorder
fong-spivak-seven-sketches-F00587	47	■ 0.982	$a \leq^{\prime} b$	$a \leq' b$	$\leq'$ (we'll write $\leq \leq \leq'$) if $a \leq b$ implies $a \leq' b$ for every $a, b \in S$. A preorder of preorder
fong-spivak-seven-sketches-F00588	47	■ 0.982	$a, b \in S$	$a, b \in S$	$\leq'$ (we'll write $\leq \leq \leq'$) if $a \leq b$ implies $a \leq' b$ for every $a, b \in S$. A preorder of preorder
fong-spivak-seven-sketches-F00589	47	■ 1.000	$\mathbf{Pos}(S) \rightarrow \mathbf{Rel}(S)$	$\mathbf{Pos}(S) \rightarrow \mathbf{Rel}(S)$	$\mathbf{Pos}(S) \rightarrow \mathbf{Rel}(S)$. This is the right adjoint of a Galois connection. Its left adjoint
fong-spivak-seven-sketches-F00590	47	■ 0.977	$\mathrm{Cl}: \mathbf{Rel}(S) \rightarrow \mathbf{Pos}(S)$	$\mathrm{Cl}: \mathbf{Rel}(S) \rightarrow \mathbf{Pos}(S)$	is a monotone map $\mathrm{Cl}: \mathbf{Rel}(S) \rightarrow \mathbf{Pos}(S)$ given by taking any relation R , writing it in
fong-spivak-seven-sketches-F00591	47	■ 1.000	$s \leq s$	$s \leq s$	infix notation using $\leq$, and taking the reflexive and transitive closure, i.e. adding 
fong-spivak-seven-sketches-F00592	47	■ 1.000	$s \leq u$	$s \leq u$	for every s and adding $s \leq u$ whenever $s \leq t$ and $t \leq u$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_1 F00593	47	■ 1.000	$s \leq t$	$s \leq t$	for every s and adding $s \leq u$ whenever $s \leq t$ and $t \leq u$.
fong-spivak-seven-sketches_1 F00594	47	■ 1.000	$t \leq u$	$t \leq u$	for every s and adding $s \leq u$ whenever $s \leq t$ and $t \leq u$.
fong-spivak-seven-sketches_1 F00595	47	■ 1.000	$S=\{1,2,3\}$	$S = \{1, 2, 3\}$	Exercise 1.125. Let $S = \{1, 2, 3\}$. Let's try to understand the adjunction discussed
fong-spivak-seven-sketches_1 F00596	47	■ 1.000	$U(\leq)$	$U(\leq)$	1. Come up with any preorder relation $\leq$ on S , and define $U(\leq)$ to be the subset
fong-spivak-seven-sketches_1 F00597	47	■ 1.000	$U(\leq):=\left\{\left(s_1, s_2\right) \mid s_1 \leq s_2\right\} \subseteq S \times S$	$U(\leq) := \{(s_1, s_2) \mid s_1 \leq s_2\} \subseteq S \times S$	$U(\leq) := \{(s_1, s_2) \mid s_1 \leq s_2\} \subseteq S \times S$ i.e. $U(\leq)$ is the image of $\leq$ under the inclusion
fong-spivak-seven-sketches_1 F00598	48	■ 1.000	$Q \subseteq S \times S$	$Q \subseteq S \times S$	2. Come up with any two binary relations $Q \subseteq S \times S$ and $Q' \subseteq S \times S$ such that
fong-spivak-seven-sketches_1 F00599	48	■ 1.000	$Q' \subseteq S \times S$	$Q' \subseteq S \times S$	2. Come up with any two binary relations $Q \subseteq S \times S$ and $Q' \subseteq S \times S$ such that
fong-spivak-seven-sketches_1 F00600	48	■ 1.000	$Q \subseteq U(\leq)$	$Q \subseteq U(\leq)$	$Q \subseteq U(\leq)$ but $Q' \not\subseteq U(\leq)$. Note that your choice of Q, Q' do not have to come
fong-spivak-seven-sketches_1 F00601	48	■ 1.000	$Q' \not\subseteq U(\leq)$	$Q' \not\subseteq U(\leq)$	$Q \subseteq U(\leq)$ but $Q' \not\subseteq U(\leq)$. Note that your choice of Q, Q' do not have to come
fong-spivak-seven-sketches_1 F00602	48	■ 1.000	Q, Q'	Q, Q'	$Q \subseteq U(\leq)$ but $Q' \not\subseteq U(\leq)$. Note that your choice of Q, Q' do not have to come
fong-spivak-seven-sketches_1 F00603	48	■ 0.973	$\mathrm{Cl}(Q) \sqsubseteq \leq$	$\mathrm{Cl}(Q) \sqsubseteq \leq$	show that $\mathrm{Cl}(Q) \sqsubseteq \leq$, where $\sqsubseteq$ is the order on $\mathbf{Pos}(S)$, defined immediately above
fong-spivak-seven-sketches_1 F00604	48	■ 0.973	$\mathrm{Pos}(S)$	$\mathrm{Pos}(S)$	show that $\mathrm{Cl}(Q) \sqsubseteq \leq$, where $\sqsubseteq$ is the order on $\mathbf{Pos}(S)$, defined immediately above
fong-spivak-seven-sketches_1 F00605	48	■ 0.986	$\mathrm{Cl}(Q') \not\subseteq \leq$	$\mathrm{Cl}(Q') \not\subseteq \leq$	4. Concretely show that $\mathrm{Cl}(Q') \not\subseteq \leq$.
fong-spivak-seven-sketches_1 F00606	48	■ 1.000	$a \otimes b$	$a \otimes b$	idea of a monoidal structure on preorders. This allows us to construct an element $a \otimes b$
fong-spivak-seven-sketches_2 F00609	53	■ 1.000	$d \in [0, \infty]$	$d \in [0, \infty]$	the question of getting from a to b has an answer $d \in [0, \infty]$, a cost.
fong-spivak-seven-sketches_2 F00610	53	■ 1.000	$I \in X$	$I \in X$	(i) an element $I \in X$, called the <i>monoidal unit</i> , and
fong-spivak-seven-sketches_2 F00611	53	■ 0.983	$\otimes: X \times X \rightarrow X$	$\otimes: X \times X \rightarrow X$	(ii) a function $\otimes: X \times X \rightarrow X$, called the <i>monoidal product</i> .
fong-spivak-seven-sketches_2 F00612	53	■ 0.997	$\otimes(x_1, x_2) =$	$\otimes(x_1, x_2) =$	These constituents must satisfy the following properties, where we write $\otimes(x_1, x_2) =$
fong-spivak-seven-sketches_2 F00613	53	■ 1.000	$x_1 \otimes x_2$	$x_1 \otimes x_2$	$x_1 \otimes x_2$:
fong-spivak-seven-sketches_2 F00614	54	■ 1.000	$x_1, x_2, y_1, y_2 \in X$	$x_1, x_2, y_1, y_2 \in X$	(a) for all $x_1, x_2, y_1, y_2 \in X$, if $x_1 \leq y_1$ and $x_2 \leq y_2$, then $x_1 \otimes x_2 \leq y_1 \otimes y_2$,
fong-spivak-seven-sketches_2 F00615	54	■ 1.000	$x_1 \leq y_1$	$x_1 \leq y_1$	(a) for all $x_1, x_2, y_1, y_2 \in X$, if $x_1 \leq y_1$ and $x_2 \leq y_2$, then $x_1 \otimes x_2 \leq y_1 \otimes y_2$,
fong-spivak-seven-sketches_2 F00616	54	■ 1.000	$x_2 \leq y_2$	$x_2 \leq y_2$	(a) for all $x_1, x_2, y_1, y_2 \in X$, if $x_1 \leq y_1$ and $x_2 \leq y_2$, then $x_1 \otimes x_2 \leq y_1 \otimes y_2$,
fong-spivak-seven-sketches_2 F00617	54	■ 1.000	$x_1 \otimes x_2 \leq y_1 \otimes y_2$	$x_1 \otimes x_2 \leq y_1 \otimes y_2$	(a) for all $x_1, x_2, y_1, y_2 \in X$, if $x_1 \leq y_1$ and $x_2 \leq y_2$, then $x_1 \otimes x_2 \leq y_1 \otimes y_2$,
fong-spivak-seven-sketches_2 F00618	54	■ 1.000	$I \otimes x = x$	$I \otimes x = x$	(b) for all $x \in X$, the equations $I \otimes x = x$ and $x \otimes I = x$ hold,
fong-spivak-seven-sketches_2 F00619	54	■ 1.000	$x \otimes I = x$	$x \otimes I = x$	(b) for all $x \in X$, the equations $I \otimes x = x$ and $x \otimes I = x$ hold,
fong-spivak-seven-sketches_2 F00620	54	■ 1.000	$x, y, z \in X$	$x, y, z \in X$	(c) for all $x, y, z \in X$, the equation $(x \otimes y) \otimes z = x \otimes (y \otimes z)$ holds, and

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-1 F00621	54	1.000	$(x \otimes y) \otimes z = x \otimes (y \otimes z)$	$(x \otimes y) \otimes z = x \otimes (y \otimes z)$	(c) for all $x, y, z \in X$, the equation $(x \otimes y) \otimes z = x \otimes (y \otimes z)$ holds, and
fong-spivak-seven-sketches-1 F00622	54	1.000	$x \otimes y = y \otimes x$	$x \otimes y = y \otimes x$	(d) for all $x, y \in X$, the equation $x \otimes y = y \otimes x$ holds.
fong-spivak-seven-sketches-1 F00623	54	0.849	$(X, \leq, I, \otimes)$	$(X, \leq, I, \otimes)$	A preorder equipped with a symmetric monoidal structure, $(X, \leq, I, \otimes)$, is called a
fong-spivak-seven-sketches-1 F00624	54	1.000	I	I	Anyone can propose a set X , an order $\leq$ on X , an element I in X , and a binary
fong-spivak-seven-sketches-1 F00625	54	0.933	$\cong$	$\cong$	Remark 2.3. It is often useful to replace $=$ with $\cong$ throughout Definition 2.2. The result is
fong-spivak-seven-sketches-1 F00626	54	0.585	$\{*\}$	$\{*\}$	units we have seen include I (as in the definition), 0 , 1 , true , false , $\{*\}$, and more.
fong-spivak-seven-sketches-1 F00627	54	0.999	$-5 \leq \sqrt{2}$	$-5 \leq \sqrt{2}$	numbers; e.g. $-5 \leq \sqrt{2}$. We propose 0 as a monoidal unit and $+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ as a
fong-spivak-seven-sketches-1 F00628	54	0.999	$+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$	$+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$	numbers; e.g. $-5 \leq \sqrt{2}$. We propose 0 as a monoidal unit and $+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ as a
fong-spivak-seven-sketches-1 F00629	54	0.902	$(\mathbb{R}, \leq, 0, +)$	$(\mathbb{R}, \leq, 0, +)$	monoidal product. Does $(\mathbb{R}, \leq, 0, +)$ satisfy the conditions of Definition 2.2?
fong-spivak-seven-sketches-1 F00630	54	1.000	$x_1 + x_2 \leq y_1 + y_2$	$x_1 + x_2 \leq y_1 + y_2$	If $x_1 \leq y_1$ and $x_2 \leq y_2$, it is true that $x_1 + x_2 \leq y_1 + y_2$. It is also true that $0 + x = x$
fong-spivak-seven-sketches-1 F00631	54	1.000	$0 + x = x$	$0 + x = x$	If $x_1 \leq y_1$ and $x_2 \leq y_2$, it is true that $x_1 + x_2 \leq y_1 + y_2$. It is also true that $0 + x = x$
fong-spivak-seven-sketches-1 F00632	54	0.989	$x + 0 = x$	$x + 0 = x$	and $x + 0 = x$, that $(x + y) + z = x + (y + z)$, and that $x + y = y + x$. Thus $(\mathbb{R}, \leq, 0, +)$
fong-spivak-seven-sketches-1 F00633	54	0.989	$(x + y) + z = x + (y + z)$	$(x + y) + z = x + (y + z)$	and $x + 0 = x$, that $(x + y) + z = x + (y + z)$, and that $x + y = y + x$. Thus $(\mathbb{R}, \leq, 0, +)$
fong-spivak-seven-sketches-1 F00634	54	0.989	$x + y = y + x$	$x + y = y + x$	and $x + 0 = x$, that $(x + y) + z = x + (y + z)$, and that $x + y = y + x$. Thus $(\mathbb{R}, \leq, 0, +)$
fong-spivak-seven-sketches-1 F00635	54	0.554	M	M	Example 2.6. A monoid consists of a set M , a function $*$: $M \times M \rightarrow M$ called the monoid
fong-spivak-seven-sketches-1 F00636	54	0.554	$M \times M \rightarrow M$	$M \times M \rightarrow M$	Example 2.6. A monoid consists of a set M , a function $*$: $M \times M \rightarrow M$ called the monoid
fong-spivak-seven-sketches-1 F00637	54	1.000	$e \in M$	$e \in M$	multiplication, and an element $e \in M$ called the monoid unit, such that, when you write
fong-spivak-seven-sketches-1 F00638	54	0.944	$*(m, n)$	$*(m, n)$	$*(m, n)$ as $m * n$, i.e. using infix notation, the equations
fong-spivak-seven-sketches-1 F00639	54	0.944	$m * n$	$m * n$	$*(m, n)$ as $m * n$, i.e. using infix notation, the equations
fong-spivak-seven-sketches-1 F00640	54	1.000	$m, n, p \in M$	$m, n, p \in M$	hold for all $m, n, p \in M$. It is called commutative if also $m * n = n * m$.
fong-spivak-seven-sketches-1 F00641	54	1.000	$m * n = n * m$	$m * n = n * m$	hold for all $m, n, p \in M$. It is called commutative if also $m * n = n * m$.
fong-spivak-seven-sketches-1 F00642	54	0.912	$\mathbf{Disc}_S$	$\mathbf{Disc}_S$	Every set S determines a discrete preorder $\mathbf{Disc}_S$ (where $m \leq n$ iff $m = n$; see
fong-spivak-seven-sketches-1 F00643	54	0.912	$m = n$	$m = n$	Every set S determines a discrete preorder $\mathbf{Disc}_S$ (where $m \leq n$ iff $m = n$; see
fong-spivak-seven-sketches-1 F00644	54	1.000	$(M, e, *)$	$(M, e, *)$	Example 1.32), and it is easy to check that if $(M, e, *)$ is a commutative monoid then
fong-spivak-seven-sketches-1 F00645	54	0.770	$(\mathbf{Disc}_M, =, e, *)$	$(\mathbf{Disc}_M, =, e, *)$	$(\mathbf{Disc}_M, =, e, *)$ is a symmetric monoidal preorder.
fong-spivak-seven-sketches-1 F00646	54	0.770	$(M, =, e, *)$	$(M, =, e, *)$	$(\mathbf{Disc}_M, =, e, *)$ is a symmetric monoidal preorder.
fong-spivak-seven-sketches-1 F00647	55	0.896	$(M, *, e)$	$(M, *, e)$	Exercise 2.8. We said it was easy to check that if $(M, *, e)$ is a commutative monoid

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F00648	55	■ 0.447	$\backslash\mathrm{left}_{\{M\}},=, *, e\backslash\mathrm{right})$	$M, =, *, e)$	then $(\mathrm{Disc}_M, =, *, e)$ is a symmetric monoidal preorder. Are we telling the truth? $\diamond$
fong-spivak-seven-sketches F00649	55	■ 1.000	H	H	Let H be the set of all poker hands, where a hand means a choice of five cards from the
fong-spivak-seven-sketches F00650	55	■ 1.000	$h \leq h^{\prime}$	$h \leq h'$	standard 52-card deck. As an order, put $h \leq h'$ if h' beats or equals h in poker.
fong-spivak-seven-sketches F00651	55	■ 1.000	$h^{\prime}$	h'	standard 52-card deck. As an order, put $h \leq h'$ if h' beats or equals h in poker.
fong-spivak-seven-sketches F00652	55	■ 1.000	$\otimes: H \times H \rightarrow H$	$\otimes: H \times H \rightarrow H$	One could propose a monoidal product $\otimes: H \times H \rightarrow H$ by assigning $h_1 \otimes h_2$ to be
fong-spivak-seven-sketches F00653	55	■ 1.000	$h_{\{1\}} \otimes h_{\{2\}}$	$h_1 \otimes h_2$	One could propose a monoidal product $\otimes: H \times H \rightarrow H$ by assigning $h_1 \otimes h_2$ to be
fong-spivak-seven-sketches F00654	55	■ 1.000	$h_{\{1\}}$	h_1	“the best hand one can form out of the ten cards in h_1 and h_2 .” If some cards are in
fong-spivak-seven-sketches F00655	55	■ 1.000	$h_{\{2\}}$	h_2	“the best hand one can form out of the ten cards in h_1 and h_2 .” If some cards are in
fong-spivak-seven-sketches F00656	55	■ 0.355	$\backslash\{2 \odot, 5 \diamond, 6 \diamond, 6 \star, 7 \star\} = \backslash\{2 \diamond, 3 \diamond, 4 \diamond, 5 \diamond, 6 \circ\}$	$\{2\odot, 5\circ, 6\circ, 6\star, 7\star\} = \{2\circ, 3\circ, 4\circ, 5\circ, 6\circ\}$	$\{2\heartsuit, 5\heartsuit, 6\heartsuit, 6\clubsuit, 7\clubsuit\} = \{2\heartsuit, 3\heartsuit, 4\heartsuit, 5\heartsuit, 6\heartsuit\}$, because the latter is the best hand you
fong-spivak-seven-sketches F00657	55	■ 1.000	$h_{\{1\}} \leq i_{\{1\}}$	$h_1 \leq i_1$	could be the case that $h_1 \leq i_1$ and $h_2 \leq i_2$, and yet <i>not</i> be the case that $h_1 \otimes h_2 \leq i_1 \otimes i_2$.
fong-spivak-seven-sketches F00658	55	■ 1.000	$h_{\{2\}} \leq i_{\{2\}}$	$h_2 \leq i_2$	could be the case that $h_1 \leq i_1$ and $h_2 \leq i_2$, and yet <i>not</i> be the case that $h_1 \otimes h_2 \leq i_1 \otimes i_2$.
fong-spivak-seven-sketches F00659	55	■ 1.000	$h_{\{1\}} \otimes h_{\{2\}} \leq i_{\{1\}} \otimes i_{\{2\}}$	$h_1 \otimes h_2 \leq i_1 \otimes i_2$	could be the case that $h_1 \leq i_1$ and $h_2 \leq i_2$, and yet <i>not</i> be the case that $h_1 \otimes h_2 \leq i_1 \otimes i_2$.
fong-spivak-seven-sketches F00660	55	■ 0.925	$h_{\{1\}} \otimes h_{\{2\}} = \{10 \star, \mathrm{J} \star, \mathrm{Q} \star, \mathrm{K} \star, \mathrm{A} \star\}$	$h_1 \otimes h_2 = \{10\star, \mathrm{J}\star, \mathrm{Q}\star, \mathrm{K}\star, \mathrm{A}\star\}$	Here, $h_1 \leq i_1$ and $h_2 \leq i_2$, but $h_1 \otimes h_2 = \{10\clubsuit, \mathrm{J}\clubsuit, \mathrm{Q}\clubsuit, \mathrm{K}\clubsuit, \mathrm{A}\clubsuit\}$ is the best possible
fong-spivak-seven-sketches F00661	55	■ 0.999	$i_{\{1\}} \otimes i_{\{2\}} = \{5 \bullet, 5 \diamond, 6 \diamond, 6 \diamond, \mathrm{Q} \diamond\}$	$i_1 \otimes i_2 = \{5\bullet, 5\circ, 6\circ, 6\circ, \mathrm{Q}\circ\}$	hand and beats $h_1 \otimes h_2 = \{5\clubsuit, 5\heartsuit, 6\heartsuit, 6\heartsuit, \mathrm{Q}\heartsuit\}$.
fong-spivak-seven-sketches F00662	55	■ 0.973	$0, 1, 2$	$0, 1, 2$	We can chain some number of these $\leq$ -relationships—say 0, 1, 2, or 3 of them—together
fong-spivak-seven-sketches F00663	56	■ 0.974	$(x \in X)$	$(x \in X)$	$(x \in X)$, relationship ($\leq$), and combination (using transitivity and $\otimes$), and so do wiring
fong-spivak-seven-sketches F00664	56	■ 0.974	$(\leq)$	$(\leq)$	$(x \in X)$, relationship ($\leq$), and combination (using transitivity and $\otimes$), and so do wiring
fong-spivak-seven-sketches F00665	56	■ 1.000	$x \otimes y$	$x \otimes y$	draw one wire labeled $x \otimes y$.
fong-spivak-seven-sketches F00666	57	■ 0.997	$x_{\{1\}} \otimes x_{\{2\}} \leq y_{\{1\}} \otimes y_{\{2\}} \otimes y_{\{3\}}$	$x_1 \otimes x_2 \leq y_1 \otimes y_2 \otimes y_3$	The validity of the second box corresponds to the inequality $x_1 \otimes x_2 \leq y_1 \otimes y_2 \otimes y_3$.
fong-spivak-seven-sketches F00667	57	■ 0.990	$4 \leq 7$	$4 \leq 7$	And here we express a couple facts about $(\mathbb{R}, \leq, 0, +)$ in this language: $4 \leq 7$ and
fong-spivak-seven-sketches F00668	57	■ 1.000	$2+5 \leq -1+5+3$	$2+5 \leq -1+5+3$	$2+5 \leq -1+5+3$.
fong-spivak-seven-sketches F00669	58	■ 1.000	$x_{\{1\}} \otimes x_{\{2\}} \leq$	$x_1 \otimes x_2 \leq$	structure (Definition 2.2). Property (a) says that if $x_1 \leq y_1$ and $x_2 \leq y_2$ then $x_1 \otimes x_2 \leq$
fong-spivak-seven-sketches F00670	58	■ 1.000	$y_{\{1\}} \otimes y_{\{2\}}$	$y_1 \otimes y_2$	$y_1 \otimes y_2$. This corresponds to the idea that stacking any two valid boxes in parallel is
fong-spivak-seven-sketches F00671	59	■ 0.985	$\mathcal{X} = (X, \leq, I, \otimes)$	$\mathcal{X} = (X, \leq, I, \otimes)$	Wiring diagrams as graphical proofs Given a monoidal preorder $\mathcal{X} = (X, \leq, I, \otimes)$, a
fong-spivak-seven-sketches F00672	59	■ 0.381	$\mathcal{X}$	$\mathcal{X}$	wiring diagram is a graphical proof of something about $\mathcal{X}$. Each box in the diagram
fong-spivak-seven-sketches F00673	59	■ 1.000	$v+w+u$	$v+w+u$	technically there is no such thing as $v+w+u$, for example. Instead, there is $(v+w)+u$
fong-spivak-seven-sketches F00674	59	■ 1.000	$(v+w)+u$	$(v+w)+u$	technically there is no such thing as $v+w+u$, for example. Instead, there is $(v+w)+u$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_60 F00675	59	■ 1.000	$v + (w + u)$	$v + (w + u)$	and $v + (w + u)$, and so on.
fong-spivak-seven-sketches_61 F00676	61	■ 1.000	$2 \mathrm{H}_{2} \mathrm{O} + 2 \mathrm{Na}$	$2\mathrm{H}_2\mathrm{O} + 2\mathrm{Na}$	The collection on the left, $2\mathrm{H}_2\mathrm{O} + 2\mathrm{Na}$ is called the <i>reactant</i> , and the collection on the
fong-spivak-seven-sketches_61 F00677	61	■ 1.000	$2 \mathrm{NaOH} + \mathrm{H}_{2}$	$2\mathrm{NaOH} + \mathrm{H}_2$	right, $2\mathrm{NaOH} + \mathrm{H}_2$ is called the <i>product</i> .
fong-spivak-seven-sketches_61 F00678	61	■ 0.997	$\in$	$\in$	of atoms and molecules, sometimes called <i>materials</i> . So we have $\mathrm{NaCl} \in \mathrm{Mat}$ and
fong-spivak-seven-sketches_61 F00679	61	■ 0.543	$4 \mathrm{H}_{2} \mathrm{O} + 6 \mathrm{Ne} \in$	$4\mathrm{H}_2\mathrm{O} + 6\mathrm{Ne} \in$	$4\mathrm{H}_2\mathrm{O} + 6\mathrm{Ne} \in \mathrm{Mat}$.
fong-spivak-seven-sketches_61 F00680	61	■ 1.000	k	k	Here k is the catalyst because it is found both in the reactant and the product of the
fong-spivak-seven-sketches_61 F00681	61	■ 1.000	$x + y \rightarrow z$	$x + y \rightarrow z$	reaction. It is said to catalyze the reaction $x + y \rightarrow z$. The idea is that the reaction
fong-spivak-seven-sketches_61 F00682	61	■ 1.000	$x + y + k \rightarrow z + k$	$x + y + k \rightarrow z + k$	corresponds to the composite reaction $x + y + k \rightarrow z + k$.
fong-spivak-seven-sketches_62 F00683	62	■ 1.000	$x \leq I$	$x \leq I$	(e) $x \leq I$ for all $x \in X$.
fong-spivak-seven-sketches_62 F00684	62	■ 1.000	$x \rightarrow 0$	$x \rightarrow 0$	chemistry section, we would write instead $x \rightarrow 0$: any x yields nothing. But this is
fong-spivak-seven-sketches_63 F00685	63	■ 1.000	10^{23}	10^{23}	manipulate—each item produced. In chemistry, when you have 10^{23} of substance A
fong-spivak-seven-sketches_64 F00686	64	■ 1.000	$x \leq x + x$	$x \leq x + x$	(f) $x \leq x + x$ for all $x \in X$.
fong-spivak-seven-sketches_64 F00687	64	■ 0.904	$=0$	$= 0$	false = 0 and true = 1, then $\wedge$ corresponds to the usual multiplication operation $*$.
fong-spivak-seven-sketches_64 F00688	64	■ 0.904	$=1$	$= 1$	false = 0 and true = 1, then $\wedge$ corresponds to the usual multiplication operation $*$.
fong-spivak-seven-sketches_64 F00689	64	■ 0.299	$(\mathbb{B}, \leq$	$= (\mathbb{B}, \leq$	preorder which we denote Bool := $(\mathbb{B}, \leq, \text{true}, \wedge)$.
fong-spivak-seven-sketches_64 F00690	64	■ 0.299	$\Lambda)$	$\Lambda)$	preorder which we denote Bool := $(\mathbb{B}, \leq, \text{true}, \wedge)$.
fong-spivak-seven-sketches_64 F00691	64	■ 0.986	$\mathcal{V} = (V, \leq, I, \otimes)$	$\mathcal{V} = (V, \leq, I, \otimes)$	monoidal preorder $\mathcal{V} = (V, \leq, I, \otimes)$ means “letting $\mathcal{V}$ structure the question of getting
fong-spivak-seven-sketches_64 F00692	64	■ 0.928	$\otimes$	$\otimes$	relation $\leq$, the monoidal unit I , and the monoidal product $\otimes$ —play a role in how
fong-spivak-seven-sketches_65 F00693	65	■ 0.182	$(\mathbb{B}, \leq$	$(\mathbb{B}, \leq$	For example, let’s look at the case of Bool = $(\mathbb{B}, \leq, \text{true}, \wedge)$. The fact that its
fong-spivak-seven-sketches_65 F00694	65	■ 0.182	$\wedge)$	$\wedge)$	For example, let’s look at the case of Bool = $(\mathbb{B}, \leq, \text{true}, \wedge)$. The fact that its
fong-spivak-seven-sketches_65 F00695	65	■ 0.666	$\mathbb{B}, \leq$	$\mathbb{B}, \leq$	<i>Exercise 2.29.</i> Let $(\mathbb{B}, \leq)$ be as above, but now consider the monoidal product to be $\vee$
fong-spivak-seven-sketches_65 F00696	65	■ 0.991	$(\mathbb{N}, \leq)$	$(\mathbb{N}, \leq)$	preorder $(\mathbb{N}, \leq)$ of natural numbers, where $\leq$ is the usual ordering ($0 \leq 1$ and $5 \leq 16$).
fong-spivak-seven-sketches_65 F00697	65	■ 0.991	$5 \leq 16$	$5 \leq 16$	preorder $(\mathbb{N}, \leq)$ of natural numbers, where $\leq$ is the usual ordering ($0 \leq 1$ and $5 \leq 16$).
fong-spivak-seven-sketches_65 F00698	65	■ 0.994	$6 + 4 = 10$	$6 + 4 = 10$	where the monoidal unit is 0 and the monoidal product is $+$, i.e. $6 + 4 = 10$. It is easy
fong-spivak-seven-sketches_65 F00699	65	■ 0.992	$*$	$*$	product is $*$, i.e. $6 * 4 = 24$. What should the monoidal unit be?
fong-spivak-seven-sketches_65 F00700	65	■ 0.992	$6 * 4 = 24$	$6 * 4 = 24$	product is $*$, i.e. $6 * 4 = 24$. What should the monoidal unit be?
fong-spivak-seven-sketches_65 F00701	65	■ 1.000	$m \mid n$	$m \mid n$	a “divisibility” order on $\mathbb{N}$: we write $m \mid n$ to mean that m divides into n without

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F00702	65	■ 1.000	$1 \mid m$	$1 \mid m$	remainder. So $1 \mid m$ for all m and $4 \nmid 12$.
fong-spivak-seven-sketches F00703	65	■ 1.000	$4 \mid 12$	$4 \mid 12$	remainder. So $1 \mid m$ for all m and $4 \nmid 12$.
fong-spivak-seven-sketches F00704	65	■ 0.995	$(\mathbb{N}, \mid)$	$(\mathbb{N}, \mid)$	There is a monoidal structure on $(\mathbb{N}, \mid)$, where the monoidal unit is 1 and the
fong-spivak-seven-sketches F00705	65	■ 0.912	$x_1 \mid y_1$	$x_1 \mid y_1$	monoidal product is $*$, i.e. $6 * 4 = 24$. Then if $x_1 \mid y_1$ and $x_2 \mid y_2$, then $(x_1 * x_2) \mid (y_1 * y_2)$.
fong-spivak-seven-sketches F00706	65	■ 0.912	$x_2 \mid y_2$	$x_2 \mid y_2$	monoidal product is $*$, i.e. $6 * 4 = 24$. Then if $x_1 \mid y_1$ and $x_2 \mid y_2$, then $(x_1 * x_2) \mid (y_1 * y_2)$.
fong-spivak-seven-sketches F00707	65	■ 0.912	$\left(x_1 * x_2\right) \mid \left(y_1 * y_2\right)$	$(x_1 * x_2) \mid (y_1 * y_2)$	monoidal product is $*$, i.e. $6 * 4 = 24$. Then if $x_1 \mid y_1$ and $x_2 \mid y_2$, then $(x_1 * x_2) \mid (y_1 * y_2)$.
fong-spivak-seven-sketches F00708	65	■ 1.000	$p_1, p_2 \in \mathbb{N}$	$p_1, p_2 \in \mathbb{N}$	Indeed, if there is some $p_1, p_2 \in \mathbb{N}$ such that $x_1 * p_1 = y_1$ and $x_2 * p_2 = y_2$, then
fong-spivak-seven-sketches F00709	65	■ 1.000	$x_1 * p_1 = y_1$	$x_1 * p_1 = y_1$	Indeed, if there is some $p_1, p_2 \in \mathbb{N}$ such that $x_1 * p_1 = y_1$ and $x_2 * p_2 = y_2$, then
fong-spivak-seven-sketches F00710	65	■ 1.000	$x_2 * p_2 = y_2$	$x_2 * p_2 = y_2$	Indeed, if there is some $p_1, p_2 \in \mathbb{N}$ such that $x_1 * p_1 = y_1$ and $x_2 * p_2 = y_2$, then
fong-spivak-seven-sketches F00711	65	■ 1.000	$\left(p_1 * p_2\right) * \left(x_1 * x_2\right) = y_1 * y_2$	$(p_1 * p_2) * (x_1 * x_2) = y_1 * y_2$	$p_1 * p_2 * (x_1 * x_2) = y_1 * y_2$.
fong-spivak-seven-sketches F00712	66	■ 1.000	$\mathrm{P}(S)$	$P(S)$	Exercise 2.35. Let S be a set and let $P(S)$ be its power set, the set of all subsets of
fong-spivak-seven-sketches F00713	66	■ 1.000	$\varnothing \subseteq S$	$\emptyset \subseteq S$	S , including the empty subset, $\emptyset \subseteq S$, and the “everything” subset, $S \subseteq S$. We can
fong-spivak-seven-sketches F00714	66	■ 1.000	$S \subseteq S$	$S \subseteq S$	S , including the empty subset, $\emptyset \subseteq S$, and the “everything” subset, $S \subseteq S$. We can
fong-spivak-seven-sketches F00715	66	■ 1.000	$A \subseteq B$	$A \subseteq B$	give $P(S)$ an order: $A \leq B$ is given by the subset relation $A \subseteq B$, as discussed in
fong-spivak-seven-sketches F00716	66	■ 1.000	$A \cap B$	$A \cap B$	unit S and monoidal product given by intersection $A \cap B$.
fong-spivak-seven-sketches F00717	66	■ 0.902	$\{\}^{\mathbb{N}}$	$\mathbb{N}$	Exercise 2.36. Let $\mathrm{Prop}^{\mathbb{N}}$ denote the set of all mathematical statements one can make
fong-spivak-seven-sketches F00718	66	■ 0.901	$\operatorname{Prop}^{\mathbb{N}}$	$\mathrm{Prop}^{\mathbb{N}}$	if and only if the other is true. For example “ n is prime” is an element of $\mathrm{Prop}^{\mathbb{N}}$, and so
fong-spivak-seven-sketches F00719	66	■ 0.993	$n=2$	$n = 2$	are “ $n = 2$ ” and “ $n \geq 11$.” The statements “ $n + 2 = 5$ ” and “ n is the least odd prime”
fong-spivak-seven-sketches F00720	66	■ 0.993	$n \geq 11$	$n \geq 11$	are “ $n = 2$ ” and “ $n \geq 11$.” The statements “ $n + 2 = 5$ ” and “ n is the least odd prime”
fong-spivak-seven-sketches F00721	66	■ 0.993	$n+2=5$	$n + 2 = 5$	are “ $n = 2$ ” and “ $n \geq 11$.” The statements “ $n + 2 = 5$ ” and “ n is the least odd prime”
fong-spivak-seven-sketches F00722	66	■ 0.992	$P, Q \in \operatorname{Prop}^{\mathbb{N}}$	$P, Q \in \mathrm{Prop}^{\mathbb{N}}$	are considered the same. Given $P, Q \in \mathrm{Prop}^{\mathbb{N}}$, we say $P \leq Q$ if for all $n \in \mathbb{N}$, whenever
fong-spivak-seven-sketches F00723	66	■ 0.992	$P \leq Q$	$P \leq Q$	are considered the same. Given $P, Q \in \mathrm{Prop}^{\mathbb{N}}$, we say $P \leq Q$ if for all $n \in \mathbb{N}$, whenever
fong-spivak-seven-sketches F00724	66	■ 1.000	$P(n)$	$P(n)$	$P(n)$ is true, so is $Q(n)$.
fong-spivak-seven-sketches F00725	66	■ 1.000	$Q(n)$	$Q(n)$	$P(n)$ is true, so is $Q(n)$.
fong-spivak-seven-sketches F00726	66	■ 1.000	$[0, \infty]$	$[0, \infty]$	Example 2.37 (Lawvere’s monoidal preorder, Cost). Let $[0, \infty]$ denote the set of non-
fong-spivak-seven-sketches F00727	66	■ 0.999	$0, 1, 15.33 \overline{3}$	$0, 1, 15.33\overline{3}$	negative real numbers—such as $0, 1, 15.33\overline{3}$, and 2π —together with ∞ . Consider the
fong-spivak-seven-sketches F00728	66	■ 0.999	2π	2π	negative real numbers—such as $0, 1, 15.33\overline{3}$, and 2π —together with ∞ . Consider the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F00729	66	■ 0.999	∞	∞	negative real numbers—such as 0, 1, 15.333, and 2π —together with ∞ . Consider the
fong-spivak-seven-sketches-F00730	66	■ 0.725	$([0, \infty], \geq)$	$([0, \infty], \geq)$	preorder $([0, \infty], \geq)$, with the usual notion of $\geq$, where of course $\infty \geq x$ for all $x \in [0, \infty]$.
fong-spivak-seven-sketches-F00731	66	■ 0.725	$\infty \geq x$	$\infty \geq x$	preorder $([0, \infty], \geq)$, with the usual notion of $\geq$, where of course $\infty \geq x$ for all $x \in [0, \infty]$.
fong-spivak-seven-sketches-F00732	66	■ 0.725	$x \in [0, \infty]$	$x \in [0, \infty]$	preorder $([0, \infty], \geq)$, with the usual notion of $\geq$, where of course $\infty \geq x$ for all $x \in [0, \infty]$.
fong-spivak-seven-sketches-F00733	66	■ 0.983	$x + \infty = \infty$	$x + \infty = \infty$	the monoidal product is $+$. In particular, $0 = \infty = x$ for any $x \in [0, \infty]$. Let's call this
fong-spivak-seven-sketches-F00734	67	■ 0.999	$(X, \leq)^{\mathrm{op}} := (X, \geq)$	$(X, \leq)^{\mathrm{op}} := (X, \geq)$	flip the order: $(X, \leq)^{\mathrm{op}} := (X, \geq)$; see Example 1.58. Proposition 2.38 says that if the
fong-spivak-seven-sketches-F00735	67	■ 0.955	$\mathcal{X} = (X, \leq)$	$\mathcal{X} = (X, \leq)$	Proposition 2.38. Suppose $\mathcal{X} = (X, \leq)$ is a preorder and $\mathcal{X}^{\mathrm{op}} = (X, \geq)$ is its opposite. If
fong-spivak-seven-sketches-F00736	67	■ 0.955	$\mathcal{X}^{\mathrm{op}} = (X, \geq)$	$\mathcal{X}^{\mathrm{op}} = (X, \geq)$	Proposition 2.38. Suppose $\mathcal{X} = (X, \leq)$ is a preorder and $\mathcal{X}^{\mathrm{op}} = (X, \geq)$ is its opposite. If
fong-spivak-seven-sketches-F00737	67	■ 0.930	$(X, \leq I, \otimes)$	$(X, \leq I, \otimes)$	$(X, \leq, I, \otimes)$ is a symmetric monoidal preorder then so is its opposite, $(X, \geq, I, \otimes)$.
fong-spivak-seven-sketches-F00738	67	■ 0.930	$(X, \geq I, \otimes)$	$(X, \geq, I, \otimes)$	$(X, \leq, I, \otimes)$ is a symmetric monoidal preorder then so is its opposite, $(X, \geq, I, \otimes)$.
fong-spivak-seven-sketches-F00739	67	■ 0.838	$x_1 \geq y_1$	$x_1 \geq y_1$	<i>Proof.</i> Let's first check monotonicity. Suppose $x_1 \geq y_1$ and $x_2 \geq y_2$ in $\mathcal{X}^{\mathrm{op}}$; we need to
fong-spivak-seven-sketches-F00740	67	■ 0.838	$x_2 \geq y_2$	$x_2 \geq y_2$	<i>Proof.</i> Let's first check monotonicity. Suppose $x_1 \geq y_1$ and $x_2 \geq y_2$ in $\mathcal{X}^{\mathrm{op}}$; we need to
fong-spivak-seven-sketches-F00741	67	■ 0.838	$\mathcal{X}^{\mathrm{op}}$	$\mathcal{X}^{\mathrm{op}}$	<i>Proof.</i> Let's first check monotonicity. Suppose $x_1 \geq y_1$ and $x_2 \geq y_2$ in $\mathcal{X}^{\mathrm{op}}$; we need to
fong-spivak-seven-sketches-F00742	67	■ 0.999	$x_1 \otimes x_2 \geq y_1 \otimes y_2$	$x_1 \otimes x_2 \geq y_1 \otimes y_2$	show that $x_1 \otimes x_2 \geq y_1 \otimes y_2$. But by definition of opposite order, we have $y_1 \leq x_1$ and
fong-spivak-seven-sketches-F00743	67	■ 0.999	$y_1 \leq x_1$	$y_1 \leq x_1$	show that $x_1 \otimes x_2 \geq y_1 \otimes y_2$. But by definition of opposite order, we have $y_1 \leq x_1$ and
fong-spivak-seven-sketches-F00744	67	■ 0.558	$y_2 \leq x_2$	$y_2 \leq x_2$	$y_2 \leq x_2$ in $\mathcal{X}$, and thus $y_1 \otimes y_2 \leq x_1 \otimes x_2$ in $\mathcal{X}$. Thus indeed $x_1 \otimes x_2 \geq y_1 \otimes y_2$ in $\mathcal{X}^{\mathrm{op}}$.
fong-spivak-seven-sketches-F00745	67	■ 0.558	$y_1 \otimes y_2 \leq x_1 \otimes x_2$	$y_1 \otimes y_2 \leq x_1 \otimes x_2$	$y_2 \leq x_2$ in $\mathcal{X}$, and thus $y_1 \otimes y_2 \leq x_1 \otimes x_2$ in $\mathcal{X}$. Thus indeed $x_1 \otimes x_2 \geq y_1 \otimes y_2$ in $\mathcal{X}^{\mathrm{op}}$.
fong-spivak-seven-sketches-F00746	67	■ 0.996	$\{ \}^{\mathrm{op}}$	$^{\mathrm{op}}$	Cost ^{op} is too.
fong-spivak-seven-sketches-F00747	67	■ 1.000	$x \cong x'$	$x \cong x'$	relation $\cong$ on X , where $x \cong x'$ iff both $x \leq x'$ and $x' \leq x$.
fong-spivak-seven-sketches-F00748	67	■ 1.000	$x \leq x'$	$x \leq x'$	relation $\cong$ on X , where $x \cong x'$ iff both $x \leq x'$ and $x' \leq x$.
fong-spivak-seven-sketches-F00749	67	■ 1.000	$x' \leq x$	$x' \leq x$	relation $\cong$ on X , where $x \cong x'$ iff both $x \leq x'$ and $x' \leq x$.
fong-spivak-seven-sketches-F00750	67	■ 1.000	$\mathcal{P} = (P, \leq_P, I_P, \otimes_P)$	$\mathcal{P} = (P, \leq_P, I_P, \otimes_P)$	Definition 2.41. Let $\mathcal{P} = (P, \leq_P, I_P, \otimes_P)$ and $\mathcal{Q} = (Q, \leq_Q, I_Q, \otimes_Q)$ be monoidal preorders.
fong-spivak-seven-sketches-F00751	67	■ 1.000	$\mathcal{Q} = (Q, \leq_Q, I_Q, \otimes_Q)$	$\mathcal{Q} = (Q, \leq_Q, I_Q, \otimes_Q)$	Definition 2.41. Let $\mathcal{P} = (P, \leq_P, I_P, \otimes_P)$ and $\mathcal{Q} = (Q, \leq_Q, I_Q, \otimes_Q)$ be monoidal preorders.
fong-spivak-seven-sketches-F00752	67	■ 0.995	$\mathcal{P}$	$\mathcal{P}$	A monoidal monotone from $\mathcal{P}$ to $\mathcal{Q}$ is a monotone map $f: (P, \leq_P) \rightarrow (Q, \leq_Q)$, satisfying
fong-spivak-seven-sketches-F00753	67	■ 1.000	$I_Q \leq_Q f(I_P)$	$I_Q \leq_Q f(I_P)$	(a) $I_Q \leq_Q f(I_P)$, and
fong-spivak-seven-sketches-F00754	67	■ 1.000	$f(p_1) \otimes_Q f(p_2) \leq_Q f(p_1 \otimes_P p_2)$	$f(p_1) \otimes_Q f(p_2) \leq_Q f(p_1 \otimes_P p_2)$	(b) $f(p_1) \otimes_Q f(p_2) \leq_Q f(p_1 \otimes_P p_2)$
fong-spivak-seven-sketches-F00755	67	■ 1.000	$p_1, p_2 \in P$	$p_1, p_2 \in P$	for all $p_1, p_2 \in P$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F00756	68	■ 1.000	$I_{\{0\}} \cong f(I_P)$	$I_Q \cong f(I_P)$	(a') $I_Q \cong f(I_P)$, and
fong-spivak-seven-sketches F00757	68	■ 1.000	$f(p_1) \otimes_Q f(p_2) \cong f(p_1 \otimes_P p_2)$	$f(p_1) \otimes_Q f(p_2) \cong f(p_1 \otimes_P p_2)$	(b') $f(p_1) \otimes_Q f(p_2) \cong f(p_1 \otimes_P p_2)$;
fong-spivak-seven-sketches F00758	68	■ 0.895	$I_{\{0\}} = f(I_P)$	$I_Q = f(I_P)$	(a'') $I_Q = f(I_P)$, and
fong-spivak-seven-sketches F00759	68	■ 0.995	$f(p_1) \otimes_Q f(p_2) = f(p_1 \otimes_P p_2)$	$f(p_1) \otimes_Q f(p_2) = f(p_1 \otimes_P p_2)$	(b'') $f(p_1) \otimes_Q f(p_2) = f(p_1 \otimes_P p_2)$.
fong-spivak-seven-sketches F00760	68	■ 1.000	$i: (\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{R}, \leq, 0, +)$	$i: (\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{R}, \leq, 0, +)$	Example 2.42. There is a monoidal monotone $i: (\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{R}, \leq, 0, +)$, where $i(n) =$
fong-spivak-seven-sketches F00761	68	■ 1.000	$i(n) =$	$i(n) =$	Example 2.42. There is a monoidal monotone $i: (\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{R}, \leq, 0, +)$, where $i(n) =$
fong-spivak-seven-sketches F00762	68	■ 1.000	$i(m) \leq i(n)$	$i(m) \leq i(n)$	n for all $n \in \mathbb{N}$. It is clearly monotonic, $m \leq n$ implies $i(m) \leq i(n)$. It is even strict
fong-spivak-seven-sketches F00763	68	■ 1.000	$i(0) = 0$	$i(0) = 0$	monoidal because $i(0) = 0$ and $i(m + n) = i(m) + i(n)$.
fong-spivak-seven-sketches F00764	68	■ 1.000	$i(m+n) = i(m) + i(n)$	$i(m + n) = i(m) + i(n)$	monoidal because $i(0) = 0$ and $i(m + n) = i(m) + i(n)$.
fong-spivak-seven-sketches F00765	68	■ 0.998	$f: (\mathbb{R}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq, 0, +)$	$f: (\mathbb{R}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq, 0, +)$	There is also a monoidal monotone $f: (\mathbb{R}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq, 0, +)$ going the other
fong-spivak-seven-sketches F00766	68	■ 1.000	$f(x) := \lfloor x \rfloor$	$f(x) := \lfloor x \rfloor$	way. Here $f(x) := \lfloor x \rfloor$ is the floor function, e.g. $f(3.14) = 3$. It is monotonic because
fong-spivak-seven-sketches F00767	68	■ 1.000	$f(3.14) = 3$	$f(3.14) = 3$	way. Here $f(x) := \lfloor x \rfloor$ is the floor function, e.g. $f(3.14) = 3$. It is monotonic because
fong-spivak-seven-sketches F00768	68	■ 1.000	$f(0) = 0$	$f(0) = 0$	$x \leq y$ implies $f(x) \leq f(y)$. Also $f(0) = 0$ and $f(x) + f(y) \leq f(x + y)$, so it is a monoidal
fong-spivak-seven-sketches F00769	68	■ 1.000	$f(x) + f(y) \leq f(x + y)$	$f(x) + f(y) \leq f(x + y)$	$x \leq y$ implies $f(x) \leq f(y)$. Also $f(0) = 0$ and $f(x) + f(y) \leq f(x + y)$, so it is a monoidal
fong-spivak-seven-sketches F00770	68	■ 1.000	$f(0.5) + f(0.5) \neq f(0.5 + 0.5)$	$f(0.5) + f(0.5) \neq f(0.5 + 0.5)$	monotone. But it is not strict or even strong because $f(0.5) + f(0.5) \neq f(0.5 + 0.5)$.
fong-spivak-seven-sketches F00771	68	■ 0.754	$=([0, \infty], \geq, 0, +)$	$=([0, \infty], \geq, 0, +)$	Recall Bool = $(\mathbb{B}, \leq, \text{true}, \wedge)$ from Example 2.27 and Cost = $([0, \infty], \geq, 0, +)$ from
fong-spivak-seven-sketches F00772	68	■ 0.905	$:= \infty$	$:= \infty$	Example 2.37. There is a monoidal monotone $g: \mathbf{Bool} \rightarrow \mathbf{Cost}$, given by $g(\text{false}) := \infty$
fong-spivak-seven-sketches F00773	68	■ 0.998	$:= 0$	$:= 0$	and $g(\text{true}) := 0$.
fong-spivak-seven-sketches F00774	68	■ 0.683	$g: (\mathbb{B}, \leq$	$g: (\mathbb{B}, \leq$	1. Check that the map $g: (\mathbb{B}, \leq, \text{true}, \wedge) \rightarrow ([0, \infty], \geq, 0, +)$ presented above indeed
fong-spivak-seven-sketches F00775	68	■ 0.683	$\wedge) \rightarrow ([0, \infty], \geq, 0, +)$	$\wedge) \rightarrow ([0, \infty], \geq, 0, +)$	1. Check that the map $g: (\mathbb{B}, \leq, \text{true}, \wedge) \rightarrow ([0, \infty], \geq, 0, +)$ presented above indeed
fong-spivak-seven-sketches F00776	68	■ 1.000	$d, u: [0, \infty] \rightarrow \mathbb{B}$	$d, u: [0, \infty] \rightarrow \mathbb{B}$	functions $d, u: [0, \infty] \rightarrow \mathbb{B}$ defined as follows:
fong-spivak-seven-sketches F00777	68	■ 1.000	u	u	4. Is u monotonic?
fong-spivak-seven-sketches F00778	69	■ 0.929	$(\mathbb{N}, \leq, 1, *)$	$(\mathbb{N}, \leq, 1, *)$	1. Is $(\mathbb{N}, \leq, 1, *)$ a monoidal preorder, where $*$ is the usual multiplication of natural
fong-spivak-seven-sketches F00779	69	■ 0.998	$(\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq$	$(\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq$	2. If not, why not? If so, does there exist a monoidal monotone $(\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq$
fong-spivak-seven-sketches F00780	69	■ 0.833	$, 1, *)$	$, 1, *)$	$, 1, *)$? If not; why not? If so, find it.
fong-spivak-seven-sketches F00781	69	■ 0.935	$(\mathbb{Z}, \leq, *, 1)$	$(\mathbb{Z}, \leq, *, 1)$	3. Is $(\mathbb{Z}, \leq, *, 1)$ a monoidal preorder?

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_69 F00782	69	■ 0.949	$\operatorname{Ob}(\mathcal{X})$	$\operatorname{Ob}(\mathcal{X})$	(i) one specifies a set $\operatorname{Ob}(\mathcal{X})$, elements of which are called <i>objects</i> ;
fong-spivak-seven-sketches_69 F00783	69	■ 1.000	$\mathcal{X}(x, y) \in V$	$\mathcal{X}(x, y) \in V$	(ii) for every two objects x, y , one specifies an element $\mathcal{X}(x, y) \in V$, called the <i>hom-</i>
fong-spivak-seven-sketches_69 F00784	69	■ 0.631	$x \in \operatorname{Ob}(X)$	$x \in \operatorname{Ob}(X)$	(a) for every object $x \in \operatorname{Ob}(\mathcal{X})$ we have $I \leq \mathcal{X}(x, x)$, and
fong-spivak-seven-sketches_69 F00785	69	■ 0.631	$I \leq \mathcal{X}(x, x)$	$I \leq \mathcal{X}(x, x)$	(a) for every object $x \in \operatorname{Ob}(\mathcal{X})$ we have $I \leq \mathcal{X}(x, x)$, and
fong-spivak-seven-sketches_69 F00786	69	■ 0.416	$x, y, z \in \operatorname{Ob}(X)$	$x, y, z \in \operatorname{Ob}(X)$	(b) for every three objects $x, y, z \in \operatorname{Ob}(\mathcal{X})$ we have $\mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$.
fong-spivak-seven-sketches_69 F00787	69	■ 0.416	$\mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$	$\mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$	(b) for every three objects $x, y, z \in \operatorname{Ob}(\mathcal{X})$, we have $\mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$.
fong-spivak-seven-sketches_69 F00788	69	■ 0.999	$\mathcal{X}(x, y)$	$\mathcal{X}(x, y)$	name for $\mathcal{X}(x, y)$ might be <i>mapping object</i> , but we use “hom” mainly because it is an important jargon word
fong-spivak-seven-sketches_70 F00789	70	■ 0.986	$\operatorname{Ob}(\mathcal{X}) = \{p, q, r, s, t\}$	$\operatorname{Ob}(\mathcal{X}) = \{p, q, r, s, t\}$	the elements of the preorder, i.e. $\operatorname{Ob}(\mathcal{X}) = \{p, q, r, s, t\}$. Next, for every pair of objects
fong-spivak-seven-sketches_70 F00790	70	■ 1.000	$t \not\leq s$	$t \not\leq s$	if otherwise. So for example, since $s \leq t$ and $t \not\leq s$, we have $\mathcal{X}(s, t) = \text{true}$ and
fong-spivak-seven-sketches_70 F00791	70	■ 1.000	$\mathcal{X}(s, t) =$	$\mathcal{X}(s, t) =$	if otherwise. So for example, since $s \leq t$ and $t \not\leq s$, we have $\mathcal{X}(s, t) = \text{true}$ and
fong-spivak-seven-sketches_70 F00792	70	■ 0.989	$\mathcal{X}(t, s) =$	$\mathcal{X}(t, s) =$	$\mathcal{X}(t, s) = \text{false}$. Recalling from Example 2.27 that the monoidal unit I of Bool is true ,
fong-spivak-seven-sketches_70 F00793	70	■ 0.598	$x(x, y)$	$x(x, y)$	(x, y) th entry is simply the hom-object $\mathcal{X}(x, y)$. So, for example, the above preorder in
fong-spivak-seven-sketches_131 F00794	131	■ 0.848	r	r	infeasible—for that pair. In other words, “yes I can provide p given r ” or “no, I cannot
fong-spivak-seven-sketches_71 F00795	71	■ 0.503	$x, y \in \operatorname{Ob}(X)$	$x, y \in \operatorname{Ob}(X)$	(ii) for every $x, y \in \operatorname{Ob}(\mathcal{X})$ an element $\mathcal{X}(x, y) \in \mathbb{B}$, i.e. either $\mathcal{X}(x, y) = \text{true}$ or
fong-spivak-seven-sketches_71 F00796	71	■ 0.503	$\mathcal{X}(x, y) \in \mathbb{B}$	$\mathcal{X}(x, y) \in \mathbb{B}$	(ii) for every $x, y \in \operatorname{Ob}(\mathcal{X})$ an element $\mathcal{X}(x, y) \in \mathbb{B}$, i.e. either $\mathcal{X}(x, y) = \text{true}$ or
fong-spivak-seven-sketches_71 F00797	71	■ 0.503	$\mathcal{X}(x, y) =$	$\mathcal{X}(x, y) =$	(ii) for every $x, y \in \operatorname{Ob}(\mathcal{X})$ an element $\mathcal{X}(x, y) \in \mathbb{B}$, i.e. either $\mathcal{X}(x, y) = \text{true}$ or
fong-spivak-seven-sketches_71 F00798	71	■ 0.958	$\mathcal{X}(x, y) =$	$\mathcal{X}(x, y) =$	$\mathcal{X}(x, y) = \text{false}$.
fong-spivak-seven-sketches_71 F00799	71	■ 0.507	$X := \operatorname{Ob}(X)$	$X := \operatorname{Ob}(X)$	objects of $\mathcal{X}$. So let’s call the preorder $(X, \leq)$, and let $X := \operatorname{Ob}(\mathcal{X})$. For the $\leq$ relation,
fong-spivak-seven-sketches_71 F00800	71	■ 0.952	$\leq \mathcal{X}(x, x)$	$\leq \mathcal{X}(x, x)$	(a) for every element $x \in X$ we have $\text{true} \leq \mathcal{X}(x, x)$,
fong-spivak-seven-sketches_71 F00801	71	■ 0.798	$\mathcal{X}(x, y) \wedge \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$	$\mathcal{X}(x, y) \wedge \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$	(b) for every three elements $x, y, z \in X$ we have $\mathcal{X}(x, y) \wedge \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$.
fong-spivak-seven-sketches_71 F00802	71	■ 0.998	$b \in$	$b \in$	For $b \in \mathbf{Bool}$, if $\text{true} \leq b$ then $b = \text{true}$, so the first statement says $\mathcal{X}(x, x) = \text{true}$,
fong-spivak-seven-sketches_71 F00803	71	■ 0.998	$\leq b$	$\leq b$	For $b \in \mathbf{Bool}$, if $\text{true} \leq b$ then $b = \text{true}$, so the first statement says $\mathcal{X}(x, x) = \text{true}$,
fong-spivak-seven-sketches_71 F00804	71	■ 0.998	$b =$	$b =$	For $b \in \mathbf{Bool}$, if $\text{true} \leq b$ then $b = \text{true}$, so the first statement says $\mathcal{X}(x, x) = \text{true}$,
fong-spivak-seven-sketches_71 F00805	71	■ 0.998	$\mathcal{X}(x, x) =$	$\mathcal{X}(x, x) =$	For $b \in \mathbf{Bool}$, if $\text{true} \leq b$ then $b = \text{true}$, so the first statement says $\mathcal{X}(x, x) = \text{true}$,
fong-spivak-seven-sketches_71 F00806	71	■ 0.691	$b \in \mathbb{B}$	$b \in \mathbb{B}$	$\text{false} \leq b$ for all $b \in \mathbb{B}$, the only way statement (b) has any force is if $\mathcal{X}(x, y) = \text{true}$
fong-spivak-seven-sketches_71 F00807	71	■ 0.943	$\mathcal{X}(y, z) =$	$\mathcal{X}(y, z) =$	and $\mathcal{X}(y, z) = \text{true}$, in which case it forces $\mathcal{X}(x, z) = \text{true}$. This condition exactly
fong-spivak-seven-sketches_71 F00808	71	■ 0.943	$\mathcal{X}(x, z) =$	$\mathcal{X}(x, z) =$	and $\mathcal{X}(y, z) = \text{true}$, in which case it forces $\mathcal{X}(x, z) = \text{true}$. This condition exactly

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_71 F00809	71	0.766	(X, d)	(X, d)	Definition 2.51. A metric space (X, d) consists of:
fong-spivak-seven-sketches_71 F00810	71	1.000	$d: X \times X \rightarrow \mathbb{R}_{\geq 0}$	$d: X \times X \rightarrow \mathbb{R}_{\geq 0}$	(ii) a function $d: X \times X \rightarrow \mathbb{R}_{\geq 0}$, where $d(x, y)$ is called the <i>distance between x and y</i> .
fong-spivak-seven-sketches_71 F00811	71	1.000	$d(x, y)$	$d(x, y)$	(ii) a function $d: X \times X \rightarrow \mathbb{R}_{\geq 0}$, where $d(x, y)$ is called the <i>distance between x and y</i> .
fong-spivak-seven-sketches_72 F00812	72	1.000	$d(x, x)=0$	$d(x, x) = 0$	(a) for every $x \in X$, we have $d(x, x) = 0$,
fong-spivak-seven-sketches_72 F00813	72	1.000	$d(x, y)=0$	$d(x, y) = 0$	(b) for every $x, y \in X$, if $d(x, y) = 0$ then $x = y$,
fong-spivak-seven-sketches_72 F00814	72	1.000	$d(x, y)=d(y, x)$	$d(x, y) = d(y, x)$	(c) for every $x, y \in X$, we have $d(x, y) = d(y, x)$, and
fong-spivak-seven-sketches_72 F00815	72	1.000	$d(x, y)+d(y, z) \geq d(x, z)$	$d(x, y) + d(y, z) \geq d(x, z)$	(d) for every $x, y, z \in X$, we have $d(x, y) + d(y, z) \geq d(x, z)$.
fong-spivak-seven-sketches_72 F00816	72	0.772	$d: X \times X \rightarrow [0, \infty] = \mathbb{R}_{\geq 0} \cup \{\infty\}$	$d: X \times X \rightarrow [0, \infty] = \mathbb{R}_{\geq 0} \cup \{\infty\}$	If we ask instead in (ii) for a function $d: X \times X \rightarrow [0, \infty] = \mathbb{R}_{\geq 0} \cup \{\infty\}$, we call (X, d)
fong-spivak-seven-sketches_72 F00817	72	0.772	X, d	X, d	If we ask instead in (ii) for a function $d: X \times X \rightarrow [0, \infty] = \mathbb{R}_{\geq 0} \cup \{\infty\}$, we call (X, d)
fong-spivak-seven-sketches_72 F00818	72	1.000	z	z	The triangle inequality says that when plotting a route from x to z , the distance is
fong-spivak-seven-sketches_72 F00819	72	1.000	$x \rightarrow y \rightarrow z$	$x \rightarrow y \rightarrow z$	always at most what you get by choosing an intermediate point y and going $x \rightarrow y \rightarrow z$.
fong-spivak-seven-sketches_72 F00820	72	1.000	$3+5 \geq 7.2$	$3 + 5 \geq 7.2$	It can be invoked three different ways in the above picture: $3 + 5 \geq 7.2$, but also
fong-spivak-seven-sketches_72 F00821	72	1.000	$5+7.2 \geq 3$	$5 + 7.2 \geq 3$	$5 + 7.2 \geq 3$ and $3 + 7.2 \geq 5$. Oh yeah, and $5 + 3 \geq 7.2$, $7.2 + 5 \geq 3$ and $7.2 + 3 \geq 5$.
fong-spivak-seven-sketches_72 F00822	72	1.000	$3+7.2 \geq 5$	$3 + 7.2 \geq 5$	$5 + 7.2 \geq 3$ and $3 + 7.2 \geq 5$. Oh yeah, and $5 + 3 \geq 7.2$, $7.2 + 5 \geq 3$ and $7.2 + 3 \geq 5$.
fong-spivak-seven-sketches_72 F00823	72	1.000	$5+3 \geq 7.2, 7.2+5 \geq 3$	$5 + 3 \geq 7.2, 7.2 + 5 \geq 3$	$5 + 7.2 \geq 3$ and $3 + 7.2 \geq 5$. Oh yeah, and $5 + 3 \geq 7.2$, $7.2 + 5 \geq 3$ and $7.2 + 3 \geq 5$.
fong-spivak-seven-sketches_72 F00824	72	1.000	$7.2+3 \geq 5$	$7.2 + 3 \geq 5$	$5 + 7.2 \geq 3$ and $3 + 7.2 \geq 5$. Oh yeah, and $5 + 3 \geq 7.2$, $7.2 + 5 \geq 3$ and $7.2 + 3 \geq 5$.
fong-spivak-seven-sketches_72 F00825	72	0.764	$d(x, y)=? d(y, x)$	$d(x, y) =? d(y, x)$	are any hills, the symmetry axiom, $d(x, y) =? d(y, x)$, fails: it's easier to get from x
fong-spivak-seven-sketches_72 F00826	72	0.689	$d(\mathrm{US})$	$d(\mathrm{US}$	distance in the worst-case scenario?" So $d(\mathrm{US}, \mathrm{Spain})$ is the distance from somewhere
fong-spivak-seven-sketches_72 F00827	72	0.689	)	)	distance in the worst-case scenario?" So $d(\mathrm{US}, \mathrm{Spain})$ is the distance from somewhere
fong-spivak-seven-sketches_72 F00828	72	0.998	$d($	$d($	It also violates another condition, because $d(\mathrm{Boston}, \mathrm{US}) = 0$. No matter where you
fong-spivak-seven-sketches_72 F00829	72	0.998	)=0	) = 0	It also violates another condition, because $d(\mathrm{Boston}, \mathrm{US}) = 0$. No matter where you
fong-spivak-seven-sketches_72 F00830	72	0.852	$U \subseteq X$	$U \subseteq X$	³ The Hausdorff distance gives a metric on the set of all subsets $U \subseteq X$ of a given metric space (X, d) .
fong-spivak-seven-sketches_73 F00831	73	0.547	$) \neq \emptyset$	$) \neq \emptyset$	$d(\mathrm{US}, \mathrm{Boston}) \neq 0$.
fong-spivak-seven-sketches_73 F00832	73	0.867	$\{r \in \mathbb{R} \mid r < 0\}$	$\{r \in \mathbb{R} \mid r < 0\}$	between two regions is often infinite, e.g. the distance between $\{r \in \mathbb{R} \mid r < 0\}$ and $\{0\}$
fong-spivak-seven-sketches_73 F00833	73	0.996	$(\mathbb{R}, d)$	$(\mathbb{R}, d)$	as subsets of $(\mathbb{R}, d)$ is infinite.
fong-spivak-seven-sketches_73 F00834	73	0.899	$x, y \in \mathrm{Ob}(\mathcal{X})$	$x, y \in \mathrm{Ob}(\mathcal{X})$	(ii) for every $x, y \in \mathrm{Ob}(\mathcal{X})$ an element $\mathcal{X}(x, y) \in [0, \infty]$.
fong-spivak-seven-sketches_73 F00835	73	0.899	$\mathcal{X}(x, y) \in [0, \infty]$	$\mathcal{X}(x, y) \in [0, \infty]$	(ii) for every $x, y \in \mathrm{Ob}(\mathcal{X})$ an element $\mathcal{X}(x, y) \in [0, \infty]$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_73 F00836	73	■ 1.000	$0 \leq d(x, x)$	$0 \geq d(x, x)$	(a) $0 \geq d(x, x)$ for all $x \in X$, and
fong-spivak-seven-sketches_73 F00837	73	■ 1.000	$d(x, x) \in [0, \infty]$	$d(x, x) \in [0, \infty]$	Since $d(x, x) \in [0, \infty]$, if $0 \geq d(x, x)$ then $d(x, x) = 0$. So the first condition is equivalent
fong-spivak-seven-sketches_73 F00838	73	■ 1.000	$d(x, y) := y - x $	$d(x, y) := y - x $	hence a Lawvere metric space structure. Namely $d(x, y) := y - x $, the absolute value
fong-spivak-seven-sketches_73 F00839	73	■ 1.000	$d(3, 7) = 4$	$d(3, 7) = 4$	of the difference. So $d(3, 7) = 4$.
fong-spivak-seven-sketches_73 F00840	73	■ 0.214	$\mathbb{R}_{\geq 0, \geq, 0, +}$	$\mathbb{R}_{\geq 0, \geq, 0, +}$	<i>Exercise 2.55.</i> Consider the symmetric monoidal preorder $(\mathbb{R}_{\geq 0}, \geq, 0, +)$, which is
fong-spivak-seven-sketches_73 F00841	73	■ 0.824	$\left(\mathbb{R}_{\geq 0, \geq, 0, +}\right)$	$(\mathbb{R}_{\geq 0}, \geq, 0, +)$	the difference between a Lawvere metric space and a $(\mathbb{R}_{\geq 0}, \geq, 0, +)$ -category in the sense
fong-spivak-seven-sketches_73 F00842	73	■ 1.000	$d(U, V) := \max(d_L(U, V), d_L(V, U))$	$d(U, V) := \max(d_L(U, V), d_L(V, U))$	formula and symmetrize it: $d(U, V) := \max(d_L(U, V), d_L(V, U))$. We happen to see the unsymmetrized
fong-spivak-seven-sketches_74 F00843	74	■ 1.000	$w \geq 0$	$w \geq 0$	are labeled with numbers $w \geq 0$ —into a Lawvere metric space. In fact, we shall
fong-spivak-seven-sketches_74 F00844	74	■ 0.985	d_X	d_X	Given a weighted graph, one forms a metric d_X on its set X of vertices by setting $d(p, q)$
fong-spivak-seven-sketches_74 F00845	74	■ 0.985	$d(p, q)$	$d(p, q)$	Given a weighted graph, one forms a metric d_X on its set X of vertices by setting $d(p, q)$
fong-spivak-seven-sketches_74 F00847	74	■ 0.999	M_G	M_G	taking G and getting a square matrix M_G , whose rows and columns are indexed by the
fong-spivak-seven-sketches_75 F00849	75	■ 0.998	d_Y	d_Y	that the more difficult “distance matrices” d_Y , such as Eq. (2.57), can be obtained from
fong-spivak-seven-sketches_75 F00850	75	■ 0.992	M_Y	M_Y	the easy graph matrices M_Y , such as Eq. (2.59), by repeating a certain sort of “matrix
fong-spivak-seven-sketches_75 F00851	75	■ 0.699	M_X	M_X	<i>Exercise 2.60.</i> Fill out the matrix M_X associated to the graph X in Eq. (2.56):
fong-spivak-seven-sketches_104 F00852	104	■ 1.000	D	D	and D' to D . That is, once we have made this decision about how F acts on objects,
fong-spivak-seven-sketches_75 F00854	75	■ 0.993	$\mathcal{M} := (\mathrm{P}(M), \subseteq, M, \cap)$	$\mathcal{M} := (\mathrm{P}(M), \subseteq, M, \cap)$	<i>Exercise 2.62.</i> Let M be a set and let $\mathcal{M} := (\mathrm{P}(M), \subseteq, M, \cap)$ be the monoidal preorder
fong-spivak-seven-sketches_75 F00855	75	■ 0.870	$\mathcal{M}$	$\mathcal{M}$	modes of transportation (e.g. car, boat, foot). Then an $\mathcal{M}$ -category $\mathcal{X}$ tells you all the
fong-spivak-seven-sketches_75 F00856	75	■ 0.966	$a, b \in \mathrm{Ob}(\mathcal{X})$	$a, b \in \mathrm{Ob}(\mathcal{X})$	modes that will get you from a all the way to b , for any two points $a, b \in \mathrm{Ob}(\mathcal{X})$.”
fong-spivak-seven-sketches_75 F00857	75	■ 0.595	$M = \{$	$M = \{$	of $M = \{\text{car}, \text{boat}, \text{foot}\}$.
fong-spivak-seven-sketches_76 F00858	76	■ 1.000	$\mathcal{W} := (\mathbb{N} \cup \{\infty\}, \leq, \infty, \min)$	$\mathcal{W} := (\mathbb{N} \cup \{\infty\}, \leq, \infty, \min)$	<i>Exercise 2.63.</i> Consider the monoidal preorder $\mathcal{W} := (\mathbb{N} \cup \{\infty\}, \leq, \infty, \min)$.
fong-spivak-seven-sketches_76 F00859	76	■ 1.000	$\mathbb{N} \cup \{\infty\}$	$\mathbb{N} \cup \{\infty\}$	1. Draw a small graph labeled by elements of $\mathbb{N} \cup \{\infty\}$.
fong-spivak-seven-sketches_76 F00860	76	■ 1.000	$\mathcal{W}$	$\mathcal{W}$	3. Prove that this matrix is the matrix of hom-objects for a $\mathcal{W}$ -category. This will
fong-spivak-seven-sketches_76 F00861	76	■ 1.000	$f: \mathcal{V} \rightarrow \mathcal{W}$	$f: \mathcal{V} \rightarrow \mathcal{W}$	base. This allows us to use a monoidal monotone $f: \mathcal{V} \rightarrow \mathcal{W}$ to construct $\mathcal{W}$ -categories
fong-spivak-seven-sketches_76 F00862	76	■ 1.000	$\mathcal{V} \rightarrow \mathcal{W}$	$\mathcal{V} \rightarrow \mathcal{W}$	Any monoidal monotone $\mathcal{V} \rightarrow \mathcal{W}$ between symmetric monoidal preorders lets us
fong-spivak-seven-sketches_76 F00863	76	■ 0.999	$\mathcal{C}$	$\mathcal{C}$	Construction 2.64. Let $f: \mathcal{V} \rightarrow \mathcal{W}$ be a monoidal monotone. Given a $\mathcal{V}$ -category $\mathcal{C}$, one
fong-spivak-seven-sketches_76 F00864	76	■ 1.000	$\mathcal{C}_f$	$\mathcal{C}_f$	forms the associated $\mathcal{W}$ -category, say $\mathcal{C}_f$ as follows.
fong-spivak-seven-sketches_76 F00865	76	■ 0.825	$\mathrm{Ob}(\mathcal{C}_f) := \mathrm{Ob}(\mathcal{C})$	$\mathrm{Ob}(\mathcal{C}_f) := \mathrm{Ob}(\mathcal{C})$	(i) We take the same objects: $\mathrm{Ob}(\mathcal{C}_f) := \mathrm{Ob}(\mathcal{C})$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_76 F00866	76	■ 0.963	$c, d \in \operatorname{Ob}(\mathcal{C})$	$c, d \in \operatorname{Ob}(\mathcal{C})$	(ii) For any $c, d \in \operatorname{Ob}(\mathcal{C})$, put $\mathcal{C}_f(c, d) := f(\mathcal{C}(c, d))$.
fong-spivak-seven-sketches_76 F00867	76	■ 0.963	$\mathcal{C}_f(c, d) := f(\mathcal{C}(c, d))$	$\mathcal{C}_f(c, d) := f(\mathcal{C}(c, d))$	(ii) For any $c, d \in \operatorname{Ob}(\mathcal{C})$, put $\mathcal{C}_f(c, d) := f(\mathcal{C}(c, d))$.
fong-spivak-seven-sketches_76 F00868	76	■ 1.000	$c \in \mathcal{C}$	$c \in \mathcal{C}$	(a) for every $c \in \mathcal{C}$, we have
fong-spivak-seven-sketches_77 F00869	77	■ 0.964	$c, d, e \in \operatorname{Ob}(\mathcal{C})$	$c, d, e \in \operatorname{Ob}(\mathcal{C})$	(b) for every $c, d, e \in \operatorname{Ob}(\mathcal{C})$ we have
fong-spivak-seven-sketches_77 F00870	77	■ 1.000	$f: [0, \infty] \rightarrow \{$	$f: [0, \infty] \rightarrow \{$	Example 2.65. As an example, consider the function $f: [0, \infty] \rightarrow \{\text{true}, \text{false}\}$ given
fong-spivak-seven-sketches_77 F00871	77	■ 0.651	$x_f \neq x_g$	$x_f \neq x_g$	$\mathcal{X}_f \neq \mathcal{X}_g$.
fong-spivak-seven-sketches_77 F00872	77	■ 0.606	$F: \mathcal{X} \rightarrow \mathcal{Y}$	$F: \mathcal{X} \rightarrow \mathcal{Y}$	Definition 2.69. Let $\mathcal{X}$ and $\mathcal{Y}$ be $\mathcal{V}$ -categories. A $\mathcal{V}$ -functor from $\mathcal{X}$ to $\mathcal{Y}$, denoted $F: \mathcal{X} \rightarrow \mathcal{Y}$,
fong-spivak-seven-sketches_77 F00873	77	■ 0.961	$F: \operatorname{Ob}(\mathcal{X}) \rightarrow \operatorname{Ob}(\mathcal{Y})$	$F: \operatorname{Ob}(\mathcal{X}) \rightarrow \operatorname{Ob}(\mathcal{Y})$	(i) a function $F: \operatorname{Ob}(\mathcal{X}) \rightarrow \operatorname{Ob}(\mathcal{Y})$
fong-spivak-seven-sketches_77 F00874	77	■ 0.365	$x_1, x_2 \in \operatorname{Ob}(\mathcal{X})$	$x_1, x_2 \in \operatorname{Ob}(\mathcal{X})$	(a) for all $x_1, x_2 \in \operatorname{Ob}(\mathcal{X})$, one has $\mathcal{X}(x_1, x_2) \leq \mathcal{Y}(F(x_1), F(x_2))$.
fong-spivak-seven-sketches_77 F00875	77	■ 0.365	$\mathcal{X}(x_1, x_2) \leq \mathcal{Y}(F(x_1), F(x_2))$	$\mathcal{X}(x_1, x_2) \leq \mathcal{Y}(F(x_1), F(x_2))$	(a) for all $x_1, x_2 \in \operatorname{Ob}(\mathcal{X})$, one has $\mathcal{X}(x_1, x_2) \leq \mathcal{Y}(F(x_1), F(x_2))$.
fong-spivak-seven-sketches_78 F00876	78	■ 0.829	$X(x_1, x_2) =$	$X(x_1, x_2) =$	orders are Bool -categories, where $\mathcal{X}(x_1, x_2) = \text{true}$ is denoted $x_1 \leq x_2$. One would hope
fong-spivak-seven-sketches_78 F00877	78	■ 0.829	$x_1 \leq x_2$	$x_1 \leq x_2$	orders are Bool -categories, where $\mathcal{X}(x_1, x_2) = \text{true}$ is denoted $x_1 \leq x_2$. One would hope
fong-spivak-seven-sketches_78 F00878	78	■ 1.000	$(X, \leq_X) \rightarrow (Y, \leq_Y)$	$(X, \leq_X) \rightarrow (Y, \leq_Y)$	and that's true. A monotone map $(X, \leq_X) \rightarrow (Y, \leq_Y)$ is a function $F: X \rightarrow Y$ such that
fong-spivak-seven-sketches_78 F00879	78	■ 1.000	$x_1, x_2 \in X$	$x_1, x_2 \in X$	for every $x_1, x_2 \in X$, if $x_1 \leq_X x_2$ then $F(x_1) \leq_Y F(x_2)$. In other words, we have
fong-spivak-seven-sketches_78 F00880	78	■ 1.000	$x_1 \leq_X x_2$	$x_1 \leq_X x_2$	for every $x_1, x_2 \in X$, if $x_1 \leq_X x_2$ then $F(x_1) \leq_Y F(x_2)$. In other words, we have
fong-spivak-seven-sketches_78 F00881	78	■ 1.000	$F(x_1) \leq_Y F(x_2)$	$F(x_1) \leq_Y F(x_2)$	for every $x_1, x_2 \in X$, if $x_1 \leq_X x_2$ then $F(x_1) \leq_Y F(x_2)$. In other words, we have
fong-spivak-seven-sketches_78 F00882	78	■ 0.996	$\mathcal{V} =$	$\mathcal{V} =$	where the above $\leq$ takes place in the enriching category $\mathcal{V} = \mathbf{Bool}$; this is exactly the
fong-spivak-seven-sketches_78 F00883	78	■ 0.990	(X, d_X)	(X, d_X)	does not increase. That is, given Lawvere metric spaces (X, d_X) and (Y, d_Y) , a Cost -
fong-spivak-seven-sketches_78 F00884	78	■ 0.990	(Y, d_Y)	(Y, d_Y)	does not increase. That is, given Lawvere metric spaces (X, d_X) and (Y, d_Y) , a Cost -
fong-spivak-seven-sketches_78 F00885	78	■ 0.982	$d_X(x_1, x_2) \geq d_Y(F(x_1), F(x_2))$	$d_X(x_1, x_2) \geq d_Y(F(x_1), F(x_2))$	$d_X(x_1, x_2) \geq d_Y(F(x_1), F(x_2))$.
fong-spivak-seven-sketches_78 F00886	78	■ 0.971	$\mathcal{X}^{\operatorname{op}}$	$\mathcal{X}^{\operatorname{op}}$	$\mathcal{V}$ -category $\mathcal{X}$ is denoted $\mathcal{X}^{\operatorname{op}}$ and is defined by
fong-spivak-seven-sketches_78 F00887	78	■ 0.784	$\operatorname{Ob}(\mathcal{X}^{\operatorname{op}}) := \operatorname{Ob}(\mathcal{X})$	$\operatorname{Ob}(\mathcal{X}^{\operatorname{op}}) := \operatorname{Ob}(\mathcal{X})$	(i) $\operatorname{Ob}(\mathcal{X}^{\operatorname{op}}) := \operatorname{Ob}(\mathcal{X})$, and
fong-spivak-seven-sketches_78 F00888	78	■ 0.959	$x, y \in \mathcal{X}$	$x, y \in \mathcal{X}$	(ii) for all $x, y \in \mathcal{X}$, we have $\mathcal{X}^{\operatorname{op}}(x, y) := \mathcal{X}(y, x)$.
fong-spivak-seven-sketches_78 F00889	78	■ 0.959	$\mathcal{X}^{\operatorname{op}}(x, y) := \mathcal{X}(y, x)$	$\mathcal{X}^{\operatorname{op}}(x, y) := \mathcal{X}(y, x)$	(ii) for all $x, y \in \mathcal{X}$, we have $\mathcal{X}^{\operatorname{op}}(x, y) := \mathcal{X}(y, x)$.
fong-spivak-seven-sketches_78 F00890	78	■ 0.334	$\mathbf{t}: \mathcal{X} \rightarrow \mathcal{X}^{\operatorname{op}}$	$\mathbf{t}: \mathcal{X} \rightarrow \mathcal{X}^{\operatorname{op}}$	A $\mathcal{V}$ -category $\mathcal{X}$ is a <i>dagger</i> $\mathcal{V}$ -category if the identity function is a $\mathcal{V}$ -functor $\mathbf{t}: \mathcal{X} \rightarrow \mathcal{X}^{\operatorname{op}}$.
fong-spivak-seven-sketches_78 F00891	78	■ 0.470	$I \leq \mathcal{X}(x, y)$	$I \leq \mathcal{X}(x, y)$	And a <i>skeletal</i> $\mathcal{V}$ -category is one in which if $I \leq \mathcal{X}(x, y)$ and $I \leq \mathcal{X}(y, x)$, then $x = y$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_78 F08892	78	■ 0.470	$I \leq X(y, x)$	$I \leq X(y, x)$	And a <i>skeletal</i> $\mathcal{V}$ -category is one in which if $I \leq \mathcal{X}(x, y)$ and $I \leq \mathcal{X}(y, x)$, then $x = y$.
fong-spivak-seven-sketches_78 F08893	78	■ 0.868	$\mathcal{Y}$	$\mathcal{Y}$	If $\mathcal{V} = (V, \leq, I, \otimes)$ is a symmetric monoidal preorder and $\mathcal{X}$ and $\mathcal{Y}$ are $\mathcal{V}$ -categories, then
fong-spivak-seven-sketches_79 F08894	79	■ 0.966	$\mathcal{X} \times \mathcal{Y}$	$\mathcal{X} \times \mathcal{Y}$	to be the $\mathcal{V}$ -category $\mathcal{X} \times \mathcal{Y}$ with
fong-spivak-seven-sketches_79 F08895	79	■ 0.382	$\operatorname{Ob}(\mathcal{X} \times \mathcal{Y}) := \operatorname{Ob}(\mathcal{X}) \times \operatorname{Ob}(\mathcal{Y})$	$\operatorname{Ob}(\mathcal{X} \times \mathcal{Y}) := \operatorname{Ob}(\mathcal{X}) \times \operatorname{Ob}(\mathcal{Y})$	(i) $\operatorname{Ob}(\mathcal{X} \times \mathcal{Y}) := \operatorname{Ob}(\mathcal{X}) \times \operatorname{Ob}(\mathcal{Y})$,
fong-spivak-seven-sketches_79 F08896	79	■ 0.711	$(\mathcal{X} \times \mathcal{Y})((x, y), (x', y')) := \mathcal{X}(x, x') \otimes \mathcal{Y}(y, y')$	$(\mathcal{X} \times \mathcal{Y})((x, y), (x', y')) := \mathcal{X}(x, x') \otimes \mathcal{Y}(y, y')$	(ii) $(\mathcal{X} \times \mathcal{Y})((x, y), (x', y')) := \mathcal{X}(x, x') \otimes \mathcal{Y}(y, y')$,
fong-spivak-seven-sketches_79 F08897	79	■ 0.334	(x', y')	(x', y')	for two objects (x, y) and (x', y') in $\operatorname{Ob}(\mathcal{X} \times \mathcal{Y})$.
fong-spivak-seven-sketches_79 F08898	79	■ 0.334	$\operatorname{Ob}(\mathcal{X} \times \mathcal{Y})$	$\operatorname{Ob}(\mathcal{X} \times \mathcal{Y})$	for two objects (x, y) and (x', y') in $\operatorname{Ob}(\mathcal{X} \times \mathcal{Y})$.
fong-spivak-seven-sketches_79 F08899	79	■ 0.603	$(x, y) \in \operatorname{Ob}(\mathcal{X} \times \mathcal{Y})$	$(x, y) \in \operatorname{Ob}(\mathcal{X} \times \mathcal{Y})$	1. Check that for every object $(x, y) \in \operatorname{Ob}(\mathcal{X} \times \mathcal{Y})$ we have $I \leq (\mathcal{X} \times \mathcal{Y})((x, y), (x, y))$.
fong-spivak-seven-sketches_79 F08900	79	■ 0.603	$I \leq (x \times y)((x, y), (x, y))$	$I \leq (x \times y)((x, y), (x, y))$	1. Check that for every object $(x, y) \in \operatorname{Ob}(\mathcal{X} \times \mathcal{Y})$ we have $I \leq (\mathcal{X} \times \mathcal{Y})((x, y), (x, y))$.
fong-spivak-seven-sketches_79 F08901	79	■ 1.000	$(x_1, y_1), (x_2, y_2)$	$(x_1, y_1), (x_2, y_2)$	2. Check that for every three objects $(x_1, y_1), (x_2, y_2)$, and (x_3, y_3) , we have
fong-spivak-seven-sketches_79 F08902	79	■ 1.000	(x_3, y_3)	(x_3, y_3)	2. Check that for every three objects $(x_1, y_1), (x_2, y_2)$, and (x_3, y_3) , we have
fong-spivak-seven-sketches_79 F08903	79	■ 1.000	$(P, \leq_P) \times (Q, \leq_Q)$	$(P, \leq_P) \times (Q, \leq_Q)$	When taking the product of two preorders $(P, \leq_P) \times (Q, \leq_Q)$, as first described in
fong-spivak-seven-sketches_79 F08904	79	■ 0.960	$(p_1, q_1) \leq (p_2, q_2)$	$(p_1, q_1) \leq (p_2, q_2)$	Example 1.56, we say that $(p_1, q_1) \leq (p_2, q_2)$ iff both $p_1 \leq p_2$ AND $q_1 \leq q_2$; the AND is
fong-spivak-seven-sketches_79 F08905	79	■ 0.960	$q_1 \leq q_2$	$q_1 \leq q_2$	Example 1.56, we say that $(p_1, q_1) \leq (p_2, q_2)$ iff both $p_1 \leq p_2$ AND $q_1 \leq q_2$; the AND is
fong-spivak-seven-sketches_79 F08906	79	■ 0.640	$x \times y$	$x \times y$	objects in $\mathcal{X} \times \mathcal{Y}$. And the distance $d_{\mathcal{X} \times \mathcal{Y}}((x, y), (x', y'))$ between any two points is given
fong-spivak-seven-sketches_79 F08907	79	■ 0.640	$d_{\mathcal{X} \times \mathcal{Y}}((x, y), (x', y'))$	$d_{\mathcal{X} \times \mathcal{Y}}((x, y), (x', y'))$	objects in $\mathcal{X} \times \mathcal{Y}$. And the distance $d_{\mathcal{X} \times \mathcal{Y}}((x, y), (x', y'))$ between any two points is given
fong-spivak-seven-sketches_79 F08908	79	■ 1.000	$d_{\mathcal{X}}(x, x') + d_{\mathcal{Y}}(y, y')$	$d_{\mathcal{X}}(x, x') + d_{\mathcal{Y}}(y, y')$	by the sum $d_{\mathcal{X}}(x, x') + d_{\mathcal{Y}}(y, y')$.
fong-spivak-seven-sketches_80 F08909	80	■ 1.000	$\mathbb{R} \times \mathbb{R}$	$\mathbb{R} \times \mathbb{R}$	Example 2.54). Form the Cost -product $\mathbb{R} \times \mathbb{R}$. What is the distance from (5,6) to
fong-spivak-seven-sketches_80 F08910	80	■ 1.000	(5,6)	(5,6)	Example 2.54). Form the Cost -product $\mathbb{R} \times \mathbb{R}$. What is the distance from (5,6) to
fong-spivak-seven-sketches_80 F08911	80	■ 0.876	(-1,4)	(-1,4)	(-1,4)? Hint: apply Definition 2.74; the answer is not $\sqrt{40}$.
fong-spivak-seven-sketches_80 F08912	80	■ 0.876	$\sqrt{40}$	$\sqrt{40}$	(-1,4)? Hint: apply Definition 2.74; the answer is not $\sqrt{40}$.
fong-spivak-seven-sketches_80 F08920	80	■ 0.482	$\mathfrak{y}$	$\mathfrak{y}$	like $\mathfrak{X}$ —for every entry of $\mathfrak{Y}$. Make sure you can see each block as the $\mathfrak{X}$ -matrix shifted
fong-spivak-seven-sketches_81 F08921	81	■ 1.000	$v, w \in V$	$v, w \in V$	monoidal closed (or just closed) if, for every two elements $v, w \in V$, there is an element
fong-spivak-seven-sketches_81 F08922	81	■ 1.000	$a, v, w \in V$	$a, v, w \in V$	for all $a, v, w \in V$.
fong-spivak-seven-sketches_81 F08923	81	■ 1.000	$v \in V$	$v \in V$	preorder is monoidal closed iff, given any $v \in V$, the map $(- \otimes v): V \rightarrow V$ given by
fong-spivak-seven-sketches_81 F08924	81	■ 1.000	$(- \otimes v): V \rightarrow V$	$(- \otimes v): V \rightarrow V$	preorder is monoidal closed iff, given any $v \in V$, the map $(- \otimes v): V \rightarrow V$ given by

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_81_F00925	81	0.747	$(v\text{-}\circlearrowright): V \rightarrowtail V$	$(v - \circ -) : V \rightarrow V$	multiplying with v has a right adjoint. We write this right adjoint $(v - \circ -): V \rightarrow V$.
fong-spivak-seven-sketches_81_F00926	81	1.000	$(-\otimes v)$	$(- \otimes v)$	1. Using Definition 2.2, show that $(- \otimes v)$ is monotone.
fong-spivak-seven-sketches_81_F00927	81	0.959	$((v\text{-}o\ w) \otimes v) \leq w$	$((v - ow) \otimes v) \leq w$	2. Supposing that $\mathcal{V}$ is closed, show that for all $v, w \in V$ we have $((v - \circ w) \otimes v) \leq w$.
fong-spivak-seven-sketches_81_F00928	81	0.702	$(v\text{-}o\text{-})$	$(v - o -)$	3. Using 2., show that $(v - \circ -)$ is monotone.
fong-spivak-seven-sketches_81_F00929	81	0.950	$x, y \in [0, \infty]$	$x, y \in [0, \infty]$	deed, for any $x, y \in [0, \infty]$, define $x - \circ y := \max(0, y - x)$. Then, for any $a, x, y \in [0, \infty]$,
fong-spivak-seven-sketches_81_F00930	81	0.950	$x \multimap y := \max(0, y - x)$	$x - \circ y := \max(0, y - x)$	deed, for any $x, y \in [0, \infty]$, define $x - \circ y := \max(0, y - x)$. Then, for any $a, x, y \in [0, \infty]$,
fong-spivak-seven-sketches_81_F00931	81	0.950	$a, x, y \in [0, \infty]$	$a, x, y \in [0, \infty]$	deed, for any $x, y \in [0, \infty]$, define $x - \circ y := \max(0, y - x)$. Then, for any $a, x, y \in [0, \infty]$,
fong-spivak-seven-sketches_82_F00932	82	0.746	$\leq p \multimap q$	$\leq p - \circ q$	as in Definition 2.79. Note that $\text{false} \leq p - \circ q$, for any p, q no matter what $- \circ$ is,
fong-spivak-seven-sketches_82_F00933	82	0.746	$\rightarrow$	$\rightarrow$	as in Definition 2.79. Note that $\text{false} \leq p - \circ q$, for any p, q no matter what $- \circ$ is,
fong-spivak-seven-sketches_82_F00934	82	0.987	$a =$	$a =$	because false is less than everything. But using $a = \text{false}$, $p = \text{true}$, and $q = \text{false}$,
fong-spivak-seven-sketches_82_F00935	82	0.987	$p =$	$p =$	because false is less than everything. But using $a = \text{false}$, $p = \text{true}$, and $q = \text{false}$,
fong-spivak-seven-sketches_82_F00936	82	0.987	$q =$	$q =$	because false is less than everything. But using $a = \text{false}$, $p = \text{true}$, and $q = \text{false}$,
fong-spivak-seven-sketches_82_F00937	82	0.999	$(a \vee p) \not\leq q$	$(a \vee p) \not\leq q$	we then get a contradiction: $(a \vee p) \not\leq q$ and yet $a \leq (p - \circ q)$.
fong-spivak-seven-sketches_82_F00938	82	0.999	$a \leq (p \multimap q)$	$a \leq (p - \circ q)$	we then get a contradiction: $(a \vee p) \not\leq q$ and yet $a \leq (p - \circ q)$.
fong-spivak-seven-sketches_82_F00939	82	1.000	c, d	c, d	say that for every two material collections c, d one can form a material collection $c - \circ d$
fong-spivak-seven-sketches_82_F00940	82	1.000	$c \rightarrowtail d$	$c \rightarrow d$	say that for every two material collections c, d one can form a material collection $c - \circ d$
fong-spivak-seven-sketches_82_F00941	82	1.000	$2 \text{H}_2 + 2 \text{Na} \rightarrowtail 2 \text{NaOH} + \text{H}_2$	$2\text{H}_2\text{O} + 2\text{Na} \rightarrow 2\text{NaOH} + \text{H}_2$	Or more down to earth, since we have the reaction $2\text{H}_2\text{O} + 2\text{Na} \rightarrow 2\text{NaOH} + \text{H}_2$, we
fong-spivak-seven-sketches_82_F00942	82	0.426	$2 \text{Na} \rightarrowtail (2 \text{NaOH} + \text{H}_2)$	$2\text{Na} \rightarrow (2\text{NaOH} + \text{H}_2)$	But it is not so far-fetched: this hypothetical new substance $(2\text{Na} - \circ (2\text{NaOH} + \text{H}_2))$
fong-spivak-seven-sketches_82_F00943	82	0.935	$\mathcal{V} = (V, \leq, I, \otimes, - \circ)$	$\mathcal{V} = (V, \leq, I, \otimes, - \circ)$	Proposition 2.87. Suppose $\mathcal{V} = (V, \leq, I, \otimes, - \circ)$ is a symmetric monoidal preorder that
fong-spivak-seven-sketches_82_F00944	82	1.000	$- \otimes v : (V, \leq) \rightarrowtail (V, \leq)$	$- \otimes v : (V, \leq) \rightarrow (V, \leq)$	(a) For every $v \in V$, the monotone map $- \otimes v : (V, \leq) \rightarrow (V, \leq)$ is left adjoint to
fong-spivak-seven-sketches_82_F00945	82	0.913	$v \rightarrowtail - : (V, \leq) \rightarrowtail (V, \leq)$	$v \rightarrow - : (V, \leq) \rightarrow (V, \leq)$	$v - \circ - : (V, \leq) \rightarrow (V, \leq)$.
fong-spivak-seven-sketches_82_F00946	82	1.000	$A \subseteq V$	$A \subseteq V$	(b) For any element $v \in V$ and set of elements $A \subseteq V$, if the join $\bigvee_{a \in A} a$ exists then
fong-spivak-seven-sketches_82_F00947	82	1.000	$\bigvee_{a \in A} a$	$\bigvee_{a \in A} a$	(b) For any element $v \in V$ and set of elements $A \subseteq V$, if the join $\bigvee_{a \in A} a$ exists then
fong-spivak-seven-sketches_82_F00948	82	1.000	$\bigvee_{a \in A} v \otimes a$	$\bigvee_{a \in A} v \otimes a$	so does $\bigvee_{a \in A} v \otimes a$ and we have
fong-spivak-seven-sketches_82_F00949	82	1.000	$v \otimes (v \multimap w) \leq w$	$v \otimes (v - \circ w) \leq w$	(c) For any $v, w \in V$, we have $v \otimes (v - \circ w) \leq w$.
fong-spivak-seven-sketches_82_F00950	82	1.000	$v \cong (I \multimap v)$	$v \cong (I - \circ v)$	(d) For any $v \in V$, we have $v \cong (I - \circ v)$.
fong-spivak-seven-sketches_82_F00951	82	1.000	$u, v, w \in V$	$u, v, w \in V$	(e) For any $u, v, w \in V$, we have $(u - \circ v) \otimes (v - \circ w) \leq (u - \circ w)$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F00952	82	■ 1.000	$(u \multimap v) \otimes (v \multimap w) \leq (u \multimap w)$	$(u \multimap v) \otimes (v \multimap w) \leq (u \multimap w)$	(e) For any $u, v, w \in V$, we have $(u \multimap v) \otimes (v \multimap w) \leq (u \multimap w)$.
fong-spivak-seven-sketches F00953	83	■ 0.748	$(v \multimap \circ -)$	$(v \multimap \circ -)$	(a) The definition of $(- \otimes v)$ being left adjoint to $(v \multimap -)$ is exactly the condition
fong-spivak-seven-sketches F00954	83	■ 0.999	$(v \multimap w) \leq (v \multimap w)$	$(v \multimap w) \leq (v \multimap w)$	in Proposition 1.107. More concretely, from reflexivity $(v \multimap w) \leq (v \multimap w)$, we
fong-spivak-seven-sketches F00955	83	■ 0.971	$(v \multimap w) \otimes v \leq w$	$(v \multimap w) \otimes v \leq w$	obtain $(v \multimap w) \otimes v \leq w$ Eq. (2.80), and we are done by symmetry, which says
fong-spivak-seven-sketches F00956	83	■ 0.988	$v \otimes (v \multimap w) = (v \multimap w) \otimes v$	$v \otimes (v \multimap w) = (v \multimap w) \otimes v$	$v \otimes (v \multimap w) = (v \multimap w) \otimes v$.
fong-spivak-seven-sketches F00957	83	■ 1.000	$v \otimes I = v \leq v$	$v \otimes I = v \leq v$	(d) Since $v \otimes I = v \leq v$, Eq. (2.80) says $v \leq (I \multimap v)$. For the other direction, we have
fong-spivak-seven-sketches F00958	83	■ 1.000	$v \leq (I \multimap v)$	$v \leq (I \multimap v)$	(d) Since $v \otimes I = v \leq v$, Eq. (2.80) says $v \leq (I \multimap v)$. For the other direction, we have
fong-spivak-seven-sketches F00959	83	■ 0.909	$(I \multimap v) = I \otimes (I \multimap v) \leq v$	$(I \multimap v) = I \otimes (I \multimap v) \leq v$	$(I \multimap v) = I \otimes (I \multimap v) \leq v$ by (c).
fong-spivak-seven-sketches F00960	83	■ 0.966	$u \otimes (u \multimap v) \otimes (v \multimap w) \leq w$	$u \otimes (u \multimap v) \otimes (v \multimap w) \leq w$	(e) To obtain this inequality, we just need $u \otimes (u \multimap v) \otimes (v \multimap w) \leq w$. But this
fong-spivak-seven-sketches F00961	83	■ 0.994	$v, w \in \mathrm{Ob}(\mathcal{V})$	$v, w \in \mathrm{Ob}(\mathcal{V})$	Remark 2.89. We can consider $\mathcal{V}$ to be enriched in itself. That is, for every $v, w \in \mathrm{Ob}(\mathcal{V})$,
fong-spivak-seven-sketches F00962	83	■ 0.899	$\mathcal{V}(v, w) := (v \multimap w) \in \mathcal{V}$	$\mathcal{V}(v, w) := (v \multimap w) \in \mathcal{V}$	we can define $\mathcal{V}(v, w) := (v \multimap w) \in \mathcal{V}$. For this to really be an enrichment, we just need
fong-spivak-seven-sketches F00963	83	■ 0.858	$I \leq \mathcal{X}(x, x) = (x \multimap x)$	$I \leq \mathcal{X}(x, x) = (x \multimap x)$	to check the two conditions of Definition 2.46. The first condition $I \leq \mathcal{X}(x, x) = (x \multimap x)$
fong-spivak-seven-sketches F00964	83	■ 1.000	$I \otimes x \leq x$	$I \otimes x \leq x$	is satisfied because $I \otimes x \leq x$. The second condition is satisfied by Proposition 2.87(e).
fong-spivak-seven-sketches F00965	83	■ 0.844	$\mathcal{V} = (V, \leq, I, \otimes, \multimap)$	$\mathcal{V} = (V, \leq, I, \otimes, \multimap)$	$\mathcal{V} = (V, \leq, I, \otimes, \multimap)$ that has all joins: $\bigvee A$ exists for every $A \subseteq V$. In particular, we often
fong-spivak-seven-sketches F00966	83	■ 0.844	$\bigvee A$	$\bigvee A$	$\mathcal{V} = (V, \leq, I, \otimes, \multimap)$ that has all joins: $\bigvee A$ exists for every $A \subseteq V$. In particular, we often
fong-spivak-seven-sketches F00967	83	■ 1.000	$0 := \bigvee \varnothing$	$0 := \bigvee \varnothing$	denote the empty join by $0 := \bigvee \varnothing$.
fong-spivak-seven-sketches F00968	83	■ 1.000	$A \subseteq [0, \infty]$	$A \subseteq [0, \infty]$	Cost is a quantale, we take an arbitrary set of elements $A \subseteq [0, \infty]$ and ask if it has a
fong-spivak-seven-sketches F00969	83	■ 0.564	$\bigvee A$	$\bigvee A$	join $\bigvee A$. To be a join, it needs to satisfy two properties:
fong-spivak-seven-sketches F00970	83	■ 0.959	$a \geq \bigvee A$	$a \geq \bigvee A$	a. $a \geq \bigvee A$ for all $a \in A$, and
fong-spivak-seven-sketches F00971	83	■ 0.969	$b \in [0, \infty]$	$b \in [0, \infty]$	b. if $b \in [0, \infty]$ is any element such that $a \geq b$ for all $a \in A$, then $\bigvee A \geq b$.
fong-spivak-seven-sketches F00972	83	■ 0.969	$a \geq b$	$a \geq b$	b. if $b \in [0, \infty]$ is any element such that $a \geq b$ for all $a \in A$, then $\bigvee A \geq b$.
fong-spivak-seven-sketches F00973	83	■ 0.969	$\bigvee A \geq b$	$\bigvee A \geq b$	b. if $b \in [0, \infty]$ is any element such that $a \geq b$ for all $a \in A$, then $\bigvee A \geq b$.
fong-spivak-seven-sketches F00974	84	■ 0.728	$A \cdot \{ \}^{\{5\}}$	$A \cdot \{ \}^{\{5\}}$	bound, of $A \cdot \{ \}^{\{5\}}$ For example, if $A = \{2, 3\}$ then $\bigvee A = 2$. We have joins for infinite sets
fong-spivak-seven-sketches F00975	84	■ 0.728	$A = \{2, 3\}$	$A = \{2, 3\}$	bound, of $A \cdot \{ \}^{\{5\}}$ For example, if $A = \{2, 3\}$ then $\bigvee A = 2$. We have joins for infinite sets
fong-spivak-seven-sketches F00976	84	■ 0.728	$\bigvee A = 2$	$\bigvee A = 2$	bound, of $A \cdot \{ \}^{\{5\}}$ For example, if $A = \{2, 3\}$ then $\bigvee A = 2$. We have joins for infinite sets
fong-spivak-seven-sketches F00977	84	■ 0.573	$B = \{2.5, 2.05, 2.005, \dots\}$	$B = \{2.5, 2.05, 2.005, \dots\}$	too: if $B = \{2.5, 2.05, 2.005, \dots\}$, its infimum is 2. Finally, in order to say that $([0, \infty], \geq)$
fong-spivak-seven-sketches F00978	84	■ 1.000	$A = \varnothing$	$A = \varnothing$	has all joins, we need a join to exist for the empty set $A = \varnothing$ too. The first condition

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-1 F00979	84	■ 0.994	$\bigvee \varnothing \geq b$	$\bigvee \emptyset \geq b$	$b \in [0, \infty]$ we have $\bigvee \emptyset \geq b$; this means $\bigvee \emptyset = \infty$.
fong-spivak-seven-sketches-1 F00980	84	■ 0.994	$\bigvee \varnothing = \infty$	$\bigvee \emptyset = \infty$	$b \in [0, \infty]$ we have $\bigvee \emptyset \geq b$; this means $\bigvee \emptyset = \infty$.
fong-spivak-seven-sketches-1 F00981	84	■ 0.998	$\vee \varnothing$	$\vee \emptyset$	1. What is $\bigvee \emptyset$, which we generally denote 0, in the case
fong-spivak-seven-sketches-1 F00982	84	■ 0.408	$\mathcal{V} = \mathbf{Bool} = (\mathbb{B}, \leq)$	$\mathcal{V} = \mathbf{Bool} = (\mathbb{B}, \leq)$	a. $\mathcal{V} = \mathbf{Bool} = (\mathbb{B}, \leq, \text{true}, \wedge)$?
fong-spivak-seven-sketches-1 F00983	84	■ 0.959	$\mathcal{V} = \mathbf{Cost} = ([0, \infty], \geq, 0, +)$	$\mathcal{V} = \mathbf{Cost} = ([0, \infty], \geq, 0, +)$	b. $\mathcal{V} = \mathbf{Cost} = ([0, \infty], \geq, 0, +)$?
fong-spivak-seven-sketches-1 F00984	84	■ 1.000	$x \vee y$	$x \vee y$	2. What is the join $x \vee y$ in the case
fong-spivak-seven-sketches-1 F00985	84	■ 0.724	$\mathcal{V} = \mathbf{Bool}$	$\mathcal{V} = \mathbf{Bool}$	a. $\mathcal{V} = \mathbf{Bool}$, and $x, y \in \mathbb{B}$ are booleans?
fong-spivak-seven-sketches-1 F00986	84	■ 0.724	$x, y \in \mathbb{B}$	$x, y \in \mathbb{B}$	a. $\mathcal{V} = \mathbf{Bool}$, and $x, y \in \mathbb{B}$ are booleans?
fong-spivak-seven-sketches-1 F00987	84	■ 0.939	$\mathbf{Bool} = (\mathbb{B}, \leq)$	$\mathbf{Bool} = (\mathbb{B}, \leq)$	Exercise 2.93. Show that $\mathbf{Bool} = (\mathbb{B}, \leq, \text{true}, \wedge)$ is a quantale.
fong-spivak-seven-sketches-1 F00988	84	■ 0.884	$P(S), \subseteq, S, \cap$	$P(S), \subseteq, S, \cap$	Exercise 2.94. Let S be a set and recall the power set monoidal preorder $(P(S), \subseteq, S, \cap)$
fong-spivak-seven-sketches-1 F00989	84	■ 1.000	$\mathcal{P} = (P, \leq)$	$\mathcal{P} = (P, \leq)$	Proposition 2.96. Let $\mathcal{P} = (P, \leq)$ be a preorder. It has all joins iff it has all meets.
fong-spivak-seven-sketches-1 F00990	84	■ 1.000	$A \subseteq \mathcal{P}$	$A \subseteq \mathcal{P}$	Suppose $\mathcal{P}$ has all joins and suppose that $A \subseteq \mathcal{P}$ is a subset for which we want
fong-spivak-seven-sketches-1 F00991	84	■ 1.000	$M_A := \{p \in P \mid p \leq a \text{ for all } a \in A\}$	$M_A := \{p \in P \mid p \leq a\}$	the meet. Consider the set $M_A := \{p \in P \mid p \leq a \text{ for all } a \in A\}$ of elements below
fong-spivak-seven-sketches-1 F00992	84	■ 1.000	$a \in A$	$a \in A$	the meet. Consider the set $M_A := \{p \in P \mid p \leq a \text{ for all } a \in A\}$ of elements below
fong-spivak-seven-sketches-1 F00993	84	■ 1.000	$m_A := \bigvee_{p \in M_A} p$	$m_A := \bigvee_{p \in M_A} p$	everything in A . Let $m_A := \bigvee_{p \in M_A} p$ be their join. We claim that m_A is a meet for A .
fong-spivak-seven-sketches-1 F00994	84	■ 1.000	m_A	m_A	everything in A . Let $m_A := \bigvee_{p \in M_A} p$ be their join. We claim that m_A is a meet for A .
fong-spivak-seven-sketches-1 F00995	84	■ 1.000	$m_A \leq a$	$m_A \leq a$	We first need to know that for any $a \in A$ we have $m_A \leq a$, but this is by definition of
fong-spivak-seven-sketches-1 F00996	84	■ 1.000	$p \in M_A$	$p \in M_A$	join: since all $p \in M_A$ satisfy $p \leq a$, so does their join $m_A \leq a$. We second need to know
fong-spivak-seven-sketches-1 F00997	84	■ 1.000	$m' \in P$	$m' \in P$	that for any $m' \in P$ with $m' \leq a$ for all $a \in A$, we have $m' \leq m$. But every such m' is
fong-spivak-seven-sketches-1 F00998	84	■ 1.000	$m' \leq a$	$m' \leq a$	that for any $m' \in P$ with $m' \leq a$ for all $a \in A$, we have $m' \leq m$. But every such m' is
fong-spivak-seven-sketches-1 F00999	84	■ 1.000	$m' \leq m$	$m' \leq m$	that for any $m' \in P$ with $m' \leq a$ for all $a \in A$, we have $m' \leq m$. But every such m' is
fong-spivak-seven-sketches-1 F01000	84	■ 1.000	m'	m'	that for any $m' \in P$ with $m' \leq a$ for all $a \in A$, we have $m' \leq m$. But every such m' is
fong-spivak-seven-sketches-1 F01001	84	■ 1.000	M_A	M_A	actually an element of M_A and m is their join, so $m' \leq m$. This completes the proof.
fong-spivak-seven-sketches-1 F01002	84	■ 0.666	$\{3.1, 3.01, 3.001, \dots\}$	$\{3.1, 3.01, 3.001, \dots\}$	largest number that is $\leq$ everything in A . For example, the infimum of $\{3.1, 3.01, 3.001, \dots\}$ is 3. But note
fong-spivak-seven-sketches-1 F01003	85	■ 1.000	$V \subseteq X$	$V \subseteq X$	and $V \subseteq X$, we can generalize the usual Hausdorff distance, on the left below, to the
fong-spivak-seven-sketches-1 F01004	85	■ 0.302	$u \in U$	$u \in U$	answers the question “can I get into V from every $u \in U$,” i.e. $\forall u \in U. \exists v \in V. u \leq v$. Or
fong-spivak-seven-sketches-1 F01005	85	■ 0.302	$\forall u \in U. \exists v \in V. u \leq v$	$\forall u \in U. \exists v \in V. u \leq v$	answers the question “can I get into V from every $u \in U$,” i.e. $\forall u \in U. \exists v \in V. u \leq v$. Or

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F01006	85	■ 0.999	$\mathcal{V}=\mathrm{P}(M)$	$\mathcal{V} = P(M)$	for another example, use $\mathcal{V} = P(M)$ with its interpretation as modes of transportation,
fong-spivak-seven-sketches F01007	85	■ 1.000	$d(U, V) \in \mathrm{P}(M)$	$d(U, V) \in P(M)$	as in Exercise 2.62. Then the Hausdorff distance $d(U, V) \in P(M)$ tells us those modes
fong-spivak-seven-sketches F01008	85	■ 1.000	$- \otimes v: V \rightarrowtail V$	$- \otimes v : V \rightarrow V$	Eq. (2.88) holds. We need to show that Eq. (2.88) holds then $- \otimes v: V \rightarrow V$ has a right
fong-spivak-seven-sketches F01009	85	■ 0.995	$v \multimap -$	$v \multimap -$	adjoint $v \multimap -$. This is just the adjoint functor theorem, Theorem 1.115. It says we can
fong-spivak-seven-sketches F01010	85	■ 1.000	$v \multimap w$	$v \multimap w$	define $v \multimap w$ to be
fong-spivak-seven-sketches F01011	85	■ 0.703	$\sum$	$\sum$	We will get a formula where joins stand in for the sum operation $\sum$, and $\otimes$ stands in
fong-spivak-seven-sketches F01012	85	■ 1.000	$\mathcal{V}=(V, \leq, \otimes, I)$	$\mathcal{V} = (V, \leq, \otimes, I)$	Definition 2.100. Let $\mathcal{V} = (V, \leq, \otimes, I)$ be a quantale. Given sets X and Y , a <i>matrix with</i>
fong-spivak-seven-sketches F01013	85	■ 1.000	$M: X \times Y \rightarrowtail V$	$M : X \times Y \rightarrow V$	<i>entries in $\mathcal{V}$</i> , or simply a $\mathcal{V}$ - <i>matrix</i> , is a function $M: X \times Y \rightarrow V$. For any $x \in X$ and
fong-spivak-seven-sketches F01014	85	■ 0.999	$M(x, y)$	$M(x, y)$	$y \in Y$, we call $M(x, y)$ the (x, y) - <i>entry</i> .
fong-spivak-seven-sketches F01015	86	■ 1.000	$N: Y \times Z \rightarrowtail V$	$N : Y \times Z \rightarrow V$	Here is how you multiply $\mathcal{V}$ -matrices $M: X \times Y \rightarrow V$ and $N: Y \times Z \rightarrow V$. Their
fong-spivak-seven-sketches F01016	86	■ 0.997	$(M * N): X \times Z \rightarrowtail V$	$(M * N) : X \times Z \rightarrow V$	product is defined to be the matrix $(M * N): X \times Z \rightarrow V$, whose entries are given by
fong-spivak-seven-sketches F01017	86	■ 0.595	$\mathcal{V}=\mathbf{Bool}$	$\mathcal{V} = \mathbf{Bool}$	<i>Example 2.102.</i> Let $\mathcal{V} = \mathbf{Bool}$. Here is an example of matrix multiplication $M * N$. Here
fong-spivak-seven-sketches F01018	86	■ 0.595	$M * N$	$M * N$	<i>Example 2.102.</i> Let $\mathcal{V} = \mathbf{Bool}$. Here is an example of matrix multiplication $M * N$. Here
fong-spivak-seven-sketches F01019	86	■ 0.996	$X=\{1,2,3\}, Y=\{1,2\}$	$X = \{1, 2, 3\}, Y = \{1, 2\}$	$X = \{1, 2, 3\}, Y = \{1, 2\}$, and $Z = \{1, 2, 3\}$, matrices $M: X \times Y \rightarrow \mathbb{B}$ and $N: Y \times Z \rightarrow \mathbb{B}$
fong-spivak-seven-sketches F01020	86	■ 0.996	$Z=\{1,2,3\}$	$Z = \{1, 2, 3\}$	$X = \{1, 2, 3\}, Y = \{1, 2\}$, and $Z = \{1, 2, 3\}$, matrices $M: X \times Y \rightarrow \mathbb{B}$ and $N: Y \times Z \rightarrow \mathbb{B}$
fong-spivak-seven-sketches F01021	86	■ 0.996	$M: X \times Y \rightarrowtail \mathbb{B}$	$M : X \times Y \rightarrow \mathbb{B}$	$X = \{1, 2, 3\}, Y = \{1, 2\}$, and $Z = \{1, 2, 3\}$, matrices $M: X \times Y \rightarrow \mathbb{B}$ and $N: Y \times Z \rightarrow \mathbb{B}$
fong-spivak-seven-sketches F01022	86	■ 0.996	$N: Y \times Z \rightarrowtail \mathbb{B}$	$N : Y \times Z \rightarrow \mathbb{B}$	$X = \{1, 2, 3\}, Y = \{1, 2\}$, and $Z = \{1, 2, 3\}$, matrices $M: X \times Y \rightarrow \mathbb{B}$ and $N: Y \times Z \rightarrow \mathbb{B}$
fong-spivak-seven-sketches F01023	86	■ 1.000	$I_X: X \times X \rightarrowtail V$	$I_X : X \times X \rightarrow V$	The identity $\mathcal{V}$ -matrix on a set X is $I_X: X \times X \rightarrow V$ given by
fong-spivak-seven-sketches F01024	86	■ 0.964	2×2	2×2	<i>Exercise 2.103.</i> Write down the 2×2 -identity matrix for each of the quantales $(\mathbb{N}, \leq, 1, *)$,
fong-spivak-seven-sketches F01025	86	■ 0.171	$([0, \infty], \geq, 0, +)$	$([0, \infty], \geq, 0, +)$	$\mathbf{Bool} = (\mathbb{B}, \leq, \text{true}, \wedge)$, and $\mathbf{Cost} = ([0, \infty], \geq, 0, +)$.
fong-spivak-seven-sketches F01026	86	■ 1.000	$I_X * M = M$	$I_X * M = M$	has $I_X * M = M$.
fong-spivak-seven-sketches F01027	86	■ 1.000	$M: W \times X \rightarrowtail V, N: X \times Y \rightarrowtail V$	$M : W \times X \rightarrow V, N : X \times Y \rightarrow V$	2. Prove the <i>associative law</i> : for any matrices $M: W \times X \rightarrow V, N: X \times Y \rightarrow V$, and
fong-spivak-seven-sketches F01028	86	■ 1.000	$P: Y \times Z \rightarrowtail V$	$P : Y \times Z \rightarrow V$	$P: Y \times Z \rightarrow V$, one has $(M * N) * P = M * (N * P)$.
fong-spivak-seven-sketches F01029	86	■ 1.000	$(M * N) * P = M * (N * P)$	$(M * N) * P = M * (N * P)$	$P: Y \times Z \rightarrow V$, one has $(M * N) * P = M * (N * P)$.
fong-spivak-seven-sketches F01030	87	■ 0.999	M_Y^2	M_Y^2	When we multiply M_Y by itself using the formula Eq. (2.101), the result M_Y^2 tells us
fong-spivak-seven-sketches F01031	87	■ 1.000	M_Y^3	M_Y^3	the length of the shortest path traversing 2 edges or fewer. Similarly M_Y^3 tells us about
fong-spivak-seven-sketches F01032	87	■ 1.000	$M_Y^2 = M_Y^3$	$M_Y^2 = M_Y^3$	One sees that the powers stabilize: $M_Y^2 = M_Y^3$; as soon as that happens one has the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_87 F01033	87	■ 1.000	$M_{\{Y\}^{\{n\}}}$	M_Y^n	matrix of distances, d_Y . Indeed M_Y^n records the lengths of the shortest path traverse n
fong-spivak-seven-sketches_87 F01034	87	■ 0.911	$M_{\{X\}^{\{2\}}}, M_{\{X\}^{\{3\}}}$	M_X^2, M_X^3	Calculate M_X^2, M_X^3 , and M_X^4 . Check that M_X^4 is what you got for the distance matrix in
fong-spivak-seven-sketches_87 F01035	87	■ 0.911	$M_{\{X\}^{\{4\}}}$	M_X^4	Calculate M_X^2, M_X^3 , and M_X^4 . Check that M_X^4 is what you got for the distance matrix in
fong-spivak-seven-sketches_89 F01036	89	■ 1.000	$\$ 2.5$	\$2.5	technology) budgets, and that the market for data integration software was \$2.5 billion
fong-spivak-seven-sketches_89 F01037	89	■ 0.798	8%	8%	in 2007 and increasing at a rate of more than 8% per year. In other words, it is a major
fong-spivak-seven-sketches_91 F01038	91	■ 0.984	$\mathcal{C} \rightarrow$	$\mathcal{C} \rightarrow$	by a ‘set-valued’ functor $\mathcal{C} \rightarrow \mathbf{Set}$, and that databases can be mapped to each other via
fong-spivak-seven-sketches_91 F01039	91	■ 1.000	$\mathcal{C} \rightarrow \mathcal{D}$	$\mathcal{C} \rightarrow \mathcal{D}$	functors $\mathcal{C} \rightarrow \mathcal{D}$. In other words, there is a relatively large overlap between database
fong-spivak-seven-sketches_93 F01040	93	■ 1.000	$\{ \}^2 \mathrm{Ob}(\mathcal{C})$	${}^2\mathrm{Ob}(\mathcal{C})$	(i) one specifies a collection ² $\mathrm{Ob}(\mathcal{C})$, elements of which are called <i>objects</i> .
fong-spivak-seven-sketches_93 F01041	93	■ 0.963	$\mathcal{C}(c, d), \{ \}^3$	$\mathcal{C}(c, d), {}^3$	(ii) for every two objects c, d , one specifies a set $\mathcal{C}(c, d)$, ³ elements of which are called
fong-spivak-seven-sketches_93 F01042	93	■ 0.964	$c \in \mathrm{Ob}(\mathcal{C})$	$c \in \mathrm{Ob}(\mathcal{C})$	(iii) for every object $c \in \mathrm{Ob}(\mathcal{C})$, one specifies a morphism $\mathrm{id}_c \in \mathcal{C}(c, c)$, called the
fong-spivak-seven-sketches_93 F01043	93	■ 0.964	$\mathrm{id}_c \in \mathcal{C}(c, c)$	$\mathrm{id}_c \in \mathcal{C}(c, c)$	(iii) for every object $c \in \mathrm{Ob}(\mathcal{C})$, one specifies a morphism $\mathrm{id}_c \in \mathcal{C}(c, c)$, called the
fong-spivak-seven-sketches_93 F01044	93	■ 0.939	$f \in \mathcal{C}(c, d)$	$f \in \mathcal{C}(c, d)$	(iv) for every three objects $c, d, e \in \mathrm{Ob}(\mathcal{C})$ and morphisms $f \in \mathcal{C}(c, d)$ and $g \in \mathcal{C}(d, e)$,
fong-spivak-seven-sketches_93 F01045	93	■ 0.939	$g \in \mathcal{C}(d, e)$	$g \in \mathcal{C}(d, e)$	(iv) for every three objects $c, d, e \in \mathrm{Ob}(\mathcal{C})$ and morphisms $f \in \mathcal{C}(c, d)$ and $g \in \mathcal{C}(d, e)$,
fong-spivak-seven-sketches_93 F01046	93	■ 0.998	$f ; g \in \mathcal{C}(c, e)$	$f ; g \in \mathcal{C}(c, e)$	one specifies a morphism $f \circ g \in \mathcal{C}(c, e)$, called <i>the composite of f and g</i> .
fong-spivak-seven-sketches_93 F01047	93	■ 0.948	$f: c \rightarrow d$	$f: c \rightarrow d$	to denote elements $f \in \mathcal{C}(c, d)$ as $f: c \rightarrow d$. Here, c is called the <i>domain</i> of f , and d is
fong-spivak-seven-sketches_93 F01048	93	■ 0.620	$\mathrm{id}_c ; f = f$	$\mathrm{id}_c ; f = f$	does nothing: $\mathrm{id}_c \circ f = f$ and $f \circ \mathrm{id}_d = f$.
fong-spivak-seven-sketches_93 F01049	93	■ 0.620	$f ; \mathrm{id}_d = f$	$f ; \mathrm{id}_d = f$	does nothing: $\mathrm{id}_c \circ f = f$ and $f \circ \mathrm{id}_d = f$.
fong-spivak-seven-sketches_93 F01050	93	■ 1.000	$f: c_0 \rightarrow c_1, g: c_1 \rightarrow c_2$	$f: c_0 \rightarrow c_1, g: c_1 \rightarrow c_2$	(b) <i>associativity</i> : for any three morphisms $f: c_0 \rightarrow c_1, g: c_1 \rightarrow c_2$, and $h: c_2 \rightarrow c_3$,
fong-spivak-seven-sketches_93 F01051	93	■ 1.000	$h: c_2 \rightarrow c_3$	$h: c_2 \rightarrow c_3$	(b) <i>associativity</i> : for any three morphisms $f: c_0 \rightarrow c_1, g: c_1 \rightarrow c_2$, and $h: c_2 \rightarrow c_3$,
fong-spivak-seven-sketches_93 F01052	93	■ 1.000	$(f ; g) ; h = f ; (g ; h)$	$(f ; g) ; h = f ; (g ; h)$	the following are equal: $(f \circ g) \circ h = f \circ (g \circ h)$. We write this composite simply
fong-spivak-seven-sketches_93 F01053	93	■ 0.947	$f ; g ; h$	$f ; g ; h$	as $f \circ g \circ h$.
fong-spivak-seven-sketches_93 F01054	93	■ 0.998	$\mathcal{C}(c, d)$	$\mathcal{C}(c, d)$	³ This set $\mathcal{C}(c, d)$ is often denoted $\mathrm{Hom}_{\mathcal{C}}(c, d)$, and called the “hom-set from c to d .” The word “hom”
fong-spivak-seven-sketches_93 F01055	93	■ 0.998	$\mathrm{Hom}_{\mathcal{C}}(c, d)$	$\mathrm{Hom}_{\mathcal{C}}(c, d)$	³ This set $\mathcal{C}(c, d)$ is often denoted $\mathrm{Hom}_{\mathcal{C}}(c, d)$, and called the “hom-set from c to d .” The word “hom”
fong-spivak-seven-sketches_94 F01056	94	■ 1.000	$p: v \rightarrow w$	$p: v \rightarrow w$	write <u>$p: v \rightarrow w$</u> . To every vertex v , there is a trivial path, containing no arrows, starting
fong-spivak-seven-sketches_94 F01057	94	■ 0.999	id_v	id_v	and ending at v ; we often denote it by id_v or simply by v . We may also concatenate
fong-spivak-seven-sketches_94 F01058	94	■ 1.000	$q: w \rightarrow x$	$q: w \rightarrow x$	paths: given $p: v \rightarrow w$ and <u>$q: w \rightarrow x$</u> , their concatenation is denoted $p \circ q$, and it goes
fong-spivak-seven-sketches_94 F01059	94	■ 1.000	$p ; q$	$p ; q$	paths: given $p: v \rightarrow w$ and $q: w \rightarrow x$, their concatenation is denoted $p \circ q$, and it goes

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches- F01060	94	■ 1.000	$v \rightarrowtail x$	$v \rightarrow x$	
fong-spivak-seven-sketches- F01061	94	■ 0.980	$(V, \leq)$	$(V, \leq)$	In Chapter 1, we used graphs to depict preorders $(V, \leq)$: the vertices form the
fong-spivak-seven-sketches- F01062	94	■ 0.999	$\operatorname{Free}(G)$	$\operatorname{Free}(G)$	Definition 3.7. For any graph $G = (V, A, s, t)$, we can define a category Free (G), called
fong-spivak-seven-sketches- F01063	94	■ 0.754	$v_{\{1\}}$	v_1	It has two objects v_1 and v_2 , and three morphisms: $\operatorname{id}_{v_1} : v_1 \rightarrow v_1$, $f_1 : v_1 \rightarrow v_2$, and
fong-spivak-seven-sketches- F01064	94	■ 0.754	$v_{\{2\}}$	v_2	It has two objects v_1 and v_2 , and three morphisms: $\operatorname{id}_{v_1} : v_1 \rightarrow v_1$, $f_1 : v_1 \rightarrow v_2$, and
fong-spivak-seven-sketches- F01065	94	■ 0.754	$\operatorname{id}_{v_{\{1\}}} : v_{\{1\}} \rightarrowtail v_{\{1\}}, f_{\{1\}} : v_{\{1\}} \rightarrowtail v_{\{2\}}$	$\operatorname{id}_{v_1} : v_1 \rightarrow v_1, f_1 : v_1 \rightarrow v_2$	It has two objects v_1 and v_2 , and three morphisms: $\operatorname{id}_{v_1} : v_1 \rightarrow v_1$, $f_1 : v_1 \rightarrow v_2$, and
fong-spivak-seven-sketches- F01066	94	■ 0.607	$\operatorname{id}_{v_{\{2\}}} : v_{\{2\}} \rightarrowtail v_{\{2\}}$	$\operatorname{id}_{v_2} : v_2 \rightarrow v_2$	$\operatorname{id}_{v_2} : v_2 \rightarrow v_2$. Here id_{v_1} is the path of length 0 starting and ending at v_1 , f_1 is the path
fong-spivak-seven-sketches- F01067	94	■ 0.607	$\operatorname{id}_{v_{\{1\}}}$	id_{v_1}	$\operatorname{id}_{v_2} : v_2 \rightarrow v_2$. Here id_{v_1} is the path of length 0 starting and ending at v_1 , f_1 is the path
fong-spivak-seven-sketches- F01068	94	■ 0.607	$v_{\{1\}}, f_{\{1\}}$	v_1, f_1	$\operatorname{id}_{v_2} : v_2 \rightarrow v_2$. Here id_{v_1} is the path of length 0 starting and ending at v_1 , f_1 is the path
fong-spivak-seven-sketches- F01069	94	■ 0.997	$f_{\{1\}}$	f_1	of length 1 consisting of just the arrow f_1 , and id_{v_2} is the length 0 path at v_2 . As our
fong-spivak-seven-sketches- F01070	94	■ 0.997	$\operatorname{id}_{v_{\{2\}}}$	id_{v_2}	of length 1 consisting of just the arrow f_1 , and id_{v_2} is the length 0 path at v_2 . As our
fong-spivak-seven-sketches- F01071	94	■ 0.562	$f_{\{1\}} = f_{\{1\}}$	$f_1 = f_1$	for v_2 . As composition is given by concatenation, we have, for example $\operatorname{id}_{v_1} \circ f_1 = f_1$,
fong-spivak-seven-sketches- F01072	94	■ 0.524	$\operatorname{id}_{v_{\{2\}}} \circ \operatorname{id}_{v_{\{2\}}} = \operatorname{id}_{v_{\{2\}}}$	$\operatorname{id}_{v_2} \circ \operatorname{id}_{v_2} = \operatorname{id}_{v_2}$	$\operatorname{id}_{v_2} \circ \operatorname{id}_{v_2} = \operatorname{id}_{v_2}$, and so on.
fong-spivak-seven-sketches- F01073	95	■ 0.621	$\mathbf{0}$	0	2. What is the category 0 ?
fong-spivak-seven-sketches- F01074	95	■ 1.000	$\mathbf{n}$	n	3. What is the formula for the number of morphisms in n for arbitrary $n \in \mathbb{N}$?
fong-spivak-seven-sketches- F01075	95	■ 0.968	$\{z, s, (s ; s), (s ; s ; s), \ldots\}$	$\{z, s, (s ; s), (s ; s ; s), \ldots\}$	$\{z, s, (s \circ s), (s \circ s \circ s), \ldots\}$, where we write z for the length 0 path on z ; it represents
fong-spivak-seven-sketches- F01076	95	■ 0.967	id_z	id_z	the morphism id_z . There is a one-to-one correspondence between Path and the natural
fong-spivak-seven-sketches- F01077	95	■ 1.000	$\mathbb{N} = \{0, 1, 2, 3, \ldots\}$	$\mathbb{N} = \{0, 1, 2, 3, \ldots\}$	numbers, $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$.
fong-spivak-seven-sketches- F01078	95	■ 0.966	$* : M \times M \rightarrowtail M$	$* : M \times M \rightarrow M$	a tuple $(M, *, e)$ where $* : M \times M \rightarrow M$ is a function and $e \in M$ is an element, and
fong-spivak-seven-sketches- F01079	95	■ 1.000	$m * 1 = m = 1 * m$	$m * 1 = m = 1 * m$	$m * 1 = m = 1 * m$ and $(m * n) * p = m * (n * p)$.
fong-spivak-seven-sketches- F01080	95	■ 1.000	$(m * n) * p = m * (n * p)$	$(m * n) * p = m * (n * p)$	$m * 1 = m = 1 * m$ and $(m * n) * p = m * (n * p)$.
fong-spivak-seven-sketches- F01081	95	■ 0.851	$M := \mathcal{C}(\bullet, \bullet)$	$M := \mathcal{C}(\bullet, \bullet)$	connection. Given a category $\mathcal{C}$ with one object, say $\bullet$, let $M := \mathcal{C}(\bullet, \bullet)$, let $e = \operatorname{id}_\bullet$, and
fong-spivak-seven-sketches- F01082	95	■ 0.851	$e = \operatorname{id}_\bullet$	$e = \operatorname{id}_\bullet$	connection. Given a category $\mathcal{C}$ with one object, say $\bullet$, let $M := \mathcal{C}(\bullet, \bullet)$, let $e = \operatorname{id}_\bullet$, and
fong-spivak-seven-sketches- F01083	95	■ 0.922	$* : \mathcal{C}(\bullet, \bullet) \times \mathcal{C}(\bullet, \bullet) \rightarrowtail \mathcal{C}(\bullet, \bullet)$	$* : \mathcal{C}(\bullet, \bullet) \times \mathcal{C}(\bullet, \bullet) \rightarrow \mathcal{C}(\bullet, \bullet)$	let $* : \mathcal{C}(\bullet, \bullet) \times \mathcal{C}(\bullet, \bullet) \rightarrow \mathcal{C}(\bullet, \bullet)$ be the composition operation $* = \circ$. The associativity
fong-spivak-seven-sketches- F01084	95	■ 0.922	$* = \circ$	$* = \circ$	let $* : \mathcal{C}(\bullet, \bullet) \times \mathcal{C}(\bullet, \bullet) \rightarrow \mathcal{C}(\bullet, \bullet)$ be the composition operation $* = \circ$. The associativity
fong-spivak-seven-sketches- F01085	95	■ 1.000	$m, n \in \mathbb{N}$	$m, n \in \mathbb{N}$	numbers $n \in \mathbb{N}$. Paths can be concatenated. Given numbers $m, n \in \mathbb{N}$, what number
fong-spivak-seven-sketches- F01086	96	■ 0.920	$f \circ h$	$f \circ h$	because $f \circ h$ and $g \circ i$ are made equal. This category is called the “commutative square.”

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_96 F01087	96	■ 0.920	$g \circ i$	$g \circ i$	because $f \circ h$ and $g \circ i$ are made equal. This category is called the “commutative square.”
fong-spivak-seven-sketches_96 F01088	96	■ 0.547	$g_{\{9\}} i$	$g_9 i$	One might say “the missing one is $g \circ i$,” but that is not quite right: $g \circ i$ is there too,
fong-spivak-seven-sketches_96 F01089	96	■ 0.795	$f ; h$	$f;h$	because it is equal to $f \circ h$. As usual, A denotes id_A , etc.
fong-spivak-seven-sketches_96 F01090	96	■ 0.795	$\operatorname{id}_{\{A\}}$	id_A	because it is equal to $f \circ h$. As usual, A denotes id_A , etc.
fong-spivak-seven-sketches_97 F01091	97	■ 0.499	$s ; s=z$	$s; s = z$	In $\mathcal{C}$ we have the equation $s \circ s = z$. But this implies $s \circ s \circ s = z \circ s = s!$ And similarly
fong-spivak-seven-sketches_97 F01092	97	■ 0.499	$s ; s ; s=z ; s=s$	$s; s; s = z; s = s$	In $\mathcal{C}$ we have the equation $s \circ s = z$. But this implies $s \circ s \circ s = z \circ s = s!$ And similarly
fong-spivak-seven-sketches_97 F01093	97	■ 0.493	$; s=z$	$; s = z$	we have $s \circ s \circ s \circ s = z \circ z = z$. The set of morphisms in $\mathcal{C}$ is in fact merely $\{z, s\}$, with
fong-spivak-seven-sketches_97 F01094	97	■ 0.493	$=z$	$= z$	we have $s \circ s \circ s \circ s = z \circ z = z$. The set of morphisms in $\mathcal{C}$ is in fact merely $\{z, s\}$, with
fong-spivak-seven-sketches_97 F01095	97	■ 0.493	$\{z, s\}$	$\{z, s\}$	we have $s \circ s \circ s \circ s = z \circ z = z$. The set of morphisms in $\mathcal{C}$ is in fact merely $\{z, s\}$, with
fong-spivak-seven-sketches_97 F01096	97	■ 0.556	$z=z$	$z = z$	composition described by $s \circ s = z \circ z = z$, and $z \circ s = s \circ z = s$. In group theory, one
fong-spivak-seven-sketches_97 F01097	97	■ 0.556	$z ; s=s ; z=s$	$z; s = s; z = s$	composition described by $s \circ s = z \circ z = z$, and $z \circ s = s \circ z = s$. In group theory, one
fong-spivak-seven-sketches_97 F01098	97	■ 1.000	$\mathbb{Z} / 2 \mathbb{Z}$	$\mathbb{Z}/2\mathbb{Z}$	would speak of a group called $\mathbb{Z}/2\mathbb{Z}$.
fong-spivak-seven-sketches_97 F01099	97	■ 1.000	$\mathcal{D}$	$\mathcal{D}$	Exercise 3.19. Write down all the morphisms in the category $\mathcal{D}$ from Example 3.18.
fong-spivak-seven-sketches_97 F01100	97	■ 1.000	$\operatorname{Ob}(\mathcal{P})=P$	$\text{Ob}(\mathcal{P}) = P$	follows. The objects of $\mathcal{P}$ are precisely the elements of P ; that is, $\text{Ob}(\mathcal{P}) = P$. As for
fong-spivak-seven-sketches_97 F01101	97	■ 1.000	$p \rightarrow q$	$p \rightarrow q$	morphisms, $\mathcal{P}$ has exactly one morphism $p \rightarrow q$ if $p \leq q$ and no morphisms $p \rightarrow q$ if
fong-spivak-seven-sketches_97 F01102	97	■ 1.000	$p \not \leq q$	$p \not\leq q$	$p \not\leq q$. The fact that $\leq$ is reflexive ensures that every object has an identity, and the fact
fong-spivak-seven-sketches_98 F01103	98	■ 1.000	$(C, \leq)$	$(C, \leq)$	$(C, \leq)$ from it by destroying the distinction between any two parallel morphisms. That
fong-spivak-seven-sketches_98 F01104	98	■ 0.985	$C:=\operatorname{Ob}(\mathcal{C})$	$C := \text{Ob}(\mathcal{C})$	is, let $C := \text{Ob}(\mathcal{C})$, and put $c_1 \leq c_2$ iff $\mathcal{C}(c_1, c_2) \neq \emptyset$. If there is one, or two, or fifty,
fong-spivak-seven-sketches_98 F01105	98	■ 0.985	$c_{\{1\}} \leq c_{\{2\}}$	$c_1 \leq c_2$	is, let $C := \text{Ob}(\mathcal{C})$, and put $c_1 \leq c_2$ iff $\mathcal{C}(c_1, c_2) \neq \emptyset$. If there is one, or two, or fifty,
fong-spivak-seven-sketches_98 F01106	98	■ 0.985	$\mathcal{C}(c_{\{1\}}, c_{\{2\}}) \neq \varnothing$	$\mathcal{C}(c_1, c_2) \neq \emptyset$	is, let $C := \text{Ob}(\mathcal{C})$, and put $c_1 \leq c_2$ iff $\mathcal{C}(c_1, c_2) \neq \emptyset$. If there is one, or two, or fifty,
fong-spivak-seven-sketches_98 F01107	98	■ 1.000	$c_{\{1\}} \rightarrow c_{\{2\}}$	$c_1 \rightarrow c_2$	or infinitely many morphisms $c_1 \rightarrow c_2$ in $\mathcal{C}$, the preorder reflection does not see the
fong-spivak-seven-sketches_99 F01108	99	■ 0.810	$\operatorname{Ob}(\mathcal{C})$	$\text{Ob}(\mathcal{C})$	(i) $\text{Ob}(\mathbf{Set})$ is the collection of all sets.
fong-spivak-seven-sketches_99 F01109	99	■ 0.998	$\mathbf{Set}(S, T)=\{f: S \rightarrow T \mid f$	$\mathbf{Set}(S, T) = \{f : S \rightarrow T \mid f$	(ii) If S and T are sets, then $\mathbf{Set}(S, T) = \{f : S \rightarrow T \mid f$ is a function}.
fong-spivak-seven-sketches_99 F01110	99	■ 0.958	$\operatorname{id}_S: S \rightarrow S$	$\text{id}_S : S \rightarrow S$	(iii) For each set S , the identity morphism is the function $\text{id}_S : S \rightarrow S$ given by
fong-spivak-seven-sketches_99 F01111	99	■ 0.908	$\operatorname{id}_S(s):=s$	$\text{id}_S(s) := s$	$\text{id}_S(s) := s$ for each $s \in S$.
fong-spivak-seven-sketches_99 F01112	99	■ 1.000	$f: S \rightarrow T$	$f : S \rightarrow T$	(iv) Given $f : S \rightarrow T$ and $g : T \rightarrow U$, their composite is the function $f \circ g : S \rightarrow U$
fong-spivak-seven-sketches_99 F01113	99	■ 1.000	$g: T \rightarrow U$	$g : T \rightarrow U$	(iv) Given $f : S \rightarrow T$ and $g : T \rightarrow U$, their composite is the function $f \circ g : S \rightarrow U$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_99 F01114	99	■ 1.000	$f : S \rightarrow U$	$f; g : S \rightarrow U$	(iv) Given $f: S \rightarrow T$ and $g: T \rightarrow U$, their composite is the function $f \circ g: S \rightarrow U$
fong-spivak-seven-sketches_99 F01115	99	■ 1.000	$(f \circ g)(s) := g(f(s))$	$(f; g)(s) := g(f(s))$	given by $(f \circ g)(s) := g(f(s))$.
fong-spivak-seven-sketches_99 F01116	99	■ 1.000	$\underline{2} = \{1, 2\}$	$\underline{2} = \{1, 2\}$	Exercise 3.25. Let $\underline{2} = \{1, 2\}$ and $\underline{3} = \{1, 2, 3\}$. These are objects in the category Set
fong-spivak-seven-sketches_99 F01117	99	■ 1.000	$\underline{3} = \{1, 2, 3\}$	$\underline{3} = \{1, 2, 3\}$	Exercise 3.25. Let $\underline{2} = \{1, 2\}$ and $\underline{3} = \{1, 2, 3\}$. These are objects in the category Set
fong-spivak-seven-sketches_99 F01118	99	■ 0.952	$\mathbf{Set}, \mathbf{Set}$	Set(2, 3)	discussed in Definition 3.24. Write down all the elements of the set Set(2, 3) ; there
fong-spivak-seven-sketches_100 F01119	100	■ 0.984	$\mathcal{C}^{\text{op}}$	$\mathcal{C}^{\text{op}}$	Example 3.27. Let $\mathcal{C}$ be a category. Its <i>opposite</i> , denoted $\mathcal{C}^{\text{op}}$, is the category with
fong-spivak-seven-sketches_100 F01120	100	■ 0.268	$\text{Ob}(\mathcal{C}^{\text{op}}) := \text{Ob}(\mathcal{C})$	$\text{Ob}(\mathcal{C}^{\text{op}}) := \text{Ob}(\mathcal{C})$	the same objects, $\text{Ob}(\mathcal{C}^{\text{op}}) := \text{Ob}(\mathcal{C})$, and for any two objects $c, d \in \text{Ob}(\mathcal{C})$, one has
fong-spivak-seven-sketches_100 F01121	100	■ 0.984	$\mathcal{C}^{\text{op}}(c, d) := \mathcal{C}(d, c)$	$\mathcal{C}^{\text{op}}(c, d) := \mathcal{C}(d, c)$	$\mathcal{C}^{\text{op}}(c, d) := \mathcal{C}(d, c)$. Identities and composition are as in $\mathcal{C}$.
fong-spivak-seven-sketches_100 F01122	100	■ 0.980	$\{\mathbf{m}, \square\}$	$\{\mathbf{m}, \square\}$	example, in the category Set , one can exchange the set $\{\mathbf{m}, \square\}$ for the set $\{0, 1\}$ and
fong-spivak-seven-sketches_100 F01123	100	■ 0.980	$\{0, 1\}$	$\{0, 1\}$	example, in the category Set , one can exchange the set $\{\mathbf{m}, \square\}$ for the set $\{0, 1\}$ and
fong-spivak-seven-sketches_100 F01124	100	■ 0.999	$a \cong b$	$a \cong b$	has a preorder with elements a, b , such that $a \leq b$ and $b \leq a$, i.e. $a \cong b$, then a and b are
fong-spivak-seven-sketches_100 F01125	100	■ 1.000	$c \leq a$	$c \leq a$	know that $c \leq a$ iff $c \leq b$, and $c \geq a$ iff $c \geq b$. The notion of isomorphism formalizes
fong-spivak-seven-sketches_100 F01126	100	■ 1.000	$c \leq b$	$c \leq b$	know that $c \leq a$ iff $c \leq b$, and $c \geq a$ iff $c \geq b$. The notion of isomorphism formalizes
fong-spivak-seven-sketches_100 F01127	100	■ 1.000	$c \geq a$	$c \geq a$	know that $c \leq a$ iff $c \leq b$, and $c \geq a$ iff $c \geq b$. The notion of isomorphism formalizes
fong-spivak-seven-sketches_100 F01128	100	■ 1.000	$c \geq b$	$c \geq b$	know that $c \leq a$ iff $c \leq b$, and $c \geq a$ iff $c \geq b$. The notion of isomorphism formalizes
fong-spivak-seven-sketches_100 F01129	100	■ 0.931	$g: B \rightarrow A$	$g: B \rightarrow A$	morphism $g: B \rightarrow A$ satisfying $f \circ g = \text{id}_A$ and $g \circ f = \text{id}_B$. In this case we call f and g
fong-spivak-seven-sketches_100 F01130	100	■ 0.931	$f \circ g = \text{id}_A$	$f; g = \text{id}_A$	morphism $g: B \rightarrow A$ satisfying $f \circ g = \text{id}_A$ and $g \circ f = \text{id}_B$. In this case we call f and g
fong-spivak-seven-sketches_100 F01131	100	■ 0.931	$g \circ f = \text{id}_B$	$g; f = \text{id}_B$	morphism $g: B \rightarrow A$ satisfying $f \circ g = \text{id}_A$ and $g \circ f = \text{id}_B$. In this case we call f and g
fong-spivak-seven-sketches_100 F01132	100	■ 1.000	$g = f^{-1}$	$g = f^{-1}$	<i>inverses</i> , and we often write $g = f^{-1}$, or equivalently $f = g^{-1}$. We also say that A and B
fong-spivak-seven-sketches_100 F01133	100	■ 1.000	$f = g^{-1}$	$f = g^{-1}$	<i>inverses</i> , and we often write $g = f^{-1}$, or equivalently $f = g^{-1}$. We also say that A and B
fong-spivak-seven-sketches_100 F01134	100	■ 1.000	$A := \{a, b, c\}$	$A := \{a, b, c\}$	Example 3.29. The set $A := \{a, b, c\}$ and the set $\underline{3} = \{1, 2, 3\}$ are isomorphic; that is,
fong-spivak-seven-sketches_100 F01135	100	■ 1.000	$f: A \rightarrow \underline{3}$	$f: A \rightarrow \underline{3}$	there exists an isomorphism $f: A \rightarrow \underline{3}$ given by $f(a) = 2, f(b) = 1, f(c) = 3$. The
fong-spivak-seven-sketches_100 F01136	100	■ 1.000	$f(a)=2, f(b)=1, f(c)=3$	$f(a) = 2, f(b) = 1, f(c) = 3$	there exists an isomorphism $f: A \rightarrow \underline{3}$ given by $f(a) = 2, f(b) = 1, f(c) = 3$. The
fong-spivak-seven-sketches_100 F01137	100	■ 0.954	$A \in$	$A \in$	understood in terms of isomorphisms in FinSet . Namely, for any finite set $A \in \mathbf{FinSet}$,
fong-spivak-seven-sketches_100 F01138	100	■ 1.000	$A \cong \underline{n}$	$A \cong \underline{n}$	its cardinality is the number $n \in \mathbb{N}$ such that there exists an isomorphism $A \cong \underline{n}$. Georg
fong-spivak-seven-sketches_100 F01139	100	■ 1.000	$f^{-1}: \underline{3} \rightarrow A$	$f^{-1}: \underline{3} \rightarrow A$	1. What is the inverse $f^{-1}: \underline{3} \rightarrow A$ of the function f given in Example 3.29?
fong-spivak-seven-sketches_100 F01140	100	■ 0.998	$A \rightarrow \underline{3}$	$A \rightarrow \underline{3}$	2. How many distinct isomorphisms are there $A \rightarrow \underline{3}$?

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F01141	100	1.000	$\mathrm{id}_{\{c\}}$	id_c	identity id_c is an isomorphism.
fong-spivak-seven-sketches-F01142	101	0.880	$\mathrm{Free}(G)$	$\mathbf{Free}(G)$	<i>Exercise 3.33.</i> Let G be a graph, and let $\mathbf{Free}(G)$ be the corresponding free category.
fong-spivak-seven-sketches-F01143	101	1.000	$f: \underline{2} \rightarrow \underline{3}$	$f: \underline{2} \rightarrow \underline{3}$	Consider the functions $f: \underline{2} \rightarrow \underline{3}$ and $g: \underline{3} \rightarrow \underline{2}$ drawn below:
fong-spivak-seven-sketches-F01144	101	1.000	$g: \underline{3} \rightarrow \underline{2}$	$g: \underline{3} \rightarrow \underline{2}$	Consider the functions $f: \underline{2} \rightarrow \underline{3}$ and $g: \underline{3} \rightarrow \underline{2}$ drawn below:
fong-spivak-seven-sketches-F01145	101	0.330	$f; g = \mathrm{id}_{\underline{2}}$	$f; g = \mathrm{id}_{\underline{2}}$	Then the reader should be able to instantly check that $f \circ g = \mathrm{id}_{\underline{2}}$ but $g \circ f \neq \mathrm{id}_{\underline{3}}$. Thus f
fong-spivak-seven-sketches-F01146	101	0.330	$g \circ f \neq \mathrm{id}_{\underline{3}}$	$g \circ f \neq \mathrm{id}_{\underline{3}}$	Then the reader should be able to instantly check that $f \circ g = \mathrm{id}_{\underline{2}}$ but $g \circ f \neq \mathrm{id}_{\underline{3}}$. Thus f
fong-spivak-seven-sketches-F01147	103	0.998	$F: \mathcal{C} \rightarrow \mathcal{D}$	$F: \mathcal{C} \rightarrow \mathcal{D}$	$F: \mathcal{C} \rightarrow \mathcal{D}$,
fong-spivak-seven-sketches-F01148	103	0.989	$F(c) \in \mathrm{Ob}(\mathcal{D})$	$F(c) \in \mathrm{Ob}(\mathcal{D})$	(i) for every object $c \in \mathrm{Ob}(\mathcal{C})$, one specifies an object $F(c) \in \mathrm{Ob}(\mathcal{D})$;
fong-spivak-seven-sketches-F01149	103	0.203	$f: c_1 \rightarrow c_2$	$f: c_1 \rightarrow c_2$	(ii) for every morphism $f: c_1 \rightarrow c_2$ in $\mathcal{C}$, one specifies a morphism $F(f): F(c_1) \rightarrow$
fong-spivak-seven-sketches-F01150	103	0.203	$\mathcal{C}_{\text{one}}$	$\mathcal{C}_{\text{one}}$	(ii) for every morphism $f: c_1 \rightarrow c_2$ in $\mathcal{C}$, one specifies a morphism $F(f): F(c_1) \rightarrow$
fong-spivak-seven-sketches-F01151	103	0.203	$F(f): F(c_1) \rightarrow$	$F(f): F(c_1) \rightarrow$	(ii) for every morphism $f: c_1 \rightarrow c_2$ in $\mathcal{C}$, one specifies a morphism $F(f): F(c_1) \rightarrow$
fong-spivak-seven-sketches-F01152	103	1.000	$F(c_2)$	$F(c_2)$	$F(c_2)$ in $\mathcal{D}$.
fong-spivak-seven-sketches-F01153	103	0.628	$F(\mathrm{id}_c) = \mathrm{id}_{F(c)}$	$F(\mathrm{id}_c) = \mathrm{id}_{F(c)}$	(a) for every object $c \in \mathrm{Ob}(\mathcal{C})$, we have $F(\mathrm{id}_c) = \mathrm{id}_{F(c)}$.
fong-spivak-seven-sketches-F01154	103	0.870	$c_1, c_2, c_3 \in \mathrm{Ob}(\mathcal{C})$	$c_1, c_2, c_3 \in \mathrm{Ob}(\mathcal{C})$	(b) for every three objects $c_1, c_2, c_3 \in \mathrm{Ob}(\mathcal{C})$ and two morphisms $f \in \mathcal{C}(c_1, c_2)$, $g \in$
fong-spivak-seven-sketches-F01155	103	0.870	$f \in \mathcal{C}(c_1, c_2), g \in$	$f \in \mathcal{C}(c_1, c_2), g \in$	(b) for every three objects $c_1, c_2, c_3 \in \mathrm{Ob}(\mathcal{C})$ and two morphisms $f \in \mathcal{C}(c_1, c_2)$, $g \in$
fong-spivak-seven-sketches-F01156	103	1.000	$\mathcal{C}(c_2, c_3)$	$\mathcal{C}(c_2, c_3)$	$\mathcal{C}(c_2, c_3)$, the equation $F(f \circ g) = F(f) \circ F(g)$ holds in $\mathcal{D}$.
fong-spivak-seven-sketches-F01157	103	1.000	$F(f \circ g) = F(f) \circ F(g)$	$F(f \circ g) = F(f) \circ F(g)$	$\mathcal{C}(c_2, c_3)$, the equation $F(f \circ g) = F(f) \circ F(g)$ holds in $\mathcal{D}$.
fong-spivak-seven-sketches-F01158	103	0.975	$F: 2 \rightarrow 3$	$F: 2 \rightarrow 3$	<i>Example 3.36.</i> For example, here we draw three functors $F: 2 \rightarrow 3$:
fong-spivak-seven-sketches-F01159	103	1.000	n_0	n_0	path to the trivial path, i.e. the identity on n_0 . In the middle diagram $F(m_0) = n_0$,
fong-spivak-seven-sketches-F01160	103	1.000	$F(m_0) = n_0$	$F(m_0) = n_0$	path to the trivial path, i.e. the identity on n_0 . In the middle diagram $F(m_0) = n_0$,
fong-spivak-seven-sketches-F01161	103	1.000	$F(f_1) = g_1$	$F(f_1) = g_1$	$F(f_1) = g_1$, and $F(m_1) = n_1$. In the right-hand diagram, $F(m_0) = n_0$, $F(m_1) = n_2$, and
fong-spivak-seven-sketches-F01162	103	1.000	$F(m_1) = n_1$	$F(m_1) = n_1$	$F(f_1) = g_1$, and $F(m_1) = n_1$. In the right-hand diagram, $F(m_0) = n_0$, $F(m_1) = n_2$, and
fong-spivak-seven-sketches-F01163	103	1.000	$F(m_0) = n_0, F(m_1) = n_2$	$F(m_0) = n_0, F(m_1) = n_2$	$F(f_1) = g_1$, and $F(m_1) = n_1$. In the right-hand diagram, $F(m_0) = n_0$, $F(m_1) = n_2$, and
fong-spivak-seven-sketches-F01164	103	0.996	$F(f_1) = g_1; g_2$	$F(f_1) = g_1; g_2$	$F(f_1) = g_1 \circ g_2$.
fong-spivak-seven-sketches-F01165	103	0.772	$2 \rightarrow 3$	$2 \rightarrow 3$	the remaining functors $2 \rightarrow 3$.
fong-spivak-seven-sketches-F01166	104	1.000	$\mathcal{F}$	$\mathcal{F}$	There are lots of functors from the free square category (let's call it $\mathcal{F}$) to the commutative
fong-spivak-seven-sketches-F01167	104	1.000	$F: \mathcal{F} \rightarrow \mathcal{C}$	$F: \mathcal{F} \rightarrow \mathcal{C}$	However, there is exactly one functor $F: \mathcal{F} \rightarrow \mathcal{C}$ that sends A' to A , B' to B , C' to C ,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_104 F01168	104	■ 1.000	$A, B^{\prime}$	A, B'	However, there is exactly one functor $F: \mathcal{F} \rightarrow \mathcal{C}$ that sends A' to A , B' to B , C' to C ,
fong-spivak-seven-sketches_104 F01169	104	■ 1.000	$B, C^{\prime}$	B, C'	However, there is exactly one functor $F: \mathcal{F} \rightarrow \mathcal{C}$ that sends A' to A , B' to B , C' to C ,
fong-spivak-seven-sketches_104 F01170	104	■ 1.000	$D^{\prime}$	D'	and D' to D . That is, once we have made this decision about how F acts on objects,
fong-spivak-seven-sketches_104 F01171	104	■ 0.994	$\mathcal{C} = \bullet \rightarrow \bullet$	$\mathcal{C} = \bullet \rightarrow \bullet$	Exercise 3.40. Consider the free categories $\mathcal{C} = \boxed{\bullet \rightarrow \bullet}$ and $\mathcal{D} = \boxed{\bullet \rightrightarrows \bullet}$. Give two
fong-spivak-seven-sketches_104 F01172	104	■ 0.994	$\mathcal{D} = \bullet \rightrightarrows \bullet$	$\mathcal{D} = \bullet \rightrightarrows \bullet$	Exercise 3.40. Consider the free categories $\mathcal{C} = \boxed{\bullet \rightarrow \bullet}$ and $\mathcal{D} = \boxed{\bullet \rightrightarrows \bullet}$. Give two
fong-spivak-seven-sketches_104 F01173	104	■ 1.000	$F, G: \mathcal{C} \rightarrow \mathcal{D}$	$F, G: \mathcal{C} \rightarrow \mathcal{D}$	functors $F, G: \mathcal{C} \rightarrow \mathcal{D}$ that act the same on objects but differently on morphisms.
fong-spivak-seven-sketches_104 F01174	104	■ 0.979	$A^{\prime}, B$	A', B	to the free square category $\mathcal{F}$, but <i>none</i> that sends A to A' , B to B' , C to C' , and D to D' .
fong-spivak-seven-sketches_104 F01175	104	■ 0.979	$B^{\prime}, C$	B', C	to the free square category $\mathcal{F}$, but <i>none</i> that sends A to A' , B to B' , C to C' , and D to D' .
fong-spivak-seven-sketches_104 F01176	104	■ 0.979	$C^{\prime}$	C'	to the free square category $\mathcal{F}$, but <i>none</i> that sends A to A' , B to B' , C to C' , and D to D' .
fong-spivak-seven-sketches_104 F01177	104	■ 0.352	$f^{\circ}; h = g^{\circ} i$	$f^{\circ}; h = g^{\circ} i$	The reason is that if F were such a functor, then since $f \circ h = g \circ i$ in $\mathcal{C}$, we would have
fong-spivak-seven-sketches_104 F01178	104	■ 1.000	$F(f; h) = F(g; i)$	$F(f; h) = F(g; i)$	$F(f \circ h) = F(g \circ i)$, but then the rules of functors would let us reason as follows:
fong-spivak-seven-sketches_104 F01179	104	■ 0.932	$f^{\prime}; h^{\prime} = g^{\prime}; i^{\prime}$	$f'; h' = g'; i'$	The resulting equation, $f' \circ h' = g' \circ i'$ does not hold in $\mathcal{F}$ because it is a free category
fong-spivak-seven-sketches_104 F01180	104	■ 0.999	$\mathcal{N}$	$\mathcal{N}$	For example, consider the preorder $(\mathbb{N}, \leq)$ considered as a category $\mathcal{N}$ with objects
fong-spivak-seven-sketches_104 F01181	104	■ 1.000	$\text{Ob}(\mathcal{N}) = \mathbb{N}$	$\text{Ob}(\mathcal{N}) = \mathbb{N}$	$\text{Ob}(\mathcal{N}) = \mathbb{N}$ and a unique morphism $m \rightarrow n$ iff $m \leq n$. A functor $F: \mathcal{N} \rightarrow \mathcal{N}$ sends
fong-spivak-seven-sketches_104 F01182	104	■ 1.000	$m \rightarrow n$	$m \rightarrow n$	$\text{Ob}(\mathcal{N}) = \mathbb{N}$ and a unique morphism $m \rightarrow n$ iff $m \leq n$. A functor $F: \mathcal{N} \rightarrow \mathcal{N}$ sends
fong-spivak-seven-sketches_104 F01183	104	■ 1.000	$F: \mathcal{N} \rightarrow \mathcal{N}$	$F: \mathcal{N} \rightarrow \mathcal{N}$	$\text{Ob}(\mathcal{N}) = \mathbb{N}$ and a unique morphism $m \rightarrow n$ iff $m \leq n$. A functor $F: \mathcal{N} \rightarrow \mathcal{N}$ sends
fong-spivak-seven-sketches_104 F01184	104	■ 1.000	$F(n) \in \mathbb{N}$	$F(n) \in \mathbb{N}$	each object $n \in \mathbb{N}$ to an object $F(n) \in \mathbb{N}$. It must send morphisms in $\mathcal{N}$ to morphisms
fong-spivak-seven-sketches_104 F01185	104	■ 1.000	$F(m) \rightarrow F(n)$	$F(m) \rightarrow F(n)$	$F(m) \rightarrow F(n)$. In other words, if $m \leq n$, then we had better have $F(m) \leq F(n)$. But as
fong-spivak-seven-sketches_104 F01186	104	■ 1.000	$F(m) \leq F(n)$	$F(m) \leq F(n)$	$F(m) \rightarrow F(n)$. In other words, if $m \leq n$, then we had better have $F(m) \leq F(n)$. But as
fong-spivak-seven-sketches_105 F01187	105	■ 1.000	$\mathbb{N} \rightarrow \mathbb{N}$	$\mathbb{N} \rightarrow \mathbb{N}$	Thus a functor $F: \mathcal{N} \rightarrow \mathcal{N}$ and a monotone map $\mathbb{N} \rightarrow \mathbb{N}$ are the same thing.
fong-spivak-seven-sketches_105 F01188	105	■ 0.873	$\text{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$	$\text{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$	1. Given any category $\mathcal{C}$, show that there exists a functor $\text{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$, known as the
fong-spivak-seven-sketches_105 F01189	105	■ 1.000	$\text{Ob}(\mathcal{C})$	$\text{Ob}(\mathcal{C})$	Note that a functor $\mathcal{C} \rightarrow \mathcal{D}$ consists of a function from $\text{Ob}(\mathcal{C})$ to $\text{Ob}(\mathcal{D})$ and for each
fong-spivak-seven-sketches_105 F01190	105	■ 1.000	$\text{Ob}(\mathcal{D})$	$\text{Ob}(\mathcal{D})$	Note that a functor $\mathcal{C} \rightarrow \mathcal{D}$ consists of a function from $\text{Ob}(\mathcal{C})$ to $\text{Ob}(\mathcal{D})$ and for each
fong-spivak-seven-sketches_105 F01191	105	■ 0.987	$c_1, c_2 \in \mathcal{C}$	$c_1, c_2 \in \mathcal{C}$	pair of objects $c_1, c_2 \in \mathcal{C}$ a function from $\mathcal{C}(c_1, c_2)$ to $\mathcal{D}(F(c_1), F(c_2))$.
fong-spivak-seven-sketches_105 F01192	105	■ 0.987	$\mathcal{C}(c_1, c_2)$	$\mathcal{C}(c_1, c_2)$	pair of objects $c_1, c_2 \in \mathcal{C}$ a function from $\mathcal{C}(c_1, c_2)$ to $\mathcal{D}(F(c_1), F(c_2))$.
fong-spivak-seven-sketches_105 F01193	105	■ 0.987	$\mathcal{D}(F(c_1), F(c_2))$	$\mathcal{D}(F(c_1), F(c_2))$	pair of objects $c_1, c_2 \in \mathcal{C}$ a function from $\mathcal{C}(c_1, c_2)$ to $\mathcal{D}(F(c_1), F(c_2))$.
fong-spivak-seven-sketches_105 F01194	105	■ 0.982	$G: \mathcal{D} \rightarrow \mathcal{E}$	$G: \mathcal{D} \rightarrow \mathcal{E}$	2. Show that given $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{D} \rightarrow \mathcal{E}$, we can define a new functor $(F \circ$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_105 F01195	105	■ 1.000	$G): \mathcal{C} \rightarrow \mathcal{E}$	$G): \mathcal{C} \rightarrow \mathcal{E}$	$G): \mathcal{C} \rightarrow \mathcal{E}$ just by composing functions.
fong-spivak-seven-sketches_105 F01196	105	■ 0.991	$F: \mathcal{C} \rightarrow$	$F: \mathcal{C} \rightarrow$	Let $\mathcal{C}$ be a category, and recall the category Set from Definition 3.24. A functor $F: \mathcal{C} \rightarrow$
fong-spivak-seven-sketches_105 F01197	105	■ 0.508	$I: \mathcal{C} \rightarrow$	$I: \mathcal{C} \rightarrow$	of the database—is given by a set-valued functor $I: \mathcal{C} \rightarrow \mathbf{Set}$. The only additional
fong-spivak-seven-sketches_105 F01198	105	■ 1.000	$c=\stackrel{\text{string}}{\circ}$	$c=\stackrel{\text{string}}{\circ}$	detail is that for any white node, such as $c=\stackrel{\text{string}}{\circ}$, we want to force I to map to the set
fong-spivak-seven-sketches_105 F01199	105	■ 0.954	id_1	id_1	functor $I: \mathcal{C} \rightarrow \mathbf{Set}$. ⁵
fong-spivak-seven-sketches_105 F01200	105	■ 0.976	$F: \mathbf{1} \rightarrow$	$F: \mathbf{1} \rightarrow$	phism id_1 , and no other morphisms. For any functor $F: \mathbf{1} \rightarrow \mathbf{Set}$ one can extract a set
fong-spivak-seven-sketches_105 F01201	105	■ 0.976	$F(1)$	$F(1)$	phism id_1 , and no other morphisms. For any functor $F: \mathbf{1} \rightarrow \mathbf{Set}$ one can extract a set
fong-spivak-seven-sketches_105 F01202	105	■ 1.000	$F_S: \mathbf{1} \rightarrow$	$F_S: \mathbf{1} \rightarrow$	$F(1)$. Show that for any set S , there is a functor $F_S: \mathbf{1} \rightarrow \mathbf{Set}$ such that $F_S(1) = S$.
fong-spivak-seven-sketches_105 F01203	105	■ 1.000	$F_S(1)=S$	$F_S(1)=S$	$F(1)$. Show that for any set S , there is a functor $F_S: \mathbf{1} \rightarrow \mathbf{Set}$ such that $F_S(1) = S$.
fong-spivak-seven-sketches_105 F01204	105	■ 1.000	$r \in I(c)$	$r \in I(c)$	$F(1)$. Show that for any set S , there is a functor $F_S: \mathbf{1} \rightarrow \mathbf{Set}$ such that $F_S(1) = S$.
fong-spivak-seven-sketches_106 F01205	106	■ 0.999	$Z:=F(z)$	$Z:=F(z)$	we call a row $r \in I(c)$.
fong-spivak-seven-sketches_106 F01206	106	■ 1.000	$S:=F(s): Z \rightarrow Z$	$S:=F(s): Z \rightarrow Z$	What is a set-valued functor $F: \mathcal{C} \rightarrow \mathbf{Set}$? It will consist of a set $Z:=F(z)$ and a function
fong-spivak-seven-sketches_106 F01207	106	■ 1.000	$S; S=S$	$S; S=S$	$S:=F(s): Z \rightarrow Z$, subject to the requirement that $S \circ S = S$. Here are some examples
fong-spivak-seven-sketches_106 F01208	106	■ 1.000	$Z=\mathbb{N}$	$Z=\mathbb{N}$	$S:=F(s): Z \rightarrow Z$, subject to the requirement that $S \circ S = S$. Here are some examples
fong-spivak-seven-sketches_106 F01209	106	■ 0.837	$13+(2 * 4)$	$13+(2 * 4)$	• $Z = \mathbb{N}$ is the set of natural numbers and S sends each number to 0. In particular,
fong-spivak-seven-sketches_106 F01210	106	■ 0.917	$Z=\mathbb{N}_{\geq 2}$	$Z=\mathbb{N}_{\geq 2}$	• Z is the set of all well-formed arithmetic expressions, such as $13+(2*4)$ or -5 , that
fong-spivak-seven-sketches_106 F01211	106	■ 0.990	$\alpha: F \Rightarrow G$	$\alpha: F \Rightarrow G$	• $Z = \mathbb{N}_{\geq 2}$, and S sends n to its smallest prime factor. The smallest prime factor of
fong-spivak-seven-sketches_107 F01213	107	■ 1.000	$\alpha_c: F(c) \rightarrow G(c)$	$\alpha_c: F(c) \rightarrow G(c)$	a natural transformation $\alpha: F \Rightarrow G$,
fong-spivak-seven-sketches_107 F01214	107	■ 0.884	α	α	(i) for each object $c \in \mathcal{C}$, one specifies a morphism $\alpha_c: F(c) \rightarrow G(c)$ in $\mathcal{D}$, called the
fong-spivak-seven-sketches_107 F01215	107	■ 1.000	$\alpha: F \rightarrow G$	$\alpha: F \rightarrow G$	c -component of α .
fong-spivak-seven-sketches_107 F01216	107	■ 1.000	α_c	α_c	A natural transformation $\alpha: F \rightarrow G$ is called a <i>natural isomorphism</i> if each component
fong-spivak-seven-sketches_107 F01217	107	■ 0.996	$D: \mathcal{J} \rightarrow \mathcal{C}$	$D: \mathcal{J} \rightarrow \mathcal{C}$	α_c is an isomorphism in $\mathcal{D}$.
fong-spivak-seven-sketches_107 F01218	107	■ 0.988	$\mathcal{J}$	$\mathcal{J}$	Definition 3.51. A diagram D in $\mathcal{C}$ is a functor $D: \mathcal{J} \rightarrow \mathcal{C}$ from any category $\mathcal{J}$, called
fong-spivak-seven-sketches_107 F01219	107	■ 0.988	$D(f)=D(f')$	$D(f)=D(f')$	Definition 3.51. A diagram D in $\mathcal{C}$ is a functor $D: \mathcal{J} \rightarrow \mathcal{C}$ from any category $\mathcal{J}$, called
fong-spivak-seven-sketches_107 F01220	107	■ 1.000	$f, f': a \rightarrow b$	$f, f': a \rightarrow b$	the indexing category of the diagram D . We say that D commutes if $D(f) = D(f')$ holds
fong-spivak-seven-sketches_107 F01221	107	■ 0.738	$\mathcal{J}^7$	$\mathcal{J}^7$	for every parallel pair of morphisms $f, f': a \rightarrow b$ in $\mathcal{J}$. ⁷
fong-spivak-seven-sketches_107 F01222	107	■ 0.738			for every parallel pair of morphisms $f, f': a \rightarrow b$ in $\mathcal{J}$. ⁷

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F01223	107	1.000	$F(c) \rightarrowtail G(d)$	$F(c) \Rightarrow G(d)$	In terms of Eq. (3.50), the only case of two parallel morphisms is that of $F(c) \Rightarrow G(d)$,
fong-spivak-seven-sketches-F01224	107	0.528	$F(f) \stackrel{\circ}{\circlearrowleft} \alpha_d = \alpha_c \circ G(f)$	$F(f) \stackrel{\circ}{\circlearrowleft} \alpha_d = \alpha_c \circ G(f)$	so to say that the diagram commutes is to say that $F(f) \circ \alpha_d = \alpha_c \circ G(f)$. This is exactly
fong-spivak-seven-sketches-F01225	108	1.000	$\alpha_1: v \rightarrowtail x$	$\alpha_1: v \rightarrow x$	transformation $\alpha: F \Rightarrow G$ is given by choosing components $\alpha_1: v \rightarrow x$ and $\alpha_2: w \rightarrow y$.
fong-spivak-seven-sketches-F01226	108	1.000	$\alpha_2: w \rightarrowtail y$	$\alpha_2: w \rightarrow y$	transformation $\alpha: F \Rightarrow G$ is given by choosing components $\alpha_1: v \rightarrow x$ and $\alpha_2: w \rightarrow y$.
fong-spivak-seven-sketches-F01227	108	1.000	$\alpha_1 = c$	$\alpha_1 = c$	We have highlighted the only choice for each in green; namely, $\alpha_1 = c$ and $\alpha_2 = g$.
fong-spivak-seven-sketches-F01228	108	1.000	$\alpha_2 = g$	$\alpha_2 = g$	We have highlighted the only choice for each in green; namely, $\alpha_1 = c$ and $\alpha_2 = g$.
fong-spivak-seven-sketches-F01229	108	0.150	$\mathcal{C} \xrightarrow[\mathcal{G}]{F} \mathcal{D}$	$\mathcal{C} \xrightarrow[\mathcal{G}]{F} \mathcal{D}$	the notation $\mathcal{C} \begin{array}{c} \xrightarrow{F} \\ \alpha \Downarrow \\ \xrightarrow{G} \end{array} \mathcal{D}?$
fong-spivak-seven-sketches-F01230	108	0.959	$A, B: \mathbf{1} \rightarrowtail$	$A, B: \mathbf{1} \rightarrow$	a set. So suppose A and B are sets considered as functors $A, B: \mathbf{1} \rightarrow \mathbf{Set}$. A natural
fong-spivak-seven-sketches-F01231	108	1.000	$\mathcal{D}^{\mathcal{C}}$	$\mathcal{D}^{\mathcal{C}}$	Definition 3.54. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. We denote by $\mathcal{D}^{\mathcal{C}}$ the category whose objects
fong-spivak-seven-sketches-F01232	108	0.678	$\mathcal{D}^{\mathcal{P}}(F, G)$	$\mathcal{D}^{\mathcal{P}}(F, G)$	are functors $F: \mathcal{C} \rightarrow \mathcal{D}$ and whose morphisms $\mathcal{D}^{\mathcal{C}}(F, G)$ are the natural transformations
fong-spivak-seven-sketches-F01233	108	0.962	$\mathcal{D}^e$	$\mathcal{D}^e$	Exercise 3.55. Let's look more deeply at how $\mathcal{D}^e$ is a category.
fong-spivak-seven-sketches-F01234	108	0.875	$F \in \mathcal{D}^{\mathcal{C}}$	$F \in \mathcal{D}^{\mathcal{C}}$	2. Propose an identity natural transformation on any object $F \in \mathcal{D}^{\mathcal{C}}$, and check that
fong-spivak-seven-sketches-F01235	108	0.774	$\mathbf{Set}^1$	$\mathbf{Set}^1$	Example 3.56. In our new language, Example 3.53 says that $\mathbf{Set}^1$ is equivalent to $\mathbf{Set}$.
fong-spivak-seven-sketches-F01236	108	0.923	$(F_0, F_1, F_2, \dots)$	$(F_0, F_1, F_2, \dots)$	sequence $(F_0, F_1, F_2, \dots)$ of natural numbers, i.e. $F_0 \leq F_1 \leq F_2 \leq \dots$. If G is another
fong-spivak-seven-sketches-F01237	108	0.923	$F_0 \leq F_1 \leq F_2 \leq \dots$	$F_0 \leq F_1 \leq F_2 \leq \dots$	sequence $(F_0, F_1, F_2, \dots)$ of natural numbers, i.e. $F_0 \leq F_1 \leq F_2 \leq \dots$. If G is another
fong-spivak-seven-sketches-F01238	109	1.000	$\alpha_n: F_n \rightarrowtail G_n$	$\alpha_n: F_n \rightarrow G_n$	ponent $\alpha_n: F_n \rightarrow G_n$ has no data, it just tells us a fact: that $F_n \leq G_n$. And the naturality
fong-spivak-seven-sketches-F01239	109	1.000	$F_n \leq G_n$	$F_n \leq G_n$	ponent $\alpha_n: F_n \rightarrow G_n$ has no data, it just tells us a fact: that $F_n \leq G_n$. And the naturality
fong-spivak-seven-sketches-F01240	109	0.987	$F \Rightarrow G$	$F \Rightarrow G$	$F \Rightarrow G$ are the same. In other words, the category $\mathcal{N}^{\mathcal{N}}$ is itself a preorder; namely the
fong-spivak-seven-sketches-F01241	109	0.987	$\mathcal{N}^{\mathcal{N}}$	$\mathcal{N}^{\mathcal{N}}$	$F \Rightarrow G$ are the same. In other words, the category $\mathcal{N}^{\mathcal{N}}$ is itself a preorder; namely the
fong-spivak-seven-sketches-F01242	109	1.000	$F, G: \mathcal{C} \rightarrowtail \mathcal{P}$	$F, G: \mathcal{C} \rightarrow \mathcal{P}$	1. For any two functors $F, G: \mathcal{C} \rightarrow \mathcal{P}$, there is at most one natural transformation
fong-spivak-seven-sketches-F01243	109	1.000	$F \rightarrowtail G$	$F \rightarrow G$	$F \rightarrow G$.
fong-spivak-seven-sketches-F01244	109	0.999	$F, G: \mathcal{P} \rightarrowtail \mathcal{C}$	$F, G: \mathcal{P} \rightarrow \mathcal{C}$	2. For any two functors $F, G: \mathcal{P} \rightarrow \mathcal{C}$, there is at most one natural transformation
fong-spivak-seven-sketches-F01245	109	0.865	$F; G \cong \text{id}_{\mathbf{Pro}}$	$F; G \cong \text{id}_{\mathbf{Pro}}$	such that there exist natural isomorphisms $F; G \cong \text{id}_{\mathbf{Pro}}$ and $G; F \cong \text{id}_{\mathbf{Bool-Cat}}$ in the
fong-spivak-seven-sketches-F01246	109	0.865	$G; F \cong \text{id}_{\mathbf{Bool-Cat}}$	$G; F \cong \text{id}_{\mathbf{Bool-Cat}}$	such that there exist natural isomorphisms $F; G \cong \text{id}_{\mathbf{Pro}}$ and $G; F \cong \text{id}_{\mathbf{Bool-Cat}}$ in the
fong-spivak-seven-sketches-F01247	109	1.000	$I, J: \mathcal{C} \rightarrowtail$	$I, J: \mathcal{C} \rightarrow$	Definition 3.60. Suppose that $\mathcal{C}$ is a database schema and $I, J: \mathcal{C} \rightarrow \mathbf{Set}$ are database
fong-spivak-seven-sketches-F01248	109	1.000	$\alpha: I \rightarrowtail$	$\alpha: I \rightarrow$	instances. An <i>instance homomorphism</i> between them is a natural transformation $\alpha: I \rightarrow$
fong-spivak-seven-sketches-F01249	109	0.490	$:=$	$:=$	J . Write $\mathcal{C}\text{-Inst} := \mathbf{Set}^{\mathcal{C}}$ to denote the functor category as defined in Definition 3.54.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_109 F01250	109	■ 0.490	$\mathcal{C}$	$\mathcal{C}$	J . Write $\mathcal{C}\text{-Inst} := \mathbf{Set}^{\mathcal{C}}$ to denote the functor category as defined in Definition 3.54.
fong-spivak-seven-sketches_110 F01251	110	■ 0.986	$I($	$I($	Here $I(\text{Arrow}) = \{a, b, c, d, e\}$, and $I(\text{Vertex}) = \{1, 2, 3, 4\}$. One can draw the instance
fong-spivak-seven-sketches_110 F01252	110	■ 0.986	$)=\{a, b, c, d, e\}$	$) = \{a, b, c, d, e\}$	Here $I(\text{Arrow}) = \{a, b, c, d, e\}$, and $I(\text{Vertex}) = \{1, 2, 3, 4\}$. One can draw the instance
fong-spivak-seven-sketches_110 F01253	110	■ 0.986	$)=\{1,2,3,4\}$	$) = \{1, 2, 3, 4\}$	Here $I(\text{Arrow}) = \{a, b, c, d, e\}$, and $I(\text{Vertex}) = \{1, 2, 3, 4\}$. One can draw the instance
fong-spivak-seven-sketches_110 F01254	110	■ 0.983	$G, H: \mathbf{Gr} \rightarrow$	$G, H: \mathbf{Gr} \rightarrow$	are called <i>graph homomorphisms</i> . Let’s unwind this. Suppose that $G, H: \mathbf{Gr} \rightarrow \mathbf{Set}$
fong-spivak-seven-sketches_110 F01255	110	■ 0.989	$G, H \in \mathbf{Gr}$	$G, H \in \mathbf{Gr}$	are functors (i.e. Gr -instances); that is, they are objects $G, H \in \mathbf{Gr}\text{-Inst}$. A morphism
fong-spivak-seven-sketches_110 F01256	110	■ 0.999	$G \rightarrow H$	$G \rightarrow H$	$G \rightarrow H$ is a natural transformation $\alpha: G \rightarrow H$ between them; what does that entail?
fong-spivak-seven-sketches_110 F01257	110	■ 0.999	$\alpha: G \rightarrow H$	$\alpha: G \rightarrow H$	$G \rightarrow H$ is a natural transformation $\alpha: G \rightarrow H$ between them; what does that entail?
fong-spivak-seven-sketches_110 F01258	110	■ 1.000	α_{Vertex}	α_{Vertex}	α_{Vertex} and α_{Arrow} to respect source and targets in G and H . The left diagram says “start
fong-spivak-seven-sketches_110 F01259	110	■ 1.000	α_{Arrow}	α_{Arrow}	α_{Vertex} and α_{Arrow} to respect source and targets in G and H . The left diagram says “start
fong-spivak-seven-sketches_111 F01262	111	■ 1.000	G, H	G, H	<i>Exercise 3.64.</i> We claim that—with G, H as in Example 3.63—there is exactly one
fong-spivak-seven-sketches_111 F01263	111	■ 0.995	$\alpha_{\text{Arrow}}(a)=d$	$\alpha_{\text{Arrow}}(a) = d$	graph homomorphism $\alpha: G \rightarrow H$ such that $\alpha_{\text{Arrow}}(a) = d$.
fong-spivak-seven-sketches_112 F01264	112	■ 0.598	$F: \mathbf{Gr} \rightarrow \mathbf{DDS}$	$F: \mathbf{Gr} \rightarrow \mathbf{DDS}$	$F: \mathbf{Gr} \rightarrow \mathbf{DDS}$ to migrate data in a way that makes sense to us, the authors. Here we
fong-spivak-seven-sketches_113 F01265	113	■ 0.384	$F_9^{\circ} I: \mathbf{Gr} \rightarrow$	$F_9^{\circ} I: \mathbf{Gr} \rightarrow$	$F_{\S} I: \mathbf{Gr} \rightarrow \mathbf{Set}$. Both F and I respect identities and composition, so $F_{\S} I$ does too. Thus
fong-spivak-seven-sketches_113 F01266	113	■ 0.384	$F_9^{\circ} I$	$F_9^{\circ} I$	$F_{\S} I: \mathbf{Gr} \rightarrow \mathbf{Set}$. Both F and I respect identities and composition, so $F_{\S} I$ does too. Thus
fong-spivak-seven-sketches_114 F01267	114	■ 0.731	$I: \mathcal{D} \rightarrow$	$I: \mathcal{D} \rightarrow$	set-valued functor $I: \mathcal{D} \rightarrow \mathbf{Set}$, we refer to the composite functor $F_{\S} I: \mathcal{C} \rightarrow \mathbf{Set}$ as the
fong-spivak-seven-sketches_114 F01268	114	■ 0.731	$F: I: \mathcal{C} \rightarrow$	$F: I: \mathcal{C} \rightarrow$	set-valued functor $I: \mathcal{D} \rightarrow \mathbf{Set}$, we refer to the composite functor $F_{\S} I: \mathcal{C} \rightarrow \mathbf{Set}$ as the
fong-spivak-seven-sketches_114 F01269	114	■ 1.000	$\alpha: I \Rightarrow J$	$\alpha: I \Rightarrow J$	Given a natural transformation $\alpha: I \Rightarrow J$, there is a natural transformation $\alpha_F: F_{\S} I \Rightarrow$
fong-spivak-seven-sketches_114 F01270	114	■ 1.000	$\alpha_F: F_9^{\circ} I \Rightarrow$	$\alpha_F: F_9^{\circ} I \Rightarrow$	Given a natural transformation $\alpha: I \Rightarrow J$, there is a natural transformation $\alpha_F: F_{\S} I \Rightarrow$
fong-spivak-seven-sketches_114 F01271	114	■ 0.314	$F_9^{\circ} J$	$F_9^{\circ} J$	$F_{\S} J$, whose component $(F_{\S} I)(c) \rightarrow (F_{\S} J)(c)$ for any $c \in \text{Ob}(\mathcal{C})$ is given by $(\alpha_F)_c := \alpha_{Fc}$.
fong-spivak-seven-sketches_114 F01272	114	■ 0.314	$(F_9^{\circ} I)(c) \rightarrow (F_9^{\circ} J)(c)$	$(F_9^{\circ} I)(c) \rightarrow (F_9^{\circ} J)(c)$	$F_{\S} J$, whose component $(F_{\S} I)(c) \rightarrow (F_{\S} J)(c)$ for any $c \in \text{Ob}(\mathcal{C})$ is given by $(\alpha_F)_c := \alpha_{Fc}$.
fong-spivak-seven-sketches_114 F01273	114	■ 0.314	$c \in \text{Ob}(\mathcal{C})$	$c \in \text{Ob}(\mathcal{C})$	$F_{\S} J$, whose component $(F_{\S} I)(c) \rightarrow (F_{\S} J)(c)$ for any $c \in \text{Ob}(\mathcal{C})$ is given by $(\alpha_F)_c := \alpha_{Fc}$.
fong-spivak-seven-sketches_114 F01274	114	■ 0.314	$(\alpha_F)_c := \alpha_{Fc}$	$(\alpha_F)_c := \alpha_{Fc}$	$F_{\S} J$, whose component $(F_{\S} I)(c) \rightarrow (F_{\S} J)(c)$ for any $c \in \text{Ob}(\mathcal{C})$ is given by $(\alpha_F)_c := \alpha_{Fc}$.
fong-spivak-seven-sketches_114 F01275	114	■ 0.338	$\Delta_F: \mathcal{D}$	$\Delta_F: \mathcal{D}$	This uses the data of F to define a functor $\Delta_F: \mathcal{D}\text{-Inst} \rightarrow \mathcal{C}\text{-Inst}$.
fong-spivak-seven-sketches_114 F01276	114	■ 1.000	Δ	Δ	in general. The relevance to databases is that the data migration functor Δ from Defi-
fong-spivak-seven-sketches_114 F01277	114	■ 0.998	$P(a, b)$	$P(a, b)$	Recall from Section 3.2.3 that in a preorder P , a hom-set $P(a, b)$ has one element when
fong-spivak-seven-sketches_114 F01278	114	■ 0.997	$Q(f(p), q) \cong P(p, g(q))$	$Q(f(p), q) \cong P(p, g(q))$	of sets $Q(f(p), q) \cong P(p, g(q))$: either both are one-element sets or both are 0-element
fong-spivak-seven-sketches_114 F01279	114	■ 1.000	$L: \mathcal{C} \rightarrow \mathcal{D}$	$L: \mathcal{C} \rightarrow \mathcal{D}$	Definition 3.70. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and $L: \mathcal{C} \rightarrow \mathcal{D}$ and $R: \mathcal{D} \rightarrow \mathcal{C}$ be functors.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_114 F01280	114	■ 1.000	$R: \mathcal{D} \rightarrow \mathcal{C}$	$R: \mathcal{D} \rightarrow \mathcal{C}$	Definition 3.70. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and $L: \mathcal{C} \rightarrow \mathcal{D}$ and $R: \mathcal{D} \rightarrow \mathcal{C}$ be functors.
fong-spivak-seven-sketches_114 F01281	114	■ 1.000	L	L	We say that L is <i>left adjoint</i> to R (and that R is <i>right adjoint</i> to L) if, for any $c \in \mathcal{C}$ and
fong-spivak-seven-sketches_114 F01282	114	■ 0.844	$d \in \mathcal{D}$	$d \in \mathcal{D}$	$d \in \mathcal{D}$, there is an isomorphism of hom-sets
fong-spivak-seven-sketches_114 F01283	114	■ 0.615	$d.^{8}$	$d.^{8}$	that is natural in c and d . ⁸
fong-spivak-seven-sketches_114 F01284	114	■ 0.998	$f: c \rightarrow R(d)$	$f: c \rightarrow R(d)$	Given a morphism $f: c \rightarrow R(d)$ in $\mathcal{C}$, its image $g := \alpha_{c,d}(f)$ is called its <i>mate</i> .
fong-spivak-seven-sketches_114 F01285	114	■ 0.998	$g:=\alpha_{c, d}(f)$	$g := \alpha_{c,d}(f)$	Given a morphism $f: c \rightarrow R(d)$ in $\mathcal{C}$, its image $g := \alpha_{c,d}(f)$ is called its <i>mate</i> .
fong-spivak-seven-sketches_114 F01286	114	■ 1.000	$g: L(c) \rightarrow d$	$g: L(c) \rightarrow d$	Similarly, the mate of $g: L(c) \rightarrow d$ is f .
fong-spivak-seven-sketches_115 F01287	115	■ 1.000	$L \dashv R$	$L \dashv R$	To denote an adjunction we write $L \dashv R$, or in diagrams,
fong-spivak-seven-sketches_115 F01288	115	■ 0.801	$B \in \mathrm{Ob}(\mathbf{Set})$	$B \in \mathrm{Ob}(\mathbf{Set})$	<i>Example 3.72.</i> Let $B \in \mathrm{Ob}(\mathbf{Set})$ be any set. There is an adjunction called ‘currying B ,’
fong-spivak-seven-sketches_115 F01289	115	■ 1.000	A, C	A, C	Abstractly we write it as on the left, but what this means is that for any sets A, C , there
fong-spivak-seven-sketches_115 F01290	115	■ 0.912	$B \rightarrow C$	$B \rightarrow C$	and C are sets. Then the set of functions $B \rightarrow C$ is also a set; let’s denote it C^B . It’s
fong-spivak-seven-sketches_115 F01291	115	■ 0.912	$C^{\{B\}}$	C^B	and C are sets. Then the set of functions $B \rightarrow C$ is also a set; let’s denote it C^B . It’s
fong-spivak-seven-sketches_115 F01292	115	■ 1.000	$10^{\{3\}}$	10^3	written this way because if C has 10 elements and B has 3 elements then C^B has 10^3
fong-spivak-seven-sketches_115 F01293	115	■ 1.000	$ \mathrm{left} C^{\{B\}}\mathrm{right} = C ^{ B }$	$ C^B = C ^{ B }$	elements, and more generally for any two finite sets $C^B = C ^{ B }$.
fong-spivak-seven-sketches_115 F01294	115	■ 1.000	$(A \times B) \rightarrow C$	$(A \times B) \rightarrow C$	dence between functions $(A \times B) \rightarrow C$ and functions $A \rightarrow C^B$. Intuitively, if I have a
fong-spivak-seven-sketches_115 F01295	115	■ 1.000	$A \rightarrow C^{\{B\}}$	$A \rightarrow C^B$	dence between functions $(A \times B) \rightarrow C$ and functions $A \rightarrow C^B$. Intuitively, if I have a
fong-spivak-seven-sketches_115 F01296	115	■ 0.996	$b \multimap c$	$b \multimap c$	objects and what we called hom-elements $b \multimap c$ in Definition 2.79.
fong-spivak-seven-sketches_115 F01297	115	■ 0.983	$- \times B$	$- \times B$	<i>Exercise 3.73.</i> In Example 3.72, we discussed an adjunction between functors $- \times B$
fong-spivak-seven-sketches_115 F01298	115	■ 1.000	$(-)^{\{B\}}$	$(-)^B$	and $(-)^B$. But we only said how these functors worked on objects: for an arbitrary set
fong-spivak-seven-sketches_115 F01299	115	■ 1.000	$X \times B$	$X \times B$	X , they return sets $X \times B$ and X^B respectively.
fong-spivak-seven-sketches_115 F01300	115	■ 1.000	$X^{\{B\}}$	X^B	X , they return sets $X \times B$ and X^B respectively.
fong-spivak-seven-sketches_115 F01301	115	■ 0.953	$\mathcal{C}^{\mathrm{op}} \times \mathcal{D} \rightarrow$	$\mathcal{C}^{\mathrm{op}} \times \mathcal{D} \rightarrow$	⁸ This naturality is between functors $\mathcal{C}^{\mathrm{op}} \times \mathcal{D} \rightarrow \mathbf{Set}$. It says that for any morphisms $f: c' \rightarrow c$ in $\mathcal{C}$
fong-spivak-seven-sketches_115 F01302	115	■ 0.953	$f: c^{\{\prime\}} \rightarrow c$	$f: c' \rightarrow c$	⁸ This naturality is between functors $\mathcal{C}^{\mathrm{op}} \times \mathcal{D} \rightarrow \mathbf{Set}$. It says that for any morphisms $f: c' \rightarrow c$ in $\mathcal{C}$
fong-spivak-seven-sketches_115 F01303	115	■ 1.000	$g: d \rightarrow d^{\{\prime\}}$	$g: d \rightarrow d'$	and $g: d \rightarrow d'$ in $\mathcal{D}$, the following diagram commutes:
fong-spivak-seven-sketches_116 F01304	116	■ 1.000	$- \times B: X \times B \rightarrow Y \times B$	$- \times B: X \times B \rightarrow Y \times B$	1. Given a morphism $f: X \rightarrow Y$, what morphism should $- \times B: X \times B \rightarrow Y \times B$
fong-spivak-seven-sketches_116 F01305	116	■ 1.000	$(-)^{\{B\}}: X^{\{B\}} \rightarrow Y^{\{B\}}$	$(-)^B: X^B \rightarrow Y^B$	2. Given a morphism $f: X \rightarrow Y$, what morphism should $(-)^B: X^B \rightarrow Y^B$ return?
fong-spivak-seven-sketches_116 F01306	116	■ 1.000	$+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$	$+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$	3. Consider the function $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, which sends $(a, b) \mapsto a + b$. Currying $+$,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_116 F01307	116	■ 1.000	$(a, b) \mapsto a+b$	$(a, b) \mapsto a + b$	3. Consider the function $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, which sends $(a, b) \mapsto a + b$. Currying $+$,
fong-spivak-seven-sketches_116 F01308	116	■ 1.000	$p: \mathbb{N} \rightarrow \mathbb{N}^{\mathbb{N}}$	$p: \mathbb{N} \rightarrow \mathbb{N}^{\mathbb{N}}$	we get a certain function $p: \mathbb{N} \rightarrow \mathbb{N}^{\mathbb{N}}$. What is $p(3)$?
fong-spivak-seven-sketches_116 F01309	116	■ 1.000	$p(3)$	$p(3)$	we get a certain function $p: \mathbb{N} \rightarrow \mathbb{N}^{\mathbb{N}}$. What is $p(3)$?
fong-spivak-seven-sketches_116 F01310	116	■ 1.000	Δ_F	Δ_F	Given $F: \mathcal{C} \rightarrow \mathcal{D}$, the data migration functor Δ_F turns $\mathcal{D}$ -instances into $\mathcal{C}$ -instances.
fong-spivak-seven-sketches_117 F01311	117	■ 1.000	Σ_F	Σ_F	and get a graph, the migration functors Σ_F and Π_F can be used to turn any graph into
fong-spivak-seven-sketches_117 F01312	117	■ 1.000	Π_F	Π_F	and get a graph, the migration functors Σ_F and Π_F can be used to turn any graph into
fong-spivak-seven-sketches_117 F01313	117	■ 0.910	$J: \mathbf{Gr} \rightarrow$	$J: \mathbf{Gr} \rightarrow$	a discrete dynamical system. That is, given an instance $J: \mathbf{Gr} \rightarrow \mathbf{Set}$, we can get
fong-spivak-seven-sketches_117 F01314	117	■ 1.000	$\Sigma_F(J)$	$\Sigma_F(J)$	instances $\Sigma_F(J)$ and $\Pi_F(J)$ on DDS . This, however, is quite technical, and we leave it
fong-spivak-seven-sketches_117 F01315	117	■ 1.000	$\Pi_F(J)$	$\Pi_F(J)$	instances $\Sigma_F(J)$ and $\Pi_F(J)$ on DDS . This, however, is quite technical, and we leave it
fong-spivak-seven-sketches_117 F01316	117	■ 1.000	(e, f)	(e, f)	The functor Π_F puts into Airline Seat the set of those pairs (e, f) where e is an
fong-spivak-seven-sketches_118 F01317	118	■ 1.000	$\Pi_F(I)$	$\Pi_F(I)$	instance I , in which case $\Pi_F(I)$ (Airline Seat) would be empty. But in other uses of these
fong-spivak-seven-sketches_118 F01318	118	■ 1.000	Δ_F, Σ_F	Δ_F, Σ_F	F between schemas and using the “induced” functors Δ_F , Σ_F , and Π_F . Indeed, in
fong-spivak-seven-sketches_118 F01319	118	■ 0.931	$\mathbf{1}$	$\mathbf{1}$	namely where the target category $\mathcal{D}$ is equal to $\mathbf{1}$; see Exercise 3.12. In this case, there
fong-spivak-seven-sketches_118 F01320	118	■ 0.911	$\mathcal{C} \rightarrow \mathbf{1}$	$\mathcal{C} \rightarrow \mathbf{1}$	is exactly one functor $\mathcal{C} \rightarrow \mathbf{1}$ for any $\mathcal{C}$; let’s denote it
fong-spivak-seven-sketches_118 F01321	118	■ 0.915	$\mathcal{C} \rightarrow \mathbf{1}$	$\mathcal{C} \rightarrow \mathbf{1}$	<i>Exercise 3.76.</i> Describe the functor $!: \mathcal{C} \rightarrow \mathbf{1}$ from Eq. (3.75). Where does it send each
fong-spivak-seven-sketches_118 F01322	118	■ 0.133	$\Sigma_!$	$\Sigma_!$	We want to consider the data migration functors $\Sigma_!: \mathcal{C}\text{-Inst} \rightarrow \mathbf{1}\text{-Inst}$ and $\Pi_!: \mathcal{C}\text{-Inst} \rightarrow$
fong-spivak-seven-sketches_118 F01323	118	■ 0.133	$\Pi_!: \mathcal{C} \rightarrow$	$\Pi_! :$	We want to consider the data migration functors $\Sigma_!: \mathcal{C}\text{-Inst} \rightarrow \mathbf{1}\text{-Inst}$ and $\Pi_!: \mathcal{C}\text{-Inst} \rightarrow$
fong-spivak-seven-sketches_118 F01324	118	■ 0.988	$\Sigma_!(I)$	$\Sigma_!(I)$	Given any schema $\mathcal{C}$ and instance $I: \mathcal{C} \rightarrow \mathbf{Set}$, we will get sets $\Sigma_!(I)$ and $\Pi_!(I)$. Thinking
fong-spivak-seven-sketches_118 F01325	118	■ 0.988	$\Pi_!(I)$	$\Pi_!(I)$	Given any schema $\mathcal{C}$ and instance $I: \mathcal{C} \rightarrow \mathbf{Set}$, we will get sets $\Sigma_!(I)$ and $\Pi_!(I)$. Thinking
fong-spivak-seven-sketches_118 F01326	118	■ 0.985	$I: \mathcal{G} \rightarrow$	$I: \mathcal{G} \rightarrow$	Here’s a sample instance $I: \mathcal{G} \rightarrow \mathbf{Set}$:
fong-spivak-seven-sketches_119 F01327	119	■ 1.000	$\mathcal{G}$	$\mathcal{G}$	<i>Exercise 3.78.</i> Note that $\mathcal{G}$ from Eq. (3.77) is isomorphic to the schema $\mathbf{Gr}$. In Sec-
fong-spivak-seven-sketches_119 F01328	119	■ 0.769	$!: \mathcal{G} \rightarrow \mathbf{1}$	$!: \mathcal{G} \rightarrow \mathbf{1}$	Now we have a unique functor $!: \mathcal{G} \rightarrow \mathbf{1}$, and we want to say what $\Sigma_!(I)$ and $\Pi_!(I)$ give
fong-spivak-seven-sketches_119 F01329	119	■ 1.000	$\Pi_!$	$\Pi_!$	Where do these facts—that $\Pi_!$ and $\Sigma_!$ act the way we said—come from? Everything
fong-spivak-seven-sketches_119 F01330	119	■ 0.788	$\mathbf{1}, \Sigma$	$\mathbf{1}, \Sigma$	reader should keep in mind that, in general and not just for functors to $\mathbf{1}$, Σ -operations
fong-spivak-seven-sketches_120 F01331	120	■ 0.999	$C \rightarrow Z$	$C \rightarrow Z$	each object C of $\mathcal{C}$, there exists a unique morphism $!: C \rightarrow Z$.
fong-spivak-seven-sketches_120 F01332	120	■ 0.539	$\{\bullet\}$	$\{\bullet\}$	Consider some such set $\{\bullet\}$. Then for any other set C we need to check that there is
fong-spivak-seven-sketches_120 F01333	120	■ 0.821	$C \rightarrow \{\bullet\}$	$C \rightarrow \{\bullet\}$	exactly one function $!: C \rightarrow \{\bullet\}$. This unique function is the one that does the only

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F01334	120	1.000	$c \in C$	$c \in C$	thing that can be done: it maps each element $c \in C$ to the element $\bullet \in \{\bullet\}$.
fong-spivak-seven-sketches-F01335	120	1.000	$\bullet \in \{\bullet\}$	$\bullet \in \{\bullet\}$	thing that can be done: it maps each element $c \in C$ to the element $\bullet \in \{\bullet\}$.
fong-spivak-seven-sketches-F01336	120	0.918	$z \in P$	$z \in P$	Exercise 3.81. Let $(P, \leq)$ be a preorder, let $z \in P$ be an element, and let $\mathcal{P}$ be the
fong-spivak-seven-sketches-F01337	120	1.000	$c \in P$	$c \in P$	only if it is a <i>top element</i> in P : that is, if and only if for all $c \in P$ we have $c \leq z$.
fong-spivak-seven-sketches-F01338	120	1.000	$c \leq z$	$c \leq z$	only if it is a <i>top element</i> in P : that is, if and only if for all $c \in P$ we have $c \leq z$.
fong-spivak-seven-sketches-F01339	120	0.924	$Z^{\prime}$	Z'	Proof. This is a simple, but powerful standard argument. Suppose Z and Z' are both
fong-spivak-seven-sketches-F01340	120	1.000	$a: Z \rightarrow Z^{\prime}$	$a: Z \rightarrow Z'$	terminal objects in some category $\mathcal{C}$. Then there exist (unique) maps $a: Z \rightarrow Z'$ and
fong-spivak-seven-sketches-F01341	120	1.000	$b: Z^{\prime} \rightarrow Z$	$b: Z' \rightarrow Z$	$b: Z' \rightarrow Z$. Composing these, we get a map $a \circ b: Z \rightarrow Z$. Now since Z is terminal, this
fong-spivak-seven-sketches-F01342	120	1.000	$a \circ b: Z \rightarrow Z$	$a \circ b: Z \rightarrow Z$	$b: Z' \rightarrow Z$. Composing these, we get a map $a \circ b: Z \rightarrow Z$. Now since Z is terminal, this
fong-spivak-seven-sketches-F01343	120	0.707	$Z \rightarrow Z$	$Z \rightarrow Z$	map $Z \rightarrow Z$ must be unique. But id_Z is also such a map. So we must have $a \circ b = \text{id}_Z$.
fong-spivak-seven-sketches-F01344	120	0.707	id_Z	id_Z	map $Z \rightarrow Z$ must be unique. But id_Z is also such a map. So we must have $a \circ b = \text{id}_Z$.
fong-spivak-seven-sketches-F01345	120	0.707	$a \circ b = \text{id}_Z$	$a \circ b = \text{id}_Z$	map $Z \rightarrow Z$ must be unique. But id_Z is also such a map. So we must have $a \circ b = \text{id}_Z$.
fong-spivak-seven-sketches-F01346	120	0.900	$b \circ a = \text{id}_{Z^{\prime}}$	$b \circ a = \text{id}_{Z'}$	Similarly, we find that $b \circ a = \text{id}_{Z'}$. Thus a is an isomorphism, with inverse b .
fong-spivak-seven-sketches-F01347	120	1.000	X, Y	X, Y	Definition 3.86. Let $\mathcal{C}$ be a category, and let X, Y be objects in $\mathcal{C}$. A <i>product</i> of X and Y is
fong-spivak-seven-sketches-F01348	120	1.000	$p_X: X \times Y \rightarrow X$	$p_X: X \times Y \rightarrow X$	an object, denoted $X \times Y$, together with morphisms $p_X: X \times Y \rightarrow X$ and $p_Y: X \times Y \rightarrow Y$
fong-spivak-seven-sketches-F01349	120	1.000	$p_Y: X \times Y \rightarrow Y$	$p_Y: X \times Y \rightarrow Y$	an object, denoted $X \times Y$, together with morphisms $p_X: X \times Y \rightarrow X$ and $p_Y: X \times Y \rightarrow Y$
fong-spivak-seven-sketches-F01350	120	1.000	$f: C \rightarrow X$	$f: C \rightarrow X$	such that for all objects C together with morphisms $f: C \rightarrow X$ and $g: C \rightarrow Y$, there
fong-spivak-seven-sketches-F01351	120	1.000	$g: C \rightarrow Y$	$g: C \rightarrow Y$	such that for all objects C together with morphisms $f: C \rightarrow X$ and $g: C \rightarrow Y$, there
fong-spivak-seven-sketches-F01352	121	1.000	$C \rightarrow X \times Y$	$C \rightarrow X \times Y$	exists a unique morphism $C \rightarrow X \times Y$, denoted $\langle f, g \rangle$, for which the following diagram
fong-spivak-seven-sketches-F01353	121	1.000	$\triangleleft f, g \triangleright$	$\langle f, g \rangle$	exists a unique morphism $C \rightarrow X \times Y$, denoted $\langle f, g \rangle$, for which the following diagram
fong-spivak-seven-sketches-F01354	121	1.000	$p_X(x, y) := x$	$p_X(x, y) := x$	$p_X(x, y) := x$ and $p_Y(x, y) := y$.
fong-spivak-seven-sketches-F01355	121	1.000	$p_Y(x, y) := y$	$p_Y(x, y) := y$	$p_X(x, y) := x$ and $p_Y(x, y) := y$.
fong-spivak-seven-sketches-F01356	121	1.000	$\triangleleft f, g \triangleright(c) := (f(c), g(c))$	$\langle f, g \rangle(c) := (f(c), g(c))$	to $X \times Y$ such that the required diagram commutes is given by $\langle f, g \rangle(c) := (f(c), g(c))$.
fong-spivak-seven-sketches-F01357	121	1.000	$\underline{6} \times \underline{4}$	$\underline{6} \times \underline{4}$	Here is a picture of the product $\underline{6} \times \underline{4}$ of sets $\underline{6}$ and $\underline{4}$.
fong-spivak-seven-sketches-F01358	121	1.000	$\underline{6}$	$\underline{6}$	Here is a picture of the product $\underline{6} \times \underline{4}$ of sets $\underline{6}$ and $\underline{4}$.
fong-spivak-seven-sketches-F01359	121	1.000	$\underline{4}$	$\underline{4}$	Here is a picture of the product $\underline{6} \times \underline{4}$ of sets $\underline{6}$ and $\underline{4}$.
fong-spivak-seven-sketches-F01360	121	0.943	$x, y \in P$	$x, y \in P$	Exercise 3.88. Let $(P, \leq)$ be a preorder, let $x, y \in P$ be elements, and let $\mathcal{P}$ be the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F01361	121	1.000	$x \wedge y$	$x \wedge y$	corresponding category. Show that the product $x \times y$ in $\mathcal{P}$ agrees with their meet $x \wedge y$
fong-spivak-seven-sketches-F01362	122	0.997	$\mathcal{C} \times \mathcal{D}$	$\mathcal{C} \times \mathcal{D}$	<i>Example 3.89.</i> Given two categories $\mathcal{C}$ and $\mathcal{D}$, their product $\mathcal{C} \times \mathcal{D}$ may be given as
fong-spivak-seven-sketches-F01363	122	1.000	(c, d)	(c, d)	follows. The objects of this category are pairs (c, d) , where c is an object of $\mathcal{C}$ and d is an
fong-spivak-seven-sketches-F01364	122	0.887	$(c, d) \rightarrow (c', d')$	$(c, d) \rightarrow (c', d')$	object of $\mathcal{D}$. Similarly, morphisms $(c, d) \rightarrow (c', d')$ are pairs (f, g) where $f: c \rightarrow c'$ is a
fong-spivak-seven-sketches-F01365	122	0.887	(f, g)	(f, g)	object of $\mathcal{D}$. Similarly, morphisms $(c, d) \rightarrow (c', d')$ are pairs (f, g) where $f: c \rightarrow c'$ is a
fong-spivak-seven-sketches-F01366	122	0.887	$f: c \rightarrow c'$	$f: c \rightarrow c'$	object of $\mathcal{D}$. Similarly, morphisms $(c, d) \rightarrow (c', d')$ are pairs (f, g) where $f: c \rightarrow c'$ is a
fong-spivak-seven-sketches-F01367	122	0.944	$(f, g), \left(f^{\prime}, g^{\prime} \right) = \left(f ; f^{\prime}, g ; g^{\prime} \right)$	$(f, g), (f', g') = (f; f', g; g')$	given by composing each entry in the pair separately, so $(f, g) \circ (f', g') = (f \circ f', g \circ g')$.
fong-spivak-seven-sketches-F01368	122	1.000	$\mathbf{1} \times \mathbf{2}$	$\mathbf{1} \times \mathbf{2}$	3. What is the product category $\mathbf{1} \times \mathbf{2}$?
fong-spivak-seven-sketches-F01369	122	0.941	$\mathcal{P} \times \mathcal{Q}$	$\mathcal{P} \times \mathcal{Q}$	4. What is the product category $\mathcal{P} \times \mathcal{Q}$ when P and Q are preorders and $\mathcal{P}$ and $\mathcal{Q}$ are
fong-spivak-seven-sketches-F01370	122	1.000	$X \stackrel{f}{\leftarrow} C \stackrel{g}{\rightarrow} Y$	$X \stackrel{f}{\leftarrow} C \stackrel{g}{\rightarrow} Y$	pair of maps $X \stackrel{f}{\leftarrow} C \stackrel{g}{\rightarrow} Y$, there exists a unique map $C \rightarrow X \times Y$ such that certain
fong-spivak-seven-sketches-F01371	122	0.623	(X, Y)	(X, Y)	we'll call Cone (X, Y) , the category of cones over X and Y in $\mathcal{C}$. We will see in Exercise 3.91
fong-spivak-seven-sketches-F01372	122	0.437	$X \stackrel{p_X}{\leftarrow} X \times Y \stackrel{p_Y}{\rightarrow} Y$	$X \stackrel{p_X}{\leftarrow} X \times Y \stackrel{p_Y}{\rightarrow} Y$	that $X \stackrel{p_X}{\leftarrow} X \times Y \stackrel{p_Y}{\rightarrow} Y$ is a terminal object in Cone (X, Y) .
fong-spivak-seven-sketches-F01373	122	0.437	$\mathbf{Cone}(X, Y)$	Cone (X, Y)	that $X \stackrel{p_X}{\leftarrow} X \times Y \stackrel{p_Y}{\rightarrow} Y$ is a terminal object in Cone (X, Y) .
fong-spivak-seven-sketches-F01374	122	0.857	$X \stackrel{f'}{\leftarrow} C' \stackrel{g'}{\rightarrow} Y$	$X \stackrel{f'}{\leftarrow} C' \stackrel{g'}{\rightarrow} Y$	$X \stackrel{f}{\leftarrow} C \stackrel{g}{\rightarrow} Y$ to $X \stackrel{f'}{\leftarrow} C' \stackrel{g'}{\rightarrow} Y$ in Cone (X, Y) is a morphism $a: C \rightarrow C'$ in $\mathcal{C}$ such that
fong-spivak-seven-sketches-F01375	122	0.857	$a: C \rightarrow C'$	$a: C \rightarrow C'$	$X \stackrel{f}{\leftarrow} C \stackrel{g}{\rightarrow} Y$ to $X \stackrel{f'}{\leftarrow} C' \stackrel{g'}{\rightarrow} Y$ in Cone (X, Y) is a morphism $a: C \rightarrow C'$ in $\mathcal{C}$ such that
fong-spivak-seven-sketches-F01376	122	0.714	$X \stackrel{p_X}{\leftarrow} X \times Y \stackrel{p_Y}{\rightarrow} Y$	$X \stackrel{p_X}{\leftarrow} X \times Y \stackrel{p_Y}{\rightarrow} Y$	<i>Exercise 3.91.</i> Check that a product $X \stackrel{p_X}{\leftarrow} X \times Y \stackrel{p_Y}{\rightarrow} Y$ is exactly the same as a terminal
fong-spivak-seven-sketches-F01377	123	0.659	$\mathfrak{J}$	$\mathfrak{J}$	$D: \mathcal{J} \rightarrow \mathcal{C}$. Here $\mathcal{J}$ is called the <i>indexing category</i> of the diagram D .
fong-spivak-seven-sketches-F01378	123	0.996	(C, c_*)	(C, c_*)	Definition 3.92. Let $D: \mathcal{J} \rightarrow \mathcal{C}$ be a diagram. A <i>cone</i> (C, c_*) over D consists of
fong-spivak-seven-sketches-F01379	123	1.000	$C \in \mathcal{C}$	$C \in \mathcal{C}$	(i) an object $C \in \mathcal{C}$;
fong-spivak-seven-sketches-F01380	123	0.915	$j \in \mathcal{J}$	$j \in \mathcal{J}$	(ii) for each object $j \in \mathcal{J}$, a morphism $c_j: C \rightarrow D(j)$.
fong-spivak-seven-sketches-F01381	123	0.915	$c_j: C \rightarrow D(j)$	$c_j: C \rightarrow D(j)$	(ii) for each object $j \in \mathcal{J}$, a morphism $c_j: C \rightarrow D(j)$.
fong-spivak-seven-sketches-F01382	123	0.672	$f: j \rightarrow k$	$f: j \rightarrow k$	(a) for each $f: j \rightarrow k$ in $\mathcal{J}$, we have $c_k = c_j \circ D(f)$.
fong-spivak-seven-sketches-F01383	123	0.672	$c_k = c_j; D(f)$	$c_k = c_j; D(f)$	(a) for each $f: j \rightarrow k$ in $\mathcal{J}$, we have $c_k = c_j \circ D(f)$.
fong-spivak-seven-sketches-F01384	123	0.997	$(C, c_*) \rightarrow (C', c'_*)$	$(C, c_*) \rightarrow (C', c'_*)$	A <i>morphism of cones</i> $(C, c_*) \rightarrow (C', c'_*)$ is a morphism $a: C \rightarrow C'$ in $\mathcal{C}$ such that for all
fong-spivak-seven-sketches-F01385	123	0.271	$c_j = a$	$c_j = a$	$j \in \mathcal{J}$ we have $c_j = a \circ c'_j$. Cones over D , and their morphisms, form a category Cone (D) .
fong-spivak-seven-sketches-F01386	123	0.271	c'_j	c'_j	$j \in \mathcal{J}$ we have $c_j = a \circ c'_j$. Cones over D , and their morphisms, form a category Cone (D) .

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F01387	123	0.271	<code>\operatorname{Cone}(D)</code>	$\operatorname{Cone}(D)$	$j \in \mathcal{J}$ we have $c_j = a \circ c'_j$. Cones over D , and their morphisms, form a category $\mathbf{Cone}(D)$.
fong-spivak-seven-sketches-F01388	123	0.742	<code>\lim (D)</code>	$\lim(D)$	The <i>limit</i> of D , denoted $\lim(D)$, is the terminal object in the category $\mathbf{Cone}(D)$. Say
fong-spivak-seven-sketches-F01389	123	0.997	<code>\lim (D)=\left(C, c_{*}\right)</code>	$\lim(D) = (C, c_*)$	it is the cone $\lim(D) = (C, c_*)$; we refer to C as the <i>limit object</i> and the map c_j for any
fong-spivak-seven-sketches-F01390	123	0.997	<code>c_{\{j\}}</code>	c_j	it is the cone $\lim(D) = (C, c_*)$; we refer to C as the <i>limit object</i> and the map c_j for any
fong-spivak-seven-sketches-F01391	123	0.995	<code>j^{\text {th }}</code>	j^{th}	$j \in \mathcal{J}$ as the j^{th} <i>projection map</i> .
fong-spivak-seven-sketches-F01392	123	0.998	<code>\mathcal{J}=\varnothing</code>	$\mathcal{J} = \emptyset$	Example 3.93. Terminal objects are limits where the indexing category is empty, $\mathcal{J} = \emptyset$.
fong-spivak-seven-sketches-F01393	123	0.198	<code>v, w</code>	v, w	v, w and no arrows, $\mathcal{J} = \begin{array}{ c c } \hline v & w \\ \hline \bullet & \bullet \\ \hline \end{array}$.
fong-spivak-seven-sketches-F01394	123	0.198	<code>\mathcal{J}=</code>	$\mathcal{J} =$	v, w and no arrows, $\mathcal{J} = \begin{array}{ c c } \hline v & w \\ \hline \bullet & \bullet \\ \hline \end{array}$.
fong-spivak-seven-sketches-F01395	123	0.198	<code>\cdot</code>	$\cdot$	v, w and no arrows, $\mathcal{J} = \begin{array}{ c c } \hline v & w \\ \hline \bullet & \bullet \\ \hline \end{array}$.
fong-spivak-seven-sketches-F01396	124	0.879	<code>V, A, s, t</code>	V, A, s, t	Theorem 3.95. Let $\mathcal{J}$ be a category presented by the finite graph (V, A, s, t) together with
fong-spivak-seven-sketches-F01397	124	0.994	<code>D: \mathcal{J} \rightarrow</code>	$D : \mathcal{J} \rightarrow$	some equations, and let $D : \mathcal{J} \rightarrow \mathbf{Set}$ be a set-valued functor. Write $V = \{v_1, \dots, v_n\}$.
fong-spivak-seven-sketches-F01398	124	0.994	<code>V=\left\{v_{1}, \ldots, v_{n}\right\}</code>	$V = \{v_1, \dots, v_n\}$	some equations, and let $D : \mathcal{J} \rightarrow \mathbf{Set}$ be a set-valued functor. Write $V = \{v_1, \dots, v_n\}$.
fong-spivak-seven-sketches-F01399	124	1.000	<code>p_{\{i\}}:\left(\lim _{\mathcal{J}} D\right) \rightarrow D\left(v_{i}\right)</code>	$p_i : \left(\lim_{\mathcal{J}} D\right) \rightarrow D(v_i)$	together with the projection maps $p_i : (\lim_{\mathcal{J}} D) \rightarrow D(v_i)$ given by $p_i(d_1, \dots, d_n) := d_i$,
fong-spivak-seven-sketches-F01400	124	1.000	<code>p_{\{i\}}\left(d_{1}, \ldots, d_{n}\right):=d_{i}</code>	$p_i(d_1, \dots, d_n) := d_i$	together with the projection maps $p_i : (\lim_{\mathcal{J}} D) \rightarrow D(v_i)$ given by $p_i(d_1, \dots, d_n) := d_i$,
fong-spivak-seven-sketches-F01401	124	0.999	<code>J</code>	J	Example 3.96. If J is the empty graph $\square$, then $n = 0$: there are no vertices. There is ex-
fong-spivak-seven-sketches-F01402	124	0.999	<code>n=0</code>	$n = 0$	Example 3.96. If J is the empty graph $\square$, then $n = 0$: there are no vertices. There is ex-
fong-spivak-seven-sketches-F01403	124	1.000	<code>D: \mathbf{1} \rightarrow</code>	$D : \mathbf{1} \rightarrow$	Exercise 3.98. If $D : \mathbf{1} \rightarrow \mathbf{Set}$ is a functor, what is the limit of D ? Compute it using
fong-spivak-seven-sketches-F01404	124	0.784	<code>\left(d_{1}, \ldots, d_{n}\right)</code>	$(d_1, \dots, d_n)$	$(d_1, \dots, d_n)$ such that $D(a)(d_i) = d_j$ for each morphism $a : i \rightarrow j$ in $\mathcal{J}$ allows us to use
fong-spivak-seven-sketches-F01405	124	0.784	<code>D(a)\left(d_{i}\right)=d_{j}</code>	$D(a)(d_i) = d_j$	$(d_1, \dots, d_n)$ such that $D(a)(d_i) = d_j$ for each morphism $a : i \rightarrow j$ in $\mathcal{J}$ allows us to use
fong-spivak-seven-sketches-F01406	124	0.784	<code>a: i \rightarrow j</code>	$a : i \rightarrow j$	$(d_1, \dots, d_n)$ such that $D(a)(d_i) = d_j$ for each morphism $a : i \rightarrow j$ in $\mathcal{J}$ allows us to use
fong-spivak-seven-sketches-F01407	124	0.986	<code>\stackrel{x}{\bullet} \rightarrow \bullet \stackrel{a}{\leftarrow} \bullet \stackrel{g}{\leftarrow} \bullet</code>	$\begin{array}{c} x \xrightarrow{f} \bullet \xrightarrow{a} \bullet \xleftarrow{g} y \\ \bullet \end{array}$	Example 3.99. If J is presented by the <i>cospan</i> graph $\begin{array}{c} \bullet \xrightarrow{f} \bullet \xleftarrow{g} \bullet \\ \bullet \end{array}$, then its limit is
fong-spivak-seven-sketches-F01408	124	0.999	<code>X \rightarrow A \leftarrow Y</code>	$X \xrightarrow{f} A \xleftarrow{g} Y$	known as a <i>pullback</i> . Given the diagram $X \xrightarrow{f} A \xleftarrow{g} Y$, the pullback is the cone shown
fong-spivak-seven-sketches-F01409	125	1.000	<code>c_{\{a\}}</code>	c_a	The fact that the diagram commutes means that the diagonal arrow c_a is in some
fong-spivak-seven-sketches-F01410	125	0.579	<code>X \times_{\mathbf{A}} Y</code>	$X \times_A Y$	naming the cone point $X \times_A Y$, and adding the $\lrcorner$ symbol, as shown to the right above.
fong-spivak-seven-sketches-F01411	125	1.000	<code>X=\underline{6}, Y=\underline{4}</code>	$X = \underline{6}, Y = \underline{4}$	Here is a picture to help us unpack the definition in $\mathbf{Set}$. We take $X = \underline{6}$, $Y = \underline{4}$, and

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_1 F01412	125	■ 1.000	$f: \underline{6} \rightarrow A$	$f: \underline{6} \rightarrow A$	The functions $f: \underline{6} \rightarrow A$ and $g: \underline{4} \rightarrow A$ are expressed in the coloring of the dots: for
fong-spivak-seven-sketches_1 F01413	125	■ 1.000	$g: \underline{4} \rightarrow A$	$g: \underline{4} \rightarrow A$	The functions $f: \underline{6} \rightarrow A$ and $g: \underline{4} \rightarrow A$ are expressed in the coloring of the dots: for
fong-spivak-seven-sketches_1 F01414	125	■ 1.000	$g(2)=g(4)=$	$g(2) = g(4) =$	example, $g(2) = g(4) =$ red, while $f(5) =$ black. The pullback selects pairs $(i, j) \in \underline{6} \times \underline{4}$
fong-spivak-seven-sketches_1 F01415	125	■ 1.000	$f(5)=$	$f(5) =$	example, $g(2) = g(4) =$ red, while $f(5) =$ black. The pullback selects pairs $(i, j) \in \underline{6} \times \underline{4}$
fong-spivak-seven-sketches_1 F01416	125	■ 1.000	$(i, j) \in \underline{6} \times \underline{4}$	$(i, j) \in \underline{6} \times \underline{4}$	example, $g(2) = g(4) =$ red, while $f(5) =$ black. The pullback selects pairs $(i, j) \in \underline{6} \times \underline{4}$
fong-spivak-seven-sketches_1 F01417	125	■ 1.000	$f(i)$	$f(i)$	such that $f(i)$ and $g(j)$ have the same color.
fong-spivak-seven-sketches_1 F01418	125	■ 1.000	$g(j)$	$g(j)$	such that $f(i)$ and $g(j)$ have the same color.
fong-spivak-seven-sketches_1 F01419	125	■ 0.516	$F^{\text{op}}: \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$	$F^{\text{op}}: \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$	Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a functor. How should we define its opposite, $F^{\text{op}}: \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$? That
fong-spivak-seven-sketches_1 F01420	126	■ 0.997	F^{op}	F^{op}	is, how should F^{op} act on objects, and how should it act on morphisms?
fong-spivak-seven-sketches_1 F01421	126	■ 0.981	$D^{\text{op}}: \mathcal{J}^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$	$D^{\text{op}}: \mathcal{J}^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$	Given a diagram $D: \mathcal{J} \rightarrow \mathcal{C}$, we may take the limit of the functor $D^{\text{op}}: \mathcal{J}^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$.
fong-spivak-seven-sketches_1 F01422	126	■ 1.000	Δ, Σ, Π	Δ, Σ, Π	have all three data migration functors Δ, Σ, Π ; see [Spi12; SW15b]. The version of this
fong-spivak-seven-sketches_1 F01423	130	■ 1.000	$\$10 \leq \$20, 5W \leq 6W$	$\$10 \leq \$20, 5W \leq 6W$	$\$10 \leq \$20, 5W \leq 6W$ and so on.
fong-spivak-seven-sketches_1 F01424	131	■ 1.000	$(p, r) \in P \times R$	$(p, r) \in P \times R$	every pair $(p, r) \in P \times R$, where P is the preorder of resources to be produced and R
fong-spivak-seven-sketches_1 F01425	131	■ 1.000	$\Phi: P \times R \rightarrow$	$\Phi: P \times R \rightarrow$	Feasibility relations hence define a function $\Phi: P \times R \rightarrow \mathbf{Bool}$. For a function
fong-spivak-seven-sketches_1 F01426	131	■ 1.000	$\Phi(p, r)=$	$\Phi(p, r) =$	(a) If $\Phi(p, r) = \mathbf{true}$ and $p' \leq p$, then $\Phi(p', r) = \mathbf{true}$.
fong-spivak-seven-sketches_1 F01427	131	■ 1.000	$p^{\text{prime}} \leq p$	$p' \leq p$	(a) If $\Phi(p, r) = \mathbf{true}$ and $p' \leq p$, then $\Phi(p', r) = \mathbf{true}$.
fong-spivak-seven-sketches_1 F01428	131	■ 1.000	$\Phi(p^{\text{prime}}, r)=$	$\Phi(p', r) =$	(a) If $\Phi(p, r) = \mathbf{true}$ and $p' \leq p$, then $\Phi(p', r) = \mathbf{true}$.
fong-spivak-seven-sketches_1 F01429	131	■ 1.000	$r \leq r^{\text{prime}}$	$r \leq r'$	(b) If $\Phi(p, r) = \mathbf{true}$ and $r \leq r'$ then $\Phi(p, r') = \mathbf{true}$.
fong-spivak-seven-sketches_1 F01430	131	■ 1.000	$\Phi(p, r^{\text{prime}})=$	$\Phi(p, r') =$	(b) If $\Phi(p, r) = \mathbf{true}$ and $r \leq r'$ then $\Phi(p, r') = \mathbf{true}$.
fong-spivak-seven-sketches_1 F01431	131	■ 1.000	$r^{\text{prime}} \geq r$	$r' \geq r$	(b) also produce p given more resources $r' \geq r$. We will see that these two conditions
fong-spivak-seven-sketches_1 F01432	131	■ 0.685	$P^{\text{op}} \times R \rightarrow$	$P^{\text{op}} \times R \rightarrow$	are formalized by requiring Φ to be a monotone map $P^{\text{op}} \times R \rightarrow \mathbf{Bool}$.
fong-spivak-seven-sketches_1 F01433	131	■ 0.516	$5W \leq 10W$	$5W \leq 10W$	wattage, if $5W \leq 10W$, it means that whenever we are provided $10W$, we implicitly also
fong-spivak-seven-sketches_1 F01434	132	■ 0.500	$X = (X, \leq)$	$X = (X, \leq)$	profunctors as feasibility relations. Recall that if $\mathcal{X} = (X, \leq)$ is a preorder, then its
fong-spivak-seven-sketches_1 F01435	132	■ 0.982	$X^{\text{op}} = (X, \geq)$	$X^{\text{op}} = (X, \geq)$	opposite $\mathcal{X}^{\text{op}} = (X, \geq)$ has $x \geq y$ iff $y \leq x$.
fong-spivak-seven-sketches_1 F01436	132	■ 0.982	$x \geq y$	$x \geq y$	opposite $\mathcal{X}^{\text{op}} = (X, \geq)$ has $x \geq y$ iff $y \leq x$.
fong-spivak-seven-sketches_1 F01437	132	■ 0.521	$X = (X, \leq_X)$	$X = (X, \leq_X)$	Definition 4.2. Let $\mathcal{X} = (X, \leq_X)$ and $\mathcal{Y} = (Y, \leq_Y)$ be preorders. A feasibility relation for $\mathcal{X}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F01438	132	0.521	$y=\left(Y, \leq_Y\right)$	$y = (Y, \leq_Y)$	Definition 4.2. Let $\mathcal{X} = (X, \leq_X)$ and $\mathcal{Y} = (Y, \leq_Y)$ be preorders. A <i>feasibility relation</i> for $\mathcal{X}$
fong-spivak-seven-sketches F01439	132	0.521	$\mathfrak{X}$	$\mathfrak{X}$	Definition 4.2. Let $\mathcal{X} = (X, \leq_X)$ and $\mathcal{Y} = (Y, \leq_Y)$ be preorders. A <i>feasibility relation</i> for $\mathcal{X}$
fong-spivak-seven-sketches F01440	132	0.775	$\Phi: \mathcal{X} \rightarrow y$	$\Phi: \mathcal{X} \rightarrow y$	We denote this by $\Phi: \mathcal{X} \twoheadrightarrow \mathcal{Y}$.
fong-spivak-seven-sketches F01441	132	1.000	$\Phi(x, y)=$	$\Phi(x, y) =$	Given $x \in X$ and $y \in Y$, if $\Phi(x, y) = \text{true}$ we say x can be obtained given y .
fong-spivak-seven-sketches F01442	132	1.000	$x^{\prime} \leq_X x$	$x' \leq_X x$	if $x' \leq_X x$ and $y \leq_Y y'$ then $\Phi(x, y) \leq_{\text{Bool}} \Phi(x', y')$. In other words, if x can be
fong-spivak-seven-sketches F01443	132	1.000	$y \leq_Y y^{\prime}$	$y \leq_Y y'$	if $x' \leq_X x$ and $y \leq_Y y'$ then $\Phi(x, y) \leq_{\text{Bool}} \Phi(x', y')$. In other words, if x can be
fong-spivak-seven-sketches F01444	132	1.000	$\Phi(x, y) \leq_{\text{Bool}} \Phi\left(x^{\prime}, y^{\prime}\right)$	$\Phi(x, y) \leq_{\text{Bool}} \Phi(x', y')$	if $x' \leq_X x$ and $y \leq_Y y'$ then $\Phi(x, y) \leq_{\text{Bool}} \Phi(x', y')$. In other words, if x can be
fong-spivak-seven-sketches F01445	132	1.000	$x^{\prime}$	x'	obtained given y , and if x' is available given x , then x' can be obtained given y . And if
fong-spivak-seven-sketches F01446	132	0.605	$\mathcal{X}^{\text{op}} \times \mathcal{Y}$	$\mathcal{X}^{\text{op}} \times \mathcal{Y}$	1. Draw the Hasse diagram for the preorder $\mathcal{X}^{\text{op}} \times \mathcal{Y}$.
fong-spivak-seven-sketches F01447	132	0.994	$\Lambda: \mathcal{X} \rightarrow y$	$\Lambda: \mathcal{X} \rightarrow y$	2. Write down a profunctor $\Lambda: \mathcal{X} \twoheadrightarrow \mathcal{Y}$ and, reading $\Lambda(x, y) = \text{true}$ as “my aunt can
fong-spivak-seven-sketches F01448	132	0.994	$\Lambda(x, y)=$	$\Lambda(x, y) =$	2. Write down a profunctor $\Lambda: \mathcal{X} \twoheadrightarrow \mathcal{Y}$ and, reading $\Lambda(x, y) = \text{true}$ as “my aunt can
fong-spivak-seven-sketches F01449	132	0.372	$X^{\text{op}} \times y$	$X^{\text{op}} \times y$	forms an upper set in $\mathcal{X}^{\text{op}} \times \mathcal{Y}$.
fong-spivak-seven-sketches F01450	132	0.637	$\Rightarrow: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	$\Rightarrow: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	joins $\vee$, and a closure operation, which we’ll write $\Rightarrow: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$. By definition, this
fong-spivak-seven-sketches F01451	132	1.000	$b, c, d \in \mathbb{B}$	$b, c, d \in \mathbb{B}$	operation satisfies the property that for all $b, c, d \in \mathbb{B}$ one has
fong-spivak-seven-sketches F01453	133	0.671	$\Phi: \text{Ob}(\mathcal{X}) \times \text{Ob}(\mathcal{Y}) \rightarrow V$	$\Phi: \text{Ob}(\mathcal{X}) \times \text{Ob}(\mathcal{Y}) \rightarrow V$	$\Phi: \text{Ob}(\mathcal{X}) \times \text{Ob}(\mathcal{Y}) \rightarrow V$ such that for any $x, x' \in \mathcal{X}$ and $y, y' \in \mathcal{Y}$ the following
fong-spivak-seven-sketches F01454	133	0.671	$x, x^{\prime} \in \mathcal{X}$	$x, x' \in \mathcal{X}$	$\Phi: \text{Ob}(\mathcal{X}) \times \text{Ob}(\mathcal{Y}) \rightarrow V$ such that for any $x, x' \in \mathcal{X}$ and $y, y' \in \mathcal{Y}$ the following
fong-spivak-seven-sketches F01455	133	0.671	$y, y^{\prime} \in \mathcal{Y}$	$y, y' \in \mathcal{Y}$	$\Phi: \text{Ob}(\mathcal{X}) \times \text{Ob}(\mathcal{Y}) \rightarrow V$ such that for any $x, x' \in \mathcal{X}$ and $y, y' \in \mathcal{Y}$ the following
fong-spivak-seven-sketches F01456	134	0.821	$\Phi: X \rightarrow Y$	$\Phi: X \rightarrow Y$	nition 4.8 in the case $\mathcal{V} = \mathbf{Bool}$. One way to imagine a Bool -profunctor $\Phi: X \twoheadrightarrow Y$ is in
fong-spivak-seven-sketches F01457	134	1.000	$W \leq N$	$W \leq N$	Both X and Y are preorders, e.g. with $W \leq N$ and $b \leq a$. With bridges coming from
fong-spivak-seven-sketches F01458	134	1.000	N	N	get from points in city X to points in city Y . For example, since there is a path from N
fong-spivak-seven-sketches F01459	134	1.000	E	E	to e and E to a , we have $\Phi(N, e) = \text{true}$ and $\Phi(E, a) = \text{true}$. On the other hand, since
fong-spivak-seven-sketches F01460	134	1.000	$\Phi(N, e)=$	$\Phi(N, e) =$	to e and E to a , we have $\Phi(N, e) = \text{true}$ and $\Phi(E, a) = \text{true}$. On the other hand, since
fong-spivak-seven-sketches F01461	134	1.000	$\Phi(E, a)=$	$\Phi(E, a) =$	to e and E to a , we have $\Phi(N, e) = \text{true}$ and $\Phi(E, a) = \text{true}$. On the other hand, since
fong-spivak-seven-sketches F01462	134	1.000	W	W	there is no path from W to d , we have $\Phi(W, d) = \text{false}$.
fong-spivak-seven-sketches F01463	134	1.000	$\Phi(W, d)=$	$\Phi(W, d) =$	there is no path from W to d , we have $\Phi(W, d) = \text{false}$.
fong-spivak-seven-sketches F01464	134	1.000	$W \leq N \leq c \leq a$	$W \leq N \leq c \leq a$	$W \leq N \leq c \leq a$, etc. This is called the <i>collage</i> of Φ ; we’ll explore this in more detail
fong-spivak-seven-sketches F01465	134	1.000	$\Phi(m, n) \in \mathbb{B}$	$\Phi(m, n) \in \mathbb{B}$	$\Phi(m, n) \in \mathbb{B}$. Fill out the Bool -matrix:

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_1467	135	1.000	$d_X * M_{\Phi} * d_Y$	$d_X * M_{\Phi} * d_Y$	The matrix of distances for the profunctor Φ will be equal to $d_X * M_{\Phi} * d_Y$. In fact, since
fong-spivak-seven-sketches_1468	135	0.835	$\Phi = M_X^3 * M_{\Phi} * M_Y^2$	$\Phi = M_X^3 * M_{\Phi} * M_Y^2$	X has four elements and Y has three, we also know that $\Phi = M_X^3 * M_{\Phi} * M_Y^2$.
fong-spivak-seven-sketches_1469	135	1.000	$M_X^3 * M_{\Phi} * M_Y^2$	$M_X^3 * M_{\Phi} * M_Y^2$	<i>Exercise 4.17.</i> Calculate $M_X^3 * M_{\Phi} * M_Y^2$, remembering to do matrix multiplication
fong-spivak-seven-sketches_1470	136	0.873	$([0, \infty], \leq)$	$([0, \infty], \leq)$	load might be the preorder $([0, \infty], \leq)$, where an element $x \in [0, \infty]$ represents the
fong-spivak-seven-sketches_1471	136	—	100, more_than_	(not rendered)	—
fong-spivak-seven-sketches_1472	136	0.965	$\Phi: (T \times E) \mapsto \$$	$\Phi: (T \times E) \mapsto \$$	$\Phi: (T \times E) \twoheadrightarrow \$$. We represent this by the box
fong-spivak-seven-sketches_1473	136	0.695	T, E	T, E	where T, E , and $\$$ are the preorders drawn below:
fong-spivak-seven-sketches_1474	136	0.695	$\$$	$\$$	where T, E , and $\$$ are the preorders drawn below:
fong-spivak-seven-sketches_1475	137	0.940	$\$ 500 \text{ \texttt{\textit{~K}}}$	$\$500 \text{ K}$	This says, for example, that a good-natured but boring movie costs \$500K to produce
fong-spivak-seven-sketches_1476	137	—	500K given \	(not rendered)	—
fong-spivak-seven-sketches_1477	137	0.928	$\$ 500$	$\$500$	given \$500K”, and “I can provide a mean and boring movie given a good-natured
fong-spivak-seven-sketches_1478	137	—	IM indicates also that "I can provide a mean and boring movie given \	(not rendered)	—
fong-spivak-seven-sketches_1479	137	0.952	$\mathcal{P}, Q$	$\mathcal{P}, Q$	two of them in series. Imagine you have cities $\mathcal{P}, Q$, and $\mathcal{R}$ and you have bridges—and
fong-spivak-seven-sketches_1480	137	0.952	$\mathcal{R}$	$\mathcal{R}$	two of them in series. Imagine you have cities $\mathcal{P}, Q$, and $\mathcal{R}$ and you have bridges—and
fong-spivak-seven-sketches_1481	138	0.497	$\Phi: \mathcal{P} \mapsto Q$	$\Phi: \mathcal{P} \mapsto Q$	hence feasibility matrices—connecting these cities, say $\Phi: \mathcal{P} \twoheadrightarrow Q$ and $\Psi: Q \twoheadrightarrow \mathcal{R}$.
fong-spivak-seven-sketches_1482	138	0.497	$\Psi: \mathcal{Q} \rightarrow \mathcal{R}$	$\Psi: \mathcal{Q} \rightarrow \mathcal{R}$	hence feasibility matrices—connecting these cities, say $\Phi: \mathcal{P} \twoheadrightarrow Q$ and $\Psi: Q \twoheadrightarrow \mathcal{R}$.
fong-spivak-seven-sketches_1483	138	1.000	Ψ	Ψ	The feasibility matrices for Φ (in blue) and Ψ (in red) are:
fong-spivak-seven-sketches_1484	138	0.223	$\Phi_9^9 \Psi$	$\Phi_9^9 \Psi$	trying to give a feasibility matrix $\Phi \circ \Psi$ for getting from $\mathcal{P}$ to $\mathcal{R}$. How should this be
fong-spivak-seven-sketches_1485	138	0.984	$p \in \mathcal{P}$	$p \in \mathcal{P}$	done? Basically, for every pair $p \in \mathcal{P}$ and $r \in \mathcal{R}$, the navigator searches through Q for a
fong-spivak-seven-sketches_1486	138	0.984	$r \in \mathcal{R}$	$r \in \mathcal{R}$	done? Basically, for every pair $p \in \mathcal{P}$ and $r \in \mathcal{R}$, the navigator searches through Q for a
fong-spivak-seven-sketches_1487	138	0.984	$\mathcal{Q}$	Q	done? Basically, for every pair $p \in \mathcal{P}$ and $r \in \mathcal{R}$, the navigator searches through Q for a
fong-spivak-seven-sketches_1489	139	0.604	$\mathcal{X}, \mathcal{Y}$	$\mathcal{X}, \mathcal{Y}$	Definition 4.21. Let $\mathcal{V}$ be a quantale, let $\mathcal{X}, \mathcal{Y}$, and $\mathcal{Z}$ be $\mathcal{V}$ -categories, and let $\Phi: \mathcal{X} \twoheadrightarrow \mathcal{Y}$
fong-spivak-seven-sketches_1490	139	0.604	$\mathcal{Z}$	$\mathcal{Z}$	Definition 4.21. Let $\mathcal{V}$ be a quantale, let $\mathcal{X}, \mathcal{Y}$, and $\mathcal{Z}$ be $\mathcal{V}$ -categories, and let $\Phi: \mathcal{X} \twoheadrightarrow \mathcal{Y}$
fong-spivak-seven-sketches_1491	139	0.604	$\Phi: \mathfrak{X} \rightarrow y$	$\Phi: \mathfrak{X} \rightarrow y$	Definition 4.21. Let $\mathcal{V}$ be a quantale, let $\mathcal{X}, \mathcal{Y}$, and $\mathcal{Z}$ be $\mathcal{V}$ -categories, and let $\Phi: \mathcal{X} \twoheadrightarrow \mathcal{Y}$
fong-spivak-seven-sketches_1492	139	0.666	$\Psi: y \rightarrow z$	$\Psi: y \rightarrow z$	and $\Psi: \mathcal{Y} \twoheadrightarrow \mathcal{Z}$ be $\mathcal{V}$ -profunctors. We define their <i>composite</i> , denoted $\Phi \circ \Psi: \mathcal{X} \twoheadrightarrow \mathcal{Z}$ by
fong-spivak-seven-sketches_1493	139	0.666	$\Phi; \Psi: \mathcal{X} \rightarrow z$	$\Phi; \Psi: \mathcal{X} \rightarrow z$	and $\Psi: \mathcal{Y} \twoheadrightarrow \mathcal{Z}$ be $\mathcal{V}$ -profunctors. We define their <i>composite</i> , denoted $\Phi \circ \Psi: \mathcal{X} \twoheadrightarrow \mathcal{Z}$ by
fong-spivak-seven-sketches_1494	139	0.405	$\Phi: X \rightarrow y$	$\Phi: X \rightarrow y$	<i>Exercise 4.22.</i> Consider the Cost -profunctors $\Phi: \mathcal{X} \twoheadrightarrow \mathcal{Y}$ and $\Psi: \mathcal{Y} \twoheadrightarrow \mathcal{Z}$ shown below:

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_1496	139	0.564	$_{{\mathcal{V}}}$	$\mathcal{V}$	Theorem 4.23. For any skeletal quantale $\mathcal{V}$, there is a category $\mathbf{Prof}_{\mathcal{V}}$ whose objects are
fong-spivak-seven-sketches_1497	139	0.426	$\mathcal{X} \rightarrowtail \mathcal{Y}$	$\mathcal{X} \rightarrow \mathcal{Y}$	$\mathcal{V}$ -categories $\mathcal{X}$, whose morphisms are $\mathcal{V}$ -profunctors $\mathcal{X} \rightarrowtail \mathcal{Y}$, and with composition
fong-spivak-seven-sketches_1498	139	0.505	$_{{\text{Bool}}}$	$\mathbf{Bool}$	Definition 4.24. We define $\mathbf{Feas} := \mathbf{Prof}_{\mathbf{Bool}}$.
fong-spivak-seven-sketches_1499	140	0.744	$\mathrm{U}_x: \mathcal{X} \rightarrowtail \mathcal{X}$	$U_x: \mathcal{X} \rightarrow \mathcal{X}$	Define the <i>unit profunctor</i> $U_{\mathcal{X}}: \mathcal{X} \rightarrowtail \mathcal{X}$ on a $\mathcal{V}$ -category $\mathcal{X}$ by the formula
fong-spivak-seven-sketches_1500	140	0.988	$\mathcal{X}(x, y) \in \mathcal{V}$	$\mathcal{X}(x, y) \in \mathcal{V}$	pair of elements $x, y \in \mathcal{X}$ an hom-object $\mathcal{X}(x, y) \in \mathcal{V}$. The unit profunctor $U_{\mathcal{X}}$ just
fong-spivak-seven-sketches_1501	140	0.988	U_x	U_x	pair of elements $x, y \in \mathcal{X}$ an hom-object $\mathcal{X}(x, y) \in \mathcal{V}$. The unit profunctor $U_{\mathcal{X}}$ just
fong-spivak-seven-sketches_1502	140	0.666	$U_X: \mathcal{X} \rightarrowtail \mathcal{X}$	$U_X: \mathcal{X} \rightarrow \mathcal{X}$	for the unit profunctor $U_{\mathcal{X}}: \mathcal{X} \rightarrowtail \mathcal{X}$.
fong-spivak-seven-sketches_1503	140	0.982	$\Phi: \mathcal{P} \rightarrowtail \mathcal{Q}$	$\Phi: \mathcal{P} \rightarrow \mathcal{Q}$	Lemma 4.27. Composing any profunctor $\Phi: \mathcal{P} \rightarrow \mathcal{Q}$ with either unit profunctor, $U_{\mathcal{P}}$ or
fong-spivak-seven-sketches_1504	140	0.982	$U_{\mathcal{P}}$	$U_{\mathcal{P}}$	Lemma 4.27. Composing any profunctor $\Phi: \mathcal{P} \rightarrow \mathcal{Q}$ with either unit profunctor, $U_{\mathcal{P}}$ or
fong-spivak-seven-sketches_1505	140	0.637	U_Q	U_Q	U_Q , returns Φ :
fong-spivak-seven-sketches_1506	140	0.296	$\mathrm{U}_{\mathcal{P}}; \Phi = \Phi$	$U_{\mathcal{P}}; \Phi = \Phi$	<i>Proof.</i> We show that $U_{\mathcal{P}} \circ \Phi = \Phi$ holds; proving $\Phi = \Phi \circ U_Q$ is similar. Fix $p \in P$ and
fong-spivak-seven-sketches_1507	140	0.296	$\Phi = \Phi$	$\Phi = \Phi$	<i>Proof.</i> We show that $U_{\mathcal{P}} \circ \Phi = \Phi$ holds; proving $\Phi = \Phi \circ U_Q$ is similar. Fix $p \in P$ and
fong-spivak-seven-sketches_1508	140	0.673	$\Phi \leq U_{\mathcal{P}}$	$\Phi \leq U_{\mathcal{P}}; \Phi$	$q \in Q$. Since $\mathcal{V}$ is skeletal, to prove the equality it's enough to show $\Phi \leq U_{\mathcal{P}} \circ \Phi$ and
fong-spivak-seven-sketches_1509	140	0.996	$\mathrm{U}_{\mathcal{P}}; \Phi \leq \Phi$	$U_{\mathcal{P}}; \Phi \leq \Phi$	$U_{\mathcal{P}} \circ \Phi \leq \Phi$. We have one direction:
fong-spivak-seven-sketches_1510	140	1.000	$\bigvee_{p_1 \in P} (\mathcal{P}(p, p_1) \otimes \Phi(p_1, q)) \leq \Phi(p, q)$	$\bigvee_{p_1 \in P} (\mathcal{P}(p, p_1) \otimes \Phi(p_1, q)) \leq \Phi(p, q)$	For the other direction, we must show $\bigvee_{p_1 \in P} (\mathcal{P}(p, p_1) \otimes \Phi(p_1, q)) \leq \Phi(p, q)$. But by
fong-spivak-seven-sketches_1511	140	1.000	$\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \leq \Phi(p, q)$	$\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \leq \Phi(p, q)$	definition of join, this holds iff $\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \leq \Phi(p, q)$ is true for each $p_1 \in P$. This
fong-spivak-seven-sketches_1512	140	1.000	$p_1 \in P$	$p_1 \in P$	definition of join, this holds iff $\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \leq \Phi(p, q)$ is true for each $p_1 \in P$. This
fong-spivak-seven-sketches_1513	140	0.696	$(=, \leq, \leq, =)$	$(=, \leq, \leq, =)$	1. Justify each of the four steps $(=, \leq, \leq, =)$ in Eq. (4.28).
fong-spivak-seven-sketches_1514	141	0.964	$\leq, \leq$	$\leq, \leq$	3. Justify each of the three steps $(=, \leq, \leq)$ in Eq. (4.29).
fong-spivak-seven-sketches_1515	141	0.567	$\Phi: \mathcal{P} \rightarrowtail \mathcal{Q}, \Psi: \mathcal{Q} \rightarrowtail \mathcal{R}$	$\Phi: \mathcal{P} \rightarrow \mathcal{Q}, \Psi: \mathcal{Q} \rightarrow \mathcal{R}$	tors $\Phi: \mathcal{P} \rightarrowtail \mathcal{Q}, \Psi: \mathcal{Q} \rightarrowtail \mathcal{R}$, and $\Upsilon: \mathcal{R} \rightarrowtail \mathcal{S}$, we have
fong-spivak-seven-sketches_1516	141	0.567	$\Upsilon: \mathcal{R} \rightarrowtail \mathcal{S}$	$\Upsilon: \mathcal{R} \rightarrow \mathcal{S}$	tors $\Phi: \mathcal{P} \rightarrowtail \mathcal{Q}, \Psi: \mathcal{Q} \rightarrowtail \mathcal{R}$, and $\Upsilon: \mathcal{R} \rightarrowtail \mathcal{S}$, we have
fong-spivak-seven-sketches_1517	142	0.426	$_{{\mathcal{C}}}$	$\mathcal{C}$	the category $\mathbf{Cospan}_{\mathcal{C}}$ in Definition 6.45. Second, we might relax what we mean by
fong-spivak-seven-sketches_1518	142	0.977	$F: \mathcal{P} \rightarrowtail \mathcal{Q}$	$F: \mathcal{P} \rightarrow \mathcal{Q}$	Definition 4.34. Let $F: \mathcal{P} \rightarrow \mathcal{Q}$ be a $\mathcal{V}$ -functor. The <i>companion</i> of F , denoted $\widehat{F}: \mathcal{P} \rightarrowtail \mathcal{Q}$
fong-spivak-seven-sketches_1519	142	0.977	$\widehat{F}: \mathcal{P} \rightarrowtail \mathcal{Q}$	$\widehat{F}: \mathcal{P} \rightarrow \mathcal{Q}$	Definition 4.34. Let $F: \mathcal{P} \rightarrow \mathcal{Q}$ be a $\mathcal{V}$ -functor. The <i>companion</i> of F , denoted $\widehat{F}: \mathcal{P} \rightarrowtail \mathcal{Q}$
fong-spivak-seven-sketches_1520	142	0.488	$\check{F}: \mathcal{Q} \mapsto \mathcal{P}$	$\check{F}: \mathcal{Q} \mapsto \mathcal{P}$	and the <i>conjoint</i> of F , denoted $\check{F}: \mathcal{Q} \mapsto \mathcal{P}$ are defined to be the following $\mathcal{V}$ -profunctors:
fong-spivak-seven-sketches_1521	142	0.995	$F(p) \in \mathcal{Q}$	$F(p) \in \mathcal{Q}$	$F(p) \in \mathcal{Q}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_142_F01522	142	■ 1.000	$\widehat{F}$	$\widehat{F}$	picture in that light, they are seeing the companion $\widehat{F}$. But now mentally reverse every
fong-spivak-seven-sketches_142_F01523	142	■ 0.979	$Q \rightarrowtail \mathcal{P}$	$Q \rightarrow \mathcal{P}$	dotted arrow, and the result would be bridges $\mathcal{Q}$ to $\mathcal{P}$. This is a profunctor $\mathcal{Q} \twoheadrightarrow \mathcal{P}$! We
fong-spivak-seven-sketches_142_F01524	142	■ 1.000	$\check{F}$	$\check{F}$	call it $\check{F}$.
fong-spivak-seven-sketches_142_F01525	142	■ 0.710	$\mathrm{id}: \mathcal{P} \rightarrowtail \mathcal{P}$	$\mathrm{id}: \mathcal{P} \rightarrow \mathcal{P}$	<i>Example 4.35.</i> For any preorder $\mathcal{P}$, there is an identity functor $\mathrm{id}: \mathcal{P} \rightarrow \mathcal{P}$. Its compan-
fong-spivak-seven-sketches_142_F01526	142	■ 0.699	$\widehat{\mathrm{id}} = \check{\mathrm{id}}: \mathcal{P} \twoheadrightarrow \mathcal{P}$	$\widehat{\mathrm{id}} = \check{\mathrm{id}}: \mathcal{P} \twoheadrightarrow \mathcal{P}$	ion and conjoint agree $\widehat{\mathrm{id}} = \check{\mathrm{id}}: \mathcal{P} \twoheadrightarrow \mathcal{P}$. The resulting profunctor is in fact the unit
fong-spivak-seven-sketches_142_F01527	142	■ 1.000	$\mathrm{U}_{\mathcal{P}}$	$\mathrm{U}_{\mathcal{P}}$	profunctor, $\mathrm{U}_{\mathcal{P}}$, as defined in Eq. (4.25).
fong-spivak-seven-sketches_143_F01528	143	■ 0.991	$\widehat{\mathrm{id}}$	$\widehat{\mathrm{id}}$	<i>Exercise 4.36.</i> Check that the companion $\widehat{\mathrm{id}}$ of $\mathrm{id}: \mathcal{P} \rightarrow \mathcal{P}$ really has the unit profunctor
fong-spivak-seven-sketches_143_F01529	143	■ 0.991	$\mathcal{P} \rightarrowtail \mathcal{P}$	$\mathcal{P} \rightarrow \mathcal{P}$	<i>Exercise 4.36.</i> Check that the companion $\widehat{\mathrm{id}}$ of $\mathrm{id}: \mathcal{P} \rightarrow \mathcal{P}$ really has the unit profunctor
fong-spivak-seven-sketches_143_F01530	143	■ 0.637	$+: \mathbb{R} \times \mathbb{R} \times \mathbb{R} \rightarrowtail \mathbb{R}$	$+: \mathbb{R} \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$	<i>Example 4.37.</i> Consider the function $+: \mathbb{R} \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$, sending a triple (a, b, c) of real
fong-spivak-seven-sketches_143_F01531	143	■ 0.979	$a+b+c \in \mathbb{R}$	$a + b + c \in \mathbb{R}$	numbers to $a + b + c \in \mathbb{R}$. This function is monotonic, because if $(a, b, c) \leq (a', b', c')$ —
fong-spivak-seven-sketches_143_F01532	143	■ 0.979	$(a, b, c) \leq (a', b', c')$	$(a, b, c) \leq (a', b', c')$	numbers to $a + b + c \in \mathbb{R}$. This function is monotonic, because if $(a, b, c) \leq (a', b', c')$ —
fong-spivak-seven-sketches_143_F01533	143	■ 1.000	$a \leq a'$	$a \leq a'$	i.e. if $a \leq a'$ and $b \leq b'$, and $c \leq c'$ —then obviously $a + b + c \leq a' + b' + c'$. Thus it has
fong-spivak-seven-sketches_143_F01534	143	■ 1.000	$b \leq b'$	$b \leq b'$	i.e. if $a \leq a'$ and $b \leq b'$, and $c \leq c'$ —then obviously $a + b + c \leq a' + b' + c'$. Thus it has
fong-spivak-seven-sketches_143_F01535	143	■ 1.000	$c \leq c'$	$c \leq c'$	i.e. if $a \leq a'$ and $b \leq b'$, and $c \leq c'$ —then obviously $a + b + c \leq a' + b' + c'$. Thus it has
fong-spivak-seven-sketches_143_F01536	143	■ 1.000	$a+b+c \leq a'^{\prime}+b'^{\prime}+c'^{\prime}$	$a + b + c \leq a' + b' + c'$	i.e. if $a \leq a'$ and $b \leq b'$, and $c \leq c'$ —then obviously $a + b + c \leq a' + b' + c'$. Thus it has
fong-spivak-seven-sketches_143_F01537	143	■ 0.955	$\widehat{+}: (\mathbb{R} \times \mathbb{R} \times \mathbb{R}) \rightarrowtail \mathbb{R}$	$\widehat{+}: (\mathbb{R} \times \mathbb{R} \times \mathbb{R}) \rightarrow \mathbb{R}$	Its companion $\widehat{+}: (\mathbb{R} \times \mathbb{R} \times \mathbb{R}) \twoheadrightarrow \mathbb{R}$ is the function that sends (a, b, c, d) to true if
fong-spivak-seven-sketches_143_F01538	143	■ 0.955	(a, b, c, d)	(a, b, c, d)	Its companion $\widehat{+}: (\mathbb{R} \times \mathbb{R} \times \mathbb{R}) \twoheadrightarrow \mathbb{R}$ is the function that sends (a, b, c, d) to true if
fong-spivak-seven-sketches_143_F01539	143	■ 1.000	$a+b+c \leq d$	$a + b + c \leq d$	$a + b + c \leq d$ and to false otherwise.
fong-spivak-seven-sketches_143_F01540	143	■ 1.000	$\check{+}$	$\check{+}$	<i>Exercise 4.38.</i> Let $+: \mathbb{R} \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be as in Example 4.37. What is its conjoint $\check{+}$?
fong-spivak-seven-sketches_143_F01541	143	■ 0.994	$G: \mathcal{Q} \rightarrowtail \mathcal{P}$	$G: \mathcal{Q} \rightarrow \mathcal{P}$	of $\mathcal{V}$ -functors $F: \mathcal{P} \rightarrow \mathcal{Q}$ and $G: \mathcal{Q} \rightarrow \mathcal{P}$ such that the following holds for all $p \in P$ and
fong-spivak-seven-sketches_143_F01542	143	■ 0.990	$\widehat{F} = \widetilde{G}$	$\widehat{F} = \widetilde{G}$	of the former equals the conjoint of the latter: $\widehat{F} = \widetilde{G}$.
fong-spivak-seven-sketches_143_F01543	143	■ 1.000	$\widehat{\mathrm{id}} = \widetilde{\mathrm{id}}$	$\widehat{\mathrm{id}} = \widetilde{\mathrm{id}}$	2. Use this to prove that $\widehat{\mathrm{id}} = \widetilde{\mathrm{id}}$, as was stated in Example 4.35.
fong-spivak-seven-sketches_143_F01544	143	■ 0.998	$\mathrm{Col}(\Phi)$	$\mathrm{Col}(\Phi)$	a $\mathcal{V}$ -profunctor. The <i>collage of</i> Φ , denoted $\mathrm{Col}(\Phi)$ is the $\mathcal{V}$ -category defined as follows:
fong-spivak-seven-sketches_143_F01545	143	■ 0.976	$\mathrm{Ob}(\mathrm{Col}(\Phi)) := \mathrm{Ob}(\mathcal{X}) \sqcup \mathrm{Ob}(\mathcal{Y})$	$\mathrm{Ob}(\mathrm{Col}(\Phi)) := \mathrm{Ob}(\mathcal{X}) \sqcup \mathrm{Ob}(\mathcal{Y})$	(i) $\mathrm{Ob}(\mathrm{Col}(\Phi)) := \mathrm{Ob}(\mathcal{X}) \sqcup \mathrm{Ob}(\mathcal{Y})$;
fong-spivak-seven-sketches_143_F01546	143	■ 0.567	$a, b \in \mathrm{Ob}(\mathrm{Col}(\Phi))$	$a, b \in \mathrm{Ob}(\mathrm{Col}(\Phi))$	(ii) For any $a, b \in \mathrm{Ob}(\mathrm{Col}(\Phi))$, define $\mathrm{Col}(\Phi)(a, b) \in \mathcal{V}$ to be
fong-spivak-seven-sketches_143_F01547	143	■ 0.567	$\mathrm{Col}(\Phi)(a, b) \in \mathcal{V}$	$\mathrm{Col}(\Phi)(a, b) \in \mathcal{V}$	(ii) For any $a, b \in \mathrm{Ob}(\mathrm{Col}(\Phi))$, define $\mathrm{Col}(\Phi)(a, b) \in \mathcal{V}$ to be

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_143 F01548	143	■ 0.994	$i_x: \mathcal{X} \rightarrow \mathbf{Col}(\Phi)$	$i_x: \mathcal{X} \rightarrow \mathbf{Col}(\Phi)$	There are obvious functors $i_x: \mathcal{X} \rightarrow \mathbf{Col}(\Phi)$ and $i_y: \mathcal{Y} \rightarrow \mathbf{Col}(\Phi)$, sending each object
fong-spivak-seven-sketches_143 F01549	143	■ 0.994	$i_y: \mathcal{Y} \rightarrow \mathbf{Col}(\Phi)$	$i_y: \mathcal{Y} \rightarrow \mathbf{Col}(\Phi)$	There are obvious functors $i_x: \mathcal{X} \rightarrow \mathbf{Col}(\Phi)$ and $i_y: \mathcal{Y} \rightarrow \mathbf{Col}(\Phi)$, sending each object
fong-spivak-seven-sketches_144 F01550	144	■ 0.600	$\Phi: x \rightarrow y$	$\Phi: x \rightarrow y$	<i>Example 4.43.</i> Consider the following picture of a Cost -profunctor $\Phi: \mathcal{X} \rightarrow \mathcal{Y}$:
fong-spivak-seven-sketches_145 F01551	145	■ 1.000	$5+3=8$	$5 + 3 = 8$	that $5 + 3 = 8$. One way to categorify $\mathbb{N}$ is to use the category FinSet of finite sets
fong-spivak-seven-sketches_145 F01552	145	■ 0.709	$\overline{5} = \{$	$\overline{5} = \{$	sets of that many elements, say $\overline{5} = \{\text{apple, banana, cherry, dragonfruit, elephant}\}$,
fong-spivak-seven-sketches_145 F01553	145	■ 0.456	$\overline{3} = \{$	$\overline{3} = \{$	$\overline{3} = \{\text{apple, tomato, cantaloupe}\}$, and $\overline{8} = \{\text{Ali, Bob, Carl, Deb, Eli, Fritz, Gem, Helen}\}$
fong-spivak-seven-sketches_145 F01554	145	■ 0.456	$\overline{8} = \{$	$\overline{8} = \{$	$\overline{3} = \{\text{apple, tomato, cantaloupe}\}$, and $\overline{8} = \{\text{Ali, Bob, Carl, Deb, Eli, Fritz, Gem, Helen}\}$
fong-spivak-seven-sketches_145 F01555	145	■ 0.895	$\sqcup$	$\sqcup$	respectively. We also replace $+$ with disjoint union of sets $\sqcup$, and the brute property of
fong-spivak-seven-sketches_145 F01556	145	■ 0.998	$\overline{5} \sqcup \overline{3} \cong \overline{8}$	$\overline{5} \sqcup \overline{3} \cong \overline{8}$	is that, having made these replacements, the analogue of $5 + 3 = 8$ is still true: $\overline{5} \sqcup \overline{3} \cong \overline{8}$.
fong-spivak-seven-sketches_145 F01557	145	■ 1.000	$\overline{5} \sqcup \overline{3}$	$\overline{5} \sqcup \overline{3}$	Thus it’s not the case that $\overline{5} \sqcup \overline{3}$ is <i>equal</i> to our chosen eight-element set $\overline{8}$, but more
fong-spivak-seven-sketches_145 F01558	145	■ 1.000	$\overline{8}$	$\overline{8}$	Thus it’s not the case that $\overline{5} \sqcup \overline{3}$ is <i>equal</i> to our chosen eight-element set $\overline{8}$, but more
fong-spivak-seven-sketches_146 F01559	146	■ 1.000	$\mathcal{P}(a, b) =$	$\mathcal{P}(a, b) =$	exists a morphism $a \rightarrow b$,” denoted $a \leq b$ or $\mathcal{P}(a, b) = \text{true}$, we instead say “here is a set
fong-spivak-seven-sketches_146 F01560	146	■ 1.000	$x_0 \leq x_1$	$x_0 \leq x_1$	whenever $x_0 \leq x_1$. Chaining these together, we could prove facts in our preorder. For
fong-spivak-seven-sketches_146 F01561	146	■ 1.000	$x_0 \leq x_3$	$x_0 \leq x_3$	provides a proof that $x_0 \leq x_3$ (the exterior box) using three facts (the interior boxes),
fong-spivak-seven-sketches_146 F01562	146	■ 1.000	$x_0 \leq x_1, x_1 \leq x_2$	$x_0 \leq x_1, x_1 \leq x_2$	$x_0 \leq x_1, x_1 \leq x_2$, and $x_2 \leq x_3$.
fong-spivak-seven-sketches_146 F01563	146	■ 1.000	$x_2 \leq x_3$	$x_2 \leq x_3$	$x_0 \leq x_1, x_1 \leq x_2$, and $x_2 \leq x_3$.
fong-spivak-seven-sketches_146 F01564	146	■ 1.000	$g: B \rightarrow C$	$g: B \rightarrow C$	Suppose given additional morphisms $g: B \rightarrow C$, and $h: C \rightarrow D$. Representing these
fong-spivak-seven-sketches_146 F01565	146	■ 1.000	$h: C \rightarrow D$	$h: C \rightarrow D$	Suppose given additional morphisms $g: B \rightarrow C$, and $h: C \rightarrow D$. Representing these
fong-spivak-seven-sketches_146 F01566	146	■ 0.918	$f; (g; h)$	$f; (g; h)$	say! We don’t know which morphism it is. Is it $f \circ (g \circ h)$? Or $(f \circ g) \circ h$? It’s good that
fong-spivak-seven-sketches_146 F01567	146	■ 0.918	$(f; g); h$	$(f; g); h$	say! We don’t know which morphism it is. Is it $f \circ (g \circ h)$? Or $(f \circ g) \circ h$? It’s good that
fong-spivak-seven-sketches_146 F01568	146	■ 0.743	$f; (g; h) = (f; g); h$	$f; (g; h) = (f; g); h$	Associativity says $f \circ (g \circ h) = (f \circ g) \circ h$, so it doesn’t matter which way we chose to try
fong-spivak-seven-sketches_147 F01569	147	■ 0.647	id_x	id_x	we will see that it is not harmful to draw id_x in any of the following three ways:
fong-spivak-seven-sketches_147 F01570	147	■ 0.926	$f = f = f$	$f = f = f$	property and the associativity property. The unitality says that $\mathrm{id}_x \circ f = f = f \circ \mathrm{id}_y$ for
fong-spivak-seven-sketches_147 F01571	147	■ 0.926	id_y	id_y	property and the associativity property. The unitality says that $\mathrm{id}_x \circ f = f = f \circ \mathrm{id}_y$ for
fong-spivak-seven-sketches_147 F01572	147	■ 1.000	$f: x \rightarrow y$	$f: x \rightarrow y$	any $f: x \rightarrow y$. In terms of diagrams this would say
fong-spivak-seven-sketches_148 F01573	148	■ 0.943	$\{1\}, \times$	$\{1\}, \times$	We will soon meet the monoidal category (Set, {1}, ×) . The monoidal product will
fong-spivak-seven-sketches_148 F01574	148	■ 1.000	$S \times T = \{(s, t) \mid s \in S, t \in T\}$	$S \times T = \{(s, t) \mid s \in S, t \in T\}$	take two sets, S and T , and return the set $S \times T = \{(s, t) \mid s \in S, t \in T\}$. But whereas

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_F01575	148	0.999	$(p \otimes q) \otimes r = p \otimes (q \otimes r)$	$(p \otimes q) \otimes r = p \otimes (q \otimes r)$	for monoidal preorders we had the brute associative property $(p \otimes q) \otimes r = p \otimes (q \otimes r)$,
fong-spivak-seven-sketches_F01576	148	1.000	$T \times U$	$T \times U$	and an element in $T \times U$, while the second contains pairs consisting of an element in
fong-spivak-seven-sketches_F01577	148	1.000	$S \times T$	$S \times T$	$S \times T$ and an element in U . The sets are not equal, but they are clearly isomorphic,
fong-spivak-seven-sketches_F01578	148	1.000	$\alpha_{s,t,u}$	$\alpha_{s,t,u}$	example, above we replaced the associativity equation with an isomorphism $\alpha_{s,t,u}$, but
fong-spivak-seven-sketches_F01579	148	0.731	$I \in \operatorname{Ob}(\mathcal{C})$	$I \in \operatorname{Ob}(\mathcal{C})$	(i) an object $I \in \operatorname{Ob}(\mathcal{C})$ called the <i>monoidal unit</i> , and
fong-spivak-seven-sketches_F01580	149	0.990	$\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$	$\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$	(ii) a functor $\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$, called the <i>monoidal product</i>
fong-spivak-seven-sketches_F01581	149	0.967	$\lambda_c: I \otimes c \cong c$	$\lambda_c: I \otimes c \cong c$	(a) $\lambda_c: I \otimes c \cong c$ for every $c \in \operatorname{Ob}(\mathcal{C})$,
fong-spivak-seven-sketches_F01582	149	0.835	$\rho_c: c \otimes I \cong c$	$\rho_c: c \otimes I \cong c$	(b) $\rho_c: c \otimes I \cong c$ for every $c \in \operatorname{Ob}(\mathcal{C})$,
fong-spivak-seven-sketches_F01583	149	0.981	$\alpha_{c,d,e}: (c \otimes d) \otimes e \cong c \otimes (d \otimes e)$	$\alpha_{c,d,e}: (c \otimes d) \otimes e \cong c \otimes (d \otimes e)$	(c) $\alpha_{c,d,e}: (c \otimes d) \otimes e \cong c \otimes (d \otimes e)$ for every $c, d, e \in \operatorname{Ob}(\mathcal{C})$, and
fong-spivak-seven-sketches_F01584	149	0.964	$\sigma_{c,d}: c \otimes d \cong d \otimes c$	$\sigma_{c,d}: c \otimes d \cong d \otimes c$	(d) $\sigma_{c,d}: c \otimes d \cong d \otimes c$ for every $c, d \in \operatorname{Ob}(\mathcal{C})$, called the <i>swap map</i> , such that $\sigma \circ \sigma = \operatorname{id}$.
fong-spivak-seven-sketches_F01585	149	0.964	$c, d \in \operatorname{Ob}(\mathcal{C})$	$c, d \in \operatorname{Ob}(\mathcal{C})$	(d) $\sigma_{c,d}: c \otimes d \cong d \otimes c$ for every $c, d \in \operatorname{Ob}(\mathcal{C})$, called the <i>swap map</i> , such that $\sigma \circ \sigma = \operatorname{id}$.
fong-spivak-seven-sketches_F01586	149	0.964	$\sigma \circ \sigma = \operatorname{id}$	$\sigma \circ \sigma = \operatorname{id}$	(d) $\sigma_{c,d}: c \otimes d \cong d \otimes c$ for every $c, d \in \operatorname{Ob}(\mathcal{C})$, called the <i>swap map</i> , such that $\sigma \circ \sigma = \operatorname{id}$.
fong-spivak-seven-sketches_F01587	149	0.999	$(\mathcal{P}, I, \otimes)$	$(\mathcal{P}, I, \otimes)$	a monoidal preorder is a monoidal category $(\mathcal{P}, I, \otimes)$ where, for every $p, q \in \mathcal{P}$, the set
fong-spivak-seven-sketches_F01588	149	0.999	$p, q \in \mathcal{P}$	$p, q \in \mathcal{P}$	a monoidal preorder is a monoidal category $(\mathcal{P}, I, \otimes)$ where, for every $p, q \in \mathcal{P}$, the set
fong-spivak-seven-sketches_F01589	149	1.000	$\mathcal{P}(p, q)$	$\mathcal{P}(p, q)$	$\mathcal{P}(p, q)$ has at most one element.
fong-spivak-seven-sketches_F01590	149	1.000	$I := \{1\}$	$I := \{1\}$	monoidal unit is some choice of singleton set, say $I := \{1\}$, and the monoidal product
fong-spivak-seven-sketches_F01591	149	0.513	$\otimes := \times$	$\otimes := \times$	is $\otimes := \times$. What it means that $\times$ is a functor is that:
fong-spivak-seven-sketches_F01592	149	0.513	$\times$	$\times$	is $\otimes := \times$. What it means that $\times$ is a functor is that:
fong-spivak-seven-sketches_F01593	149	0.537	$(S, T) \in \operatorname{Ob}(\mathcal{C})$	$(S, T) \in \operatorname{Ob}(\mathcal{C})$	• For any pair of objects, i.e. sets, $(S, T) \in \operatorname{Ob}(\mathbf{Set} \times \mathbf{Set})$, one obtains a set $(S \times T) \in$
fong-spivak-seven-sketches_F01594	149	0.537	$(S \times T) \in$	$(S \times T) \in$	• For any pair of objects, i.e. sets, $(S, T) \in \operatorname{Ob}(\mathbf{Set} \times \mathbf{Set})$, one obtains a set $(S \times T) \in$
fong-spivak-seven-sketches_F01595	149	0.988	$\{(s, t) \mid s \in S, t \in T\}$	$\{(s, t) \mid s \in S, t \in T\}$	$\operatorname{Ob}(\mathbf{Set})$. We know what it is: the set of pairs $\{(s, t) \mid s \in S, t \in T\}$.
fong-spivak-seven-sketches_F01596	149	1.000	$f: S \rightarrow S'$	$f: S \rightarrow S'$	• For any pair of morphisms, i.e. functions, $f: S \rightarrow S'$ and $g: T \rightarrow T'$, one obtains a
fong-spivak-seven-sketches_F01597	149	1.000	$g: T \rightarrow T'$	$g: T \rightarrow T'$	• For any pair of morphisms, i.e. functions, $f: S \rightarrow S'$ and $g: T \rightarrow T'$, one obtains a
fong-spivak-seven-sketches_F01598	149	1.000	$(f \times g): (S \times T) \rightarrow (S' \times T')$	$(f \times g): (S \times T) \rightarrow (S' \times T')$	function $(f \times g): (S \times T) \rightarrow (S' \times T')$. It works pointwise: $(f \times g)(s, t) := (f(s), g(t))$.
fong-spivak-seven-sketches_F01599	149	1.000	$(f \times g)(s, t) := (f(s), g(t))$	$(f \times g)(s, t) := (f(s), g(t))$	function $(f \times g): (S \times T) \rightarrow (S' \times T')$. It works pointwise: $(f \times g)(s, t) := (f(s), g(t))$.
fong-spivak-seven-sketches_F01600	149	0.441	$\operatorname{id}_S \times \operatorname{id}_T = \operatorname{id}_{S \times T}$	$\operatorname{id}_S \times \operatorname{id}_T = \operatorname{id}_{S \times T}$	• These should preserve identities: $\operatorname{id}_S \times \operatorname{id}_T = \operatorname{id}_{S \times T}$ for any sets S, T .
fong-spivak-seven-sketches_F01601	149	1.000	$S \xrightarrow{f} S' \xrightarrow{f'} S''$	$S \xrightarrow{f} S' \xrightarrow{f'} S''$	• These should preserve composition: for any functions $S \xrightarrow{f} S' \xrightarrow{f'} S''$ and $T \xrightarrow{g}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_F01602	149	1.000	$T \xrightarrow{g}$	$T \xrightarrow{g}$	• These should preserve composition: for any functions $S \xrightarrow{f} S' \xrightarrow{f'} S''$ and $T \xrightarrow{g}$
fong-spivak-seven-sketches_F01603	149	1.000	$T^{\{\prime\}} \xrightarrow{g^{\{\prime\}}} T^{\{\prime\prime\}}$	$T' \xrightarrow{g'} T''$	$T' \xrightarrow{g'} T''$, one has
fong-spivak-seven-sketches_F01604	150	1.000	$\{1\} \times S \cong S$	$\{1\} \times S \cong S$	The four conditions, (a), (b), (c), and (d) give isomorphisms $\{1\} \times S \cong S$, etc. These
fong-spivak-seven-sketches_F01605	150	0.742	$\{(1, s) \mid s \in S\} \rightarrow S$	$\{(1, s) \mid s \in S\} \rightarrow S$	maps are obvious in the case of Set , e.g. the function $\{(1, s) \mid s \in S\} \rightarrow S$ sending $(1, s)$
fong-spivak-seven-sketches_F01606	150	0.742	$1, s$	$1, s$	maps are obvious in the case of Set , e.g. the function $\{(1, s) \mid s \in S\} \rightarrow S$ sending $(1, s)$
fong-spivak-seven-sketches_F01607	150	0.984	$1, \times$	$1, \times$	<i>Exercise 4.50.</i> Consider the monoidal category $(\mathbf{Set}, 1, \times)$, together with the diagram
fong-spivak-seven-sketches_F01608	150	0.755	$A=B=C=D=F=G=\mathbb{Z}$	$A = B = C = D = F = G = \mathbb{Z}$	Suppose that $A = B = C = D = F = G = \mathbb{Z}$ and $E = \mathbb{B} = \{\text{true}, \text{false}\}$, and
fong-spivak-seven-sketches_F01609	150	0.755	$E=\mathbb{B}=$	$E = \mathbb{B} =$	Suppose that $A = B = C = D = F = G = \mathbb{Z}$ and $E = \mathbb{B} = \{\text{true}, \text{false}\}$, and
fong-spivak-seven-sketches_F01610	150	0.976	$f_C(a)= a , f_D(a)=a * 5, g_E(d, b)=$	$f_C(a) = a , f_D(a) = a * 5, g_E(d, b) =$	suppose that $f_C(a) = a , f_D(a) = a * 5, g_E(d, b) =$ “ $d \leq b$,” $g_F(d, b) = d - b$, and
fong-spivak-seven-sketches_F01611	150	0.976	$d \leq b$	$d \leq b$	suppose that $f_C(a) = a , f_D(a) = a * 5, g_E(d, b) =$ “ $d \leq b$,” $g_F(d, b) = d - b$, and
fong-spivak-seven-sketches_F01612	150	0.976	$g_F(d, b)=d-b$	$g_F(d, b) = d - b$	suppose that $f_C(a) = a , f_D(a) = a * 5, g_E(d, b) =$ “ $d \leq b$,” $g_F(d, b) = d - b$, and
fong-spivak-seven-sketches_F01613	150	1.000	$h(c, e)=$	$h(c, e) =$	$h(c, e) =$ if e then c else $1 - c$.
fong-spivak-seven-sketches_F01614	150	1.000	$1 - c$	$1 - c$	$h(c, e) =$ if e then c else $1 - c$.
fong-spivak-seven-sketches_F01615	150	1.000	$g_E(5, 3)$	$g_E(5, 3)$	1. What are $g_E(5, 3)$ and $g_F(5, 3)$?
fong-spivak-seven-sketches_F01616	150	1.000	$g_F(5, 3)$	$g_F(5, 3)$	1. What are $g_E(5, 3)$ and $g_F(5, 3)$?
fong-spivak-seven-sketches_F01617	150	1.000	$g_E(3, 5)$	$g_E(3, 5)$	2. What are $g_E(3, 5)$ and $g_F(3, 5)$?
fong-spivak-seven-sketches_F01618	150	1.000	$g_F(3, 5)$	$g_F(3, 5)$	2. What are $g_E(3, 5)$ and $g_F(3, 5)$?
fong-spivak-seven-sketches_F01619	150	0.992	$h(5)$	$h(5)$	3. What is $h(5, \text{true})$?
fong-spivak-seven-sketches_F01620	150	0.758	$h(-5)$	$h(-5)$	4. What is $h(-5, \text{true})$?
fong-spivak-seven-sketches_F01621	150	1.000	$A \times B \rightarrow G \times F$	$A \times B \rightarrow G \times F$	The whole diagram now defines a function $A \times B \rightarrow G \times F$; call it q .
fong-spivak-seven-sketches_F01622	150	1.000	$q_G(-2, 3)$	$q_G(-2, 3)$	6. What are $q_G(-2, 3)$ and $q_F(-2, 3)$?
fong-spivak-seven-sketches_F01623	150	1.000	$q_F(-2, 3)$	$q_F(-2, 3)$	6. What are $q_G(-2, 3)$ and $q_F(-2, 3)$?
fong-spivak-seven-sketches_F01624	150	1.000	$q_G(2, 3)$	$q_G(2, 3)$	7. What are $q_G(2, 3)$ and $q_F(2, 3)$?
fong-spivak-seven-sketches_F01625	150	1.000	$q_F(2, 3)$	$q_F(2, 3)$	7. What are $q_G(2, 3)$ and $q_F(2, 3)$?
fong-spivak-seven-sketches_F01626	150	0.542	$X(x, y) \in \mathcal{V}$	$X(x, y) \in \mathcal{V}$	(ii) for every pair $x, y \in \text{Ob}(\mathcal{X})$, one specifies an object $\mathcal{X}(x, y) \in \mathcal{V}$, called the <i>hom-</i>
fong-spivak-seven-sketches_F01627	150	0.356	$x \in \text{Ob}(X)$	$x \in \text{Ob}(X)$	(iii) for every $x \in \text{Ob}(\mathcal{X})$, one specifies a morphism $\text{id}_x: I \rightarrow \mathcal{X}(x, x)$ in $\mathcal{V}$, called the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_150 F01628	150	0.356	$\mathrm{id}_x: I \rightarrow X(x, x)$	$\mathrm{id}_x: I \rightarrow X(x, x)$	(iii) for every $x \in \mathrm{Ob}(\mathcal{X})$, one specifies a morphism $\mathrm{id}_x: I \rightarrow X(x, x)$ in $\mathcal{V}$, called the
fong-spivak-seven-sketches_151 F01629	151	0.608	$x, y, z \in \mathrm{Ob}(X)$	$x, y, z \in \mathrm{Ob}(X)$	(iv) for each $x, y, z \in \mathrm{Ob}(\mathcal{X})$, one specifies a morphism $\mathfrak{z}: X(x, y) \otimes X(y, z) \rightarrow X(x, z)$,
fong-spivak-seven-sketches_151 F01630	151	0.608	$9: \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \rightarrow \mathfrak{X}(x, z)$	$9: \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \rightarrow \mathfrak{X}(x, z)$	(iv) for each $x, y, z \in \mathrm{Ob}(\mathcal{X})$, one specifies a morphism $\mathfrak{z}: X(x, y) \otimes X(y, z) \rightarrow X(x, z)$,
fong-spivak-seven-sketches_151 F01631	151	1.000	$\mathcal{V} = ($	$\mathcal{V} = ($	<i>Exercise 4.52.</i> Recall from Example 4.49 that $\mathcal{V} = (\mathbf{Set}, \{1\}, \times)$ is a symmetric monoidal
fong-spivak-seven-sketches_151 F01632	151	1.000	$, \{1\}, \times)$	$, \{1\}, \times)$	<i>Exercise 4.52.</i> Recall from Example 4.49 that $\mathcal{V} = (\mathbf{Set}, \{1\}, \times)$ is a symmetric monoidal
fong-spivak-seven-sketches_151 F01633	151	0.961	$\mathrm{id}_x: I \rightarrow \mathcal{X}(x, x)$	$\mathrm{id}_x: I \rightarrow \mathcal{X}(x, x)$	$\mathrm{id}_x: I \rightarrow X(x, x)$ for the object x simply becomes the requirement (a) that $I \leq X(x, x)$ is
fong-spivak-seven-sketches_151 F01634	151	0.961	$I \leq \mathcal{X}(x, x)$	$I \leq \mathcal{X}(x, x)$	$\mathrm{id}_x: I \rightarrow X(x, x)$ for the object x simply becomes the requirement (a) that $I \leq X(x, x)$ is
fong-spivak-seven-sketches_153 F01635	153	0.644	$(\mathcal{C}, I, \otimes)$	$(\mathcal{C}, I, \otimes)$	Definition 4.58. Let $(\mathcal{C}, I, \otimes)$ be a symmetric monoidal category, and $c \in \mathrm{Ob}(\mathcal{C})$ an
fong-spivak-seven-sketches_153 F01636	153	0.999	$c^* \in \mathrm{Ob}(\mathcal{C})$	$c^* \in \mathrm{Ob}(\mathcal{C})$	(i) an object $c^* \in \mathrm{Ob}(\mathcal{C})$, called the <i>dual of c</i> ,
fong-spivak-seven-sketches_153 F01637	153	1.000	$\eta_c: I \rightarrow c^* \otimes c$	$\eta_c: I \rightarrow c^* \otimes c$	(ii) a morphism $\eta_c: I \rightarrow c^* \otimes c$, called the <i>unit for c</i> ,
fong-spivak-seven-sketches_153 F01638	153	1.000	$\epsilon_c: c \otimes c^* \rightarrow I$	$\epsilon_c: c \otimes c^* \rightarrow I$	(iii) a morphism $\epsilon_c: c \otimes c^* \rightarrow I$, called the <i>counit for c</i> .
fong-spivak-seven-sketches_153 F01639	153	0.516	c^*	c^*	If for every object $c \in \mathrm{Ob}(\mathcal{C})$ there exists a dual c^* for c , then we say that $(\mathcal{C}, I, \otimes)$ is
fong-spivak-seven-sketches_153 F01640	153	0.997	$\xrightarrow{c}$	$\xrightarrow{c}$	wire labeled c^* , i.e. $\xrightarrow{c}$ is the same as $\xleftarrow{c}$. The cup and cap discussed above are in fact
fong-spivak-seven-sketches_153 F01641	153	0.997	$\xleftarrow{c^*}$	$\xleftarrow{c^*}$	wire labeled c^* , i.e. $\xrightarrow{c}$ is the same as $\xleftarrow{c}$. The cup and cap discussed above are in fact
fong-spivak-seven-sketches_154 F01642	154	0.877	ϵ_{sound}	ϵ_{sound}	Note that the pictures in Eq. (4.57) correspond to ϵ_{sound} and η_{sound^*} .
fong-spivak-seven-sketches_154 F01643	154	0.877	η_{sound^*}	η_{sound^*}	Note that the pictures in Eq. (4.57) correspond to ϵ_{sound} and η_{sound^*} .
fong-spivak-seven-sketches_154 F01644	154	0.879	$\mathcal{C}(b \otimes c, d) \cong \mathcal{C}(b, c \multimap d)$	$\mathcal{C}(b \otimes c, d) \cong \mathcal{C}(b, c \multimap d)$	object $c \multimap d$ and an isomorphism $\mathcal{C}(b \otimes c, d) \cong \mathcal{C}(b, c \multimap d)$, natural in b . While we will
fong-spivak-seven-sketches_154 F01645	154	1.000	c'	c'	2. if c^* and c' are both duals to c then there is an isomorphism $c^* \cong c'$; and
fong-spivak-seven-sketches_154 F01646	154	1.000	$c^* \cong c'$	$c^* \cong c'$	2. if c^* and c' are both duals to c then there is an isomorphism $c^* \cong c'$; and
fong-spivak-seven-sketches_154 F01647	154	1.000	$c \cong c^{**}$	$c \cong c^{**}$	3. there is an isomorphism between c and its double-dual, $c \cong c^{**}$.
fong-spivak-seven-sketches_154 F01648	154	0.947	$c^* \otimes d$	$c^* \otimes d$	To prove 1., the key idea is that for any c and d , the object $c \multimap d$ is given by $c^* \otimes d$,
fong-spivak-seven-sketches_154 F01649	154	0.999	$\mathcal{C}(b \otimes c, d) \cong \mathcal{C}(b, c \rightarrow d)$	$\mathcal{C}(b \otimes c, d) \cong \mathcal{C}(b, c \rightarrow d)$	and the natural isomorphism $\mathcal{C}(b \otimes c, d) \cong \mathcal{C}(b, c \multimap d)$ is given by precomposing with
fong-spivak-seven-sketches_154 F01650	154	0.899	$\mathrm{id}_b \otimes \eta_c$	$\mathrm{id}_b \otimes \eta_c$	$\mathrm{id}_b \otimes \eta_c$.
fong-spivak-seven-sketches_154 F01651	154	1.000	$\alpha: A \rightarrow B$	$\alpha: A \rightarrow B$	look at than to write down formally. ² If in addition to the corelation $\alpha: A \rightarrow B$ above
fong-spivak-seven-sketches_154 F01652	154	1.000	$\beta: B \rightarrow C$	$\beta: B \rightarrow C$	we have another corelation $\beta: B \rightarrow C$
fong-spivak-seven-sketches_155 F01653	155	0.999	$\beta \circ \alpha$	$\beta \circ \alpha$	Then the composite $\beta \circ \alpha$ of our two corelations is given by

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_155 F01654	155	■ 1.000	$\backslash\mathrm{beta}$	β	one to the other staying within equivalence classes of either α or β .
fong-spivak-seven-sketches_155 F01655	155	■ 0.873	$(\backslash\mathrm{varnothing}, \backslash\mathrm{sqcup})$	$(\emptyset, \sqcup)$	$(\emptyset, \sqcup)$. This monoidal category is compact closed, with every finite set its own
fong-spivak-seven-sketches_155 F01656	155	■ 1.000	$A \backslash\mathrm{sqcup} A := \{(a, 1), (a, 2) \mid a \in A\}$	$A \sqcup A := \{(a, 1), (a, 2) \mid a \in A\}$	$A \sqcup A := \{(a, 1), (a, 2) \mid a \in A\}$ where each part simply consists of the two elements $(a, 1)$
fong-spivak-seven-sketches_155 F01657	155	■ 1.000	$(a, 1)$	$(a, 1)$	$A \sqcup A := \{(a, 1), (a, 2) \mid a \in A\}$ where each part simply consists of the two elements $(a, 1)$
fong-spivak-seven-sketches_155 F01658	155	■ 1.000	$(a, 2)$	$(a, 2)$	and $(a, 2)$ for each $a \in A$. The unit on a finite set A is the corelation $\eta_A: \emptyset \rightarrow A \sqcup A$
fong-spivak-seven-sketches_155 F01659	155	■ 1.000	$\backslash\mathrm{eta}_{\{A\}}: \backslash\mathrm{varnothing} \rightarrowtail A \backslash\mathrm{sqcup} A$	$\eta_A: \emptyset \rightarrow A \sqcup A$	and $(a, 2)$ for each $a \in A$. The unit on a finite set A is the corelation $\eta_A: \emptyset \rightarrow A \sqcup A$
fong-spivak-seven-sketches_155 F01660	155	■ 1.000	$\backslash\mathrm{epsilon}_{\{A\}}: A \backslash\mathrm{sqcup} A \rightarrowtail \backslash\mathrm{varnothing}$	$\epsilon_A: A \sqcup A \rightarrow \emptyset$	$\epsilon_A: A \sqcup A \rightarrow \emptyset$ specied by this same equivalence relation.
fong-spivak-seven-sketches_155 F01661	155	■ 0.997	$\backslash\mathrm{varnothing} \rightarrowtail \backslash\mathrm{underline{3}} \backslash\mathrm{sqcup} \backslash\mathrm{underline{3}}$	$\emptyset \rightarrow 3 \sqcup 3$	1. Draw a picture of the unit corelation $\emptyset \rightarrow 3 \sqcup 3$.
fong-spivak-seven-sketches_155 F01662	155	■ 1.000	$\backslash\mathrm{underline{3}} \backslash\mathrm{sqcup} \backslash\mathrm{underline{3}} \rightarrowtail \backslash\mathrm{varnothing}$	$3 \sqcup 3 \rightarrow \emptyset$	2. Draw a picture of the counit corelation $3 \sqcup 3 \rightarrow \emptyset$.
fong-spivak-seven-sketches_155 F01663	155	■ 0.993	$(\backslash\mathrm{mathcal{V}}, I, \otimes)$	$(\mathcal{V}, I, \otimes)$	extra work to prove that for any skeletal (unital, commutative) quantale $(\mathcal{V}, I, \otimes)$ the
fong-spivak-seven-sketches_155 F01664	155	■ 0.562	$\backslash\mathrm{operatorname{Prof}}_{\backslash\mathrm{mathcal{V}}}$	$\mathrm{Prof}_{\mathcal{V}}$	profunctor category $\mathrm{Prof}_{\mathcal{V}}$ of Theorem 4.23 is compact closed, so we'll discuss this
fong-spivak-seven-sketches_155 F01665	155	■ 0.863	$\backslash\mathrm{alpha} ; \backslash\mathrm{beta}$	$\alpha; \beta$	$\alpha \circ \beta$ on $A \sqcup C$. To do so requires three steps: (i) consider α and β as relations on $A \sqcup B \sqcup C$, (ii) take the
fong-spivak-seven-sketches_155 F01666	155	■ 0.863	$A \backslash\mathrm{sqcup} C$	$A \sqcup C$	$\alpha \circ \beta$ on $A \sqcup C$. To do so requires three steps: (i) consider α and β as relations on $A \sqcup B \sqcup C$, (ii) take the
fong-spivak-seven-sketches_155 F01667	155	■ 0.863	$A \backslash\mathrm{sqcup} B \backslash\mathrm{sqcup} C$	$A \sqcup B \sqcup C$	$\alpha \circ \beta$ on $A \sqcup C$. To do so requires three steps: (i) consider α and β as relations on $A \sqcup B \sqcup C$, (ii) take the
fong-spivak-seven-sketches_155 F01668	155	■ 1.000	$\backslash\mathrm{alpha} \backslash\mathrm{subseq}(A \backslash\mathrm{sqcup} B) \times (A \backslash\mathrm{sqcup} B)$	$\alpha \subseteq (A \sqcup B) \times (A \sqcup B)$	formal description. Note that as binary relations, we have $\alpha \subseteq (A \sqcup B) \times (A \sqcup B)$, and $\beta \subseteq (B \sqcup C) \times (B \sqcup C)$.
fong-spivak-seven-sketches_155 F01669	155	■ 1.000	$\backslash\mathrm{beta} \backslash\mathrm{subseq}(B \backslash\mathrm{sqcup} C) \times (B \backslash\mathrm{sqcup} C)$	$\beta \subseteq (B \sqcup C) \times (B \sqcup C)$	formal description. Note that as binary relations, we have $\alpha \subseteq (A \sqcup B) \times (A \sqcup B)$, and $\beta \subseteq (B \sqcup C) \times (B \sqcup C)$.
fong-spivak-seven-sketches_155 F01670	155	■ 1.000	$\backslash\mathrm{iota}_{A \backslash\mathrm{sqcup} B}: A \backslash\mathrm{sqcup} B \rightarrowtail A \backslash\mathrm{sqcup} B \backslash\mathrm{sqcup} C, \backslash\mathrm{iota}_{B \backslash\mathrm{sqcup} C}: B \backslash\mathrm{sqcup} C \rightarrowtail A \backslash\mathrm{sqcup} B \backslash\mathrm{sqcup} C$	$\iota_{A \sqcup B}: A \sqcup B \rightarrow A \sqcup B \sqcup C, \iota_{B \sqcup C}: B \sqcup C \rightarrow A \sqcup B \sqcup C$	We also have three inclusions: $\iota_{A \sqcup B}: A \sqcup B \rightarrow A \sqcup B \sqcup C, \iota_{B \sqcup C}: B \sqcup C \rightarrow A \sqcup B \sqcup C$, and $\iota_{A \sqcup C}: A \sqcup C \rightarrow$
fong-spivak-seven-sketches_155 F01671	155	■ 1.000	$\backslash\mathrm{iota}_{A \backslash\mathrm{sqcup} C}: A \backslash\mathrm{sqcup} C \rightarrowtail$	$\iota_{A \sqcup C}: A \sqcup C \rightarrow$	We also have three inclusions: $\iota_{A \sqcup B}: A \sqcup B \rightarrow A \sqcup B \sqcup C, \iota_{B \sqcup C}: B \sqcup C \rightarrow A \sqcup B \sqcup C$, and $\iota_{A \sqcup C}: A \sqcup C \rightarrow$
fong-spivak-seven-sketches_156 F01672	156	■ 1.000	$\{ \}_\backslash\mathrm{mathcal{V}}$	$\mathcal{V}$	Monoidal products in $\mathrm{Prof}_{\mathcal{V}}$ are just product categories. In terms of wiring diagrams,
fong-spivak-seven-sketches_156 F01673	156	■ 0.383	$\backslash\mathrm{Phi}: X_{\{1\}} \mapsto X_{\{2\}}$	$\Phi: X_1 \mapsto X_2$	$\mathrm{Prof}_{\mathcal{V}}$ are $\mathcal{V}$ -profunctors. So given $\mathcal{V}$ -profunctors $\Phi: \mathcal{X}_1 \mapsto \mathcal{X}_2$ and $\Psi: \mathcal{Y}_1 \mapsto \mathcal{Y}_2$, one
fong-spivak-seven-sketches_156 F01674	156	■ 0.383	$\backslash\mathrm{Psi}: y_{\{1\}} \rightarrowtail y_{\{2\}}$	$\Psi: y_1 \rightarrow y_2$	$\mathrm{Prof}_{\mathcal{V}}$ are $\mathcal{V}$ -profunctors. So given $\mathcal{V}$ -profunctors $\Phi: \mathcal{X}_1 \mapsto \mathcal{X}_2$ and $\Psi: \mathcal{Y}_1 \mapsto \mathcal{Y}_2$, one
fong-spivak-seven-sketches_156 F01675	156	■ 0.262	$(\backslash\mathrm{Phi} \times \backslash\mathrm{Psi}): \backslash\mathrm{mathfrak{X}}_{\{1\}} \times y_{\{1\}} \rightarrowtail x_{\{2\}} \times y_{\{2\}}$	$(\Phi \times \Psi): \mathfrak{X}_1 \times y_1 \rightarrow x_2 \times y_2$	defines a $\mathcal{V}$ -profunctor $(\Phi \times \Psi): \mathfrak{X}_1 \times \mathfrak{Y}_1 \mapsto \mathfrak{X}_2 \times \mathfrak{Y}_2$ by
fong-spivak-seven-sketches_156 F01676	156	■ 0.857	$\backslash\mathrm{mathbf{P r o f}}_{\backslash\mathrm{text{Bool}}}$	$\mathrm{Prof}_{\mathrm{Bool}}$	<i>Exercise 4.64.</i> Interpret the monoidal products in $\mathrm{Prof}_{\mathrm{Bool}}$ in terms of feasibility.
fong-spivak-seven-sketches_156 F01677	156	■ 1.000	$\backslash\mathrm{Phi} \times \backslash\mathrm{Psi}$	$\Phi \times \Psi$	sense as the monoidal product of resource preorders $\mathcal{X}$ and $\mathcal{Y}$ and why $\Phi \times \Psi$ makes
fong-spivak-seven-sketches_156 F01678	156	■ 0.593	$(1, 1) = I$	$(1, 1) = I$	$\mathcal{V}$ -category $\mathbf{1}$; it has one object, say 1 , and $(1, 1) = I$ is the monoidal unit of $\mathcal{V}$. We take
fong-spivak-seven-sketches_156 F01679	156	■ 0.467	$X \times \backslash\mathrm{mathbf{1}} \rightarrowtail X$	$X \times \mathbf{1} \rightarrow X$	phisms $\mathcal{X} \times \mathbf{1} \mapsto \mathcal{X}$ and $\mathbf{1} \times \mathcal{X} \mapsto \mathcal{X}$ in $\mathrm{Prof}_{\mathcal{V}}$, for any $\mathcal{V}$ -category $\mathcal{X}$. What are they?
fong-spivak-seven-sketches_156 F01680	156	■ 0.467	$\backslash\mathrm{mathbf{1}} \times X \rightarrowtail X$	$\mathbf{1} \times X \rightarrow X$	phisms $\mathcal{X} \times \mathbf{1} \mapsto \mathcal{X}$ and $\mathbf{1} \times \mathcal{X} \mapsto \mathcal{X}$ in $\mathrm{Prof}_{\mathcal{V}}$, for any $\mathcal{V}$ -category $\mathcal{X}$. What are they?

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_161681	157	0.894	$\mathcal{X}^{\text{op}}$	$\mathcal{X}^{\text{op}}$	Duals are easy: for every $\mathcal{V}$ -category $\mathcal{X}$, its dual is its opposite category $\mathcal{X}^{\text{op}}$ (see
fong-spivak-seven-sketches_161682	157	0.225	$\eta_x: \mathbf{1} \rightarrow \mathcal{X}^{\text{op}} \times \mathcal{X}$	$\eta_x: \mathbf{1} \rightarrow \mathcal{X}^{\text{op}} \times \mathcal{X}$	$\mathcal{V}$ -profunctor $\eta_{\mathcal{X}}: \mathbf{1} \rightarrow \mathcal{X}^{\text{op}} \times \mathcal{X}$. By definition, this is a $\mathcal{V}$ -functor
fong-spivak-seven-sketches_161683	157	0.981	$\eta_x(1, x, x') := \mathcal{X}(x, x')$	$\eta_x(1, x, x') := \mathcal{X}(x, x')$	we define it by $\eta_{\mathcal{X}}(1, x, x') := \mathcal{X}(x, x')$. Similarly, the counit is the profunctor $\epsilon_{\mathcal{X}}: (\mathcal{X} \times$
fong-spivak-seven-sketches_161684	157	0.981	$\epsilon_x: (\mathcal{X} \times$	$\epsilon_x: (\mathcal{X} \times$	we define it by $\eta_{\mathcal{X}}(1, x, x') := \mathcal{X}(x, x')$. Similarly, the counit is the profunctor $\epsilon_{\mathcal{X}}: (\mathcal{X} \times$
fong-spivak-seven-sketches_161685	157	0.710	$\mathcal{X}^{\text{op}}) \rightarrow \mathbf{1}$	$\mathcal{X}^{\text{op}}) \rightarrow \mathbf{1}$	$\mathcal{X}^{\text{op}}) \rightarrow \mathbf{1}$, defined by $\epsilon_{\mathcal{X}}(x, x', 1) := \mathcal{X}(x, x')$.
fong-spivak-seven-sketches_161686	157	0.710	$\epsilon_{\mathcal{X}}(x, x', 1) := \mathcal{X}(x, x')$	$\epsilon_{\mathcal{X}}(x, x', 1) := \mathcal{X}(x, x')$	$\mathcal{X}^{\text{op}}) \rightarrow \mathbf{1}$, defined by $\epsilon_{\mathcal{X}}(x, x', 1) := \mathcal{X}(x, x')$.
fong-spivak-seven-sketches_161687	160	1.000	$u = 15x$	$u = 15x$	produces two output signals: $u = 15x$ (upper right) and $v = 3x + 21y$ (lower right).
fong-spivak-seven-sketches_161688	160	1.000	$v = 3x + 21y$	$v = 3x + 21y$	produces two output signals: $u = 15x$ (upper right) and $v = 3x + 21y$ (lower right).
fong-spivak-seven-sketches_161689	160	1.000	$x + 7y$	$x + 7y$	adds the second copy of x to the lower signal to get $x + 7y$, and so on.
fong-spivak-seven-sketches_161690	161	0.608	$(\mathcal{C}, 0, +)$	$(\mathcal{C}, 0, +)$	Definition 5.2. A <i>prop</i> is a symmetric strict monoidal category $(\mathcal{C}, 0, +)$ for which
fong-spivak-seven-sketches_161691	161	0.950	$\text{Ob}(\mathcal{C}) = \mathbb{N}$	$\text{Ob}(\mathcal{C}) = \mathbb{N}$	$\text{Ob}(\mathcal{C}) = \mathbb{N}$, the monoidal unit is $0 \in \mathbb{N}$, and the monoidal product on objects is
fong-spivak-seven-sketches_161692	161	0.950	$0 \in \mathbb{N}$	$0 \in \mathbb{N}$	$\text{Ob}(\mathcal{C}) = \mathbb{N}$, the monoidal unit is $0 \in \mathbb{N}$, and the monoidal product on objects is
fong-spivak-seven-sketches_161693	161	0.994	$\mathcal{C}(m, n)$	$\mathcal{C}(m, n)$	(i) a set $\mathcal{C}(m, n)$ of morphisms $m \rightarrow n$, for $m, n \in \mathbb{N}$.
fong-spivak-seven-sketches_161694	161	0.655	$\text{id}_n: n \rightarrow n$	$\text{id}_n: n \rightarrow n$	(ii) for all $n \in \mathbb{N}$, an identity map $\text{id}_n: n \rightarrow n$.
fong-spivak-seven-sketches_161695	161	0.999	$\sigma_{m,n}: m+n \rightarrow n+m$	$\sigma_{m,n}: m+n \rightarrow n+m$	(iii) for all $m, n \in \mathbb{N}$, a symmetry map $\sigma_{m,n}: m+n \rightarrow n+m$.
fong-spivak-seven-sketches_161696	161	1.000	$f: m \rightarrow n$	$f: m \rightarrow n$	(iv) a composition rule: given $f: m \rightarrow n$ and $g: n \rightarrow p$, a map $(f \circ g): m \rightarrow p$.
fong-spivak-seven-sketches_161697	161	1.000	$g: n \rightarrow p$	$g: n \rightarrow p$	(iv) a composition rule: given $f: m \rightarrow n$ and $g: n \rightarrow p$, a map $(f \circ g): m \rightarrow p$.
fong-spivak-seven-sketches_161698	161	1.000	$(f; g): m \rightarrow p$	$(f; g): m \rightarrow p$	(iv) a composition rule: given $f: m \rightarrow n$ and $g: n \rightarrow p$, a map $(f \circ g): m \rightarrow p$.
fong-spivak-seven-sketches_161699	161	1.000	$f: m \rightarrow m'$	$f: m \rightarrow m'$	(v) a monoidal product on morphisms: given $f: m \rightarrow m'$ and $g: n \rightarrow n'$, a map
fong-spivak-seven-sketches_161700	161	1.000	$g: n \rightarrow n'$	$g: n \rightarrow n'$	(v) a monoidal product on morphisms: given $f: m \rightarrow m'$ and $g: n \rightarrow n'$, a map
fong-spivak-seven-sketches_161701	161	0.963	$(f+g): m+n \rightarrow m'+n'$	$(f+g): m+n \rightarrow m'+n'$	$f+g: m+n \rightarrow m'+n'$.
fong-spivak-seven-sketches_161702	162	1.000	$\underline{m} = \{1, \dots, m\}$	$\underline{m} = \{1, \dots, m\}$	from $\underline{m} = \{1, \dots, m\}$ to $\underline{n} = \{1, \dots, n\}$. (The identities, symmetries, and composition
fong-spivak-seven-sketches_161703	162	1.000	$\underline{n} = \{1, \dots, n\}$	$\underline{n} = \{1, \dots, n\}$	from $\underline{m} = \{1, \dots, m\}$ to $\underline{n} = \{1, \dots, n\}$. (The identities, symmetries, and composition
fong-spivak-seven-sketches_161704	162	1.000	$f+g: m+n \rightarrow m'+n'$	$f+g: m+n \rightarrow m'+n'$	functions: that is, given $f: m \rightarrow m'$ and $g: n \rightarrow n'$, we define $f+g: m+n \rightarrow m'+n'$
fong-spivak-seven-sketches_161705	162	1.000	$f: 3 \rightarrow 2$	$f: 3 \rightarrow 2$	1. Draw a morphism $f: 3 \rightarrow 2$ and a morphism $g: 2 \rightarrow 4$ in FinSet .
fong-spivak-seven-sketches_161706	162	1.000	$g: 2 \rightarrow 4$	$g: 2 \rightarrow 4$	1. Draw a morphism $f: 3 \rightarrow 2$ and a morphism $g: 2 \rightarrow 4$ in FinSet .
fong-spivak-seven-sketches_161707	162	1.000	$f+g$	$f+g$	2. Draw $f+g$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F01708	162	■ 0.998	$\sigma_{m,n}$	$\sigma_{m,n}$	5. Choose $m, n \in \mathbb{N}$, and draw the symmetry map $\sigma_{m,n}$ in FinSet ?
fong-spivak-seven-sketches F01709	162	■ 1.000	$\underline{m} \rightarrow \underline{n}$	$\underline{m} \rightarrow \underline{n}$	bijections $\underline{m} \rightarrow \underline{n}$. Note that in this case morphisms $m \rightarrow n$ only exist when $m = n$;
fong-spivak-seven-sketches F01710	162	■ 0.987	$m \neq n$	$m \neq n$	when $m \neq n$ the homset $\mathbf{Bij}(m, n)$ is empty. Since Bij is a subcategory of FinSet , we
fong-spivak-seven-sketches F01711	162	■ 0.987	$\mathbf{Bij}(m, n)$	$\mathbf{Bij}(m, n)$	when $m \neq n$ the homset $\mathbf{Bij}(m, n)$ is empty. Since Bij is a subcategory of FinSet , we
fong-spivak-seven-sketches F01712	162	■ 0.987	$\mathbf{Bij}$	Bij	when $m \neq n$ the homset $\mathbf{Bij}(m, n)$ is empty. Since Bij is a subcategory of FinSet , we
fong-spivak-seven-sketches F01713	162	■ 1.000	$\underline{m} \sqcup \underline{n}$	$\underline{m} \sqcup \underline{n}$	are partitions on $\underline{m} \sqcup \underline{n}$ (see Example 4.61), is a prop.
fong-spivak-seven-sketches F01714	162	■ 0.952	$R \subseteq \underline{m} \times \underline{n}$	$R \subseteq \underline{m} \times \underline{n}$	Example 5.8. There is a prop Rel for which morphisms $m \rightarrow n$ are relations, $R \subseteq \underline{m} \times \underline{n}$.
fong-spivak-seven-sketches F01715	162	■ 1.000	$S \subseteq \underline{n} \times \underline{p}$	$S \subseteq \underline{n} \times \underline{p}$	The composition of R with $S \subseteq \underline{n} \times \underline{p}$ is
fong-spivak-seven-sketches F01716	163	■ 0.934	$F(n)=n$	$F(n) = n$	(a) F is identity-on-objects, i.e. $F(n) = n$ for all $n \in \text{Ob}(\mathcal{C}) = \text{Ob}(\mathcal{D}) = \mathbb{N}$, and
fong-spivak-seven-sketches F01717	163	■ 0.934	$n \in \text{Ob}(\mathcal{C}) = \text{Ob}(\mathcal{D}) = \mathbb{N}$	$n \in \text{Ob}(\mathcal{C}) = \text{Ob}(\mathcal{D}) = \mathbb{N}$	(a) F is identity-on-objects, i.e. $F(n) = n$ for all $n \in \text{Ob}(\mathcal{C}) = \text{Ob}(\mathcal{D}) = \mathbb{N}$, and
fong-spivak-seven-sketches F01718	163	■ 0.977	$f: m_1 \rightarrow m_2$	$f: m_1 \rightarrow m_2$	(b) for all $f: m_1 \rightarrow m_2$ and $g: n_1 \rightarrow n_2$ in $\mathcal{C}$, we have $F(f) + F(g) = F(f + g)$ in $\mathcal{D}$.
fong-spivak-seven-sketches F01719	163	■ 0.977	$g: n_1 \rightarrow n_2$	$g: n_1 \rightarrow n_2$	(b) for all $f: m_1 \rightarrow m_2$ and $g: n_1 \rightarrow n_2$ in $\mathcal{C}$, we have $F(f) + F(g) = F(f + g)$ in $\mathcal{D}$.
fong-spivak-seven-sketches F01720	163	■ 0.977	$F(f)+F(g)=F(f+g)$	$F(f) + F(g) = F(f + g)$	(b) for all $f: m_1 \rightarrow m_2$ and $g: n_1 \rightarrow n_2$ in $\mathcal{C}$, we have $F(f) + F(g) = F(f + g)$ in $\mathcal{D}$.
fong-spivak-seven-sketches F01721	163	■ 0.966	i	i	Example 5.12. The inclusion $i: \mathbf{Bij} \rightarrow \mathbf{FinSet}$ is a prop functor. Perhaps more interest-
fong-spivak-seven-sketches F01722	163	■ 0.503	$F:$	$F:$	ingly, there is a prop functor $F: \mathbf{FinSet} \rightarrow \mathbf{Rel}_{\mathbf{Fin}}$. It sends a function $f: \underline{m} \rightarrow \underline{n}$ to the
fong-spivak-seven-sketches F01723	163	■ 0.503	$\mathbf{Rel}_{\mathbf{Fin}}$	Rel _{Fin}	ingly, there is a prop functor $F: \mathbf{FinSet} \rightarrow \mathbf{Rel}_{\mathbf{Fin}}$. It sends a function $f: \underline{m} \rightarrow \underline{n}$ to the
fong-spivak-seven-sketches F01724	163	■ 0.503	$f: \underline{m} \rightarrow \underline{n}$	$f: \underline{m} \rightarrow \underline{n}$	ingly, there is a prop functor $F: \mathbf{FinSet} \rightarrow \mathbf{Rel}_{\mathbf{Fin}}$. It sends a function $f: \underline{m} \rightarrow \underline{n}$ to the
fong-spivak-seven-sketches F01725	163	■ 1.000	$F(f) := \{(i, j) \mid f(i) = j\} \subseteq \underline{m} \times \underline{n}$	$F(f) := \{(i, j) \mid f(i) = j\} \subseteq \underline{m} \times \underline{n}$	relation $F(f) := \{(i, j) \mid f(i) = j\} \subseteq \underline{m} \times \underline{n}$.
fong-spivak-seven-sketches F01726	163	■ 0.616	ι	ι	Definition 5.13. For $m, n \in \mathbb{N}$, an (m, n) -port graph (V, in, out, ι) is specified by
fong-spivak-seven-sketches F01727	163	■ 0.835	$V \rightarrow \mathbb{N}$	$V \rightarrow \mathbb{N}$	(ii) functions $in, out: V \rightarrow \mathbb{N}$, where $in(v)$ and $out(v)$ are called the <i>in degree</i> and <i>out</i>
fong-spivak-seven-sketches F01728	163	■ 0.835	$in(v)$	$in(v)$	(ii) functions $in, out: V \rightarrow \mathbb{N}$, where $in(v)$ and $out(v)$ are called the <i>in degree</i> and <i>out</i>
fong-spivak-seven-sketches F01729	163	■ 0.835	$out(v)$	$out(v)$	(ii) functions $in, out: V \rightarrow \mathbb{N}$, where $in(v)$ and $out(v)$ are called the <i>in degree</i> and <i>out</i>
fong-spivak-seven-sketches F01730	163	■ 0.793	$\iota: \underline{m} \sqcup O \xrightarrow{\cong} I \sqcup \underline{n}$	$\iota: \underline{m} \sqcup O \xrightarrow{\cong} I \sqcup \underline{n}$	(iii) a bijection $\iota: \underline{m} \sqcup O \xrightarrow{\cong} I \sqcup \underline{n}$, where $I = \{(v, i) \mid v \in V, 1 \leq i \leq in(v)\}$ is the set of
fong-spivak-seven-sketches F01731	163	■ 0.793	$I = \{(v, i) \mid v \in V, 1 \leq i \leq in(v)\}$	$I = \{(v, i) \mid v \in V, 1 \leq i \leq in(v)\}$	(iii) a bijection $\iota: \underline{m} \sqcup O \xrightarrow{\cong} I \sqcup \underline{n}$, where $I = \{(v, i) \mid v \in V, 1 \leq i \leq in(v)\}$ is the set of
fong-spivak-seven-sketches F01732	163	■ 1.000	$O = \{(v, i) \mid v \in V, 1 \leq i \leq out(v)\}$	$O = \{(v, i) \mid v \in V, 1 \leq i \leq out(v)\}$	vertex inputs, and $O = \{(v, i) \mid v \in V, 1 \leq i \leq out(v)\}$ is the set of vertex outputs.
fong-spivak-seven-sketches F01733	163	■ 0.933	$e_{v,j}^{u,i}: u \rightarrow v$	$e_{v,j}^{u,i}: u \rightarrow v$	construct the graph with vertices V and with an arrow $e_{v,j}^{u,i}: u \rightarrow v$ for every $i, j \in \mathbb{N}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F01734	163	0.933	$i, j \in \mathbb{N}$	$i, j \in \mathbb{N}$	construct the graph with vertices V and with an arrow $e_{v,j}^{u,i}: u \rightarrow v$ for every $i, j \in \mathbb{N}$
fong-spivak-seven-sketches-F01735	163	1.000	$\iota(u, i) = (v, j)$	$\iota(u, i) = (v, j)$	such that $\iota(u, i) = (v, j)$; call it the <i>internal flow graph</i> . If the internal flow graph is
fong-spivak-seven-sketches-F01736	163	1.000	$R_{\{1\}} + R_{\{2\}}$	$R_1 + R_2$	³ The monoidal product $R_1 + R_2$ of relations $R_1 \subseteq \underline{m_1} \times \underline{n_1}$ and $R_2 \subseteq \underline{m_2} \times \underline{n_2}$ is given by $R_1 \sqcup R_2 \subseteq$
fong-spivak-seven-sketches-F01737	163	1.000	$R_{\{1\}} \subseteq \underline{m_{\{1\}}} \times \underline{n_{\{1\}}}$	$R_1 \subseteq \underline{m_1} \times \underline{n_1}$	³ The monoidal product $R_1 + R_2$ of relations $R_1 \subseteq \underline{m_1} \times \underline{n_1}$ and $R_2 \subseteq \underline{m_2} \times \underline{n_2}$ is given by $R_1 \sqcup R_2 \subseteq$
fong-spivak-seven-sketches-F01738	163	1.000	$R_{\{2\}} \subseteq \underline{m_{\{2\}}} \times \underline{n_{\{2\}}}$	$R_2 \subseteq \underline{m_2} \times \underline{n_2}$	³ The monoidal product $R_1 + R_2$ of relations $R_1 \subseteq \underline{m_1} \times \underline{n_1}$ and $R_2 \subseteq \underline{m_2} \times \underline{n_2}$ is given by $R_1 \sqcup R_2 \subseteq$
fong-spivak-seven-sketches-F01739	163	1.000	$R_{\{1\}} \sqcup R_{\{2\}} \subseteq$	$R_1 \sqcup R_2 \subseteq$	³ The monoidal product $R_1 + R_2$ of relations $R_1 \subseteq \underline{m_1} \times \underline{n_1}$ and $R_2 \subseteq \underline{m_2} \times \underline{n_2}$ is given by $R_1 \sqcup R_2 \subseteq$
fong-spivak-seven-sketches-F01740	163	0.420	$\left(\underline{m_{\{1\}}} \times \underline{n_{\{1\}}} \right) \sqcup \left(\underline{m_{\{2\}}} \times \underline{n_{\{2\}}} \right) \subseteq \left(\underline{m_{\{1\}}} \sqcup \underline{m_{\{2\}}} \right) \times \left(\underline{n_{\{1\}}} \sqcup \underline{n_{\{2\}}} \right)$	$(\underline{m_1} \times \underline{n_1}) \sqcup (\underline{m_2} \times \underline{n_2}) \subseteq (\underline{m_1} \sqcup \underline{m_2}) \times (\underline{n_1} \sqcup \underline{n_2})$	$(\underline{m_1} \times \underline{n_1}) \sqcup (\underline{m_2} \times \underline{n_2}) \subseteq (\underline{m_1} \sqcup \underline{m_2}) \times (\underline{n_1} \sqcup \underline{n_2})$.
fong-spivak-seven-sketches-F01741	164	0.999	$m=2$	$m = 2$	<i>Example 5.14.</i> Here is an example of a (2,3)-port graph, i.e. with $m = 2$ and $n = 3$:
fong-spivak-seven-sketches-F01742	164	0.999	$n=3$	$n = 3$	<i>Example 5.14.</i> Here is an example of a (2,3)-port graph, i.e. with $m = 2$ and $n = 3$:
fong-spivak-seven-sketches-F01743	164	1.000	$V = \{a, b, c\}$	$V = \{a, b, c\}$	outer box, and three on the right. The vertex set is $V = \{a, b, c\}$ and, for example
fong-spivak-seven-sketches-F01744	164	0.994	$\operatorname{in}(a) = 1$	$\operatorname{in}(a) = 1$	$\operatorname{in}(a) = 1$ and $\operatorname{out}(a) = 3$, so we draw one port on the left-hand side and three ports on
fong-spivak-seven-sketches-F01745	164	0.994	$\operatorname{out}(a) = 3$	$\operatorname{out}(a) = 3$	$\operatorname{in}(a) = 1$ and $\operatorname{out}(a) = 3$, so we draw one port on the left-hand side and three ports on
fong-spivak-seven-sketches-F01746	164	0.996	(m, n)	(m, n)	A category PG whose morphisms are port graphs. Given an (m, n) -port graph
fong-spivak-seven-sketches-F01747	164	0.179	$(V$	$(V$	$(V, \operatorname{in}, \operatorname{out}, \iota)$ and an (n, p) -port graph $(V', \operatorname{in}', \operatorname{out}', \iota')$, we may compose them to produce
fong-spivak-seven-sketches-F01748	164	0.179	$\iota)$	$\iota)$	$(V, \operatorname{in}, \operatorname{out}, \iota)$ and an (n, p) -port graph $(V', \operatorname{in}', \operatorname{out}', \iota')$, we may compose them to produce
fong-spivak-seven-sketches-F01749	164	0.179	(n, p)	(n, p)	$(V, \operatorname{in}, \operatorname{out}, \iota)$ and an (n, p) -port graph $(V', \operatorname{in}', \operatorname{out}', \iota')$, we may compose them to produce
fong-spivak-seven-sketches-F01750	164	0.179	$V^{\prime}$	V'	$(V, \operatorname{in}, \operatorname{out}, \iota)$ and an (n, p) -port graph $(V', \operatorname{in}', \operatorname{out}', \iota')$, we may compose them to produce
fong-spivak-seven-sketches-F01751	164	0.179	$\iota^{\prime}$	ι'	$(V, \operatorname{in}, \operatorname{out}, \iota)$ and an (n, p) -port graph $(V', \operatorname{in}', \operatorname{out}', \iota')$, we may compose them to produce
fong-spivak-seven-sketches-F01752	164	0.147	(m, p)	(m, p)	an (m, p) -port graph $(V \sqcup V', [\operatorname{in}, \operatorname{in}'], [\operatorname{out}, \operatorname{out}'], \iota'')$. Here $[\operatorname{in}, \operatorname{in}']$ denotes the function
fong-spivak-seven-sketches-F01753	164	0.147	$\left(V \sqcup V^{\prime}, [$	$(V \sqcup V', [$	an (m, p) -port graph $(V \sqcup V', [\operatorname{in}, \operatorname{in}'], [\operatorname{out}, \operatorname{out}'], \iota'')$. Here $[\operatorname{in}, \operatorname{in}']$ denotes the function
fong-spivak-seven-sketches-F01754	164	0.147	$], [$	$], [$	an (m, p) -port graph $(V \sqcup V', [\operatorname{in}, \operatorname{in}'], [\operatorname{out}, \operatorname{out}'], \iota'')$. Here $[\operatorname{in}, \operatorname{in}']$ denotes the function
fong-spivak-seven-sketches-F01755	164	0.147	$\left.], \iota^{\prime \prime} \right)$	$], \iota'')$	an (m, p) -port graph $(V \sqcup V', [\operatorname{in}, \operatorname{in}'], [\operatorname{out}, \operatorname{out}'], \iota'')$. Here $[\operatorname{in}, \operatorname{in}']$ denotes the function
fong-spivak-seven-sketches-F01756	164	0.777	$V \sqcup V^{\prime} \rightarrow \mathbb{N}$	$V \sqcup V' \rightarrow \mathbb{N}$	$V \sqcup V' \rightarrow \mathbb{N}$ which maps elements of V according to in , and elements of V' according
fong-spivak-seven-sketches-F01757	164	0.191	$\operatorname{in}^{\prime}$	in'	to in' , and similarly for $[\operatorname{out}, \operatorname{out}']$. The bijection $\iota'': \underline{m} \sqcup O \sqcup O' \rightarrow I \sqcup I' \sqcup \underline{p}$ is defined
fong-spivak-seven-sketches-F01758	164	0.191	$\iota^{\prime \prime}: \underline{m} \sqcup O \sqcup O' \rightarrow I \sqcup I' \sqcup \underline{p}$	$\iota'': \underline{m} \sqcup O \sqcup O' \rightarrow I \sqcup I' \sqcup \underline{p}$	to in' , and similarly for $[\operatorname{out}, \operatorname{out}']$. The bijection $\iota'': \underline{m} \sqcup O \sqcup O' \rightarrow I \sqcup I' \sqcup \underline{p}$ is defined

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_F01786	167	■ 1.000	$A=\{a\}$	$A = \{a\}$	1. What are the elements of the free monoid on the set $A = \{a\}$?
fong-spivak-seven-sketches_F01787	167	■ 1.000	$\{a\}$	$\{a\}$	2. Can you find a well-known monoid that is isomorphic to the free monoid on $\{a\}$?
fong-spivak-seven-sketches_F01788	167	■ 0.999	G, s, t	G, s, t	Definition 5.25. A <i>prop signature</i> is a tuple (G, s, t) , where G is a set and $s, t: G \rightarrow \mathbb{N}$
fong-spivak-seven-sketches_F01789	167	■ 0.999	$s, t: G \rightarrow \mathbb{N}$	$s, t: G \rightarrow \mathbb{N}$	Definition 5.25. A <i>prop signature</i> is a tuple (G, s, t) , where G is a set and $s, t: G \rightarrow \mathbb{N}$
fong-spivak-seven-sketches_F01790	167	■ 1.000	$g \in G$	$g \in G$	are functions; each element $g \in G$ is called a <i>generator</i> and $s(g), t(g) \in \mathbb{N}$ are called its
fong-spivak-seven-sketches_F01791	167	■ 1.000	$s(g), t(g) \in \mathbb{N}$	$s(g), t(g) \in \mathbb{N}$	are functions; each element $g \in G$ is called a <i>generator</i> and $s(g), t(g) \in \mathbb{N}$ are called its
fong-spivak-seven-sketches_F01792	167	■ 0.860	s, t	s, t	<i>in-arity and out-arity</i> . We often denote (G, s, t) simply by G , taking s, t to be implicit.
fong-spivak-seven-sketches_F01793	167	■ 0.577	$\Gamma = (V$	$\Gamma = (V$	A G -labeling of a port graph $\Gamma = (V, in, out, \iota)$ is a function $\ell: V \rightarrow G$ such that the
fong-spivak-seven-sketches_F01794	167	■ 0.577	$\ell: V \rightarrow G$	$\ell: V \rightarrow G$	A G -labeling of a port graph $\Gamma = (V, in, out, \iota)$ is a function $\ell: V \rightarrow G$ such that the
fong-spivak-seven-sketches_F01795	167	■ 0.999	$s(\ell(v)) = \operatorname{in}(v)$	$s(\ell(v)) = \operatorname{in}(v)$	arities agree: $s(\ell(v)) = \operatorname{in}(v)$ and $t(\ell(v)) = \operatorname{out}(v)$ for each $v \in V$.
fong-spivak-seven-sketches_F01796	167	■ 0.999	$t(\ell(v)) = \operatorname{out}(v)$	$t(\ell(v)) = \operatorname{out}(v)$	arities agree: $s(\ell(v)) = \operatorname{in}(v)$ and $t(\ell(v)) = \operatorname{out}(v)$ for each $v \in V$.
fong-spivak-seven-sketches_F01797	167	■ 1.000	$(m, n) \in \mathbb{N} \times \mathbb{N}$	$(m, n) \in \mathbb{N} \times \mathbb{N}$	⁴ The arity of a prop morphism is a pair $(m, n) \in \mathbb{N} \times \mathbb{N}$, where m is the number of inputs and n is the
fong-spivak-seven-sketches_F01798	168	■ 0.852	ℓ'	ℓ'	port graphs $\mathbf{PG}$ (see Eq. (5.17)); the labelings (the ℓ 's) are just carried along.
fong-spivak-seven-sketches_F01799	168	■ 1.000	ℓ	ℓ	the right. In wiring diagrams, we depict the labeling function $\ell: V \rightarrow G$ by using ℓ to
fong-spivak-seven-sketches_F01800	168	■ 1.000	$G = \{f: 1 \rightarrow 1, g: 2 \rightarrow 2, h: 2 \rightarrow 1\}$	$G = \{f: 1 \rightarrow 1, g: 2 \rightarrow 2, h: 2 \rightarrow 1\}$	with the same generator. For example, if $G = \{f: 1 \rightarrow 1, g: 2 \rightarrow 2, h: 2 \rightarrow 1\}$, then the
fong-spivak-seven-sketches_F01801	168	■ 0.987	$3 \rightarrow 2$	$3 \rightarrow 2$	following is a morphism $3 \rightarrow 2$ in $\mathbf{Free}(G)$:
fong-spivak-seven-sketches_F01802	168	■ 1.000	$V \rightarrow \varnothing$	$V \rightarrow \varnothing$	must have a labelling function of the form $V \rightarrow \varnothing$, and hence we must have $V = \varnothing$;
fong-spivak-seven-sketches_F01803	168	■ 1.000	$V = \varnothing$	$V = \varnothing$	must have a labelling function of the form $V \rightarrow \varnothing$, and hence we must have $V = \varnothing$;
fong-spivak-seven-sketches_F01804	168	■ 1.000	(n, m)	(n, m)	see Exercise 1.25. Thus the only morphisms (n, m) are those given by port graphs
fong-spivak-seven-sketches_F01805	168	■ 0.997	$(\varnothing, !, !, \sigma)$	$(\varnothing, !, !, \sigma)$	$(\varnothing, !, !, \sigma)$, where $\sigma: n \rightarrow m$ is a bijection.
fong-spivak-seven-sketches_F01806	168	■ 0.997	$\sigma: n \rightarrow m$	$\sigma: n \rightarrow m$	$(\varnothing, !, !, \sigma)$, where $\sigma: n \rightarrow m$ is a bijection.
fong-spivak-seven-sketches_F01807	168	■ 0.985	$(m, n) \in \mathbb{N}^2$	$(m, n) \in \mathbb{N}^2$	i.e. having one generating morphism for each $(m, n) \in \mathbb{N}^2$. Show that $\mathbf{Free}(G)$ is the
fong-spivak-seven-sketches_F01808	168	■ 0.886	$\operatorname{prop} \mathcal{C}$	$\operatorname{prop} \mathcal{C}$	for any prop $\mathcal{C}$, the prop functors $\mathbf{Free}(G) \rightarrow \mathcal{C}$ are in one-to-one correspondence with
fong-spivak-seven-sketches_F01809	168	■ 0.886	$\operatorname{Free}(G) \rightarrow \mathcal{C}$	$\operatorname{Free}(G) \rightarrow \mathcal{C}$	for any prop $\mathcal{C}$, the prop functors $\mathbf{Free}(G) \rightarrow \mathcal{C}$ are in one-to-one correspondence with
fong-spivak-seven-sketches_F01810	168	■ 1.000	$G \rightarrow \mathcal{C}$	$G \rightarrow \mathcal{C}$	functions $G \rightarrow \mathcal{C}$ that send each $g \in G$ to a morphism $s(g) \rightarrow t(g)$ in $\mathcal{C}$.
fong-spivak-seven-sketches_F01811	168	■ 1.000	$s(g) \rightarrow t(g)$	$s(g) \rightarrow t(g)$	functions $G \rightarrow \mathcal{C}$ that send each $g \in G$ to a morphism $s(g) \rightarrow t(g)$ in $\mathcal{C}$.
fong-spivak-seven-sketches_F01812	169	■ 0.995	$e: m \rightarrow n$	$e: m \rightarrow n$	G -generated prop expression, or simply expression $e: m \rightarrow n$, where $m, n \in \mathbb{N}$, inductively

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_F01813	169	■ 0.976	$\mathrm{id}_0: 0 \rightarrow 0$	$\mathrm{id}_0: 0 \rightarrow 0$	• The empty morphism $\mathrm{id}_0: 0 \rightarrow 0$, the identity morphism $\mathrm{id}_1: 1 \rightarrow 1$, and the
fong-spivak-seven-sketches_F01814	169	■ 0.976	$\mathrm{id}_1: 1 \rightarrow 1$	$\mathrm{id}_1: 1 \rightarrow 1$	• The empty morphism $\mathrm{id}_0: 0 \rightarrow 0$, the identity morphism $\mathrm{id}_1: 1 \rightarrow 1$, and the
fong-spivak-seven-sketches_F01815	169	■ 1.000	$\sigma: 2 \rightarrow 2$	$\sigma: 2 \rightarrow 2$	symmetry $\sigma: 2 \rightarrow 2$ are expressions. ⁵
fong-spivak-seven-sketches_F01816	169	■ 1.000	$g: s(g) \rightarrow t(g)$	$g: s(g) \rightarrow t(g)$	• the generators $g \in G$ are expressions $q: s(q) \rightarrow t(q)$.
fong-spivak-seven-sketches_F01817	169	■ 1.000	$\alpha: m \rightarrow n$	$\alpha: m \rightarrow n$	• if $\alpha: m \rightarrow n$ and $\beta: p \rightarrow q$ are expressions, then $\alpha + \beta: m + p \rightarrow n + q$ is an
fong-spivak-seven-sketches_F01818	169	■ 1.000	$\beta: p \rightarrow q$	$\beta: p \rightarrow q$	• if $\alpha: m \rightarrow n$ and $\beta: p \rightarrow q$ are expressions, then $\alpha + \beta: m + p \rightarrow n + q$ is an
fong-spivak-seven-sketches_F01819	169	■ 1.000	$\alpha + \beta: m + p \rightarrow n + q$	$\alpha + \beta: m + p \rightarrow n + q$	• if $\alpha: m \rightarrow n$ and $\beta: p \rightarrow q$ are expressions, then $\alpha + \beta: m + p \rightarrow n + q$ is an
fong-spivak-seven-sketches_F01820	169	■ 1.000	$\beta: n \rightarrow p$	$\beta: n \rightarrow p$	• if $\alpha: m \rightarrow n$ and $\beta: n \rightarrow p$ are expressions, then $\alpha \circ \beta: m \rightarrow p$ is an expression.
fong-spivak-seven-sketches_F01821	169	■ 1.000	$\alpha; \beta: m \rightarrow p$	$\alpha; \beta: m \rightarrow p$	• if $\alpha: m \rightarrow n$ and $\beta: n \rightarrow p$ are expressions, then $\alpha \circ \beta: m \rightarrow p$ is an expression.
fong-spivak-seven-sketches_F01822	169	■ 1.000	$\operatorname{Expr}(G)$	$\operatorname{Expr}(G)$	We write $\operatorname{Expr}(G)$ for the set of expressions in G . If $e: m \rightarrow n$ is an expression, we refer
fong-spivak-seven-sketches_F01823	169	■ 1.000	$f: 1 \rightarrow 1$	$f: 1 \rightarrow 1$	• $f: 1 \rightarrow 1$,
fong-spivak-seven-sketches_F01824	169	■ 0.963	$f: \mathrm{id}_1: 1 \rightarrow 1$	$f: \mathrm{id}_1: 1 \rightarrow 1$	• $f \circ \mathrm{id}_1: 1 \rightarrow 1$,
fong-spivak-seven-sketches_F01825	169	■ 0.997	$h + \mathrm{id}_1: 3 \rightarrow 2$	$h + \mathrm{id}_1: 3 \rightarrow 2$	• $h + \mathrm{id}_1: 3 \rightarrow 2$, and
fong-spivak-seven-sketches_F01826	169	■ 0.551	$\left(h + \mathrm{id}_1\right); \sigma; g; \sigma: 3 \rightarrow 2$	$(h + \mathrm{id}_1); \sigma; g; \sigma: 3 \rightarrow 2$	• $(h + \mathrm{id}_1) \circ \sigma \circ g \circ \sigma: 3 \rightarrow 2$
fong-spivak-seven-sketches_F01827	169	■ 0.981	$f; \mathrm{id}_1$	$f; \mathrm{id}_1$	phism. For example, we want to consider $f \circ \mathrm{id}_1$ and f to be the same morphism,
fong-spivak-seven-sketches_F01828	169	■ 0.708	$f; \mathrm{id}_1 = f$	$f; \mathrm{id}_1 = f$	since the unitality axiom for categories says $f \circ \mathrm{id}_1 = f$. Nonetheless, we only consider
fong-spivak-seven-sketches_F01829	169	■ 0.923	σ	σ	⁵ One can think of σ as the “swap” icon $\times: 2 \rightarrow 2$
fong-spivak-seven-sketches_F01830	169	■ 0.923	$\mathrm{X}: 2 \rightarrow 2$	$\mathrm{X}: 2 \rightarrow 2$	⁵ One can think of σ as the “swap” icon $\times: 2 \rightarrow 2$
fong-spivak-seven-sketches_F01831	170	■ 0.975	$\left(g + \mathrm{id}_1\right) \circ \left(\mathrm{id}_1 + \sigma\right) \circ \left(\mathrm{id}_1 + g\right) \circ \left(h + \mathrm{id}_1\right)$	$(g + \mathrm{id}_1) \circ (\mathrm{id}_1 + \sigma) \circ (\mathrm{id}_1 + g) \circ (h + \mathrm{id}_1)$	Here the result is $(g + \mathrm{id}_1) \circ (\mathrm{id}_1 + \sigma) \circ (\mathrm{id}_1 + g) \circ (h + \mathrm{id}_1)$.
fong-spivak-seven-sketches_F01832	170	■ 1.000	$G = \{f: 1 \rightarrow 1, g: 2 \rightarrow$	$G = \{f: 1 \rightarrow 1, g: 2 \rightarrow$	Exercise 5.32. Consider again the free prop on generators $G = \{f: 1 \rightarrow 1, g: 2 \rightarrow$
fong-spivak-seven-sketches_F01833	170	■ 0.285	$2, h: 2 \rightarrow 1\}$	$2, h: 2 \rightarrow 1\}$	$2, h: 2 \rightarrow 1\}$. Draw a picture of $(f + \mathrm{id}_1 + \mathrm{id}_1) \circ (\sigma + \mathrm{id}_1) \circ (\mathrm{id}_1 + h) \circ \sigma \circ g$, where $\sigma: 2 \rightarrow 2$
fong-spivak-seven-sketches_F01834	170	■ 0.285	$\left(f + \mathrm{id}_1 + \mathrm{id}_1\right) \circ \left(\sigma + \mathrm{id}_1\right) \circ \left(\mathrm{id}_1 + h\right) \circ \sigma \circ g$	$(f + \mathrm{id}_1 + \mathrm{id}_1) \circ (\sigma + \mathrm{id}_1) \circ (\mathrm{id}_1 + h) \circ \sigma \circ g$	$2, h: 2 \rightarrow 1\}$. Draw a picture of $(f + \mathrm{id}_1 + \mathrm{id}_1) \circ (\sigma + \mathrm{id}_1) \circ (\mathrm{id}_1 + h) \circ \sigma \circ g$, where $\sigma: 2 \rightarrow 2$
fong-spivak-seven-sketches_F01835	170	■ 0.817	G, s, t, E	G, s, t, E	Rough Definition 5.33. A presentation (G, s, t, E) for a prop is a set G , functions
fong-spivak-seven-sketches_F01836	170	■ 1.000	$E \subseteq \operatorname{Expr}(G) \times \operatorname{Expr}(G)$	$E \subseteq \operatorname{Expr}(G) \times \operatorname{Expr}(G)$	$s, t: G \rightarrow \mathbb{N}$, and a set $E \subseteq \operatorname{Expr}(G) \times \operatorname{Expr}(G)$ of pairs of G -generated prop expressions,
fong-spivak-seven-sketches_F01837	170	■ 1.000	e_1	e_1	such that e_1 and e_2 have the same arity for each $(e_1, e_2) \in E$. We refer to G as the set of

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_170 F01838	170	■ 1.000	$e_{\{2\}}$	e_2	such that e_1 and e_2 have the same arity for each $(e_1, e_2) \in E$. We refer to G as the set of
fong-spivak-seven-sketches_170 F01839	170	■ 1.000	$\left(e_{\{1\}}, e_{\{2\}}\right) \in E$	$(e_1, e_2) \in E$	such that e_1 and e_2 have the same arity for each $(e_1, e_2) \in E$. We refer to G as the set of
fong-spivak-seven-sketches_170 F01840	170	■ 0.936	(G, s, t, E)	(G, s, t, E)	The prop $\mathcal{G}$ presented by the presentation (G, s, t, E) is the prop whose morphisms
fong-spivak-seven-sketches_170 F01841	170	■ 1.000	$e_{\{1\}}=e_{\{2\}}$	$e_1 = e_2$	are elements in $\text{Expr}(G)$, quotiented by both the equations $e_1 = e_2$ where $(e_1, e_2) \in E$,
fong-spivak-seven-sketches_171 F01842	171	■ 1.000	$g \in G, f(g)$	$g \in G, f(g)$	• for all $g \in G$, $f(g)$ is a morphism $s(g) \rightarrow t(g)$, and
fong-spivak-seven-sketches_171 F01843	171	■ 1.000	$f\left(e_{\{1\}}\right)=f\left(e_{\{2\}}\right)$	$f(e_1) = f(e_2)$	• for all $(e_1, e_2) \in E$, we have that $f(e_1) = f(e_2)$ in $\mathcal{C}$, where $f(e)$ denotes the
fong-spivak-seven-sketches_171 F01844	171	■ 1.000	$f(e)$	$f(e)$	• for all $(e_1, e_2) \in E$, we have that $f(e_1) = f(e_2)$ in $\mathcal{C}$, where $f(e)$ denotes the
fong-spivak-seven-sketches_171 F01845	171	■ 0.686	(G, s, t)	(G, s, t)	Exercise 5.35. Is it the case that the free prop on generators (G, s, t) , defined in
fong-spivak-seven-sketches_171 F01846	171	■ 0.793	$G, s, t, \varnothing$	$G, s, t, \emptyset$	Definition 5.25, is the same thing as the prop presented by $(G, s, t, \emptyset)$, having no
fong-spivak-seven-sketches_171 F01847	171	■ 0.997	$R, 0, +, 1, *$	$R, 0, +, 1, *$	Definition 5.36. A rig is a tuple $(R, 0, +, 1, *)$, where R is a set, $0, 1 \in R$ are elements,
fong-spivak-seven-sketches_171 F01848	171	■ 0.997	$0, 1 \in R$	$0, 1 \in R$	Definition 5.36. A rig is a tuple $(R, 0, +, 1, *)$, where R is a set, $0, 1 \in R$ are elements,
fong-spivak-seven-sketches_171 F01849	171	■ 0.775	$R \times R \rightarrow R$	$R \times R \rightarrow R$	and $+, *: R \times R \rightarrow R$ are functions, such that
fong-spivak-seven-sketches_171 F01850	171	■ 0.924	$R, +, 0$	$R, +, 0$	(a) $(R, +, 0)$ is a commutative monoid,
fong-spivak-seven-sketches_171 F01851	171	■ 0.911	$R, *, 1$	$R, *, 1$	(b) $(R, *, 1)$ is a monoid, ⁸ and
fong-spivak-seven-sketches_171 F01852	171	■ 1.000	$a \cdot (b+c)=a \cdot b+a \cdot c$	$a \cdot (b+c) = a \cdot b + a \cdot c$	(c) $a \cdot (b+c) = a \cdot b + a \cdot c$ and $(a+b) \cdot c = a \cdot c + b \cdot c$ for all $a, b, c \in R$.
fong-spivak-seven-sketches_171 F01853	171	■ 1.000	$(a+b) \cdot c=a \cdot c+b \cdot c$	$(a+b) \cdot c = a \cdot c + b \cdot c$	(c) $a \cdot (b+c) = a \cdot b + a \cdot c$ and $(a+b) \cdot c = a \cdot c + b \cdot c$ for all $a, b, c \in R$.
fong-spivak-seven-sketches_171 F01854	171	■ 1.000	$a, b, c \in R$	$a, b, c \in R$	(c) $a \cdot (b+c) = a \cdot b + a \cdot c$ and $(a+b) \cdot c = a \cdot c + b \cdot c$ for all $a, b, c \in R$.
fong-spivak-seven-sketches_171 F01855	171	■ 1.000	$a \cdot 0=0=0 \cdot a$	$a \cdot 0 = 0 = 0 \cdot a$	(d) $a \cdot 0 = 0 = 0 \cdot a$ for all $a \in R$.
fong-spivak-seven-sketches_171 F01856	171	■ 1.000	$a \in R$	$a \in R$	(d) $a \cdot 0 = 0 = 0 \cdot a$ for all $a \in R$.
fong-spivak-seven-sketches_171 F01857	171	■ 0.800	$(\mathbb{N}, 0, +, 1, *)$	$(\mathbb{N}, 0, +, 1, *)$	Example 5.37. The natural numbers form a rig $(\mathbb{N}, 0, +, 1, *)$.
fong-spivak-seven-sketches_171 F01858	171	■ 0.952	$(R, *, 1)$	$(R, *, 1)$	⁸ Note that we did not demand that $(R, *, 1)$ be commutative; we will see a naturally-arising example
fong-spivak-seven-sketches_172 F01859	172	■ 0.737	$V, 0, \vee, I, \otimes$	$V, 0, \vee, I, \otimes$	Example 5.39. Any quantale $\mathcal{V} = (V, \leq, I, \otimes)$ determines a rig $(V, 0, \vee, I, \otimes)$, where $0 =$
fong-spivak-seven-sketches_172 F01860	172	■ 0.737	$0=$	$0 =$	Example 5.39. Any quantale $\mathcal{V} = (V, \leq, I, \otimes)$ determines a rig $(V, 0, \vee, I, \otimes)$, where $0 =$
fong-spivak-seven-sketches_172 F01861	172	■ 0.971	$\bigvee \varnothing$	$\bigvee \emptyset$	$\bigvee \emptyset$ is the empty join. See Definition 2.79.
fong-spivak-seven-sketches_172 F01862	172	■ 1.000	$\operatorname{Mat}_{\{n\}}(R)$	$\text{Mat}_n(R)$	Example 5.40. If R is a rig and $n \in \mathbb{N}$ is any natural number, then the set $\text{Mat}_n(R)$ of
fong-spivak-seven-sketches_172 F01863	172	■ 1.000	$(n \times n)$	$(n \times n)$	$(n \times n)$ -matrices in R forms a rig. A matrix $M \in \text{Mat}_n(R)$ is a function $M: \underline{n} \times \underline{n} \rightarrow$
fong-spivak-seven-sketches_172 F01864	172	■ 1.000	$M \in \operatorname{Mat}_{\{n\}}(R)$	$M \in \text{Mat}_n(R)$	$(n \times n)$ -matrices in R forms a rig. A matrix $M \in \text{Mat}_n(R)$ is a function $M: \underline{n} \times \underline{n} \rightarrow$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_F01865	172	1.000	$M: \underline{n} \times \underline{n} \rightarrow$	$M: \underline{n} \times \underline{n} \rightarrow$	$(n \times n)$ -matrices in R forms a rig. A matrix $M \in \text{Mat}_n(R)$ is a function $M: \underline{n} \times \underline{n} \rightarrow$
fong-spivak-seven-sketches_F01866	172	1.000	$M+N$	$M + N$	R . Addition $M + N$ of matrices is given by $(M + N)(i, j) := M(i, j) + N(i, j)$ and
fong-spivak-seven-sketches_F01867	172	1.000	$(M+N)(i, j) := M(i, j) + N(i, j)$	$(M + N)(i, j) := M(i, j) + N(i, j)$	R . Addition $M + N$ of matrices is given by $(M + N)(i, j) := M(i, j) + N(i, j)$ and
fong-spivak-seven-sketches_F01868	172	0.977	$(M * N)(i, j) := \sum_{k \in \underline{n}} M(i, k) * N(k, j)$	$(M * N)(i, j) := \sum_{k \in \underline{n}} M(i, k) * N(k, j)$	multiplication $M * N$ is given by $(M * N)(i, j) := \sum_{k \in \underline{n}} M(i, k) * N(k, j)$. The 0-matrix is
fong-spivak-seven-sketches_F01869	172	1.000	$0(i, j) := 0$	$0(i, j) := 0$	$0(i, j) := 0$ for all $i, j \in \underline{n}$. Note that $\text{Mat}_n(R)$ is generally not commutative.
fong-spivak-seven-sketches_F01870	172	1.000	$i, j \in \underline{n}$	$i, j \in \underline{n}$	$0(i, j) := 0$ for all $i, j \in \underline{n}$. Note that $\text{Mat}_n(R)$ is generally not commutative.
fong-spivak-seven-sketches_F01871	172	1.000	$1 \in \text{Mat}_n(R)$	$1 \in \text{Mat}_n(R)$	its multiplicative identity $1 \in \text{Mat}_n(R)$?
fong-spivak-seven-sketches_F01872	172	1.000	$\text{Mat}_n(\mathbb{N})$	$\text{Mat}_n(\mathbb{N})$	that $\text{Mat}_n(\mathbb{N})$ is not commutative, where $\mathbb{N}$ is as in Example 5.37.
fong-spivak-seven-sketches_F01873	172	0.993	$\mathbb{R}, 0, +, 1, *$	$\mathbb{R}, 0, +, 1, *$	Example 5.42. Any ring forms a rig. In particular, the real numbers $(\mathbb{R}, 0, +, 1, *)$ are a
fong-spivak-seven-sketches_F01874	172	1.000	$M=R$	$M = R$	R -module M . We will not discuss this here, keeping to the simpler requirement that $M = R$.
fong-spivak-seven-sketches_F01875	173	1.000	$R=\mathbb{N}$	$R = \mathbb{N}$	For example, consider the rig $R = \mathbb{N}$ from Example 5.37, where the scalars are the
fong-spivak-seven-sketches_F01876	173	1.000	$a=15 \ x$	$a = 15x$	$a = 15x$ and $b = 3x + 21y$.
fong-spivak-seven-sketches_F01877	173	1.000	$b=3 \ x+21 \ y$	$b = 3x + 21y$	$a = 15x$ and $b = 3x + 21y$.
fong-spivak-seven-sketches_F01878	174	0.999	$15 \ x$	$15x$	finally amplified by 3. Thus, if x is the first input signal, then this contributes $15x$ to the
fong-spivak-seven-sketches_F01879	174	1.000	$3 \ x=x+2 \ x$	$3x = x + 2x$	there are two paths to it from each input. The top input thus contributes $3x = x + 2x$,
fong-spivak-seven-sketches_F01880	174	1.000	$21 \ y$	$21y$	$21y$.
fong-spivak-seven-sketches_F01881	174	1.000	(x, y, z)	(x, y, z)	the movement of a particle. For example, consider a particle whose position (x, y, z) in
fong-spivak-seven-sketches_F01882	175	1.000	$s, t: G_R \rightarrow \mathbb{N}$	$s, t: G_R \rightarrow \mathbb{N}$	and let $s, t: G_R \rightarrow \mathbb{N}$ be given by the number of dangling wires on the left and right of
fong-spivak-seven-sketches_F01883	175	0.686	$\text{Free}(G_R)$	$\text{Free}(G_R)$	prop $\text{Free}(G_R)$ on this set G_R of generators. We define $\text{SFG}_R := \text{Free}(G_R)$.
fong-spivak-seven-sketches_F01884	175	0.686	G_R	G_R	prop $\text{Free}(G_R)$ on this set G_R of generators. We define $\text{SFG}_R := \text{Free}(G_R)$.
fong-spivak-seven-sketches_F01885	175	0.686	$\text{Free}(G_R)$	$\text{SFG}_R := \text{Free}(G_R)$	prop $\text{Free}(G_R)$ on this set G_R of generators. We define $\text{SFG}_R := \text{Free}(G_R)$.
fong-spivak-seven-sketches_F01886	175	0.896	$\text{Free}(G_R)$	$\text{Free}(G_R)$	port graphs, morphisms in $\text{Free}(G_R)$ should be drawn slightly differently. For example,
fong-spivak-seven-sketches_F01887	176	0.978	$(m \times n)$	$(m \times n)$	An $(m \times n)$ -matrix M with values in R is a function $M: (\underline{m} \times \underline{n}) \rightarrow R$. Given an
fong-spivak-seven-sketches_F01888	176	0.978	$M: (\underline{m} \times \underline{n}) \rightarrow R$	$M: (\underline{m} \times \underline{n}) \rightarrow R$	An $(m \times n)$ -matrix M with values in R is a function $M: (\underline{m} \times \underline{n}) \rightarrow R$. Given an
fong-spivak-seven-sketches_F01889	176	0.815	$(n \times p)$	$(n \times p)$	$(m \times n)$ -matrix M and an $(n \times p)$ -matrix N , their <i>composite</i> is the $(m \times p)$ -matrix $M \circ N$
fong-spivak-seven-sketches_F01890	176	0.815	$(m \times p)$	$(m \times p)$	$(m \times n)$ -matrix M and an $(n \times p)$ -matrix N , their <i>composite</i> is the $(m \times p)$ -matrix $M \circ N$
fong-spivak-seven-sketches_F01891	176	0.815	$M \circ N$	$M \circ N$	$(m \times n)$ -matrix M and an $(n \times p)$ -matrix N , their <i>composite</i> is the $(m \times p)$ -matrix $M \circ N$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_176 F01892	176	■ 0.958	$a \in \underline{m}$	$a \in \underline{m}$	defined as follows for any $a \in \underline{m}$ and $c \in \underline{p}$:
fong-spivak-seven-sketches_176 F01893	176	■ 0.958	$c \in \underline{p}$	$c \in \underline{p}$	defined as follows for any $a \in \underline{m}$ and $c \in \underline{p}$:
fong-spivak-seven-sketches_176 F01894	176	■ 0.871	$\sum_{b \in \underline{n}}$	$\sum_{b \in \underline{n}}$	Here the $\sum_{b \in \underline{n}}$ just means repeated addition (using the rig R 's $+$ operation), as usual.
fong-spivak-seven-sketches_176 F01895	176	■ 1.000	$A \ v$	Av	by multiplication in the order Av , where v is a column vector. In keeping with our
fong-spivak-seven-sketches_176 F01896	176	■ 1.000	$v \ ; \ A$	$v; A$	composition convention, we use the opposite order, $v \circ A$, where v is a row vector. See
fong-spivak-seven-sketches_176 F01897	176	■ 1.000	$A: m \rightarrow n$	$A: m \rightarrow n$	monoidal product is given by the direct sum of matrices: given matrices $A: m \rightarrow n$
fong-spivak-seven-sketches_176 F01898	176	■ 1.000	$b: p \rightarrow q$	$b: p \rightarrow q$	and $b: p \rightarrow q$, we define $A + B: m + p \rightarrow n + q$ to be the block matrix
fong-spivak-seven-sketches_176 F01899	176	■ 1.000	$A+B: m+p \rightarrow n+q$	$A + B: m + p \rightarrow n + q$	and $b: p \rightarrow q$, we define $A + B: m + p \rightarrow n + q$ to be the block matrix
fong-spivak-seven-sketches_176 F01900	176	■ 1.000	$m \times q$	$m \times q$	where each 0 represents a matrix of zeros of the appropriate dimension ($m \times q$ and
fong-spivak-seven-sketches_176 F01901	176	■ 1.000	$n \times p$	$n \times p$	$n \times p$). We refer to any combination of multiplication and direct sum as a <i>interconnection</i>
fong-spivak-seven-sketches_176 F01902	176	■ 1.000	$A+B$	$A + B$	What is the direct sum matrix $A + B$?
fong-spivak-seven-sketches_177 F01903	177	■ 1.000	1×1	1×1	matrix. For example, we interpret amplification by $a \in R$ as the 1×1 matrix $(a): 1 \rightarrow 1$:
fong-spivak-seven-sketches_177 F01904	177	■ 1.000	$(a): 1 \rightarrow 1$	$(a): 1 \rightarrow 1$	matrix. For example, we interpret amplification by $a \in R$ as the 1×1 matrix $(a): 1 \rightarrow 1$:
fong-spivak-seven-sketches_177 F01905	177	■ 1.000	$x \in R$	$x \in R$	it is an operation that takes an input $x \in R$ and returns $a * x$. Similarly, we can interpret
fong-spivak-seven-sketches_177 F01906	177	■ 1.000	$a * x$	$a * x$	it is an operation that takes an input $x \in R$ and returns $a * x$. Similarly, we can interpret
fong-spivak-seven-sketches_177 F01907	177	■ 0.897	$\supset$	$\supset$	$\supset$ as the 2×1 matrix $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$: it is an operation that takes a row vector consisting of two
fong-spivak-seven-sketches_177 F01908	177	■ 0.897	2×1	2×1	$\supset$ as the 2×1 matrix $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$: it is an operation that takes a row vector consisting of two
fong-spivak-seven-sketches_177 F01909	177	■ 0.897	$\binom{1}{1}$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$\supset$ as the 2×1 matrix $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$: it is an operation that takes a row vector consisting of two
fong-spivak-seven-sketches_177 F01910	177	■ 1.000	$x+y$	$x + y$	inputs, x and y , and returns $x + y$. Here is a table showing the interpretation of each
fong-spivak-seven-sketches_221 F01911	221	■ 0.984	$1 \rightarrow 1$	$1 \rightarrow 1$	to get another morphism $\underline{1} \rightarrow \underline{1}$. How do we do this? There are two parts: to get
fong-spivak-seven-sketches_313 F01912	313	■ 0.395	$2 \rightarrow 1$	$2 \rightarrow 1$	2. The behavior $B(\supset)$ of the reversed copy icon, $\supset: 2 \rightarrow 1$ is the relation $\{(x, y, z) \in R^3 \mid x = y =$
fong-spivak-seven-sketches_177 F01915	177	■ 1.000	$0 \times n$	$0 \times n$	dimensions. In linear algebra it is unusual to consider matrices of the form $0 \times n$ or $n \times 0$
fong-spivak-seven-sketches_177 F01916	177	■ 1.000	$n \times 0$	$n \times 0$	dimensions. In linear algebra it is unusual to consider matrices of the form $0 \times n$ or $n \times 0$
fong-spivak-seven-sketches_177 F01917	177	■ 0.749	0×3	0×3	you can multiply a 0×3 matrix by a $3 \times n$ matrix for any n , but you can not multiply it
fong-spivak-seven-sketches_177 F01918	177	■ 0.749	$3 \times n$	$3 \times n$	you can multiply a 0×3 matrix by a $3 \times n$ matrix for any n , but you can not multiply it
fong-spivak-seven-sketches_177 F01919	177	■ 1.000	$2 \times n$	$2 \times n$	by a $2 \times n$ matrix.
fong-spivak-seven-sketches_178 F01920	178	■ 0.999	$S: \mathbf{SFG}_{\{R\}} \rightarrow \mathbf{Mat}(R)$	$S: \mathbf{SFG}_R \rightarrow \mathbf{Mat}(R)$	Theorem 5.53. There is a prop functor $S: \mathbf{SFG}_R \rightarrow \mathbf{Mat}(R)$ that sends the generators

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F01921	178	■ 1.000	$S(g)$	$S(g)$	We have now constructed a matrix $S(g)$ from any signal flow graph g . But how can
fong-spivak-seven-sketches-F01922	178	■ 1.000	$m \times n$	$m \times n$	and for the generators in Definition 5.45, the associated matrix has dimension $m \times n$,
fong-spivak-seven-sketches-F01923	178	■ 1.000	(i, j)	(i, j)	where m is the number of inputs and n the number of outputs, with (i, j) th entry
fong-spivak-seven-sketches-F01924	178	■ 1.000	$\mathrm{id}_0: 0 \rightarrow 0, \mathrm{id}_1: 1 \rightarrow 1, \sigma: 2 \rightarrow 2$	$\mathrm{id}_0: 0 \rightarrow 0, \mathrm{id}_1: 1 \rightarrow 1, \sigma: 2 \rightarrow 2$	built from the morphisms $\mathrm{id}_0: 0 \rightarrow 0, \mathrm{id}_1: 1 \rightarrow 1, \sigma: 2 \rightarrow 2$, and the generators in G_R ,
fong-spivak-seven-sketches-F01925	178	■ 1.000	$(\alpha + \beta): (m + p) \rightarrow (n + q)$	$(\alpha + \beta): (m + p) \rightarrow (n + q)$	• if $\alpha: m \rightarrow n$ and $\beta: p \rightarrow q$ are expressions, then $(\alpha + \beta): (m + p) \rightarrow (n + q)$ is an
fong-spivak-seven-sketches-F01926	178	■ 0.999	$\mathrm{id}_0, \mathrm{id}_1$	$\mathrm{id}_0, \mathrm{id}_1$	is true when g is equal to $\mathrm{id}_0, \mathrm{id}_1$, and σ .
fong-spivak-seven-sketches-F01927	178	■ 0.682	$0 \rightarrow 0$	$0 \rightarrow 0$	$(): 0 \rightarrow 0$. The morphisms id_1, σ , and $a \in R$ map to the identity matrix, the swap
fong-spivak-seven-sketches-F01928	178	■ 0.682	id_1, σ	id_1, σ	$(): 0 \rightarrow 0$. The morphisms id_1, σ , and $a \in R$ map to the identity matrix, the swap
fong-spivak-seven-sketches-F01929	178	■ 0.712	(a)	(a)	matrix, and the scalar matrix (a) respectively:
fong-spivak-seven-sketches-F01930	178	■ 0.848	$n = p$	$n = p$	holds for (i) $\alpha \circ \beta$ (when $n = p$) and for (ii) $\alpha + \beta$ (in general).
fong-spivak-seven-sketches-F01931	178	■ 0.848	$\alpha + \beta$	$\alpha + \beta$	holds for (i) $\alpha \circ \beta$ (when $n = p$) and for (ii) $\alpha + \beta$ (in general).
fong-spivak-seven-sketches-F01932	179	■ 1.000	$\beta: n \rightarrow q$	$\beta: n \rightarrow q$	Here $\alpha: m \rightarrow n$ and $\beta: n \rightarrow q$ are signal flow graphs, assumed to obey the proposition.
fong-spivak-seven-sketches-F01933	179	■ 0.994	$j \in \underline{n}$	$j \in \underline{n}$	k —of the amplification along that path. So let’s also fix some $j \in \underline{n}$, and consider paths
fong-spivak-seven-sketches-F01934	179	■ 0.779	$\alpha /$	$\alpha /$	some j th output of α /input of β , the amplification that i contributes to k is
fong-spivak-seven-sketches-F01935	179	■ 0.606	$S(\alpha); S(\beta)$	$S(\alpha); S(\beta)$	This is exactly the formula for matrix multiplication, which is composition $S(\alpha) \circ S(\beta)$
fong-spivak-seven-sketches-F01936	179	■ 0.358	$\mathrm{Mat}(R)$	$\mathrm{Mat}(R)$	in the prop $\mathbf{Mat}(R)$; see Definition 5.50. So $\alpha \circ \beta$ obeys the proposition when α and β
fong-spivak-seven-sketches-F01937	179	■ 0.684	$m + i$	$m + i$	of $\alpha + \beta$, the i th input of β is the $(m + i)$ th output of $\alpha + \beta$, and the j th output of β is the
fong-spivak-seven-sketches-F01938	179	■ 0.999	$(n + j)$	$(n + j)$	$(n + j)$ th output of $\alpha + \beta$. This means that the matrix with (i, j) th entry describing the
fong-spivak-seven-sketches-F01939	179	■ 1.000	$S(\alpha) + S(\beta) = S(\alpha + \beta)$	$S(\alpha) + S(\beta) = S(\alpha + \beta)$	amplification of the i th input that contributes to the j th output is $S(\alpha) + S(\beta) = S(\alpha + \beta)$,
fong-spivak-seven-sketches-F01940	181	■ 0.989	$M \in \mathrm{Mat}(R)$	$M \in \mathrm{Mat}(R)$	Proposition 5.56. Given any matrix $M \in \mathbf{Mat}(R)$, there exists a signal flow graph
fong-spivak-seven-sketches-F01941	181	■ 1.000	$g \in \mathbf{SFG}_R$	$g \in \mathbf{SFG}_R$	$g \in \mathbf{SFG}_R$ such that such that $S(g) = M$.
fong-spivak-seven-sketches-F01942	181	■ 1.000	$S(g) = M$	$S(g) = M$	$g \in \mathbf{SFG}_R$ such that such that $S(g) = M$.
fong-spivak-seven-sketches-F01943	181	■ 0.890	$M \in \mathbf{Mat}(R)$	$M \in \mathbf{Mat}(R)$	<i>Proof sketch.</i> Let $M \in \mathbf{Mat}(R)$ be an $(m \times n)$ -matrix. We want a signal flow graph g
fong-spivak-seven-sketches-F01944	181	■ 1.000	$S(g)(i, j)$	$S(g)(i, j)$	such that $S(g) = M$. In particular, to compute $S(g)(i, j)$, we know that we can simply
fong-spivak-seven-sketches-F01945	181	■ 1.000	$M(i, j)$	$M(i, j)$	and that this path has exactly one scalar multiplication icon, namely $M(i, j)$.
fong-spivak-seven-sketches-F01946	181	■ 0.939	$\left(\begin{array}{cc} a & b \\ c & d \end{array} \right)$	$\left(\begin{array}{cc} a & b \\ c & d \end{array} \right)$	from just considering the case of 2×2 -matrices. Suppose M is the 2×2 -matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F01947	181	■ 1.000	$\left(\begin{smallmatrix} 0 \\ 1 \\ 2 \end{smallmatrix}\right)$	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$	1. $\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$
fong-spivak-seven-sketches F01948	181	■ 1.000	$\left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	2. $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$
fong-spivak-seven-sketches F01949	181	■ 1.000	$\left(\begin{smallmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{smallmatrix}\right)$	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$	3. $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$
fong-spivak-seven-sketches F01950	182	■ 1.000	$\mathbf{SFG}_R$	$\mathbf{SFG}_R$	the same as the set of generators for $\mathbf{SFG}_R$; see Definition 5.45.
fong-spivak-seven-sketches F01951	182	■ 1.000	$a, b \in R$	$a, b \in R$	We have the following equations for any $a, b \in R$:
fong-spivak-seven-sketches F01952	183	■ 0.997	$\binom{0}{6}$	$\begin{pmatrix} 0 \\ 6 \end{pmatrix}$	Example 5.61. Both of the signal flow graphs below represent the same matrix, $\begin{pmatrix} 0 \\ 6 \end{pmatrix}$:
fong-spivak-seven-sketches F01953	183	■ 0.549	$R = (\mathbb{N}, 0, +, 1, *)$	$R = (\mathbb{N}, 0, +, 1, *)$	1. Let $R = (\mathbb{N}, 0, +, 1, *)$. By examining the presentation of $\mathbf{Mat}(R)$ in Theorem 5.60,
fong-spivak-seven-sketches F01954	183	■ 1.000	$R = \mathbb{N} / 3\mathbb{N}$	$R = \mathbb{N} / 3\mathbb{N}$	2. Now suppose the rig is $R = \mathbb{N} / 3\mathbb{N}$; if you do not know what this means, just
fong-spivak-seven-sketches F01955	184	■ 0.931	M, μ, η	M, μ, η	Definition 5.65. A monoid object (M, μ, η) in a symmetric monoidal category $(\mathcal{C}, I, \otimes)$ is
fong-spivak-seven-sketches F01956	184	■ 0.931	$\mathcal{C}, I, \otimes$	$\mathcal{C}, I, \otimes$	Definition 5.65. A monoid object (M, μ, η) in a symmetric monoidal category $(\mathcal{C}, I, \otimes)$ is
fong-spivak-seven-sketches F01957	184	■ 1.000	$\mu: M \otimes M \rightarrow M$	$\mu: M \otimes M \rightarrow M$	an object M of $\mathcal{C}$ together with morphisms $\mu: M \otimes M \rightarrow M$ and $\eta: I \rightarrow M$ such that
fong-spivak-seven-sketches F01958	184	■ 1.000	$\eta: I \rightarrow M$	$\eta: I \rightarrow M$	an object M of $\mathcal{C}$ together with morphisms $\mu: M \otimes M \rightarrow M$ and $\eta: I \rightarrow M$ such that
fong-spivak-seven-sketches F01959	184	■ 0.939	$(\mu \otimes \mathrm{id}) \circ \mu = (\mathrm{id} \otimes \mu) \circ \mu$	$(\mu \otimes \mathrm{id}); \mu = (\mathrm{id} \otimes \mu); \mu$	(a) $\mu \otimes \mathrm{id} \circ \mu = (\mathrm{id} \otimes \mu) \circ \mu$ and
fong-spivak-seven-sketches F01960	184	■ 1.000	$(\eta \otimes \mathrm{id}) \circ \mu = \mathrm{id} = (\mathrm{id} \otimes \eta) \circ \mu$	$(\eta \otimes \mathrm{id}); \mu = \mathrm{id} = (\mathrm{id} \otimes \eta); \mu$	(b) $\eta \otimes \mathrm{id} \circ \mu = \mathrm{id} = (\mathrm{id} \otimes \eta) \circ \mu$.
fong-spivak-seven-sketches F01961	184	■ 0.959	$\sigma_{M,M}; \mu = \mu$	$\sigma_{M,M}; \mu = \mu$	(c) $\sigma_{M,M} \circ \mu = \mu$.
fong-spivak-seven-sketches F01962	184	■ 1.000	$\sigma_{M,M}$	$\sigma_{M,M}$	where $\sigma_{M,M}$ is the swap map on M in $\mathcal{C}$. We often denote it simply by σ .
fong-spivak-seven-sketches F01963	184	■ 1.000	$\mu: M \times M \rightarrow M$	$\mu: M \times M \rightarrow M$	Example 2.6; see also Example 3.13. That is, it is a set M , a function $\mu: M \times M \rightarrow M$,
fong-spivak-seven-sketches F01964	184	■ 0.935	$\eta(1) \in M$	$\eta(1) \in M$	which we denote by infix notation $*$, and an element $\eta(1) \in M$, which we denote by e ,
fong-spivak-seven-sketches F01965	184	■ 1.000	$(a * b) * c = a * (b * c)$	$(a * b) * c = a * (b * c)$	satisfying $a * b * c = a * (b * c)$ and $a * e = a = e * a$.
fong-spivak-seven-sketches F01966	184	■ 1.000	$a * e = a = e * a$	$a * e = a = e * a$	satisfying $(a * b) * c = a * (b * c)$ and $a * e = a = e * a$.
fong-spivak-seven-sketches F01967	184	■ 0.784	$\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$	$\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$	1. Show that if $\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is defined by $\mu(a, b) = a * b$ and if $\eta \in \mathbb{R}$ is defined to
fong-spivak-seven-sketches F01968	184	■ 0.784	$\mu(a, b) = a * b$	$\mu(a, b) = a * b$	1. Show that if $\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is defined by $\mu(a, b) = a * b$ and if $\eta \in \mathbb{R}$ is defined to
fong-spivak-seven-sketches F01969	184	■ 0.784	$\eta \in \mathbb{R}$	$\eta \in \mathbb{R}$	1. Show that if $\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is defined by $\mu(a, b) = a * b$ and if $\eta \in \mathbb{R}$ is defined to

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_F01970	184	■ 1.000	$\eta=1$	$\eta = 1$	be $\eta = 1$, then $(\mathbb{R}, *, 1)$ satisfies all three conditions of Definition 5.65.
fong-spivak-seven-sketches_F01971	184	■ 1.000	$(\mathbb{R}, *, 1)$	$(\mathbb{R}, *, 1)$	be $\eta = 1$, then $(\mathbb{R}, *, 1)$ satisfies all three conditions of Definition 5.65.
fong-spivak-seven-sketches_F01972	184	■ 0.995	$\mu(a, b)=a+b$	$\mu(a, b) = a + b$	2. Show that if $\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is defined by $\mu(a, b) = a + b$ and if $\eta \in \mathbb{R}$ is defined to
fong-spivak-seven-sketches_F01973	184	■ 0.985	$\eta=0$	$\eta = 0$	be $\eta = 0$, then $(\mathbb{R}, +, 0)$ satisfies all three conditions of Definition 5.65.
fong-spivak-seven-sketches_F01974	184	■ 0.985	$(\mathbb{R}, +, 0)$	$(\mathbb{R}, +, 0)$	be $\eta = 0$, then $(\mathbb{R}, +, 0)$ satisfies all three conditions of Definition 5.65.
fong-spivak-seven-sketches_F01975	184	■ 0.457	$\mu=>-$	$\mu = > -$	Example 5.68. Graphically, we can depict $\mu = >-$ and $\eta = -$. Then axioms (a), (b), and
fong-spivak-seven-sketches_F01976	184	■ 0.457	$\eta=\sigma$	$\eta = \sigma$	Example 5.68. Graphically, we can depict $\mu = >-$ and $\eta = -$. Then axioms (a), (b), and
fong-spivak-seven-sketches_F01977	184	■ 0.315	(R)	(R)	following: the triple $(1, >, -)$ is a commutative monoid object in the prop $\mathbf{Mat}(R)$.
fong-spivak-seven-sketches_F01978	185	■ 0.665	$U: \operatorname{Mat}(R) \rightarrow$	$U : \operatorname{Mat}(R) \rightarrow$	Exercise 5.69. For any rig R , there is a functor $U: \mathbf{Mat}(R) \rightarrow \mathbf{Set}$, sending the object
fong-spivak-seven-sketches_F01979	185	■ 0.999	$R^{\{n\}}$	R^n	$n \in \mathbb{N}$ to the set R^n , and sending a morphism (matrix) $M: m \rightarrow n$ to the function
fong-spivak-seven-sketches_F01980	185	■ 0.999	$M: m \rightarrow n$	$M : m \rightarrow n$	$n \in \mathbb{N}$ to the set R^n , and sending a morphism (matrix) $M: m \rightarrow n$ to the function
fong-spivak-seven-sketches_F01981	185	■ 1.000	$R^{\{m\}} \rightarrow R^{\{n\}}$	$R^m \rightarrow R^n$	$R^m \rightarrow R^n$ given by vector-matrix multiplication.
fong-spivak-seven-sketches_F01982	185	■ 0.881	$\{1\}$	$\{1\}$	is $\{1\}$, a set with one element, and the monoidal product is $\times$ (see Example 4.49).
fong-spivak-seven-sketches_F01983	185	■ 0.828	(M, μ, η)	(M, μ, η)	2. Show that if (M, μ, η) is a monoid object in $\mathbf{Mat}(R)$ then $(U(M), U(\mu), U(\eta))$ is
fong-spivak-seven-sketches_F01984	185	■ 0.828	$(U(M), U(\mu), U(\eta))$	$(U(M), U(\mu), U(\eta))$	2. Show that if (M, μ, η) is a monoid object in $\mathbf{Mat}(R)$ then $(U(M), U(\mu), U(\eta))$ is
fong-spivak-seven-sketches_F01985	185	■ 0.960	$\operatorname{prop} \operatorname{Mat}(R)$	$\operatorname{prop} \operatorname{Mat}(R)$	in the prop $\mathbf{Mat}(R)$. If $R = \mathbb{R}$ is the rig of real numbers, this means that we have
fong-spivak-seven-sketches_F01986	185	■ 0.960	$R=\mathbb{R}$	$R = \mathbb{R}$	in the prop $\mathbf{Mat}(R)$. If $R = \mathbb{R}$ is the rig of real numbers, this means that we have
fong-spivak-seven-sketches_F01987	185	■ 0.212	$\mathbf{Mat}(R)$	$\mathbf{Mat}(R)$	Example 5.70. The triple $(1, \prec, \dashv)$ in $\mathbf{Mat}(R)$ forms a commutative monoid object in
fong-spivak-seven-sketches_F01988	185	■ 0.157	$\operatorname{Mat}(R)^{\operatorname{op}}$	$\operatorname{Mat}(R)^{\operatorname{op}}$	$\mathbf{Mat}(R)^{\operatorname{op}}$. We hence also say that $(1, \prec, \dashv)$ forms a co-commutative comonoid object in
fong-spivak-seven-sketches_F01989	185	■ 0.245	$\varnothing, \mathbb{N}$	$\varnothing, \mathbb{N}$	$(\mathbf{CMon}, \otimes, \mathbb{N})$ of commutative monoids with tensor product.
fong-spivak-seven-sketches_F01990	185	■ 1.000	$\mu: 2 \rightarrow 1$	$\mu : 2 \rightarrow 1$	Remark 5.74. If we present a prop $\mathcal{M}$ using two generators $\mu: 2 \rightarrow 1$ and $\eta: 0 \rightarrow 1$,
fong-spivak-seven-sketches_F01991	185	■ 1.000	$\eta: 0 \rightarrow 1$	$\eta : 0 \rightarrow 1$	Remark 5.74. If we present a prop $\mathcal{M}$ using two generators $\mu: 2 \rightarrow 1$ and $\eta: 0 \rightarrow 1$,
fong-spivak-seven-sketches_F01992	185	■ 1.000	$\mathcal{M} \rightarrow \mathcal{C}$	$\mathcal{M} \rightarrow \mathcal{C}$	in $\mathcal{C}$ correspond to strict monoidal functors $\mathcal{M} \rightarrow \mathcal{C}$. This sort of idea leads to the study
fong-spivak-seven-sketches_F01993	186	■ 0.998	$g: m \rightarrow n$	$g : m \rightarrow n$	The behavioral approach. A signal flow graph $g: m \rightarrow n$ takes an input $x \in R^m$ and
fong-spivak-seven-sketches_F01994	186	■ 0.998	$x \in R^m$	$x \in R^m$	The behavioral approach. A signal flow graph $g: m \rightarrow n$ takes an input $x \in R^m$ and
fong-spivak-seven-sketches_F01995	186	■ 1.000	$y \in R^n$	$y \in R^n$	gives an output $y \in R^n$. In fact, since this is all we care about, we might just think
fong-spivak-seven-sketches_F01996	186	■ 1.000	$\operatorname{B}(g) \subseteq R^m \times R^n$	$\operatorname{B}(g) \subseteq R^m \times R^n$	(x, y) . We'll call this set the behavior of g and denote it $\operatorname{B}(g) \subseteq R^m \times R^n$. For example,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_F01997	186	■ 1.000	$(x, (x, x))$	$(x, (x, x))$	copy-behavior. Similarly, $(x, (x, x))$ is copy behavior for every $x \in R$, thus we have
fong-spivak-seven-sketches_F01998	186	■ 0.954	$g^{\text{\texttt{op}}}$	g^{op}	g^{op} , we must have $g^{\text{op}}: n \rightarrow m$. Thus if $B(g) \subseteq R^m \times R^n$ then we need $B(g^{\text{op}}) \subseteq R^n \times R^m$.
fong-spivak-seven-sketches_F01999	186	■ 0.954	$g^{\text{\texttt{op}}}: n \rightarrowtail m$	$g^{\text{op}}: n \rightarrow m$	g^{op} , we must have $g^{\text{op}}: n \rightarrow m$. Thus if $B(g) \subseteq R^m \times R^n$ then we need $B(g^{\text{op}}) \subseteq R^n \times R^m$.
fong-spivak-seven-sketches_F02000	186	■ 0.954	$\mathrm{B}(\text{\texttt{op}}) \subseteq R^n \times R^m$	$B(g^{\text{op}}) \subseteq R^n \times R^m$	g^{op} , we must have $g^{\text{op}}: n \rightarrow m$. Thus if $B(g) \subseteq R^m \times R^n$ then we need $B(g^{\text{op}}) \subseteq R^n \times R^m$.
fong-spivak-seven-sketches_F02001	186	■ 0.996	(b, a)	(b, a)	One simple way to do this is to replace each (a, b) with (b, a) , so we would have
fong-spivak-seven-sketches_F02002	186	■ 0.854	$\mathrm{B}(\neg C)$	$B(\neg C)$	1. What is the behavior $B(\neg C)$ of the reversed addition icon $\neg C: 1 \rightarrow 2$?
fong-spivak-seven-sketches_F02003	186	■ 0.854	$\neg C: 1 \rightarrowtail 2$	$\neg C: 1 \rightarrow 2$	1. What is the behavior $B(\neg C)$ of the reversed addition icon $\neg C: 1 \rightarrow 2$?
fong-spivak-seven-sketches_F02004	186	■ 0.833	$\mathrm{B}(>)$	$B(>)$	2. What is the behavior $B(>)$ of the reversed copy icon, $>: 2 \rightarrow 1$?
fong-spivak-seven-sketches_F02005	186	■ 0.833	$>: 2 \rightarrowtail 1$	$>: 2 \rightarrow 1$	2. What is the behavior $B(>)$ of the reversed copy icon, $>: 2 \rightarrow 1$?
fong-spivak-seven-sketches_F02006	187	■ 1.000	$B \subseteq R^m \times R^n$	$B \subseteq R^m \times R^n$	of a signal flow graph $m \rightarrow n$ is a subset $B \subseteq R^m \times R^n$, i.e. a relation. Why not try to
fong-spivak-seven-sketches_F02007	187	■ 1.000	$N: n \rightarrowtail p$	$N: n \rightarrow p$	flow graphs with matrices $M: m \rightarrow n$ and $N: n \rightarrow p$, we see that their behaviors are
fong-spivak-seven-sketches_F02008	187	■ 1.000	$B_1 := \{(x, Mx) \mid x \in R^m\}$	$B_1 := \{(x, Mx) \mid x \in R^m\}$	the relations $B_1 := \{(x, Mx) \mid x \in R^m\}$ and $B_2 := \{(y, Ny) \mid y \in R^n\}$, while the behavior
fong-spivak-seven-sketches_F02009	187	■ 1.000	$B_2 := \{(y, Ny) \mid y \in R^n\}$	$B_2 := \{(y, Ny) \mid y \in R^n\}$	the relations $B_1 := \{(x, Mx) \mid x \in R^m\}$ and $B_2 := \{(y, Ny) \mid y \in R^n\}$, while the behavior
fong-spivak-seven-sketches_F02010	187	■ 0.999	$\{(x, x; M; N) \mid x \in R^m\}$	$\{(x, x; M; N) \mid x \in R^m\}$	of $M \circ N$ is the relation $\{(x, x \circ M \circ N) \mid x \in R^m\}$. This is a case of relation composition.
fong-spivak-seven-sketches_F02011	187	■ 0.881	$B_1 \subseteq R^m \times R^n$	$B_1 \subseteq R^m \times R^n$	Given relations $B_1 \subseteq R^m \times R^n$ and $B_2 \subseteq R^n \times R^p$, their composite $B_1 \circ B_2 \subseteq R^m \times R^p$ is
fong-spivak-seven-sketches_F02012	187	■ 0.881	$B_2 \subseteq R^n \times R^p$	$B_2 \subseteq R^n \times R^p$	Given relations $B_1 \subseteq R^m \times R^n$ and $B_2 \subseteq R^n \times R^p$, their composite $B_1 \circ B_2 \subseteq R^m \times R^p$ is
fong-spivak-seven-sketches_F02013	187	■ 0.881	$B_1; B_2 \subseteq R^m \times R^p$	$B_1; B_2 \subseteq R^m \times R^p$	Given relations $B_1 \subseteq R^m \times R^n$ and $B_2 \subseteq R^n \times R^p$, their composite $B_1 \circ B_2 \subseteq R^m \times R^p$ is
fong-spivak-seven-sketches_F02014	187	■ 0.953	$\mathbf{Rel}_R$	$\mathbf{Rel}_R$	Definition 5.79. Let R be a rig. We define the prop $\mathbf{Rel}_R$ of R -relations to have subsets
fong-spivak-seven-sketches_F02015	187	■ 1.000	$C \subseteq R^p \times R^q$	$C \subseteq R^p \times R^q$	prop $\mathbf{Rel}_R$. If $B \subseteq R^m \times R^n$ and $C \subseteq R^p \times R^q$ are morphisms in $\mathbf{Rel}_R$, write down $B + C$
fong-spivak-seven-sketches_F02016	187	■ 1.000	$B + C$	$B + C$	prop $\mathbf{Rel}_R$. If $B \subseteq R^m \times R^n$ and $C \subseteq R^p \times R^q$ are morphisms in $\mathbf{Rel}_R$, write down $B + C$
fong-spivak-seven-sketches_F02017	187	■ 0.755	$g \in G_R$	$g \in G_R$	g^{op} for the mirror image of g , for each $g \in G_R$. So let's write $G_R^{\text{op}} := \{g^{\text{op}} \mid g \in G_R\}$ for
fong-spivak-seven-sketches_F02018	187	■ 0.755	$G_R^{\text{\texttt{op}}} := \{g^{\text{\texttt{op}}} \mid g \in G_R\}$	$G_R^{\text{op}} := \{g^{\text{op}} \mid g \in G_R\}$	g^{op} for the mirror image of g , for each $g \in G_R$. So let's write $G_R^{\text{op}} := \{g^{\text{op}} \mid g \in G_R\}$ for
fong-spivak-seven-sketches_F02019	187	■ 0.998	$\mathbf{SFG}_R^+$	$\mathbf{SFG}_R^+$	We call a morphism in the prop $\mathbf{SFG}_R^+$ a <i>(non-simplified) signal flow graph</i> : these extend
fong-spivak-seven-sketches_F02020	188	■ 1.000	$g: m \rightarrowtail n, h: \ell \rightarrowtail n$	$g: m \rightarrow n, h: \ell \rightarrow n$	<i>Exercise 5.82.</i> Let $g: m \rightarrowtail n, h: \ell \rightarrowtail n$ be signal flow graphs. Note that $h^{\text{op}}: n \rightarrow \ell$ is
fong-spivak-seven-sketches_F02021	188	■ 1.000	$h^{\text{\texttt{op}}}: n \rightarrowtail \ell$	$h^{\text{op}}: n \rightarrow \ell$	<i>Exercise 5.82.</i> Let $g: m \rightarrowtail n, h: \ell \rightarrowtail n$ be signal flow graphs. Note that $h^{\text{op}}: n \rightarrow \ell$ is
fong-spivak-seven-sketches_F02022	188	■ 0.467	$g_9^{\circ}(h^{\text{\texttt{op}}})$	$g_9^{\circ}(h^{\text{op}})$	a signal flow graph, and we can form the composite $g \circ (h^{\text{op}})$:
fong-spivak-seven-sketches_F02023	188	■ 0.274	$g^{\circ}(h^{\text{\texttt{op}}}) \subseteq R^m \times R^{\ell}$	$g^{\circ}(h^{\text{op}}) \subseteq R^m \times R^{\ell}$	Show that the behavior of $g \circ (h^{\text{op}}) \subseteq R^m \times R^{\ell}$ is equal to

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F02824	188	0.807	$g: m \rightarrow n, h: m \rightarrow p$	$g: m \rightarrow n, h: m \rightarrow p$	Exercise 5.83. Let $g: m \rightarrow n, h: m \rightarrow p$ be signal flow graphs. Note that $(g^{\text{op}}): n \rightarrow m$
fong-spivak-seven-sketches-F02825	188	0.807	$\left(g^{\mathrm{op}}\right): n \rightarrow m$	$(g^{\text{op}}): n \rightarrow m$	Exercise 5.83. Let $g: m \rightarrow n, h: m \rightarrow p$ be signal flow graphs. Note that $(g^{\text{op}}): n \rightarrow m$
fong-spivak-seven-sketches-F02826	188	0.856	$g^{\mathrm{op}}; h$	$g^{\text{op}}; h$	is a signal flow graph, and we can form the composite $g^{\text{op}} \circ h$
fong-spivak-seven-sketches-F02827	188	0.963	$S(g) \in \operatorname{Mat}(R)$	$S(g) \in \operatorname{Mat}(R)$	$S(g) \in \mathbf{Mat}(R)$ be the associated $(m \times n)$ -matrix (see Theorem 5.53).
fong-spivak-seven-sketches-F02828	188	0.994	$b_1, b_2 \in \operatorname{B}(g)$	$b_1, b_2 \in \operatorname{B}(g)$	subspace. That is, if $b_1, b_2 \in \operatorname{B}(g)$ then so are $b_1 + b_2$ and $r * b_1$, for any $r \in R$. $\diamond$
fong-spivak-seven-sketches-F02829	188	0.994	$b_1 + b_2$	$b_1 + b_2$	subspace. That is, if $b_1, b_2 \in \operatorname{B}(g)$ then so are $b_1 + b_2$ and $r * b_1$, for any $r \in R$. $\diamond$
fong-spivak-seven-sketches-F02830	188	0.994	$r * b_1$	$r * b_1$	subspace. That is, if $b_1, b_2 \in \operatorname{B}(g)$ then so are $b_1 + b_2$ and $r * b_1$, for any $r \in R$. $\diamond$
fong-spivak-seven-sketches-F02831	188	0.994	$r \in R$	$r \in R$	subspace. That is, if $b_1, b_2 \in \operatorname{B}(g)$ then so are $b_1 + b_2$ and $r * b_1$, for any $r \in R$. $\diamond$
fong-spivak-seven-sketches-F02832	189	0.848	$\mathbf{LinRel}_R$	$\mathbf{LinRel}_R$	$\mathbf{LinRel}_R$. That is, one might want to show that there is a sub-prop of the prop $\mathbf{Rel}_R$
fong-spivak-seven-sketches-F02833	189	0.989	$(x, y) \in B$	$(x, y) \in B$	that B is linear. In other words, where for any $(x, y) \in B$ and $r \in R$, the element
fong-spivak-seven-sketches-F02834	189	0.730	$(r * x, r * y) \in R^m \times R^n$	$(r * x, r * y) \in R^m \times R^n$	$(r * x, r * y) \in R^m \times R^n$ is in B , and for any $(x', y') \in B$, the element $(x + x', y + y')$ is in B .
fong-spivak-seven-sketches-F02835	189	0.730	$\left(x^{\prime}, y^{\prime}\right) \in B$	$(x', y') \in B$	$(r * x, r * y) \in R^m \times R^n$ is in B , and for any $(x', y') \in B$, the element $(x + x', y + y')$ is in B .
fong-spivak-seven-sketches-F02836	189	0.730	$\left(x+x^{\prime}, y+y^{\prime}\right)$	$(x + x', y + y')$	$(r * x, r * y) \in R^m \times R^n$ is in B , and for any $(x', y') \in B$, the element $(x + x', y + y')$ is in B .
fong-spivak-seven-sketches-F02837	189	0.512	$G_R \sqcup G_R^{\mathrm{op}}$	$G_R \sqcup G_R^{\text{op}}$	set is $G_R \sqcup G_R^{\text{op}}$ as in Eq. (5.81) and whose equations include those from Theorem 5.60,
fong-spivak-seven-sketches-F02838	189	1.000	η_1	η_1	Eq. (5.86) serve as the η_1 and ϵ_1 from Definition 4.58. Using monoidal products of these
fong-spivak-seven-sketches-F02839	189	1.000	ϵ_1	ϵ_1	Eq. (5.86) serve as the η_1 and ϵ_1 from Definition 4.58. Using monoidal products of these
fong-spivak-seven-sketches-F02840	190	1.000	$n = n^*$	$n = n^*$	is dual to itself, $n = n^*$.
fong-spivak-seven-sketches-F02841	190	1.000	$\mathbb{R}[s, s^{-1}]$	$\mathbb{R}[s, s^{-1}]$	$\mathbb{R}[s, s^{-1}]$ of polynomials in s and s^{-1} with coefficients in $\mathbb{R}$ and where $ss^{-1} = s^{-1}s = 1$.
fong-spivak-seven-sketches-F02842	190	1.000	s^{-1}	s^{-1}	$\mathbb{R}[s, s^{-1}]$ of polynomials in s and s^{-1} with coefficients in $\mathbb{R}$ and where $ss^{-1} = s^{-1}s = 1$.
fong-spivak-seven-sketches-F02843	190	1.000	$ss^{-1} = s^{-1}s = 1$	$ss^{-1} = s^{-1}s = 1$	$\mathbb{R}[s, s^{-1}]$ of polynomials in s and s^{-1} with coefficients in $\mathbb{R}$ and where $ss^{-1} = s^{-1}s = 1$.
fong-spivak-seven-sketches-F02844	190	1.000	$a * u + b * v$	$a * u + b * v$	will take some linear combination $a * u + b * v$ of the desired speed and actual speed,
fong-spivak-seven-sketches-F02845	190	1.000	$\{(F, u, v) \mid u = v\}$	$\{(F, u, v) \mid u = v\}$	this signal flow graph close to the relation $\{(F, u, v) \mid u = v\}$? By phrasing problems in
fong-spivak-seven-sketches-F02846	195	0.872	$\mathcal{C}'$	$\mathcal{C}'$	nomenon of the category $\mathcal{C}$. Our goal is to obtain $\mathcal{C}$'s colimits more directly, as a kind
fong-spivak-seven-sketches-F02847	195	0.444	$\operatorname{Cospan}_{\mathcal{C}}$	$\operatorname{Cospan}_{\mathcal{C}}$	that of <i>cospans in</i> $\mathcal{C}$, denoted $\mathbf{Cospan}_{\mathcal{C}}$ —that can conveniently package $\mathcal{C}$'s colimits in
fong-spivak-seven-sketches-F02848	195	1.000	$s_{m, n}: m \rightarrow n$	$s_{m, n}: m \rightarrow n$	(m, n) , an icon $s_{m, n}: m \rightarrow n$. These icons, known as <i>spiders</i> , ³ are drawn as follows:
fong-spivak-seven-sketches-F02849	197	1.000	$\varnothing \in \mathcal{C}$	$\varnothing \in \mathcal{C}$	Definition 6.1. Let $\mathcal{C}$ be a category. An <i>initial object</i> in $\mathcal{C}$ is an object $\varnothing \in \mathcal{C}$ such that for
fong-spivak-seven-sketches-F02850	197	1.000	$!_T: \varnothing \rightarrow T$	$!_T: \varnothing \rightarrow T$	each object T in $\mathcal{C}$ there exists a unique morphism $!_T: \varnothing \rightarrow T$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_197 F02051	197	0.844	$A, \leq$	$A, \leq$	1. $(A, \leq)$ has no initial object.
fong-spivak-seven-sketches_197 F02052	197	0.994	$(A, \leq)$	$(A, \leq)$	2. $(A, \leq)$ has exactly one initial object.
fong-spivak-seven-sketches_197 F02053	197	1.000	$\varnothing \rightarrow T$	$\varnothing \rightarrow T$	is a unique function $\varnothing \rightarrow T$, since $\varnothing$ has no elements.
fong-spivak-seven-sketches_197 F02054	197	0.615	$T \in \mathrm{Ob}(\mathcal{C})$	$T \in \mathrm{Ob}(\mathcal{C})$	show that for each object $T \in \mathrm{Ob}(\mathcal{C})$ (i.e. for both $T = A$ and $T = B$) there is a unique
fong-spivak-seven-sketches_197 F02055	197	0.615	$T=A$	$T = A$	show that for each object $T \in \mathrm{Ob}(\mathcal{C})$ (i.e. for both $T = A$ and $T = B$) there is a unique
fong-spivak-seven-sketches_197 F02056	197	0.615	$T=B$	$T = B$	show that for each object $T \in \mathrm{Ob}(\mathcal{C})$ (i.e. for both $T = A$ and $T = B$) there is a unique
fong-spivak-seven-sketches_197 F02057	197	0.999	$\varnothing = ? A$	$\varnothing = ? A$	morphism $\varnothing \rightarrow T$. Trying the case $\varnothing = ? A$ this condition fails when $T = B$: there are
fong-spivak-seven-sketches_197 F02058	197	1.000	$\varnothing = ? B$	$\varnothing = ? B$	two morphisms $A \rightarrow B$, not one. And trying the case $\varnothing = ? B$ this condition fails when
fong-spivak-seven-sketches_197 F02059	197	1.000	$B \rightarrow A$	$B \rightarrow A$	$T = A$: there are zero morphisms $B \rightarrow A$, not one.
fong-spivak-seven-sketches_197 F02060	197	0.374	$\boxed{\bullet}$	$\boxed{\bullet}$	1. $\boxed{\bullet}$
fong-spivak-seven-sketches_197 F02061	197	0.816	$\stackrel{a}{\bullet} \rightarrow \stackrel{b}{\bullet} \rightarrow \stackrel{c}{\bullet}$	$\stackrel{a}{\bullet} \rightarrow \stackrel{b}{\bullet} \rightarrow \stackrel{c}{\bullet}$	2. $\left \begin{array}{c} a \\ \bullet \end{array} \rightarrow \begin{array}{c} b \\ \bullet \end{array} \rightarrow \begin{array}{c} c \\ \bullet \end{array} \right.$
fong-spivak-seven-sketches_197 F02062	197	0.981	$\stackrel{a}{\bullet} \bullet \stackrel{b}{\bullet}$	$\stackrel{a}{\bullet} \bullet \stackrel{b}{\bullet}$	3. $\left \begin{array}{cc} a & b \\ \bullet & \bullet \end{array} \right.$
fong-spivak-seven-sketches_197 F02063	197	0.985	$(R, 0_R, +_R, 1_R, *_R)$	$(R, 0_R, +_R, 1_R, *_R)$	$(R, 0_R, +_R, 1_R, *_R)$ to $(S, 0_S, +_S, 1_S, *_S)$ is a function $f: R \rightarrow S$ such that $f(0_R) = 0_S$,
fong-spivak-seven-sketches_197 F02064	197	0.985	$(S, 0_S, +_S, 1_S, *_S)$	$(S, 0_S, +_S, 1_S, *_S)$	$(R, 0_R, +_R, 1_R, *_R)$ to $(S, 0_S, +_S, 1_S, *_S)$ is a function $f: R \rightarrow S$ such that $f(0_R) = 0_S$,
fong-spivak-seven-sketches_197 F02065	197	0.985	$f: R \rightarrow S$	$f: R \rightarrow S$	$(R, 0_R, +_R, 1_R, *_R)$ to $(S, 0_S, +_S, 1_S, *_S)$ is a function $f: R \rightarrow S$ such that $f(0_R) = 0_S$,
fong-spivak-seven-sketches_197 F02066	197	0.985	$f(0_R) = 0_S$	$f(0_R) = 0_S$	$(R, 0_R, +_R, 1_R, *_R)$ to $(S, 0_S, +_S, 1_S, *_S)$ is a function $f: R \rightarrow S$ such that $f(0_R) = 0_S$,
fong-spivak-seven-sketches_197 F02067	197	0.979	$f(r_1 +_R r_2) = f(r_1) +_S f(r_2)$	$f(r_1 +_R r_2) = f(r_1) +_S f(r_2)$	$f(r_1 +_R r_2) = f(r_1) +_S f(r_2)$, etc.
fong-spivak-seven-sketches_198 F02068	198	1.000	c_1	c_1	Exercise 6.10. Let $\mathcal{C}$ be a category, and suppose that c_1 and c_2 are initial objects. Find
fong-spivak-seven-sketches_198 F02069	198	1.000	c_2	c_2	Exercise 6.10. Let $\mathcal{C}$ be a category, and suppose that c_1 and c_2 are initial objects. Find
fong-spivak-seven-sketches_198 F02070	198	0.978	$\iota_A: A \rightarrow$	$\iota_A: A \rightarrow$	is an object, which we denote $A + B$, together with a pair of morphisms $(\iota_A: A \rightarrow$
fong-spivak-seven-sketches_198 F02071	198	0.977	$A + B, \iota_B: B \rightarrow A + B$	$A + B, \iota_B: B \rightarrow A + B$	$A + B, \iota_B: B \rightarrow A + B)$ such that for all objects T and pairs of morphisms $(f: A \rightarrow$
fong-spivak-seven-sketches_198 F02072	198	0.977	$(f: A \rightarrow$	$(f: A \rightarrow$	$A + B, \iota_B: B \rightarrow A + B)$ such that for all objects T and pairs of morphisms $(f: A \rightarrow$
fong-spivak-seven-sketches_198 F02073	198	0.994	$T, g: B \rightarrow T$	$T, g: B \rightarrow T$	$T, g: B \rightarrow T)$, there exists a unique morphism $[f, g]: A + B \rightarrow T$ such that the following
fong-spivak-seven-sketches_198 F02074	198	0.994	$[f, g]: A + B \rightarrow T$	$[f, g]: A + B \rightarrow T$	$T, g: B \rightarrow T)$, there exists a unique morphism $[f, g]: A + B \rightarrow T$ such that the following
fong-spivak-seven-sketches_198 F02075	198	0.999	$[f, g]$	$[f, g]$	We call $[f, g]$ the <i>copairing</i> of f and g .
fong-spivak-seven-sketches_198 F02076	198	1.000	$\iota_A: A \rightarrow A \sqcup B$	$\iota_A: A \rightarrow A \sqcup B$	the disjoint union $A \sqcup B$ together with the inclusion functions $\iota_A: A \rightarrow A \sqcup B$ and

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_199 F02077	199	1.000	$\iota_B: B \rightarrow A \sqcup B$	$\iota_B: B \rightarrow A \sqcup B$	$\iota_B: B \rightarrow A \sqcup B$
fong-spivak-seven-sketches_199 F02078	199	1.000	$f: A \rightarrow T$	$f: A \rightarrow T$	Suppose we have functions $f: A \rightarrow T$ and $g: B \rightarrow T$ for some other set T ,
fong-spivak-seven-sketches_199 F02079	199	1.000	$g: B \rightarrow T$	$g: B \rightarrow T$	Suppose we have functions $f: A \rightarrow T$ and $g: B \rightarrow T$ for some other set T ,
fong-spivak-seven-sketches_199 F02080	199	0.591	$[f, g]: A \sqcup B \rightarrow T$	$[f, g]: A \sqcup B \rightarrow T$	$[f, g]: A \sqcup B \rightarrow T$ such that $\iota_A \circ [f, g] = f$ and $\iota_B \circ [f, g] = g$. What is it? Any element
fong-spivak-seven-sketches_199 F02081	199	0.591	ι_A	ι_A	$[f, g]: A \sqcup B \rightarrow T$ such that $\iota_A \circ [f, g] = f$ and $\iota_B \circ [f, g] = g$. What is it? Any element
fong-spivak-seven-sketches_199 F02082	199	0.591	$[f, g]=f$	$[f, g] = f$	$[f, g]: A \sqcup B \rightarrow T$ such that $\iota_A \circ [f, g] = f$ and $\iota_B \circ [f, g] = g$. What is it? Any element
fong-spivak-seven-sketches_199 F02083	199	0.591	ι_B	ι_B	$[f, g]: A \sqcup B \rightarrow T$ such that $\iota_A \circ [f, g] = f$ and $\iota_B \circ [f, g] = g$. What is it? Any element
fong-spivak-seven-sketches_199 F02084	199	0.591	$[f, g]=g$	$[f, g] = g$	$[f, g]: A \sqcup B \rightarrow T$ such that $\iota_A \circ [f, g] = f$ and $\iota_B \circ [f, g] = g$. What is it? Any element
fong-spivak-seven-sketches_199 F02085	199	0.976	$x \in A \sqcup B$	$x \in A \sqcup B$	$x \in A \sqcup B$ is either ‘from A ’ or ‘from B ’, i.e. either there is some $a \in A$ with $x = \iota_A(a)$ or
fong-spivak-seven-sketches_199 F02086	199	0.976	$x=\iota_A(a)$	$x = \iota_A(a)$	$x \in A \sqcup B$ is either ‘from A ’ or ‘from B ’, i.e. either there is some $a \in A$ with $x = \iota_A(a)$ or
fong-spivak-seven-sketches_199 F02087	199	1.000	$b \in B$	$b \in B$	there is some $b \in B$ with $x = \iota_B(b)$. By Eq. (6.12), we must have:
fong-spivak-seven-sketches_199 F02088	199	1.000	$x=\iota_B(b)$	$x = \iota_B(b)$	there is some $b \in B$ with $x = \iota_B(b)$. By Eq. (6.12), we must have:
fong-spivak-seven-sketches_199 F02089	199	1.000	$T=\{a, b, c, \ldots, z\}$	$T = \{a, b, c, \ldots, z\}$	Exercise 6.16. Suppose $T = \{a, b, c, \ldots, z\}$ is the set of letters in the alphabet, and let
fong-spivak-seven-sketches_199 F02090	199	1.000	$f($	$f($	element of A to the first letter of its label, e.g. $f(\text{apple}) = a$. Let $g: B \rightarrow T$ be the
fong-spivak-seven-sketches_199 F02091	199	1.000	$)=a$	$) = a$	element of A to the first letter of its label, e.g. $f(\text{apple}) = a$. Let $g: B \rightarrow T$ be the
fong-spivak-seven-sketches_199 F02092	199	0.999	$g($	$g($	function sending each element of B to the last letter of its label, e.g. $g(\text{apple}) = e$. Write
fong-spivak-seven-sketches_199 F02093	199	0.999	$)=e$	$) = e$	function sending each element of B to the last letter of its label, e.g. $g(\text{apple}) = e$. Write
fong-spivak-seven-sketches_199 F02094	199	1.000	$[f, g](x)$	$[f, g](x)$	down the function $[f, g](x)$ for all eight elements of $A \sqcup B$.
fong-spivak-seven-sketches_199 F02095	199	1.000	$f: A \rightarrow C, g: B \rightarrow C$	$f: A \rightarrow C, g: B \rightarrow C$	Exercise 6.17. Let $f: A \rightarrow C, g: B \rightarrow C$, and $h: C \rightarrow D$ be morphisms in a category $\mathcal{C}$
fong-spivak-seven-sketches_199 F02096	199	0.538	$\iota_A \circ [f, g]=f$	$\iota_A \circ [f, g] = f$	1. $\iota_A \circ [f, g] = f$.
fong-spivak-seven-sketches_199 F02097	199	0.982	$\iota_B \circ [f, g]=g$	$\iota_B \circ [f, g] = g$	2. $\iota_B \circ [f, g] = g$.
fong-spivak-seven-sketches_199 F02098	199	1.000	$[f, g] \circ h=[f \circ h, g \circ h]$	$[f, g] \circ h = [f \circ h, g \circ h]$	3. $[f, g] \circ h = [f \circ h, g \circ h]$.
fong-spivak-seven-sketches_199 F02099	199	1.000	$\left[\iota_A, \iota_B\right]=\mathrm{id}_{A+B}$	$[\iota_A, \iota_B] = \mathrm{id}_{A+B}$	4. $[\iota_A, \iota_B] = \mathrm{id}_{A+B}$.
fong-spivak-seven-sketches_199 F02100	199	0.914	$(\mathcal{C},+, \varnothing)$	$(\mathcal{C}, +, \varnothing)$	denoted $\varnothing$. Then $(\mathcal{C}, +, \varnothing)$ is a symmetric monoidal category (recall Definition 4.45). In
fong-spivak-seven-sketches_199 F02101	199	0.827	$\mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$	$\mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$	1. Show that $+$ extends to a functor $\mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$. In particular, how does it act on
fong-spivak-seven-sketches_199 F02102	199	0.771	$\mathcal{C} \times \mathcal{C}$	$\mathcal{C} \times \mathcal{C}$	morphisms in $\mathcal{C} \times \mathcal{C}$?
fong-spivak-seven-sketches_199 F02103	199	1.000	$A+\varnothing \rightarrow A$	$A + \varnothing \rightarrow A$	are isomorphisms $A + \varnothing \rightarrow A$ and $\varnothing + A \rightarrow A$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F02104	199	1.000	$\varnothing \rightarrow A$	$\varnothing \rightarrow A$	are isomorphisms $A \rightarrow \varnothing$ and $\varnothing \rightarrow A$.
fong-spivak-seven-sketches-F02105	199	1.000	$(A+B) \rightarrow A+(B+C)$	$(A+B) \rightarrow A+(B+C)$	a) $A+B \rightarrow A+(B+C)$.
fong-spivak-seven-sketches-F02106	199	1.000	$A \rightarrow B \rightarrow A$	$A \rightarrow B \rightarrow A$	b) $A \rightarrow B \rightarrow A$.
fong-spivak-seven-sketches-F02107	200	1.000	$f: A \rightarrow X$	$f: A \rightarrow X$	Definition 6.19. Let $\mathcal{C}$ be a category and let $f: A \rightarrow X$ and $g: A \rightarrow Y$ be morphisms
fong-spivak-seven-sketches-F02108	200	1.000	$g: A \rightarrow Y$	$g: A \rightarrow Y$	Definition 6.19. Let $\mathcal{C}$ be a category and let $f: A \rightarrow X$ and $g: A \rightarrow Y$ be morphisms
fong-spivak-seven-sketches-F02109	200	0.786	$X+_A Y$	$X+_A Y$	in $\mathcal{C}$ that have a common domain. The <i>pushout</i> $X+_A Y$ is the colimit of the diagram
fong-spivak-seven-sketches-F02110	200	0.857	$X+_A Y$	$X+_A Y$	In more detail, a pushout consists of (i) an object $X+_A Y$ and (ii) morphisms $\iota_X: X \rightarrow$
fong-spivak-seven-sketches-F02111	200	0.857	$\iota_X: X \rightarrow$	$\iota_X: X \rightarrow$	In more detail, a pushout consists of (i) an object $X+_A Y$ and (ii) morphisms $\iota_X: X \rightarrow$
fong-spivak-seven-sketches-F02112	200	0.905	$\iota_Y: Y \rightarrow X+_A Y$	$\iota_Y: Y \rightarrow X+_A Y$	$X+_A Y$ and $\iota_Y: Y \rightarrow X+_A Y$ satisfying (a) and (b) below.
fong-spivak-seven-sketches-F02113	200	0.371	$\ulcorner$	$\ulcorner$	commutes. (We will explain the ‘ $\ulcorner$ ’ symbol below.)
fong-spivak-seven-sketches-F02114	200	1.000	$x: X \rightarrow T, y: Y \rightarrow T$	$x: X \rightarrow T, y: Y \rightarrow T$	(b) For all objects T and morphisms $x: X \rightarrow T, y: Y \rightarrow T$, if the diagram
fong-spivak-seven-sketches-F02115	200	0.933	$t: X+_A Y \rightarrow T$	$t: X+_A Y \rightarrow T$	commutes, then there exists a unique morphism $t: X+_A Y \rightarrow T$ such that
fong-spivak-seven-sketches-F02116	201	0.989	ι_X	ι_X	We further call ι_X the <i>pushout of g along f</i> , and similarly ι_Y the <i>pushout of f along g</i> .
fong-spivak-seven-sketches-F02117	201	0.989	ι_Y	ι_Y	We further call ι_X the <i>pushout of g along f</i> , and similarly ι_Y the <i>pushout of f along g</i> .
fong-spivak-seven-sketches-F02118	201	1.000	$B \leftarrow A \rightarrow C$	$B \leftarrow A \rightarrow C$	pushout of a diagram $B \leftarrow A \rightarrow C$ is equal to the coproduct $B \sqcup C$: namely, both are
fong-spivak-seven-sketches-F02119	201	1.000	$B \sqcup C$	$B \sqcup C$	pushout of a diagram $B \leftarrow A \rightarrow C$ is equal to the coproduct $B \sqcup C$: namely, both are
fong-spivak-seven-sketches-F02120	201	1.000	$B \vee C$	$B \vee C$	equal to the join $B \vee C$.
fong-spivak-seven-sketches-F02121	201	0.867	$i: A \rightarrow A'$	$i: A \rightarrow A'$	$i: A \rightarrow A'$ and $j: X \rightarrow X'$, we can take X' to be the pushout $X+_A A'$, i.e. the following
fong-spivak-seven-sketches-F02122	201	0.867	$j: X \rightarrow X'$	$j: X \rightarrow X'$	$i: A \rightarrow A'$ and $j: X \rightarrow X'$, we can take X' to be the pushout $X+_A A'$, i.e. the following
fong-spivak-seven-sketches-F02123	201	0.867	X'	X'	$i: A \rightarrow A'$ and $j: X \rightarrow X'$, we can take X' to be the pushout $X+_A A'$, i.e. the following
fong-spivak-seven-sketches-F02124	201	0.867	$X+_A A'$	$X+_A A'$	$i: A \rightarrow A'$ and $j: X \rightarrow X'$, we can take X' to be the pushout $X+_A A'$, i.e. the following
fong-spivak-seven-sketches-F02125	201	0.847	$f' := i^{-1} \circ f \circ j$	$f' := i^{-1} \circ f \circ j$	where $f' := i^{-1} \circ f \circ j$. To see this, observe that if there is any object T such that the
fong-spivak-seven-sketches-F02126	201	0.997	$f; x = i; a$	$f; x = i; a$	then $f \circ x = i \circ a$, and so we are forced to take $x': X \rightarrow T$ to be $x' := j^{-1} \circ x$. This makes
fong-spivak-seven-sketches-F02127	201	0.997	$x': X \rightarrow T$	$x': X \rightarrow T$	then $f \circ x = i \circ a$, and so we are forced to take $x': X \rightarrow T$ to be $x' := j^{-1} \circ x$. This makes
fong-spivak-seven-sketches-F02128	201	0.997	$x' := j^{-1} \circ x$	$x' := j^{-1} \circ x$	then $f \circ x = i \circ a$, and so we are forced to take $x': X \rightarrow T$ to be $x' := j^{-1} \circ x$. This makes
fong-spivak-seven-sketches-F02129	201	0.560	$f'; x' = i^{-1}; f; j; j^{-1}; x = i^{-1}; i; a = a$	$f'; x' = i^{-1}; f; j; j^{-1}; x = i^{-1}; i; a = a$	because $f' \circ x' = i^{-1} \circ f \circ j \circ j^{-1} \circ x = i^{-1} \circ i \circ a = a$.
fong-spivak-seven-sketches-F02130	201	0.998	S	S	Exercise 6.24. For any set S , we have the discrete category Discs , with S as objects

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_02131	202	1.000	$\{f(a) \sim g(a) \mid a \in A\}$	$\{f(a) \sim g(a) \mid a \in A\}$	relation $\{f(a) \sim g(a) \mid a \in A\}$.
fong-spivak-seven-sketches_02132	202	1.000	$f(a)$	$f(a)$	a connection between $f(a)$ in X and $g(a)$ in Y . The pushout is the set of connected
fong-spivak-seven-sketches_02133	202	1.000	$g(a)$	$g(a)$	a connection between $f(a)$ in X and $g(a)$ in Y . The pushout is the set of connected
fong-spivak-seven-sketches_02134	202	1.000	$f: \underline{4} \rightarrow \underline{5}$	$f: \underline{4} \rightarrow \underline{5}$	Exercise 6.26. What is the pushout of the functions $f: \underline{4} \rightarrow \underline{5}$ and $g: \underline{4} \rightarrow \underline{3}$ pictured
fong-spivak-seven-sketches_02135	202	1.000	$g: \underline{4} \rightarrow \underline{3}$	$g: \underline{4} \rightarrow \underline{3}$	Exercise 6.26. What is the pushout of the functions $f: \underline{4} \rightarrow \underline{5}$ and $g: \underline{4} \rightarrow \underline{3}$ pictured
fong-spivak-seven-sketches_02136	202	1.000	$X, Y \in$	$X, Y \in$	Example 6.27. Suppose a category $\mathcal{C}$ has an initial object $\emptyset$. For any two objects $X, Y \in$
fong-spivak-seven-sketches_02137	202	0.785	$f: \varnothing \rightarrow X$	$f: \emptyset \rightarrow X$	$\text{Ob } \mathcal{C}$, there is a unique morphism $f: \emptyset \rightarrow X$ and a unique morphism $g: \emptyset \rightarrow Y$; this
fong-spivak-seven-sketches_02138	202	0.785	$g: \varnothing \rightarrow Y$	$g: \emptyset \rightarrow Y$	$\text{Ob } \mathcal{C}$, there is a unique morphism $f: \emptyset \rightarrow X$ and a unique morphism $g: \emptyset \rightarrow Y$; this
fong-spivak-seven-sketches_02139	202	0.884	$X \stackrel{f}{\leftarrow} \varnothing \stackrel{g}{\rightarrow} Y$	$X \stackrel{f}{\leftarrow} \emptyset \stackrel{g}{\rightarrow} Y$	The diagram $X \stackrel{f}{\leftarrow} \emptyset \stackrel{g}{\rightarrow} Y$ has a pushout in $\mathcal{C}$ iff X and Y have a coproduct in $\mathcal{C}$,
fong-spivak-seven-sketches_02140	202	1.000	$X+Y$	$X + Y$	coproduct $X + Y$; then the diagram to the left
fong-spivak-seven-sketches_02141	202	0.897	$X+Y \rightarrow T$	$X + Y \rightarrow T$	is a unique map $X + Y \rightarrow T$ making the diagram as in Eq. (6.21) commute (why? ²).
fong-spivak-seven-sketches_02142	202	0.615	$X+\varnothing Y \cong X+Y$	$X + \emptyset Y \cong X + Y$	This shows that $X + Y$ is a pushout, $X +_{\emptyset} Y \cong X + Y$.
fong-spivak-seven-sketches_02143	202	0.986	$X+\varnothing Y$	$X + \emptyset Y$	Similarly, if a pushout $X +_{\emptyset} Y$ exists, then it satisfies the universal property of the
fong-spivak-seven-sketches_02144	202	0.991	$A=X=Y=\mathbb{N}$	$A = X = Y = \mathbb{N}$	Example 6.29. Let $A = X = Y = \mathbb{N}$. Consider the functions $f: A \rightarrow X$ and $g: A \rightarrow Y$
fong-spivak-seven-sketches_02145	203	1.000	$f(a):=\lfloor a / 2 \rfloor$	$f(a) := \lfloor a/2 \rfloor$	given by the ‘floor’ functions, $f(a) := \lfloor a/2 \rfloor$ and $g(a) := \lfloor (a+1)/2 \rfloor$.
fong-spivak-seven-sketches_02146	203	1.000	$g(a):=\lfloor (a+1) / 2 \rfloor$	$g(a) := \lfloor (a+1)/2 \rfloor$	given by the ‘floor’ functions, $f(a) := \lfloor a/2 \rfloor$ and $g(a) := \lfloor (a+1)/2 \rfloor$.
fong-spivak-seven-sketches_02147	203	1.000	$x: X \rightarrow T$	$x: X \rightarrow T$	If T is any other set and we have maps $x: X \rightarrow T$ and $y: Y \rightarrow T$ that commute with
fong-spivak-seven-sketches_02148	203	1.000	$y: Y \rightarrow T$	$y: Y \rightarrow T$	If T is any other set and we have maps $x: X \rightarrow T$ and $y: Y \rightarrow T$ that commute with
fong-spivak-seven-sketches_02149	203	1.000	$f \cong x=g \cong y$	$f \cong x = g \cong y$	f and g , i.e. $f \cong x = g \cong y$, then this commutativity implies that
fong-spivak-seven-sketches_02150	203	0.781	$f(1)=0$	$f(1) = 0$	In other words, Y ’s 0 and X ’s 0 go to the same place in T , say t . But since $f(1) = 0$
fong-spivak-seven-sketches_02151	203	0.686	$g(1)=1$	$g(1) = 1$	and $g(1) = 1$, we also have that $t = x(0) = x(f(1)) = y(g(1)) = y(1)$. This means Y ’s 1
fong-spivak-seven-sketches_02152	203	0.686	$t=x(0)=x(f(1))=y(g(1))=y(1)$	$t = x(0) = x(f(1)) = y(g(1)) = y(1)$	and $g(1) = 1$, we also have that $t = x(0) = x(f(1)) = y(g(1)) = y(1)$. This means Y ’s 1
fong-spivak-seven-sketches_02153	203	1.000	$g(2)=1$	$g(2) = 1$	goes to t also. But since $g(2) = 1$ and $f(2) = 1$, we also have that $t = g(1) = y(g(2)) =$
fong-spivak-seven-sketches_02154	203	1.000	$f(2)=1$	$f(2) = 1$	goes to t also. But since $g(2) = 1$ and $f(2) = 1$, we also have that $t = g(1) = y(g(2)) =$
fong-spivak-seven-sketches_02155	203	1.000	$t=g(1)=y(g(2))=$	$t = g(1) = y(g(2)) =$	goes to t also. But since $g(2) = 1$ and $f(2) = 1$, we also have that $t = g(1) = y(g(2)) =$
fong-spivak-seven-sketches_02156	203	0.900	$x(f(2))=x(1)$	$x(f(2)) = x(1)$	$x(f(2)) = x(1)$, which means that X ’s 1 also goes to t . One can keep repeating this
fong-spivak-seven-sketches_02157	203	0.996	$X \sqcup_A Y \cong \{1\}$	$X \sqcup_A Y \cong \{1\}$	induction, one can prove that the pushout is in fact a 1-element set, $X \sqcup_A Y \cong \{1\}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_203 F02158	203	0.832	$j \in \operatorname{Ob} \mathcal{J}$	$j \in \operatorname{Ob} \mathcal{J}$	must have finitely many objects too, because each object $j \in \operatorname{Ob} \mathcal{J}$ has its own identity morphism id_j .
fong-spivak-seven-sketches_203 F02159	203	0.832	id_j	id_j	must have finitely many objects too, because each object $j \in \operatorname{Ob} \mathcal{J}$ has its own identity morphism id_j .
fong-spivak-seven-sketches_204 F02160	204	0.996	$\operatorname{colim}_{\mathcal{J}} D$	$\operatorname{colim}_{\mathcal{J}} D$	Definition 6.30. We say that a category $\mathcal{C}$ has <i>finite colimits</i> if a colimit, $\operatorname{colim}_{\mathcal{J}} D$, exists
fong-spivak-seven-sketches_204 F02161	204	0.890	$\mathbf{0} \rightarrow \mathcal{C}$	$\mathbf{0} \rightarrow \mathcal{C}$	$! : \mathbf{0} \rightarrow \mathcal{C}$, where $\mathbf{0}$ is the category with no objects and no morphisms, and $!$ is the unique
fong-spivak-seven-sketches_205 F02162	205	1.000	$(v, d) \sim (w, e)$	$(v, d) \sim (w, e)$	generated by putting $(v, d) \sim (w, e)$ if there is an arrow $a : v \rightarrow w$ in J such that
fong-spivak-seven-sketches_205 F02163	205	1.000	$a : v \rightarrow w$	$a : v \rightarrow w$	generated by putting $(v, d) \sim (w, e)$ if there is an arrow $a : v \rightarrow w$ in J such that
fong-spivak-seven-sketches_205 F02164	205	0.996	$D(a)(d) = e$	$D(a)(d) = e$	$D(a)(d) = e$. Then this set, together with the functions $\iota_v : D(v) \rightarrow \operatorname{colim}_{\mathcal{J}} D$ given by
fong-spivak-seven-sketches_205 F02165	205	0.996	$\iota_v : D(v) \rightarrow \operatorname{colim}_{\mathcal{J}} D$	$\iota_v : D(v) \rightarrow \operatorname{colim}_{\mathcal{J}} D$	$D(a)(d) = e$. Then this set, together with the functions $\iota_v : D(v) \rightarrow \operatorname{colim}_{\mathcal{J}} D$ given by
fong-spivak-seven-sketches_205 F02166	205	1.000	$d \in D(v)$	$d \in D(v)$	sending $d \in D(v)$ to its equivalence class, constitutes a colimit of D .
fong-spivak-seven-sketches_205 F02167	205	0.654	$\mathcal{J} =$	$\mathcal{J} =$	<i>Example 6.39.</i> A coproduct is a colimit on the graph $\mathcal{J} = \begin{array}{cc} v_1 & v_2 \\ \bullet & \bullet \end{array}$. A functor $D : \mathcal{J} \rightarrow \mathbf{Set}$
fong-spivak-seven-sketches_205 F02168	205	0.654	$D : \mathcal{J} \rightarrow$	$D : \mathcal{J} \rightarrow$	<i>Example 6.39.</i> A coproduct is a colimit on the graph $\mathcal{J} = \begin{array}{cc} v_1 & v_2 \\ \bullet & \bullet \end{array}$. A functor $D : \mathcal{J} \rightarrow \mathbf{Set}$
fong-spivak-seven-sketches_205 F02169	205	1.000	$X := D(v_1)$	$X := D(v_1)$	can be identified with a choice of two sets, $X := D(v_1)$ and $Y := D(v_2)$. Since there are
fong-spivak-seven-sketches_205 F02170	205	1.000	$Y := D(v_2)$	$Y := D(v_2)$	can be identified with a choice of two sets, $X := D(v_1)$ and $Y := D(v_2)$. Since there are
fong-spivak-seven-sketches_206 F02171	206	0.716	$\mathbf{1} = \boxed{\bullet}$	$\mathbf{1} = \boxed{\bullet}$	<i>Example 6.40.</i> If $\mathcal{J}$ is the category $\mathbf{1} = \boxed{\bullet}$, the formula in Theorem 6.37 yields the set
fong-spivak-seven-sketches_206 F02172	206	1.000	$D(v)$	$D(v)$	This is isomorphic to the set $D(v)$. In other words, if X is a set considered as a diagram
fong-spivak-seven-sketches_206 F02173	206	0.824	$X : \mathbf{1} \rightarrow$	$X : \mathbf{1} \rightarrow$	$X : \mathbf{1} \rightarrow \mathbf{Set}$, then its colimit (like its limit) is just X again.
fong-spivak-seven-sketches_206 F02174	206	1.000	$X \xleftarrow{f} N \xrightarrow{g} Y$	$X \xleftarrow{f} N \xrightarrow{g} Y$	diagram $X \xleftarrow{f} N \xrightarrow{g} Y$ —agree with the description we gave in Example 6.25.
fong-spivak-seven-sketches_206 F02175	206	0.994	$X \xrightarrow[g]{f} Y$	$X \xrightarrow[g]{f} Y$	Consider some diagram $X \xrightarrow[g]{f} Y$ on this graph in $\mathbf{Set}$. The coequalizer of this
fong-spivak-seven-sketches_206 F02176	206	1.000	$y \sim y'$	$y \sim y'$	declaring $y \sim y'$ whenever there exists x in X such that $f(x) = y$ and $g(x) = y'$.
fong-spivak-seven-sketches_206 F02177	206	1.000	$f(x) = y$	$f(x) = y$	declaring $y \sim y'$ whenever there exists x in X such that $f(x) = y$ and $g(x) = y'$.
fong-spivak-seven-sketches_206 F02178	206	1.000	$g(x) = y'$	$g(x) = y'$	declaring $y \sim y'$ whenever there exists x in X such that $f(x) = y$ and $g(x) = y'$.
fong-spivak-seven-sketches_206 F02179	206	0.777	$f, g : X \rightarrow Y$	$f, g : X \rightarrow Y$	circles $\bullet$, the blue and red arrows respectively define functions $f, g : X \rightarrow Y$. Let's
fong-spivak-seven-sketches_206 F02180	206	1.000	$A \rightarrow N \leftarrow B$	$A \rightarrow N \leftarrow B$	common object $A \rightarrow N \leftarrow B$. The common object N is called the <i>apex</i> of the cospan
fong-spivak-seven-sketches_207 F02181	207	0.749	$N \xleftarrow{g} B \xrightarrow{h} P$	$N \xleftarrow{g} B \xrightarrow{h} P$	Then, if a pushout of $N \xleftarrow{g} B \xrightarrow{h} P$ exists in $\mathcal{C}$, as shown on the left, we can extract a
fong-spivak-seven-sketches_207 F02182	207	1.000	$C \in \operatorname{Ob} \mathcal{C}$	$C \in \operatorname{Ob} \mathcal{C}$	need an identity on every object $C \in \operatorname{Ob} \mathcal{C}$; but that's not hard: use $C \rightarrow C \leftarrow C$ where
fong-spivak-seven-sketches_207 F02183	207	1.000	$C \rightarrow C \leftarrow C$	$C \rightarrow C \leftarrow C$	need an identity on every object $C \in \operatorname{Ob} \mathcal{C}$; but that's not hard: use $C \rightarrow C \leftarrow C$ where

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_207 F02184	207	■ 1.000	$A \rightarrowtail B \rightarrowtail C$	$A \rightarrow B \rightarrow C$	so we don't know that every two sequential morphisms $A \rightarrow B \rightarrow C$ can be composed.
fong-spivak-seven-sketches_207 F02185	207	■ 1.000	$A \rightarrowtail P \lleftarrowtail B$	$A \rightarrow P \leftarrow B$	between their respective apexes. That is, the cospan $A \rightarrow P \leftarrow B$ and $A \rightarrow P' \leftarrow B$
fong-spivak-seven-sketches_207 F02186	207	■ 1.000	$A \rightarrowtail P^{\{\prime\}} \lleftarrowtail B$	$A \rightarrow P' \leftarrow B$	between their respective apexes. That is, the cospan $A \rightarrow P \leftarrow B$ and $A \rightarrow P' \leftarrow B$
fong-spivak-seven-sketches_207 F02187	207	■ 1.000	$P \cong P^{\{\prime\}}$	$P \cong P'$	in the diagram shown left below are equivalent iff there is an isomorphism $P \cong P'$
fong-spivak-seven-sketches_207 F02188	207	■ 0.761	$\{ \}_{\text{C}}$	$\mathcal{C}$	business: $\mathbf{Cospan}_{\mathcal{C}}$ will form a category. But in fact, we are very close to getting more.
fong-spivak-seven-sketches_207 F02189	207	■ 0.743	$\operatorname{Cospan}_{\mathcal{C}}$	$\mathbf{Cospan}_{\mathcal{C}}$	If we also demand that $\mathcal{C}$ has an initial object $\emptyset$ as well, then we can upgrade $\mathbf{Cospan}_{\mathcal{C}}$
fong-spivak-seven-sketches_208 F02190	208	■ 0.839	$\mathcal{C}$	$\mathcal{C}$	$\mathbf{Cospan}_{\mathcal{C}}$ with the same objects as $\mathcal{C}$, i.e. $\operatorname{Ob}(\mathbf{Cospan}_{\mathcal{C}}) = \operatorname{Ob}(\mathcal{C})$, where the morphisms
fong-spivak-seven-sketches_208 F02191	208	■ 0.839	$\operatorname{Ob}(\mathcal{C})$	$\operatorname{Ob}(\mathcal{C})$	$\mathbf{Cospan}_{\mathcal{C}}$ with the same objects as $\mathcal{C}$, i.e. $\operatorname{Ob}(\mathbf{Cospan}_{\mathcal{C}}) = \operatorname{Ob}(\mathcal{C})$, where the morphisms
fong-spivak-seven-sketches_208 F02192	208	■ 0.839	$\operatorname{Ob}(\mathcal{C}) = \operatorname{Ob}(\mathcal{C})$	$\operatorname{Ob}(\mathcal{C}) = \operatorname{Ob}(\mathcal{C})$	$\mathbf{Cospan}_{\mathcal{C}}$ with the same objects as $\mathcal{C}$, i.e. $\operatorname{Ob}(\mathbf{Cospan}_{\mathcal{C}}) = \operatorname{Ob}(\mathcal{C})$, where the morphisms
fong-spivak-seven-sketches_208 F02193	208	■ 0.370	$\mathcal{C}, \emptyset, +$	$\mathcal{C}, \emptyset, +$	There is a symmetric monoidal structure on this category, denoted $(\mathbf{Cospan}_{\mathcal{C}}, \emptyset, +)$.
fong-spivak-seven-sketches_208 F02194	208	■ 1.000	$A + \emptyset \cong A$	$A + \emptyset \cong A$	coproduct. The coherence isomorphisms, e.g. $A + \emptyset \cong A$, can be defined in a similar
fong-spivak-seven-sketches_208 F02195	208	■ 0.699	$\mathcal{C}, \emptyset, +$	$\mathcal{C}, \emptyset, +$	It is a straightforward but time-consuming exercise to verify that $(\mathbf{Cospan}_{\mathcal{C}}, \emptyset, +)$
fong-spivak-seven-sketches_208 F02196	208	■ 0.955	$\mathbf{FinSet}$	$\mathbf{FinSet}$	symmetric monoidal category $\mathbf{Cospan}_{\mathbf{FinSet}}$. What does it look like? It looks a lot like
fong-spivak-seven-sketches_208 F02197	208	■ 1.000	$\mathbf{FinSet}$	$\mathbf{FinSet}$	The objects of $\mathbf{Cospan}_{\mathbf{FinSet}}$ are finite sets; here let's draw them as collections of $\bullet$'s.
fong-spivak-seven-sketches_208 F02198	208	■ 0.958	$A \rightarrowtail N \lleftarrowtail B$	$A \xrightarrow{f} N \xleftarrow{g} B$	six element set. Below are two depictions of a cospan $A \xrightarrow{f} N \xleftarrow{g} B$
fong-spivak-seven-sketches_208 F02199	208	■ 1.000	$g(b)$	$g(b)$	arrows from each $a \in A$ to $f(a)$ and each $b \in B$ to $g(b)$. In the depiction on the right, we
fong-spivak-seven-sketches_209 F02200	209	■ 0.904	$A+B \rightarrowtail B+C$	$A + B \rightarrow B + C$	Draw their monoidal product as a morphism $A + B \rightarrow B + C$ in $\mathbf{Cospan}_{\mathbf{FinSet}}$.
fong-spivak-seven-sketches_209 F02201	209	■ 0.904	$\operatorname{Cospan}_{\mathbf{FinSet}}$	$\mathbf{Cospan}_{\mathbf{FinSet}}$	Draw their monoidal product as a morphism $A + B \rightarrow B + C$ in $\mathbf{Cospan}_{\mathbf{FinSet}}$.
fong-spivak-seven-sketches_210 F02202	210	■ 0.992	(X, μ, η)	(X, μ, η)	To make this precise, recall from Definition 5.65 that a commutative monoid (X, μ, η)
fong-spivak-seven-sketches_210 F02205	210	■ 0.963	$X \otimes X$	$X \otimes X$	where $\bowtie$ is the symmetry on $X \otimes X$. A cocommutative comonoid (X, δ, ϵ) is an
fong-spivak-seven-sketches_210 F02206	210	■ 0.963	(X, δ, ϵ)	(X, δ, ϵ)	where $\bowtie$ is the symmetry on $X \otimes X$. A cocommutative comonoid (X, δ, ϵ) is an
fong-spivak-seven-sketches_210 F02207	210	■ 1.000	$\delta: X \rightarrowtail X \otimes X, \epsilon: X \rightarrowtail I$	$\delta: X \rightarrow X \otimes X, \epsilon: X \rightarrow I$	object X with maps $\delta: X \rightarrowtail X \otimes X, \epsilon: X \rightarrowtail I$, obeying the mirror images of the laws in
fong-spivak-seven-sketches_210 F02208	210	■ 1.000	μ, η, δ	μ, η, δ	comonoid, and consider a wiring diagram built only from the icons μ, η, δ , and ϵ ,
fong-spivak-seven-sketches_210 F02209	210	■ 1.000	ϵ	ϵ	comonoid, and consider a wiring diagram built only from the icons μ, η, δ , and ϵ ,
fong-spivak-seven-sketches_210 F02210	210	■ 1.000	μ	μ	how the morphisms μ and δ , the merger and the splitter, interact.
fong-spivak-seven-sketches_210 F02211	210	■ 1.000	δ	δ	how the morphisms μ and δ , the merger and the splitter, interact.
fong-spivak-seven-sketches_210 F02212	210	■ 0.953	$(\mathcal{C}, \otimes, I)$	$(\mathcal{C}, \otimes, I)$	Definition 6.52. Let X be an object in a symmetric monoidal category $(\mathcal{C}, \otimes, I)$. A Frobe-

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_1 F02213	210	0.574	$(\mu, \eta, \delta, \epsilon)$	$(\mu, \eta, \delta, \epsilon)$	<i>nus structure</i> on X consists of a 4-tuple $(\mu, \eta, \delta, \epsilon)$ such that (X, μ, η) is a commutative
fong-spivak-seven-sketches_1 F02214	211	1.000	$X^{\otimes m} \rightarrow X^{\otimes n}$	$X^{\otimes m} \rightarrow X^{\otimes n}$	With these two equations, it turns out that two morphisms $X^{\otimes m} \rightarrow X^{\otimes n}$ —defined
fong-spivak-seven-sketches_1 F02215	211	0.961	μ, δ	μ, δ	by composing and tensoring identities on X and the noteworthy morphisms μ, δ , etc.—
fong-spivak-seven-sketches_1 F02216	211	0.508	$X, \mu, \eta, \delta, \epsilon$	$X, \mu, \eta, \delta, \epsilon$	Definition 6.54. Let $(X, \mu, \eta, \delta, \epsilon)$ be a Frobenius monoid in a monoidal category
fong-spivak-seven-sketches_1 F02217	211	0.999	$s_{m,n}: X^{\otimes m} \rightarrow X^{\otimes n}$	$s_{m,n}: X^{\otimes m} \rightarrow X^{\otimes n}$	$(\mathcal{C}, I, \otimes)$. Let $m, n \in \mathbb{N}$. Define $s_{m,n}: X^{\otimes m} \rightarrow X^{\otimes n}$ to be the following morphism
fong-spivak-seven-sketches_1 F02218	211	0.799	$(m-1) \mu$	$(m-1)\mu$	It can be written formally as $(m-1) \mu$ ’s followed by $(n-1) \delta$ ’s, with special cases when
fong-spivak-seven-sketches_1 F02219	211	0.799	$(n-1) \delta$	$(n-1)\delta$	It can be written formally as $(m-1) \mu$ ’s followed by $(n-1) \delta$ ’s, with special cases when
fong-spivak-seven-sketches_1 F02220	211	1.000	$m=0$	$m=0$	$m=0$ or $n=0$.
fong-spivak-seven-sketches_1 F02221	211	0.892	$s_{m,n}$	$s_{m,n}$	We call $s_{m,n}$ the <i>spider of type (m, n)</i> , and can draw it more simply as the icon
fong-spivak-seven-sketches_1 F02222	211	1.000	$\mu, \eta, \delta, \epsilon, \sigma$	$\mu, \eta, \delta, \epsilon, \sigma$	talking, if a wire is spiderable, it means that we have a bunch of morphisms $\mu, \eta, \delta, \epsilon, \sigma$
fong-spivak-seven-sketches_1 F02223	211	0.781	$(X, \mu, \eta, \delta, \epsilon)$	$(X, \mu, \eta, \delta, \epsilon)$	Theorem 6.55. Let $(X, \mu, \eta, \delta, \epsilon)$ be a Frobenius monoid in a monoidal category $(\mathcal{C}, I, \otimes)$.
fong-spivak-seven-sketches_1 F02224	211	1.000	$f: X^{\otimes m} \rightarrow X^{\otimes n}$	$f: X^{\otimes m} \rightarrow X^{\otimes n}$	Suppose that we have a map $f: X^{\otimes m} \rightarrow X^{\otimes n}$ each constructed from spiders and the
fong-spivak-seven-sketches_1 F02225	211	1.000	$\sigma: X^{\otimes 2} \rightarrow X^{\otimes 2}$	$\sigma: X^{\otimes 2} \rightarrow X^{\otimes 2}$	symmetry map $\sigma: X^{\otimes 2} \rightarrow X^{\otimes 2}$ using composition and the monoidal product, and such
fong-spivak-seven-sketches_1 F02226	211	1.000	$f=s_{m,n}$	$f=s_{m,n}$	$f=s_{m,n}$.
fong-spivak-seven-sketches_1 F02227	211	1.000	$X \otimes X \rightarrow X \otimes X \otimes X$	$X \otimes X \rightarrow X \otimes X \otimes X$	morphisms $X \otimes X \rightarrow X \otimes X \otimes X$ in the following list are necessarily equal?
fong-spivak-seven-sketches_1 F02228	212	0.937	$G:=\{\mu, \eta, \delta, \epsilon\}$	$G := \{\mu, \eta, \delta, \epsilon\}$	Theorem 6.58. Consider the four-element set $G := \{\mu, \eta, \delta, \epsilon\}$ and define $in, out: G \rightarrow$
fong-spivak-seven-sketches_1 F02229	212	0.937	$G \rightarrow$	$G \rightarrow$	Theorem 6.58. Consider the four-element set $G := \{\mu, \eta, \delta, \epsilon\}$ and define $in, out: G \rightarrow$
fong-spivak-seven-sketches_1 F02230	212	0.976	(G, E)	(G, E)	Then the free prop on (G, E) is equivalent, as a symmetric monoidal category, ^a to
fong-spivak-seven-sketches_1 F02231	213	0.985	$X, \mu_X, \delta_X, \eta_X, \epsilon_X$	$X, \mu_X, \delta_X, \eta_X, \epsilon_X$	which each object X is equipped with a Frobenius structure $(X, \mu_X, \delta_X, \eta_X, \epsilon_X)$ such
fong-spivak-seven-sketches_1 F02232	214	0.991	$\eta_I=\mathrm{id}_I=\epsilon_I$	$\eta_I = \mathrm{id}_I = \epsilon_I$	for all objects X, Y , and such that $\eta_I = \mathrm{id}_I = \epsilon_I$.
fong-spivak-seven-sketches_1 F02233	214	0.978	$\mu_X, \delta_X, \eta_X, \epsilon_X$	$\mu_X, \delta_X, \eta_X, \epsilon_X$	Frobenius morphisms $\mu_X, \delta_X, \eta_X, \epsilon_X$ for each object X are constructed using the uni-
fong-spivak-seven-sketches_1 F02234	214	0.794	$\underline{1}$	$\underline{1}$	ject $\underline{1}$ in $\mathbf{Cospan}_{\mathbf{FinSet}}$ using both the function and wiring depictions as in Example 6.46.
fong-spivak-seven-sketches_1 F02235	214	0.792	$\varnothing, \sqcup$	$\varnothing, \sqcup$	<i>Example 6.64.</i> Recall the monoidal category $(\mathbf{Corel}, \varnothing, \sqcup)$ from Example 4.61; its objects
fong-spivak-seven-sketches_1 F02236	214	0.578	$\mu_X: X \sqcup X \rightarrow X$	$\mu_X: X \sqcup X \rightarrow X$	corelation $\mu_X: X \sqcup X \rightarrow X$ such that two elements of $X \sqcup X \sqcup X$ are equivalent if and
fong-spivak-seven-sketches_1 F02237	214	0.578	$X \sqcup X \sqcup X$	$X \sqcup X \sqcup X$	corelation $\mu_X: X \sqcup X \rightarrow X$ such that two elements of $X \sqcup X \sqcup X$ are equivalent if and
fong-spivak-seven-sketches_1 F02238	214	1.000	$\delta_X: X \rightarrow X \sqcup X$	$\delta_X: X \rightarrow X \sqcup X$	only if they come from the same underlying element of X . Define $\delta_X: X \rightarrow X \sqcup X$ in
fong-spivak-seven-sketches_1 F02239	214	1.000	$\eta_X: \varnothing \rightarrow X$	$\eta_X: \varnothing \rightarrow X$	the same way, and define $\eta_X: \varnothing \rightarrow X$ and $\epsilon_X: X \rightarrow \varnothing$ such that no two elements of

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_202240	214	■ 1.000	$\epsilon_X: X \rightarrow \varnothing$	$\epsilon_X: X \rightarrow \varnothing$	the same way, and define $\eta_X: \varnothing \rightarrow X$ and $\epsilon_X: X \rightarrow \varnothing$ such that no two elements of
fong-spivak-seven-sketches_202241	214	■ 1.000	$X = \varnothing \sqcup X = X \sqcup \varnothing$	$X = \varnothing \sqcup X = X \sqcup \varnothing$	$X = \varnothing \sqcup X = X \sqcup \varnothing$ are equivalent.
fong-spivak-seven-sketches_202242	214	■ 1.000	$(X, \mu_X, \eta_X, \delta_X, \epsilon_X)$	$(X, \mu_X, \eta_X, \delta_X, \epsilon_X)$	These maps define a special commutative Frobenius monoid $(X, \mu_X, \eta_X, \delta_X, \epsilon_X)$,
fong-spivak-seven-sketches_202243	216	■ 0.384	$(\mathcal{C}, I_{\mathcal{C}}, \otimes_{\mathcal{C}})$	$(\mathcal{C}, I_{\mathcal{C}}, \otimes_{\mathcal{C}})$	Rough Definition 6.68. Let $(\mathcal{C}, I_{\mathcal{C}}, \otimes_{\mathcal{C}})$ and $(\mathcal{D}, I_{\mathcal{D}}, \otimes_{\mathcal{D}})$ be symmetric monoidal cate-
fong-spivak-seven-sketches_202244	216	■ 0.384	$(\mathcal{D}, I_{\mathcal{D}}, \otimes_{\mathcal{D}})$	$(\mathcal{D}, I_{\mathcal{D}}, \otimes_{\mathcal{D}})$	Rough Definition 6.68. Let $(\mathcal{C}, I_{\mathcal{C}}, \otimes_{\mathcal{C}})$ and $(\mathcal{D}, I_{\mathcal{D}}, \otimes_{\mathcal{D}})$ be symmetric monoidal cate-
fong-spivak-seven-sketches_202245	216	■ 0.994	(F, φ)	(F, φ)	gories. To specify a <i>symmetric monoidal functor</i> (F, φ) between them,
fong-spivak-seven-sketches_202246	216	■ 0.984	$\varphi_I: I_{\mathcal{D}} \rightarrow F(I_{\mathcal{C}})$	$\varphi_I: I_{\mathcal{D}} \rightarrow F(I_{\mathcal{C}})$	(ii) one specifies a morphism $\varphi_I: I_{\mathcal{D}} \rightarrow F(I_{\mathcal{C}})$.
fong-spivak-seven-sketches_202247	216	■ 0.899	$c_1, c_2 \in \text{Ob}(\mathcal{C})$	$c_1, c_2 \in \text{Ob}(\mathcal{C})$	(iii) for each $c_1, c_2 \in \text{Ob}(\mathcal{C})$, one specifies a morphism
fong-spivak-seven-sketches_202248	216	■ 1.000	φ	φ	We call the various maps φ <i>coherence maps</i> . We require the coherence maps to obey
fong-spivak-seven-sketches_202249	216	■ 1.000	φ_I	φ_I	monoidal structures on $\mathcal{C}$ and $\mathcal{D}$. If φ_I and φ_{c_1, c_2} are isomorphisms for all c_1, c_2 , we say
fong-spivak-seven-sketches_202250	216	■ 1.000	φ_{c_1, c_2}	φ_{c_1, c_2}	monoidal structures on $\mathcal{C}$ and $\mathcal{D}$. If φ_I and φ_{c_1, c_2} are isomorphisms for all c_1, c_2 , we say
fong-spivak-seven-sketches_202251	216	■ 1.000	c_1, c_2	c_1, c_2	monoidal structures on $\mathcal{C}$ and $\mathcal{D}$. If φ_I and φ_{c_1, c_2} are isomorphisms for all c_1, c_2 , we say
fong-spivak-seven-sketches_202252	216	■ 1.000	$S \in \mathbf{Set}$	$S \in$	sending a set $S \in \mathbf{Set}$ to its set of subsets $P(S) := \{R \subseteq S\}$. It acts on morphisms by
fong-spivak-seven-sketches_202253	216	■ 1.000	$\mathbf{P}(S) := \{R \subseteq S\}$	$P(S) := \{R \subseteq S\}$	sending a set $S \in \mathbf{Set}$ to its set of subsets $P(S) := \{R \subseteq S\}$. It acts on morphisms by
fong-spivak-seven-sketches_202254	216	■ 1.000	$\text{im}_f: \mathbf{P}(S) \rightarrow \mathbf{P}(T)$	$\text{im}_f: P(S) \rightarrow P(T)$	sending a function $f: S \rightarrow T$ to the image map $\text{im}_f: P(S) \rightarrow P(T)$, which maps $R \subseteq S$
fong-spivak-seven-sketches_202255	216	■ 1.000	$R \subseteq S$	$R \subseteq S$	sending a function $f: S \rightarrow T$ to the image map $\text{im}_f: P(S) \rightarrow P(T)$, which maps $R \subseteq S$
fong-spivak-seven-sketches_202256	216	■ 1.000	$\{f(r) \mid r \in R\} \subseteq T$	$\{f(r) \mid r \in R\} \subseteq T$	to $\{f(r) \mid r \in R\} \subseteq T$.
fong-spivak-seven-sketches_202257	216	■ 1.000	$(\{1\}, \times)$	$(\{1\}, \times)$	Now consider the symmetric monoidal structure $(\{1\}, \times)$ on $\mathbf{Set}$ from Example 4.49.
fong-spivak-seven-sketches_202258	216	■ 1.000	$\varphi_I: \{1\} \rightarrow$	$\varphi_I: \{1\} \rightarrow$	To make $\mathbf{P}$ a symmetric monoidal functor, we need to specify a function $\varphi_I: \{1\} \rightarrow$
fong-spivak-seven-sketches_202259	216	■ 0.997	$\mathbf{P}(\{1\})$	$P(\{1\})$	$P(\{1\})$ and for all sets S and T , a functor $\varphi_{S, T}: P(S) \times P(T) \rightarrow P(S \times T)$. One possibility is
fong-spivak-seven-sketches_202260	216	■ 0.997	$\varphi_{S, T}: \mathbf{P}(S) \times \mathbf{P}(T) \rightarrow \mathbf{P}(S \times T)$	$\varphi_{S, T}: P(S) \times P(T) \rightarrow P(S \times T)$	$P(\{1\})$ and for all sets S and T , a functor $\varphi_{S, T}: P(S) \times P(T) \rightarrow P(S \times T)$. One possibility is
fong-spivak-seven-sketches_202261	216	■ 1.000	$\varphi_I(1)$	$\varphi_I(1)$	to define $\varphi_I(1)$ to be the maximal subset $\{1\} \subseteq \{1\}$, and given subsets $A \subseteq S$ and $B \subseteq T$,
fong-spivak-seven-sketches_202262	216	■ 1.000	$\{1\} \subseteq \{1\}$	$\{1\} \subseteq \{1\}$	to define $\varphi_I(1)$ to be the maximal subset $\{1\} \subseteq \{1\}$, and given subsets $A \subseteq S$ and $B \subseteq T$,
fong-spivak-seven-sketches_202263	216	■ 1.000	$A \subseteq S$	$A \subseteq S$	to define $\varphi_I(1)$ to be the maximal subset $\{1\} \subseteq \{1\}$, and given subsets $A \subseteq S$ and $B \subseteq T$,
fong-spivak-seven-sketches_202264	216	■ 1.000	$B \subseteq T$	$B \subseteq T$	to define $\varphi_I(1)$ to be the maximal subset $\{1\} \subseteq \{1\}$, and given subsets $A \subseteq S$ and $B \subseteq T$,
fong-spivak-seven-sketches_202265	216	■ 1.000	$\varphi_{S, T}(A, B)$	$\varphi_{S, T}(A, B)$	to define $\varphi_{S, T}(A, B)$ to be the product subset $A \times B \subseteq S \times T$. With these definitions,
fong-spivak-seven-sketches_202266	216	■ 1.000	$A \times B \subseteq S \times T$	$A \times B \subseteq S \times T$	to define $\varphi_{S, T}(A, B)$ to be the product subset $A \times B \subseteq S \times T$. With these definitions,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F02267	216	■ 0.957	(P, φ)	(P, φ)	(P, φ) is a symmetric monoidal functor.
fong-spivak-seven-sketches-F02268	216	■ 1.000	$\varphi_{S, T}$	$\varphi_{S, T}$	<i>Exercise 6.70.</i> Check that the maps $\varphi_{S, T}$ defined in Example 6.69 are natural in S and
fong-spivak-seven-sketches-F02269	217	■ 1.000	$2 \ \Omega$	2Ω	as a resistor of resistance 2Ω). For this circuit to be a morphism in some category, i.e.
fong-spivak-seven-sketches-F02270	217	■ 1.000	$A \rightarrowtail N$	$A \rightarrow N$	of the functions $A \rightarrow N$ and $B \rightarrow N$ are indicated by the grey arrows. This forms a
fong-spivak-seven-sketches-F02271	217	■ 1.000	$B \rightarrowtail N$	$B \rightarrow N$	of the functions $A \rightarrow N$ and $B \rightarrow N$ are indicated by the grey arrows. This forms a
fong-spivak-seven-sketches-F02272	218	■ 1.000	$(F, \varphi) : (\mathcal{C}, +) \rightarrowtail$	$(F, \varphi) : (\mathcal{C}, +) \longrightarrow$	Definition 6.75. Let $\mathcal{C}$ be a category with finite colimits, and $(F, \varphi) : (\mathcal{C}, +) \longrightarrow$
fong-spivak-seven-sketches-F02273	218	■ 1.000	$\times)$	$\times)$	Definition 6.75. Let $\mathcal{C}$ be a category with finite colimits, and $(F, \varphi) : (\mathcal{C}, +) \longrightarrow (\mathbf{Set}, \times)$
fong-spivak-seven-sketches-F02274	218	■ 0.672	$A \rightarrowtail N \stackrel{o}{\leftarrow} B$	$A \xrightarrow{i} N \xleftarrow{o} B$	$A \xrightarrow{i} N \xleftarrow{o} B$ in $\mathcal{C}$ together with an element $s \in F(N)$. ⁵ We call (F, φ) the <i>decoration</i>
fong-spivak-seven-sketches-F02275	218	■ 0.672	$s \in F(N)$.	$s \in F(N)$. ⁵	$A \xrightarrow{i} N \xleftarrow{o} B$ in $\mathcal{C}$ together with an element $s \in F(N)$. ⁵ We call (F, φ) the <i>decoration</i>
fong-spivak-seven-sketches-F02276	218	■ 0.596	$\mathcal{C} =$	$\mathcal{C} =$	The intuition here is to use $\mathcal{C} = \mathbf{FinSet}$, and, for each object $N \in \mathbf{FinSet}$, the functor
fong-spivak-seven-sketches-F02277	218	■ 0.596	$N \in$	$N \in$	The intuition here is to use $\mathcal{C} = \mathbf{FinSet}$, and, for each object $N \in \mathbf{FinSet}$, the functor
fong-spivak-seven-sketches-F02278	218	■ 1.000	$F(N)$	$F(N)$	available decorations on N nodes, taken from the set $F(N)$.
fong-spivak-seven-sketches-F02279	218	■ 0.980	$(A \rightarrowtail N \stackrel{g}{\leftarrow} B, s)$	$(A \xrightarrow{f} N \xleftarrow{g} B, s)$	Let $(A \xrightarrow{f} N \xleftarrow{g} B, s)$ and $(B \xrightarrow{h} P \xleftarrow{k} C, t)$ represent decorated cospans. Their
fong-spivak-seven-sketches-F02280	218	■ 0.980	$(B \rightarrowtail P \stackrel{k}{\leftarrow} C, t)$	$(B \xrightarrow{h} P \xleftarrow{k} C, t)$	Let $(A \xrightarrow{f} N \xleftarrow{g} B, s)$ and $(B \xrightarrow{h} P \xleftarrow{k} C, t)$ represent decorated cospans. Their
fong-spivak-seven-sketches-F02281	218	■ 1.000	$B \rightarrowtail P \stackrel{k}{\leftarrow} C$	$B \xrightarrow{h} P \xleftarrow{k} C$	composite is represented by the composite of the cospans $A \xrightarrow{f} N \xleftarrow{g} B$ and $B \xrightarrow{h} P \xleftarrow{k} C$,
fong-spivak-seven-sketches-F02282	219	■ 0.998	$F(N +_B P)$	$F(N +_B P)$	paired with the following element of $F(N +_B P)$:
fong-spivak-seven-sketches-F02283	219	■ 0.875	F	F	tor $(F, \varphi) : (\mathcal{C}, +) \longrightarrow (\mathbf{Set}, \times)$, there is a hypergraph category $\mathbf{Cospan}_F$ whose objects
fong-spivak-seven-sketches-F02284	219	■ 0.581	C	$\mathbf{C}$	$\mathbf{Cospan}_{\mathbf{C}}$.
fong-spivak-seven-sketches-F02285	219	■ 0.876	e	$\mathbf{e}$	<i>Exercise 6.78.</i> Suppose you’re worried that the notation $\mathbf{Cospan}_{\mathbf{C}}$ looks like the
fong-spivak-seven-sketches-F02286	219	■ 0.579	Cospan_F	$\mathbf{Cospan}_F$	notation $\mathbf{Cospan}_F$, even though they’re very different. An expert tells you “they’re not
fong-spivak-seven-sketches-F02287	219	■ 1.000	$F(c) := \{*\}$	$F(c) := \{*\}$	so different; one is a special case of the other. Just use the constant functor $F(c) := \{*\}$.”
fong-spivak-seven-sketches-F02288	219	■ 1.000	$\mathbb{R}^+$	$\mathbb{R}^+$	$\mathbb{R}^+$. But we write $\mathbf{C}$ this way to remind us that it consists of circuit components. If we
fong-spivak-seven-sketches-F02289	219	■ 1.000	$\ell : A \rightarrowtail C$	$\ell : A \rightarrow C$	source and target functions, together with a function $\ell : A \rightarrow C$ labeling each edge with
fong-spivak-seven-sketches-F02290	219	■ 1.000	$V = \{1, 2\}, A = \{e\}, s(e) = 1$	$V = \{1, 2\}, A = \{e\}, s(e) = 1$	For example, we might have the simple case of $V = \{1, 2\}, A = \{e\}, s(e) = 1$,
fong-spivak-seven-sketches-F02291	219	■ 0.562	$t(e) = 2$	$t(e) = 2$	$t(e) = 2$ —so e is an edge from 1 to 2—and $\ell(e) = 3\Omega$. This represents a resistor with
fong-spivak-seven-sketches-F02292	219	■ 0.562	$\ell(e) = 3 \ \Omega$	$\ell(e) = 3\Omega$	$t(e) = 2$ —so e is an edge from 1 to 2—and $\ell(e) = 3\Omega$. This represents a resistor with

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F02293	219	■ 1.000	$3 \ \Omega$	3Ω	resistance 3Ω :
fong-spivak-seven-sketches-F02294	220	■ 1.000	$V=\{1,2\}, \ A=\{e\}$	$V = \{1, 2\}, A = \{e\}$	resistor could also be represented by the ‘reversed graph’, with data $V = \{1, 2\}, A = \{e\}$,
fong-spivak-seven-sketches-F02295	220	■ 1.000	$s(e)=2, \ t(e)=1$	$s(e) = 2, t(e) = 1$	$s(e) = 2, t(e) = 1$, and $\ell(e) = 3F$.
fong-spivak-seven-sketches-F02296	220	■ 1.000	$\ell(e)=3 \ F$	$\ell(e) = 3F$	$s(e) = 2, t(e) = 1$, and $\ell(e) = 3F$.
fong-spivak-seven-sketches-F02297	220	■ 0.983	$V, \ A, \ s, \ t, \ \ell$	V, A, s, t, ℓ	<i>Exercise 6.79.</i> Write a tuple (V, A, s, t, ℓ) that represents the circuit in Eq. (6.71).
fong-spivak-seven-sketches-F02298	220	■ 0.951	ψ	ψ	(Circ, ψ) . We start by defining the functor part
fong-spivak-seven-sketches-F02299	220	■ 1.000	$f: V \rightarrow V^{\prime}$	$f : V \rightarrow V'$	On morphisms, Circ sends a function $f: V \rightarrow V'$ to the function
fong-spivak-seven-sketches-F02300	220	■ 0.693	$c \in \operatorname{Circ}(\underline{4})$	$c \in \text{Circ}(\underline{4})$	<i>Exercise 6.80.</i> To understand this functor better, let $c \in \text{Circ}(\underline{4})$ be the circuit
fong-spivak-seven-sketches-F02301	220	■ 1.000	$f: \underline{4} \rightarrow \underline{3}$	$f : \underline{4} \rightarrow \underline{3}$	and let $f: \underline{4} \rightarrow \underline{3}$ be the function
fong-spivak-seven-sketches-F02302	220	■ 1.000	$\operatorname{Circ}(f)(c)$	$\text{Circ}(f)(c)$	Draw a picture of the circuit $\text{Circ}(f)(c)$.
fong-spivak-seven-sketches-F02303	220	■ 1.000	$\psi_{V, V^{\prime}}$	$\psi_{V, V'}$	coherence maps $\psi_{V, V'}$ for finite sets V, V' , we define
fong-spivak-seven-sketches-F02304	220	■ 1.000	$V, V^{\prime}$	V, V'	coherence maps $\psi_{V, V'}$ for finite sets V, V' , we define
fong-spivak-seven-sketches-F02305	220	■ 1.000	$V+V^{\prime}$	$V + V'$	considers them together as a circuit on the disjoint union of vertices $V + V'$.
fong-spivak-seven-sketches-F02306	220	■ 1.000	$\psi_{\underline{2}, \underline{2}}(b, s) \in \operatorname{Circ}(\underline{2}+\underline{2})=\operatorname{Circ}(\underline{4})$	$\psi_{\underline{2}, \underline{2}}(b, s) \in \text{Circ}(\underline{2} + \underline{2}) = \text{Circ}(\underline{4})$	$\psi_{\underline{2}, \underline{2}}(b, s) \in \text{Circ}(\underline{2} + \underline{2}) = \text{Circ}(\underline{4})$ should be, and draw a picture.
fong-spivak-seven-sketches-F02307	221	■ 0.339	$($	$($	$(\text{Circ}, \psi): (\mathbf{FinSet}, +) \rightarrow (\mathbf{Set}, \times)$, we can construct our promised hypergraph category
fong-spivak-seven-sketches-F02308	221	■ 0.339	$\psi): ($	$\psi) : ($	$(\text{Circ}, \psi): (\mathbf{FinSet}, +) \rightarrow (\mathbf{Set}, \times)$, we can construct our promised hypergraph category
fong-spivak-seven-sketches-F02309	221	■ 0.339	$) \rightarrow ($	$) \rightarrow ($	$(\text{Circ}, \psi): (\mathbf{FinSet}, +) \rightarrow (\mathbf{Set}, \times)$, we can construct our promised hypergraph category
fong-spivak-seven-sketches-F02310	221	■ 0.945	$\{ \}_{\text{Circ}}$	Circ	Our notation for this category is $\mathbf{Cospan}_{\text{Circ}}$. Following Theorem 6.77, the objects
fong-spivak-seven-sketches-F02311	221	■ 0.147	$\underline{2}$	$\underline{2}$	collections of white circles $\circ$. For example, we’ll represent the object $\underline{2}$ of $\mathbf{Cospan}_{\text{Circ}}$ as
fong-spivak-seven-sketches-F02312	221	■ 0.147	$_{\text{Circ}}$	Circ	collections of white circles $\circ$. For example, we’ll represent the object $\underline{2}$ of $\mathbf{Cospan}_{\text{Circ}}$ as
fong-spivak-seven-sketches-F02313	221	■ 0.999	$\underline{m} \rightarrow \underline{p} \leftarrow \underline{n}$	$\underline{m} \rightarrow \underline{p} \leftarrow \underline{n}$	that is, $\text{cospan } \underline{m} \rightarrow \underline{p} \leftarrow \underline{n}$ together with an element c of $\text{Circ}(\underline{p})$. As an example,
fong-spivak-seven-sketches-F02314	221	■ 0.999	$\operatorname{Circ}(\underline{p})$	$\text{Circ}(\underline{p})$	that is, $\text{cospan } \underline{m} \rightarrow \underline{p} \leftarrow \underline{n}$ together with an element c of $\text{Circ}(\underline{p})$. As an example,
fong-spivak-seven-sketches-F02315	221	■ 1.000	$\underline{1} \xrightarrow{i_1} \underline{2} \xleftarrow{i_2} \underline{1}$	$\underline{1} \xrightarrow{i_1} \underline{2} \xleftarrow{i_2} \underline{1}$	consider the cospan $\underline{1} \xrightarrow{i_1} \underline{2} \xleftarrow{i_2} \underline{1}$ where $i_1(1) = 1$ and $i_2(1) = 2$, equipped with the
fong-spivak-seven-sketches-F02316	221	■ 1.000	$i_1(1)=1$	$i_1(1) = 1$	consider the cospan $\underline{1} \xrightarrow{i_1} \underline{2} \xleftarrow{i_2} \underline{1}$ where $i_1(1) = 1$ and $i_2(1) = 2$, equipped with the
fong-spivak-seven-sketches-F02317	221	■ 1.000	$i_2(1)=2$	$i_2(1) = 2$	consider the cospan $\underline{1} \xrightarrow{i_1} \underline{2} \xleftarrow{i_2} \underline{1}$ where $i_1(1) = 1$ and $i_2(1) = 2$, equipped with the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F02318	221	0.760	$(V, A, s, t, \ell) \in \operatorname{Circ}(\underline{2})$	$(V, A, s, t, \ell) \in \operatorname{Circ}(2)$	some C -circuit $(V, A, s, t, \ell) \in \operatorname{Circ}(2)$ What is it?
fong-spivak-seven-sketches-F02319	221	0.578	$\underline{1} \rightarrow \underline{1}$	$\underline{1} \rightarrow \underline{1}$	Since this and the circuit in (6.83) are both morphisms $\underline{1} \rightarrow \underline{1}$, we may compose them
fong-spivak-seven-sketches-F02320	221	1.000	$\operatorname{Circ}(\left[\iota_N, \iota_P\right])(\psi_{N,P}(s, t))$	$\operatorname{Circ}([\iota_N, \iota_P])(\psi_{N,P}(s, t))$	new decoration, we use the formula $\operatorname{Circ}([\iota_N, \iota_P])(\psi_{N,P}(s, t))$ from (6.76). Again, this is
fong-spivak-seven-sketches-F02321	222	0.962	$\underline{1} = \{\circ\}$	$\underline{1} = \{\circ\}$	(We simply ignore the decorations for now.) If we pushout over the common set $\underline{1} = \{\circ\}$,
fong-spivak-seven-sketches-F02322	222	0.307	$\psi_{2,2}, \underline{2}$	$\psi_{2,2}$	new decoration. You told us what the map $\psi_{2,2}$ does in Exercise 6.82. It takes the two
fong-spivak-seven-sketches-F02323	222	0.834	$\operatorname{Circ}(\underline{4})$	$\operatorname{Circ}(\underline{4})$	in $\operatorname{Circ}(\underline{4})$. So this is what the $\psi_{N,P}(s, t)$ part means. What does the $[\iota_N, \iota_P]$ mean? Recall
fong-spivak-seven-sketches-F02324	222	0.834	$\psi_{N,P}(s, t)$	$\psi_{N,P}(s, t)$	in $\operatorname{Circ}(\underline{4})$. So this is what the $\psi_{N,P}(s, t)$ part means. What does the $[\iota_N, \iota_P]$ mean? Recall
fong-spivak-seven-sketches-F02325	222	0.834	$[\iota_N, \iota_P]$	$[\iota_N, \iota_P]$	in $\operatorname{Circ}(\underline{4})$. So this is what the $\psi_{N,P}(s, t)$ part means. What does the $[\iota_N, \iota_P]$ mean? Recall
fong-spivak-seven-sketches-F02326	222	0.963	$\operatorname{Cosp}_{\operatorname{Circ}}$	$\operatorname{Cosp}_{\operatorname{Circ}}$	diagram as morphisms in $\operatorname{Cosp}_{\operatorname{Circ}}$, and compute their composite. Does it match the
fong-spivak-seven-sketches-F02327	222	0.930	$A \rightarrow M \leftarrow B$	$A \rightarrow M \leftarrow B$	we again just work in $\operatorname{Cosp}_{\operatorname{FinSet}}$: given two cospans $A \rightarrow M \leftarrow B$ and $C \rightarrow N \leftarrow D$,
fong-spivak-seven-sketches-F02328	222	0.930	$C \rightarrow N \leftarrow D$	$C \rightarrow N \leftarrow D$	we again just work in $\operatorname{Cosp}_{\operatorname{FinSet}}$: given two cospans $A \rightarrow M \leftarrow B$ and $C \rightarrow N \leftarrow D$,
fong-spivak-seven-sketches-F02329	222	1.000	$A + C \rightarrow M + N \leftarrow B + D$	$A + C \rightarrow M + N \leftarrow B + D$	we take their coproduct cospan $A + C \rightarrow M + N \leftarrow B + D$. And for decorations, we
fong-spivak-seven-sketches-F02330	222	0.997	$\psi_{M,N} : \operatorname{Circ}(M) \times \operatorname{Circ}(N) \rightarrow \operatorname{Circ}(M + N)$	$\psi_{M,N} : \operatorname{Circ}(M) \times \operatorname{Circ}(N) \rightarrow \operatorname{Circ}(M + N)$	use the map $\psi_{M,N} : \operatorname{Circ}(M) \times \operatorname{Circ}(N) \rightarrow \operatorname{Circ}(M + N)$. So, for example, suppose we
fong-spivak-seven-sketches-F02331	223	1.000	$\eta : 0 \rightarrow 2$	$\eta : 0 \rightarrow 2$	Exercise 6.88. Write x for the open circuit in (6.87). Also define cospans $\eta : 0 \rightarrow 2$ and
fong-spivak-seven-sketches-F02332	223	0.915	$\eta : 2 \rightarrow 0$	$\eta : 2 \rightarrow 0$	$\eta : 2 \rightarrow 0$ as follows:
fong-spivak-seven-sketches-F02333	223	0.967	$(\underline{1}, \emptyset, !, !, !) \in \operatorname{Circ}(\underline{1})$	$(\underline{1}, \emptyset, !, !, !) \in \operatorname{Circ}(\underline{1})$	where each of these are decorated by the empty circuit $(\underline{1}, \emptyset, !, !, !) \in \operatorname{Circ}(\underline{1})$.
fong-spivak-seven-sketches-F02334	223	0.782	$\eta ; x ; \epsilon$	$\eta ; x ; \epsilon$	Compute the composite $\eta ; x ; \epsilon$ in $\operatorname{Cosp}_{\operatorname{Circ}}$. This is a morphism $\underline{0} \rightarrow \underline{0}$; we call
fong-spivak-seven-sketches-F02335	223	0.782	$\underline{0} \rightarrow \underline{0}$	$\underline{0} \rightarrow \underline{0}$	Compute the composite $\eta ; x ; \epsilon$ in $\operatorname{Cosp}_{\operatorname{Circ}}$. This is a morphism $\underline{0} \rightarrow \underline{0}$; we call
fong-spivak-seven-sketches-F02336	223	0.574	$\mathcal{O}$	$\mathcal{O}$	structures from operad functors $\mathcal{O} \rightarrow \mathbf{Set}$.
fong-spivak-seven-sketches-F02337	225	1.000	f, g, h, k	f, g, h, k	The cost is that the ‘boxes’ f, g, h, k in Eq. (6.90) no longer have a left/right dis-
fong-spivak-seven-sketches-F02338	225	0.997	$\mathcal{M} \rightarrow$	$\mathcal{M} \rightarrow$	called the theory of monoids as a prop $\mathcal{M}$, and define monoids using functors $\mathcal{M} \rightarrow \mathbf{Set}$;
fong-spivak-seven-sketches-F02339	225	0.992	$\operatorname{FinSet} \rightarrow$	$\operatorname{FinSet} \rightarrow$	$\operatorname{Cosp}_{\operatorname{FinSet}} \rightarrow \mathbf{Set}$.
fong-spivak-seven-sketches-F02340	225	0.725	$t_1, \dots, t_n, t$	$t_1, \dots, t_n, t$	(ii) for each tuple $(t_1, \dots, t_n, t)$ of types, one specifies a set $\mathcal{O}(t_1, \dots, t_n; t)$, whose
fong-spivak-seven-sketches-F02341	225	0.725	$\mathcal{O}(t_1, \dots, t_n; t)$	$\mathcal{O}(t_1, \dots, t_n; t)$	(ii) for each tuple $(t_1, \dots, t_n, t)$ of types, one specifies a set $\mathcal{O}(t_1, \dots, t_n; t)$, whose
fong-spivak-seven-sketches-F02342	225	0.623	$(t_1, \dots, t_n; t)$	$(t_1, \dots, t_n; t)$	elements are called <i>operations of arity</i> $(t_1, \dots, t_n; t)$;
fong-spivak-seven-sketches-F02343	225	0.988	$(s_1, \dots, s_m, t_i)$	$(s_1, \dots, s_m, t_i)$	(iii) for each pair of tuples $(s_1, \dots, s_m, t_i)$ and $(t_1, \dots, t_n, t)$, one specifies a function
fong-spivak-seven-sketches-F02344	225	0.988	$(t_1, \dots, t_n, t)$	$(t_1, \dots, t_n, t)$	(iii) for each pair of tuples $(s_1, \dots, s_m, t_i)$ and $(t_1, \dots, t_n, t)$, one specifies a function

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches- F02345	225	0.739	$\mathrm{id}_{\mathrm{t}} \in O(\mathrm{t} ; \mathrm{t})$	$\mathrm{id}_t \in O(t; t)$	(iv) for each type t , one specifies an operation $\mathrm{id}_t \in O(t; t)$ called the <i>identity operation</i> .
fong-spivak-seven-sketches- F02346	225	1.000	$m+n-1$	$m + n - 1$	operation of arity $m + n - 1$: we have m arguments to fill in m , and the remaining $n - 1$
fong-spivak-seven-sketches- F02347	225	1.000	$n-1$	$n - 1$	operation of arity $m + n - 1$: we have m arguments to fill in m , and the remaining $n - 1$
fong-spivak-seven-sketches- F02348	226	0.943	$\mathrm{circ}_{\mathrm{i}}$	$\circ_i$	the substitution function $\circ_i$, which says we obtain the operation $f \circ_i g \in \mathcal{O}(m + n - 1)$.
fong-spivak-seven-sketches- F02349	226	0.943	$f \mathrm{circ}_{\mathrm{i}} g \in \mathcal{O}(m+n-1)$	$f \circ_i g \in \mathcal{O}(m + n - 1)$	the substitution function $\circ_i$, which says we obtain the operation $f \circ_i g \in \mathcal{O}(m + n - 1)$.
fong-spivak-seven-sketches- F02350	226	0.835	$X_{\mathrm{1}} \times \cdots \times X_{\mathrm{n}} \rightarrow Y$	$X_1 \times \cdots \times X_n \rightarrow Y$	(ii) Functions $X_1 \times \cdots \times X_n \rightarrow Y$ as operations of arity $(X_1, \dots, X_n; Y)$.
fong-spivak-seven-sketches- F02351	226	0.835	$X_{\mathrm{1}}, \ldots, X_{\mathrm{n}} ; Y$	$X_1, \dots, X_n; Y$	(ii) Functions $X_1 \times \cdots \times X_n \rightarrow Y$ as operations of arity $(X_1, \dots, X_n; Y)$.
fong-spivak-seven-sketches- F02352	227	0.985	$f \in \operatorname{Set}(W_{\mathrm{1}}, \ldots, W_{\mathrm{m}} ; X_{\mathrm{i}} \mathrm{right}) , g \in \operatorname{Set}(X_{\mathrm{1}}, \ldots, X_{\mathrm{n}} ; Y \mathrm{right})$	$f \in \operatorname{Set}(W_1, \dots, W_m; X_i), g \in \operatorname{Set}(X_1, \dots, X_n; Y)$	where $f \in \operatorname{Set}(W_1, \dots, W_m; X_i), g \in \operatorname{Set}(X_1, \dots, X_n; Y)$, and hence $g \circ_i f$ is a
fong-spivak-seven-sketches- F02353	227	0.985	$g \mathrm{circ}_{\mathrm{i}} f$	$g \circ_i f$	where $f \in \operatorname{Set}(W_1, \dots, W_m; X_i), g \in \operatorname{Set}(X_1, \dots, X_n; Y)$, and hence $g \circ_i f$ is a
fong-spivak-seven-sketches- F02354	227	0.994	$\operatorname{id}_{\mathrm{X}} \in \operatorname{Set}(X ; X)$	$\mathrm{id}_X \in \operatorname{Set}(X; X)$	(iv) Identities $\mathrm{id}_X \in \operatorname{Set}(X; X)$ are given by the identity function $\mathrm{id}_X: X \rightarrow X$.
fong-spivak-seven-sketches- F02355	227	0.994	$\operatorname{id}_{\mathrm{X}}: X \rightarrow X$	$\mathrm{id}_X: X \rightarrow X$	(iv) Identities $\mathrm{id}_X \in \operatorname{Set}(X; X)$ are given by the identity function $\mathrm{id}_X: X \rightarrow X$.
fong-spivak-seven-sketches- F02356	227	1.000	$a \in \mathbb{N}$	$a \in \mathbb{N}$	(i) Natural numbers $a \in \mathbb{N}$ as types.
fong-spivak-seven-sketches- F02357	227	0.998	$\underline{a_{\mathrm{1}}} + \cdots + \underline{a_{\mathrm{n}}} \rightarrow \underline{p} \leftarrow \underline{b}$	$\underline{a_1} + \cdots + \underline{a_n} \rightarrow \underline{p} \leftarrow \underline{b}$	(ii) Cospans $\underline{a_1} + \cdots + \underline{a_n} \rightarrow \underline{p} \leftarrow \underline{b}$ of finite sets as operations of arity $(a_1, \dots, a_n; b)$.
fong-spivak-seven-sketches- F02358	227	0.998	$\left(a_{\mathrm{1}}, \ldots, a_{\mathrm{n}} ; b \mathrm{right})$	$(a_1, \dots, a_n; b)$	(ii) Cospans $\underline{a_1} + \cdots + \underline{a_n} \rightarrow \underline{p} \leftarrow \underline{b}$ of finite sets as operations of arity $(a_1, \dots, a_n; b)$.
fong-spivak-seven-sketches- F02359	227	0.180	$\operatorname{id}_{\mathrm{a}} \in \operatorname{Set}(a ; a)$	$\mathrm{id}_a \in \operatorname{Set}(a; a)$	(iv) Identities $\mathrm{id}_a \in \operatorname{Set}(a; a)$ just given by the identity cospan $\underline{a} \xrightarrow{\mathrm{id}_a} \underline{a} \xleftarrow{\mathrm{id}_a} \underline{a}$.
fong-spivak-seven-sketches- F02360	227	0.180	$\operatorname{cospan} \underline{a} \xrightarrow{\mathrm{id}_a} \underline{a} \stackrel{\mathrm{id}_a}{\mathrm{a}} \underline{a}$	$\operatorname{cospan} \underline{a} \xrightarrow{\mathrm{id}_a} \underline{a} \stackrel{\mathrm{id}_a}{\mathrm{a}} \underline{a}$	(iv) Identities $\mathrm{id}_a \in \operatorname{Set}(a; a)$ just given by the identity cospan $\underline{a} \xrightarrow{\mathrm{id}_a} \underline{a} \xleftarrow{\mathrm{id}_a} \underline{a}$.
fong-spivak-seven-sketches- F02361	227	0.894	$_ \{ \text{FinSet} \} , 0, +$	$\operatorname{FinSet}, 0, +$	This is the operadic analogue of the monoidal category $(\operatorname{Cospan}_{\operatorname{FinSet}}, 0, +)$.
fong-spivak-seven-sketches- F02362	227	0.844	$\underline{3}, \underline{3}, \underline{4}, \underline{2}, \underline{3}$	$\underline{3}, \underline{3}, \underline{4}, \underline{2}, \underline{3}$	This is an operation of arity $(\underline{3}, \underline{3}, \underline{4}, \underline{2}; \underline{3})$. Why? The circles marked f and g have 3
fong-spivak-seven-sketches- F02363	227	0.809	$3, 3, 4, 2$	$3, 3, 4, 2$	ports, h has 4 ports, k has 2 ports, and the outer circle has 3 ports: $3, 3, 4, 2; 3$.
fong-spivak-seven-sketches- F02364	227	1.000	$\underline{3} + \underline{3} + \underline{4} + \underline{2} \rightarrow \underline{p} \stackrel{b}{\leftarrow} \underline{3}$	$\underline{3} + \underline{3} + \underline{4} + \underline{2} \xrightarrow{a} \underline{p} \xleftarrow{b} \underline{3}$	arity is, by (ii), a cospan $\underline{3} + \underline{3} + \underline{4} + \underline{2} \xrightarrow{a} \underline{p} \xleftarrow{b} \underline{3}$. In the diagram above, the apex $\underline{p}$ is the
fong-spivak-seven-sketches- F02365	227	1.000	$\underline{p}$	$\underline{p}$	arity is, by (ii), a cospan $\underline{3} + \underline{3} + \underline{4} + \underline{2} \xrightarrow{a} \underline{p} \xleftarrow{b} \underline{3}$. In the diagram above, the apex $\underline{p}$ is the
fong-spivak-seven-sketches- F02366	227	0.590	$\underline{7}$	$\underline{7}$	set $\underline{7}$, because there are 7 nodes $\bullet$ in the diagram. The function a sends each port on
fong-spivak-seven-sketches- F02367	228	0.986	$f \in \operatorname{Cospan}(2, 2 ; 2)$	$f \in \operatorname{Cospan}(2, 2; 2)$	1. Consider the following cospan $f \in \operatorname{Cospan}(2, 2; 2)$:
fong-spivak-seven-sketches- F02368	228	1.000	$g \in \operatorname{Cospan}(2, 2, 2 ; 0)$	$g \in \operatorname{Cospan}(2, 2, 2; 0)$	2. Draw the wiring diagram corresponding to the following cospan $g \in \operatorname{Cospan}(2, 2, 2; 0)$:

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches- F02369	228	■ 1.000	$g \circ_1 f$	$g \circ_1 f$	3. Compute the cospan $g \circ_1 f$. What is its arity?
fong-spivak-seven-sketches- F02370	228	■ 0.461	$\mathcal{O}_{\mathcal{C}}$	$\mathcal{O}_{\mathcal{C}}$	Definition 6.97. For any symmetric monoidal category $(\mathcal{C}, I, \otimes)$, there is an operad $\mathcal{O}_{\mathcal{C}}$,
fong-spivak-seven-sketches- F02371	228	■ 0.535	$C_1 \otimes \cdots \otimes C_n \rightarrow D$	$C_1 \otimes \cdots \otimes C_n \rightarrow D$	(ii) morphisms $C_1 \otimes \cdots \otimes C_n \rightarrow D$ in $\mathcal{C}$ as the operations of arity $(C_1, \dots, C_n; D)$.
fong-spivak-seven-sketches- F02372	228	■ 0.535	$C_1, \dots, C_n ; D$	$C_1, \dots, C_n; D$	(ii) morphisms $C_1 \otimes \cdots \otimes C_n \rightarrow D$ in $\mathcal{C}$ as the operations of arity $(C_1, \dots, C_n; D)$.
fong-spivak-seven-sketches- F02373	228	■ 0.639	$\operatorname{id}_a \in \mathcal{O}_{\mathcal{C}}(a; a)$	$\operatorname{id}_a \in \mathcal{O}_{\mathcal{C}}(a; a)$	(iv) identities $\operatorname{id}_a \in \mathcal{O}_{\mathcal{C}}(a; a)$ defined by id_a .
fong-spivak-seven-sketches- F02374	228	■ 0.639	id_a	id_a	(iv) identities $\operatorname{id}_a \in \mathcal{O}_{\mathcal{C}}(a; a)$ defined by id_a .
fong-spivak-seven-sketches- F02375	229	■ 1.000	$F: \mathcal{O} \rightarrow \mathcal{P}$	$F: \mathcal{O} \rightarrow \mathcal{P}$	and U respectively. To specify an operad functor $F: \mathcal{O} \rightarrow \mathcal{P}$,
fong-spivak-seven-sketches- F02376	229	■ 1.000	$f: T \rightarrow U$	$f: T \rightarrow U$	(i) one specifies a function $f: T \rightarrow U$.
fong-spivak-seven-sketches- F02377	229	■ 1.000	$F: \mathcal{O} \rightarrow$	$F: \mathcal{O} \rightarrow$	Definition 6.99. An <i>algebra</i> for an operad $\mathcal{O}$ is an operad functor $F: \mathcal{O} \rightarrow \mathbf{Set}$.
fong-spivak-seven-sketches- F02378	229	■ 0.998	$F(t)$	$F(t)$	operad $\mathcal{O}$ and an algebra $F: \mathcal{O} \rightarrow \mathbf{Set}$ will take a type t and return a set $F(t)$ of fillers
fong-spivak-seven-sketches- F02379	229	■ 1.000	$f \in \mathcal{O}(t_1, \dots, t_n; t)$	$f \in \mathcal{O}(t_1, \dots, t_n; t)$	for box t . Moreover, given an operation (i.e., a wiring diagram) $f \in \mathcal{O}(t_1, \dots, t_n; t)$, we
fong-spivak-seven-sketches- F02380	229	■ 1.000	$F(f)$	$F(f)$	get a function $F(f)$ that takes an element of each set $F(t_i)$, and returns an element of
fong-spivak-seven-sketches- F02381	229	■ 1.000	$F(t_i)$	$F(t_i)$	get a function $F(f)$ that takes an element of each set $F(t_i)$, and returns an element of
fong-spivak-seven-sketches- F02382	229	■ 1.000	$t_1, \dots, t_n$	$t_1, \dots, t_n$	$F(t)$. For example, it takes n circuits with interface $\mathbb{F}_2^{n+1}$ respectively, and returns
fong-spivak-seven-sketches- F02383	229	■ 0.854	$T = \operatorname{Ob}(\mathbf{Circ})$	$T = \operatorname{Ob}(\mathbf{Circ})$	<i>Example 6.100.</i> For electric circuits, the types are again finite sets, $T = \operatorname{Ob}(\mathbf{FinSet})$,
fong-spivak-seven-sketches- F02384	229	■ 1.000	$\mathbf{Circ}(t)$	$\mathbf{Circ}(t)$	set $\mathbf{Circ}(t)$ of fillers, namely the set of electric circuits with that t -marked terminals. As
fong-spivak-seven-sketches- F02385	229	■ 0.967	$\operatorname{Circ}(\varphi) \in \operatorname{Set}(\operatorname{Circ}(2), \operatorname{Circ}(2), \operatorname{Circ}(2); \operatorname{Circ}(0))$	$\operatorname{Circ}(\varphi) \in \operatorname{Set}(\operatorname{Circ}(2), \operatorname{Circ}(2), \operatorname{Circ}(2); \operatorname{Circ}(0))$	drawn case, we would get a morphism $\operatorname{Circ}(\varphi) \in \operatorname{Set}(\operatorname{Circ}(2), \operatorname{Circ}(2), \operatorname{Circ}(2); \operatorname{Circ}(0))$,
fong-spivak-seven-sketches- F02386	233	■ 1.000	$\in \mathbb{R}$	$\in \mathbb{R}$	some $\text{safe} \in \mathbb{R}$ from other objects. To do so, several components must interact: a
fong-spivak-seven-sketches- F02387	233	■ 1.000	$S^{\prime}$	$S^{\prime}$	sensor that approximates the real distance by an internal variable $S^{\prime}$, a controller that
fong-spivak-seven-sketches- F02388	234	■ 1.000	$S \geq$	$S \geq$	In order for the system to ensure $S \geq \text{safe}$, we need each of the components to
fong-spivak-seven-sketches- F02389	234	■ 1.000	$S <$	$S <$	the action signal A , e.g. that if at any time $S < \text{safe}$, then within one second it will
fong-spivak-seven-sketches- F02390	235	■ 1.000	$A =$	$A =$	emit the signal $A = \text{go}$. The motor maintains a relationship between A and S , e.g. that
fong-spivak-seven-sketches- F02391	236	■ 1.000	$[a, b]$	$[a, b]$	• S (wire): The behavior of S over a time-interval $[a, b]$ is that of all continuous
fong-spivak-seven-sketches- F02392	236	■ 1.000	$[a, b] \rightarrow \mathbb{R}$	$[a, b] \rightarrow \mathbb{R}$	real-valued functions $[a, b] \rightarrow \mathbb{R}$.
fong-spivak-seven-sketches- F02393	236	■ 0.809	$\{ (S^{\prime}, A) \mid Eq. (7.2) \}$	$\{ (S^{\prime}, A) \mid Eq. (7.2) \}$	• controller (box): the relation $\{ (S^{\prime}, A) \mid Eq. (7.2) \}$, i.e. all behavioral pairs $(S^{\prime}, A)$
fong-spivak-seven-sketches- F02394	236	■ 0.809	$(S^{\prime}, A)$	$(S^{\prime}, A)$	• controller (box): the relation $\{ (S^{\prime}, A) \mid Eq. (7.2) \}$, i.e. all behavioral pairs $(S^{\prime}, A)$
fong-spivak-seven-sketches- F02395	236	■ 0.982	$\in P$	$\in P$	to the set $\mathbb{B} = \{\text{false}, \text{true}\}$ of truth values, where $\text{Brendan} \in P$ maps to true because

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F02396	237	■ 0.994	$\mathcal{E}$	$\mathcal{E}$	Set in many ways. Here are some facts that are true of any topos $\mathcal{E}$:
fong-spivak-seven-sketches-F02397	237	■ 1.000	$1 \rightarrow \Omega$	$1 \xrightarrow{\text{true}} \Omega$	5. $\mathcal{E}$ has a subobject classifier $1 \xrightarrow{\text{true}} \Omega$.
fong-spivak-seven-sketches-F02398	237	■ 0.736	$\mathcal{E} =$	$\mathcal{E} =$	In particular, since Set is a topos, all of the above facts are true for $\mathcal{E} = \mathbf{Set}$. Our first
fong-spivak-seven-sketches-F02399	237	■ 0.999	(B, C, E, F)	(B, C, E, F)	(B, C, E, F) is a pullback. But in the right-hand square, does the corner symbol mean
fong-spivak-seven-sketches-F02400	237	■ 0.939	(A, B, D, E)	(A, B, D, E)	that (A, B, D, E) is a pullback or that (A, C, D, F) is a pullback? It's ambiguous, but as
fong-spivak-seven-sketches-F02401	237	■ 0.939	(A, C, D, F)	(A, C, D, F)	that (A, B, D, E) is a pullback or that (A, C, D, F) is a pullback? It's ambiguous, but as
fong-spivak-seven-sketches-F02402	237	■ 0.980	$\left(B, C, B^{\prime}, C^{\prime}\right)$	(B, C, B', C')	Proposition 7.3. In the commutative diagram below, suppose that the (B, C, B', C')
fong-spivak-seven-sketches-F02403	237	■ 0.574	$\left(A, B, A^{\prime}, B^{\prime}\right)$	(A, B, A', B')	Then the (A, B, A', B') square is a pullback iff the (A, C, A', C') rectangle is a pullback.
fong-spivak-seven-sketches-F02404	237	■ 0.574	$\left(A, C, A^{\prime}, C^{\prime}\right)$	(A, C, A', C')	Then the (A, B, A', B') square is a pullback iff the (A, C, A', C') rectangle is a pullback.
fong-spivak-seven-sketches-F02405	238	■ 1.000	$i: B^{\prime} \rightarrow B$	$i: B' \rightarrow B$	phism. That is, suppose that $i: B' \rightarrow B$ is an isomorphism and $f: A \rightarrow B$ is any
fong-spivak-seven-sketches-F02406	238	■ 1.000	$i^{\prime}$	i'	morphism. Show that i' is an isomorphism, in the following diagram:
fong-spivak-seven-sketches-F02407	238	■ 1.000	$f^{\prime}$	f'	—
fong-spivak-seven-sketches-F02408	239	■ 1.000	$f: C \rightarrow D$	$f: C \rightarrow D$	epi-mono factorizations are. We say that a morphism $f: C \rightarrow D$ in $\mathcal{E}$ has an epi-mono
fong-spivak-seven-sketches-F02409	239	■ 1.000	$\operatorname{im}(f)$	$\operatorname{im}(f)$	factorization if it has an ‘image’; that is, there is an object $\operatorname{im}(f)$, an epimorphism
fong-spivak-seven-sketches-F02410	239	■ 0.865	$C \rightarrow \operatorname{im}(f)$	$C \rightarrow \operatorname{im}(f)$	$C \twoheadrightarrow \operatorname{im}(f)$, and a monomorphism $\operatorname{im}(f) \hookrightarrow D$, whose composite is f .
fong-spivak-seven-sketches-F02411	239	■ 0.865	$\operatorname{im}(f) \rightarrow D$	$\operatorname{im}(f) \hookrightarrow D$	$C \twoheadrightarrow \operatorname{im}(f)$, and a monomorphism $\operatorname{im}(f) \hookrightarrow D$, whose composite is f .
fong-spivak-seven-sketches-F02412	239	■ 1.000	$\operatorname{im}(f) =$	$\operatorname{im}(f) =$	function $f: C \rightarrow D$ may be factored as a surjective function onto its image $\operatorname{im}(f) =$
fong-spivak-seven-sketches-F02413	239	■ 1.000	$\{f(c) \mid c \in C\}$	$\{f(c) \mid c \in C\}$	$\{f(c) \mid c \in C\}$, followed by the inclusion of this image into the codomain D . Moreover,
fong-spivak-seven-sketches-F02414	239	■ 1.000	$f: \underline{3} \rightarrow \underline{3}$	$f: \underline{3} \rightarrow \underline{3}$	<i>Exercise 7.9.</i> Factor the following function $f: \underline{3} \rightarrow \underline{3}$ as an epimorphism followed by
fong-spivak-seven-sketches-F02415	239	■ 1.000	$f=(e ; m)$	$f=(e ; m)$	epimorphism e and a monomorphism m such that $f=(e \circ m)$ is their composite.
fong-spivak-seven-sketches-F02416	239	■ 1.000	$C, D \in \mathcal{C}$	$C, D \in \mathcal{C}$	closure means that for any two objects $C, D \in \mathcal{C}$, there is a ‘hom-object’ $D^C \in \mathcal{C}$ and a
fong-spivak-seven-sketches-F02417	239	■ 1.000	$D^C \in \mathcal{C}$	$D^C \in \mathcal{C}$	closure means that for any two objects $C, D \in \mathcal{C}$, there is a ‘hom-object’ $D^C \in \mathcal{C}$ and a
fong-spivak-seven-sketches-F02418	239	■ 0.998	$A \in \mathcal{C}$	$A \in \mathcal{C}$	natural isomorphism for any $A \in \mathcal{C}$:
fong-spivak-seven-sketches-F02419	239	■ 1.000	$f(-,-): A \times C \rightarrow D$	$f(-,-): A \times C \rightarrow D$	some game $f(-,-): A \times C \rightarrow D$: for whatever action $a \in A$ that you take and action
fong-spivak-seven-sketches-F02420	239	■ 1.000	$f(a, c)$	$f(a, c)$	$c \in C$ that I take, $f(a, c)$ is some value in D . Since you’re self-centered but loving, you
fong-spivak-seven-sketches-F02421	239	■ 0.993	$f(a,-): C \rightarrow D$	$f(a,-): C \rightarrow D$	do a , you make a game $f(a,-): C \rightarrow D$ for me alone. In the formalism, D^C represents
fong-spivak-seven-sketches-F02422	239	■ 0.993	D^C	D^C	do a , you make a game $f(a,-): C \rightarrow D$ for me alone. In the formalism, D^C represents

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_1 F02423	239	■ 0.962	$v, w, x \in V$	$v, w, x \in V$	tion 2.90—and suppose it satisfies the following for all $v, w, x \in V$:
fong-spivak-seven-sketches_1 F02424	239	■ 1.000	$v \leq I$	$v \leq I$	• $v \leq I$,
fong-spivak-seven-sketches_1 F02425	239	■ 1.000	$v \otimes w \leq v$	$v \otimes w \leq v$	• $v \otimes w \leq v$ and $v \otimes w \leq w$, and
fong-spivak-seven-sketches_1 F02426	239	■ 1.000	$v \otimes w \leq w$	$v \otimes w \leq w$	• $v \otimes w \leq v$ and $v \otimes w \leq w$, and
fong-spivak-seven-sketches_1 F02427	239	■ 1.000	$x \leq v$	$x \leq v$	• if $x \leq v$ and $x \leq w$ then $x \leq v \otimes w$.
fong-spivak-seven-sketches_1 F02428	239	■ 1.000	$x \leq w$	$x \leq w$	• if $x \leq v$ and $x \leq w$ then $x \leq v \otimes w$.
fong-spivak-seven-sketches_1 F02429	239	■ 1.000	$x \leq v \otimes w$	$x \leq v \otimes w$	• if $x \leq v$ and $x \leq w$ then $x \leq v \otimes w$.
fong-spivak-seven-sketches_1 F02430	240	■ 0.924	$X \mapsto Y$	$X \mapsto Y$	are injections, and any injection $X \hookrightarrow Y$ is isomorphic to a subset of Y . This gives
fong-spivak-seven-sketches_1 F02431	240	■ 0.925	$\Omega \in \mathcal{E}$	$\Omega \in \mathcal{E}$	minimal object 1. A <i>subobject classifier</i> in $\mathcal{E}$ consists of an object $\Omega \in \mathcal{E}$, together with a
fong-spivak-seven-sketches_1 F02432	240	■ 1.000	$1 \rightarrow \Omega$	$1 \rightarrow \Omega$	monomorphism $\text{true}: 1 \rightarrow \Omega$, satisfying the following property: for any objects X and
fong-spivak-seven-sketches_1 F02433	240	■ 0.899	$m: X \rightarrowtail Y$	$m: X \rightarrowtail Y$	Y and monomorphism $m: X \rightarrowtail Y$ in $\mathcal{E}$, there is a unique morphism $\ulcorner m \urcorner: Y \rightarrow \Omega$ such
fong-spivak-seven-sketches_1 F02434	240	■ 0.899	$\ulcorner m \urcorner: Y \rightarrow \Omega$	$\ulcorner m \urcorner: Y \rightarrow \Omega$	Y and monomorphism $m: X \rightarrowtail Y$ in $\mathcal{E}$, there is a unique morphism $\ulcorner m \urcorner: Y \rightarrow \Omega$ such
fong-spivak-seven-sketches_1 F02435	240	■ 1.000	$\ulcorner m \urcorner$	$\ulcorner m \urcorner$	We refer to $\ulcorner m \urcorner$ as the <i>characteristic map</i> of m , or we say that $\ulcorner m \urcorner$ <i>classifies</i> m . Conversely,
fong-spivak-seven-sketches_1 F02436	240	■ 1.000	$p: Y \rightarrow \Omega$	$p: Y \rightarrow \Omega$	given any map $p: Y \rightarrow \Omega$, we denote the pullback of true as on the right of Eq. (7.13).
fong-spivak-seven-sketches_1 F02437	240	■ 1.000	$Y \rightarrow \Omega$	$Y \rightarrow \Omega$	A <i>predicate</i> on Y is a morphism $Y \rightarrow \Omega$.
fong-spivak-seven-sketches_1 F02438	241	■ 0.999	$1 \rightarrow$	$1 \rightarrow$	generically called $\text{true}: 1 \rightarrow \Omega$; in the case of Set it is played by the function $1 \rightarrow$
fong-spivak-seven-sketches_1 F02439	241	■ 0.980	$m: X \hookrightarrowtail Y$	$m: X \hookrightarrowtail Y$	tion 7.12 says that for any injective function $m: X \hookrightarrowtail Y$ between sets, we are supposed
fong-spivak-seven-sketches_1 F02440	241	■ 0.998	$\ulcorner m \urcorner: Y \rightarrow \{\text{true}, \text{false}\}$	$\ulcorner m \urcorner: Y \rightarrow \{\text{true}, \text{false}\}$	to be able to find a characteristic function $\ulcorner m \urcorner: Y \rightarrow \{\text{true}, \text{false}\}$ with some sort of
fong-spivak-seven-sketches_1 F02441	241	■ 1.000	$\ulcorner m \urcorner(y)$	$\ulcorner m \urcorner(y)$	In other words, if we think of X as a subobject of Y , then we make $\ulcorner m \urcorner(y)$ equal to
fong-spivak-seven-sketches_1 F02442	241	■ 1.000	$y \in X$	$y \in X$	true iff $y \in X$.
fong-spivak-seven-sketches_1 F02443	241	■ 1.000	$p: Y \rightarrow \mathbb{B}$	$p: Y \rightarrow \mathbb{B}$	$p: Y \rightarrow \mathbb{B}$, and vice versa. How it works is encoded in Definition 7.12, but the basic
fong-spivak-seven-sketches_1 F02444	241	■ 1.000	$\{Y \mid p\}$	$\{Y \mid p\}$	This might help explain our abstract notation $\{Y \mid p\}$ in Eq. (7.13).
fong-spivak-seven-sketches_1 F02445	241	■ 0.998	$X = \mathbb{N} = \{0, 1, 2, \dots\}$	$X = \mathbb{N} = \{0, 1, 2, \dots\}$	Exercise 7.16. Let $X = \mathbb{N} = \{0, 1, 2, \dots\}$ and $Y = \mathbb{Z} = \{\dots, -1, 0, 1, 2, \dots\}$; we have
fong-spivak-seven-sketches_1 F02446	241	■ 0.998	$Y = \mathbb{Z} = \{\dots, -1, 0, 1, 2, \dots\}$	$Y = \mathbb{Z} = \{\dots, -1, 0, 1, 2, \dots\}$	Exercise 7.16. Let $X = \mathbb{N} = \{0, 1, 2, \dots\}$ and $Y = \mathbb{Z} = \{\dots, -1, 0, 1, 2, \dots\}$; we have
fong-spivak-seven-sketches_1 F02447	241	■ 0.985	$\ulcorner m \urcorner: Y \rightarrow \mathbb{B}$	$\ulcorner m \urcorner: Y \rightarrow \mathbb{B}$	$\ulcorner m \urcorner: Y \rightarrow \mathbb{B}$, as in Definition 7.12.
fong-spivak-seven-sketches_1 F02448	241	■ 1.000	$\ulcorner m \urcorner(-5) \in \mathbb{B}$	$\ulcorner m \urcorner(-5) \in \mathbb{B}$	1. What is $\ulcorner m \urcorner(-5) \in \mathbb{B}$?
fong-spivak-seven-sketches_1 F02449	241	■ 1.000	$\ulcorner m \urcorner(0) \in \mathbb{B}$	$\ulcorner m \urcorner(0) \in \mathbb{B}$	2. What is $\ulcorner m \urcorner(0) \in \mathbb{B}$?
fong-spivak-seven-sketches_1 F02450	241	■ 0.915	$\text{id}_{\mathbb{N}}: \mathbb{N} \rightarrow \mathbb{N}$	$\text{id}_{\mathbb{N}}: \mathbb{N} \rightarrow \mathbb{N}$	1. Consider the identity function $\text{id}_{\mathbb{N}}: \mathbb{N} \rightarrow \mathbb{N}$. It is an injection, so it has a charac-

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-1 F02451	241	0.746	$\left\langle \ulcorner \operatorname{id}_{\mathbb{N}} \urcorner \right\rangle : \mathbb{N} \rightarrow \mathbb{B}$	$\ulcorner \operatorname{id}_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B}$	teristic function $\ulcorner \operatorname{id}_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B}$. Give a concrete description of $\ulcorner \operatorname{id}_{\mathbb{N}} \urcorner$, i.e. its exact
fong-spivak-seven-sketches-1 F02452	241	0.746	$\left\langle \ulcorner \operatorname{id}_{\mathbb{N}} \urcorner \right\rangle$	$\ulcorner \operatorname{id}_{\mathbb{N}} \urcorner$	teristic function $\ulcorner \operatorname{id}_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B}$. Give a concrete description of $\ulcorner \operatorname{id}_{\mathbb{N}} \urcorner$, i.e. its exact
fong-spivak-seven-sketches-1 F02453	242	0.990	$!_{\mathbb{N}} : \emptyset \rightarrow \mathbb{N}$	$!_{\mathbb{N}} : \emptyset \rightarrow \mathbb{N}$	2. Consider the unique function $!_{\mathbb{N}} : \emptyset \rightarrow \mathbb{N}$ from the empty set. Give a concrete
fong-spivak-seven-sketches-1 F02454	242	0.470	$\ulcorner !_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B}$	$\ulcorner !_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B}$	description of $\ulcorner !_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B}$.
fong-spivak-seven-sketches-1 F02455	242	1.000	$1 : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	$1 : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	Obtaining the AND operation. Consider the function $1 : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$ picking out the
fong-spivak-seven-sketches-1 F02456	242	0.257	$\ulcorner 1 \urcorner$	$\ulcorner 1 \urcorner$	$\ulcorner (\text{true}, \text{true}) \urcorner : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$. What function is it? By Eq. (7.15) the only element of $\mathbb{B} \times \mathbb{B}$
fong-spivak-seven-sketches-1 F02457	242	0.257	$\mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	$\mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	$\ulcorner (\text{true}, \text{true}) \urcorner : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$. What function is it? By Eq. (7.15) the only element of $\mathbb{B} \times \mathbb{B}$
fong-spivak-seven-sketches-1 F02458	242	0.257	$\mathbb{B} \times \mathbb{B}$	$\mathbb{B} \times \mathbb{B}$	$\ulcorner (\text{true}, \text{true}) \urcorner : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$. What function is it? By Eq. (7.15) the only element of $\mathbb{B} \times \mathbb{B}$
fong-spivak-seven-sketches-1 F02459	242	0.990	$\ulcorner ($	$\ulcorner ($	that can be sent to true is (true, true). Thus $\ulcorner (\text{true}, \text{true}) \urcorner(P, Q) \in \mathbb{B}$ must be given
fong-spivak-seven-sketches-1 F02460	242	0.990	$\urcorner(P, Q) \in \mathbb{B}$	$\urcorner(P, Q) \in \mathbb{B}$	that can be sent to true is (true, true). Thus $\ulcorner (\text{true}, \text{true}) \urcorner(P, Q) \in \mathbb{B}$ must be given
fong-spivak-seven-sketches-1 F02462	242	1.000	$P \wedge Q$	$P \wedge Q$	This is exactly the truth table for the AND of P and Q , i.e. for $P \wedge Q$. In other words,
fong-spivak-seven-sketches-1 F02463	242	0.965	$\urcorner = \wedge$	$\urcorner = \wedge$	$\ulcorner (\text{true}, \text{true}) \urcorner = \wedge$. Note that this defines $\wedge$ as a function $\wedge : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$, and we use
fong-spivak-seven-sketches-1 F02464	242	0.965	$\wedge : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	$\wedge : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	$\ulcorner (\text{true}, \text{true}) \urcorner = \wedge$. Note that this defines $\wedge$ as a function $\wedge : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$, and we use
fong-spivak-seven-sketches-1 F02465	242	1.000	$x \wedge y := \wedge(x, y)$	$x \wedge y := \wedge(x, y)$	the usual infix notation $x \wedge y := \wedge(x, y)$.
fong-spivak-seven-sketches-1 F02466	242	1.000	$\vee : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	$\vee : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	OR of P and Q , i.e. that of the function $\vee : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$ defining OR, is:
fong-spivak-seven-sketches-1 F02468	242	1.000	$m : X \subseteq \mathbb{B} \times \mathbb{B}$	$m : X \subseteq \mathbb{B} \times \mathbb{B}$	$m : X \subseteq \mathbb{B} \times \mathbb{B}$, what subset would X be? By Eq. (7.15), X should be the set of $y \in Y$
fong-spivak-seven-sketches-1 F02469	242	0.801	$\mathbb{B} \times \{$	$\mathbb{B} \times \{$	think of X as the union of $\{\text{true}\} \times \mathbb{B}$ and $\mathbb{B} \times \{\text{true}\}$ —a colimit of limits involving the
fong-spivak-seven-sketches-1 F02470	242	1.000	$\Omega \times \Omega$	$\Omega \times \Omega$	subobject of $\Omega \times \Omega$, and hence classify a map $\Omega \times \Omega \rightarrow \Omega$, in any topos $\mathcal{E}$.
fong-spivak-seven-sketches-1 F02471	242	1.000	$\Omega \times \Omega \rightarrow \Omega$	$\Omega \times \Omega \rightarrow \Omega$	subobject of $\Omega \times \Omega$, and hence classify a map $\Omega \times \Omega \rightarrow \Omega$, in any topos $\mathcal{E}$.
fong-spivak-seven-sketches-1 F02472	243	1.000	$\neg : \mathbb{B} \rightarrow \mathbb{B}$	$\neg : \mathbb{B} \rightarrow \mathbb{B}$	function $\neg : \mathbb{B} \rightarrow \mathbb{B}$ is the characteristic function of some thing, (*?*)).
fong-spivak-seven-sketches-1 F02473	243	1.000	$P \Rightarrow Q$	$P \Rightarrow Q$	<i>Exercise 7.20.</i> Given two booleans P, Q , define $P \Rightarrow Q$ to mean $P = (P \wedge Q)$.
fong-spivak-seven-sketches-1 F02474	243	1.000	$P = (P \wedge Q)$	$P = (P \wedge Q)$	<i>Exercise 7.20.</i> Given two booleans P, Q , define $P \Rightarrow Q$ to mean $P = (P \wedge Q)$.
fong-spivak-seven-sketches-1 F02475	243	1.000	$m : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	$m : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	3. What is the characteristic function $m : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$ for $P \Rightarrow Q$?
fong-spivak-seven-sketches-1 F02476	243	1.000	$E := \{n \in \mathbb{N} \mid n$	$E := \{n \in \mathbb{N} \mid n$	<i>Exercise 7.21.</i> Consider the sets $E := \{n \in \mathbb{N} \mid n$ is even $\}$, $P := \{n \in \mathbb{N} \mid n$ is prime $\}$,
fong-spivak-seven-sketches-1 F02477	243	1.000	$\}$, $P := \{n \in \mathbb{N} \mid n$	$\}, P := \{n \in \mathbb{N} \mid n$	<i>Exercise 7.21.</i> Consider the sets $E := \{n \in \mathbb{N} \mid n$ is even $\}$, $P := \{n \in \mathbb{N} \mid n$ is prime $\}$,
fong-spivak-seven-sketches-1 F02478	243	1.000	$T := \{n \in \mathbb{N} \mid n \geq 10\}$	$T := \{n \in \mathbb{N} \mid n \geq 10\}$	and $T := \{n \in \mathbb{N} \mid n \geq 10\}$. Each is a subset of $\mathbb{N}$, so defines a function $\mathbb{N} \rightarrow \mathbb{B}$.
fong-spivak-seven-sketches-1 F02479	243	1.000	$\mathbb{N} \rightarrow \mathbb{B}$	$\mathbb{N} \rightarrow \mathbb{B}$	and $T := \{n \in \mathbb{N} \mid n \geq 10\}$. Each is a subset of $\mathbb{N}$, so defines a function $\mathbb{N} \rightarrow \mathbb{B}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_1 F02480	243	■ 1.000	$\ulcorner E \urcorner(17)$	$\ulcorner E \urcorner(17)$	1. What is $\ulcorner E \urcorner(17)$?
fong-spivak-seven-sketches_1 F02481	243	■ 1.000	$\ulcorner P \urcorner(17)$	$\ulcorner P \urcorner(17)$	2. What is $\ulcorner P \urcorner(17)$?
fong-spivak-seven-sketches_1 F02482	243	■ 1.000	$\ulcorner T \urcorner(17)$	$\ulcorner T \urcorner(17)$	3. What is $\ulcorner T \urcorner(17)$?
fong-spivak-seven-sketches_1 F02483	243	■ 0.995	$(\ulcorner E \urcorner \wedge \ulcorner P \urcorner) \vee \ulcorner T \urcorner$	$(\ulcorner E \urcorner \wedge \ulcorner P \urcorner) \vee \ulcorner T \urcorner$	4. Name the smallest three elements in the set classified by $(\ulcorner E \urcorner \wedge \ulcorner P \urcorner) \vee \ulcorner T \urcorner$. $\diamond$
fong-spivak-seven-sketches_1 F02484	243	■ 1.000	$\wedge, \vee, \Rightarrow: \Omega \times \Omega \rightarrow \Omega$	$\wedge, \vee, \Rightarrow: \Omega \times \Omega \rightarrow \Omega$	how to understand logic in terms of Ω : there are various maps $\wedge, \vee, \Rightarrow: \Omega \times \Omega \rightarrow \Omega$
fong-spivak-seven-sketches_1 F02485	243	■ 1.000	$\neg: \Omega \rightarrow \Omega$	$\neg: \Omega \rightarrow \Omega$	and $\neg: \Omega \rightarrow \Omega$ etc., which serve as logical connectives. These are operations on truth
fong-spivak-seven-sketches_1 F02486	244	■ 1.000	$\mathbb{R}^2$	$\mathbb{R}^2$	of the plane $\mathbb{R}^2$ as a space, we might examine only points and regions in it. But if we
fong-spivak-seven-sketches_1 F02487	244	■ 0.729	$\mathcal{C}^{\text{op}} \rightarrow$	$\mathcal{C}^{\text{op}} \rightarrow$	functors $\mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$. ⁵ Such functors are called <i>presheaves on $\mathcal{C}$</i> .
fong-spivak-seven-sketches_1 F02488	244	■ 0.988	$P(c)$	$P(c)$	To each object $c \in \mathcal{C}$, we refer to the set $P(c)$ as <i>the set of sections of P over c</i> . To each
fong-spivak-seven-sketches_1 F02489	244	■ 1.000	$P(f): P(c) \rightarrow P(c')$	$P(f): P(c) \rightarrow P(c')$	morphism $f: c' \rightarrow c$, we refer to the function $P(f): P(c) \rightarrow P(c')$ as the <i>restriction map</i>
fong-spivak-seven-sketches_1 F02490	244	■ 1.000	$s \in P(c)$	$s \in P(c)$	<i>along f</i> . For any section $s \in P(c)$, we may denote $P(f)(s) \in P(c')$, i.e. its restriction
fong-spivak-seven-sketches_1 F02491	244	■ 1.000	$P(f)(s) \in P(c')$	$P(f)(s) \in P(c')$	<i>along f</i> . For any section $s \in P(c)$, we may denote $P(f)(s) \in P(c')$, i.e. its restriction
fong-spivak-seven-sketches_1 F02492	244	■ 1.000	$s _f$	$s _f$	<i>along f</i> , by $s _f$.
fong-spivak-seven-sketches_1 F02493	245	■ 1.000	$\alpha: P \rightarrow Q$	$\alpha: P \rightarrow Q$	If P and Q are presheaves, a <i>morphism $\alpha: P \rightarrow Q$</i> between them is a natural trans-
fong-spivak-seven-sketches_1 F02494	245	■ 0.852	$\mathbf{ArShp}^{\text{op}} \rightarrow$	$\mathbf{ArShp}^{\text{op}} \rightarrow$	A presheaf on $\mathbf{ArShp}$ is a functor $I: \mathbf{ArShp}^{\text{op}} \rightarrow \mathbf{Set}$, which is a database instance on
fong-spivak-seven-sketches_1 F02495	245	■ 0.532	$\mathbf{ArShp}^{\text{op}}$	$\mathbf{ArShp}^{\text{op}}$	$\mathbf{ArShp}^{\text{op}}$. Note that $\mathbf{ArShp}^{\text{op}}$ is what we called Gr in Section 3.3.5; there we showed
fong-spivak-seven-sketches_1 F02496	245	■ 0.532	$\mathbf{ArShp}^{\text{op}}$	$\mathbf{ArShp}^{\text{op}}$	$\mathbf{ArShp}^{\text{op}}$. Note that $\mathbf{ArShp}^{\text{op}}$ is what we called Gr in Section 3.3.5; there we showed
fong-spivak-seven-sketches_1 F02497	246	■ 1.000	$P(X) = \{U \subseteq X\}$	$P(X) = \{U \subseteq X\}$	Definition 7.25. Let X be a set, and let $P(X) = \{U \subseteq X\}$ denote its set of subsets. A
fong-spivak-seven-sketches_1 F02498	246	■ 0.672	$\mathbf{Op} \subseteq P(X)$	$\mathbf{Op} \subseteq P(X)$	<i>topology</i> on X is a subset $\mathbf{Op} \subseteq P(X)$, elements of which we call <i>open sets</i> , ⁶ satisfying the
fong-spivak-seven-sketches_1 F02499	246	■ 1.000	$X \in \mathbf{Op}$	$X \subseteq X$	(a) Whole set: the subset $X \subseteq X$ is open, i.e. $X \in \mathbf{Op}$.
fong-spivak-seven-sketches_1 F02500	246	■ 1.000	$X \in \mathbf{Op}$	$X \in \mathbf{Op}$	(a) Whole set: the subset $X \subseteq X$ is open, i.e. $X \in \mathbf{Op}$.
fong-spivak-seven-sketches_1 F02501	246	■ 0.871	$U, V \in \mathbf{Op}$	$U, V \in \mathbf{Op}$	(b) Binary intersections: if $U, V \in \mathbf{Op}$ then $(U \cap V) \in \mathbf{Op}$.
fong-spivak-seven-sketches_1 F02502	246	■ 0.871	$(U \cap V) \in \mathbf{Op}$	$(U \cap V) \in \mathbf{Op}$	(b) Binary intersections: if $U, V \in \mathbf{Op}$ then $(U \cap V) \in \mathbf{Op}$.
fong-spivak-seven-sketches_1 F02503	246	■ 0.743	$U_i \in \mathbf{Op}$	$U_i \in \mathbf{Op}$	(c) Arbitrary unions: if I is a set and if we are given an open set $U_i \in \mathbf{Op}$ for each i ,
fong-spivak-seven-sketches_1 F02504	246	■ 0.999	$\left(\bigcup_{i \in I} U_i\right) \in \mathbf{Op}$	$\left(\bigcup_{i \in I} U_i\right) \in \mathbf{Op}$	then their union is also open, $(\bigcup_{i \in I} U_i) \in \mathbf{Op}$. We interpret the particular case
fong-spivak-seven-sketches_1 F02505	246	■ 1.000	$I = \emptyset$	$I = \emptyset$	where $I = \emptyset$ to mean that the empty set is open: $\emptyset \in \mathbf{Op}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_1 F02506	246	■ 1.000	$\varnothing \in \mathbf{Op}$	$\emptyset \in \mathbf{Op}$	where $I = \emptyset$ to mean that the empty set is open: $\emptyset \in \mathbf{Op}$.
fong-spivak-seven-sketches_1 F02507	246	■ 1.000	$U = \bigcup_{i \in I} U_i$	$U = \bigcup_{i \in I} U_i$	If $U = \bigcup_{i \in I} U_i$, we say that $(U_i)_{i \in I}$ covers U .
fong-spivak-seven-sketches_1 F02508	246	■ 1.000	$\left(U_i\right)_{i \in I}$	$(U_i)_{i \in I}$	If $U = \bigcup_{i \in I} U_i$, we say that $(U_i)_{i \in I}$ covers U .
fong-spivak-seven-sketches_1 F02509	246	■ 0.630	$(X, \mathbf{Op})$	$(X, \mathbf{Op})$	A pair $(X, \mathbf{Op})$, where X is a set and $\mathbf{Op}$ is a topology on X , is called a <i>topological</i>
fong-spivak-seven-sketches_1 F02510	246	■ 0.630	$\mathbf{Op}$	$\mathbf{Op}$	A pair $(X, \mathbf{Op})$, where X is a set and $\mathbf{Op}$ is a topology on X , is called a <i>topological</i>
fong-spivak-seven-sketches_1 F02511	246	■ 0.914	$X, \mathbf{Op}_X$	$X, \mathbf{Op}_X$	A <i>continuous function</i> between topological spaces $(X, \mathbf{Op}_X)$ and $(Y, \mathbf{Op}_Y)$ is a function
fong-spivak-seven-sketches_1 F02512	246	■ 0.914	$Y, \mathbf{Op}_Y$	$Y, \mathbf{Op}_Y$	A <i>continuous function</i> between topological spaces $(X, \mathbf{Op}_X)$ and $(Y, \mathbf{Op}_Y)$ is a function
fong-spivak-seven-sketches_1 F02513	246	■ 0.650	$U \in \mathbf{Op}_Y$	$U \in \mathbf{Op}_Y$	$f: X \rightarrow Y$ such that for every $U \in \mathbf{Op}_Y$, the preimage $f^{-1}(U)$ is in $\mathbf{Op}_X$.
fong-spivak-seven-sketches_1 F02514	246	■ 0.650	$f^{-1}(U)$	$f^{-1}(U)$	$f: X \rightarrow Y$ such that for every $U \in \mathbf{Op}_Y$, the preimage $f^{-1}(U)$ is in $\mathbf{Op}_X$.
fong-spivak-seven-sketches_1 F02515	246	■ 0.650	$\mathbf{Op}_X$	$\mathbf{Op}_X$	$f: X \rightarrow Y$ such that for every $U \in \mathbf{Op}_Y$, the preimage $f^{-1}(U)$ is in $\mathbf{Op}_X$.
fong-spivak-seven-sketches_1 F02516	246	■ 1.000	$\epsilon \in \mathbb{R}$	$\epsilon \in \mathbb{R}$	Example 7.26. The usual topology $\mathbf{Op}$ on $\mathbb{R}^2$ is based on ‘ ϵ -balls.’ For any $\epsilon \in \mathbb{R}$ with
fong-spivak-seven-sketches_1 F02517	246	■ 1.000	$\epsilon > 0$	$\epsilon > 0$	$\epsilon > 0$, and any point $p = (x, y) \in \mathbb{R}^2$, define the ϵ -ball centered at p to be:
fong-spivak-seven-sketches_1 F02518	246	■ 1.000	$p = (x, y) \in \mathbb{R}^2$	$p = (x, y) \in \mathbb{R}^2$	$\epsilon > 0$, and any point $p = (x, y) \in \mathbb{R}^2$, define the ϵ -ball centered at p to be:
fong-spivak-seven-sketches_1 F02519	246	■ 1.000	$U \in \mathbf{Op}$	$U \in \mathbf{Op}$	⁶ In other words, we refer to a subset $U \subseteq X$ as <i>open</i> if $U \in \mathbf{Op}$.
fong-spivak-seven-sketches_1 F02520	247	■ 0.999	$B(x, y; \epsilon)$	$B(x, y; \epsilon)$	In other words, $B(x, y; \epsilon)$ is the set of all points within ϵ of (x, y) .
fong-spivak-seven-sketches_1 F02521	247	■ 0.974	$U \subseteq \mathbb{R}^2$	$U \subseteq \mathbb{R}^2$	For an arbitrary subset $U \subseteq \mathbb{R}^2$, we call it open and put it in $\mathbf{Op}$ if, for every $(x, y) \in U$
fong-spivak-seven-sketches_1 F02522	247	■ 0.974	$\in U$	$\in U$	For an arbitrary subset $U \subseteq \mathbb{R}^2$, we call it open and put it in $\mathbf{Op}$ if, for every $(x, y) \in U$
fong-spivak-seven-sketches_1 F02523	247	■ 1.000	$B(x, y; \epsilon) \subseteq U$	$B(x, y; \epsilon) \subseteq U$	there exists a (small enough) $\epsilon > 0$ such that $B(x, y; \epsilon) \subseteq U$.
fong-spivak-seven-sketches_1 F02524	247	■ 1.000	$p = (x, y) \in U$	$p = (x, y) \in U$	an open set $U \subseteq \mathbb{R}^2$, a point $p = (x, y) \in U$,
fong-spivak-seven-sketches_1 F02525	247	■ 1.000	$d(x_1, x_2) := x_1 - x_2 $	$d(x_1, x_2) := x_1 - x_2 $	Exercise 7.27. Consider the set $\mathbb{R}$. It is a metric space with $d(x_1, x_2) := x_1 - x_2 $.
fong-spivak-seven-sketches_1 F02526	247	■ 1.000	$x \in \mathbb{R}$	$x \in \mathbb{R}$	for each $x \in \mathbb{R}$, define $B(x, \epsilon)$.
fong-spivak-seven-sketches_1 F02527	247	■ 1.000	$B(x, \epsilon)$	$B(x, \epsilon)$	for each $x \in \mathbb{R}$, define $B(x, \epsilon)$.
fong-spivak-seven-sketches_1 F02528	247	■ 1.000	$U \subseteq \mathbb{R}$	$U \subseteq \mathbb{R}$	2. When is an arbitrary subset $U \subseteq \mathbb{R}$ called open, in analogy with Example 7.26?
fong-spivak-seven-sketches_1 F02529	247	■ 1.000	U_1, U_2	U_1, U_2	3. Find three open sets U_1, U_2 , and U in $\mathbb{R}$, such that $(U_i)_{i \in \{1,2\}}$ covers U .
fong-spivak-seven-sketches_1 F02530	247	■ 1.000	$\left(U_i\right)_{i \in \{1,2\}}$	$(U_i)_{i \in \{1,2\}}$	3. Find three open sets U_1, U_2 , and U in $\mathbb{R}$, such that $(U_i)_{i \in \{1,2\}}$ covers U .
fong-spivak-seven-sketches_1 F02531	247	■ 0.803	$\mathbf{Op}_{\text{crse}} = (\varnothing, X)$	$\mathbf{Op}_{\text{crse}} = (\emptyset, X)$	possible: $\mathbf{Op}_{\text{crse}} = (\emptyset, X)$. There is also a ‘finest’ topology, having as many open sets as

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_1 F02532	247	■ 0.592	$\mathbf{0\ p}_{\{\text{fine}\}}=\mathrm{P}(X)$	$\mathbf{Op}_{\text{fine}} = \mathrm{P}(X)$	possible: $\mathbf{Op}_{\text{fine}} = \mathrm{P}(X)$. The latter, $(X, \mathrm{P}(X))$ is called the <i>discrete space on the set X</i> .
fong-spivak-seven-sketches_1 F02533	247	■ 0.592	$(X, \mathrm{P}(X))$	$(X, \mathrm{P}(X))$	possible: $\mathbf{Op}_{\text{fine}} = \mathrm{P}(X)$. The latter, $(X, \mathrm{P}(X))$ is called the <i>discrete space on the set X</i> .
fong-spivak-seven-sketches_1 F02534	247	■ 0.995	$\mathbf{0\ p}_{\{\text{crse}\}}$	$\mathbf{Op}_{\text{crse}}$	1. Verify that for any set X , what we called $\mathbf{Op}_{\text{crse}}$ in Example 7.28 really is a topology,
fong-spivak-seven-sketches_1 F02535	247	■ 0.998	$\mathbf{0\ p}_{\{\text{fine}\}}$	$\mathbf{Op}_{\text{fine}}$	2. Verify also that $\mathbf{Op}_{\text{fine}}$ really is a topology.
fong-spivak-seven-sketches_1 F02536	247	■ 0.762	$\left(Y, \mathbf{0\ p}_{\{Y\}}\right)$	$(Y, \mathbf{Op}_Y)$	3. Show that if $(X, \mathrm{P}(X))$ is discrete and $(Y, \mathbf{Op}_Y)$ is any topological space, then every
fong-spivak-seven-sketches_1 F02537	247	■ 1.000	$X \rightarrowtail Y$	$X \rightarrow Y$	function $X \rightarrow Y$ is continuous.
fong-spivak-seven-sketches_1 F02538	247	■ 1.000	$X=\{1,2\}$	$X = \{1, 2\}$	Example 7.30. There are four topologies possible on $X = \{1, 2\}$. Two are $\mathbf{Op}_{\text{crse}}$ and
fong-spivak-seven-sketches_1 F02539	247	■ 1.000	$d\left((x, y), \left(x^{\prime}, y^{\prime}\right)\right):=\sqrt{\left(x-x^{\prime}\right)^2+\left(y-y^{\prime}\right)^2}$	$d\left((x, y),\left(x^{\prime}, y^{\prime}\right)\right):=\sqrt{\left(x-x^{\prime}\right)^2+\left(y-y^{\prime}\right)^2}$	⁷ Here, $d((x, y), (x', y')) := \sqrt{(x - x')^2 + (y - y')^2}$ is the usual ‘Euclidean distance’ between two points.
fong-spivak-seven-sketches_1 F02540	248	■ 0.817	$\mathbf{0\ p}_{\{\text{fine}\}}$	$\mathbf{Op}_{\text{fine}}$	$\mathbf{Op}_{\text{fine}}$ from Example 7.28. The other two are:
fong-spivak-seven-sketches_1 F02541	248	■ 0.943	$\{1,2\}, \mathbf{0\ p}_{\{1\}}$	$\{1, 2\}, \mathbf{Op}_1$	The two topological spaces $(\{1, 2\}, \mathbf{Op}_1)$ and $(\{1, 2\}, \mathbf{Op}_2)$ are isomorphic; either one
fong-spivak-seven-sketches_1 F02542	248	■ 0.943	$\{1,2\}, \mathbf{0\ p}_{\{2\}}$	$\{1, 2\}, \mathbf{Op}_2$	The two topological spaces $(\{1, 2\}, \mathbf{Op}_1)$ and $(\{1, 2\}, \mathbf{Op}_2)$ are isomorphic; either one
fong-spivak-seven-sketches_1 F02543	248	■ 0.931	$(\mathbf{0\ p}, \subseteq)$	$(\mathbf{Op}, \subseteq)$	$(X, \mathbf{Op})$, the set $\mathbf{Op}$ has the structure of a preorder using the subset relation, $(\mathbf{Op}, \subseteq)$.
fong-spivak-seven-sketches_1 F02544	248	■ 1.000	$U \subseteq U$	$U \subseteq U$	It is reflexive because $U \subseteq U$ for any $U \in \mathbf{Op}$, and it is transitive because if $U \subseteq V$ and
fong-spivak-seven-sketches_1 F02545	248	■ 1.000	$U \subseteq V$	$U \subseteq V$	It is reflexive because $U \subseteq U$ for any $U \in \mathbf{Op}$, and it is transitive because if $U \subseteq V$ and
fong-spivak-seven-sketches_1 F02546	248	■ 1.000	$V \subseteq W$	$V \subseteq W$	$V \subseteq W$ then $U \subseteq W$.
fong-spivak-seven-sketches_1 F02547	248	■ 1.000	$U \subseteq W$	$U \subseteq W$	$V \subseteq W$ then $U \subseteq W$.
fong-spivak-seven-sketches_1 F02548	248	■ 1.000	U, V	U, V	category: its objects are the open sets U and for any U, V the set of morphisms $\mathbf{Op}(U, V)$
fong-spivak-seven-sketches_1 F02549	248	■ 1.000	$\mathbf{0\ p}(U, V)$	$\mathbf{Op}(U, V)$	category: its objects are the open sets U and for any U, V the set of morphisms $\mathbf{Op}(U, V)$
fong-spivak-seven-sketches_1 F02550	248	■ 1.000	$U \nsubseteq V$	$U \not\subseteq V$	is empty if $U \not\subseteq V$ and it has one element if $U \subseteq V$.
fong-spivak-seven-sketches_1 F02551	248	■ 0.761	$X, \mathbf{0\ p}_{\{1\}}$	$X, \mathbf{Op}_1$	Exercise 7.31. Recall the Sierpinski space, say $(X, \mathbf{Op}_1)$ from Example 7.30.
fong-spivak-seven-sketches_1 F02552	248	■ 0.686	$X, \mathbf{0\ p}$	$X, \mathbf{Op}$	Exercise 7.32. Given any topological space $(X, \mathbf{Op})$, any subset $Y \subseteq X$ can be given the
fong-spivak-seven-sketches_1 F02553	248	■ 0.686	$Y \subseteq X$	$Y \subseteq X$	Exercise 7.32. Given any topological space $(X, \mathbf{Op})$, any subset $Y \subseteq X$ can be given the
fong-spivak-seven-sketches_1 F02554	248	■ 0.955	$\mathbf{0\ p}_{\{? \cap \gamma\}}$	$\mathbf{Op}_{? \cap \gamma}$	subspace topology, call it $\mathbf{Op}_{? \cap \gamma}$. This topology defines any $A \subseteq Y$ to be open, $A \in \mathbf{Op}_{? \cap \gamma}$,
fong-spivak-seven-sketches_1 F02555	248	■ 0.955	$A \subseteq Y$	$A \subseteq Y$	subspace topology, call it $\mathbf{Op}_{? \cap \gamma}$. This topology defines any $A \subseteq Y$ to be open, $A \in \mathbf{Op}_{? \cap \gamma}$,
fong-spivak-seven-sketches_1 F02556	248	■ 0.955	$A \in \mathbf{0\ p}_{\{? \cap \gamma\}}$	$A \in \mathbf{Op}_{? \cap \gamma}$	subspace topology, call it $\mathbf{Op}_{? \cap \gamma}$. This topology defines any $A \subseteq Y$ to be open, $A \in \mathbf{Op}_{? \cap \gamma}$,
fong-spivak-seven-sketches_1 F02557	248	■ 1.000	$B \in \mathbf{0\ p}$	$B \in \mathbf{Op}$	if there is an open set $B \in \mathbf{Op}$ such that $A = B \cap Y$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-1 F02558	248	■ 1.000	$A=B \cap Y$	$A = B \cap Y$	if there is an open set $B \in \mathbf{Op}$ such that $A = B \cap Y$.
fong-spivak-seven-sketches-1 F02559	248	■ 1.000	$Y \in \mathbf{Op}_{? \cap Y}$	$Y \in \mathbf{Op}_{? \cap Y}$	1. Find a $B \in \mathbf{Op}$ that shows that the whole set Y is open, i.e. $Y \in \mathbf{Op}_{? \cap Y}$.
fong-spivak-seven-sketches-1 F02560	248	■ 1.000	$\mathbf{Op}_{? \cap Y}$	$\mathbf{Op}_{? \cap Y}$	2. Show that $\mathbf{Op}_{? \cap Y}$ is a topology in the sense of Definition 7.25. ⁸
fong-spivak-seven-sketches-1 F02561	248	■ 1.000	$Y \hookrightarrow X$	$Y \hookrightarrow X$	3. Show that the inclusion function $Y \hookrightarrow X$ is a continuous function.
fong-spivak-seven-sketches-1 F02562	248	■ 0.555	$\mathbf{Op}, \subseteq$	$\mathbf{Op}, \subseteq$	Remark 7.33. Suppose $(X, \mathbf{Op})$ is a topological space, and consider the preorder $(\mathbf{Op}, \subseteq)$
fong-spivak-seven-sketches-1 F02563	248	■ 0.983	$(\mathbf{Op}, \subseteq, X, \cap)$	$(\mathbf{Op}, \subseteq, X, \cap)$	of open sets. It turns out that $(\mathbf{Op}, \subseteq, X, \cap)$ is always a quantale in the sense of
fong-spivak-seven-sketches-1 F02564	248	■ 0.638	(X, Op)	(X, Op)	If $(X, \mathbf{Op})$ is a topological space and $\mathcal{V}$ the corresponding quantale as in Remark 7.33,
fong-spivak-seven-sketches-1 F02565	248	■ 0.997	$U_i \subseteq X$	$U_i \subseteq X$	⁸ Hint 1: for any set I , collection of sets $(U_i)_{i \in I}$ with $U_i \subseteq X$, and set $V \subseteq X$, one has $(\bigcup_{i \in I} U_i) \cap V =$
fong-spivak-seven-sketches-1 F02566	248	■ 0.997	$\left(\bigcup_{i \in I} U_i\right) \cap V =$	$\left(\bigcup_{i \in I} U_i\right) \cap V =$	⁸ Hint 1: for any set I , collection of sets $(U_i)_{i \in I}$ with $U_i \subseteq X$, and set $V \subseteq X$, one has $(\bigcup_{i \in I} U_i) \cap V =$
fong-spivak-seven-sketches-1 F02567	248	■ 0.798	$\bigcup_{i \in I} (U_i \cap V)$	$\bigcup_{i \in I} (U_i \cap V)$	$\bigcup_{i \in I} (U_i \cap V)$. Hint 2: for any $U, V, W \subseteq X$, one has $(U \cap W) \cap (V \cap W) = (U \cap V) \cap W$.
fong-spivak-seven-sketches-1 F02568	248	■ 0.798	$U, V, W \subseteq X$	$U, V, W \subseteq X$	$\bigcup_{i \in I} (U_i \cap V)$. Hint 2: for any $U, V, W \subseteq X$, one has $(U \cap W) \cap (V \cap W) = (U \cap V) \cap W$.
fong-spivak-seven-sketches-1 F02569	248	■ 0.798	$(U \cap W) \cap (V \cap W) = (U \cap V) \cap W$	$(U \cap W) \cap (V \cap W) = (U \cap V) \cap W$	$\bigcup_{i \in I} (U_i \cap V)$. Hint 2: for any $U, V, W \subseteq X$, one has $(U \cap W) \cap (V \cap W) = (U \cap V) \cap W$.
fong-spivak-seven-sketches-1 F02570	249	■ 0.770	$P: \mathbf{Op}^{\mathrm{op}} \rightarrow$	$P: \mathbf{Op}^{\mathrm{op}} \rightarrow$	$P: \mathbf{Op}^{\mathrm{op}} \rightarrow \mathbf{Set}$. Thus to every open set $U \in \mathbf{Op}$ we have a set $P(U)$, called the set
fong-spivak-seven-sketches-1 F02571	249	■ 0.770	$P(U)$	$P(U)$	$P: \mathbf{Op}^{\mathrm{op}} \rightarrow \mathbf{Set}$. Thus to every open set $U \in \mathbf{Op}$ we have a set $P(U)$, called the set
fong-spivak-seven-sketches-1 F02572	249	■ 1.000	$V \subseteq U$	$V \subseteq U$	of sections over U , and to every inclusion of open sets $V \subseteq U$ we have a function
fong-spivak-seven-sketches-1 F02573	249	■ 1.000	$P(U) \rightarrow P(V)$	$P(U) \rightarrow P(V)$	$P(U) \rightarrow P(V)$ called the <i>restriction</i> . If $s \in P(U)$ is a section over U , we may denote its
fong-spivak-seven-sketches-1 F02574	249	■ 1.000	$s \in P(U)$	$s \in P(U)$	$P(U) \rightarrow P(V)$ called the <i>restriction</i> . If $s \in P(U)$ is a section over U , we may denote its
fong-spivak-seven-sketches-1 F02575	249	■ 1.000	$s _V$	$s _V$	restriction to V by $s _V$. Recall that we say a collection of open sets $(U_i)_{i \in I}$ covers an open
fong-spivak-seven-sketches-1 F02576	249	■ 0.613	$P: \mathbf{Op}^{\mathrm{op}} \rightarrow$	$P: \mathbf{Op}^{\mathrm{op}} \rightarrow$	Definition 7.35. Let $(X, \mathbf{Op})$ be a topological space, and let $P: \mathbf{Op}^{\mathrm{op}} \rightarrow \mathbf{Set}$ be a
fong-spivak-seven-sketches-1 F02577	249	■ 1.000	$U_i \in \mathbf{Op}$	$U_i \in \mathbf{Op}$	Let $(U_i)_{i \in I}$ be a collection of open sets $U_i \in \mathbf{Op}$ covering U . A <i>matching family</i> $(s_i)_{i \in I}$
fong-spivak-seven-sketches-1 F02578	249	■ 1.000	$(s_i)_{i \in I}$	$(s_i)_{i \in I}$	Let $(U_i)_{i \in I}$ be a collection of open sets $U_i \in \mathbf{Op}$ covering U . A <i>matching family</i> $(s_i)_{i \in I}$
fong-spivak-seven-sketches-1 F02579	249	■ 1.000	$s_i \in P(U_i)$	$s_i \in P(U_i)$	of P -sections over $(U_i)_{i \in I}$ consists of a section $s_i \in P(U_i)$ for each $i \in I$, such that for every
fong-spivak-seven-sketches-1 F02580	249	■ 1.000	$i \in I$	$i \in I$	of P -sections over $(U_i)_{i \in I}$ consists of a section $s_i \in P(U_i)$ for each $i \in I$, such that for every
fong-spivak-seven-sketches-1 F02581	249	■ 1.000	$i, j \in I$	$i, j \in I$	$i, j \in I$ we have
fong-spivak-seven-sketches-1 F02582	249	■ 1.000	$s _{U_i} = s_i$	$s _{U_i} = s_i$	gluing, or <i>glued section</i> , of the matching family if $s _{U_i} = s_i$ holds for all $i \in I$.
fong-spivak-seven-sketches-1 F02583	249	■ 0.992	$\mathbf{Shv}(X, \mathbf{Op})$	$\mathbf{Shv}(X, \mathbf{Op})$	We denote by $\mathbf{Shv}(X, \mathbf{Op})$ the category of sheaves on X .
fong-spivak-seven-sketches-1 F02584	249	■ 1.000	$U = \emptyset \subseteq X$	$U = \emptyset \subseteq X$	we take $U = \emptyset \subseteq X$ and $I = \emptyset$, then the empty collection of subsets (one for each $i \in I$,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F02585	249	■ 0.999	$P(\varnothing) \cong \{()\}$	$P(\varnothing) \cong \{()\}$	rarely sufficient) condition is that $P(\varnothing) \cong \{()\}$, i.e. $P(\varnothing)$ must be a set with one element.
fong-spivak-seven-sketches-F02586	249	■ 0.999	$P(\varnothing)$	$P(\varnothing)$	rarely sufficient) condition is that $P(\varnothing) \cong \{()\}$, i.e. $P(\varnothing)$ must be a set with one element.
fong-spivak-seven-sketches-F02587	250	■ 0.990	$f\left(a_1\right)=f\left(a_2\right)=a, f\left(b_1\right)=b$	$f\left(a_1\right)=f\left(a_2\right)=a, f\left(b_1\right)=b$	the element of Y immediately below it. For example, $f(a_1) = f(a_2) = a$, $f(b_1) = b$, and
fong-spivak-seven-sketches-F02588	250	■ 0.999	$f^{-1}(y) \subseteq X$	$f^{-1}(y) \subseteq X$	For each point $y \in Y$, the preimage set $f^{-1}(y) \subseteq X$ above it is often called the <i>fiber over</i>
fong-spivak-seven-sketches-F02589	250	■ 1.000	$f^{\prime}: X \rightarrow Y$	$f^{\prime}: X \rightarrow Y$	4. Gave an example of a function $f^{\prime}: X \rightarrow Y$ for which every fiber has either one or
fong-spivak-seven-sketches-F02590	250	■ 0.886	$\operatorname{Sec}_f: \mathbf{Op}(Y)^{\mathrm{op}} \rightarrow$	$\operatorname{Sec}_f: \mathbf{Op}(Y)^{\mathrm{op}} \rightarrow$	we begin by building a presheaf—i.e. a functor $\operatorname{Sec}_f: \mathbf{Op}(Y)^{\mathrm{op}} \rightarrow \mathbf{Set}$ —and then we’ll
fong-spivak-seven-sketches-F02591	250	■ 1.000	Sec_f	Sec_f	Define the presheaf Sec_f on an arbitrary subset $U \subseteq Y$ by:
fong-spivak-seven-sketches-F02592	250	■ 1.000	$U \subseteq Y$	$U \subseteq Y$	Define the presheaf Sec_f on an arbitrary subset $U \subseteq Y$ by:
fong-spivak-seven-sketches-F02593	250	■ 1.000	$\operatorname{Sec}_f(U)$	$\operatorname{Sec}_f(U)$	One might describe $\operatorname{Sec}_f(U)$ as the set of all ways to pick a ‘cross-section’ of the f
fong-spivak-seven-sketches-F02594	250	■ 1.000	$s \in \operatorname{Sec}_f(U)$	$s \in \operatorname{Sec}_f(U)$	arrangement over U . That is, an element $s \in \operatorname{Sec}_f(U)$ is a choice of one element per
fong-spivak-seven-sketches-F02595	250	■ 0.999	$U=\{a, b\}$	$U=\{a, b\}$	As an example, let’s say $U=\{a, b\}$. How many such s ’s are there in $\operatorname{Sec}_f(U)$? To
fong-spivak-seven-sketches-F02596	251	■ 0.996	$V_1=\{a, b, c\}$	$V_1=\{a, b, c\}$	1. Let $V_1=\{a, b, c\}$. Draw all the sections over it, i.e. all elements of $\operatorname{Sec}_f(V_1)$, as
fong-spivak-seven-sketches-F02597	251	■ 0.996	$\operatorname{Sec}_f\left(V_1\right)$	$\operatorname{Sec}_f\left(V_1\right)$	1. Let $V_1=\{a, b, c\}$. Draw all the sections over it, i.e. all elements of $\operatorname{Sec}_f(V_1)$, as
fong-spivak-seven-sketches-F02598	251	■ 1.000	$V_2=\{a, b, c, d\}$	$V_2=\{a, b, c, d\}$	2. Let $V_2=\{a, b, c, d\}$. Again draw all the sections, $\operatorname{Sec}_f(V_2)$.
fong-spivak-seven-sketches-F02599	251	■ 1.000	$\operatorname{Sec}_f\left(V_2\right)$	$\operatorname{Sec}_f\left(V_2\right)$	2. Let $V_2=\{a, b, c, d\}$. Again draw all the sections, $\operatorname{Sec}_f(V_2)$.
fong-spivak-seven-sketches-F02600	251	■ 0.999	$V_3=\{a, b, d, e\}$	$V_3=\{a, b, d, e\}$	3. Let $V_3=\{a, b, d, e\}$. How many sections (elements of $\operatorname{Sec}_f(V_3)$) are there?
fong-spivak-seven-sketches-F02601	251	■ 0.999	$\operatorname{Sec}_f\left(V_3\right)$	$\operatorname{Sec}_f\left(V_3\right)$	3. Let $V_3=\{a, b, d, e\}$. How many sections (elements of $\operatorname{Sec}_f(V_3)$) are there?
fong-spivak-seven-sketches-F02602	251	■ 0.974	Sec_f	Sec_f	Sec_f on objects, so you are half way to understanding Sec_f as a presheaf. That is, as
fong-spivak-seven-sketches-F02603	251	■ 1.000	$V=\{a\}$	$V=\{a\}$	restriction maps are easy: if $V \subseteq U$, say $V=\{a\}$ and $U=\{a, b\}$, then given a section s
fong-spivak-seven-sketches-F02604	251	■ 1.000	$\operatorname{Sec}_f(\{a, b, c\})$	$\operatorname{Sec}_f(\{a, b, c\})$	1. Write out the sets of sections $\operatorname{Sec}_f(\{a, b, c\})$ and $\operatorname{Sec}_f(\{a, c\})$.
fong-spivak-seven-sketches-F02605	251	■ 1.000	$\operatorname{Sec}_f(\{a, c\})$	$\operatorname{Sec}_f(\{a, c\})$	1. Write out the sets of sections $\operatorname{Sec}_f(\{a, b, c\})$ and $\operatorname{Sec}_f(\{a, c\})$.
fong-spivak-seven-sketches-F02606	251	■ 1.000	$U_1=\{a, b\}$	$U_1=\{a, b\}$	sheaf condition, consider the set $U_1=\{a, b\}$ and $U_2=\{b, e\}$. These cover the set
fong-spivak-seven-sketches-F02607	251	■ 1.000	$U_2=\{b, e\}$	$U_2=\{b, e\}$	sheaf condition, consider the set $U_1=\{a, b\}$ and $U_2=\{b, e\}$. These cover the set
fong-spivak-seven-sketches-F02608	251	■ 1.000	$U=\{a, b, e\}=U_1 \cup U_2$	$U=\{a, b, e\}=U_1 \cup U_2$	$U=\{a, b, e\}=U_1 \cup U_2$. By Definition 7.35, a matching family for this cover consists of
fong-spivak-seven-sketches-F02609	251	■ 1.000	U_1	U_1	a section over U_1 and a section over U_2 that agree on the overlap set, $U_1 \cap U_2=\{b\}$.
fong-spivak-seven-sketches-F02610	251	■ 1.000	U_2	U_2	a section over U_1 and a section over U_2 that agree on the overlap set, $U_1 \cap U_2=\{b\}$.
fong-spivak-seven-sketches-F02611	251	■ 1.000	$U_1 \cap U_2=\{b\}$	$U_1 \cap U_2=\{b\}$	a section over U_1 and a section over U_2 that agree on the overlap set, $U_1 \cap U_2=\{b\}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F02612	252	1.000	$s_{\{1\}} \in \operatorname{Sec}_f(\operatorname{left}(U_{\{1\}}))$	$s_1 \in \operatorname{Sec}_f(U_1)$	So consider $s_1 \in \operatorname{Sec}_f(U_1)$ and $s_2 \in \operatorname{Sec}_f(U_2)$ shown below.
fong-spivak-seven-sketches-F02613	252	1.000	$s_{\{2\}} \in \operatorname{Sec}_f(\operatorname{left}(U_{\{2\}}))$	$s_2 \in \operatorname{Sec}_f(U_2)$	So consider $s_1 \in \operatorname{Sec}_f(U_1)$ and $s_2 \in \operatorname{Sec}_f(U_2)$ shown below.
fong-spivak-seven-sketches-F02614	252	1.000	$g_{\{1\}}$	g_1	Since sections g_1 and g_2 agree on the overlap—they both send b to b_2 —the two sections
fong-spivak-seven-sketches-F02615	252	1.000	$g_{\{2\}}$	g_2	Since sections g_1 and g_2 agree on the overlap—they both send b to b_2 —the two sections
fong-spivak-seven-sketches-F02616	252	1.000	$b_{\{2\}}$	b_2	Since sections g_1 and g_2 agree on the overlap—they both send b to b_2 —the two sections
fong-spivak-seven-sketches-F02617	252	1.000	$U=\{a, b, e\}$	$U = \{a, b, e\}$	shown in Eq. (7.43) can be glued to form a single section over $U = \{a, b, e\}$:
fong-spivak-seven-sketches-F02618	252	0.968	$\operatorname{left}(s_{\{1\}}, s_{\{2\}})$	(s_1, s_2)	2. For your answer (s_1, s_2) in part 1, can you find a section $s \in \operatorname{Sec}_f(U_1 \cup U_2)$ such
fong-spivak-seven-sketches-F02619	252	0.968	$s \in \operatorname{Sec}_f(\operatorname{left}(U_{\{1\}} \cup U_{\{2\}}))$	$s \in \operatorname{Sec}_f(U_1 \cup U_2)$	2. For your answer (s_1, s_2) in part 1, can you find a section $s \in \operatorname{Sec}_f(U_1 \cup U_2)$ such
fong-spivak-seven-sketches-F02620	252	1.000	$\operatorname{left}.s\operatorname{right} _{U_{\{1\}}}=s_{\{1\}}$	$s _{U_1} = s_1$	that $s _{U_1} = s_1$ and $s _{U_2} = s_2$?
fong-spivak-seven-sketches-F02621	252	1.000	$\operatorname{left}.s\operatorname{right} _{U_{\{2\}}}=s_{\{2\}}$	$s _{U_2} = s_2$	that $s _{U_1} = s_1$ and $s _{U_2} = s_2$?
fong-spivak-seven-sketches-F02622	252	0.683	$h_{\{1\}} \in \operatorname{Sec}_f(\operatorname{left}(U_{\{1\}}))$	$h_1 \in \operatorname{Sec}_f(U_1)$	3. Find a section $h_1 \in \operatorname{Sec}_f(U_1)$ and a section $h_2 \in \operatorname{Sec}_f(U_2)$ that <i>do</i> agree on the
fong-spivak-seven-sketches-F02623	252	0.683	$h_{\{2\}} \in \operatorname{Sec}_f(\operatorname{left}(U_{\{2\}}))$	$h_2 \in \operatorname{Sec}_f(U_2)$	3. Find a section $h_1 \in \operatorname{Sec}_f(U_1)$ and a section $h_2 \in \operatorname{Sec}_f(U_2)$ that <i>do</i> agree on the
fong-spivak-seven-sketches-F02624	252	0.683	d o	<i>do</i>	3. Find a section $h_1 \in \operatorname{Sec}_f(U_1)$ and a section $h_2 \in \operatorname{Sec}_f(U_2)$ that <i>do</i> agree on the
fong-spivak-seven-sketches-F02625	252	0.975	$h \in \operatorname{Sec}_f(\operatorname{left}(U_{\{1\}} \cup U_{\{2\}}))$	$h \in \operatorname{Sec}_f(U_1 \cup U_2)$	4. Can you find a section $h \in \operatorname{Sec}_f(U_1 \cup U_2)$ such that $h _{U_1} = h_1$ and $h _{U_2} = h_2$? $\diamond$
fong-spivak-seven-sketches-F02626	252	0.975	$\operatorname{left}.h\operatorname{right} _{U_{\{1\}}}=h_{\{1\}}$	$h _{U_1} = h_1$	4. Can you find a section $h \in \operatorname{Sec}_f(U_1 \cup U_2)$ such that $h _{U_1} = h_1$ and $h _{U_2} = h_2$? $\diamond$
fong-spivak-seven-sketches-F02627	252	0.975	$\operatorname{left}.h\operatorname{right} _{U_{\{2\}}}=h_{\{2\}}$	$h _{U_2} = h_2$	4. Can you find a section $h \in \operatorname{Sec}_f(U_1 \cup U_2)$ such that $h _{U_1} = h_1$ and $h _{U_2} = h_2$? $\diamond$
fong-spivak-seven-sketches-F02628	252	0.984	$f:\operatorname{left}(X, \operatorname{mathbf{0} p}_{\{X\}})\operatorname{right}\operatorname{arrow}\operatorname{left}(Y, \operatorname{mathbf{0} p}_{\{Y\}})$	$f : (X, \mathbf{Op}_X) \rightarrow (Y, \mathbf{Op}_Y)$	Example 7.45. Let $f: (X, \mathbf{Op}_X) \rightarrow (Y, \mathbf{Op}_Y)$ be a continuous function. Consider the
fong-spivak-seven-sketches-F02629	252	0.901	$\operatorname{operatorname{Sec}}_f:\operatorname{mathbf{0} p}_{\{Y\}}^{\operatorname{op}}\operatorname{right}\operatorname{arrow}$	$\operatorname{Sec}_f : \mathbf{Op}_Y^{\operatorname{op}} \rightarrow$	functor $\operatorname{Sec}_f : \mathbf{Op}_Y^{\operatorname{op}} \rightarrow$ Set given by
fong-spivak-seven-sketches-F02630	253	0.623	$\operatorname{mathbf{0} p}_{\{Y\}}$	$\mathbf{Op}_Y$	The morphisms of $\mathbf{Op}_Y$ are inclusions $V \subseteq U$. Given $g: U \rightarrow X$ and $V \subseteq U$, what we
fong-spivak-seven-sketches-F02631	253	0.623	$g: U \operatorname{right}\operatorname{arrow} X$	$g: U \rightarrow X$	The morphisms of $\mathbf{Op}_Y$ are inclusions $V \subseteq U$. Given $g: U \rightarrow X$ and $V \subseteq U$, what we
fong-spivak-seven-sketches-F02632	253	0.996	T M	TM	space TM whose points are pairs (m, v) , where $m \in M$ is a point in the manifold and
fong-spivak-seven-sketches-F02633	253	0.996	(m, v)	(m, v)	space TM whose points are pairs (m, v) , where $m \in M$ is a point in the manifold and
fong-spivak-seven-sketches-F02634	253	0.996	$m \in M$	$m \in M$	space TM whose points are pairs (m, v) , where $m \in M$ is a point in the manifold and
fong-spivak-seven-sketches-F02635	253	1.000	$\pi: T M \operatorname{right}\operatorname{arrow}$	$\pi : TM \rightarrow$	The tangent bundle TM on a manifold M comes with a continuous map $\pi: TM \rightarrow$
fong-spivak-seven-sketches-F02636	253	0.984	(m, v) \mapsto m	$(m, v) \mapsto m$	M back down to the manifold, sending $(m, v) \mapsto m$. One might say that π “forgets the
fong-spivak-seven-sketches-F02637	253	0.984	π	π	M back down to the manifold, sending $(m, v) \mapsto m$. One might say that π “forgets the

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fong-spivak-seven-sketches_253_F02638	253	■ 0.960	Sec_{π}	Sec_{π}	defines a sheaf Sec_{π} . It could be called the sheaf of ‘tangent vector sections on M' , but
fong-spivak-seven-sketches_253_F02639	253	■ 0.960	$M^{\prime}$	M'	defines a sheaf Sec_{π} . It could be called the sheaf of ‘tangent vector sections on M' , but
fong-spivak-seven-sketches_253_F02640	253	■ 1.000	$U \subseteq M$	$U \subseteq M$	we spoke of the sheaf of wind velocities on Earth, above. Given an open subset $U \subseteq M$,
fong-spivak-seven-sketches_253_F02641	253	■ 1.000	$v \in \operatorname{Sec}_{\pi}(U)$	$v \in \mathrm{Sec}_{\pi}(U)$	an element $v \in \mathrm{Sec}_{\pi}(U)$ is called a vector field over U because it continuously assigns
fong-spivak-seven-sketches_253_F02642	253	■ 1.000	$v(u)$	$v(u)$	a tangent vector $v(u)$ to each point $u \in U$. The tangent vector at u tells us the velocity
fong-spivak-seven-sketches_253_F02643	253	■ 1.000	$U \neq M$	$U \neq M$	possible to prove that for every open set U , as long as $U \neq M$, there is a vector field
fong-spivak-seven-sketches_254_F02644	254	■ 1.000	$U \neq X$	$U \neq X$	namely that hair can be well combed whenever $U \neq X$ but cannot be well combed when
fong-spivak-seven-sketches_254_F02645	254	■ 1.000	$U=X$	$U = X$	$U = X$ —can be thought of as a generative effect, and can be measured by cohomology
fong-spivak-seven-sketches_254_F02646	254	■ 0.988	$\boldsymbol{\operatorname{Shv}}(M)$	$\mathbf{Shv}(M)$	that we can consider the category $\mathbf{Shv}(M)$ of sheaves on M ; in fact, such categories are
fong-spivak-seven-sketches_254_F02647	254	■ 0.485	X, Op	$X, \mathbf{Op}$	<i>Example 7.48.</i> For every topological space $(X, \mathbf{Op})$, we have the topos of sheaves on it.
fong-spivak-seven-sketches_254_F02648	254	■ 0.928	$\{1, 2\}, \mathbf{Op}_1$	$\{1, 2\}, \mathbf{Op}_1$	<i>Exercise 7.49.</i> Consider the Sierpinski space $(\{1, 2\}, \mathbf{Op}_1)$ from Example 7.30.
fong-spivak-seven-sketches_254_F02649	254	■ 0.896	$X=\mathbf{1}$	$X = \mathbf{1}$	one-point space $X = \mathbf{1}$ with its unique topology, we find that the category $\mathbf{Set}$ is a topos,
fong-spivak-seven-sketches_255_F02650	255	■ 1.000	$\wedge, \vee, \neg, \rightarrow, \exists, \forall$	$\wedge, \vee, \neg, \Rightarrow, \exists, \forall$	topological space. There, the same logical symbols $\wedge, \vee, \neg, \Rightarrow, \exists, \forall$ become operations
fong-spivak-seven-sketches_255_F02651	255	■ 0.975	$\Omega=\mathbb{B}$	$\Omega = \mathbb{B}$	of booleans $\Omega = \mathbb{B}$. In a sheaf topos $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$, the object $\Omega \in \mathcal{E}$ is a sheaf, not
fong-spivak-seven-sketches_255_F02652	255	■ 0.975	$\mathcal{E}=\mathbf{Shv}(X, \mathbf{Op})$	$\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$	of booleans $\Omega = \mathbb{B}$. In a sheaf topos $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$, the object $\Omega \in \mathcal{E}$ is a sheaf, not
fong-spivak-seven-sketches_255_F02653	255	■ 0.854	$\Omega: \mathbf{Op}^{\mathrm{op}} \rightarrow \mathbf{Set}$	$\Omega : \mathbf{Op}^{\mathrm{op}} \rightarrow$	$\Omega: \mathbf{Op}^{\mathrm{op}} \rightarrow \mathbf{Set}$; it assigns a set $\Omega(U)$ to each open $U \subseteq X$ and comes with a restriction
fong-spivak-seven-sketches_255_F02654	255	■ 0.854	$\Omega(U)$	$\Omega(U)$	$\Omega: \mathbf{Op}^{\mathrm{op}} \rightarrow \mathbf{Set}$; it assigns a set $\Omega(U)$ to each open $U \subseteq X$ and comes with a restriction
fong-spivak-seven-sketches_255_F02655	255	■ 1.000	$\Omega(U) \rightarrow \Omega(V)$	$\Omega(U) \rightarrow \Omega(V)$	map $\Omega(U) \rightarrow \Omega(V)$ whenever $V \subseteq U$. So in our quest to understand Ω , we first ask the
fong-spivak-seven-sketches_256_F02656	256	■ 1.000	$X=\{1\}$	$X = \{1\}$	<i>Exercise 7.52.</i> Let $X = \{1\}$ be the one point space. We said above that its subobject
fong-spivak-seven-sketches_256_F02657	256	■ 1.000	$W \subseteq V \subseteq U$	$W \subseteq V \subseteq U$	that whenever $W \subseteq V \subseteq U$, the restriction map $\Omega(U) \rightarrow \Omega(V)$ followed by the
fong-spivak-seven-sketches_256_F02658	256	■ 1.000	$\Omega(V) \rightarrow \Omega(W)$	$\Omega(V) \rightarrow \Omega(W)$	restriction map $\Omega(V) \rightarrow \Omega(W)$ is the same as the restriction map $\Omega(U) \rightarrow \Omega(W)$.
fong-spivak-seven-sketches_256_F02659	256	■ 1.000	$\Omega(U) \rightarrow \Omega(W)$	$\Omega(U) \rightarrow \Omega(W)$	restriction map $\Omega(V) \rightarrow \Omega(W)$ is the same as the restriction map $\Omega(U) \rightarrow \Omega(W)$.
fong-spivak-seven-sketches_256_F02660	256	■ 1.000	$V_i \in \Omega(U_i)$	$V_i \in \Omega(U_i)$	suppose that we have a cover $U = \bigcup_{i \in I} U_i$, and suppose given an element $V_i \in \Omega(U_i)$,
fong-spivak-seven-sketches_256_F02661	256	■ 1.000	$V_i \subseteq U_i$	$V_i \subseteq U_i$	i.e. an open set $V_i \subseteq U_i$, for each $i \in I$. Suppose further that for all $i, j \in I$, it is the
fong-spivak-seven-sketches_256_F02662	256	■ 1.000	$V_i \cap U_j = V_j \cap U_i$	$V_i \cap U_j = V_j \cap U_i$	case that $V_i \cap U_j = V_j \cap U_i$, i.e. that the elements form a matching family. Define
fong-spivak-seven-sketches_256_F02663	256	■ 0.995	$V := \bigcup_{i \in I} V_i$	$V := \bigcup_{i \in I} V_i$	$V := \bigcup_{i \in I} V_i$; it is an open subset of U , so we can consider V as an element of $\Omega(U)$.
fong-spivak-seven-sketches_256_F02664	256	■ 1.000	$(V_i)_{i \in I}$	$(V_i)_{i \in I}$	The following verifies that V is indeed a gluing for the $(V_i)_{i \in I}$:
fong-spivak-seven-sketches_256_F02665	256	■ 1.000	$V \cap U_j = V_j$	$V \cap U_j = V_j$	In other words $V \cap U_j = V_j$ for any $j \in I$. So our Ω has been upgraded from presheaf

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F02666	256	1.000	$j \in I$	$j \in I$	In other words $V \cap U_j = V_j$ for any $j \in I$. So our Ω has been upgraded from presheaf
fong-spivak-seven-sketches-F02667	256	1.000	$\{1\} \rightarrow \Omega$	$\{1\} \rightarrow \Omega$	in good standing, we also need to give a sheaf morphism $\text{true}: \{1\} \rightarrow \Omega$. Here
fong-spivak-seven-sketches-F02668	256	0.623	$\{1\}: \mathbf{Op}^{\text{op}} \rightarrow$	$\{1\}: \mathbf{Op}^{\text{op}} \rightarrow$	$\{1\}: \mathbf{Op}^{\text{op}} \rightarrow$ Set is the terminal sheaf; it maps every open set to the terminal, one
fong-spivak-seven-sketches-F02669	256	1.000	$\{1\} = \{1\}(U) \rightarrow \Omega(U)$	$\{1\} = \{1\}(U) \rightarrow \Omega(U)$	the sheaf morphism that assigns, for every $U \in \mathbf{Op}$ the function $\{1\} = \{1\}(U) \rightarrow \Omega(U)$
fong-spivak-seven-sketches-F02670	256	0.997	$1 \mapsto U$	$1 \mapsto U$	sending $1 \mapsto U$, the largest open set $U \subseteq U$. From now on we denote $\{1\}$ simply as 1.
fong-spivak-seven-sketches-F02671	256	1.000	$U = \varnothing$	$U = \varnothing$	If this U is everything, $U = X$, then P is really true; if U is nothing, $U = \varnothing$, then P is
fong-spivak-seven-sketches-F02672	257	1.000	$(V, V; A)$	$(V, V; A)$	true: $1 \rightarrow \Omega$ sends that loop to $(V, V; A)$.
fong-spivak-seven-sketches-F02673	257	1.000	$i: H \subseteq G$	$i: H \subseteq G$	Given any graph G and subgraph $i: H \subseteq G$, we need to construct a graph homo-
fong-spivak-seven-sketches-F02674	257	0.931	$\ulcorner H \urcorner$	$\ulcorner H \urcorner$	morphism $\ulcorner H \urcorner: G \rightarrow \Omega$ classifying H . The idea is that for each part of G , we decide
fong-spivak-seven-sketches-F02675	257	0.931	$G \rightarrow \Omega$	$G \rightarrow \Omega$	morphism $\ulcorner H \urcorner: G \rightarrow \Omega$ classifying H . The idea is that for each part of G , we decide
fong-spivak-seven-sketches-F02676	257	1.000	G^{prime}	G''	is its source in H ? is its target in G'' ? both? neither? Based on the answers to these
fong-spivak-seven-sketches-F02677	257	0.589	$(V, \emptyset; \emptyset), (\emptyset, V; \emptyset), (V, V; \emptyset)$	$(V, 0; 0), (0, V; 0), (V, V; 0)$	questions we send a to $(V, 0; 0), (0, V; 0), (V, V; 0)$, or $(0, 0; 0)$, respectively.
fong-spivak-seven-sketches-F02678	257	0.589	$\emptyset; \emptyset$	$0; 0$	questions we send a to $(V, 0; 0), (0, V; 0), (V, V; 0)$, or $(0, 0; 0)$, respectively.
fong-spivak-seven-sketches-F02679	257	1.000	$H \subseteq G$	$H \subseteq G$	<i>Exercise 7.55.</i> Consider the subgraph $H \subseteq G$ shown here:
fong-spivak-seven-sketches-F02680	257	0.907	$\ulcorner H \urcorner: G \rightarrow \Omega$	$\ulcorner H \urcorner: G \rightarrow \Omega$	Find the graph homomorphism $\ulcorner H \urcorner: G \rightarrow \Omega$ classifying it. See Example 7.54 .
fong-spivak-seven-sketches-F02681	257	0.615	$U, V \in \Omega(X)$	$U, V \in \Omega(X)$	values $U, V \in \Omega(X)$, then their conjunction ‘ U AND V ’ is their intersection, and their
fong-spivak-seven-sketches-F02682	257	1.000	$\cap$	$\cap$	These formulas are easy to remember, because $\wedge$ looks like $\cap$ and $\vee$ looks like $\cup$. The
fong-spivak-seven-sketches-F02683	257	1.000	$\cup$	$\cup$	These formulas are easy to remember, because $\wedge$ looks like $\cap$ and $\vee$ looks like $\cup$. The
fong-spivak-seven-sketches-F02684	257	1.000	$U \Rightarrow V$	$U \Rightarrow V$	implication $U \Rightarrow V$ is the largest open set R such that $R \cap U \subseteq V$, i.e.
fong-spivak-seven-sketches-F02685	257	1.000	$R \cap U \subseteq V$	$R \cap U \subseteq V$	implication $U \Rightarrow V$ is the largest open set R such that $R \cap U \subseteq V$, i.e.
fong-spivak-seven-sketches-F02686	257	0.668	$\neg U :=$	$\neg U :=$	Finally, the negation of U is given by $\neg U := (U \Rightarrow \text{false})$, and this turns out
fong-spivak-seven-sketches-F02687	257	0.668	$U \Rightarrow$	$U \Rightarrow$	Finally, the negation of U is given by $\neg U := (U \Rightarrow \text{false})$, and this turns out
fong-spivak-seven-sketches-F02688	257	1.000	$R \cap U = \varnothing$	$R \cap U = \varnothing$	$R \cap U = \varnothing$, i.e. the union of all open sets in the complement of U . If you know topology,
fong-spivak-seven-sketches-F02689	257	1.000	$\neg U$	$\neg U$	you might recognize that $\neg U$ is the ‘interior of the complement of U .’
fong-spivak-seven-sketches-F02690	258	1.000	$X = \mathbb{R}$	$X = \mathbb{R}$	<i>Example 7.58.</i> Consider the real line $X = \mathbb{R}$ as a topological space (see Exercise 7.27).
fong-spivak-seven-sketches-F02691	258	1.000	$U = \{x \in \mathbb{R} \mid x < 3\}$	$U = \{x \in \mathbb{R} \mid x < 3\}$	Let $U, V \in \Omega(X)$ be the open sets $U = \{x \in \mathbb{R} \mid x < 3\}$ and $V = \{x \in \mathbb{R} \mid -4 < x < 4\}$.
fong-spivak-seven-sketches-F02692	258	1.000	$V = \{x \in \mathbb{R} \mid -4 < x < 4\}$	$V = \{x \in \mathbb{R} \mid -4 < x < 4\}$	Let $U, V \in \Omega(X)$ be the open sets $U = \{x \in \mathbb{R} \mid x < 3\}$ and $V = \{x \in \mathbb{R} \mid -4 < x < 4\}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F02693	258	■ 1.000	$U=(-\infty, 3)$	$U = (-\infty, 3)$	Using interval notation, $U = (-\infty, 3)$ and $V = (-4, 4)$. Then
fong-spivak-seven-sketches-F02694	258	■ 1.000	$V=(-4, 4)$	$V = (-4, 4)$	Using interval notation, $U = (-\infty, 3)$ and $V = (-4, 4)$. Then
fong-spivak-seven-sketches-F02695	258	■ 1.000	$U \wedge V=(-4, 3)$	$U \wedge V = (-4, 3)$	• $U \wedge V = (-4, 3)$.
fong-spivak-seven-sketches-F02696	258	■ 1.000	$U \vee V=(-\infty, 4)$	$U \vee V = (-\infty, 4)$	• $U \vee V = (-\infty, 4)$.
fong-spivak-seven-sketches-F02697	258	■ 1.000	$\neg U=(3, \infty)$	$\neg U = (3, \infty)$	• $\neg U = (3, \infty)$.
fong-spivak-seven-sketches-F02698	258	■ 1.000	$\neg V=(-\infty, -4) \cup (4, \infty)$	$\neg V = (-\infty, -4) \cup (4, \infty)$	• $\neg V = (-\infty, -4) \cup (4, \infty)$.
fong-spivak-seven-sketches-F02699	258	■ 0.999	$(U \Rightarrow V)=(-4, \infty)$	$(U \Rightarrow V) = (-4, \infty)$	• $(U \Rightarrow V) = (-4, \infty)$
fong-spivak-seven-sketches-F02700	258	■ 1.000	$(V \Rightarrow U)=U$	$(V \Rightarrow U) = U$	• $V \Rightarrow U = U$
fong-spivak-seven-sketches-F02701	258	■ 1.000	$U=\mathbb{R}-\{0\}$	$U = \mathbb{R} - \{0\}$	subset $U = \mathbb{R} - \{0\}$.
fong-spivak-seven-sketches-F02702	258	■ 1.000	$\neg \neg U$	$\neg \neg U$	2. What open subset is $\neg \neg U$?
fong-spivak-seven-sketches-F02703	258	■ 1.000	$U \subseteq \neg \neg U$	$U \subseteq \neg \neg U$	3. Is it true that $U \subseteq \neg \neg U$?
fong-spivak-seven-sketches-F02704	258	■ 1.000	$\neg \neg U \subseteq U$	$\neg \neg U \subseteq U$	4. Is it true that $\neg \neg U \subseteq U$?
fong-spivak-seven-sketches-F02705	258	■ 0.942	$\wedge U)=U$	$\wedge U) = U$	$U \in \mathbf{Op}$, we have $(\text{true} \wedge U) = U$. Which open set is it?
fong-spivak-seven-sketches-F02706	258	■ 0.877	$\vee U)=$	$\vee U) =$	2. Other things we should expect from true include $(\text{true} \vee U) = \text{true}$ and $(U \Rightarrow$
fong-spivak-seven-sketches-F02707	258	■ 0.877	$(U \Rightarrow$	$(U \Rightarrow$	2. Other things we should expect from true include $(\text{true} \vee U) = \text{true}$ and $(U \Rightarrow$
fong-spivak-seven-sketches-F02708	258	■ 0.559	$)=$	$) =$	$\text{true}) = \text{true}$ and $(\text{true} \Rightarrow U) = U$. Do these hold for your answer to 1?
fong-spivak-seven-sketches-F02709	258	■ 0.559	$\Rightarrow U)=U$	$\Rightarrow U) = U$	$\text{true}) = \text{true}$ and $(\text{true} \Rightarrow U) = U$. Do these hold for your answer to 1?
fong-spivak-seven-sketches-F02710	258	■ 0.974	$\vee U)=U$	$\vee U) = U$	$U \in \mathbf{Op}$, we have $(\text{false} \vee U) = U$. Which open set is it?
fong-spivak-seven-sketches-F02711	258	■ 1.000	$\wedge U$	$\wedge U$	4. Other things we should expect from false include $(\text{false} \wedge U) = \text{false}$ and
fong-spivak-seven-sketches-F02712	258	■ 1.000	$\pi: E \rightarrow X$	$\pi : E \rightarrow X$	Example 7.61. For a vector bundle $\pi: E \rightarrow X$ over a space X , the corresponding sheaf
fong-spivak-seven-sketches-F02713	258	■ 0.978	Sec_{π}	Sec_{π}	is Sec_{π} corresponding to its sections: to each open set $i_U: U \subseteq X$, we associate the set
fong-spivak-seven-sketches-F02714	258	■ 0.978	$i_U: U \subseteq X$	$i_U : U \subseteq X$	is Sec_{π} corresponding to its sections: to each open set $i_U: U \subseteq X$, we associate the set
fong-spivak-seven-sketches-F02715	258	■ 0.999	$s: U \rightarrow E$	$s : U \rightarrow E$	of functions $s: U \rightarrow E$ for which $s \circ \pi = i_U$. For example, in the case of the tangent
fong-spivak-seven-sketches-F02716	258	■ 0.999	$s; \pi=i_U$	$s; \pi = i_U$	of functions $s: U \rightarrow E$ for which $s \circ \pi = i_U$. For example, in the case of the tangent
fong-spivak-seven-sketches-F02717	258	■ 1.000	$\pi: TM \rightarrow M$	$\pi : TM \rightarrow M$	bundle $\pi: TM \rightarrow M$ (see Example 7.46), the corresponding sheaf, call it $\mathbf{VF}$, associates
fong-spivak-seven-sketches-F02718	258	■ 0.991	$\operatorname{VF}(U)$	$\operatorname{VF}(U)$	to each U the set $\operatorname{VF}(U)$ of vector fields on U .
fong-spivak-seven-sketches-F02719	258	■ 0.996	$\operatorname{VF} \rightarrow \Omega$	$\operatorname{VF} \rightarrow \Omega$	fields. For example, one could have a predicate $\operatorname{Grad}: \operatorname{VF} \rightarrow \Omega$ that asks for the largest

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F02720	258	■ 0.991	$\operatorname{Grad}(v)$	$\operatorname{Grad}(v)$	subspace $\operatorname{Grad}(v)$ on which a given vector field v comes from the gradient of some
fong-spivak-seven-sketches-F02721	259	■ 0.585	$p: S \rightarrow \Omega$	$p: S \rightarrow \Omega$	In a topos $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$, a predicate is a sheaf morphism $p: S \rightarrow \Omega$ where $S \in \mathcal{E}$ is
fong-spivak-seven-sketches-F02722	259	■ 0.585	$S \in \mathcal{E}$	$S \in \mathcal{E}$	In a topos $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$, a predicate is a sheaf morphism $p: S \rightarrow \Omega$ where $S \in \mathcal{E}$ is
fong-spivak-seven-sketches-F02723	259	■ 1.000	$p(U): S(U) \rightarrow \Omega(U)$	$p(U): S(U) \rightarrow \Omega(U)$	we get a function $p(U): S(U) \rightarrow \Omega(U)$ for any open set $U \subseteq X$. In the above example—
fong-spivak-seven-sketches-F02724	259	■ 0.972	$\operatorname{Bob} \in S(U)$	$\operatorname{Bob} \in S(U)$	come and go over time), and $\operatorname{Bob} \in S(U)$ is a person existing over a time U , and p is the
fong-spivak-seven-sketches-F02725	259	■ 0.987	$p(\operatorname{Bob})$	$p(\operatorname{Bob})$	predicate “likes the weather,” then $p(\operatorname{Bob})$ is the set of times during which Bob likes
fong-spivak-seven-sketches-F02726	259	■ 0.979	$s \in S(U)$	$s \in S(U)$	weather,” acts on sections $s \in S(U)$, say $s = \operatorname{Bob}$. But by Definition 7.12, any predicate
fong-spivak-seven-sketches-F02727	259	■ 0.979	$s =$	$s =$	weather,” acts on sections $s \in S(U)$, say $s = \operatorname{Bob}$. But by Definition 7.12, any predicate
fong-spivak-seven-sketches-F02728	259	■ 1.000	$\{S \mid p\} \subseteq S$	$\{S \mid p\} \subseteq S$	$p: S \rightarrow \Omega$ also defines a subobject of $\{S \mid p\} \subseteq S$. Describe the sections of this subsheaf.
fong-spivak-seven-sketches-F02729	259	■ 1.000	$ \Omega^E = \mathcal{E}(S, \Omega)$	$ \Omega^E = \mathcal{E}(S, \Omega)$	set of S -predicates $ \Omega^E = \mathcal{E}(S, \Omega)$ is naturally given the structure of a poset, which we
fong-spivak-seven-sketches-F02730	259	■ 0.917	$p, q: S \rightarrow \Omega$	$p, q: S \rightarrow \Omega$	Given two predicates $p, q: S \rightarrow \Omega$, we say that $p \leq^S q$ if the first implies the second.
fong-spivak-seven-sketches-F02731	259	■ 0.917	$p \leq^S q$	$p \leq^S q$	Given two predicates $p, q: S \rightarrow \Omega$, we say that $p \leq^S q$ if the first implies the second.
fong-spivak-seven-sketches-F02732	259	■ 0.997	$p(s) \subseteq U$	$p(s) \subseteq U$	$p(s) \subseteq U$ and $q(s) \subseteq U$. We say that $p \leq^S q$ if $p(s) \subseteq q(s)$ for all $U \in \mathbf{Op}$ and $s \in S(U)$.
fong-spivak-seven-sketches-F02733	259	■ 0.997	$q(s) \subseteq U$	$q(s) \subseteq U$	$p(s) \subseteq U$ and $q(s) \subseteq U$. We say that $p \leq^S q$ if $p(s) \subseteq q(s)$ for all $U \in \mathbf{Op}$ and $s \in S(U)$.
fong-spivak-seven-sketches-F02734	259	■ 0.997	$p(s) \subseteq q(s)$	$p(s) \subseteq q(s)$	$p(s) \subseteq U$ and $q(s) \subseteq U$. We say that $p \leq^S q$ if $p(s) \subseteq q(s)$ for all $U \in \mathbf{Op}$ and $s \in S(U)$.
fong-spivak-seven-sketches-F02735	259	■ 0.999	$\leq^S$	$\leq^S$	We often drop the superscript from $\leq^S$ and simply write $\leq$. In formal logic notation,
fong-spivak-seven-sketches-F02736	260	■ 1.000	$S = 1$	$S = 1$	In particular, if $S = 1$ is the terminal object, we denote $ \Omega^S $ by $ \Omega $, and refer to elements
fong-spivak-seven-sketches-F02737	260	■ 1.000	$ \Omega^S $	$ \Omega^S $	In particular, if $S = 1$ is the terminal object, we denote $ \Omega^S $ by $ \Omega $, and refer to elements
fong-spivak-seven-sketches-F02738	260	■ 1.000	$ \Omega $	$ \Omega $	In particular, if $S = 1$ is the terminal object, we denote $ \Omega^S $ by $ \Omega $, and refer to elements
fong-spivak-seven-sketches-F02739	260	■ 1.000	$p \in \Omega $	$p \in \Omega $	$p \in \Omega $ as <i>propositions</i> . They are just morphisms $p: 1 \rightarrow \Omega$.
fong-spivak-seven-sketches-F02740	260	■ 1.000	$p: 1 \rightarrow \Omega$	$p: 1 \rightarrow \Omega$	$p \in \Omega $ as <i>propositions</i> . They are just morphisms $p: 1 \rightarrow \Omega$.
fong-spivak-seven-sketches-F02741	260	■ 1.000	$p = q$	$p = q$	$p = q$. The reason is that for any subsets $U, V \subseteq X$, if $U \subseteq V$ and $V \subseteq U$ then $U = V$.
fong-spivak-seven-sketches-F02742	260	■ 1.000	$U, V \subseteq X$	$U, V \subseteq X$	$p = q$. The reason is that for any subsets $U, V \subseteq X$, if $U \subseteq V$ and $V \subseteq U$ then $U = V$.
fong-spivak-seven-sketches-F02743	260	■ 1.000	$U = V$	$U = V$	$p = q$. The reason is that for any subsets $U, V \subseteq X$, if $U \subseteq V$ and $V \subseteq U$ then $U = V$.
fong-spivak-seven-sketches-F02744	260	■ 0.976	$S \in \mathbf{Shv}(X)$	$S \in \mathbf{Shv}(X)$	<i>Exercise 7.64.</i> Give an example of a space X , a sheaf $S \in \mathbf{Shv}(X)$, and two predicates
fong-spivak-seven-sketches-F02745	260	■ 1.000	$p(s) \vdash_{s:S} q(s)$	$p(s) \vdash_{s:S} q(s)$	$p, q: S \rightarrow \Omega$ for which $p(s) \vdash_{s:S} q(s)$ holds. You do not have to be formal.
fong-spivak-seven-sketches-F02746	260	■ 1.000	$\wedge, \vee, \Rightarrow, \neg$	$\wedge, \vee, \Rightarrow, \neg$	All of the logical symbols (true, false, $\wedge, \vee, \Rightarrow, \neg$) from Section 7.4.2 make sense

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F02747	260	■ 1.000	$(p \wedge q) : S \rightarrow \Omega$	$(p \wedge q) : S \rightarrow \Omega$	in any such poset $ \Omega^S $. For any two predicates $p, q : S \rightarrow \Omega$, we define $(p \wedge q) : S \rightarrow \Omega$
fong-spivak-seven-sketches-F02748	260	■ 1.000	$(p \wedge q)(s) := p(s) \wedge q(s)$	$(p \wedge q)(s) := p(s) \wedge q(s)$	by $(p \wedge q)(s) := p(s) \wedge q(s)$, and similarly for $\vee$. Thus one says that these operations
fong-spivak-seven-sketches-F02749	260	■ 1.000	$n+1$	$n + 1$	exists.’ Each takes in a predicate of $n + 1$ variables and returns a predicate of n variables.
fong-spivak-seven-sketches-F02750	260	■ 1.000	$S, T \in \mathbf{Shv}(\mathbf{S h v}(X, \mathbf{0 p}))$	$S, T \in \mathbf{Shv}(X, \mathbf{Op})$	Example 7.65. Suppose we have two sheaves $S, T \in \mathbf{Shv}(X, \mathbf{Op})$ and a predicate $p : S \times$
fong-spivak-seven-sketches-F02751	260	■ 1.000	$p : S \times$	$p : S \times$	Example 7.65. Suppose we have two sheaves $S, T \in \mathbf{Shv}(X, \mathbf{Op})$ and a predicate $p : S \times$
fong-spivak-seven-sketches-F02752	260	■ 1.000	$T \rightarrow \Omega$	$T \rightarrow \Omega$	$T \rightarrow \Omega$. Let’s say T represents what’s considered newsworthy and S is again the set
fong-spivak-seven-sketches-F02753	260	■ 1.000	$t \in T(U)$	$t \in T(U)$	of people. So for a subset of time U , a section $t \in T(U)$ is something that’s considered
fong-spivak-seven-sketches-F02754	260	■ 1.000	$p(s, t)$	$p(s, t)$	of times on which $p(s, t)$ is true.
fong-spivak-seven-sketches-F02755	260	■ 0.876	$T, \ldots$	$T, \dots$	“For all t in $T, \dots$ is worried about t ” is itself a predicate on just one variable, S ,
fong-spivak-seven-sketches-F02756	260	■ 1.000	$t^{\prime}$	t''	t'' is also a predicate on S , which we denote $\exists(t : T). p(s, t)$. If we apply this predicate
fong-spivak-seven-sketches-F02757	260	■ 1.000	$\exists(t : T) . p(s, t)$	$\exists(t : T). p(s, t)$	t'' is also a predicate on S , which we denote $\exists(t : T). p(s, t)$. If we apply this predicate
fong-spivak-seven-sketches-F02758	261	■ 1.000	$p : \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{B}$	$p : \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{B}$	Exercise 7.66. In the topos Set , where $\Omega = \mathbb{B}$, consider the predicate $p : \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{B}$
fong-spivak-seven-sketches-F02759	261	■ 0.999	$\forall(z : \mathbb{Z}) . p(n, z)$	$\forall(z : \mathbb{Z}). p(n, z)$	1. What is the set of $n \in \mathbb{N}$ for which the predicate $\forall(z : \mathbb{Z}). p(n, z)$ holds?
fong-spivak-seven-sketches-F02760	261	■ 1.000	$\exists(z : \mathbb{Z}) . p(n, z)$	$\exists(z : \mathbb{Z}). p(n, z)$	2. What is the set of $n \in \mathbb{N}$ for which the predicate $\exists(z : \mathbb{Z}). p(n, z)$ holds?
fong-spivak-seven-sketches-F02761	261	■ 0.999	$z \in \mathbb{Z}$	$z \in \mathbb{Z}$	3. What is the set of $z \in \mathbb{Z}$ for which the predicate $\forall(n : \mathbb{N}). p(n, z)$ holds?
fong-spivak-seven-sketches-F02762	261	■ 0.999	$\forall(n : \mathbb{N}) . p(n, z)$	$\forall(n : \mathbb{N}). p(n, z)$	3. What is the set of $z \in \mathbb{Z}$ for which the predicate $\forall(n : \mathbb{N}). p(n, z)$ holds?
fong-spivak-seven-sketches-F02763	261	■ 0.998	$\exists(n : \mathbb{N}) . p(n, z)$	$\exists(n : \mathbb{N}). p(n, z)$	4. What is the set of $z \in \mathbb{Z}$ for which the predicate $\exists(n : \mathbb{N}). p(n, z)$ holds?
fong-spivak-seven-sketches-F02764	261	■ 1.000	$\forall(t$	$\forall(t$	So given p , we have a universally- and an existentially-quantified predicate $\forall(t :$
fong-spivak-seven-sketches-F02765	261	■ 1.000	$S \rightarrow \Omega$	$S \rightarrow \Omega$	morphisms $S \rightarrow \Omega$ or, equivalently, as subsheaves of S ?
fong-spivak-seven-sketches-F02766	261	■ 1.000	$p : S \times T \rightarrow \Omega$	$p : S \times T \rightarrow \Omega$	Universal quantification. Given a predicate $p : S \times T \rightarrow \Omega$, the universally-quantified
fong-spivak-seven-sketches-F02767	261	■ 0.997	$\forall(t : T) . p(s, t)$	$\forall(t : T). p(s, t)$	predicate $\forall(t : T). p(s, t)$ takes a section $s \in S(U)$, for any open set U , and returns a
fong-spivak-seven-sketches-F02768	261	■ 1.000	$V \in \Omega(U)$	$V \in \Omega(U)$	certain open set $V \in \Omega(U)$. Namely, it returns the largest open set $V \subseteq U$ for which
fong-spivak-seven-sketches-F02769	261	■ 1.000	$p _V(t) = V$	$p(s _V, t) = V$	$p(s _V, t) = V$ holds for all $t \in T(V)$.
fong-spivak-seven-sketches-F02770	261	■ 1.000	$t \in T(V)$	$t \in T(V)$	$p(s _V, t) = V$ holds for all $t \in T(V)$.
fong-spivak-seven-sketches-F02771	261	■ 0.926	$p(s, t) =$	$p(s, t) =$	above definition to the example $p(s, t) =$ “person s is worried about news t ” from
fong-spivak-seven-sketches-F02772	261	■ 1.000	$T(V)$	$T(V)$	Example 7.65. Here, $T(V)$ is the set of items that are in the news throughout the
fong-spivak-seven-sketches-F02773	261	■ 1.000	$p^{\prime} : S \rightarrow \Omega^T$	$p' : S \rightarrow \Omega^T$	where $p' : S \rightarrow \Omega^T$ is the currying of $S \times T \rightarrow \Omega$ and true^T is the currying of the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F02774	261	■ 1.000	$S \times T \rightarrow \Omega$	$S \times T \rightarrow \Omega$	where $p': S \rightarrow \Omega^T$ is the currying of $S \times T \rightarrow \Omega$ and true^T is the currying of the
fong-spivak-seven-sketches-F02775	261	■ 1.000	$\{ \}^{\{T\}}$	T	where $p': S \rightarrow \Omega^T$ is the currying of $S \times T \rightarrow \Omega$ and true^T is the currying of the
fong-spivak-seven-sketches-F02776	261	■ 0.982	$1 \times T \rightarrow \Omega$	$1 \times T \xrightarrow{!} 1 \xrightarrow{\text{true}} \Omega$	composite $1 \times T \xrightarrow{!} 1 \xrightarrow{\text{true}} \Omega$. See Eq. (7.10).
fong-spivak-seven-sketches-F02777	261	■ 1.000	$V = \bigcup_i V_i$	$V = \bigcup_i V_i$	a certain open set $V \in \Omega(U)$, namely the union $V = \bigcup_i V_i$ of all the open sets V_i for
fong-spivak-seven-sketches-F02778	261	■ 1.000	V_i	V_i	a certain open set $V \in \Omega(U)$, namely the union $V = \bigcup_i V_i$ of all the open sets V_i for
fong-spivak-seven-sketches-F02779	261	■ 1.000	$t_i \in T(V_i)$	$t_i \in T(V_i)$	which there exists some $t_i \in T(V_i)$ satisfying $p(s _{V_i}, t_i) = V_i$. If the result is U itself, you
fong-spivak-seven-sketches-F02780	261	■ 1.000	$p(s _{V_i}, t_i) = V_i$	$p(s _{V_i}, t_i) = V_i$	which there exists some $t_i \in T(V_i)$ satisfying $p(s _{V_i}, t_i) = V_i$. If the result is U itself, you
fong-spivak-seven-sketches-F02781	261	■ 1.000	$p(t)$	$p(t)$	might be tempted to think “ah, so there exists some $t \in T(U)$ satisfying $p(t)$,” but that
fong-spivak-seven-sketches-F02782	261	■ 0.516	$U = \bigcup U_i$	$U = \bigcup U_i$	is not necessarily so. There is just a cover of $U = \bigcup U_i$ and local sections $t_i \in T(U_i)$,
fong-spivak-seven-sketches-F02783	261	■ 0.516	$t_i \in T(U_i)$	$t_i \in T(U_i)$	is not necessarily so. There is just a cover of $U = \bigcup U_i$ and local sections $t_i \in T(U_i)$,
fong-spivak-seven-sketches-F02784	262	■ 0.951	t'	t'	<i>Exercise 7.68.</i> Apply the above definition to the “person s is worried about news t ”
fong-spivak-seven-sketches-F02785	262	■ 1.000	$\{(s, t) \in S \times T \mid p(s, t)\} \subseteq S \times T$	$\{(s, t) \in S \times T \mid p(s, t)\} \subseteq S \times T$	with the subobject classified by p , namely $\{(s, t) \in S \times T \mid p(s, t)\} \subseteq S \times T$, compose
fong-spivak-seven-sketches-F02786	262	■ 0.982	$\pi_S: S \times T \rightarrow S$	$\pi_S: S \times T \rightarrow S$	with the projection $\pi_S: S \times T \rightarrow S$ as on the upper right; then take the epi-mono
fong-spivak-seven-sketches-F02787	262	■ 0.983	$(\Omega^S , \leq^S)$	$(\Omega^S , \leq^S)$	poset $(\Omega^S , \leq^S)$ of predicates on S , where $ \Omega^S = \mathcal{E}(S, \Omega)$ is just the set of morphisms
fong-spivak-seven-sketches-F02788	262	■ 0.983	$ \Omega^S = \mathcal{E}(S, \Omega)$	$ \Omega^S = \mathcal{E}(S, \Omega)$	poset $(\Omega^S , \leq^S)$ of predicates on S , where $ \Omega^S = \mathcal{E}(S, \Omega)$ is just the set of morphisms
fong-spivak-seven-sketches-F02789	262	■ 0.999	$\mathbf{Shv}(X)$	$\mathbf{Shv}(X)$	Definition 7.69. A modality in $\mathbf{Shv}(X)$ is a sheaf morphism $j: \Omega \rightarrow \Omega$ satisfying three
fong-spivak-seven-sketches-F02790	262	■ 0.999	$j: \Omega \rightarrow \Omega$	$j: \Omega \rightarrow \Omega$	Definition 7.69. A modality in $\mathbf{Shv}(X)$ is a sheaf morphism $j: \Omega \rightarrow \Omega$ satisfying three
fong-spivak-seven-sketches-F02791	262	■ 1.000	$p, q \in \Omega(U)$	$p, q \in \Omega(U)$	properties for all $U \subseteq X$ and $p, q \in \Omega(U)$:
fong-spivak-seven-sketches-F02792	262	■ 1.000	$(j; j)(p) \leq j(p)$	$(j; j)(p) \leq j(p)$	(b) $(j; j)(p) \leq j(p)$; and
fong-spivak-seven-sketches-F02793	262	■ 1.000	$j(p \wedge q) = j(p) \wedge j(q)$	$j(p \wedge q) = j(p) \wedge j(q)$	(c) $j(p \wedge q) = j(p) \wedge j(q)$.
fong-spivak-seven-sketches-F02794	262	■ 1.000	$p \in \Omega(U)$	$p \in \Omega(U)$	holds for all $U \subseteq X$ and $p \in \Omega(U)$. Show that for all $q \in \Omega(U)$ we have $j(j(q)) \leq j(q)$ iff
fong-spivak-seven-sketches-F02795	262	■ 1.000	$q \in \Omega(U)$	$q \in \Omega(U)$	holds for all $U \subseteq X$ and $p \in \Omega(U)$. Show that for all $q \in \Omega(U)$ we have $j(j(q)) \leq j(q)$ iff
fong-spivak-seven-sketches-F02796	262	■ 1.000	$j(j(q)) \leq j(q)$	$j(j(q)) \leq j(q)$	holds for all $U \subseteq X$ and $p \in \Omega(U)$. Show that for all $q \in \Omega(U)$ we have $j(j(q)) \leq j(q)$ iff
fong-spivak-seven-sketches-F02797	262	■ 1.000	$j(j(q)) = j(q)$	$j(j(q)) = j(q)$	$j(j(q)) = j(q)$.
fong-spivak-seven-sketches-F02798	262	■ 1.000	$\Omega \rightarrow \Omega$	$\Omega \rightarrow \Omega$	(a) the sheaf morphism $\Omega \rightarrow \Omega$ given by sending q to $p \Rightarrow q$ is a modality.
fong-spivak-seven-sketches-F02799	262	■ 1.000	$p \vee q$	$p \vee q$	(b) the sheaf morphism $\Omega \rightarrow \Omega$ given by sending q to $p \vee q$ is a modality.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F02800	262	■ 0.997	$(q \Rightarrow p) \Rightarrow p$	$(q \Rightarrow p) \Rightarrow p$	(c) the sheaf morphism $\Omega \rightarrow \Omega$ given by sending q to $(q \Rightarrow p) \Rightarrow p$ is a modality.
fong-spivak-seven-sketches-F02801	263	■ 1.000	$p(s)$	$p(s)$	$p(s)$, and what does it mean?
fong-spivak-seven-sketches-F02802	263	■ 1.000	$j(p(s))$	$j(p(s))$	3. What sort of thing is $j(p(s))$ and what does it mean?
fong-spivak-seven-sketches-F02803	263	■ 1.000	$p(s) \leq j(p(s))$	$p(s) \leq j(p(s))$	4. Is it true that $p(s) \leq j(p(s))$? Explain briefly.
fong-spivak-seven-sketches-F02804	263	■ 1.000	$j(j(p(s)))=j(p(s))$	$j(j(p(s))) = j(p(s))$	5. Is it true that $j(j(p(s))) = j(p(s))$? Explain briefly.
fong-spivak-seven-sketches-F02805	263	■ 1.000	$q: S \rightarrow \Omega$	$q: S \rightarrow \Omega$	6. Choose another predicate $q: S \rightarrow \Omega$. Is it true that $j(p \wedge q) = j(p) \wedge j(q)$? Explain
fong-spivak-seven-sketches-F02806	264	■ 0.500	$s \in S(c)$	$s \in S(c)$	Is it true that for every table $c \in \text{Ob}(\mathcal{C})$ and every row $s \in S(c)$ there exists
fong-spivak-seven-sketches-F02807	264	■ 1.000	$t \in T(c)$	$t \in T(c)$	a row $t \in T(c)$ such that $f(s) = t$?
fong-spivak-seven-sketches-F02808	264	■ 1.000	$f(s)=t$	$f(s) = t$	a row $t \in T(c)$ such that $f(s) = t$?
fong-spivak-seven-sketches-F02809	264	■ 1.000	$\left(U_{\{i\}} \subseteq U\right)_{i \in I}$	$(U_i \subseteq U)_{i \in I}$	bundle S , there exists an open covering of $(U_i \subseteq U)_{i \in I}$ of U and a section
fong-spivak-seven-sketches-F02810	264	■ 1.000	$f(\text{left}(t_{\{i\}})=\text{left.s}\text{right} _{U_{\{i\}}})$	$f(t_i) = s _{U_i}$	$t_i \in T(U_i)$ of the bundle T for each $i \in I$, such that $f(t_i) = s _{U_i}$ is the
fong-spivak-seven-sketches-F02811	264	■ 1.000	$U_{\{i\}}$	U_i	restriction of s to U_i ?
fong-spivak-seven-sketches-F02812	264	■ 0.912	$\boldsymbol{\operatorname{Shv}}(\llbracket \mathbb{R} \rrbracket)$	$\mathbf{Shv}(\llbracket \mathbb{R} \rrbracket)$	intervals of time. In other words, we will be interested in the category $\mathbf{Shv}(\mathbb{R})$ of
fong-spivak-seven-sketches-F02813	264	■ 0.773	$\mathbb{I} \subseteq \mathbb{R}$	$\mathbb{I} \subseteq \mathbb{R}$	and $\mathbf{Op}$ is a topology on X ; see Definition 7.25. The set of points for $\mathbb{I}$ is that of all
fong-spivak-seven-sketches-F02814	264	■ 1.000	$a < b$	$a < b$	For $a < b$ in $\mathbb{R}$, let $o_{[a,b]}$ denote the set $o_{[a,b]} := \{[d, u] \in \mathbb{I} \mid a < d \leq u < b\}$; these are
fong-spivak-seven-sketches-F02815	264	■ 1.000	$o_{\{[a, b]\}}$	$o_{[a,b]}$	For $a < b$ in $\mathbb{R}$, let $o_{[a,b]}$ denote the set $o_{[a,b]} := \{[d, u] \in \mathbb{I} \mid a < d \leq u < b\}$; these are
fong-spivak-seven-sketches-F02816	264	■ 1.000	$o_{\{[a, b]\}} := \{[d, u] \in \mathbb{I} \mid a < d \leq u < b\}$	$o_{[a,b]} := \{[d, u] \in \mathbb{I} \mid a < d \leq u < b\}$	For $a < b$ in $\mathbb{R}$, let $o_{[a,b]}$ denote the set $o_{[a,b]} := \{[d, u] \in \mathbb{I} \mid a < d \leq u < b\}$; these are
fong-spivak-seven-sketches-F02817	264	■ 1.000	$o_{\{[0,5]\}}$	$o_{[0,5]}$	Thus for example, $o_{[0,5]}$ is an open set: it contains every $[d, u]$ contained in the
fong-spivak-seven-sketches-F02818	264	■ 1.000	$[d, u]$	$[d, u]$	Thus for example, $o_{[0,5]}$ is an open set: it contains every $[d, u]$ contained in the
fong-spivak-seven-sketches-F02819	264	■ 1.000	$\{x \in \mathbb{I} \mid 0 < x < 5\}$	$\{x \in \mathbb{I} \mid 0 < x < 5\}$	open interval $\{x \in \mathbb{I} \mid 0 < x < 5\}$. Similarly $o_{[4,8]}$ is an open set, but note that
fong-spivak-seven-sketches-F02820	264	■ 1.000	$o_{\{[4,8]\}}$	$o_{[4,8]}$	open interval $\{x \in \mathbb{I} \mid 0 < x < 5\}$. Similarly $o_{[4,8]}$ is an open set, but note that
fong-spivak-seven-sketches-F02821	264	■ 0.965	$o_{\{[0,5]\}} \cup o_{\{[4,8]\}} \neq o_{\{[0,8]\}}$	$o_{[0,5]} \cup o_{[4,8]} \neq o_{[0,8]}$	$o_{[0,5]} \cup o_{[4,8]} \neq o_{[0,8]}$. Indeed, the interval $[2, 6]$ is in the right-hand side but not the left.
fong-spivak-seven-sketches-F02822	265	■ 1.000	$[2,6] \in o_{\{[0,8]\}}$	$[2,6] \in o_{[0,8]}$	1. Explain why $[2,6] \in o_{[0,8]}$.
fong-spivak-seven-sketches-F02823	265	■ 0.996	$[2,6] \notin o_{\{[0,5]\}} \cup o_{\{[4,8]\}}$	$[2,6] \notin o_{[0,5]} \cup o_{[4,8]}$	2. Explain why $[2,6] \notin o_{[0,5]} \cup o_{[4,8]}$.
fong-spivak-seven-sketches-F02824	265	■ 0.964	$\mathbb{R} \subseteq \mathbb{I}$	$\mathbb{R} \subseteq \mathbb{I}$	Exercise 7.77. Show that a subset $U \subseteq \mathbb{R}$ is open in the subspace topology of $\mathbb{R} \subseteq \mathbb{I}$
fong-spivak-seven-sketches-F02825	265	■ 1.000	$U \cap \mathbb{R}$	$U \cap \mathbb{R}$	iff $U \cap \mathbb{R}$ is open in the usual topology on $\mathbb{R}$ defined in Example 7.26.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches- F02826	265	■ 0.280	$\mathbf{BT} = \mathbf{Shv}(\mathbb{I}\mathbb{R}, \mathbf{Op})$	$\mathbf{BT} = \mathbf{Shv}(\mathbb{I}\mathbb{R}, \mathbf{Op})$	We cannot go into much depth about the sheaf topos $\mathbf{BT} = \mathbf{Shv}(\mathbb{I}\mathbb{R}, \mathbf{Op})$, for reasons
fong-spivak-seven-sketches- F02827	265	■ 0.591	$\mathbb{I}\mathbb{R}, \mathbf{Op}$	$\mathbb{I}\mathbb{R}, \mathbf{Op}$	What is a sheaf on $\mathbb{I}\mathbb{R}$? A sheaf S on the interval domain $(\mathbb{I}\mathbb{R}, \mathbf{Op})$ is a functor
fong-spivak-seven-sketches- F02828	265	■ 0.864	$S : \mathbf{Op}^{\mathbf{op}} \rightarrow \mathbf{Set}$	$S : \mathbf{Op}^{\mathbf{op}} \rightarrow$	$S : \mathbf{Op}^{\mathbf{op}} \rightarrow \mathbf{Set}$: it assigns to each open set U a set $S(U)$; how should we interpret this?
fong-spivak-seven-sketches- F02829	265	■ 0.864	$S(U)$	$S(U)$	$S : \mathbf{Op}^{\mathbf{op}} \rightarrow \mathbf{Set}$: it assigns to each open set U a set $S(U)$; how should we interpret this?
fong-spivak-seven-sketches- F02830	265	■ 0.997	(s_i)	(s_i)	V . If $U = \bigcup_{i \in I} U_i$ and we can find matching U_i -events (s_i) for each $i \in I$, then the sheaf
fong-spivak-seven-sketches- F02831	265	■ 0.835	$U \subseteq \mathbb{I}\mathbb{R}$	$U \subseteq \mathbb{I}\mathbb{R}$	We said in Section 7.5.1 that every open set $U \subseteq \mathbb{I}\mathbb{R}$ can be written as the union of
fong-spivak-seven-sketches- F02832	265	■ 1.000	$S(o_{[a,b]})$	$S(o_{[a,b]})$	basic open sets $o_{[a,b]}$. This implies that any sheaf S is determined by its values $S(o_{[a,b]})$
fong-spivak-seven-sketches- F02833	265	■ 0.753	$A \in \mathbf{Shv}(\mathbb{I}\mathbb{R})$	$A \in \mathbf{Shv}(\mathbb{I}\mathbb{R})$	<i>Example 7.78.</i> For any set A there is a sheaf $A \in \mathbf{Shv}(\mathbb{I}\mathbb{R})$ that assigns to each open set U
fong-spivak-seven-sketches- F02834	265	■ 1.000	$A(U) := A$	$A(U) := A$	the set $A(U) := A$. This allows us to refer to integers, or real numbers, or letters of an
fong-spivak-seven-sketches- F02835	265	■ 1.000	$7 \in \mathbb{N}$	$7 \in \mathbb{N}$	alphabet, as though they were behaviors. What sort of behavior is $7 \in \mathbb{N}$? It is the sort
fong-spivak-seven-sketches- F02836	266	■ 0.881	$(X, \mathbf{Op}_X)$	$(X, \mathbf{Op}_X)$	<i>Example 7.79.</i> Fix any topological space $(X, \mathbf{Op}_X)$. Then there is a sheaf F_X of local
fong-spivak-seven-sketches- F02837	266	■ 0.881	F_X	F_X	<i>Example 7.79.</i> Fix any topological space $(X, \mathbf{Op}_X)$. Then there is a sheaf F_X of local
fong-spivak-seven-sketches- F02838	266	■ 0.614	$U \in \mathbf{Op}_{\mathbb{R}}$	$U \in \mathbf{Op}_{\mathbb{R}}$	functions from $\mathbb{I}\mathbb{R}$ to X . That is, for any open set $U \in \mathbf{Op}_{\mathbb{R}}$, we assign the set $F_X(U) :=$
fong-spivak-seven-sketches- F02839	266	■ 0.614	$F_X(U) :=$	$F_X(U) :=$	functions from $\mathbb{I}\mathbb{R}$ to X . That is, for any open set $U \in \mathbf{Op}_{\mathbb{R}}$, we assign the set $F_X(U) :=$
fong-spivak-seven-sketches- F02840	266	■ 0.875	$\{f : U \rightarrow X \mid f$	$\{f : U \rightarrow X \mid f$	$\{f : U \rightarrow X \mid f$ is continuous}. There is also the sheaf G_X of local functions on the
fong-spivak-seven-sketches- F02841	266	■ 0.875	G_X	G_X	$\{f : U \rightarrow X \mid f$ is continuous}. There is also the sheaf G_X of local functions on the
fong-spivak-seven-sketches- F02842	266	■ 0.960	$\mathbb{R} \subseteq \mathbb{I}\mathbb{R}$	$\mathbb{R} \subseteq \mathbb{I}\mathbb{R}$	subspace $\mathbb{R} \subseteq \mathbb{I}\mathbb{R}$. That is, for any open set $U \in \mathbf{Op}_{\mathbb{I}\mathbb{R}}$, we assign the set $G_X(U) :=$
fong-spivak-seven-sketches- F02843	266	■ 0.960	$G_X(U) :=$	$G_X(U) :=$	subspace $\mathbb{R} \subseteq \mathbb{I}\mathbb{R}$. That is, for any open set $U \in \mathbf{Op}_{\mathbb{I}\mathbb{R}}$, we assign the set $G_X(U) :=$
fong-spivak-seven-sketches- F02844	266	■ 0.994	$\{f : U \cap \mathbb{R} \rightarrow X \mid f$	$\{f : U \cap \mathbb{R} \rightarrow X \mid f$	$\{f : U \cap \mathbb{R} \rightarrow X \mid f$ is continuous}.
fong-spivak-seven-sketches- F02845	266	■ 0.995	$R \subseteq \mathbb{I}\mathbb{R}$	$R \subseteq \mathbb{I}\mathbb{R}$	$(X, \mathbf{Op}_X)$ and any subset $R \subseteq \mathbb{I}\mathbb{R}$ of the interval domain. Define $H_X(U) := \{f : U \cap R \rightarrow$
fong-spivak-seven-sketches- F02846	266	■ 0.995	$H_X(U) := \{f : U \cap R \rightarrow$	$H_X(U) := \{f : U \cap R \rightarrow$	$(X, \mathbf{Op}_X)$ and any subset $R \subseteq \mathbb{I}\mathbb{R}$ of the interval domain. Define $H_X(U) := \{f : U \cap R \rightarrow$
fong-spivak-seven-sketches- F02847	266	■ 1.000	$X \mid f$	$X \mid f$	$X \mid f$ is continuous}.
fong-spivak-seven-sketches- F02848	266	■ 1.000	H_X	H_X	1. Is H_X a presheaf? If not, why not; if so, what are the restriction maps?
fong-spivak-seven-sketches- F02849	266	■ 0.715	$(d, d+3)$	$(d, d+3)$	idea is straightforward: for any interval of time, say a three-day interval $(d, d+3)$, let
fong-spivak-seven-sketches- F02850	266	■ 1.000	$L(d, d+3)$	$L(d, d+3)$	$L(d, d+3)$ denote the set of all possible behaviors a person who is in love could possibly
fong-spivak-seven-sketches- F02851	266	■ 0.585	$(\mathbb{I}\mathbb{R}, \mathbf{Op})$	$(\mathbb{I}\mathbb{R}, \mathbf{Op})$	$\mathbf{BT}$ is the topos of sheaves on the space $(\mathbb{I}\mathbb{R}, \mathbf{Op})$, as defined in Section 7.5.1. As
fong-spivak-seven-sketches- F02852	267	■ 1.000	$(0, 5)$	$(0, 5)$	Suppose Bob likes the weather throughout the interval $(0, 5)$ and throughout the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_267 F02853	267	■ 0.975	(0,8)	(0,8)	not imply that it's been true on (0,8): there may have been a change of shift, where
fong-spivak-seven-sketches_267 F02854	267	■ 0.833	(4,8)	(4,8)	This might be true on (0,5) and true on (4,8) but not on (0,8). In order to capture the
fong-spivak-seven-sketches_267 F02855	267	■ 1.000	$\theta \in \mathbb{R}_{\geq 0}$	$\theta \in \mathbb{R}_{\geq 0}$	are behavior types: the altimeter is a variable whose value $\theta \in \mathbb{R}_{\geq 0}$ is changing
fong-spivak-seven-sketches_267 F02856	267	■ 0.925	$\rightarrow \Omega$	$\rightarrow \Omega$	$p: \text{dials} \times \text{thrusters} \rightarrow \Omega$. While we have not written out a formal language for p ,
fong-spivak-seven-sketches_268 F02857	268	■ 1.000	$p(D, T)$	$p(D, T)$	one could imagine the predicate $p(D, T)$ for $D: \text{dials}$ and $T: \text{thrusters}$ as
fong-spivak-seven-sketches_268 F02858	268	■ 0.997	$@_t$	$@_t$	Here $@_t$ is a modality, as we discussed in Definition 7.69; in fact it turns out to be one
fong-spivak-seven-sketches_268 F02859	268	■ 0.988	$@_t(q)$	$@_t(q)$	statement $@_t(q)$ says that q is true in some small enough neighborhood around t . So
fong-spivak-seven-sketches_268 F02860	268	■ 0.998	$D \in$	$D \in$	Given an actual playing-out-of-events over a time period U , i.e. actual section $D \in$
fong-spivak-seven-sketches_268 F02861	268	■ 0.962	(U)	(U)	$\text{dials}(U)$ and section $T \in \text{thrusters}(U)$, the predicate Eq. (7.82) will hold on certain
fong-spivak-seven-sketches_268 F02862	268	■ 0.962	$T \in$	$T \in$	$\text{dials}(U)$ and section $T \in \text{thrusters}(U)$, the predicate Eq. (7.82) will hold on certain
fong-spivak-seven-sketches_271 F02863	271	■ 1.000	$f(x)=x+5$	$f(x) = x + 5$	order-preserving: Take $f(x) = x + 5$; if $x \leq y$ then $x + 5 \leq y + 5$, so f is order-preserving. Take
fong-spivak-seven-sketches_271 F02864	271	■ 1.000	$x+5 \leq y+5$	$x + 5 \leq y + 5$	order-preserving: Take $f(x) = x + 5$; if $x \leq y$ then $x + 5 \leq y + 5$, so f is order-preserving. Take
fong-spivak-seven-sketches_271 F02865	271	■ 0.416	$g(x) := -x$	$g(x) := -x$	$g(x) := -x$; even though $1 \leq 2$, the required inequality $-1 \leq -2$ does not hold, so g is not
fong-spivak-seven-sketches_271 F02866	271	■ 0.416	$1 \leq 2$	$1 \leq 2$	$g(x) := -x$; even though $1 \leq 2$, the required inequality $-1 \leq -2$ does not hold, so g is not
fong-spivak-seven-sketches_271 F02867	271	■ 0.416	$-1 \leq -2$	$-1 \leq -2$	$g(x) := -x$; even though $1 \leq 2$, the required inequality $-1 \leq -2$ does not hold, so g is not
fong-spivak-seven-sketches_271 F02868	271	■ 0.986	$f(x) := x+5$	$f(x) := x + 5$	metric-preserving: Take $f(x) := x + 5$; for any x, y we have $ x - y = (x + 5) - (y + 5) $ by the rules of
fong-spivak-seven-sketches_271 F02869	271	■ 0.986	$ x-y = (x+5) - (y+5) $	$ x - y = (x + 5) - (y + 5) $	metric-preserving: Take $f(x) := x + 5$; for any x, y we have $ x - y = (x + 5) - (y + 5) $ by the rules of
fong-spivak-seven-sketches_271 F02870	271	■ 1.000	$g(x) := 2 * x$	$g(x) := 2 * x$	arithmetic, so $ x - y = f(x) - f(y) $, meaning f preserves metric. Take $g(x) := 2 * x$; then with
fong-spivak-seven-sketches_271 F02871	271	■ 1.000	$x=1$	$x = 1$	$x = 1$ and $y = 2$ we have $ x - y = 1$ but $ 2x - 2y = 2$, so g does not preserve the metric.
fong-spivak-seven-sketches_271 F02872	271	■ 1.000	$y=2$	$y = 2$	$x = 1$ and $y = 2$ we have $ x - y = 1$ but $ 2x - 2y = 2$, so g does not preserve the metric.
fong-spivak-seven-sketches_271 F02873	271	■ 1.000	$ x-y =1$	$ x - y = 1$	$x = 1$ and $y = 2$ we have $ x - y = 1$ but $ 2x - 2y = 2$, so g does not preserve the metric.
fong-spivak-seven-sketches_271 F02874	271	■ 1.000	$ 2x - 2y = 2$	$ 2x - 2y = 2$	$x = 1$ and $y = 2$ we have $ x - y = 1$ but $ 2x - 2y = 2$, so g does not preserve the metric.
fong-spivak-seven-sketches_271 F02875	271	■ 1.000	$f(x) := 3 * x$	$f(x) := 3 * x$	addition-preserving: Take $f(x) := 3 * x$; for any x, y we have $3 * (x + y) = (3 * x) + (3 * y)$, so f preserves
fong-spivak-seven-sketches_271 F02876	271	■ 1.000	$3 * (x+y) = (3 * x) + (3 * y)$	$3 * (x + y) = (3 * x) + (3 * y)$	addition-preserving: Take $f(x) := 3 * x$; for any x, y we have $3 * (x + y) = (3 * x) + (3 * y)$, so f preserves
fong-spivak-seven-sketches_271 F02877	271	■ 1.000	$g(x) := x+1$	$g(x) := x + 1$	addition. Take $g(x) := x + 1$; then with $x = 0$ and $y = 0$, we have $g(x + y) = 1$, but $g(x) + g(y) = 2$,
fong-spivak-seven-sketches_271 F02878	271	■ 1.000	$x=0$	$x = 0$	addition. Take $g(x) := x + 1$; then with $x = 0$ and $y = 0$, we have $g(x + y) = 1$, but $g(x) + g(y) = 2$,
fong-spivak-seven-sketches_271 F02879	271	■ 1.000	$y=0$	$y = 0$	addition. Take $g(x) := x + 1$; then with $x = 0$ and $y = 0$, we have $g(x + y) = 1$, but $g(x) + g(y) = 2$,
fong-spivak-seven-sketches_271 F02880	271	■ 1.000	$g(x+y)=1$	$g(x + y) = 1$	addition. Take $g(x) := x + 1$; then with $x = 0$ and $y = 0$, we have $g(x + y) = 1$, but $g(x) + g(y) = 2$,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F02881	271	■ 1.000	$g(x)+g(y)=2$	$g(x) + g(y) = 2$	addition. Take $g(x) := x + 1$; then with $x = 0$ and $y = 0$, we have $g(x + y) = 1$, but $g(x) + g(y) = 2$,
fong-spivak-seven-sketches-F02882	272	■ 1.000	$A=(12)(3)(4)$	$A = (12)(3)(4)$	For the remaining parts, we choose $A = (12)(3)(4)$ and $B = (13)(2)(4)$.
fong-spivak-seven-sketches-F02883	272	■ 1.000	$B=(13)(2)(4)$	$B = (13)(2)(4)$	For the remaining parts, we choose $A = (12)(3)(4)$ and $B = (13)(2)(4)$.
fong-spivak-seven-sketches-F02884	272	■ 1.000	$A \vee B=(123)(4)$	$A \vee B = (123)(4)$	3. $A \vee B \equiv (123)(4)$
fong-spivak-seven-sketches-F02885	272	■ 0.622	$\{1,2,3\} \cup \{1\}=\{1,2,3\}$	$\{1,2,3\} \cup \{1\} = \{1,2,3\}$	2. The union of $\{1,2,3\}$ and $\{1\}$ is $\{1,2,3\} \cup \{1\} = \{1,2,3\}$.
fong-spivak-seven-sketches-F02886	272	■ 0.992	$\{h,1\} \times \{1,2,3\}$	$\{h,1\} \times \{1,2,3\}$	3. The six elements of $\{h,1\} \times \{1,2,3\}$ are
fong-spivak-seven-sketches-F02887	272	■ 1.000	$\{h,1\} \sqcup \{1,2,3\}$	$\{h,1\} \sqcup \{1,2,3\}$	4. The five elements of $\{h,1\} \sqcup \{1,2,3\}$ are
fong-spivak-seven-sketches-F02888	272	■ 1.000	$\{h,1\} \cup \{1,2,3\}$	$\{h,1\} \cup \{1,2,3\}$	5. The four elements of $\{h,1\} \cup \{1,2,3\}$ are
fong-spivak-seven-sketches-F02889	272	■ 1.000	$0 \notin \{n \in \mathbb{Z} \mid n \geq 1\}$	$0 \notin \{n \in \mathbb{Z} \mid n \geq 1\}$	2. This is false: $0 \in \mathbb{N}$ but $0 \notin \{n \in \mathbb{Z} \mid n \geq 1\}$.
fong-spivak-seven-sketches-F02890	273	■ 0.994	$p_1^{\wedge \prime}, p_2^{\wedge \prime} \in P^{\wedge \prime}$	$p_1', p_2' \in P'$	1. Given $p \in P$, suppose we had $p_1', p_2' \in P'$ such that $A_p = A_{p_1}'$ and $A_p = A_{p_2}'$. Well then $A_{p_1}' = A_{p_2}'$,
fong-spivak-seven-sketches-F02891	273	■ 0.994	$A_{\{p\}}=A_{\{p_1^{\wedge \prime}\}}^{\wedge \prime}$	$A_p = A_{p_1}'$	1. Given $p \in P$, suppose we had $p_1', p_2' \in P'$ such that $A_p = A_{p_1}'$ and $A_p = A_{p_2}'$. Well then $A_{p_1}' = A_{p_2}'$,
fong-spivak-seven-sketches-F02892	273	■ 0.994	$A_{\{p\}}=A_{\{p_2^{\wedge \prime}\}}^{\wedge \prime}$	$A_p = A_{p_2}'$	1. Given $p \in P$, suppose we had $p_1', p_2' \in P'$ such that $A_p = A_{p_1}'$ and $A_p = A_{p_2}'$. Well then $A_{p_1}' = A_{p_2}'$,
fong-spivak-seven-sketches-F02893	273	■ 0.994	$A_{\{p_1^{\wedge \prime}\}}=A_{\{p_2^{\wedge \prime}\}}$	$A_{p_1}' = A_{p_2}'$	1. Given $p \in P$, suppose we had $p_1', p_2' \in P'$ such that $A_p = A_{p_1}'$ and $A_p = A_{p_2}'$. Well then $A_{p_1}' = A_{p_2}'$,
fong-spivak-seven-sketches-F02894	273	■ 1.000	$A_{\{p_1^{\wedge \prime}\}} \cap A_{\{p_2^{\wedge \prime}\}}=A_{\{p_1^{\wedge \prime}\}}$	$A_{p_1}' \cap A_{p_2}' = A_{p_1}'$	so in particular $A_{p_1}' \cap A_{p_2}' = A_{p_1}'$. By the definition of partition (1.14), $A_{p_1}' \neq \emptyset$, and yet if $p_1 \neq p_2$
fong-spivak-seven-sketches-F02895	273	■ 1.000	$A_{\{p_1^{\wedge \prime}\}} \neq \varnothing$	$A_{p_1}' \neq \emptyset$	so in particular $A_{p_1}' \cap A_{p_2}' = A_{p_1}'$. By the definition of partition (1.14), $A_{p_1}' \neq \emptyset$, and yet if $p_1 \neq p_2$
fong-spivak-seven-sketches-F02896	273	■ 1.000	$p_1 \neq p_2$	$p_1 \neq p_2$	so in particular $A_{p_1}' \cap A_{p_2}' = A_{p_1}'$. By the definition of partition (1.14), $A_{p_1}' \neq \emptyset$, and yet if $p_1 \neq p_2$
fong-spivak-seven-sketches-F02897	273	■ 0.998	$A_{\{p_1^{\wedge \prime}\}} \cap A_{\{p_2^{\wedge \prime}\}}=\varnothing$	$A_{p_1}' \cap A_{p_2}' = \emptyset$	then $A_{p_1}' \cap A_{p_2}' = \emptyset$. This can't be, so we must have $p_1' = p_2'$, as desired.
fong-spivak-seven-sketches-F02898	273	■ 0.998	$p_1^{\wedge \prime}=p_2^{\wedge \prime}$	$p_1' = p_2'$	then $A_{p_1}' \cap A_{p_2}' = \emptyset$. This can't be, so we must have $p_1' = p_2'$, as desired.
fong-spivak-seven-sketches-F02899	273	■ 1.000	$A_{\{p^{\wedge \prime}\}}^{\wedge \prime} \neq \varnothing$	$A_{p'}' \neq \emptyset$	2. Suppose given $p' \in P'$; we want to show that there is a $p \in P$ such that $A_p = A_{p'}'$. Since $A_{p'}' \neq \emptyset$
fong-spivak-seven-sketches-F02900	273	■ 1.000	$a \in A_{\{p^{\wedge \prime}\}}^{\wedge \prime}$	$a \in A_{p'}'$	is nonempty by definition, we can pick some $a \in A_{p'}'$; since $A_{p'}' \subseteq A$, we have $a \in A$. Finally,
fong-spivak-seven-sketches-F02901	273	■ 1.000	$A_{\{p^{\wedge \prime}\}}^{\wedge \prime} \subseteq A$	$A_{p'}' \subseteq A$	is nonempty by definition, we can pick some $a \in A_{p'}'$; since $A_{p'}' \subseteq A$, we have $a \in A$. Finally,
fong-spivak-seven-sketches-F02902	273	■ 1.000	$p^{\wedge \prime \prime} \in P^{\wedge \prime \prime}$	$p'' \in P'$	$A_p = A_{p'}'$. By assumption there is some $p'' \in P'$ with $A_p = A_{p''}'$, so now $a \in A_{p''}'$ and $a \in A_{p'}'$,
fong-spivak-seven-sketches-F02903	273	■ 1.000	$A_{\{p\}}=A_{\{p^{\wedge \prime \prime}\}}^{\wedge \prime \prime}$	$A_p = A_{p''}'$	$A_p = A_{p'}'$. By assumption there is some $p'' \in P'$ with $A_p = A_{p''}'$, so now $a \in A_{p''}'$ and $a \in A_{p'}'$,
fong-spivak-seven-sketches-F02904	273	■ 1.000	$a \in A_{\{p^{\wedge \prime \prime}\}}^{\wedge \prime \prime}$	$a \in A_{p''}'$	$A_p = A_{p'}'$. By assumption there is some $p'' \in P'$ with $A_p = A_{p''}'$, so now $a \in A_{p''}'$ and $a \in A_{p'}'$,
fong-spivak-seven-sketches-F02905	273	■ 1.000	$a \in A_{\{p^{\wedge \prime \prime}\}}^{\wedge \prime \prime} \cap A_{\{p^{\wedge \prime \prime}\}}^{\wedge \prime \prime}$	$a \in A_{p'}' \cap A_{p''}'$	so $a \in A_{p'}' \cap A_{p''}'$. Again by definition, having a nonempty intersection means $p' = p''$. So we
fong-spivak-seven-sketches-F02906	273	■ 1.000	$p^{\wedge \prime \prime}=p^{\wedge \prime \prime}$	$p' = p''$	so $a \in A_{p'}' \cap A_{p''}'$. Again by definition, having a nonempty intersection means $p' = p''$. So we
fong-spivak-seven-sketches-F02907	273	■ 1.000	$A_{\{p\}}=A_{\{p^{\wedge \prime \prime}\}}$	$A_p = A_{p'}$	conclude that $A_p = A_{p'}$.

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fong-spivak-seven-sketches_275 F02936	275	■ 1.000	$\mathrm{U}(X)=\mathrm{P}(X)$	$U(X) = P(X)$	exactly the power set, $U(X) = P(X)$.
fong-spivak-seven-sketches_276 F02937	276	■ 1.000	$0 \leq \cdots \leq 3$	$0 \leq \cdots \leq 3$	With $X = \{0,1,2\}$, the Hasse diagram for $P(X)$, the preorder $0 \leq \cdots \leq 3$, and the cardinality map
fong-spivak-seven-sketches_276 F02938	276	■ 1.000	$q \in \uparrow p$	$q \in \uparrow p$	1. Let $q \in \uparrow p$, and suppose $q \leq q'$. Since $q \in \uparrow p$, we have $p \leq q$. Thus by transitivity $p \leq q'$, so
fong-spivak-seven-sketches_276 F02939	276	■ 1.000	$p \leq q^{\{\prime\}}$	$p \leq q'$	1. Let $q \in \uparrow p$, and suppose $q \leq q'$. Since $q \in \uparrow p$, we have $p \leq q$. Thus by transitivity $p \leq q'$, so
fong-spivak-seven-sketches_276 F02940	276	■ 0.991	$q^{\{\prime\}} \in \uparrow p$	$q' \in \uparrow p$	$q' \in \uparrow p$. Thus $\uparrow p$ is an upper set.
fong-spivak-seven-sketches_276 F02941	276	■ 0.589	$q \leq^{\mathrm{op}} p$	$q \leq^{\mathrm{op}} p$	2. Suppose $p \leq q$ in P ; this means that $q \leq^{\mathrm{op}} p$ in P^{op} . We must show that $\uparrow q \subseteq \uparrow p$. Take any
fong-spivak-seven-sketches_276 F02942	276	■ 0.589	$\uparrow q \subseteq \uparrow p$	$\uparrow q \subseteq \uparrow p$	2. Suppose $p \leq q$ in P ; this means that $q \leq^{\mathrm{op}} p$ in P^{op} . We must show that $\uparrow q \subseteq \uparrow p$. Take any
fong-spivak-seven-sketches_276 F02943	276	■ 0.997	$q^{\{\prime\}} \in \uparrow q$	$q' \in \uparrow q$	$q' \in \uparrow q$. Then $q \leq q'$, so by transitivity $p \leq q'$, and hence $q' \in \uparrow p$. Thus $\uparrow q \subseteq \uparrow p$.
fong-spivak-seven-sketches_276 F02944	276	■ 0.997	$p \not\leq p^{\{\prime\}}$	$p \not\leq p'$	if $p \not\leq p'$ then $\uparrow(p') \not\subseteq \uparrow(p)$. This is straightforward, since by reflexivity we always have $p' \in \uparrow(p')$,
fong-spivak-seven-sketches_276 F02945	276	■ 0.997	$\uparrow(p^{\{\prime\}}) \not\subseteq \uparrow(p)$	$\uparrow(p') \not\subseteq \uparrow(p)$	if $p \not\leq p'$ then $\uparrow(p') \not\subseteq \uparrow(p)$. This is straightforward, since by reflexivity we always have $p' \in \uparrow(p')$,
fong-spivak-seven-sketches_276 F02946	276	■ 0.997	$p^{\{\prime\}} \in \uparrow(p^{\{\prime\}})$	$p' \in \uparrow(p')$	if $p \not\leq p'$ then $\uparrow(p') \not\subseteq \uparrow(p)$. This is straightforward, since by reflexivity we always have $p' \in \uparrow(p')$,
fong-spivak-seven-sketches_276 F02947	276	■ 0.998	$p^{\{\prime\}} \notin \uparrow(p)$	$p' \notin \uparrow(p)$	but if $p \not\leq p'$, then $p' \notin \uparrow(p)$, so $\uparrow(p') \not\subseteq \uparrow(p)$.
fong-spivak-seven-sketches_276 F02948	276	■ 0.290	$\uparrow: P^{\mathrm{op}} \rightarrow U(P)$	$\uparrow: P^{\mathrm{op}} \rightarrow U(P)$	4. The map $\uparrow: P^{\mathrm{op}} \rightarrow U(P)$ can be depicted:
fong-spivak-seven-sketches_276 F02949	276	■ 1.000	$p_1 \leq_P p_2$	$p_1 \leq_P p_2$	function $f: P \rightarrow Q$ is monotone, i.e. that if $p_1 \leq_P p_2$ then $f(p_1) \leq_Q f(p_2)$. But in P we have $p_1 \leq_P p_2$
fong-spivak-seven-sketches_276 F02950	276	■ 1.000	$f(p_1) \leq_Q f(p_2)$	$f(p_1) \leq_Q f(p_2)$	function $f: P \rightarrow Q$ is monotone, i.e. that if $p_1 \leq_P p_2$ then $f(p_1) \leq_Q f(p_2)$. But in P we have $p_1 \leq_P p_2$
fong-spivak-seven-sketches_276 F02951	276	■ 1.000	$f(p_1) = f(p_2)$	$f(p_1) = f(p_2)$	iff $p_1 = p_2$; that's what discrete means. If $p_1 \leq_P p_2$ then $p_1 = p_2$, so $f(p_1) = f(p_2)$, so $f(p_1) \leq f(p_2)$.
fong-spivak-seven-sketches_276 F02952	276	■ 1.000	$f(p_1) \leq f(p_2)$	$f(p_1) \leq f(p_2)$	iff $p_1 = p_2$; that's what discrete means. If $p_1 \leq_P p_2$ then $p_1 = p_2$, so $f(p_1) = f(p_2)$, so $f(p_1) \leq f(p_2)$.
fong-spivak-seven-sketches_276 F02953	276	■ 1.000	$X=\mathbb{Z}=\{\ldots,-2,-1,0,1,\ldots\}$	$X = \mathbb{Z} = \{\ldots, -2, -1, 0, 1, \ldots\}$	Let $X = \mathbb{Z} = \{\ldots, -2, -1, 0, 1, \ldots\}$ be the set of all integers, and let $Y = \{n, z, p\}$; let $f: X \rightarrow Y$ send
fong-spivak-seven-sketches_276 F02954	276	■ 1.000	$Y=\{n, z, p\}$	$Y = \{n, z, p\}$	Let $X = \mathbb{Z} = \{\ldots, -2, -1, 0, 1, \ldots\}$ be the set of all integers, and let $Y = \{n, z, p\}$; let $f: X \rightarrow Y$ send
fong-spivak-seven-sketches_276 F02955	276	■ 1.000	$P:=(n\ z)(p)$	$P := (nz)(p)$	We consider two partitions of Y , namely $P := (nz)(p)$ and $Q := (np)(z)$. Technically, these are notation
fong-spivak-seven-sketches_276 F02956	276	■ 1.000	$Q:=(n\ p)(z)$	$Q := (np)(z)$	We consider two partitions of Y , namely $P := (nz)(p)$ and $Q := (np)(z)$. Technically, these are notation
fong-spivak-seven-sketches_276 F02957	276	■ 1.000	$\{\{n, z\}, \{p\}\}$	$\{\{n, z\}, \{p\}\}$	for $\{\{n, z\}, \{p\}\}$ and $\{\{n, p\}, \{z\}\}$ as sets of disjoint subsets whose union is Y . Their pulled back
fong-spivak-seven-sketches_276 F02958	276	■ 1.000	$\{\{n, p\}, \{z\}\}$	$\{\{n, p\}, \{z\}\}$	for $\{\{n, z\}, \{p\}\}$ and $\{\{n, p\}, \{z\}\}$ as sets of disjoint subsets whose union is Y . Their pulled back
fong-spivak-seven-sketches_276 F02959	276	■ 1.000	$f^*P=(\ldots,-2,-1,0)(1,2,\ldots)$	$f^*P = (\ldots, -2, -1, 0)(1, 2, \ldots)$	partitions are $f^*P = (\ldots, -2, -1, 0)(1, 2, \ldots)$ and $f^*Q = (0)(\ldots, -2, -1, 1, 2, \ldots)$, or technically
fong-spivak-seven-sketches_276 F02960	276	■ 1.000	$f^*Q=(0)(\ldots,-2,-1,1,2,\ldots)$	$f^*Q = (0)(\ldots, -2, -1, 1, 2, \ldots)$	partitions are $f^*P = (\ldots, -2, -1, 0)(1, 2, \ldots)$ and $f^*Q = (0)(\ldots, -2, -1, 1, 2, \ldots)$, or technically
fong-spivak-seven-sketches_277 F02961	277	■ 1.000	$(P, \leq_P), (Q, \leq_Q)$	$(P, \leq_P), (Q, \leq_Q)$	We have preorders $(P, \leq_P), (Q, \leq_Q)$, and $(R, \leq_R)$, and we have monotone maps $f: P \rightarrow Q$ and $g: Q \rightarrow R$.
fong-spivak-seven-sketches_277 F02962	277	■ 0.544	id_P	id_P	1. To see that id_P is monotone, we need to show that if $p_1 \leq_P p_2$ then $\mathrm{id}_P(p_1) \leq \mathrm{id}_P(p_2)$. But
fong-spivak-seven-sketches_277 F02963	277	■ 0.544	$\mathrm{id}_P(p_1) \leq \mathrm{id}_P(p_2)$	$\mathrm{id}_P(p_1) \leq \mathrm{id}_P(p_2)$	1. To see that id_P is monotone, we need to show that if $p_1 \leq_P p_2$ then $\mathrm{id}_P(p_1) \leq \mathrm{id}_P(p_2)$. But
fong-spivak-seven-sketches_277 F02964	277	■ 1.000	$\mathrm{id}_P(p)=p$	$\mathrm{id}_P(p) = p$	$\mathrm{id}_P(p) = p$ for all $p \in P$, so this is clear.

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fong-spivak-seven-sketches- F02965	277	■ 1.000	$q_{\{1\}} \leq_{\{Q\}} q_{\{2\}}$	$q_1 \leq_Q q_2$	2. We have that $p_1 \leq_P p_2$ implies $f(p_1) \leq_Q f(p_2)$ and that $q_1 \leq_Q q_2$ implies $g(q_1) \leq_R g(q_2)$. By
fong-spivak-seven-sketches- F02966	277	■ 1.000	$g(\text{left}(q_{\{1\}})\text{right}) \leq_{\{R\}} g(\text{left}(q_{\{2\}})\text{right})$	$g(q_1) \leq_R g(q_2)$	2. We have that $p_1 \leq_P p_2$ implies $f(p_1) \leq_Q f(p_2)$ and that $q_1 \leq_Q q_2$ implies $g(q_1) \leq_R g(q_2)$. By
fong-spivak-seven-sketches- F02967	277	■ 1.000	$g(\text{left}(f(\text{left}(p_{\{1\}})\text{right})\text{right}) \leq_{\{R\}} g(\text{left}(f(\text{left}(p_{\{2\}})\text{right})\text{right})$	$g(f(p_1)) \leq_R g(f(p_2))$	substitution, $p_1 \leq_P p_2$ implies $g(f(p_1)) \leq_R g(f(p_2))$ which is exactly what is required for $(f \circ g)$
fong-spivak-seven-sketches- F02968	277	■ 1.000	$(f ; g)$	$(f;g)$	substitution, $p_1 \leq_P p_2$ implies $g(f(p_1)) \leq_R g(f(p_2))$ which is exactly what is required for $(f \circ g)$
fong-spivak-seven-sketches- F02969	277	■ 1.000	$p_{\{2\}} \leq p_{\{1\}}$	$p_2 \leq p_1$	i.e. $p_1 \leq p_2$ and $p_2 \leq p_1$ implies $p_1 = p_2$. And suppose it is dagger, i.e. $p_1 \leq p_2$ implies $p_2 \leq p_1$. Well
fong-spivak-seven-sketches- F02970	277	■ 0.323	$\operatorname{map} \Phi: \operatorname{Prt}(\{\bullet, *, \circ\}) \rightarrow$	$\operatorname{map} \Phi: \operatorname{Prt}(\{\bullet, *, \circ\}) \rightarrow$	not $\bullet$ was in the same partition as $*$. We need to see that it's actually a monotone map $\Phi: \operatorname{Prt}(\{\bullet, *, \circ\}) \rightarrow$
fong-spivak-seven-sketches- F02971	277	■ 0.998	$\Phi(P)=$	$\Phi(P) =$	$\mathbb{B}$. So suppose P, Q are partitions with $P \leq Q$; we need to show that if $\Phi(P) = \text{true}$ then $\Phi(Q) = \text{true}$.
fong-spivak-seven-sketches- F02972	277	■ 0.998	$\Phi(Q)=$	$\Phi(Q) =$	$\mathbb{B}$. So suppose P, Q are partitions with $P \leq Q$; we need to show that if $\Phi(P) = \text{true}$ then $\Phi(Q) = \text{true}$.
fong-spivak-seven-sketches- F02973	277	■ 0.997	$x \sim_{\{P\}} y$	$x \sim_P y$	stuff together. Technically, $x \sim_P y$ implies $x \sim_Q y$ for all $x, y \in \{\bullet, *, \circ\}$. Applying this to $\bullet, *$ gives the
fong-spivak-seven-sketches- F02974	277	■ 0.997	$x \sim_{\{Q\}} y$	$x \sim_Q y$	stuff together. Technically, $x \sim_P y$ implies $x \sim_Q y$ for all $x, y \in \{\bullet, *, \circ\}$. Applying this to $\bullet, *$ gives the
fong-spivak-seven-sketches- F02975	277	■ 0.997	$x, y \in \{\bullet, *, \circ\}$	$x, y \in \{\bullet, *, \circ\}$	stuff together. Technically, $x \sim_P y$ implies $x \sim_Q y$ for all $x, y \in \{\bullet, *, \circ\}$. Applying this to $\bullet, *$ gives the
fong-spivak-seven-sketches- F02976	277	■ 0.997	$\bullet, *$	$\bullet, *$	stuff together. Technically, $x \sim_P y$ implies $x \sim_Q y$ for all $x, y \in \{\bullet, *, \circ\}$. Applying this to $\bullet, *$ gives the
fong-spivak-seven-sketches- F02977	277	■ 0.642	$U \mapsto f^{-1}(U)$	$U \mapsto f^{-1}(U)$	Given a function $f: P \rightarrow Q$, we have $f^*: \mathcal{U}(Q) \rightarrow \mathcal{U}(P)$ given by $U \mapsto f^{-1}(U)$. But upper sets in Q are
fong-spivak-seven-sketches- F02978	277	■ 1.000	$f^*(U)$	$f^*(U)$	classified by monotone maps $u: Q \rightarrow \mathbb{B}$, and similarly for P ; our job is to show that $f^*(U)$ is given by
fong-spivak-seven-sketches- F02979	277	■ 1.000	$q \mapsto$	$q \mapsto$	Given an upper set $U \subseteq Q$, let $u: Q \rightarrow \mathbb{B}$ be the corresponding monotone map, which sends $q \mapsto \text{true}$
fong-spivak-seven-sketches- F02980	277	■ 0.890	$p \mapsto$	$p \mapsto$	iff $q \in U$. Then $(f \circ u): P \rightarrow \mathbb{B}$ sends $p \mapsto \text{true}$ iff $f(p) \in U$; it corresponds to the upper set
fong-spivak-seven-sketches- F02981	277	■ 0.890	$f(p) \in U$	$f(p) \in U$	iff $q \in U$. Then $(f \circ u): P \rightarrow \mathbb{B}$ sends $p \mapsto \text{true}$ iff $f(p) \in U$; it corresponds to the upper set
fong-spivak-seven-sketches- F02982	277	■ 1.000	$\{p \in P \mid f(p) \in U\}$	$\{p \in P \mid f(p) \in U\}$	$\{p \in P \mid f(p) \in U\}$ which is exactly $f^{-1}(U)$.
fong-spivak-seven-sketches- F02983	277	■ 1.000	$S=\text{left}\{\text{left}.\frac{1}{n+1} \text{right}\text{rvert}, n \in \mathbb{N}\}$	$S = \left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\}$	1. 0 is a lower bound for $S = \{\frac{1}{n+1} \mid n \in \mathbb{N}\}$ because $0 \leq \frac{1}{n+1}$ for any $n \in \mathbb{N}$.
fong-spivak-seven-sketches- F02984	277	■ 1.000	$0 \leq \frac{1}{n+1}$	$0 \leq \frac{1}{n+1}$	1. 0 is a lower bound for $S = \{\frac{1}{n+1} \mid n \in \mathbb{N}\}$ because $0 \leq \frac{1}{n+1}$ for any $n \in \mathbb{N}$.
fong-spivak-seven-sketches- F02985	277	■ 1.000	$b \leq 0$	$b \leq 0$	2. Suppose that b is a lower bound for S ; we want to see that $b \leq 0$. If one believes to the contrary
fong-spivak-seven-sketches- F02986	277	■ 1.000	$0 < b$	$0 < b$	that $0 < b$, then consider $1/b$; it is a real number so we can find a natural number n that's bigger
fong-spivak-seven-sketches- F02987	277	■ 1.000	$1 / b$	$1/b$	that $0 < b$, then consider $1/b$; it is a real number so we can find a natural number n that's bigger
fong-spivak-seven-sketches- F02988	277	■ 1.000	$1 / b < n < n + 1$	$1/b < n < n + 1$	$1/b < n < n + 1$. This implies $1 < b(n + 1)$ and hence $\frac{1}{n+1} < b$, but that is a contradiction of b
fong-spivak-seven-sketches- F02989	277	■ 1.000	$1 < b(n + 1)$	$1 < b(n + 1)$	$1/b < n < n + 1$. This implies $1 < b(n + 1)$ and hence $\frac{1}{n+1} < b$, but that is a contradiction of b
fong-spivak-seven-sketches- F02990	277	■ 1.000	$\frac{1}{n+1} < b$	$\frac{1}{n+1} < b$	$1/b < n < n + 1$. This implies $1 < b(n + 1)$ and hence $\frac{1}{n+1} < b$, but that is a contradiction of b
fong-spivak-seven-sketches- F02991	277	■ 1.000	$a=p$	$a = p$	$q \leq p$. But the only $a \in A$ is $a = p$, so both are obvious.
fong-spivak-seven-sketches- F02992	277	■ 1.000	$4 \wedge 6=2$	$4 \wedge 6 = 2$	both are 1 and 2, and 2 is higher, so $4 \wedge 6 = 2$. Similar reasoning shows that $4 \vee 6 = 12$. The meet is the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F02993	277	■ 1.000	$4 \vee 6 = 12$	$4 \vee 6 = 12$	both are 1 and 2, and 2 is higher, so $4 \wedge 6 = 2$. Similar reasoning shows that $4 \vee 6 = 12$. The meet is the
fong-spivak-seven-sketches-F02994	278	■ 1.000	$a \leq a \vee b$	$a \leq a \vee b$	Since f is monotone, the facts that $a \leq a \vee b$ and $b \leq a \vee b$ imply that $f(a) \leq f(a \vee b)$ and $f(b) \leq f(a \vee b)$.
fong-spivak-seven-sketches-F02995	278	■ 1.000	$b \leq a \vee b$	$b \leq a \vee b$	Since f is monotone, the facts that $a \leq a \vee b$ and $b \leq a \vee b$ imply that $f(a) \leq f(a \vee b)$ and $f(b) \leq f(a \vee b)$.
fong-spivak-seven-sketches-F02996	278	■ 1.000	$f(a) \leq f(a \vee b)$	$f(a) \leq f(a \vee b)$	Since f is monotone, the facts that $a \leq a \vee b$ and $b \leq a \vee b$ imply that $f(a) \leq f(a \vee b)$ and $f(b) \leq f(a \vee b)$.
fong-spivak-seven-sketches-F02997	278	■ 1.000	$f(b) \leq f(a \vee b)$	$f(b) \leq f(a \vee b)$	Since f is monotone, the facts that $a \leq a \vee b$ and $b \leq a \vee b$ imply that $f(a) \leq f(a \vee b)$ and $f(b) \leq f(a \vee b)$.
fong-spivak-seven-sketches-F02998	278	■ 0.771	$\lfloor -/3 \rfloor$	$\lfloor -/3 \rfloor$	By analogy with Example 1.97, the right adjoint for $(3 \times -)$ should be $\lfloor -/3 \rfloor$. But to prove this is correct,
fong-spivak-seven-sketches-F02999	278	■ 1.000	$r \in \mathbb{R}$	$r \in \mathbb{R}$	we must show that for any any $r \in \mathbb{R}$ and $z \in \mathbb{Z}$ we have $z \leq \lfloor r/3 \rfloor$ iff $3 * z \leq r$.
fong-spivak-seven-sketches-F03000	278	■ 1.000	$z \leq \lfloor r / 3 \rfloor$	$z \leq \lfloor r/3 \rfloor$	we must show that for any any $r \in \mathbb{R}$ and $z \in \mathbb{Z}$ we have $z \leq \lfloor r/3 \rfloor$ iff $3 * z \leq r$.
fong-spivak-seven-sketches-F03001	278	■ 1.000	$3 * z \leq r$	$3 * z \leq r$	we must show that for any any $r \in \mathbb{R}$ and $z \in \mathbb{Z}$ we have $z \leq \lfloor r/3 \rfloor$ iff $3 * z \leq r$.
fong-spivak-seven-sketches-F03002	278	■ 1.000	$r / 3$	$r/3$	Suppose the largest integer below $r/3$ is $z' := \lfloor r/3 \rfloor$. Then $z \leq z'$ implies $3 * z \leq 3 * z' \leq 3 * r/3 = r$, giving
fong-spivak-seven-sketches-F03003	278	■ 1.000	$z^{\{\text{prime}\}} := \lfloor r / 3 \rfloor$	$z' := \lfloor r/3 \rfloor$	Suppose the largest integer below $r/3$ is $z' := \lfloor r/3 \rfloor$. Then $z \leq z'$ implies $3 * z \leq 3 * z' \leq 3 * r/3 = r$, giving
fong-spivak-seven-sketches-F03004	278	■ 1.000	$z \leq z^{\{\text{prime}\}}$	$z \leq z'$	Suppose the largest integer below $r/3$ is $z' := \lfloor r/3 \rfloor$. Then $z \leq z'$ implies $3 * z \leq 3 * z' \leq 3 * r/3 = r$, giving
fong-spivak-seven-sketches-F03005	278	■ 1.000	$3 * z \leq 3 * z^{\{\text{prime}\}} \leq 3 * r / 3 = r$	$3 * z \leq 3 * z' \leq 3 * r/3 = r$	Suppose the largest integer below $r/3$ is $z' := \lfloor r/3 \rfloor$. Then $z \leq z'$ implies $3 * z \leq 3 * z' \leq 3 * r/3 = r$, giving
fong-spivak-seven-sketches-F03006	278	■ 0.943	$z = 3 * z / 3 \leq r / 3$	$z = 3 * z/3 \leq r/3$	one direction. For the other, suppose $3 * z \leq r$. Then dividing both sides by 3, we have $z = 3 * z/3 \leq r/3$.
fong-spivak-seven-sketches-F03007	278	■ 1.000	$\lfloor r / 3 \rfloor$	$\lfloor r/3 \rfloor$	Since z is an integer below $r/3$ it is below $\lfloor r/3 \rfloor$ because $\lfloor r/3 \rfloor$ is the greatest integer below $r/3$, and we
fong-spivak-seven-sketches-F03008	278	■ 1.000	$\{(p, q) \mid 1 \leq p \leq 3$	$\{(p, q) \mid 1 \leq p \leq 3$	1. We need to check that for all nine pairs $\{(p, q) \mid 1 \leq p \leq 3$ and $1 \leq q \leq 3\}$ we have $f(p) \leq q$ iff
fong-spivak-seven-sketches-F03009	278	■ 1.000	$1 \leq q \leq 3\}$	$1 \leq q \leq 3\}$	1. We need to check that for all nine pairs $\{(p, q) \mid 1 \leq p \leq 3$ and $1 \leq q \leq 3\}$ we have $f(p) \leq q$ iff
fong-spivak-seven-sketches-F03010	278	■ 1.000	$p = q = 1$	$p = q = 1$	When $p = q = 1$ we have $f(p) = 1$ and $g(q) = 2$, so both $f(p) = 1 \leq 1 = q$ and $p = 1 \leq g(q)$;
fong-spivak-seven-sketches-F03011	278	■ 1.000	$f(p) = 1$	$f(p) = 1$	When $p = q = 1$ we have $f(p) = 1$ and $g(q) = 2$, so both $f(p) = 1 \leq 1 = q$ and $p = 1 \leq g(q)$;
fong-spivak-seven-sketches-F03012	278	■ 1.000	$g(q) = 2$	$g(q) = 2$	When $p = q = 1$ we have $f(p) = 1$ and $g(q) = 2$, so both $f(p) = 1 \leq 1 = q$ and $p = 1 \leq g(q)$;
fong-spivak-seven-sketches-F03013	278	■ 1.000	$f(p) = 1 \leq 1 = q$	$f(p) = 1 \leq 1 = q$	When $p = q = 1$ we have $f(p) = 1$ and $g(q) = 2$, so both $f(p) = 1 \leq 1 = q$ and $p = 1 \leq g(q)$;
fong-spivak-seven-sketches-F03014	278	■ 1.000	$p = 1 \leq g(q)$	$p = 1 \leq g(q)$	When $p = q = 1$ we have $f(p) = 1$ and $g(q) = 2$, so both $f(p) = 1 \leq 1 = q$ and $p = 1 \leq g(q)$;
fong-spivak-seven-sketches-F03015	278	■ 0.616	(p, q)	(p, q)	it works! Same sort of story happens when (p, q) is (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), and (3, 3). A
fong-spivak-seven-sketches-F03016	278	■ 0.616	$(1, 2), (1, 3), (2, 1), (2, 2), (2, 3)$	$(1, 2), (1, 3), (2, 1), (2, 2), (2, 3)$	it works! Same sort of story happens when (p, q) is $(1, 2), (1, 3), (2, 1), (2, 2), (2, 3)$, and (3, 3). A
fong-spivak-seven-sketches-F03017	278	■ 0.616	$(3, 3)$	$(3, 3)$	it works! Same sort of story happens when (p, q) is (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), and (3, 3). A
fong-spivak-seven-sketches-F03018	278	■ 1.000	$p = 3, q = 1$	$p = 3, q = 1$	different story happens for $p = 3, q = 1$ and $p = 3, q = 2$. In those cases $f(p) = 3$ and $g(q) = 2$,
fong-spivak-seven-sketches-F03019	278	■ 1.000	$p = 3, q = 2$	$p = 3, q = 2$	different story happens for $p = 3, q = 1$ and $p = 3, q = 2$. In those cases $f(p) = 3$ and $g(q) = 2$,
fong-spivak-seven-sketches-F03020	278	■ 1.000	$f(p) = 3$	$f(p) = 3$	different story happens for $p = 3, q = 1$ and $p = 3, q = 2$. In those cases $f(p) = 3$ and $g(q) = 2$,
fong-spivak-seven-sketches-F03021	278	■ 1.000	$f(p) \not\leq q$	$f(p) \not\leq q$	and neither inequality holds: $f(p) \not\leq q$ and $p \not\leq g(q)$. But that's fine, we still have $f(p) \leq q$ iff

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches- F03022	278	■ 1.000	$p \not\leq g(q)$	$p \not\leq g(q)$	and neither inequality holds: $f(p) \not\leq q$ and $p \not\leq g(q)$. But that's fine, we still have $f(p) \leq q$ iff
fong-spivak-seven-sketches- F03023	278	■ 1.000	$f(2) \not\leq 1$	$f(2) \not\leq 1$	Here f is <i>not</i> left adjoint to g because $f(2) \not\leq 1$ but $2 \leq g(1)$.
fong-spivak-seven-sketches- F03024	278	■ 1.000	$2 \leq g(1)$	$2 \leq g(1)$	Here f is <i>not</i> left adjoint to g because $f(2) \not\leq 1$ but $2 \leq g(1)$.
fong-spivak-seven-sketches- F03025	278	■ 1.000	$C(r) := \lceil r / 3 \rceil$	$C(r) := \lceil r/3 \rceil$	happens. Writing $C(r) := \lceil r/3 \rceil$, then for all $z \in \mathbb{Z}$ and $r \in \mathbb{R}$ we have $L(z) \leq r$ iff $z \leq C(r)$
fong-spivak-seven-sketches- F03026	278	■ 1.000	$L(z) \leq r$	$L(z) \leq r$	happens. Writing $C(r) := \lceil r/3 \rceil$, then for all $z \in \mathbb{Z}$ and $r \in \mathbb{R}$ we have $L(z) \leq r$ iff $z \leq C(r)$
fong-spivak-seven-sketches- F03027	278	■ 1.000	$z \leq C(r)$	$z \leq C(r)$	happens. Writing $C(r) := \lceil r/3 \rceil$, then for all $z \in \mathbb{Z}$ and $r \in \mathbb{R}$ we have $L(z) \leq r$ iff $z \leq C(r)$
fong-spivak-seven-sketches- F03028	278	■ 1.000	$z=1$	$z = 1$	by definition of adjunction. So take $z = 1$ and $r = .01$; then $\lceil r/3 \rceil = 1$ so $z \leq C(r)$, and hence
fong-spivak-seven-sketches- F03029	278	■ 1.000	$r=.01$	$r = .01$	by definition of adjunction. So take $z = 1$ and $r = .01$; then $\lceil r/3 \rceil = 1$ so $z \leq C(r)$, and hence
fong-spivak-seven-sketches- F03030	278	■ 1.000	$\lceil r / 3 \rceil = 1$	$\lceil r/3 \rceil = 1$	by definition of adjunction. So take $z = 1$ and $r = .01$; then $\lceil r/3 \rceil = 1$ so $z \leq C(r)$, and hence
fong-spivak-seven-sketches- F03031	278	■ 1.000	$L(1) \leq 0.01$	$L(1) \leq 0.01$	$L(z) \leq r$, i.e. $L(1) \leq 0.01$. In the same way $L(1) \leq r$ for all $r > 0$, so $L(1) \leq 0$. By definition of
fong-spivak-seven-sketches- F03032	278	■ 1.000	$L(1) \leq r$	$L(1) \leq r$	$L(z) \leq r$, i.e. $L(1) \leq 0.01$. In the same way $L(1) \leq r$ for all $r > 0$, so $L(1) \leq 0$. By definition of
fong-spivak-seven-sketches- F03033	278	■ 1.000	$r>0$	$r > 0$	$L(z) \leq r$, i.e. $L(1) \leq 0.01$. In the same way $L(1) \leq r$ for all $r > 0$, so $L(1) \leq 0$. By definition of
fong-spivak-seven-sketches- F03034	278	■ 1.000	$L(1) \leq 0$	$L(1) \leq 0$	$L(z) \leq r$, i.e. $L(1) \leq 0.01$. In the same way $L(1) \leq r$ for all $r > 0$, so $L(1) \leq 0$. By definition of
fong-spivak-seven-sketches- F03035	278	■ 1.000	$1 \leq C(0) = \lceil 0 / 3 \rceil = 0$	$1 \leq C(0) = \lceil 0/3 \rceil = 0$	adjunction $1 \leq C(0) = \lceil 0/3 \rceil = 0$, a contradiction.
fong-spivak-seven-sketches- F03036	278	■ 1.000	$c_{\{1\}}, c_{\{2\}}, c_{\{3\}}, c_{\{4\}}$	c_1, c_2, c_3, c_4	Example 1.102. Take c_1, c_2, c_3, c_4 to be the following partitions:
fong-spivak-seven-sketches- F03037	279	■ 0.995	$g_{\{!\}} \left(c_{\{1\}} \right), g_{\{!\}} \left(c_{\{2\}} \right), g_{\{!\}} \left(c_{\{3\}} \right)$	$g_!(c_1), g_!(c_2), g_!(c_3)$	Then the induced partitions $g_!(c_1), g_!(c_2), g_!(c_3)$, and $g_!(c_4)$ on T are:
fong-spivak-seven-sketches- F03038	279	■ 0.995	$g_{\{!\}} \left(c_{\{4\}} \right)$	$g_!(c_4)$	Then the induced partitions $g_!(c_1), g_!(c_2), g_!(c_3)$, and $g_!(c_4)$ on T are:
fong-spivak-seven-sketches- F03039	279	■ 1.000	$S = \{1, 2, 3, 4\}$	$S = \{1, 2, 3, 4\}$	Here are the partitions on $S = \{1, 2, 3, 4\}$ induced via g^* by the five partitions on $T = \{12, 3, 4\}$:
fong-spivak-seven-sketches- F03040	279	■ 1.000	$f: P \rightarrowtail Q: g$	$f: P \rightarrowtail Q: g$	Suppose P and Q are preorders, and that $f: P \rightarrowtail Q: g$ are monotone maps.
fong-spivak-seven-sketches- F03041	279	■ 1.000	$g(q) \leq g(q)$	$g(q) \leq g(q)$	$q \in Q$. Then starting with the reflexivity fact $g(q) \leq g(q)$, the definition with $p := g(q)$ gives
fong-spivak-seven-sketches- F03042	279	■ 1.000	$p:=g(q)$	$p := g(q)$	$q \in Q$. Then starting with the reflexivity fact $g(q) \leq g(q)$, the definition with $p := g(q)$ gives
fong-spivak-seven-sketches- F03043	279	■ 1.000	$f(p) \leq f(g(q))$	$f(p) \leq f(g(q))$	sides, we have $f(p) \leq f(g(q))$, and then by transitivity $f(g(q)) \leq q$ implies $f(p) \leq q$, as desired.
fong-spivak-seven-sketches- F03044	280	■ 1.000	$g, g^{\prime}: Q \rightarrowtail P$	$g, g': Q \rightarrowtail P$	1. Suppose that $f: P \rightarrowtail Q$ has two right adjoints, $g, g': Q \rightarrowtail P$. We want to show that $g(q) \cong g'(q)$
fong-spivak-seven-sketches- F03045	280	■ 1.000	$g(q) \leq g^{\prime}(q)$	$g(q) \leq g'(q)$	for all $q \in Q$. We will prove $g(q) \leq g'(q)$; the inequality $g'(q) \leq g(q)$ is similar. To do this, we use
fong-spivak-seven-sketches- F03046	280	■ 1.000	$g^{\prime}(q) \leq g(q)$	$g'(q) \leq g(q)$	for all $q \in Q$. We will prove $g(q) \leq g'(q)$; the inequality $g'(q) \leq g(q)$ is similar. To do this, we use
fong-spivak-seven-sketches- F03047	280	■ 1.000	$p \leq g^{\prime}(f(p))$	$p \leq g'(f(p))$	the fact that $p \leq g'(f(p))$ and $f(g(q)) \leq q$ for all p, q by Eq. (1.108). Then the trick is to reason as
fong-spivak-seven-sketches- F03048	280	■ 0.999	$j:=\vee A$	$j := \vee A$	Suppose $f: P \rightarrowtail Q$ is left adjoint to $g: Q \rightarrowtail P$. Let $A \subseteq P$ be any subset and let $j := \vee A$ be its join.
fong-spivak-seven-sketches- F03049	280	■ 1.000	$f(a) \leq f(j)$	$f(a) \leq f(j)$	Then since f is monotone $f(a) \leq f(j)$ for all $a \in A$, so $f(j)$ is an upper bound for the set $f(A)$. We
fong-spivak-seven-sketches- F03050	280	■ 1.000	$f(j)$	$f(j)$	Then since f is monotone $f(a) \leq f(j)$ for all $a \in A$, so $f(j)$ is an upper bound for the set $f(A)$. We

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches- F03051	280	■ 1.000	$f(a) \leq b$	$f(a) \leq b$	$f(a) \leq b$ for all $a \in A$. Then by definition of adjunction, we also have $a \leq g(b)$ for all $a \in A$. By
fong-spivak-seven-sketches- F03052	280	■ 1.000	$a \leq g(b)$	$a \leq g(b)$	$f(a) \leq b$ for all $a \in A$. Then by definition of adjunction, we also have $a \leq g(b)$ for all $a \in A$. By
fong-spivak-seven-sketches- F03053	280	■ 1.000	$j \leq g(b)$	$j \leq g(b)$	definition of join, we have $j \leq g(b)$. Again by definition of adjunction $f(j) \leq b$, as desired.
fong-spivak-seven-sketches- F03054	280	■ 1.000	$f(j) \leq b$	$f(j) \leq b$	definition of join, we have $j \leq g(b)$. Again by definition of adjunction $f(j) \leq b$, as desired.
fong-spivak-seven-sketches- F03055	281	■ 1.000	$f(p)$	$f(p)$	that $f(g(q)) \leq q$, with $f(p)$ substituted for q : this gives $f(g(f(p))) \leq f(p)$, and then we apply g
fong-spivak-seven-sketches- F03058	281	■ 0.999	$B_{\{1\}} := \{a, b\}$	$B_1 := \{a, b\}$	1. Let $B_1 := \{a, b\}$ and $B_2 := \{c\}$. Then $f^*(B_1) = \{a_1\}$ and $f^*(B_2) = \{c_1, c_2\}$.
fong-spivak-seven-sketches- F03059	281	■ 0.999	$B_{\{2\}} := \{c\}$	$B_2 := \{c\}$	1. Let $B_1 := \{a, b\}$ and $B_2 := \{c\}$. Then $f^*(B_1) = \{a_1\}$ and $f^*(B_2) = \{c_1, c_2\}$.
fong-spivak-seven-sketches- F03060	281	■ 0.999	$f^*\left(B_{\{1\}}\right) = \{a_1\}$	$f^*(B_1) = \{a_1\}$	1. Let $B_1 := \{a, b\}$ and $B_2 := \{c\}$. Then $f^*(B_1) = \{a_1\}$ and $f^*(B_2) = \{c_1, c_2\}$.
fong-spivak-seven-sketches- F03061	281	■ 0.999	$f^*\left(B_{\{2\}}\right) = \{c_1, c_2\}$	$f^*(B_2) = \{c_1, c_2\}$	1. Let $B_1 := \{a, b\}$ and $B_2 := \{c\}$. Then $f^*(B_1) = \{a_1\}$ and $f^*(B_2) = \{c_1, c_2\}$.
fong-spivak-seven-sketches- F03062	281	■ 1.000	$A_{\{1\}} := \varnothing$	$A_1 := \emptyset$	2. Let $A_1 := \emptyset$ and $A_2 := \{a_1, c_1\}$. Then $f_!(A_1) = \emptyset$ and $f_!(A_2) = \{a, c\}$.
fong-spivak-seven-sketches- F03063	281	■ 1.000	$A_{\{2\}} := \left\{a_{\{1\}}, c_{\{1\}}\right\}$	$A_2 := \{a_1, c_1\}$	2. Let $A_1 := \emptyset$ and $A_2 := \{a_1, c_1\}$. Then $f_!(A_1) = \emptyset$ and $f_!(A_2) = \{a, c\}$.
fong-spivak-seven-sketches- F03064	281	■ 1.000	$f_{\{!\}}\left(A_{\{1\}}\right) = \varnothing$	$f_!(A_1) = \emptyset$	2. Let $A_1 := \emptyset$ and $A_2 := \{a_1, c_1\}$. Then $f_!(A_1) = \emptyset$ and $f_!(A_2) = \{a, c\}$.
fong-spivak-seven-sketches- F03065	281	■ 1.000	$f_{\{!\}}\left(A_{\{2\}}\right) = \{a, c\}$	$f_!(A_2) = \{a, c\}$	2. Let $A_1 := \emptyset$ and $A_2 := \{a_1, c_1\}$. Then $f_!(A_1) = \emptyset$ and $f_!(A_2) = \{a, c\}$.
fong-spivak-seven-sketches- F03066	281	■ 1.000	$A_{\{1\}}$	A_1	3. With the same A_1 and A_2 , we compute $f_*(A_1) = \{b\}$ and $f_*(A_2) = \{a, b\}$.
fong-spivak-seven-sketches- F03067	281	■ 1.000	$A_{\{2\}}$	A_2	3. With the same A_1 and A_2 , we compute $f_*(A_1) = \{b\}$ and $f_*(A_2) = \{a, b\}$.
fong-spivak-seven-sketches- F03068	281	■ 1.000	$f_{\{*\}}\left(A_{\{1\}}\right) = \{b\}$	$f_*(A_1) = \{b\}$	3. With the same A_1 and A_2 , we compute $f_*(A_1) = \{b\}$ and $f_*(A_2) = \{a, b\}$.
fong-spivak-seven-sketches- F03069	281	■ 1.000	$f_{\{*\}}\left(A_{\{2\}}\right) = \{a, b\}$	$f_*(A_2) = \{a, b\}$	3. With the same A_1 and A_2 , we compute $f_*(A_1) = \{b\}$ and $f_*(A_2) = \{a, b\}$.
fong-spivak-seven-sketches- F03070	281	■ 1.000	$g(f(p))$	$g(f(p))$	1. It is part of the definition of adjunction (Proposition 1.107) that $p \leq g(f(p))$, and of course $g(f(p))$
fong-spivak-seven-sketches- F03071	281	■ 1.000	$(f ; g)(p)$	$(f;g)(p)$	and $(f \circ g)(p)$ mean the same thing.
fong-spivak-seven-sketches- F03072	281	■ 1.000	$g(f(g(f(p)))) \leq g(f(p))$	$g(f(g(f(p)))) \leq g(f(p))$	2. We want to show that $g(f(g(f(p)))) \leq g(f(p))$ and $g(f(p)) \leq g(f(g(f(p))))$ for all p . The latter
fong-spivak-seven-sketches- F03073	281	■ 1.000	$g(f(p)) \leq g(f(g(f(p))))$	$g(f(p)) \leq g(f(g(f(p))))$	2. We want to show that $g(f(g(f(p)))) \leq g(f(p))$ and $g(f(p)) \leq g(f(g(f(p))))$ for all p . The latter
fong-spivak-seven-sketches- F03074	281	■ 1.000	$p^{\prime} \leq g\left(f\left(p^{\prime}\right)\right)$	$p' \leq g(f(p'))$	is just the fact that $p' \leq g(f(p'))$ for any p' , applied with $g(f(p))$ in place of p' . The former uses
fong-spivak-seven-sketches- F03075	281	■ 1.000	$f(g(f(p))) \leq f(p)$	$f(g(f(p))) \leq f(p)$	that $f(g(q)) \leq q$, with $f(p)$ substituted for q : this gives $f(g(f(p))) \leq f(p)$, and then we apply g
fong-spivak-seven-sketches- F03076	281	■ 0.911	$a \ b$	ab	We denote tuples (a, b) by ab for space reasons. So the relation $\{(1, 1), (1, 2), (2, 1)\}$ will be denoted
fong-spivak-seven-sketches- F03077	281	■ 0.911	$\{(1, 1), (1, 2), (2, 1)\}$	$\{(1, 1), (1, 2), (2, 1)\}$	We denote tuples (a, b) by ab for space reasons. So the relation $\{(1, 1), (1, 2), (2, 1)\}$ will be denoted
fong-spivak-seven-sketches- F03078	281	■ 1.000	$S := \{1, 2, 3\}$	$S := \{1, 2, 3\}$	Let $S := \{1, 2, 3\}$.
fong-spivak-seven-sketches- F03079	281	■ 1.000	$1 \leq 1, 2 \leq 2$	$1 \leq 1, 2 \leq 2$	1. Let $\leq$ be the preorder with $1 \leq 2$, and of course $1 \leq 1, 2 \leq 2$, and $3 \leq 3$. Then $U(\leq) =$
fong-spivak-seven-sketches- F03080	281	■ 1.000	$3 \leq 3$	$3 \leq 3$	1. Let $\leq$ be the preorder with $1 \leq 2$, and of course $1 \leq 1, 2 \leq 2$, and $3 \leq 3$. Then $U(\leq) =$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F03081	281	1.000	$U(\leq) =$	$U(\leq) =$	1. Let $\leq$ be the preorder with $1 \leq 2$, and of course $1 \leq 1$, $2 \leq 2$, and $3 \leq 3$. Then $U(\leq) =$
fong-spivak-seven-sketches-F03082	281	0.999	$\{(1,1), (1,2), (2,2), (3,3)\}$	$\{(1,1), (1,2), (2,2), (3,3)\}$	$\{(1,1), (1,2), (2,2), (3,3)\}$.
fong-spivak-seven-sketches-F03083	281	1.000	$Q := \{(1,1)\}$	$Q := \{(1,1)\}$	2. Let $Q := \{(1,1)\}$ and $Q' := \{(2,1)\}$.
fong-spivak-seven-sketches-F03084	281	1.000	$Q^{\{\prime\}} := \{(2,1)\}$	$Q' := \{(2,1)\}$	2. Let $Q := \{(1,1)\}$ and $Q' := \{(2,1)\}$.
fong-spivak-seven-sketches-F03085	281	0.825	$\mathrm{Cl}(Q)$	$\mathrm{Cl}(Q)$	3. The closure $\mathrm{Cl}(Q)$ of Q is the smallest preorder containing $(1,1)$, which is $\mathrm{Cl}(Q) = \{(1,1), (2,2), (3,3)\}$.
fong-spivak-seven-sketches-F03086	281	0.825	$(1,1)$	$(1,1)$	3. The closure $\mathrm{Cl}(Q)$ of Q is the smallest preorder containing $(1,1)$, which is $\mathrm{Cl}(Q) = \{(1,1), (2,2), (3,3)\}$.
fong-spivak-seven-sketches-F03087	281	0.825	$\mathrm{Cl}(Q) = \{(1,1), (2,2), (3,3)\}$	$\mathrm{Cl}(Q) = \{(1,1), (2,2), (3,3)\}$	3. The closure $\mathrm{Cl}(Q)$ of Q is the smallest preorder containing $(1,1)$, which is $\mathrm{Cl}(Q) = \{(1,1), (2,2), (3,3)\}$.
fong-spivak-seven-sketches-F03088	281	0.793	$\mathrm{Cl}(Q^{\{\prime\}}) = \{(1,1), (2,1), (2,2), (3,3)\}$	$\mathrm{Cl}(Q') = \{(1,1), (2,1), (2,2), (3,3)\}$	Similarly, $\mathrm{Cl}(Q') = \{(1,1), (2,1), (2,2), (3,3)\}$. It is easy to see that $\mathrm{Cl}(Q) \sqsubseteq \leq$ because every or-
fong-spivak-seven-sketches-F03089	281	0.999	$(2,1)$	$(2,1)$	4. It is easy to see that $\mathrm{Cl}(Q') \not\sqsubseteq \leq$ because the ordered pair $(2,1)$ is in $\mathrm{Cl}(Q')$ but is not in $\leq$.
fong-spivak-seven-sketches-F03090	281	0.999	$\mathrm{Cl}(Q^{\{\prime\}})$	$\mathrm{Cl}(Q')$	4. It is easy to see that $\mathrm{Cl}(Q') \not\sqsubseteq \leq$ because the ordered pair $(2,1)$ is in $\mathrm{Cl}(Q')$ but is not in $\leq$.
fong-spivak-seven-sketches-F03091	282	1.000	$x_{\{1\}} = -1, x_{\{2\}} = 0, y_{\{1\}} = -1$	$x_1 = -1, x_2 = 0, y_1 = -1$	The expert is right! The proposal violates property (a) when $x_1 = -1$, $x_2 = 0$, $y_1 = -1$, and $y_2 = 1$.
fong-spivak-seven-sketches-F03092	282	1.000	$y_{\{2\}} = 1$	$y_2 = 1$	The expert is right! The proposal violates property (a) when $x_1 = -1$, $x_2 = 0$, $y_1 = -1$, and $y_2 = 1$.
fong-spivak-seven-sketches-F03093	282	0.901	$-1 \leq -1$	$-1 \leq -1$	Indeed $-1 \leq -1$ and $0 \leq 1$, but $-1 * 0 = 0 \not\leq -1 = -1 * 1$.
fong-spivak-seven-sketches-F03094	282	0.901	$-1 * 0 = 0 \not\leq -1 = -1 * 1$	$-1 * 0 = 0 \not\leq -1 = -1 * 1$	Indeed $-1 \not\leq -1$ and $0 \leq 1$, but $-1 * 0 = 0 \not\leq -1 = -1 * 1$.
fong-spivak-seven-sketches-F03095	282	0.940	$(\mathrm{Disc}(M), =, *, e)$	$(\mathrm{Disc}(M), =, *, e)$	To check that $(\mathrm{Disc}(M), =, *, e)$ is a symmetric monoidal preorder, we need to check our proposed data
fong-spivak-seven-sketches-F03096	282	1.000	$x_{\{1\}} \otimes x_{\{2\}} = x_{\{1\}} \otimes x_{\{2\}}$	$x_1 \otimes x_2 = x_1 \otimes x_2$	obeys conditions (a)-(d) of Definition 2.2. Condition (a) just states the tautology that $x_1 \otimes x_2 = x_1 \otimes x_2$,
fong-spivak-seven-sketches-F03097	282	1.000	$u \leq u, v \leq v$	$u \leq u, v \leq v$	2. We use reflexivity when we assert that $u \leq u$, $v \leq v$ and $z \leq z$, and use transitivity to assert that
fong-spivak-seven-sketches-F03098	282	1.000	$z \leq z$	$z \leq z$	2. We use reflexivity when we assert that $u \leq u$, $v \leq v$ and $z \leq z$, and use transitivity to assert that
fong-spivak-seven-sketches-F03099	282	1.000	$t + u \leq y + z$	$t + u \leq y + z$	the above sequence of inequalities implies the single inequality $t + u \leq y + z$.
fong-spivak-seven-sketches-F03100	282	1.000	$z \rightarrow w$	$z \rightarrow w$	Condition (a), monotonicity, says that if $x \rightarrow y$ and $z \rightarrow w$ are reactions, then $x + z \rightarrow y + w$ is a
fong-spivak-seven-sketches-F03101	282	1.000	$x + z \rightarrow y + w$	$x + z \rightarrow y + w$	Condition (a), monotonicity, says that if $x \rightarrow y$ and $z \rightarrow w$ are reactions, then $x + z \rightarrow y + w$ is a
fong-spivak-seven-sketches-F03102	282	0.991	$1 \mid 1$	$1 \mid 1$	This proposal is not monotonic: we have $1 \mid 1$ and $1 \mid 2$, but $(1 + 1) \nmid (1 + 2)$.
fong-spivak-seven-sketches-F03103	282	0.991	$1 \mid 2$	$1 \mid 2$	This proposal is not monotonic: we have $1 \mid 1$ and $1 \mid 2$, but $(1 + 1) \nmid (1 + 2)$.
fong-spivak-seven-sketches-F03104	282	0.991	$(1+1) \nmid (1+2)$	$(1 + 1) \nmid (1 + 2)$	This proposal is not monotonic: we have $1 \mid 1$ and $1 \mid 2$, but $(1 + 1) \nmid (1 + 2)$.
fong-spivak-seven-sketches-F03105	283	0.936	$z \leq w$	$z \leq w$	2. We need to show that (a) if $x \leq y$ and $z \leq w$, then $\min(x, z) \leq \min(y, w)$, (b) $\min(x, \mathbf{yes}) = x =$
fong-spivak-seven-sketches-F03106	283	0.936	$\min(x, z) \leq \min(y, w)$	$\min(x, z) \leq \min(y, w)$	2. We need to show that (a) if $x \leq y$ and $z \leq w$, then $\min(x, z) \leq \min(y, w)$, (b) $\min(x, \mathbf{yes}) = x =$
fong-spivak-seven-sketches-F03107	283	0.936	$\min(x$	$\min(x$	2. We need to show that (a) if $x \leq y$ and $z \leq w$, then $\min(x, z) \leq \min(y, w)$, (b) $\min(x, \mathbf{yes}) = x =$
fong-spivak-seven-sketches-F03108	283	0.936	$= x =$	$= x =$	2. We need to show that (a) if $x \leq y$ and $z \leq w$, then $\min(x, z) \leq \min(y, w)$, (b) $\min(x, \mathbf{yes}) = x =$
fong-spivak-seven-sketches-F03109	283	0.948	$\min($	$\min($	$\min(\mathbf{yes}, x)$, (c) $\min(\min(x, y), z) = \min(x, \min(y, z))$, and (d) $\min(x, y) = \min(y, z)$. The most

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F03110	283	■ 0.948	$x)$	$x)$	min(yes, x), (c) $\min(\min(x, y), z) = \min(x, \min(y, z))$, and (d) $\min(x, y) = \min(y, z)$. The most
fong-spivak-seven-sketches-F03111	283	■ 0.948	$\min(\min(x, y), z) = \min(x, \min(y, z))$	$\min(\min(x, y), z) = \min(x, \min(y, z))$	min(yes, x), (c) $\min(\min(x, y), z) = \min(x, \min(y, z))$, and (d) $\min(x, y) = \min(y, z)$. The most
fong-spivak-seven-sketches-F03112	283	■ 0.948	$\min(x, y) = \min(y, z)$	$\min(x, y) = \min(y, z)$	min(yes, x), (c) $\min(\min(x, y), z) = \min(x, \min(y, z))$, and (d) $\min(x, y) = \min(y, z)$. The most
fong-spivak-seven-sketches-F03113	283	■ 0.991	$(\mathrm{P}(S), \leq, S, \cap)$	$(P(S), \leq, S, \cap)$	Yes, $(P(S), \leq, S, \cap)$ is a symmetric monoidal preorder.
fong-spivak-seven-sketches-F03114	283	■ 1.000	$(P \wedge Q)(n)$	$(P \wedge Q)(n)$	makes the statement $P \wedge Q$, where $(P \wedge Q)(n)$ is true if $P(n)$ and $Q(n)$ are true, and false otherwise.
fong-spivak-seven-sketches-F03115	283	■ 0.833	$\{ \}^{\mathbb{N}}, \leq$	$\mathbb{N}, \leq$	Then $(\mathbf{Prop}^{\mathbb{N}}, \leq, \mathbf{true}, \wedge)$ forms a symmetric monoidal preorder.
fong-spivak-seven-sketches-F03116	283	■ 0.957	$n + 10 \leq 1$	$n + 10 \leq 1$	as “ $n + 10 \leq 1$ ” or “ n is made of cheese.” We can also define $\vee$ such that $(P \vee Q)(n)$ is true if and only if
fong-spivak-seven-sketches-F03117	283	■ 0.957	$(P \vee Q)(n)$	$(P \vee Q)(n)$	as “ $n + 10 \leq 1$ ” or “ n is made of cheese.” We can also define $\vee$ such that $(P \vee Q)(n)$ is true if and only if
fong-spivak-seven-sketches-F03118	283	■ 0.561	$\left(\mathbf{Prop}^{\mathbb{N}}, \leq \right)$	$\left(\mathbf{Prop}^{\mathbb{N}}, \leq \right)$	at least one of $P(n)$ and $Q(n)$ is true. Then $(\mathbf{Prop}^{\mathbb{N}}, \leq, \mathbf{false}, \vee)$ forms a symmetric monoidal preorder.
fong-spivak-seven-sketches-F03119	283	■ 0.561	$\left(\vee \right)$	$\vee$	at least one of $P(n)$ and $Q(n)$ is true. Then $(\mathbf{Prop}^{\mathbb{N}}, \leq, \mathbf{false}, \vee)$ forms a symmetric monoidal preorder.
fong-spivak-seven-sketches-F03120	283	■ 0.676	X^{op}	X^{op}	Unitality and associativity have nothing to do with the order in $\mathcal{X}^{\mathrm{op}}$: they simply state that $I \otimes x =$
fong-spivak-seven-sketches-F03121	283	■ 0.676	$I \otimes x =$	$I \otimes x =$	Unitality and associativity have nothing to do with the order in $\mathcal{X}^{\mathrm{op}}$: they simply state that $I \otimes x =$
fong-spivak-seven-sketches-F03122	283	■ 0.964	$x = x \otimes I$	$x = x \otimes I$	$x = x \otimes I$, and $(x \otimes y) \otimes z = x \otimes (y \otimes z)$. Since these are true in $\mathcal{X}$, they are true in $\mathcal{X}^{\mathrm{op}}$. Symmetry is
fong-spivak-seven-sketches-F03123	283	■ 0.506	$y \otimes x$	$y \otimes x$	slightly trickier, since in only asks that $x \otimes y$ is equivalent to $y \otimes x$. Nonetheless, this is still true in $\mathcal{X}^{\mathrm{op}}$
fong-spivak-seven-sketches-F03124	283	■ 0.506	x^{op}	x^{op}	slightly trickier, since in only asks that $x \otimes y$ is equivalent to $y \otimes x$. Nonetheless, this is still true in $\mathcal{X}^{\mathrm{op}}$
fong-spivak-seven-sketches-F03125	283	■ 0.978	$(x \otimes y) \cong (y \otimes x)$	$(x \otimes y) \cong (y \otimes x)$	because it is true in $\mathcal{X}$. Indeed the fact that $(x \otimes y) \cong (y \otimes x)$ in $\mathcal{X}$ means that $(x \otimes y) \leq (y \otimes x)$ and
fong-spivak-seven-sketches-F03126	283	■ 0.978	$(x \otimes y) \leq (y \otimes x)$	$(x \otimes y) \leq (y \otimes x)$	because it is true in $\mathcal{X}$. Indeed the fact that $(x \otimes y) \cong (y \otimes x)$ in $\mathcal{X}$ means that $(x \otimes y) \leq (y \otimes x)$ and
fong-spivak-seven-sketches-F03127	283	■ 0.927	$(y \otimes x) \leq (x \otimes y)$	$(y \otimes x) \leq (x \otimes y)$	$(y \otimes x) \leq (x \otimes y)$ in $\mathcal{X}$, which respectively imply that $(y \otimes x) \leq (x \otimes y)$ and $(x \otimes y) \leq (y \otimes x)$ in $\mathcal{X}^{\mathrm{op}}$, and
fong-spivak-seven-sketches-F03128	283	■ 0.768	$\mathbf{Cost}^{\mathrm{op}}$	$\mathbf{Cost}^{\mathrm{op}}$	1. The preorder $\mathbf{Cost}^{\mathrm{op}}$ has underlying set $[0, \infty]$, and the usual increasing order on real numbers
fong-spivak-seven-sketches-F03129	283	■ 0.999	$= \infty \geq 0 = g$	$= \infty \geq 0 = g$	1. The map g is monotonic as $g(\mathbf{false}) = \infty \geq 0 = g(\mathbf{true})$, satisfies condition (a) since $0 \geq 0 =$
fong-spivak-seven-sketches-F03130	283	■ 0.999	$0 \geq 0 =$	$0 \geq 0 =$	1. The map g is monotonic as $g(\mathbf{false}) = \infty \geq 0 = g(\mathbf{true})$, satisfies condition (a) since $0 \geq 0 =$
fong-spivak-seven-sketches-F03131	284	■ 0.966	$f(n) = 1$	$f(n) = 1$	2. The map $f(n) = 1$ for all $n \in \mathbb{N}$ defines a monoidal monotone $f: (\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq, 1, *)$. (In
fong-spivak-seven-sketches-F03132	284	■ 0.966	$f: (\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq, 1, *)$	$f: (\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq, 1, *)$	2. The map $f(n) = 1$ for all $n \in \mathbb{N}$ defines a monoidal monotone $f: (\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq, 1, *)$. (In
fong-spivak-seven-sketches-F03133	284	■ 0.523	$\mathbb{Z}, \leq, *, 1$	$\mathbb{Z}, \leq, *, 1$	3. $(\mathbb{Z}, \leq, *, 1)$ is not a monoidal preorder because $*$ is not monotone. Indeed $-1 \leq 0$ but $(-1 * -1) \not\leq$
fong-spivak-seven-sketches-F03134	284	■ 0.523	$-1 \leq 0$	$-1 \leq 0$	3. $(\mathbb{Z}, \leq, *, 1)$ is not a monoidal preorder because $*$ is not monotone. Indeed $-1 \leq 0$ but $(-1 * -1) \not\leq$
fong-spivak-seven-sketches-F03135	284	■ 0.523	$-1 * -1$	$-1 * -1$	3. $(\mathbb{Z}, \leq, *, 1)$ is not a monoidal preorder because $*$ is not monotone. Indeed $-1 \leq 0$ but $(-1 * -1) \not\leq$
fong-spivak-seven-sketches-F03136	284	■ 0.523	$\not\subseteq$	$\not\subseteq$	3. $(\mathbb{Z}, \leq, *, 1)$ is not a monoidal preorder because $*$ is not monotone. Indeed $-1 \leq 0$ but $(-1 * -1) \not\leq$
fong-spivak-seven-sketches-F03137	284	■ 0.998	$(0 * 0)$	$(0 * 0)$	$(0 * 0)$.
fong-spivak-seven-sketches-F03138	284	■ 0.654	X_P	X_P	a Bool -category $\mathcal{X}_P$ with P as its set of objects, and with $\mathcal{X}_P(p, q) := \mathbf{true}$ if $p \leq q$, and $\mathcal{X}_P(p, q) :=$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03139	284	0.654	$X_{\{P\}}(p, q) :=$	$X_P(p, q) :=$	a Bool -category $\mathcal{X}_P$ with P as its set of objects, and with $\mathcal{X}_P(p, q) := \text{true}$ if $p \leq q$, and $\mathcal{X}_P(p, q) :=$
fong-spivak-seven-sketches F03140	284	0.880	$\operatorname{Ob}\{0b\}\left(X_{\{P\}}\right)=P$	$\operatorname{Ob}\left(X_P\right)=P$	we construct a preorder with underlying set $\operatorname{Ob}\left(\mathcal{X}_P\right)=P$, and with $p \leq q$ if and only if $\mathcal{X}_P(p, q) =$
fong-spivak-seven-sketches F03141	284	0.880	$X_{\{P\}}(p, q)=$	$X_P(p, q) =$	we construct a preorder with underlying set $\operatorname{Ob}\left(\mathcal{X}_P\right)=P$, and with $p \leq q$ if and only if $\mathcal{X}_P(p, q) =$
fong-spivak-seven-sketches F03142	284	0.335	$\left(\operatorname{Ob}\{0b\}\left(\operatorname{mathscr{X}}\right), \leq\right)$	$(\operatorname{Ob}(\mathcal{X}), \leq)$	2. Let $\mathcal{X}$ be a Bool -category. By the proof of Theorem 2.49, we construct a preorder $(\operatorname{Ob}(\mathcal{X}), \leq)$,
fong-spivak-seven-sketches F03143	284	0.948	$\operatorname{Ob}\{0b\}(X)$	$\operatorname{Ob}(X)$	1., we construct a Bool -category $\mathcal{X}'$ whose set of objects is $\operatorname{Ob}(\mathcal{X})$, and such that $\mathcal{X}'(x, y) = \text{true}$
fong-spivak-seven-sketches F03144	284	0.948	$X^{\{\text{prime}\}}(x, y)=$	$X'(x, y) =$	1., we construct a Bool -category $\mathcal{X}'$ whose set of objects is $\operatorname{Ob}(\mathcal{X})$, and such that $\mathcal{X}'(x, y) = \text{true}$
fong-spivak-seven-sketches F03145	284	0.600	$\left(\operatorname{mathrm{Ob}}(X), \leq\right)$	$(\operatorname{Ob}(X), \leq)$	if and only if $x \leq y$ in $(\operatorname{Ob}(\mathcal{X}), \leq)$. But by construction, this means $\mathcal{X}'(x, y) = \mathcal{X}(x, y)$. So we get
fong-spivak-seven-sketches F03146	284	0.600	$X^{\{\text{prime}\}}(x, y)=\operatorname{mathcal{X}}(x, y)$	$X'(x, y) = \mathcal{X}(x, y)$	if and only if $x \leq y$ in $(\operatorname{Ob}(\mathcal{X}), \leq)$. But by construction, this means $\mathcal{X}'(x, y) = \mathcal{X}(x, y)$. So we get
fong-spivak-seven-sketches F03147	284	0.563	$\left(\operatorname{mathbb{R}}_{\geq 0}, \geq, 0,+\right)$	$(\mathbb{R}_{\geq 0}, \geq, 0, +)$	a category enriched over $(\mathbb{R}_{\geq 0}, \geq, 0, +)$ is that in the latter, infinite distances are not allowed between
fong-spivak-seven-sketches F03149	284	0.890	$\min \left(\operatorname{mathcal{X}}(x, y), \operatorname{mathcal{X}}(y, z)\right) \leq \operatorname{mathcal{X}}(x, z)$	$\min(\mathcal{X}(x, y), \mathcal{X}(y, z)) \leq \mathcal{X}(x, z)$	to no, maybe, or yes. Moreover, we must have $\mathcal{X}(x, x) = \text{yes}$ and $\min(\mathcal{X}(x, y), \mathcal{X}(y, z)) \leq \mathcal{X}(x, z)$ for all
fong-spivak-seven-sketches F03150	284	1.000	x, y, z	x, y, z	x, y, z . So a NMY -category can be thought of as set of points together with an statement—no, maybe,
fong-spivak-seven-sketches F03151	285	0.978	$X(C, D)$	$X(C, D)$	For example, to compute the hom-object $\mathcal{X}(C, D)$, we notice that there are two paths: $C \rightarrow A \rightarrow$
fong-spivak-seven-sketches F03152	285	0.978	$C \rightarrowtail A \rightarrowtail$	$C \rightarrow A \rightarrow$	For example, to compute the hom-object $\mathcal{X}(C, D)$, we notice that there are two paths: $C \rightarrow A \rightarrow$
fong-spivak-seven-sketches F03153	285	1.000	$B \rightarrowtail D$	$B \rightarrow D$	$B \rightarrow D$ and $C \rightarrow D$. For the first path, the intersection is the set $\{\text{boat}\}$. For the second path, the
fong-spivak-seven-sketches F03154	285	1.000	$C \rightarrowtail D$	$C \rightarrow D$	$B \rightarrow D$ and $C \rightarrow D$. For the first path, the intersection is the set $\{\text{boat}\}$. For the second path, the
fong-spivak-seven-sketches F03155	285	0.946	$\operatorname{mathcal{X}}(x, y) \cap \operatorname{mathcal{X}}(y, z) \subseteq_{\text{seteq}} \operatorname{mathcal{X}}(x, z)$	$\mathcal{X}(x, y) \cap \mathcal{X}(y, z) \subseteq \mathcal{X}(x, z)$	we ensure that $\mathcal{X}(x, y) \cap \mathcal{X}(y, z) \subseteq \mathcal{X}(x, z)$ for all x, y, z .
fong-spivak-seven-sketches F03156	285	1.000	$(x, y)^{\{\text{th}\}}$	$(x, y)^{\text{th}}$	2. The matrix M with $(x, y)^{\text{th}}$ entry equal to the maximum, taken over all paths p from x to y , of
fong-spivak-seven-sketches F03158	285	1.000	$\min (M(x, y), M(y, z)) \leq M(x, y)$	$\min(M(x, y), M(y, z)) \leq M(x, y)$	monoidal unit ∞ , and because $\min(M(x, y), M(y, z)) \leq M(x, y)$ for all $x, y, z \in \{A, B, C\}$.
fong-spivak-seven-sketches F03159	285	1.000	$x, y, z \in\{A, B, C\}$	$x, y, z \in\{A, B, C\}$	monoidal unit ∞ , and because $\min(M(x, y), M(y, z)) \leq M(x, y)$ for all $x, y, z \in \{A, B, C\}$.
fong-spivak-seven-sketches F03160	286	0.900	$d(A, B)=d(B, A)=5$	$d(A, B)=d(B, A)=5$	2. Let $\mathcal{X}$ be the Lawvere metric space with two objects A and B , such that $d(A, B)=d(B, A)=5$.
fong-spivak-seven-sketches F03161	286	0.210	$x_{\{f\}}=$	$x_f =$	Then we have $\mathcal{X}_f = \begin{array}{ c c } \hline A & B \\ \hline \bullet & \bullet \\ \hline \end{array}$ while $\mathcal{X}_g = \begin{array}{ c } \hline A \longrightarrow B \\ \hline \bullet \longrightarrow \bullet \\ \hline \end{array}$.
fong-spivak-seven-sketches F03162	286	0.210	$x_{\{g\}}=\stackrel{A}{\operatorname{bullet}} \rightarrowtail$	$x_g = \overset{A}{\bullet} \longrightarrow$	Then we have $\mathcal{X}_f = \begin{array}{ c c } \hline A & B \\ \hline \bullet & \bullet \\ \hline \end{array}$ while $\mathcal{X}_g = \begin{array}{ c } \hline A \longrightarrow B \\ \hline \bullet \longrightarrow \bullet \\ \hline \end{array}$.
fong-spivak-seven-sketches F03163	286	1.000	$d(x, y) \leq d(y, x)$	$d(x, y) \leq d(y, x)$	so for all x, y we must have $d(x, y) \leq d(y, x)$. But this means also that $d(y, x) \leq d(x, y)$, and so
fong-spivak-seven-sketches F03164	286	1.000	$d(y, x) \leq d(x, y)$	$d(y, x) \leq d(x, y)$	so for all x, y we must have $d(x, y) \leq d(y, x)$. But this means also that $d(y, x) \leq d(x, y)$, and so
fong-spivak-seven-sketches F03165	286	1.000	$d(y, x)=0$	$d(y, x)=0$	Now let's consider the skeletality condition. This says that if $d(x, y)=0$ and $d(y, x)=0$, then
fong-spivak-seven-sketches F03166	286	0.893	$(x, y) \in \operatorname{mathcal{X}} \times \operatorname{mathcal{Y}}$	$(x, y) \in \mathcal{X} \times \mathcal{Y}$	1. Let $(x, y) \in \mathcal{X} \times \mathcal{Y}$. Since $\mathcal{X}$ and $\mathcal{Y}$ are $\mathcal{V}$ -categories, we have $I \leq \mathcal{X}(x, x)$ and $I \leq \mathcal{Y}(y, y)$. Thus
fong-spivak-seven-sketches F03167	286	0.893	$I \leq \operatorname{mathfrak{X}}(x, x)$	$I \leq \mathfrak{X}(x, x)$	1. Let $(x, y) \in \mathcal{X} \times \mathcal{Y}$. Since $\mathcal{X}$ and $\mathcal{Y}$ are $\mathcal{V}$ -categories, we have $I \leq \mathcal{X}(x, x)$ and $I \leq \mathcal{Y}(y, y)$. Thus
fong-spivak-seven-sketches F03168	286	0.893	$I \leq y(y, y)$	$I \leq y(y, y)$	1. Let $(x, y) \in \mathcal{X} \times \mathcal{Y}$. Since $\mathcal{X}$ and $\mathcal{Y}$ are $\mathcal{V}$ -categories, we have $I \leq \mathcal{X}(x, x)$ and $I \leq \mathcal{Y}(y, y)$. Thus

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches- F03169	286	■ 0.999	$I = I \otimes I \leq X(x, x) \otimes y(y, y) = (\mathcal{X} \times \mathcal{Y})(x, y)$	$I = I \otimes I \leq X(x, x) \otimes y(y, y) = (\mathcal{X} \times \mathcal{Y})(x, y)$	$I = I \otimes I \leq \mathcal{X}(x, x) \otimes \mathcal{Y}(y, y) = (\mathcal{X} \times \mathcal{Y})(x, y)$.
fong-spivak-seven-sketches- F03170	286	■ 0.955	$y \left(y_1, y_2 \right) \otimes \mathcal{X} \left(x_2, x_3 \right) = \mathcal{X} \left(x_2, x_3 \right) \otimes y \left(y_1, y_2 \right)$	$y(y_1, y_2) \otimes \mathcal{X}(x_2, x_3) = \mathcal{X}(x_2, x_3) \otimes y(y_1, y_2)$	3. In particular, we use the symmetry, to conclude that $\mathcal{Y}(y_1, y_2) \otimes \mathcal{X}(x_2, x_3) = \mathcal{X}(x_2, x_3) \otimes \mathcal{Y}(y_1, y_2)$.
fong-spivak-seven-sketches- F03171	286	■ 0.990	$(\mathbb{R} \times \mathbb{R})((5, 6), (-1, 4)) = \mathbb{R}(5, -1) + \mathbb{R}(6, 4) = 6 + 2 = 8$	$(\mathbb{R} \times \mathbb{R})((5, 6), (-1, 4)) = \mathbb{R}(5, -1) + \mathbb{R}(6, 4) = 6 + 2 = 8$	We just apply Definition 2.74(ii): $(\mathbb{R} \times \mathbb{R})((5, 6), (-1, 4)) = \mathbb{R}(5, -1) + \mathbb{R}(6, 4) = 6 + 2 = 8$.
fong-spivak-seven-sketches- F03172	286	■ 1.000	$u \leq u^{\prime}$	$u \leq u'$	1. The function $-\otimes v: V \rightarrow V$ is monotone, because if $u \leq u'$ then $u \otimes v \leq u' \otimes v$ by the monotonicity
fong-spivak-seven-sketches- F03173	286	■ 1.000	$u \otimes v \leq u^{\prime} \otimes v$	$u \otimes v \leq u' \otimes v$	1. The function $-\otimes v: V \rightarrow V$ is monotone, because if $u \leq u'$ then $u \otimes v \leq u' \otimes v$ by the monotonicity
fong-spivak-seven-sketches- F03174	286	■ 0.996	$a := (v \multimap w)$	$a := (v \multimap w)$	2. Let $a := (v \multimap w)$ in Eq. (2.80). The right hand side is thus $(v \multimap w) \leq (v \multimap w)$, which is true by
fong-spivak-seven-sketches- F03175	286	■ 0.995	$(v \multimap u) \otimes v \leq u \leq u^{\prime}$	$(v \multimap u) \otimes v \leq u \leq u'$	3. Let $u \leq u'$. Then, using 2., $(v \multimap u) \otimes v \leq u \leq u'$. Applying Eq. (2.80), we thus have
fong-spivak-seven-sketches- F03176	286	■ 1.000	$(v \multimap u) \leq (v \multimap u^{\prime})$	$(v \multimap u) \leq (v \multimap u')$	$(v \multimap u) \leq (v \multimap u')$. This shows that the map $(v \multimap -): V \rightarrow V$ is monotone.
fong-spivak-seven-sketches- F03177	286	■ 1.000	$(v \multimap -): V \rightarrow V$	$(v \multimap -): V \rightarrow V$	$(v \multimap u) \leq (v \multimap u')$. This shows that the map $(v \multimap -): V \rightarrow V$ is monotone.
fong-spivak-seven-sketches- F03178	286	■ 0.859	$(v \multimap -)$	$(v \multimap -)$	not know $(- \otimes v)$ and $(v \multimap -)$ are monotone maps. We proved this, however, in items 1 and 3
fong-spivak-seven-sketches- F03179	287	■ 1.000	$x \Rightarrow y$	$x \Rightarrow y$	We need to find the hom-element. This is given by implication. That is, define the function $x \Rightarrow y$ by
fong-spivak-seven-sketches- F03180	287	■ 0.989	$(a \wedge v) \leq w$	$(a \wedge v) \leq w$	Then $(a \wedge v) \leq w$ if and only if $a \leq (v \Rightarrow w)$. Indeed, if $v = \text{false}$ then $a \wedge \text{false} = \text{false}$, and so the
fong-spivak-seven-sketches- F03181	287	■ 0.989	$a \leq (v \Rightarrow w)$	$a \leq (v \Rightarrow w)$	Then $(a \wedge v) \leq w$ if and only if $a \leq (v \Rightarrow w)$. Indeed, if $v = \text{false}$ then $a \wedge \text{false} = \text{false}$, and so the
fong-spivak-seven-sketches- F03182	287	■ 0.989	$v =$	$v =$	Then $(a \wedge v) \leq w$ if and only if $a \leq (v \Rightarrow w)$. Indeed, if $v = \text{false}$ then $a \wedge \text{false} = \text{false}$, and so the
fong-spivak-seven-sketches- F03183	287	■ 0.989	$a \wedge$	$a \wedge$	Then $(a \wedge v) \leq w$ if and only if $a \leq (v \Rightarrow w)$. Indeed, if $v = \text{false}$ then $a \wedge \text{false} = \text{false}$, and so the
fong-spivak-seven-sketches- F03184	287	■ 0.879	$= a$	$= a$	$v = \text{true}$, then $a \wedge \text{true} = a$ so the left hand side says $a \leq w$. But $(\text{true} \Rightarrow w) = w$, so the right hand
fong-spivak-seven-sketches- F03185	287	■ 0.879	$a \leq w$	$a \leq w$	$v = \text{true}$, then $a \wedge \text{true} = a$ so the left hand side says $a \leq w$. But $(\text{true} \Rightarrow w) = w$, so the right hand
fong-spivak-seven-sketches- F03186	287	■ 0.879	$\Rightarrow w) = w$	$\Rightarrow w) = w$	$v = \text{true}$, then $a \wedge \text{true} = a$ so the left hand side says $a \leq w$. But $(\text{true} \Rightarrow w) = w$, so the right hand
fong-spivak-seven-sketches- F03187	287	■ 0.980	$(\mathcal{P}(S), \subseteq, S, \cap)$	$(\mathcal{P}(S), \subseteq, S, \cap)$	Yes, the powerset monoidal preorder $(\mathcal{P}(S), \subseteq, S, \cap)$ is a quantale. The hom-object $B \multimap C$ is given by
fong-spivak-seven-sketches- F03188	287	■ 0.980	$B \multimap C$	$B \multimap C$	Yes, the powerset monoidal preorder $(\mathcal{P}(S), \subseteq, S, \cap)$ is a quantale. The hom-object $B \multimap C$ is given by
fong-spivak-seven-sketches- F03189	287	■ 1.000	$\bar{B} \cup C$	$\bar{B} \cup C$	$\bar{B} \cup C$, where $\bar{B}$ is the <i>complement</i> of B : it contains all elements of S not contained in B . To see that this
fong-spivak-seven-sketches- F03190	287	■ 1.000	$\bar{B}$	$\bar{B}$	$\bar{B} \cup C$, where $\bar{B}$ is the <i>complement</i> of B : it contains all elements of S not contained in B . To see that this
fong-spivak-seven-sketches- F03191	287	■ 1.000	$(A \cap B) \subseteq C$	$(A \cap B) \subseteq C$	satisfies Eq. (2.80), note that if $(A \cap B) \subseteq C$, then
fong-spivak-seven-sketches- F03192	287	■ 1.000	$A \subseteq (\bar{B} \cup C)$	$A \subseteq (\bar{B} \cup C)$	On the other hand, if $A \subseteq (\bar{B} \cup C)$, then
fong-spivak-seven-sketches- F03193	287	■ 0.558	$(\bigvee \varnothing) =$	$(\bigvee \varnothing) =$	1a. In Bool , $(\bigvee \varnothing) = \text{false}$, the least element.
fong-spivak-seven-sketches- F03194	287	■ 0.946	$(\bigvee \varnothing) = \infty$	$(\bigvee \varnothing) = \infty$	1b. In Cost , $(\bigvee \varnothing) = \infty$. This is because we use the opposite order $\geq$ on $[0, \infty]$, so $\bigvee \varnothing$ is the greatest
fong-spivak-seven-sketches- F03195	287	■ 0.974	$(\bigvee \varnothing) = 0$	$(\bigvee \varnothing) = 0$	$(\bigvee \varnothing) = 0$, is a bit confusing! Beware!
fong-spivak-seven-sketches- F03196	287	■ 1.000	$\min(x, y)$	$\min(x, y)$	2b. In Cost , $x \vee y$ is the minimum $\min(x, y)$. Again because we use the opposite order on $[0, \infty]$, the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches- F03197	287	■ 1.000	$v \in V, \emptyset \otimes v = 0$	$v \in V, 0 \otimes v = 0$	1. We first use Proposition 2.87 (2) and symmetry to show that for all $v \in V, 0 \otimes v = 0$.
fong-spivak-seven-sketches- F03198	289	■ 0.999	$\text{\textbackslash left}(v, a_{\{1\}}, \text{\textbackslash dots}, a_{\{n\}}\text{\textbackslash right})$	$(v, a_1, \dots, a_n)$	path in G to be a tuple of the form $(v, a_1, \dots, a_n)$ where $v \in V$ is a vertex, $s(a_1) = v$, and $t(a_i) = s(a_{i+1})$
fong-spivak-seven-sketches- F03199	289	■ 0.999	$s\text{\textbackslash left}(a_{\{1\}}\text{\textbackslash right})=v$	$s(a_1) = v$	path in G to be a tuple of the form $(v, a_1, \dots, a_n)$ where $v \in V$ is a vertex, $s(a_1) = v$, and $t(a_i) = s(a_{i+1})$
fong-spivak-seven-sketches- F03200	289	■ 0.999	$t\text{\textbackslash left}(a_{\{i\}}\text{\textbackslash right})=s\text{\textbackslash left}(a_{\{i+1\}}\text{\textbackslash right})$	$t(a_i) = s(a_{i+1})$	path in G to be a tuple of the form $(v, a_1, \dots, a_n)$ where $v \in V$ is a vertex, $s(a_1) = v$, and $t(a_i) = s(a_{i+1})$
fong-spivak-seven-sketches- F03201	289	■ 1.000	$i \in \{1, \text{\textbackslash dots}, n-1\}$	$i \in \{1, \dots, n-1\}$	for all $i \in \{1, \dots, n-1\}$; the length of this path is n , and this definition makes sense for any $n \in \mathbb{N}$. We
fong-spivak-seven-sketches- F03202	289	■ 1.000	$s(p) := v$	$s(p) := v$	say that the source of p is $s(p) := v$ and the target of p is defined to be
fong-spivak-seven-sketches- F03203	289	■ 1.000	$p = \text{\textbackslash left}(v, a_{\{1\}}, \text{\textbackslash dots}, a_{\{m\}}\text{\textbackslash right})$	$p = (v, a_1, \dots, a_m)$	Two paths $p = (v, a_1, \dots, a_m)$ and $q = (w, b_1, \dots, b_n)$ can be concatenated if $t(p) = s(q)$, in which case
fong-spivak-seven-sketches- F03204	289	■ 1.000	$q = \text{\textbackslash left}(w, b_{\{1\}}, \text{\textbackslash dots}, b_{\{n\}}\text{\textbackslash right})$	$q = (w, b_1, \dots, b_n)$	Two paths $p = (v, a_1, \dots, a_m)$ and $q = (w, b_1, \dots, b_n)$ can be concatenated if $t(p) = s(q)$, in which case
fong-spivak-seven-sketches- F03205	289	■ 1.000	$t(p) = s(q)$	$t(p) = s(q)$	Two paths $p = (v, a_1, \dots, a_m)$ and $q = (w, b_1, \dots, b_n)$ can be concatenated if $t(p) = s(q)$, in which case
fong-spivak-seven-sketches- F03206	289	■ 0.870	$p = (v)$	$p = (v)$	We are now ready to check unitality and associativity. A path p is an identity in $\mathbf{Free}(G)$ iff $p = (v)$
fong-spivak-seven-sketches- F03207	289	■ 0.551	(v)	(v)	for some $v \in V$. It is easy to see from the above that (v) and $(w, b_1, \dots, b_n)$ can be concatenated iff
fong-spivak-seven-sketches- F03208	289	■ 0.551	$\text{\textbackslash left}(w, b_{\{1\}}, \text{\textbackslash dots}, b_{\{n\}}\text{\textbackslash right})$	$(w, b_1, \dots, b_n)$	for some $v \in V$. It is easy to see from the above that (v) and $(w, b_1, \dots, b_n)$ can be concatenated iff
fong-spivak-seven-sketches- F03209	289	■ 1.000	$v = w$	$v = w$	$v = w$, in which case the result is $(w, b_1, \dots, b_n)$. Similarly $(v, a_1, \dots, a_m)$ and (w) can be concatenated
fong-spivak-seven-sketches- F03210	289	■ 1.000	$\text{\textbackslash left}(v, a_{\{1\}}, \text{\textbackslash dots}, a_{\{m\}}\text{\textbackslash right})$	$(v, a_1, \dots, a_m)$	$v = w$, in which case the result is $(w, b_1, \dots, b_n)$. Similarly $(v, a_1, \dots, a_m)$ and (w) can be concatenated
fong-spivak-seven-sketches- F03211	289	■ 1.000	(w)	(w)	$v = w$, in which case the result is $(w, b_1, \dots, b_n)$. Similarly $(v, a_1, \dots, a_m)$ and (w) can be concatenated
fong-spivak-seven-sketches- F03212	289	■ 1.000	$w = t\text{\textbackslash left}(a_{\{m\}}\text{\textbackslash right})$	$w = t(a_m)$	iff $w = t(a_m)$, in which case the result is $(v, a_1, \dots, a_m)$. Finally, for associativity with p and q as above
fong-spivak-seven-sketches- F03213	289	■ 0.996	$r = \text{\textbackslash left}(x, c_{\{1\}}, \text{\textbackslash dots}, c_{\{0\}}\text{\textbackslash right})$	$r = (x, c_1, \dots, c_0)$	and $r = (x, c_1, \dots, c_0)$, the formula readily reads that whichever way they are concatenated, $(p \circ q) \circ r$
fong-spivak-seven-sketches- F03214	289	■ 0.996	$(p ; q) ; r$	$(p; q); r$	and $r = (x, c_1, \dots, c_0)$, the formula readily reads that whichever way they are concatenated, $(p \circ q) \circ r$
fong-spivak-seven-sketches- F03215	289	■ 0.952	$p ; (q ; r)$	$p; (q; r)$	or $p \circ (q \circ r)$, the result is
fong-spivak-seven-sketches- F03216	289	■ 0.543	$\text{\textbackslash boxtimes}$	$\boxtimes$	id_{v_2} . We write $\boxtimes$ when the composite does not make sense (i.e. when the target of the first morphism
fong-spivak-seven-sketches- F03218	292	■ 1.000	$f_{\{2\}}$	f_2	The other way reverses f_1 and f_2 .
fong-spivak-seven-sketches- F03221	289	■ 1.000	$1+2+\cdots+n$	$1 + 2 + \dots + n$	is $1 + 2 + \dots + n$. This makes sense because (and the proof strategy would be to verify that) the
fong-spivak-seven-sketches- F03222	289	■ 0.818	$n-i$	$n - i$	above graph has n paths of length 0, it has $n - 1$ paths of length 1, and so on: it has $n - i$ paths
fong-spivak-seven-sketches- F03223	289	■ 1.000	$0 \leq i \leq n$	$0 \leq i \leq n$	of length i for every $0 \leq i \leq n$.
fong-spivak-seven-sketches- F03224	290	■ 0.999	$m+n$	$m + n$	a path of length n results in a path of length $m + n$.
fong-spivak-seven-sketches- F03225	290	■ 0.663	$h=j=A$	$h = j = A$	Note that $A \circ f \circ h = j = A \circ g \circ i$.
fong-spivak-seven-sketches- F03226	290	■ 0.485	z, s, s	z, s, s	There are four morphisms in $\mathcal{D}$, shown below, namely $z, s, s \circ s$, and $s \circ s \circ s$:
fong-spivak-seven-sketches- F03227	290	■ 0.485	$s ; s$	$s; s$	There are four morphisms in $\mathcal{D}$, shown below, namely $z, s, s \circ s$, and $s \circ s \circ s$:
fong-spivak-seven-sketches- F03228	291	■ 1.000	$(f(1), f(2))$	$(f(1), f(2))$	A function $f: \underline{2} \rightarrow \underline{3}$ can be described as an ordered pair $(f(1), f(2))$. The nine such functions are

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fong-spivak-seven-sketches F03229	291	■ 0.985	n!	$n!$	number of isomorphisms between them is n -factorial, often denoted $n!$. So for example there are
fong-spivak-seven-sketches F03230	291	■ 0.999	5 * 4 * 3 * 2 * 1=120	$5 * 4 * 3 * 2 * 1 = 120$	$5 * 4 * 3 * 2 * 1 = 120$ isomorphisms between $\{1, 2, 3, 4, 5\}$ and $\{a, b, c, d, e\}$.
fong-spivak-seven-sketches F03231	291	■ 0.999	\{1,2,3,4,5\}	$\{1, 2, 3, 4, 5\}$	$5 * 4 * 3 * 2 * 1 = 120$ isomorphisms between $\{1, 2, 3, 4, 5\}$ and $\{a, b, c, d, e\}$.
fong-spivak-seven-sketches F03232	291	■ 0.999	\{a, b, c, d, e\}	$\{a, b, c, d, e\}$	$5 * 4 * 3 * 2 * 1 = 120$ isomorphisms between $\{1, 2, 3, 4, 5\}$ and $\{a, b, c, d, e\}$.
fong-spivak-seven-sketches F03233	291	■ 0.566	f: c \rightarrow c	$f : c \rightarrow c$	We have to show that for any object $c \in \mathcal{C}$, the identity id_c has an inverse, i.e. a morphism $f : c \rightarrow c$
fong-spivak-seven-sketches F03234	291	■ 0.761	\mathrm{id}_{\mathrm{c}}=\mathrm{id}_{\mathrm{c}}	$\mathrm{id}_c = \mathrm{id}_c$	such that $f \circ \mathrm{id}_c = \mathrm{id}_c$ and $\mathrm{id}_c \circ f = \mathrm{id}_c$. Take $f = \mathrm{id}_c$; this works.
fong-spivak-seven-sketches F03235	291	■ 0.761	f=\mathrm{id}_{\mathrm{c}}	$f = \mathrm{id}_c$	such that $f \circ \mathrm{id}_c = \mathrm{id}_c$ and $\mathrm{id}_c \circ f = \mathrm{id}_c$. Take $f = \mathrm{id}_c$; this works.
fong-spivak-seven-sketches F03236	291	■ 1.000	s^{n}	s^n	morphism is of the form s^n for some $n \in \mathbb{N}$ and composing it with s gives s^{n+1} , which is never
fong-spivak-seven-sketches F03237	291	■ 1.000	s^{n+1}	s^{n+1}	morphism is of the form s^n for some $n \in \mathbb{N}$ and composing it with s gives s^{n+1} , which is never
fong-spivak-seven-sketches F03238	291	■ 1.000	s^{\{0\}} . p	$s^0.p$	
fong-spivak-seven-sketches F03239	291	■ 0.803	p ; q=\mathrm{id}	$p; q = \mathrm{id}$	is 0. In order for p to be an isomorphism, there must be some q such that $p \circ q = \mathrm{id}$ and $q \circ p = \mathrm{id}$, in
fong-spivak-seven-sketches F03240	291	■ 0.803	q ; p=\mathrm{id}	$q; p = \mathrm{id}$	is 0. In order for p to be an isomorphism, there must be some q such that $p \circ q = \mathrm{id}$ and $q \circ p = \mathrm{id}$, in
fong-spivak-seven-sketches F03241	291	■ 0.961	\mathbf{2} \rightarrow \mathbf{3}	$\mathbf{2} \rightarrow \mathbf{3}$	The other three functors $\mathbf{2} \rightarrow \mathbf{3}$ are shown here:
fong-spivak-seven-sketches F03242	292	■ 1.000	g^{\prime} ; i^{\prime} \mapsto f ; h	$g'; i' \mapsto f; h$	If one of these seems different from the rest, it's probably $g' \circ i' \mapsto f \circ h$. But note that in fact also
fong-spivak-seven-sketches F03243	292	■ 0.546	g^{\prime} ; i^{\prime} \mapsto g ; i	$g'; i' \mapsto g; i$	$g' \circ i' \mapsto g \circ i$ because $g \circ i = f \circ h$, so it's not an outlier after all.
fong-spivak-seven-sketches F03244	292	■ 0.546	g ; i=f ; h	$g; i = f; h$	$g' \circ i' \mapsto g \circ i$ because $g \circ i = f \circ h$, so it's not an outlier after all.
fong-spivak-seven-sketches F03245	292	■ 0.449	F, G	F, G	We need to give two functors F, G from $\bullet \xrightarrow{f} \bullet$ to $\bullet \xrightarrow[f_2]{f_1} \bullet$ whose on-objects parts are the same and
fong-spivak-seven-sketches F03246	292	■ 0.449	\stackrel{a}{\rightarrow} \stackrel{b}{\rightarrow}	$\xrightarrow{a} \xrightarrow{b}$	We need to give two functors F, G from $\bullet \xrightarrow{f} \bullet$ to $\bullet \xrightarrow[f_2]{f_1} \bullet$ whose on-objects parts are the same and
fong-spivak-seven-sketches F03247	292	■ 0.449	\stackrel{a^{\prime}}{\underset{f_1}{\rightarrow}} \stackrel{b^{\prime}}{\underset{f_2}{\rightarrow}}	$\xrightarrow[f_1]{a'} \xrightarrow[f_2]{b'}$	We need to give two functors F, G from $\bullet \xrightarrow{f} \bullet$ to $\bullet \xrightarrow[f_2]{f_1} \bullet$ whose on-objects parts are the same and
fong-spivak-seven-sketches F03248	292	■ 0.958	\operatorname{id}_{\mathrm{C}}: \mathrm{C} \rightarrow \mathrm{C}	$\mathrm{id}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}$	1. Let $\mathcal{C}$ be a category. Then defining $\mathrm{id}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}$ by $\mathrm{id}_{\mathcal{C}}(x) = x$ for every object and morphism in
fong-spivak-seven-sketches F03249	292	■ 0.958	\operatorname{id}_{\mathrm{C}}(x)=x	$\mathrm{id}_{\mathcal{C}}(x) = x$	1. Let $\mathcal{C}$ be a category. Then defining $\mathrm{id}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}$ by $\mathrm{id}_{\mathcal{C}}(x) = x$ for every object and morphism in
fong-spivak-seven-sketches F03250 [conf 0.098]	292	■ 0.098	\operatorname{identities} \mathrm{id}_{\mathrm{C}} \left(\mathrm{id}_{\mathrm{C}} \right) = \mathrm{id}_{\mathrm{C}} = \mathrm{id}_{\mathrm{id}_{\mathrm{C}}(c)}	identities $\mathrm{id}_{\mathcal{C}}(\mathrm{id}_{\mathcal{C}}) = \mathrm{id}_{\mathcal{C}} = \mathrm{id}_{\mathrm{id}_{\mathcal{C}}(c)}$	$\mathcal{C}$ is a functor because it preserves identities $\mathrm{id}_{\mathcal{C}}(\mathrm{id}_{\mathcal{C}}) = \mathrm{id}_{\mathcal{C}} = \mathrm{id}_{\mathrm{id}_{\mathcal{C}}(c)}$ for each object $c \in \mathrm{Ob}(\mathcal{C})$,
fong-spivak-seven-sketches F03251	292	■ 0.995	\operatorname{id}_{\mathrm{C}}(f ; g)=f ; g=\operatorname{id}_{\mathrm{C}}(f) ; \operatorname{id}_{\mathrm{C}}(g)	$\mathrm{id}_{\mathcal{C}}(f; g) = f; g = \mathrm{id}_{\mathcal{C}}(f); \mathrm{id}_{\mathcal{C}}(g)$	and it preserves composition $\mathrm{id}_{\mathcal{C}}(f \circ g) = f \circ g = \mathrm{id}_{\mathcal{C}}(f) \circ \mathrm{id}_{\mathcal{C}}(g)$ for each pair of composable
fong-spivak-seven-sketches F03252	292	■ 0.995	(F ; G)\left(\mathrm{id}_{\mathrm{c}}\right)=	$(F; G)(\mathrm{id}_c) =$	it preserves preserves identities and compositions. If $c \in \mathcal{C}$ is an object then $(F \circ G)(\mathrm{id}_c) =$
fong-spivak-seven-sketches F03253	292	■ 0.515	G\left(F\left(\mathrm{id}_{\mathrm{c}}\right)\right)=G\left(\mathrm{id}_{F(c)}\right)=\operatorname{id}_{G(F(c))}	$G(F(\mathrm{id}_c)) = G(\mathrm{id}_{F(c)}) = \mathrm{id}_{G(F(c))}$	$G(F(\mathrm{id}_c)) = G(\mathrm{id}_{F(c)}) = \mathrm{id}_{G(F(c))}$ because F and G preserve identities. If f, g are composable
fong-spivak-seven-sketches F03254	292	■ 0.874	\operatorname{id}_{\mathrm{C}}	$\mathrm{id}_{\mathcal{C}}$	and we pre-compose it as above with $\mathrm{id}_{\mathcal{C}}$; it is easy to see that the result will again be F , and

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches- F03255	292	■ 0.989	$\mathrm{id}_{\{\mathrm{id}_{\mathcal{D}}\}}$	$\mathrm{id}_{\mathcal{D}}$	similarly if we post-compose F with $\mathrm{id}_{\mathcal{D}}$. This gives unitality, and associativity is just as easy,
fong-spivak-seven-sketches- F03256	292	■ 1.000	$H: \mathcal{E} \rightarrow \mathcal{F}$	$H: \mathcal{E} \rightarrow \mathcal{F}$	though more wordy. Given F as above and $G: \mathcal{D} \rightarrow \mathcal{E}$ and $H: \mathcal{E} \rightarrow \mathcal{F}$, we need to show that
fong-spivak-seven-sketches- F03257	292	■ 0.991	$(F; G); H = F; (G; H)$	$(F; G); H = F; (G; H)$	$(F \circ G) \circ H = F \circ (G \circ H)$. It's a simple application of the definition: for any $x \in \mathcal{C}$, be it an object
fong-spivak-seven-sketches- F03258	292	■ 0.991	$x \in \mathcal{C}$	$x \in \mathcal{C}$	$(F \circ G) \circ H = F \circ (G \circ H)$. It's a simple application of the definition: for any $x \in \mathcal{C}$, be it an object
fong-spivak-seven-sketches- F03259	292	■ 0.996	$F_S \circ \mathrm{id}_1 = \mathrm{id}_S$	$F_S(\mathrm{id}_1) = \mathrm{id}_S$	Let $S \in \mathbf{Set}$ be a set. Define $F_S: \mathbf{1} \rightarrow \mathbf{Set}$ by $F_S(1) = S$ and $F_S(\mathrm{id}_1) = \mathrm{id}_S$. With this definition, F_S
fong-spivak-seven-sketches- F03260	292	■ 0.996	F_S	F_S	Let $S \in \mathbf{Set}$ be a set. Define $F_S: \mathbf{1} \rightarrow \mathbf{Set}$ by $F_S(1) = S$ and $F_S(\mathrm{id}_1) = \mathrm{id}_S$. With this definition, F_S
fong-spivak-seven-sketches- F03261	292	■ 1.000	$D(z)$	$D(z)$	1. Data on this schema, i.e. a set-valued functor, assigns a set $D(z)$ and a function $D(s): D(z) \rightarrow D(z)$,
fong-spivak-seven-sketches- F03262	292	■ 1.000	$D(s): D(z) \rightarrow D(z)$	$D(s): D(z) \rightarrow D(z)$	1. Data on this schema, i.e. a set-valued functor, assigns a set $D(z)$ and a function $D(s): D(z) \rightarrow D(z)$,
fong-spivak-seven-sketches- F03263	292	■ 0.999	D_z	D_z	of the set D_z :
fong-spivak-seven-sketches- F03264	293	■ 1.000	$D(c)$	$D(c)$	2. We could make $D(c)$ the set of people at a “secret Santa” Christmas party, where everyone gives
fong-spivak-seven-sketches- F03265	293	■ 0.995	$D(b)$	$D(b)$	a gift to someone, possibly themselves. Take $D(b)$ to be the set of gifts, g the giver function (each
fong-spivak-seven-sketches- F03266	293	■ 0.997	$D(a)$	$D(a)$	gift is given by a person), and h the receiver function (each gift is received by a person), $D(a)$ is
fong-spivak-seven-sketches- F03267	293	■ 0.999	$d(f): D(a) \rightarrow D(b)$	$d(f): D(a) \rightarrow D(b)$	the set of people who give a gift to themselves, and $d(f): D(a) \rightarrow D(b)$ is the inclusion.
fong-spivak-seven-sketches- F03268	293	■ 0.568	$F, G, H \in \mathcal{D}^{\mathcal{C}}$	$F, G, H \in \mathcal{D}^{\mathcal{C}}$	1. The expert packs so much information in so little space! Suppose given three objects $F, G, H \in \mathcal{D}^{\mathcal{C}}$,
fong-spivak-seven-sketches- F03269	293	■ 1.000	$F, G, H: \mathcal{C} \rightarrow \mathcal{D}$	$F, G, H: \mathcal{C} \rightarrow \mathcal{D}$	these are functors $F, G, H: \mathcal{C} \rightarrow \mathcal{D}$. Morphisms $\alpha: F \rightarrow G$ and $\beta: G \rightarrow H$ are natural transfor-
fong-spivak-seven-sketches- F03270	293	■ 1.000	$\beta: G \rightarrow H$	$\beta: G \rightarrow H$	these are functors $F, G, H: \mathcal{C} \rightarrow \mathcal{D}$. Morphisms $\alpha: F \rightarrow G$ and $\beta: G \rightarrow H$ are natural transfor-
fong-spivak-seven-sketches- F03271	293	■ 0.877	$c \in \mathcal{C}, \alpha$	$c \in \mathcal{C}, \alpha$	So for each $c \in \mathcal{C}$, α has a component $\alpha_c: F(c) \rightarrow G(c)$ and β has a component $\beta_c: G(c) \rightarrow H(c)$
fong-spivak-seven-sketches- F03272	293	■ 0.877	$\beta_c: G(c) \rightarrow H(c)$	$\beta_c: G(c) \rightarrow H(c)$	So for each $c \in \mathcal{C}$, α has a component $\alpha_c: F(c) \rightarrow G(c)$ and β has a component $\beta_c: G(c) \rightarrow H(c)$
fong-spivak-seven-sketches- F03273	293	■ 0.872	$(\alpha; \beta)_c := (\alpha_c; \beta_c)$	$(\alpha; \beta)_c := (\alpha_c; \beta_c)$	in $\mathcal{D}$. The expert has told us to define $(\alpha \circ \beta)_c := (\alpha_c \circ \beta_c)$, and indeed that is a morphism
fong-spivak-seven-sketches- F03274	293	■ 1.000	$F(c) \rightarrow H(c)$	$F(c) \rightarrow H(c)$	$F(c) \rightarrow H(c)$.
fong-spivak-seven-sketches- F03275	293	■ 0.995	id_F	id_F	2. We propose that the identity natural transformation id_F on a functor $F: \mathcal{C} \rightarrow \mathcal{D}$ has as its
fong-spivak-seven-sketches- F03276	293	■ 0.660	$(\mathrm{id}_F)_c := \mathrm{id}_{F(c)}$	$(\mathrm{id}_F)_c := \mathrm{id}_{F(c)}$	c -component the morphism $(\mathrm{id}_F)_c := \mathrm{id}_{F(c)}$ in $\mathcal{D}$, for any c . The naturality square
fong-spivak-seven-sketches- F03277	293	■ 1.000	$\beta: F \rightarrow G$	$\beta: F \rightarrow G$	obviously commutes for any $f: c \rightarrow c'$. And it is unital: post-composing id_F with any $\beta: F \rightarrow G$
fong-spivak-seven-sketches- F03278	293	■ 0.992	$\alpha: E \rightarrow F$	$\alpha: E \rightarrow F$	(and similarly for precomposing with any $\alpha: E \rightarrow F$) results in a natural transformation $\mathrm{id}_F \circ \beta$
fong-spivak-seven-sketches- F03279	293	■ 0.992	$\mathrm{id}_F; \beta$	$\mathrm{id}_F; \beta$	(and similarly for precomposing with any $\alpha: E \rightarrow F$) results in a natural transformation $\mathrm{id}_F \circ \beta$
fong-spivak-seven-sketches- F03280	293	■ 0.896	$(\mathrm{id}_F)_c; \beta_c = (\mathrm{id}_{F(c)}; \beta_c) = \beta_c$	$(\mathrm{id}_F)_c; \beta_c = (\mathrm{id}_{F(c)}; \beta_c) = \beta_c$	with components $(\mathrm{id}_F)_c \circ \beta_c = (\mathrm{id}_{F(c)} \circ \beta_c) = \beta_c$, and this is just β as desired.
fong-spivak-seven-sketches- F03281	293	■ 1.000	$\alpha, \beta: F \rightarrow G$	$\alpha, \beta: F \rightarrow G$	1. Suppose that $F, G: \mathcal{C} \rightarrow \mathcal{P}$ are functors and $\alpha, \beta: F \rightarrow G$ are natural transformations; we need to
fong-spivak-seven-sketches- F03282	293	■ 1.000	$\alpha = \beta$	$\alpha = \beta$	show that $\alpha = \beta$. It suffices to check that $\alpha_c = \beta_c$ for each object $c \in \mathrm{Ob}(\mathcal{C})$. But α_c and β_c are

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fong-spivak-seven-sketches- F03283	293	■ 1.000	$\alpha_{\{c\}}=\beta_{\{c\}}$	$\alpha_c = \beta_c$	show that $\alpha = \beta$. It suffices to check that $\alpha_c = \beta_c$ for each object $c \in \text{Ob}(\mathcal{C})$. But α_c and β_c are
fong-spivak-seven-sketches- F03284	293	■ 1.000	$\beta_{\{c\}}$	β_c	show that $\alpha = \beta$. It suffices to check that $\alpha_c = \beta_c$ for each object $c \in \text{Ob}(\mathcal{C})$. But α_c and β_c are
fong-spivak-seven-sketches- F03285	293	■ 0.998	$F(c) \rightarrowtail G(c)$	$F(c) \rightarrow G(c)$	morphisms $F(c) \rightarrowtail G(c)$ in $\mathcal{P}$, which is a preorder, and the definition of a preorder—considered
fong-spivak-seven-sketches- F03286	293	■ 0.421	$\mathcal{P}:=\mathbf{1}$	$\mathcal{P} := \mathbf{1}$	2. This is false. Let $\mathcal{P} := \mathbf{1}$, let $\mathcal{C} := \left \bullet \begin{array}{c} \xrightarrow{f_1} \\ \xrightarrow{f_2} \end{array} \bullet \right $, let $F(1) := a$, let $G(1) := b$, let $\alpha_1 := f_1$, and let $\beta_1 := g_2$.
fong-spivak-seven-sketches- F03287	293	■ 0.421	$\mathcal{C}:=\vec{\bullet} \xrightarrow[f_2]{f_1} \bullet$ $\stackrel{\bullet}{\rightarrowtail} \{\bullet\}$	$\mathcal{C} := \vec{\bullet} \xrightarrow[f_2]{f_1} \bullet$	2. This is false. Let $\mathcal{P} := \mathbf{1}$, let $\mathcal{C} := \left \bullet \begin{array}{c} \xrightarrow{f_1} \\ \xrightarrow{f_2} \end{array} \bullet \right $, let $F(1) := a$, let $G(1) := b$, let $\alpha_1 := f_1$, and let $\beta_1 := g_2$.
fong-spivak-seven-sketches- F03288	293	■ 0.421	$F(1):=a$	$F(1) := a$	2. This is false. Let $\mathcal{P} := \mathbf{1}$, let $\mathcal{C} := \left \bullet \begin{array}{c} \xrightarrow{f_1} \\ \xrightarrow{f_2} \end{array} \bullet \right $, let $F(1) := a$, let $G(1) := b$, let $\alpha_1 := f_1$, and let $\beta_1 := g_2$.
fong-spivak-seven-sketches- F03289	293	■ 0.421	$G(1):=b$	$G(1) := b$	2. This is false. Let $\mathcal{P} := \mathbf{1}$, let $\mathcal{C} := \left \bullet \begin{array}{c} \xrightarrow{f_1} \\ \xrightarrow{f_2} \end{array} \bullet \right $, let $F(1) := a$, let $G(1) := b$, let $\alpha_1 := f_1$, and let $\beta_1 := g_2$.
fong-spivak-seven-sketches- F03290	293	■ 0.421	$\alpha_{\{1\}}:=f_{\{1\}}$	$\alpha_1 := f_1$	2. This is false. Let $\mathcal{P} := \mathbf{1}$, let $\mathcal{C} := \left \bullet \begin{array}{c} \xrightarrow{f_1} \\ \xrightarrow{f_2} \end{array} \bullet \right $, let $F(1) := a$, let $G(1) := b$, let $\alpha_1 := f_1$, and let $\beta_1 := g_2$.
fong-spivak-seven-sketches- F03291	293	■ 0.421	$\beta_{\{1\}}:=g_{\{2\}}$	$\beta_1 := g_2$	2. This is false. Let $\mathcal{P} := \mathbf{1}$, let $\mathcal{C} := \left \bullet \begin{array}{c} \xrightarrow{f_1} \\ \xrightarrow{f_2} \end{array} \bullet \right $, let $F(1) := a$, let $G(1) := b$, let $\alpha_1 := f_1$, and let $\beta_1 := g_2$.
fong-spivak-seven-sketches- F03292	294	■ 1.000	$\alpha_{\{\text{Arrow}\}}(b)=e$	$\alpha_{\text{Arrow}}(b) = e$	1. We have $\alpha_{\text{Arrow}}(b) = e$ and $\alpha_{\text{Vertex}}(1) = 4$, $\alpha_{\text{Vertex}}(2) = 5$, and $\alpha_{\text{Vertex}}(3) = 5$.
fong-spivak-seven-sketches- F03293	294	■ 1.000	$\alpha_{\{\text{Vertex}\}}(1)=4$, $\alpha_{\{\text{Vertex}\}}(2)=5$	$\alpha_{\text{Vertex}}(1) = 4, \alpha_{\text{Vertex}}(2) = 5$	1. We have $\alpha_{\text{Arrow}}(b) = e$ and $\alpha_{\text{Vertex}}(1) = 4$, $\alpha_{\text{Vertex}}(2) = 5$, and $\alpha_{\text{Vertex}}(3) = 5$.
fong-spivak-seven-sketches- F03294	294	■ 1.000	$\alpha_{\{\text{Vertex}\}}(3)=5$	$\alpha_{\text{Vertex}}(3) = 5$	1. We have $\alpha_{\text{Arrow}}(b) = e$ and $\alpha_{\text{Vertex}}(1) = 4$, $\alpha_{\text{Vertex}}(2) = 5$, and $\alpha_{\text{Vertex}}(3) = 5$.
fong-spivak-seven-sketches- F03295	295	■ 1.000	$X \times B \rightarrowtail Y \times B$	$X \times B \rightarrow Y \times B$	1. Given an arbitrary function $f: X \rightarrow Y$, we need a function $X \times B \rightarrow Y \times B$. We suggest the
fong-spivak-seven-sketches- F03296	295	■ 1.000	$f \times B$	$f \times B$	function which might be denoted $f \times B$; it sends (x, b) to $(f(x), b)$. This assignment is functorial:
fong-spivak-seven-sketches- F03297	295	■ 1.000	(x, b)	(x, b)	function which might be denoted $f \times B$; it sends (x, b) to $(f(x), b)$. This assignment is functorial:
fong-spivak-seven-sketches- F03298	295	■ 1.000	$(f(x), b)$	$(f(x), b)$	function which might be denoted $f \times B$; it sends (x, b) to $(f(x), b)$. This assignment is functorial:
fong-spivak-seven-sketches- F03299	295	■ 0.988	$\mathrm{id}_{X \times B}$	$\mathrm{id}_{X \times B}$	applied to id_X it returns $\mathrm{id}_{X \times B}$ and it preserves composition.
fong-spivak-seven-sketches- F03300	295	■ 1.000	$X^B \rightarrowtail Y^B$	$X^B \rightarrow Y^B$	2. Given a function $f: X \rightarrow Y$, we need a function $X^B \rightarrow Y^B$. The canonical function would
fong-spivak-seven-sketches- F03301	295	■ 0.988	f^B	f^B	be denoted f^B ; it sends a function $g: B \rightarrow X$ to the composite $(g \circ f): B \rightarrow X \rightarrow Y$. This is
fong-spivak-seven-sketches- F03302	295	■ 0.988	$g: B \rightarrowtail X$	$g: B \rightarrow X$	be denoted f^B ; it sends a function $g: B \rightarrow X$ to the composite $(g \circ f): B \rightarrow X \rightarrow Y$. This is
fong-spivak-seven-sketches- F03303	295	■ 0.988	$(g \circ f): B \rightarrowtail X \rightarrowtail Y$	$(g \circ f): B \rightarrow X \rightarrow Y$	be denoted f^B ; it sends a function $g: B \rightarrow X$ to the composite $(g \circ f): B \rightarrow X \rightarrow Y$. This is
fong-spivak-seven-sketches- F03304	295	■ 0.986	$f^B(\mathrm{id}_X)=\mathrm{id}_{X^B}$	$f^B(\mathrm{id}_X) = \mathrm{id}_{X^B}$	functorial: applied to id_X it sends g to g , i.e. $f^B(\mathrm{id}_X) = \mathrm{id}_{X^B}$, and applied to the composite
fong-spivak-seven-sketches- F03305	295	■ 1.000	$\left(f_{\{1\}} ; f_{\{2\}}\right): X \rightarrowtail Y \rightarrowtail Z$	$(f_1; f_2): X \rightarrow Y \rightarrow Z$	$f_1 \circ f_2: X \rightarrow Y \rightarrow Z$, we have
fong-spivak-seven-sketches- F03306	295	■ 1.000	$g \in X^B$	$g \in X^B$	for any $g \in X^B$.
fong-spivak-seven-sketches- F03307	295	■ 0.999	$\mathbb{N}^{\mathbb{N}}$	$\mathbb{N}^{\mathbb{N}}$	3. If $p: \mathbb{N} \rightarrow \mathbb{N}^{\mathbb{N}}$ is the result of currying $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, then $p(3)$ is an element of $\mathbb{N}^{\mathbb{N}}$, i.e. we have

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_295 F03308	295	1.000	$p(3) \colon \mathbb{N} \rightarrow \mathbb{N}$	$p(3) \colon \mathbb{N} \rightarrow \mathbb{N}$	$p(3) \colon \mathbb{N} \rightarrow \mathbb{N}$; what function is it? It is the function that adds three. That is $p(3)(n) := n + 3$.
fong-spivak-seven-sketches_295 F03309	295	1.000	$p(3)(n) := n + 3$	$p(3)(n) := n + 3$	$p(3) \colon \mathbb{N} \rightarrow \mathbb{N}$; what function is it? It is the function that adds three. That is $p(3)(n) := n + 3$.
fong-spivak-seven-sketches_295 F03310	295	0.533	$! \colon \mathcal{C} \rightarrow \mathbf{1}$	$! \colon \mathcal{C} \rightarrow \mathbf{1}$	The functor  from Eq. (3.75) sends each object $c \in \mathcal{C}$ to the unique object $1 \in \mathbf{1}$ and sends each
fong-spivak-seven-sketches_295 F03311	295	0.533	$1 \in \mathbf{1}$	$1 \in \mathbf{1}$	The functor $! \colon \mathcal{C} \rightarrow \mathbf{1}$ from Eq. (3.75) sends each object $c \in \mathcal{C}$ to the unique object $1 \in \mathbf{1}$ and sends each
fong-spivak-seven-sketches_295 F03312	295	0.590	$\mathrm{B} = \mathrm{Bob}, \mathrm{3} = \mathrm{Em_3}$	$B = \text{Bob}, 3 = \text{Em_3}$	Here it is, with names and emails shortened (e.g. $B = \text{Bob}, 3 = \text{Em_3}$):
fong-spivak-seven-sketches_295 F03313	295	0.999	$c \rightarrow z$	$c \rightarrow z$	An object z is terminal in some category $\mathcal{C}$ if, for every $c \in \mathcal{C}$ there exists a unique morphism $c \rightarrow z$.
fong-spivak-seven-sketches_296 F03314	296	0.940	$\mathcal{C} \in$	$\mathcal{C} \in$	any object (category) $\mathcal{C} \in \mathbf{Cat}$.
fong-spivak-seven-sketches_296 F03315	296	0.877	$2V := \bullet \bullet$	$2V := \bullet \bullet$	Consider the graph $2V := \boxed{\bullet \bullet}$ with two vertices and no arrows, and let $\mathcal{C} = \mathbf{Free}(2V)$; it has two
fong-spivak-seven-sketches_296 F03316	296	0.877	$\mathcal{C} = \operatorname{Free}(2V)$	$\mathcal{C} = \mathbf{Free}(2V)$	Consider the graph $2V := \boxed{\bullet \bullet}$ with two vertices and no arrows, and let $\mathcal{C} = \mathbf{Free}(2V)$; it has two
fong-spivak-seven-sketches_296 F03317	296	1.000	$z \in \mathcal{P}$	$z \in \mathcal{P}$	A product of x and y in $\mathcal{P}$ is an object $z \in \mathcal{P}$ equipped with maps $z \rightarrow x$ and $z \rightarrow y$ such that for any
fong-spivak-seven-sketches_296 F03318	296	1.000	$z \rightarrow x$	$z \rightarrow x$	A product of x and y in $\mathcal{P}$ is an object $z \in \mathcal{P}$ equipped with maps $z \rightarrow x$ and $z \rightarrow y$ such that for any
fong-spivak-seven-sketches_296 F03319	296	1.000	$z \rightarrow y$	$z \rightarrow y$	A product of x and y in $\mathcal{P}$ is an object $z \in \mathcal{P}$ equipped with maps $z \rightarrow x$ and $z \rightarrow y$ such that for any
fong-spivak-seven-sketches_296 F03320	296	1.000	$z^{\prime}$	$z^{\prime}$	other object $z^{\prime}$ and maps $z^{\prime} \rightarrow x$ and $z^{\prime} \rightarrow y$, there is a unique morphism $z^{\prime} \rightarrow z$ making the evident
fong-spivak-seven-sketches_296 F03321	296	1.000	$z^{\prime} \rightarrow x$	$z^{\prime} \rightarrow x$	other object $z^{\prime}$ and maps $z^{\prime} \rightarrow x$ and $z^{\prime} \rightarrow y$, there is a unique morphism $z^{\prime} \rightarrow z$ making the evident
fong-spivak-seven-sketches_296 F03322	296	1.000	$z^{\prime} \rightarrow y$	$z^{\prime} \rightarrow y$	other object $z^{\prime}$ and maps $z^{\prime} \rightarrow x$ and $z^{\prime} \rightarrow y$, there is a unique morphism $z^{\prime} \rightarrow z$ making the evident
fong-spivak-seven-sketches_296 F03323	296	1.000	$z^{\prime} \rightarrow z$	$z^{\prime} \rightarrow z$	other object $z^{\prime}$ and maps $z^{\prime} \rightarrow x$ and $z^{\prime} \rightarrow y$, there is a unique morphism $z^{\prime} \rightarrow z$ making the evident
fong-spivak-seven-sketches_296 F03324	296	1.000	$z \leq x$	$z \leq x$	diagrams commute. Thus the above becomes: a product of x and y in $\mathcal{P}$ is an object z with $z \leq x$ and
fong-spivak-seven-sketches_296 F03325	296	1.000	$z \leq y$	$z \leq y$	$z \leq y$ such that for any other $z^{\prime}$, if $z^{\prime} \leq x$ and $z^{\prime} \leq y$ then $z^{\prime} \leq z$. This is exactly the definition of meet,
fong-spivak-seven-sketches_296 F03326	296	1.000	$z^{\prime} \leq x$	$z^{\prime} \leq x$	$z \leq y$ such that for any other $z^{\prime}$, if $z^{\prime} \leq x$ and $z^{\prime} \leq y$ then $z^{\prime} \leq z$. This is exactly the definition of meet,
fong-spivak-seven-sketches_296 F03327	296	1.000	$z^{\prime} \leq y$	$z^{\prime} \leq y$	$z \leq y$ such that for any other $z^{\prime}$, if $z^{\prime} \leq x$ and $z^{\prime} \leq y$ then $z^{\prime} \leq z$. This is exactly the definition of meet,
fong-spivak-seven-sketches_296 F03328	296	1.000	$z^{\prime} \leq z$	$z^{\prime} \leq z$	$z \leq y$ such that for any other $z^{\prime}$, if $z^{\prime} \leq x$ and $z^{\prime} \leq y$ then $z^{\prime} \leq z$. This is exactly the definition of meet,
fong-spivak-seven-sketches_296 F03329	296	0.867	$z = x \wedge y$	$z = x \wedge y$	$z = x \wedge y$.
fong-spivak-seven-sketches_296 F03330	296	0.989	$(\mathrm{id}_c, \mathrm{id}_d)$	$(\mathrm{id}_c, \mathrm{id}_d)$	1. The identity morphism on the object (c, d) in the product category $\mathcal{C} \times \mathcal{D}$ is $(\mathrm{id}_c, \mathrm{id}_d)$.
fong-spivak-seven-sketches_296 F03331	296	0.621	$\left(\left(\left(f_1, g_1 \right) \stackrel{\circ}{\mathfrak{g}} \left(f_2, g_2 \right) \right) \stackrel{\circ}{\mathfrak{g}} \left(f_3, g_3 \right) = \left(f_1, g_1 \right) \stackrel{\circ}{\mathfrak{g}} \left(\left(f_2, g_2 \right) \stackrel{\circ}{\mathfrak{g}} \left(f_3, g_3 \right) \right) \right)$	$\left(\left(\left(f_1, g_1 \right) \stackrel{\circ}{\mathfrak{g}} \left(f_2, g_2 \right) \right) \stackrel{\circ}{\mathfrak{g}} \left(f_3, g_3 \right) = \left(f_1, g_1 \right) \stackrel{\circ}{\mathfrak{g}} \left(\left(f_2, g_2 \right) \stackrel{\circ}{\mathfrak{g}} \left(f_3, g_3 \right) \right) \right)$	We want to check that $((f_1, g_1) \mathbin{\mathfrak{g}} (f_2, g_2)) \mathbin{\mathfrak{g}} (f_3, g_3) = (f_1, g_1) \mathbin{\mathfrak{g}} ((f_2, g_2) \mathbin{\mathfrak{g}} (f_3, g_3))$. But composition in
fong-spivak-seven-sketches_296 F03332	296	0.400	$\left(f_1 \right)$	(f_1)	a product category is given component-wise. That means the left-hand side is $((f_1 \mathbin{\mathfrak{g}} f_2) \mathbin{\mathfrak{g}} f_3, (g_1 \mathbin{\mathfrak{g}}$
fong-spivak-seven-sketches_296 F03333	296	0.400	$\left. f_2 \right)$	$f_2)$	a product category is given component-wise. That means the left-hand side is $((f_1 \mathbin{\mathfrak{g}} f_2) \mathbin{\mathfrak{g}} f_3, (g_1 \mathbin{\mathfrak{g}}$
fong-spivak-seven-sketches_296 F03334	296	0.400	f_3	f_3	a product category is given component-wise. That means the left-hand side is $((f_1 \mathbin{\mathfrak{g}} f_2) \mathbin{\mathfrak{g}} f_3, (g_1 \mathbin{\mathfrak{g}}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches- F03335 [conf 0.030]	296	0.030	$\left.\left(g_2\right) \stackrel{\circ}{\circ} g_3\right)$	$g_2) \overset{\circ}{9} g_3)$	$g_2) \circ g_3)$, whereas the right-hand side is $(f_1 \circ (f_2 \circ f_3), g_1 \circ (g_2 \circ g_3))$, and these are equal because
fong-spivak-seven-sketches- F03336 [conf 0.030]	296	0.030	$\left(f_1 ; \left(f_2 ; f_3\right), g_1\right) \right.$	$(f_1 ; (f_2 ; f_3), g_1$	$g_2) \circ g_3)$, whereas the right-hand side is $(f_1 \circ (f_2 \circ f_3), g_1 \circ (g_2 \circ g_3))$, and these are equal because
fong-spivak-seven-sketches- F03337 [conf 0.030]	296	0.030	$\left(g_2 ; g_3\right)$	$(g_2 ; g_3)$	$g_2) \circ g_3)$, whereas the right-hand side is $(f_1 \circ (f_2 \circ f_3), g_1 \circ (g_2 \circ g_3))$, and these are equal because
fong-spivak-seven-sketches- F03338	296	0.997	$(1,2)$	$(1,2)$	3. The product category $\mathbf{1} \times \mathbf{2}$ has two objects $(1,1)$ and $(1,2)$ and one non-identity morphism
fong-spivak-seven-sketches- F03339	296	0.929	$(1,1) \rightarrow(1,2)$	$(1,1) \rightarrow(1,2)$	$(1,1) \rightarrow(1,2)$. It is not hard to see that it looks the same as 2. In fact, for any $\mathcal{C}$ there is an
fong-spivak-seven-sketches- F03340	296	1.000	$\mathbf{1} \times \mathcal{C} \cong \mathcal{C}$	$\mathbf{1} \times \mathcal{C} \cong \mathcal{C}$	isomorphism of categories $\mathbf{1} \times \mathcal{C} \cong \mathcal{C}$.
fong-spivak-seven-sketches- F03341	296	1.000	$X=P \times Q$	$X=P \times Q$	4. Let P and Q be preorders, let $X=P \times Q$ be their product preorder as defined in Example 1.56,
fong-spivak-seven-sketches- F03342	296	0.980	$\mathcal{X}=\mathcal{P} \times \mathcal{Q}$	$\mathcal{X}=\mathcal{P} \times \mathcal{Q}$	and let $\mathcal{P}, \mathcal{Q}$, and $\mathcal{X}$ be the corresponding categories. Then $\mathcal{X}=\mathcal{P} \times \mathcal{Q}$.
fong-spivak-seven-sketches- F03343	296	0.998	$X \stackrel{p_X}{\longleftarrow} Z \stackrel{p_Y}{\longrightarrow} Y$	$X \xleftarrow{p_X} Z \xrightarrow{p_Y} Y$	A product of X and Y is an object Z equipped with morphisms $X \xleftarrow{p_X} Z \xrightarrow{p_Y} Y$ such that for any other
fong-spivak-seven-sketches- F03344	296	0.998	$X \stackrel{p_X}{\longleftarrow} Z^{\prime} \stackrel{p_Y}{\longrightarrow} Y \stackrel{\Upsilon}{\longrightarrow}$	$X \xleftarrow{p_X'} Z' \xrightarrow{p_Y'} Y$	object Z' equipped with morphisms $X \xleftarrow{p_X'} Z' \xrightarrow{p_Y'} Y$, there is a unique morphism $f: Z' \rightarrow Z$ making
fong-spivak-seven-sketches- F03345	296	0.998	$f: Z^{\prime} \rightarrow Z$	$f: Z' \rightarrow Z$	object Z' equipped with morphisms $X \xleftarrow{p_X'} Z' \xrightarrow{p_Y'} Y$, there is a unique morphism $f: Z' \rightarrow Z$ making
fong-spivak-seven-sketches- F03346	296	1.000	$f ; p_X=p_X^{\prime}$	$f ; p_X = p_X'$	the triangles commute, $f \circ p_X = p_X'$ and $f \circ p_Y = p_Y'$. But “an object equipped with morphisms to X
fong-spivak-seven-sketches- F03347	296	1.000	$f ; p_Y=p_Y^{\prime}$	$f ; p_Y = p_Y'$	the triangles commute, $f \circ p_X = p_X'$ and $f \circ p_Y = p_Y'$. But “an object equipped with morphisms to X
fong-spivak-seven-sketches- F03348	296	0.894	$Y^{\prime \prime}$	Y''	and Y'' is exactly the definition of an object in $\mathbf{Cone}(X, Y)$, and a morphism f making the triangles
fong-spivak-seven-sketches- F03349	296	1.000	$Z \in \mathbf{Cone}(X, Y)$	$Z \in \mathbf{Cone}(X, Y)$	product of X and Y is an object $Z \in \mathbf{Cone}(X, Y)$ such that for any other object Z' there is a unique
fong-spivak-seven-sketches- F03350	296	0.439	$Z^{\prime} \rightarrow Z$	$Z' \rightarrow Z$	morphism $Z' \rightarrow Z$ in $\mathbf{Cone}(X, Y)$. This is exactly the definition of Z being terminal in $\mathbf{Cone}(X, Y)$.
fong-spivak-seven-sketches- F03351	296	0.521	$v_1 \bullet v_2$	$\overset{v_1}{\bullet} v_2$	Suppose $\mathcal{J}$ is the graph $\boxed{\overset{v_1}{\bullet} \bullet \overset{v_2}{\bullet}}$ and $D: \mathcal{J} \rightarrow \mathbf{Set}$ is given by two sets, $D(v_1)=A$ and $D(v_2)=B$ for sets
fong-spivak-seven-sketches- F03352	296	0.521	$D\left(v_1\right)=A$	$D\left(v_1\right)=A$	Suppose $\mathcal{J}$ is the graph $\boxed{\overset{v_1}{\bullet} \bullet \overset{v_2}{\bullet}}$ and $D: \mathcal{J} \rightarrow \mathbf{Set}$ is given by two sets, $D(v_1)=A$ and $D(v_2)=B$ for sets
fong-spivak-seven-sketches- F03353	296	0.521	$D\left(v_2\right)=B$	$D\left(v_2\right)=B$	Suppose $\mathcal{J}$ is the graph $\boxed{\overset{v_1}{\bullet} \bullet \overset{v_2}{\bullet}}$ and $D: \mathcal{J} \rightarrow \mathbf{Set}$ is given by two sets, $D(v_1)=A$ and $D(v_2)=B$ for sets
fong-spivak-seven-sketches- F03354	297	0.274	$F^{\mathrm{op}}: \mathcal{C}^{\mathrm{op}} \rightarrow \mathcal{D}^{\mathrm{op}}$	$F^{\mathrm{op}}: \mathcal{C}^{\mathrm{op}} \rightarrow \mathcal{D}^{\mathrm{op}}$	Given a functor $F: \mathcal{C} \rightarrow \mathcal{D}$, we define its opposite $F^{\mathrm{op}}: \mathcal{C}^{\mathrm{op}} \rightarrow \mathcal{D}^{\mathrm{op}}$ as follows. For each object $c \in$
fong-spivak-seven-sketches- F03355	297	0.274	$c \in$	$c \in$	Given a functor $F: \mathcal{C} \rightarrow \mathcal{D}$, we define its opposite $F^{\mathrm{op}}: \mathcal{C}^{\mathrm{op}} \rightarrow \mathcal{D}^{\mathrm{op}}$ as follows. For each object $c \in$
fong-spivak-seven-sketches- F03356	297	0.904	$\mathrm{Ob}\left(\mathcal{C}^{\mathrm{op}}\right)=\mathrm{Ob}(\mathcal{C})$	$\mathrm{Ob}\left(\mathcal{C}^{\mathrm{op}}\right)=\mathrm{Ob}(\mathcal{C})$	$\mathrm{Ob}\left(\mathcal{C}^{\mathrm{op}}\right)=\mathrm{Ob}(\mathcal{C})$, put $F^{\mathrm{op}}(c):=F(c)$. For each morphism $f: c_1 \rightarrow c_2$ in $\mathcal{C}^{\mathrm{op}}$, we have a corresponding
fong-spivak-seven-sketches- F03357	297	0.904	$F^{\mathrm{op}}(c):=F(c)$	$F^{\mathrm{op}}(c):=F(c)$	$\mathrm{Ob}\left(\mathcal{C}^{\mathrm{op}}\right)=\mathrm{Ob}(\mathcal{C})$, put $F^{\mathrm{op}}(c):=F(c)$. For each morphism $f: c_1 \rightarrow c_2$ in $\mathcal{C}^{\mathrm{op}}$, we have a corresponding
fong-spivak-seven-sketches- F03358	297	1.000	$f^{\prime}: c_2 \rightarrow c_1$	$f': c_2 \rightarrow c_1$	morphism $f': c_2 \rightarrow c_1$ in $\mathcal{C}$ and thus a morphism $F\left(f^{\prime}\right): F\left(c_2\right) \rightarrow F\left(c_1\right)$ in $\mathcal{D}$, and thus a morphism
fong-spivak-seven-sketches- F03359	297	1.000	$F\left(f^{\prime}\right): F\left(c_2\right) \rightarrow F\left(c_1\right)$	$F\left(f^{\prime}\right): F\left(c_2\right) \rightarrow F\left(c_1\right)$	morphism $f': c_2 \rightarrow c_1$ in $\mathcal{C}$ and thus a morphism $F\left(f^{\prime}\right): F\left(c_2\right) \rightarrow F\left(c_1\right)$ in $\mathcal{D}$, and thus a morphism
fong-spivak-seven-sketches- F03360	297	0.951	$F\left(f^{\prime}\right)^{\prime}: F^{\mathrm{op}}\left(c_1\right) \rightarrow F^{\mathrm{op}}\left(c_2\right)$	$F\left(f^{\prime}\right)^{\prime}: F^{\mathrm{op}}\left(c_1\right) \rightarrow F^{\mathrm{op}}\left(c_2\right)$	$F\left(f^{\prime}\right)^{\prime}: F^{\mathrm{op}}\left(c_1\right) \rightarrow F^{\mathrm{op}}\left(c_2\right)$. Hence we can define $F^{\mathrm{op}}(f):=F\left(f^{\prime}\right)^{\prime}$. Note that the primes $(-)^{\prime}$ are pretty
fong-spivak-seven-sketches- F03361	297	0.951	$F^{\mathrm{op}}(f):=F\left(f^{\prime}\right)^{\prime}$	$F^{\mathrm{op}}(f):=F\left(f^{\prime}\right)^{\prime}$	$F\left(f^{\prime}\right)^{\prime}: F^{\mathrm{op}}\left(c_1\right) \rightarrow F^{\mathrm{op}}\left(c_2\right)$. Hence we can define $F^{\mathrm{op}}(f):=F\left(f^{\prime}\right)^{\prime}$. Note that the primes $(-)^{\prime}$ are pretty

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_2022_F03362	297	0.951	$\left(-\{\}\right)^{\prime}$	$(-')$	$F(f')': F^{\mathrm{op}}(c_1) \rightarrow F^{\mathrm{op}}(c_2)$. Hence we can define $F^{\mathrm{op}}(f) := F(f')'$. Note that the primes $(-')$ are pretty
fong-spivak-seven-sketches_2022_F03363	297	0.815	F^{op}	F^{op}	It is easy to check that our definition of F^{op} is functorial: it sends identities to identities and composites
fong-spivak-seven-sketches_2022_F03364	298	0.666	$\mathcal{X}^{\mathrm{op}} \times \dagger$	$\mathcal{X}^{\mathrm{op}} \times \dagger$	1. The Hasse diagram for $\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}$ is shown here (ignore the colors):
fong-spivak-seven-sketches_2022_F03365	298	0.299	$\Lambda: \mathcal{X} \mapsto \mathcal{Y}$	$\Lambda: \mathcal{X} \mapsto \mathcal{Y}$	2. There is a profunctor $\Lambda: \mathcal{X} \nrightarrow \mathcal{Y}$, i.e. a functor $\mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \rightarrow \mathbb{B}$, such that, in the picture above, blue
fong-spivak-seven-sketches_2022_F03366	298	0.299	$\mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \rightarrow \mathbb{B}$	$\mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \rightarrow \mathbb{B}$	2. There is a profunctor $\Lambda: \mathcal{X} \nrightarrow \mathcal{Y}$, i.e. a functor $\mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \rightarrow \mathbb{B}$, such that, in the picture above, blue
fong-spivak-seven-sketches_2022_F03367	298	0.301	$x^{\mathrm{op}} \times y$	$x^{\mathrm{op}} \times y$	The preorder $\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}$ describes tasks in decreasing difficulty. For example, (we hope) it is easier for
fong-spivak-seven-sketches_2022_F03368	298	1.000	Λ	Λ	profunctor Λ describes possible states of knowledge for my aunt: she can describe monoids without
fong-spivak-seven-sketches_2022_F03369	298	0.348	$\Phi: \mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \rightarrow \mathcal{V}$	$\Phi: \mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \rightarrow \mathcal{V}$	Recall from Definition 2.41 that a $\mathcal{V}$ -functor $\Phi: \mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \rightarrow \mathcal{V}$ is a function $\Phi: \mathrm{Ob}(\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}) \rightarrow \mathrm{Ob}(\mathcal{V})$
fong-spivak-seven-sketches_2022_F03370	298	0.348	$\Phi: \mathrm{Ob}(\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}) \rightarrow \mathrm{Ob}(\mathcal{V})$	$\Phi: \mathrm{Ob}(\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}) \rightarrow \mathrm{Ob}(\mathcal{V})$	Recall from Definition 2.41 that a $\mathcal{V}$ -functor $\Phi: \mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \rightarrow \mathcal{V}$ is a function $\Phi: \mathrm{Ob}(\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}) \rightarrow \mathrm{Ob}(\mathcal{V})$
fong-spivak-seven-sketches_2022_F03371	298	0.932	$x^{\mathrm{op}} \times y$	$x^{\mathrm{op}} \times y$	such that for all (x, y) and (x', y') in $\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}$ we have
fong-spivak-seven-sketches_2022_F03372	299	0.997	$\Phi: (T \times E) \rightarrow \$$	$\Phi: (T \times E) \rightarrow \$$	Yes, this is valid: it just means that the profunctor $\Phi: (T \times E) \nrightarrow \$$ does not relate (good-natured,
fong-spivak-seven-sketches_2022_F03373	299	0.895	$\Phi(($	$\Phi(($	funny) to any element of $\$$. More formally, it means that $\Phi(\text{good-natured, funny}, p) = \text{false}$ for
fong-spivak-seven-sketches_2022_F03374	299	0.895	$), p) =$	$), p) =$	funny) to any element of $\$$. More formally, it means that $\Phi(\text{good-natured, funny}, p) = \text{false}$ for
fong-spivak-seven-sketches_2022_F03375	299	1.000	$p \in \$ = \{\$100 \text{ K}, \$500 \text{ K}, \$1 \text{ M}\}$	$p \in \$ = \{\$100 \text{ K}, \$500 \text{ K}, \$1 \text{ M}\}$	all $p \in \$ = \{\$100\text{K}, \$500\text{K}, \$1\text{M}\}$. We can interpret this to mean that it is not feasible to produce a
fong-spivak-seven-sketches_2022_F03376	299	0.712	$(\Phi; \Psi)(-, -) = \mathcal{X}(-, D) + 9 + z(r, -)$	$(\Phi; \Psi)(-, -) = \mathcal{X}(-, D) + 9 + z(r, -)$	go through the bridges from D to y and y to r , so in this case $(\Phi \sharp \Psi)(-, -) = \mathcal{X}(-, D) + 9 + \mathcal{Z}(r, -)$. This
fong-spivak-seven-sketches_2022_F03377	300	0.432	U_X	U_X	We choose the Cost -category $\mathcal{X}$ from Eq. (2.56). The unit profunctor U_X on $\mathcal{X}$ is described by the bridge
fong-spivak-seven-sketches_2022_F03378	300	0.705	$I \leq \mathcal{P}(p, p)$	$I \leq \mathcal{P}(p, p)$	of $\otimes$ (Definition 2.2(a)) applied to the inequalities $I \leq \mathcal{P}(p, p)$ (the identity law for $\mathcal{P}$ at p ,
fong-spivak-seven-sketches_2022_F03379	300	0.993	$\Phi(p, q) \leq \Phi(p, q)$	$\Phi(p, q) \leq \Phi(p, q)$	Definition 2.46(a)) and $\Phi(p, q) \leq \Phi(p, q)$ (reflexivity of preorder $\mathcal{V}$, Definition 1.30(a)). The third
fong-spivak-seven-sketches_2022_F03380	300	1.000	$x \leq x \vee y$	$x \leq x \vee y$	step uses the definition of join: for all x and y , since any $x \leq x$, we have $x \leq x \vee y$. The final
fong-spivak-seven-sketches_2022_F03381	300	0.998	$I =$	$I =$	2. Note that in Bool , $I = \text{true}$. Since the identity law at p says $\text{true} \leq \mathcal{P}(p, p)$, and true is the
fong-spivak-seven-sketches_2022_F03382	300	0.998	$\leq \mathcal{P}(p, p)$	$\leq \mathcal{P}(p, p)$	2. Note that in Bool , $I = \text{true}$. Since the identity law at p says $\text{true} \leq \mathcal{P}(p, p)$, and true is the
fong-spivak-seven-sketches_2022_F03383	300	0.972	$\mathcal{P}(p, p) =$	$\mathcal{P}(p, p) =$	largest element of the preorder Bool , we thus have $\mathcal{P}(p, p) = \text{true}$ for all p . This shows that the
fong-spivak-seven-sketches_2022_F03384	300	1.000	$\Phi(p, q)$	$\Phi(p, q)$	side of the inequality is equal to $\Phi(p, q)$. We split into two cases. Suppose $\Phi(p, q) = \text{true}$. Then,
fong-spivak-seven-sketches_2022_F03385	300	1.000	$\Phi(p, q) =$	$\Phi(p, q) =$	side of the inequality is equal to $\Phi(p, q)$. We split into two cases. Suppose $\Phi(p, q) = \text{true}$. Then,
fong-spivak-seven-sketches_2022_F03386	300	0.825	$\mathcal{P}^{\mathrm{op}} \times \mathcal{Q} \rightarrow$	$\mathcal{P}^{\mathrm{op}} \times \mathcal{Q} \rightarrow$	Next, suppose $\Phi(p, q) = \text{false}$. Note that since Φ is a monotone map $\mathcal{P}^{\mathrm{op}} \times \mathcal{Q} \rightarrow \mathbf{Bool}$, if $p \leq p_1$
fong-spivak-seven-sketches_2022_F03387	300	0.825	$p \leq p_1$	$p \leq p_1$	Next, suppose $\Phi(p, q) = \text{false}$. Note that since Φ is a monotone map $\mathcal{P}^{\mathrm{op}} \times \mathcal{Q} \rightarrow \mathbf{Bool}$, if $p \leq p_1$
fong-spivak-seven-sketches_2022_F03388	300	1.000	$\Phi(p_1, q) \leq \Phi(p, q)$	$\Phi(p_1, q) \leq \Phi(p, q)$	in $\mathcal{P}$, then $\Phi(p_1, q) \leq \Phi(p, q)$ in Bool . Thus if $\mathcal{P}(p, p_1) = \text{true}$ then $\Phi(p_1, q) = \Phi(p, q) = \text{false}$.
fong-spivak-seven-sketches_2022_F03389	300	1.000	$\mathcal{P}(p, p_1) =$	$\mathcal{P}(p, p_1) =$	in $\mathcal{P}$, then $\Phi(p_1, q) \leq \Phi(p, q)$ in Bool . Thus if $\mathcal{P}(p, p_1) = \text{true}$ then $\Phi(p_1, q) = \Phi(p, q) = \text{false}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03390	300	1.000	$\Phi\!\left(p_{\!1},\,q\right)=\Phi(p,\,q)=$	$\Phi\left(p_1,q\right)=\Phi(p,q)=$	in $\mathcal{P}$, then $\Phi(p_1,q)\leq\Phi(p,q)$ in Bool . Thus if $\mathcal{P}(p,p_1)=\text{true}$ then $\Phi(p_1,q)=\Phi(p,q)=\text{false}$.
fong-spivak-seven-sketches F03391	300	1.000	$\Phi\!\left(p_{\!1},\,q\right)=$	$\Phi\left(p_1,q\right)=$	This implies that for all $p_1\in\mathcal{P}$, we have either $\mathcal{P}(p,p_1)=\text{false}$ or $\Phi(p_1,q)=\text{false}$, and hence
fong-spivak-seven-sketches F03392	300	0.772	$\bigvee_{p_{\!1}\in\mathcal{P}}\mathcal{P}\!\left(p,\,p_{\!1}\right)\wedge\Phi\!\left(p_{\!1},\,q\right)=\bigvee_{p_{\!1}\in\mathcal{P}}\mathcal{P}\!\left(p,\,p_{\!1}\right)\wedge\Phi\!\left(p_{\!1},\,q\right)=$	$\bigvee_{p_1\in\mathcal{P}}\mathcal{P}\left(p,p_1\right)\wedge\Phi\left(p_1,q\right)=\bigvee_{p_1\in\mathcal{P}}$	$\bigvee_{p_1\in\mathcal{P}}\mathcal{P}(p,p_1)\wedge\Phi(p_1,q)=\bigvee_{p_1\in\mathcal{P}}\text{false}=\text{false}$.
fong-spivak-seven-sketches F03393	300	0.644	$\Phi(p,\,q)=\bigvee_{p_{\!1}\in\mathcal{P}}\mathcal{P}\!\left(p,\,p_{\!1}\right)\wedge\Phi\!\left(p_{\!1},\,q\right)=$	$\Phi(p,q)=\bigvee_{p_1\in\mathcal{P}}\mathcal{P}\left(p,p_1\right)\wedge\Phi\left(p_1,q\right)$	Thus, in either case, we see that $\Phi(p,q)=\bigvee_{p_1\in\mathcal{P}}\mathcal{P}(p,p_1)\wedge\Phi(p_1,q)$, as required.
fong-spivak-seven-sketches F03394	301	1.000	$v\otimes I=v$	$v\otimes I=v$	3. The first equation is unitality in monoidal categories, $v\otimes I=v$ for any $v\in V$. The second is that
fong-spivak-seven-sketches F03395	301	0.999	$I\leq Q(q,\,q)$	$I\leq Q(q,q)$	$I\leq Q(q,q)$ by unitality of enriched categories, see Definition 2.46, together with monotonicity of
fong-spivak-seven-sketches F03396	301	1.000	$v_{\!1}\leq v_{\!2}$	$v_1\leq v_2$	monoidal product: $v_1\leq v_2$ implies $v\otimes v_1\leq v\otimes v_2$. The third was shown in Exercise 4.9.
fong-spivak-seven-sketches F03397	301	1.000	$v\otimes v_{\!1}\leq v\otimes v_{\!2}$	$v\otimes v_1\leq v\otimes v_2$	monoidal product: $v_1\leq v_2$ implies $v\otimes v_1\leq v\otimes v_2$. The third was shown in Exercise 4.9.
fong-spivak-seven-sketches F03398	301	0.830	$\widehat{\mathrm{id}}:\mathcal{P}\mapsto\mathcal{P}$	$\widehat{\mathrm{id}}:\mathcal{P}\mapsto\mathcal{P}$	This is very straightforward. We wish to check $\widehat{\mathrm{id}}:\mathcal{P}\rightarrow\mathcal{P}$ has the formula $\widehat{\mathrm{id}}(p,q)=\mathcal{P}(p,q)$. By
fong-spivak-seven-sketches F03399	301	0.830	$\widehat{\mathrm{id}}(p,\,q)=\mathcal{P}(p,\,q)$	$\widehat{\mathrm{id}}(p,q)=\mathcal{P}(p,q)$	This is very straightforward. We wish to check $\widehat{\mathrm{id}}:\mathcal{P}\rightarrow\mathcal{P}$ has the formula $\widehat{\mathrm{id}}(p,q)=\mathcal{P}(p,q)$. By
fong-spivak-seven-sketches F03400	301	0.963	$\widehat{\mathrm{id}}(p,\,q):=\mathcal{P}(\mathrm{id}(p),\,q)=\mathcal{P}(p,\,q)$	$\widehat{\mathrm{id}}(p,q):=\mathcal{P}(\mathrm{id}(p),q)=\mathcal{P}(p,q)$	Definition 4.34, $\widehat{\mathrm{id}}(p,q):=\mathcal{P}(\mathrm{id}(p),q)=\mathcal{P}(p,q)$. So they’re the same.
fong-spivak-seven-sketches F03401	301	0.771	$\check{+}:\mathbb{R}\rightarrow\mathbb{R}\times\mathbb{R}\times\mathbb{R}$	$\check{+}:\mathbb{R}\rightarrow\mathbb{R}\times\mathbb{R}\times\mathbb{R}$	The conjoint $\check{+}:\mathbb{R}\rightarrow\mathbb{R}\times\mathbb{R}\times\mathbb{R}$ sends (a,b,c,d) to $\mathbb{R}(a,b+c+d)$, which is true if $a\leq b+c+d$, and
fong-spivak-seven-sketches F03402	301	0.771	$\mathbb{R}(a,\,b+c+d)$	$\mathbb{R}(a,b+c+d)$	The conjoint $\check{+}:\mathbb{R}\rightarrow\mathbb{R}\times\mathbb{R}\times\mathbb{R}$ sends (a,b,c,d) to $\mathbb{R}(a,b+c+d)$, which is true if $a\leq b+c+d$, and
fong-spivak-seven-sketches F03403	301	0.771	$a\leq b+c+d$	$a\leq b+c+d$	The conjoint $\check{+}:\mathbb{R}\rightarrow\mathbb{R}\times\mathbb{R}\times\mathbb{R}$ sends (a,b,c,d) to $\mathbb{R}(a,b+c+d)$, which is true if $a\leq b+c+d$, and
fong-spivak-seven-sketches F03404	301	0.976	$\widehat{F}(p,\,q)=Q(F(p),\,q)$	$\widehat{F}(p,q)=Q(F(p),q)$	1. By Definition 4.34, $\widehat{F}(p,q)=Q(F(p),q)$ and $\check{G}(p,q)=Q(p,G(q))$. Since $\mathcal{V}$ is skeletal, F and G are
fong-spivak-seven-sketches F03405	301	0.976	$\check{G}(p,\,q)=Q(p,\,G(q))$	$\check{G}(p,q)=Q(p,G(q))$	1. By Definition 4.34, $\widehat{F}(p,q)=Q(F(p),q)$ and $\check{G}(p,q)=Q(p,G(q))$. Since $\mathcal{V}$ is skeletal, F and G are
fong-spivak-seven-sketches F03406	301	0.992	$Q(F(p),\,q)=Q(p,\,G(q))$	$Q(F(p),q)=Q(p,G(q))$	$\mathcal{V}$ adjoints if and only if $Q(F(p),q)=Q(p,G(q))$. Thus F and G are adjoints if and only if $\widehat{F}=\check{G}$.
fong-spivak-seven-sketches F03407	301	0.992	$\widehat{F}=\check{G}$	$\widehat{F}=\check{G}$	$\mathcal{V}$ adjoints if and only if $Q(F(p),q)=Q(p,G(q))$. Thus F and G are adjoints if and only if $\widehat{F}=\check{G}$.
fong-spivak-seven-sketches F03408	301	0.898	$\widehat{\mathrm{id}}=\widehat{\mathrm{id}}$	$\widehat{\mathrm{id}}=\widehat{\mathrm{id}}$	$\widehat{\mathrm{id}}=\widehat{\mathrm{id}}$.
fong-spivak-seven-sketches F03409	301	1.000	$\otimes:\mathcal{P}\times\mathcal{P}\rightarrow\mathcal{P}$	$\otimes:\mathcal{P}\times\mathcal{P}\rightarrow\mathcal{P}$	for merely a function $\otimes:\mathcal{P}\times\mathcal{P}\rightarrow\mathcal{P}$, while Definition 4.45 calls for a <i>functor</i> .
fong-spivak-seven-sketches F03410	302	1.000	$\left(x_{\!1},\,x_{\!2}\right)\leq\left(y_{\!1},\,y_{\!2}\right)$	$(x_1,x_2)\leq(y_1,y_2)$	says that if $(x_1,x_2)\leq(y_1,y_2)$ in $\mathcal{P}\otimes\mathcal{P}$, then we must have $x_1\otimes x_2\leq y_1\otimes y_2$ in $\mathcal{P}$. So it is also the case
fong-spivak-seven-sketches F03411	302	1.000	$\mathcal{P}\otimes\mathcal{P}$	$\mathcal{P}\otimes\mathcal{P}$	says that if $(x_1,x_2)\leq(y_1,y_2)$ in $\mathcal{P}\otimes\mathcal{P}$, then we must have $x_1\otimes x_2\leq y_1\otimes y_2$ in $\mathcal{P}$. So it is also the case
fong-spivak-seven-sketches F03412	302	1.000	$g_{\!E}(5,3)=$	$g_E(5,3)=$	1. $g_E(5,3)=\text{false}$, $g_F(5,3)=2$.
fong-spivak-seven-sketches F03413	302	1.000	$g_{\!F}(5,3)=2$	$g_F(5,3)=2$	1. $g_E(5,3)=\text{false}$, $g_F(5,3)=2$.
fong-spivak-seven-sketches F03414	302	1.000	$g_{\!E}(3,5)=$	$g_E(3,5)=$	2. $g_E(3,5)=\text{true}$, $g_F(3,5)=-2$.
fong-spivak-seven-sketches F03415	302	1.000	$g_{\!F}(3,5)=-2$	$g_F(3,5)=-2$	2. $g_E(3,5)=\text{true}$, $g_F(3,5)=-2$.
fong-spivak-seven-sketches F03416	302	0.988	$)=5$	$)=5$	3. $h(5,\text{true})=5$.
fong-spivak-seven-sketches F03417	302	0.990	$)=-5$	$)=-5$	4. $h(-5,\text{true})=-5$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03418	302	■ 0.999	)=6	) = 6	5. $h(-5, \text{false}) = 6$.
fong-spivak-seven-sketches F03419	302	■ 1.000	$q_G(-2, 3) = 2, q_F(-2, 3) = -13$	$q_G(-2, 3) = 2, q_F(-2, 3) = -13$	6. $q_G(-2, 3) = 2, q_F(-2, 3) = -13$.
fong-spivak-seven-sketches F03420	302	■ 1.000	$q_G(2, 3) = -1, q_F(2, 3) = 7$	$q_G(2, 3) = -1, q_F(2, 3) = 7$	7. $q_G(2, 3) = -1, q_F(2, 3) = 7$
fong-spivak-seven-sketches F03421	302	■ 0.963	$\mathcal{C}(x, y)$	$\mathcal{C}(x, y)$	a collection of objects and (ii) asks for, for all pairs of objects x, y , a set $\mathcal{C}(x, y)$ of morphisms. Moreover,
fong-spivak-seven-sketches F03422	302	■ 0.430	$\mathrm{id}_x: I \rightarrow \mathcal{C}(x, x)$	$\mathrm{id}_x: I \rightarrow \mathcal{C}(x, x)$	recall that the monoidal unit I is the one element set $\{1\}$. This means a morphism $\mathrm{id}_x: I \rightarrow \mathcal{C}(x, x)$ is a
fong-spivak-seven-sketches F03423	302	■ 0.650	$\mathrm{id}_x: \{1\} \rightarrow \mathcal{C}(x, x)$	$\mathrm{id}_x: \{1\} \rightarrow \mathcal{C}(x, x)$	function $\mathrm{id}_x: \{1\} \rightarrow \mathcal{C}(x, x)$. This is the same data as simply an element $\mathrm{id}_x = \mathrm{id}_x(1) \in \mathcal{C}(x, x)$; we call
fong-spivak-seven-sketches F03424	302	■ 0.650	$\mathrm{id}_x = \mathrm{id}_x(1) \in \mathcal{C}(x, x)$	$\mathrm{id}_x = \mathrm{id}_x(1) \in \mathcal{C}(x, x)$	function $\mathrm{id}_x: \{1\} \rightarrow \mathcal{C}(x, x)$. This is the same data as simply an element $\mathrm{id}_x = \mathrm{id}_x(1) \in \mathcal{C}(x, x)$; we call
fong-spivak-seven-sketches F03425	302	■ 0.975	$\{ \}_9: \mathcal{C}(x, y) \otimes \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$	$g: \mathcal{C}(x, y) \otimes \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$	this data the identity morphism on x . Finally, a morphism $g: \mathcal{C}(x, y) \otimes \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$ is a function
fong-spivak-seven-sketches F03426	302	■ 0.300	$\mathcal{C}(x, y) \times \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$	$\mathcal{C}(x, y) \times \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$	$\mathcal{C}(x, y) \times \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$; this is exactly the composite required in Definition 3.6 (iv).
fong-spivak-seven-sketches F03427	302	■ 0.701	$I \rightarrow \mathcal{X}(x, x)$	$I \rightarrow \mathcal{X}(x, x)$	An identity element in a Cost -category $\mathcal{X}$ is a morphism $I \rightarrow \mathcal{X}(x, x)$ in Cost = $([0, \infty], \geq, 0, +)$, and
fong-spivak-seven-sketches F03428	302	■ 0.937	$0 \leq \mathcal{X}(x, x)$	$0 \geq \mathcal{X}(x, x)$	hence the condition that $0 \geq \mathcal{X}(x, x)$. This implies that $\mathcal{X}(x, x) = 0$. In terms of distances, we interpret
fong-spivak-seven-sketches F03429	302	■ 0.937	$\mathcal{X}(x, x) = 0$	$\mathcal{X}(x, x) = 0$	hence the condition that $0 \geq \mathcal{X}(x, x)$. This implies that $\mathcal{X}(x, x) = 0$. In terms of distances, we interpret
fong-spivak-seven-sketches F03431	303	■ 0.891	$x \in \mathcal{X}$	$x \in \mathcal{X}$	$x \in \mathcal{X}$ and $y \in \mathcal{Y}$, with $(x, y) \leq (x', y')$ iff $x \leq x'$ and $y \leq y'$. That is, if x is available given x' and y is
fong-spivak-seven-sketches F03432	303	■ 0.891	$y \in \mathcal{Y}$	$y \in \mathcal{Y}$	$x \in \mathcal{X}$ and $y \in \mathcal{Y}$, with $(x, y) \leq (x', y')$ iff $x \leq x'$ and $y \leq y'$. That is, if x is available given x' and y is
fong-spivak-seven-sketches F03433	303	■ 0.891	$(x, y) \leq (x', y')$	$(x, y) \leq (x', y')$	$x \in \mathcal{X}$ and $y \in \mathcal{Y}$, with $(x, y) \leq (x', y')$ iff $x \leq x'$ and $y \leq y'$. That is, if x is available given x' and y is
fong-spivak-seven-sketches F03434	303	■ 0.891	$y \leq y'$	$y \leq y'$	$x \in \mathcal{X}$ and $y \in \mathcal{Y}$, with $(x, y) \leq (x', y')$ iff $x \leq x'$ and $y \leq y'$. That is, if x is available given x' and y is
fong-spivak-seven-sketches F03435	303	■ 0.917	$\Phi: x_1 \rightarrow x_2$	$\Phi: x_1 \rightarrow x_2$	Given two profunctors $\Phi: \mathcal{X}_1 \rightarrow \mathcal{X}_2$ and $\Psi: \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$, the profunctor $\Phi \times \Psi$ represents their conjunction,
fong-spivak-seven-sketches F03436	303	■ 0.998	y_1	y_1	i.e. AND. In other words, if y_1 can be obtained given x_1 AND y_2 can be obtained given x_2 , then (y_1, y_2)
fong-spivak-seven-sketches F03437	303	■ 0.998	x_1	x_1	i.e. AND. In other words, if y_1 can be obtained given x_1 AND y_2 can be obtained given x_2 , then (y_1, y_2)
fong-spivak-seven-sketches F03438	303	■ 0.998	y_2	y_2	i.e. AND. In other words, if y_1 can be obtained given x_1 AND y_2 can be obtained given x_2 , then (y_1, y_2)
fong-spivak-seven-sketches F03439	303	■ 0.998	x_2	x_2	i.e. AND. In other words, if y_1 can be obtained given x_1 AND y_2 can be obtained given x_2 , then (y_1, y_2)
fong-spivak-seven-sketches F03440	303	■ 0.998	(y_1, y_2)	(y_1, y_2)	i.e. AND. In other words, if y_1 can be obtained given x_1 AND y_2 can be obtained given x_2 , then (y_1, y_2)
fong-spivak-seven-sketches F03441	303	■ 1.000	(x_1, x_2)	(x_1, x_2)	can be obtained given (x_1, x_2) .
fong-spivak-seven-sketches F03442	303	■ 0.670	$\mathcal{X} \times \mathbf{1} \rightarrow \mathcal{X}$	$\mathcal{X} \times \mathbf{1} \rightarrow \mathcal{X}$	The profunctor $\mathcal{X} \times \mathbf{1} \rightarrow \mathcal{X}$ defined by the functor $\alpha: (\mathcal{X} \times \mathbf{1})^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V}$ that maps $\alpha((x, 1), y) := \mathcal{X}(x, y)$
fong-spivak-seven-sketches F03443	303	■ 0.670	$\alpha: (\mathcal{X} \times \mathbf{1})^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V}$	$\alpha: (\mathcal{X} \times \mathbf{1})^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V}$	The profunctor $\mathcal{X} \times \mathbf{1} \rightarrow \mathcal{X}$ defined by the functor $\alpha: (\mathcal{X} \times \mathbf{1})^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V}$ that maps $\alpha((x, 1), y) := \mathcal{X}(x, y)$
fong-spivak-seven-sketches F03444	303	■ 0.670	$\alpha((x, 1), y) := \mathcal{X}(x, y)$	$\alpha((x, 1), y) := \mathcal{X}(x, y)$	The profunctor $\mathcal{X} \times \mathbf{1} \rightarrow \mathcal{X}$ defined by the functor $\alpha: (\mathcal{X} \times \mathbf{1})^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V}$ that maps $\alpha((x, 1), y) := \mathcal{X}(x, y)$
fong-spivak-seven-sketches F03445	303	■ 0.948	$\alpha^{-1}: x \mapsto x \times \mathbf{1}$	$\alpha^{-1}: x \mapsto x \times \mathbf{1}$	is an isomorphism. It has inverse $\alpha^{-1}: \mathcal{X} \rightarrow \mathcal{X} \times \mathbf{1}$ defined by $\alpha^{-1}(x, (y, 1)) := \mathcal{X}(x, y)$. To see that
fong-spivak-seven-sketches F03446	303	■ 0.948	$\alpha^{-1}(x, (y, 1)) := x(x, y)$	$\alpha^{-1}(x, (y, 1)) := x(x, y)$	is an isomorphism. It has inverse $\alpha^{-1}: \mathcal{X} \rightarrow \mathcal{X} \times \mathbf{1}$ defined by $\alpha^{-1}(x, (y, 1)) := \mathcal{X}(x, y)$. To see that
fong-spivak-seven-sketches F03447	303	■ 0.943	$\alpha^{-1}; \alpha = U_X$	$\alpha^{-1}; \alpha = U_X$	$\alpha^{-1} \circ \alpha = U_X$, note first that the unit law for $\mathcal{X}$ at z and the definition of join imply

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03448	303	0.552	$\mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$	$\mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$	while composition says $\mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$ and hence
fong-spivak-seven-sketches F03449	303	0.473	$\alpha^{-1} = \mathrm{U}_{X \times 1}$	$\alpha^{-1} = \mathrm{U}_{X \times 1}$	Similarly we can show $\alpha \circ \alpha^{-1} = \mathrm{U}_{X \times 1}$, and hence that α is an isomorphism $X \times 1 \rightarrowtail X$.
fong-spivak-seven-sketches F03450	303	0.556	$\beta((1, x), y) := \mathcal{X}(x, y)$	$\beta((1, x), y) := \mathcal{X}(x, y)$	Moreover, we can similarly show that $\beta((1, x), y) := \mathcal{X}(x, y)$ defines an isomorphism $1 \times X \rightarrowtail X$.
fong-spivak-seven-sketches F03451	303	0.556	$\beta: \mathbf{1} \times \mathcal{X} \mapsto X$	$\beta: \mathbf{1} \times \mathcal{X} \mapsto X$	Moreover, we can similarly show that $\beta((1, x), y) := \mathcal{X}(x, y)$ defines an isomorphism $1 \times X \rightarrowtail X$.
fong-spivak-seven-sketches F03452	304	0.979	α^{-1}	α^{-1}	is itself the identity (ie. the unit profunctor on X), where α and α^{-1} are the isomorphisms defined in
fong-spivak-seven-sketches F03453	304	0.571	$\Phi(x, y)$	$\Phi(x, y)$	Freely using the distributivity of $\otimes$ over $\vee$, the value $\Phi(x, y)$ of this composite at $(x, y) \in X^{\mathrm{op}} \times X$ is
fong-spivak-seven-sketches F03454	304	0.571	$(x, y) \in \mathcal{X}^{\mathrm{op}} \times \mathcal{X}$	$(x, y) \in \mathcal{X}^{\mathrm{op}} \times \mathcal{X}$	Freely using the distributivity of $\otimes$ over $\vee$, the value $\Phi(x, y)$ of this composite at $(x, y) \in \mathcal{X}^{\mathrm{op}} \times \mathcal{X}$ is
fong-spivak-seven-sketches F03455	304	0.857	$\mathrm{U}_{\mathcal{X}}(a, b) = \mathcal{X}(a, b)$	$\mathrm{U}_{\mathcal{X}}(a, b) = \mathcal{X}(a, b)$	with the unit profunctor $\mathrm{U}_{\mathcal{X}}(a, b) = \mathcal{X}(a, b)$ is the identity.
fong-spivak-seven-sketches F03456	305	1.000	$(f; g)(i) = g(f(i))$	$(f; g)(i) = g(f(i))$	given by $(f; g)(i) = g(f(i))$ for all $1 \leq i \leq m$.
fong-spivak-seven-sketches F03457	305	1.000	$1 \leq i \leq m$	$1 \leq i \leq m$	given by $(f; g)(i) = g(f(i))$ for all $1 \leq i \leq m$.
fong-spivak-seven-sketches F03458	305	0.562	$\mathrm{id}_m: m \rightarrowtail m$	$\mathrm{id}_m: m \rightarrowtail m$	4. The identity $\mathrm{id}_m: m \rightarrowtail m$ is given by $\mathrm{id}_m(i) = i$ for all $1 \leq i \leq m$. Here is a picture of id_2 and id_8 :
fong-spivak-seven-sketches F03459	305	0.562	$\mathrm{id}_m(i) = i$	$\mathrm{id}_m(i) = i$	4. The identity $\mathrm{id}_m: m \rightarrowtail m$ is given by $\mathrm{id}_m(i) = i$ for all $1 \leq i \leq m$. Here is a picture of id_2 and id_8 :
fong-spivak-seven-sketches F03460	305	0.562	id_2	id_2	4. The identity $\mathrm{id}_m: m \rightarrowtail m$ is given by $\mathrm{id}_m(i) = i$ for all $1 \leq i \leq m$. Here is a picture of id_2 and id_8 :
fong-spivak-seven-sketches F03461	305	0.562	id_8	id_8	4. The identity $\mathrm{id}_m: m \rightarrowtail m$ is given by $\mathrm{id}_m(i) = i$ for all $1 \leq i \leq m$. Here is a picture of id_2 and id_8 :
fong-spivak-seven-sketches F03462	305	1.000	$\sigma_{3,5}: 8 \rightarrowtail 8$	$\sigma_{3,5}: 8 \rightarrowtail 8$	5. Here is a picture of the symmetry $\sigma_{3,5}: 8 \rightarrowtail 8$:
fong-spivak-seven-sketches F03463	305	1.000	$m_1 \leq n_1$	$m_1 \leq n_1$	denoted $m \leq n$, and with the property that whenever $m_1 \leq n_1$ and $m_2 \leq n_2$ hold then $m_1 + m_2 \leq n_1 + n_2$
fong-spivak-seven-sketches F03464	305	1.000	$m_2 \leq n_2$	$m_2 \leq n_2$	denoted $m \leq n$, and with the property that whenever $m_1 \leq n_1$ and $m_2 \leq n_2$ hold then $m_1 + m_2 \leq n_1 + n_2$
fong-spivak-seven-sketches F03465	305	1.000	$m_1 + m_2 \leq n_1 + n_2$	$m_1 + m_2 \leq n_1 + n_2$	denoted $m \leq n$, and with the property that whenever $m_1 \leq n_1$ and $m_2 \leq n_2$ hold then $m_1 + m_2 \leq n_1 + n_2$
fong-spivak-seven-sketches F03466	305	0.970	$d \in \mathbb{N}$	$d \in \mathbb{N}$	2. Take $\leq$ to be the usual order, $m \leq n$ iff there exists $d \in \mathbb{N}$ with $d + m = n$.
fong-spivak-seven-sketches F03467	305	0.970	$d + m = n$	$d + m = n$	2. Take $\leq$ to be the usual order, $m \leq n$ iff there exists $d \in \mathbb{N}$ with $d + m = n$.
fong-spivak-seven-sketches F03468	305	0.996	$m = n + d$	$m = n + d$	3. Take $\leq$ to be the reverse of the usual order, $m \leq n$ iff there exists $d \in \mathbb{N}$ with $m = n + d$.
fong-spivak-seven-sketches F03469	305	0.999	$q \in \mathbb{N}$	$q \in \mathbb{N}$	5. (Non-example.) Take $\leq$ to be the division order, $m \leq n$ iff there exists $q \in \mathbb{N}$ with $m * q = d$. This
fong-spivak-seven-sketches F03470	305	0.999	$m * q = d$	$m * q = d$	5. (Non-example.) Take $\leq$ to be the division order, $m \leq n$ iff there exists $q \in \mathbb{N}$ with $m * q = d$. This
fong-spivak-seven-sketches F03471	305	0.932	$3 \preceq 3$	$3 \preceq 3$	$3 \preceq 3$ but not $5 \preceq^? 7$.
fong-spivak-seven-sketches F03472	305	0.932	$5 \leq^? 7$	$5 \leq^? 7$	$3 \preceq 3$ but not $5 \preceq^? 7$.
fong-spivak-seven-sketches F03473	305	1.000	$\mathbf{Bij}(m, n) := \{f: \underline{m} \rightarrowtail \underline{n} \mid f$	$\mathbf{Bij}(m, n) := \{f: \underline{m} \rightarrowtail \underline{n} \mid f$	1. $\mathbf{Bij}(m, n) := \{f: \underline{m} \rightarrowtail \underline{n} \mid f$ is a bijection}. Note that $\mathbf{Bij}(m, n) = \emptyset$ if $m \neq n$ and it has $n!$
fong-spivak-seven-sketches F03474	305	1.000	$\mathbf{Bij}(m, n) = \emptyset$	$\mathbf{Bij}(m, n) = \emptyset$	1. $\mathbf{Bij}(m, n) := \{f: \underline{m} \rightarrowtail \underline{n} \mid f$ is a bijection}. Note that $\mathbf{Bij}(m, n) = \emptyset$ if $m \neq n$ and it has $n!$
fong-spivak-seven-sketches F03475	305	1.000	$n \rightarrowtail n$	$n \rightarrowtail n$	2. The identity map $n \rightarrowtail n$ is the bijection $\underline{n} \rightarrowtail \underline{n}$ sending $i \mapsto i$.
fong-spivak-seven-sketches F03476	305	1.000	$\underline{n} \rightarrowtail \underline{n}$	$\underline{n} \rightarrowtail \underline{n}$	2. The identity map $n \rightarrowtail n$ is the bijection $\underline{n} \rightarrowtail \underline{n}$ sending $i \mapsto i$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03477	305	■ 1.000	$i \mapsto i$	$i \mapsto i$	2. The identity map $n \rightarrow n$ is the bijection $\underline{n} \rightarrow \underline{n}$ sending $i \mapsto i$.
fong-spivak-seven-sketches F03478	306	■ 1.000	$m+n \rightarrowtail n+m$	$m + n \rightarrow n + m$	3. The symmetry map $m + n \rightarrow n + m$ is the bijection $\sigma_{m,n}: \underline{m + n} \rightarrow \underline{n + m}$ given by
fong-spivak-seven-sketches F03479	306	■ 1.000	$\sigma_{m,n}: \underline{m+n} \rightarrowtail \underline{n+m}$	$\sigma_{m,n}: \underline{m + n} \rightarrow \underline{n + m}$	3. The symmetry map $m + n \rightarrow n + m$ is the bijection $\sigma_{m,n}: \underline{m + n} \rightarrow \underline{n + m}$ given by
fong-spivak-seven-sketches F03480	306	■ 1.000	$n \rightarrowtail p$	$n \rightarrow p$	4. Composition of bijections $m \rightarrow n$ and $n \rightarrow p$ is just their composition as functions, which
fong-spivak-seven-sketches F03481	306	■ 1.000	$(f+g):(m+n) \rightarrowtail$	$(f + g): (m + n) \rightarrow$	5. Given bijections $f: m \rightarrow m'$ and $g: n \rightarrow n'$, their monoidal product $(f + g): (m + n) \rightarrow$
fong-spivak-seven-sketches F03482	306	■ 1.000	$\left(m^{\prime}+n^{\prime}\right)$	$(m' + n')$	$(m' + n')$ is given by
fong-spivak-seven-sketches F03483	306	■ 0.817	$\operatorname{Corel}(m, n)$	$\operatorname{Corel}(m, n)$	1. Corel (m, n) is the set of equivalence relations on $\underline{m + n}$.
fong-spivak-seven-sketches F03484	306	■ 0.817	$\underline{m+n}$	$\underline{m + n}$	1. Corel (m, n) is the set of equivalence relations on $\underline{m + n}$.
fong-spivak-seven-sketches F03485	306	■ 1.000	$i \simtail j$	$i \sim j$	relation, i.e. where $i \sim j$ iff $i = j$.
fong-spivak-seven-sketches F03486	306	■ 1.000	$i=j$	$i = j$	relation, i.e. where $i \sim j$ iff $i = j$.
fong-spivak-seven-sketches F03487	306	■ 0.981	$m+n+n$	$m + n + n + m$	3. The symmetry map $\sigma_{m,n}$, as an equivalence relation on $\underline{m + n + n + m}$ is “the obvious
fong-spivak-seven-sketches F03488	306	■ 0.999	$ i-j =m+n+n$	$ i - j = m + n + n$	• $ i - j = m + n + n$, or
fong-spivak-seven-sketches F03489	306	■ 0.969	$m+1 \leqtail i \leqtail m+n+n$	$m + 1 \leq i \leq m + n + n$	• $m + 1 \leq i \leq m + n + n$ and $m + 1 \leq j \leq m + n + n$ and $ i - j = n$,
fong-spivak-seven-sketches F03490	306	■ 0.969	$m+1 \leqtail j \leqtail m+n+n$	$m + 1 \leq j \leq m + n + n$	• $m + 1 \leq i \leq m + n + n$ and $m + 1 \leq j \leq m + n + n$ and $ i - j = n$,
fong-spivak-seven-sketches F03491	306	■ 0.969	$ i-j =n$	$ i - j = n$	• $m + 1 \leq i \leq m + n + n$ and $m + 1 \leq j \leq m + n + n$ and $ i - j = n$.
fong-spivak-seven-sketches F03492	306	■ 1.000	$\dot{\sim}$	$\sim$	4. Composition of an equivalence relation $\sim$ on $\underline{m + n}$ and an equivalence relation $\sim$ on $n + p$
fong-spivak-seven-sketches F03493	306	■ 1.000	$n+p$	$n + p$	4. Composition of an equivalence relation $\sim$ on $\underline{m + n}$ and an equivalence relation $\sim$ on $n + p$
fong-spivak-seven-sketches F03494	306	■ 0.624	$\simeqtail$	$\simeq$	is the equivalence relation $\simeq$ on $m + p$ given by $i \simeq k$ iff there exists $j \in \underline{n}$ with $i \sim j$ and $\overline{j \sim k}$.
fong-spivak-seven-sketches F03495	306	■ 0.624	$m+p$	$m + p$	is the equivalence relation $\simeq$ on $m + p$ given by $i \simeq k$ iff there exists $j \in \underline{n}$ with $i \sim j$ and $\overline{j \sim k}$.
fong-spivak-seven-sketches F03496	306	■ 0.624	$i \simeqtail k$	$i \simeq k$	is the equivalence relation $\simeq$ on $m + p$ given by $i \simeq k$ iff there exists $j \in \underline{n}$ with $i \sim j$ and $\overline{j \sim k}$.
fong-spivak-seven-sketches F03497	306	■ 0.624	$\overline{j} \dot{\sim} k$	$\overline{j} \sim k$	is the equivalence relation $\simeq$ on $m + p$ given by $i \simeq k$ iff there exists $j \in \underline{n}$ with $i \sim j$ and $\overline{j \sim k}$.
fong-spivak-seven-sketches F03498	306	■ 0.990	$\overline{m+n}$	$\overline{m + n}$	5. Given equivalence relations $\sim$ on $\underline{m + n}$ and $\sim'$ on $\underline{m' + n'}$, we need an equivalence relation
fong-spivak-seven-sketches F03499	306	■ 0.990	$\sim^{\prime}$	$\sim'$	5. Given equivalence relations $\sim$ on $\underline{m + n}$ and $\sim'$ on $\underline{m' + n'}$, we need an equivalence relation
fong-spivak-seven-sketches F03500	306	■ 0.990	$m^{\prime}+n^{\prime}$	$m' + n'$	5. Given equivalence relations $\sim$ on $\underline{m + n}$ and $\sim'$ on $\underline{m' + n'}$, we need an equivalence relation
fong-spivak-seven-sketches F03501	306	■ 0.931	$\left(\sim+\sim^{\prime}\right)$	$(\sim + \sim')$	$(\sim + \sim')$ on $\underline{m + n + m' + n'}$. We take it to be “the obvious thing,” namely “using $\sim$ on the
fong-spivak-seven-sketches F03502	306	■ 0.931	$m+n+m^{\prime}+n^{\prime}$	$m + n + m' + n'$	$(\sim + \sim')$ on $\underline{m + n + m' + n'}$. We take it to be “the obvious thing,” namely “using $\sim$ on the
fong-spivak-seven-sketches F03503	306	■ 1.000	$i \leqtail m+n$	$i \leq m + n$	• $i \leq m + n$ and $j \leq m + n$ and $i \sim j$, or
fong-spivak-seven-sketches F03504	306	■ 1.000	$j \leqtail m+n$	$j \leq m + n$	• $i \leq m + n$ and $j \leq m + n$ and $i \sim j$, or

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03505	306	■ 1.000	$m+n+1 \leq i$	$m + n + 1 \leq i$	$m + n + 1 \leq i$ and $m + n + 1 \leq j$ and $i \sim' j$.
fong-spivak-seven-sketches F03506	306	■ 1.000	$m+n+1 \leq j$	$m + n + 1 \leq j$	$m + n + 1 \leq i$ and $m + n + 1 \leq j$ and $i \sim' j$.
fong-spivak-seven-sketches F03507	306	■ 1.000	$i \sim^{\prime} j$	$i \sim' j$	$m + n + 1 \leq i$ and $m + n + 1 \leq j$ and $i \sim' j$.
fong-spivak-seven-sketches F03508	306	■ 1.000	$\boldsymbol{\operatorname{Rel}}(m, n)$	$\mathbf{Rel}(m, n)$	1. $\mathbf{Rel}(m, n)$ is the set of relations on the set $\underline{m} \times \underline{n}$, i.e. the set of subsets of $\underline{m} \times \underline{n}$, i.e. its
fong-spivak-seven-sketches F03509	306	■ 1.000	$\underline{m} \times \underline{n}$	$\underline{m} \times \underline{n}$	1. $\mathbf{Rel}(m, n)$ is the set of relations on the set $\underline{m} \times \underline{n}$, i.e. the set of subsets of $\underline{m} \times \underline{n}$, i.e. its
fong-spivak-seven-sketches F03510	306	■ 1.000	$\{(i, j) \in \underline{n} \times \underline{n} \mid i = j\}$	$\{(i, j) \in \underline{n} \times \underline{n} \mid i = j\}$	2. The identity map $n \rightarrow n$ is the subset $\{(i, j) \in \underline{n} \times \underline{n} \mid i = j\}$.
fong-spivak-seven-sketches F03511	306	■ 0.999	$(i, j) \in (\underline{m+n}) \times (\underline{n+m})$	$(i, j) \in (\underline{m+n}) \times (\underline{n+m})$	3. The symmetry map $m+n \rightarrow n+m$ is the subset of pairs $(i, j) \in (\underline{m+n}) \times (\underline{n+m})$ such that
fong-spivak-seven-sketches F03512	306	■ 0.999	$i \leq m$	$i \leq m$	$i \leq m$ and $m+1 \leq j$ and $i+m = j$, or
fong-spivak-seven-sketches F03513	306	■ 0.999	$m+1 \leq j$	$m+1 \leq j$	$i \leq m$ and $m+1 \leq j$ and $i+m = j$, or
fong-spivak-seven-sketches F03514	306	■ 0.999	$i+m=j$	$i+m=j$	$i \leq m$ and $m+1 \leq j$ and $i+m = j$, or
fong-spivak-seven-sketches F03515	306	■ 1.000	$m+1 \leq i$	$m+1 \leq i$	$m+1 \leq i$ and $j \leq m$ and $j+m = i$.
fong-spivak-seven-sketches F03516	306	■ 1.000	$j \leq m$	$j \leq m$	$m+1 \leq i$ and $j \leq m$ and $j+m = i$.
fong-spivak-seven-sketches F03517	306	■ 1.000	$j+m=i$	$j+m=i$	$m+1 \leq i$ and $j \leq m$ and $j+m = i$.
fong-spivak-seven-sketches F03518	306	■ 0.986	$R^{\prime} \subseteq \underline{m}^{\prime} \times \underline{n}^{\prime}$	$R' \subseteq \underline{m}' \times \underline{n}'$	5. Given a relation $R \subseteq \underline{m} \times \underline{n}$ and a relation $R' \subseteq \underline{m}' \times \underline{n}'$, we need a relation $(R + R') \subseteq$
fong-spivak-seven-sketches F03519	306	■ 0.986	$(R + R^{\prime}) \subseteq$	$(R + R') \subseteq$	5. Given a relation $R \subseteq \underline{m} \times \underline{n}$ and a relation $R' \subseteq \underline{m}' \times \underline{n}'$, we need a relation $(R + R') \subseteq$
fong-spivak-seven-sketches F03520	306	■ 0.999	$\underline{m+m^{\prime}} \times \underline{n+n^{\prime}}$	$\underline{m+m'} \times \underline{n+n'}$	$\underline{m+m'} \times \underline{n+n'}$. As stated in the example (footnote), this can be given by a universal
fong-spivak-seven-sketches F03521	306	■ 0.324	$R_{\{1\}} \sqcup R_{\{2\}} \subseteq \left(\underline{m}_{\{1\}} \times \underline{n}_{\{1\}} \right) \sqcup \left(\underline{m}_{\{2\}} \times \underline{n}_{\{2\}} \right) \subseteq \left(\underline{m}_{\{1\}} \sqcup \underline{m}_{\{2\}} \right) \times \left(\underline{n}_{\{1\}} \sqcup \underline{n}_{\{2\}} \right)$	$R_1 \sqcup R_2 \subseteq (\underline{m}_1 \times \underline{n}_1) \sqcup (\underline{m}_2 \times \underline{n}_2) \subseteq (\underline{m}_1 \sqcup \underline{m}_2) \times (\underline{n}_1 \sqcup \underline{n}_2)$	Given by $R_1 \sqcup R_2 \subseteq (\underline{m}_1 \times \underline{n}_1) \sqcup (\underline{m}_2 \times \underline{n}_2) \subseteq (\underline{m}_1 \sqcup \underline{m}_2) \times (\underline{n}_1 \sqcup \underline{n}_2)$.
fong-spivak-seven-sketches F03522	307	■ 1.000	$\leq_P$	$\leq_P$	We have a relation $R \subseteq P \times P$ which generates a preorder $\leq_P$ on P , we have an arbitrary preorder
fong-spivak-seven-sketches F03523	307	■ 1.000	$x \leq_P y$	$x \leq_P y$	i.e. that for every $x \leq_P y$ we have $f(x) \leq_Q f(y)$. By definition of P being the reflexive, transitive
fong-spivak-seven-sketches F03524	307	■ 1.000	$x_{\{0\}}, \ldots, x_{\{n\}}$	$x_0, \dots, x_n$	closure of R , we have $x \leq_P y$ iff there exists $n \in \mathbb{N}$ and $x_0, \dots, x_n$ in P with $x_0 = x$ and $x_n = y$ and
fong-spivak-seven-sketches F03525	307	■ 1.000	$x_{\{0\}}=x$	$x_0 = x$	closure of R , we have $x \leq_P y$ iff there exists $n \in \mathbb{N}$ and $x_0, \dots, x_n$ in P with $x_0 = x$ and $x_n = y$ and
fong-spivak-seven-sketches F03526	307	■ 1.000	$x_{\{n\}}=y$	$x_n = y$	closure of R , we have $x \leq_P y$ iff there exists $n \in \mathbb{N}$ and $x_0, \dots, x_n$ in P with $x_0 = x$ and $x_n = y$ and
fong-spivak-seven-sketches F03527	307	■ 1.000	$R(x_{\{i\}}, x_{\{i+1\}})$	$R(x_i, x_{i+1})$	$R(x_i, x_{i+1})$ for each $0 \leq i \leq n-1$. (The case $n=0$ handles reflexivity.) But then by assumption,
fong-spivak-seven-sketches F03528	307	■ 1.000	$0 \leq i \leq n-1$	$0 \leq i \leq n-1$	$R(x_i, x_{i+1})$ for each $0 \leq i \leq n-1$. (The case $n=0$ handles reflexivity.) But then by assumption,
fong-spivak-seven-sketches F03529	307	■ 1.000	$f(x_{\{i\}}) \leq_Q f(x_{\{i+1\}})$	$f(x_i) \leq_Q f(x_{i+1})$	$R(x_i, x_{i+1})$ implies $f(x_i) \leq_Q f(x_{i+1})$ for each i . By induction on i we show that $f(x_0) \leq_Q f(x_i)$
fong-spivak-seven-sketches F03530	307	■ 1.000	$f(x_{\{0\}}) \leq_Q f(x_{\{i\}})$	$f(x_0) \leq_Q f(x_i)$	$R(x_i, x_{i+1})$ implies $f(x_i) \leq_Q f(x_{i+1})$ for each i . By induction on i we show that $f(x_0) \leq_Q f(x_i)$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03531	307	■ 0.994	$n=1, \ x_{\{0\}}=x$	$n = 1, x_0 = x$	above description is with $n = 1$, $x_0 = x$, and $x_n = y$, which we said implies $x \leq_P y$.) Since f is
fong-spivak-seven-sketches F03532	307	■ 1.000	$R(g(a), \ g(b))$	$R(g(a), g(b))$	1. If $R(g(a), g(b))$ holds for all $a \leq_Q b$ then g is monotone, because $R(x, y)$ implies $x \leq_P y$.
fong-spivak-seven-sketches F03533	307	■ 1.000	$a \ \leq_{\{Q\}} b$	$a \leq_Q b$	1. If $R(g(a), g(b))$ holds for all $a \leq_Q b$ then g is monotone, because $R(x, y)$ implies $x \leq_P y$.
fong-spivak-seven-sketches F03534	307	■ 1.000	$(g(a), \ g(b)) \ \notin R$	$(g(a), g(b)) \notin R$	and $\boxed{g(a), g(b)} \notin R$. Indeed, take $Q := \{1\}$ to be the free preorder on one element, and take
fong-spivak-seven-sketches F03535	307	■ 1.000	$Q:=\{1\}$	$Q := \{1\}$	and $(g(a), g(b)) \notin R$. Indeed, take $Q := \{1\}$ to be the free preorder on one element, and take
fong-spivak-seven-sketches F03536	307	■ 1.000	$P:=\{1\}$	$P := \{1\}$	$P := \{1\}$ with $R = \emptyset$. Then the unique function $g: Q \rightarrow P$ is monotone (because $\leq_P$ is reflexive
fong-spivak-seven-sketches F03537	307	■ 1.000	$R=\varnothing$	$R = \emptyset$	$P := \{1\}$ with $R = \emptyset$. Then the unique function $g: Q \rightarrow P$ is monotone (because $\leq_P$ is reflexive
fong-spivak-seven-sketches F03538	307	■ 1.000	$(g(1), \ g(1)) \ \notin R$	$(g(1), g(1)) \notin R$	even though R is empty), and yet $(g(1), g(1)) \notin R$.
fong-spivak-seven-sketches F03539	307	■ 1.000	$\operatorname{Mor}(\mathcal{C}) \rightarrow \operatorname{Ob}(\mathcal{C})$	$\operatorname{Mor}(\mathcal{C}) \rightarrow \operatorname{Ob}(\mathcal{C})$	1. To give a function $\operatorname{Mor}(\mathcal{C}) \rightarrow \operatorname{Ob}(\mathcal{C})$ means that for every element $\operatorname{Mor}(\mathcal{C})$ we need to give exactly
fong-spivak-seven-sketches F03540	307	■ 0.992	$q \in \operatorname{Mor}(\mathcal{C})$	$q \in \operatorname{Mor}(\mathcal{C})$	one element of $\operatorname{Ob}(\mathcal{C})$. So for dom we take any $q \in \operatorname{Mor}(\mathcal{C})$, view it as a morphism $q: y \rightarrow z$, and
fong-spivak-seven-sketches F03541	307	■ 0.992	$q: y \rightarrow z$	$q: y \rightarrow z$	one element of $\operatorname{Ob}(\mathcal{C})$. So for dom we take any $q \in \operatorname{Mor}(\mathcal{C})$, view it as a morphism $\boxed{y \rightarrow z}$, and
fong-spivak-seven-sketches F03542	307	■ 0.991	$\operatorname{cod}(q) := z$	$\operatorname{cod}(q) := z$	send it to its domain y . Similarly for cod: we put $\operatorname{cod}(q) := z$.
fong-spivak-seven-sketches F03543	307	■ 1.000	$F: \mathcal{G} \rightarrow \mathcal{C}$	$F: \mathcal{G} \rightarrow \mathcal{C}$	2. Suppose first that we are given a functor $F: \mathcal{G} \rightarrow \mathcal{C}$. On objects we have a function $\operatorname{Ob}(\mathcal{G}) \rightarrow \operatorname{Ob}(\mathcal{C})$,
fong-spivak-seven-sketches F03544	307	■ 1.000	$\operatorname{Ob}(\mathcal{G}) \rightarrow \operatorname{Ob}(\mathcal{C})$	$\operatorname{Ob}(\mathcal{G}) \rightarrow \operatorname{Ob}(\mathcal{C})$	2. Suppose first that we are given a functor $F: \mathcal{G} \rightarrow \mathcal{C}$. On objects we have a function $\operatorname{Ob}(\mathcal{G}) \rightarrow \operatorname{Ob}(\mathcal{C})$,
fong-spivak-seven-sketches F03545	307	■ 0.985	$\operatorname{Ob}(\mathcal{G}) = V$	$\operatorname{Ob}(\mathcal{G}) = V$	and this defines f since $\operatorname{Ob}(\mathcal{G}) = V$. On morphisms, first note that the arrows of graph G are ex-
fong-spivak-seven-sketches F03546	307	■ 0.970	$\operatorname{Mor}(\mathcal{G})$	$\operatorname{Mor}(\mathcal{G})$	actly the length=1 paths in G , whereas $\operatorname{Mor}(\mathcal{G})$ is the set of all paths in G , so we have an inclusion
fong-spivak-seven-sketches F03547	307	■ 1.000	$A \subseteq \operatorname{Mor}(\mathcal{G})$	$A \subseteq \operatorname{Mor}(\mathcal{G})$	$A \subseteq \operatorname{Mor}(\mathcal{G})$. The functor F provides a function $\operatorname{Mor}(\mathcal{G}) \rightarrow \operatorname{Mor}(\mathcal{C})$, which we can restrict to A
fong-spivak-seven-sketches F03548	307	■ 1.000	$\operatorname{Mor}(\mathcal{G}) \rightarrow \operatorname{Mor}(\mathcal{C})$	$\operatorname{Mor}(\mathcal{G}) \rightarrow \operatorname{Mor}(\mathcal{C})$	$A \subseteq \operatorname{Mor}(\mathcal{G})$. The functor F provides a function $\operatorname{Mor}(\mathcal{G}) \rightarrow \operatorname{Mor}(\mathcal{C})$, which we can restrict to A
fong-spivak-seven-sketches F03549	307	■ 1.000	$\operatorname{dom}(F(r)) = F(\operatorname{dom}(r))$	$\operatorname{dom}(F(r)) = F(\operatorname{dom}(r))$	to obtain $g: A \rightarrow \operatorname{Mor}(\mathcal{C})$. All functors satisfy $\operatorname{dom}(F(r)) = F(\operatorname{dom}(r))$ and $\operatorname{cod}(F(r)) = F(\operatorname{cod}(r))$
fong-spivak-seven-sketches F03550	307	■ 1.000	$\operatorname{cod}(F(r)) = F(\operatorname{cod}(r))$	$\operatorname{cod}(F(r)) = F(\operatorname{cod}(r))$	to obtain $g: A \rightarrow \operatorname{Mor}(\mathcal{C})$. All functors satisfy $\operatorname{dom}(F(r)) = F(\operatorname{dom}(r))$ and $\operatorname{cod}(F(r)) = F(\operatorname{cod}(r))$
fong-spivak-seven-sketches F03551	307	■ 1.000	$r: w \rightarrow x$	$r: w \rightarrow x$	for any $r: w \rightarrow x$. In particular when $r \in A$ is an arrow we have $\operatorname{dom}(r) = s(r)$ and $\operatorname{cod}(r) = t(r)$.
fong-spivak-seven-sketches F03552	307	■ 1.000	$r \in A$	$r \in A$	for any $r: w \rightarrow x$. In particular when $r \in A$ is an arrow we have $\operatorname{dom}(r) = s(r)$ and $\operatorname{cod}(r) = t(r)$.
fong-spivak-seven-sketches F03553	307	■ 1.000	$\operatorname{dom}(r) = s(r)$	$\operatorname{dom}(r) = s(r)$	for any $r: w \rightarrow x$. In particular when $r \in A$ is an arrow we have $\operatorname{dom}(r) = s(r)$ and $\operatorname{cod}(r) = t(r)$.
fong-spivak-seven-sketches F03554	307	■ 1.000	$\operatorname{cod}(r) = t(r)$	$\operatorname{cod}(r) = t(r)$	for any $r: w \rightarrow x$. In particular when $r \in A$ is an arrow we have $\operatorname{dom}(r) = s(r)$ and $\operatorname{cod}(r) = t(r)$.
fong-spivak-seven-sketches F03555	308	■ 0.561	$f: V \rightarrow \operatorname{Ob}(\mathcal{C})$	$f: V \rightarrow \operatorname{Ob}(\mathcal{C})$	Suppose second that we are given a pair of functions (f, g) where $f: V \rightarrow \operatorname{Ob}(\mathcal{C})$ and $g: A \rightarrow$
fong-spivak-seven-sketches F03556	308	■ 0.561	$g: A \rightarrow$	$g: A \rightarrow$	Suppose second that we are given a pair of functions (f, g) where $f: V \rightarrow \operatorname{Ob}(\mathcal{C})$ and $g: A \rightarrow$
fong-spivak-seven-sketches F03557	308	■ 1.000	$p := (v_{\{0\}}, \ a_{\{1\}}, \ a_{\{2\}}, \ \ldots, \ a_{\{n\}})$	$p := (v_0, a_1, a_2, \dots, a_n)$	objects by f . An arbitrary morphism in $\mathcal{G}$ is a path $\boxed{p := (v_0, a_1, a_2, \dots, a_n)}$ in G , where $v_0 \in V$,
fong-spivak-seven-sketches F03558	308	■ 1.000	$v_{\{0\}} \in V$	$v_0 \in V$	objects by f . An arbitrary morphism in $\mathcal{G}$ is a path $p := (v_0, a_1, a_2, \dots, a_n)$ in G , where $v_0 \in V$,
fong-spivak-seven-sketches F03559	308	■ 1.000	$a_{\{i\}} \in A, \ v_{\{0\}} = s(a_{\{1\}})$	$a_i \in A, v_0 = s(a_1)$	$a_i \in A, v_0 = s(a_1)$, and $t(a_i) = s(a_{i+1})$ for all $1 \leq i \leq n - 1$. Then $g(a_i)$ is a morphism in $\mathcal{C}$ whose

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03560	308	■ 1.000	$1 \leq i \leq n-1$	$1 \leq i \leq n-1$	$a_i \in A$, $v_0 = s(a_1)$, and $t(a_i) = s(a_{i+1})$ for all $1 \leq i \leq n-1$. Then $g(a_i)$ is a morphism in $\mathcal{C}$ whose
fong-spivak-seven-sketches F03561	308	■ 1.000	$g\left(a_{\mathrm{i}}\right)$	$g\left(a_i\right)$	$a_i \in A$, $v_0 = s(a_1)$, and $t(a_i) = s(a_{i+1})$ for all $1 \leq i \leq n-1$. Then $g(a_i)$ is a morphism in $\mathcal{C}$ whose
fong-spivak-seven-sketches F03562	308	■ 1.000	$f\left(v_{\mathrm{0}}\right)$	$f\left(v_0\right)$	domain is $f(v_0)$ and the morphisms $g(a_i)$ and $g(a_{i+1})$ are composable for every $1 \leq i \leq n-1$.
fong-spivak-seven-sketches F03563	308	■ 1.000	$g\left(a_{\mathrm{i}+1}\right)$	$g\left(a_{i+1}\right)$	domain is $f(v_0)$ and the morphisms $g(a_i)$ and $g(a_{i+1})$ are composable for every $1 \leq i \leq n-1$.
fong-spivak-seven-sketches F03564	308	■ 0.605	$F(p) := \operatorname{id}_{f\left(v_{\mathrm{0}}\right)} ; g\left(a_{\mathrm{1}}\right) ; \ldots ; g\left(a_{\mathrm{n}}\right)$	$F(p) := \operatorname{id}_{f(v_0)} ; g(a_1) ; \dots ; g(a_n)$	We then take $F(p) := \operatorname{id}_{f(v_0)} \circ g(a_1) \circ \dots \circ g(a_n)$ to be the composite. It is easy to check that this is
fong-spivak-seven-sketches F03565	308	■ 1.000	$\left(f^{\mathrm{\prime}}, g^{\mathrm{\prime}}\right)$	$\left(f', g'\right)$	them into a functor $F: \mathcal{G} \rightarrow \mathcal{C}$ and then extract the new pair of functions (f', g') , then $f = f'$ and
fong-spivak-seven-sketches F03566	308	■ 1.000	$f=f^{\mathrm{\prime}}$	$f=f'$	them into a functor $F: \mathcal{G} \rightarrow \mathcal{C}$ and then extract the new pair of functions (f', g') , then $f = f'$ and
fong-spivak-seven-sketches F03567	308	■ 0.992	$g=g^{\mathrm{\prime}}$	$g=g'$	$g = g'$. Finally, given a functor $F: \mathcal{G} \rightarrow \mathcal{C}$, we extract the pair of functions (f, g) as above and
fong-spivak-seven-sketches F03568	308	■ 1.000	$F^{\mathrm{\prime}}: \operatorname{\mathcal{G}} \rightarrow \operatorname{\mathcal{C}}$	$F': \mathcal{G} \rightarrow \mathcal{C}$	then turn them into a new functor $F': \mathcal{G} \rightarrow \mathcal{C}$. It is clear that F and F' act the same on objects,
fong-spivak-seven-sketches F03569	308	■ 1.000	$F^{\mathrm{\prime}}$	F'	then turn them into a new functor $F': \mathcal{G} \rightarrow \mathcal{C}$. It is clear that F and F' act the same on objects,
fong-spivak-seven-sketches F03570	308	■ 0.499	$U\left(\operatorname{\mathcal{C}}\right) \in$	$U(\mathcal{C}) \in$	3. $(\operatorname{Mor}(\mathcal{C}), \operatorname{Ob}(\mathcal{C}), \operatorname{dom}, \operatorname{cod})$ is a graph; let's denote it $U(\mathcal{C}) \in \mathbf{Grph}$. We have functors Free : $\mathbf{Grph} \rightleftarrows$
fong-spivak-seven-sketches F03571	308	■ 1.000	$a^{\mathrm{i}} * a^{\mathrm{j}}=a^{\mathrm{i}+\mathrm{j}}$	$a^i * a^j = a^{i+j}$	$a^i * a^j = a^{i+j}$
fong-spivak-seven-sketches F03572	308	■ 1.000	$a^{\mathrm{i}} \mapsto i$	$a^i \mapsto i$	2. This is isomorphic to $\mathbb{N}$, by sending $a^i \mapsto i$.
fong-spivak-seven-sketches F03573	308	■ 0.560	$\operatorname{prop}\operatorname{Free}(G, s, t)$	$\operatorname{prop}\operatorname{Free}(G, s, t)$	We have two props: the prop of port graphs and the free prop Free (G, s, t) where
fong-spivak-seven-sketches F03574	308	■ 0.963	(Γ, ℓ)	(Γ, ℓ)	By Definition 5.25, a morphism $m \rightarrow n$ in Free (G) is a G -labeled port graph, i.e. a pair (Γ, ℓ) , where
fong-spivak-seven-sketches F03575	308	■ 0.997	$\ell(v) \in G$	$\ell(v) \in G$	right-hand ports—an arity—and $\ell(v) \in G$ is supposed to have the correct arity; precisely, $s(\ell(v)) = \operatorname{in}(v)$
fong-spivak-seven-sketches F03576	308	■ 0.997	$s(\ell(v))=\operatorname{in}(v)$	$s(\ell(v)) = \operatorname{in}(v)$	right-hand ports—an arity—and $\ell(v) \in G$ is supposed to have the correct arity; precisely, $s(\ell(v)) = \operatorname{in}(v)$
fong-spivak-seven-sketches F03577	308	■ 1.000	Γ	Γ	graph Γ , as desired.
fong-spivak-seven-sketches F03578	309	■ 0.491	$\left(f+\operatorname{id}_{\mathrm{1}}+\operatorname{id}_{\mathrm{1}}\right) \stackrel{\circ}{\circ} \left(\sigma+\operatorname{id}_{\mathrm{1}}\right) \stackrel{\circ}{\circ} \left(\operatorname{id}_{\mathrm{1}}+h\right) \stackrel{\circ}{\circ} \sigma \stackrel{\circ}{\circ} g$	$(f + \operatorname{id}_1 + \operatorname{id}_1) \stackrel{\circ}{\circ} (\sigma + \operatorname{id}_1) \stackrel{\circ}{\circ} (\operatorname{id}_1 + h) \stackrel{\circ}{\circ} \sigma \stackrel{\circ}{\circ} g$	Here is a picture of $(f + \operatorname{id}_1 + \operatorname{id}_1) \circ (\sigma + \operatorname{id}_1) \circ (\operatorname{id}_1 + h) \circ \sigma \circ g$, in the free prop on generators $G = \{f: 1 \rightarrow$
fong-spivak-seven-sketches F03579	309	■ 0.491	$G=\{f: 1 \rightarrow$	$G = \{f: 1 \rightarrow$	Here is a picture of $(f + \operatorname{id}_1 + \operatorname{id}_1) \circ (\sigma + \operatorname{id}_1) \circ (\operatorname{id}_1 + h) \circ \sigma \circ g$, in the free prop on generators $G = \{f: 1 \rightarrow$
fong-spivak-seven-sketches F03580	309	■ 0.867	$1, g: 2 \rightarrow 2, h: 2 \rightarrow 1\}$:	$1, g: 2 \rightarrow 2, h: 2 \rightarrow 1\}$:	$1, g: 2 \rightarrow 2, h: 2 \rightarrow 1\}$:
fong-spivak-seven-sketches F03581	309	■ 0.996	$\{\{1\},\{2\},\{3\}\}$	$\{\{1\}, \{2\}, \{3\}\}$	example the quotient of $S = \{1, 2, 3\}$ by the trivial equivalence relation is the set $\{\{1\}, \{2\}, \{3\}\}$. It is
fong-spivak-seven-sketches F03582	309	■ 0.992	$(R, 0, +, 1, *)$	$(R, 0, +, 1, *)$	1. If $(R, 0, +, 1, *)$ is a rig, then the multiplicative identity $1 \in \operatorname{Mat}_n(R)$ is the usual n -by- n identity
fong-spivak-seven-sketches F03583	309	■ 1.000	$n=4$	$n = 4$	of R). So for $n=4$ it is:
fong-spivak-seven-sketches F03584	309	■ 1.000	$A, B \in \operatorname{Mat}_{\mathrm{2}}\left(\operatorname{\mathbb{N}}\right)$	$A, B \in \operatorname{Mat}_2(\mathbb{N})$	2. We choose $n = 2$ and hence need to find two elements $A, B \in \operatorname{Mat}_2(\mathbb{N})$ such that $A * B \neq B * A$.
fong-spivak-seven-sketches F03585	309	■ 1.000	$A * B \neq B * A$	$A * B \neq B * A$	2. We choose $n = 2$ and hence need to find two elements $A, B \in \operatorname{Mat}_2(\mathbb{N})$ such that $A * B \neq B * A$.
fong-spivak-seven-sketches F03586	309	■ 0.737	$(A * B)(1, 1)=$	$(A * B)(1, 1) =$	One can calculate from the multiplication formula (recalled in Example 5.40) says $(A * B)(1, 1) =$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03587	309	■ 0.999	$0 * 0 + 1 * 1 = 1$	$0 * 0 + 1 * 1 = 1$	$0 * 0 + 1 * 1 = 1$ and $(B * A)(1, 1) = 0 * 0 + 0 * 0 = 0$, which are not equal.
fong-spivak-seven-sketches F03588	309	■ 0.999	$(B * A)(1, 1) = 0 * 0 + 0 * 0 = 0$	$(B * A)(1, 1) = 0 * 0 + 0 * 0 = 0$	$0 * 0 + 1 * 1 = 1$ and $(B * A)(1, 1) = 0 * 0 + 0 * 0 = 0$, which are not equal.
fong-spivak-seven-sketches F03589	309	■ 0.991	$(16 \ x + 4 \ y, \ x + 4 \ y)$	$(16x + 4y, x + 4y)$	the resulting output signal is $(16x + 4y, x + 4y)$.
fong-spivak-seven-sketches F03590	309	■ 1.000	$A = \left(\begin{array}{lll} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array} \right)$	$A = \left(\begin{array}{lll} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array} \right)$	The monoidal product of $A = \left(\begin{array}{lll} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array} \right)$ and $B = \left(\begin{array}{llll} 2 & 5 & 6 & 1 \end{array} \right)$ is
fong-spivak-seven-sketches F03591	309	■ 1.000	$B = \left(\begin{array}{llll} 2 & 5 & 6 & 1 \end{array} \right)$	$B = \left(\begin{array}{llll} 2 & 5 & 6 & 1 \end{array} \right)$	The monoidal product of $A = \left(\begin{array}{lll} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array} \right)$ and $B = \left(\begin{array}{llll} 2 & 5 & 6 & 1 \end{array} \right)$ is
fong-spivak-seven-sketches F03592	310	■ 1.000	$c_{\{n\}}$	c_n	• For the first layer g_1 , take the monoidal product of m copies of c_n ,
fong-spivak-seven-sketches F03593	310	■ 0.999	$s_{\{a\}}: 1 \rightarrow 1$	$s_a: 1 \rightarrow 1$	where $s_a: 1 \rightarrow 1$ is the signal flow graph generator “scalar multiplication by a .” This layer
fong-spivak-seven-sketches F03594	310	■ 0.993	$g_{\{3\}}: m \times n \rightarrow m \times n$	$g_3: m \times n \rightarrow m \times n$	signal flow graph $g_3: m \times n \rightarrow m \times n$, that is the composite and monoidal product of swap and
fong-spivak-seven-sketches F03595	310	■ 1.000	$(i-1) \ m + j$	$(i-1)m + j$	identity maps, such that the $(i-1)m + j$ th input is sent to the $(j-1)n + i$ th output, for all $1 \leq i \leq n$
fong-spivak-seven-sketches F03596	310	■ 1.000	$(j-1) \ n + i$	$(j-1)n + i$	identity maps, such that the $(i-1)m + j$ th input is sent to the $(j-1)n + i$ th output, for all $1 \leq i \leq n$
fong-spivak-seven-sketches F03597	310	■ 1.000	$1 \leq i \leq n$	$1 \leq i \leq n$	identity maps, such that the $(i-1)m + j$ th input is sent to the $(j-1)n + i$ th output, for all $1 \leq i \leq n$
fong-spivak-seven-sketches F03598	310	■ 1.000	$1 \leq j \leq m$	$1 \leq j \leq m$	and $1 \leq j \leq m$.
fong-spivak-seven-sketches F03599	311	■ 0.999	$a_{\{m\}}$	a_m	where a_m is the signal flow graph that adds m inputs to produce a single output:
fong-spivak-seven-sketches F03600 [conf 0.087]	311	■ 0.087	$g = g_{\{1\}} \ g_{\{2\}} \ g_{\{3\}} \ g_{\{4\}}: m \rightarrow$	$g = g_1 g_2 g_3 g_4: m \rightarrow$	Using Proposition 5.54, it is a straightforward but tedious calculation to show that $g = g_1 \circ g_2 \circ g_3 \circ g_4: m \rightarrow$
fong-spivak-seven-sketches F03601	312	■ 0.729	$\sigma_{M,M}: \mu = \mu$	$\sigma_{M,M}: \mu = \mu$	(c) $\sigma_{M,M} \circ \mu = \mu$.
fong-spivak-seven-sketches F03602	312	■ 1.000	$R^{\{0\}} = \{()\}$	$R^0 = \{()\}$	particular, $R^0 = \{()\}$ consists of a single vector of dimension 0.
fong-spivak-seven-sketches F03603	312	■ 0.977	$\operatorname{prop}(\operatorname{Mat}(R))$	$\operatorname{prop}(\operatorname{Mat}(R))$	1. U preserves the monoidal unit because 0 is the monoidal unit of any prop ($\operatorname{Mat}(R)$ is a prop),
fong-spivak-seven-sketches F03604	312	■ 0.874	$R^{\{0\}}$	R^0	$\{1\}$ is the monoidal unit of Set , and R^0 is canonically isomorphic to $\{1\}$. U also preserves the
fong-spivak-seven-sketches F03605	312	■ 0.874	$\{1\} \ . \ U$	$\{1\}.U$	$\{1\}$ is the monoidal unit of Set , and R^0 is canonically isomorphic to $\{1\}$. U also preserves the
fong-spivak-seven-sketches F03606	312	■ 1.000	$R^{\{m\}} \times R^{\{n\}} \cong R^{\{m+n\}}$	$R^m \times R^n \cong R^{m+n}$	monoidal product because there is a canonical isomorphism $R^m \times R^n \cong R^{m+n}$.
fong-spivak-seven-sketches F03607	312	■ 0.889	(m, μ, η)	(m, μ, η)	2. A monoid object in $\operatorname{Mat}(R)$ is a tuple (m, μ, η) where $m \in \mathbb{N}$, $\mu: m + m \rightarrow m$, and $\eta: 0 \rightarrow m$
fong-spivak-seven-sketches F03608	312	■ 0.889	$m \in \mathbb{N}, \mu: m + m \rightarrow m$	$m \in \mathbb{N}, \mu: m + m \rightarrow m$	2. A monoid object in $\operatorname{Mat}(R)$ is a tuple (m, μ, η) where $m \in \mathbb{N}$, $\mu: m + m \rightarrow m$, and $\eta: 0 \rightarrow m$
fong-spivak-seven-sketches F03609	312	■ 0.889	$\eta: 0 \rightarrow m$	$\eta: 0 \rightarrow m$	2. A monoid object in $\operatorname{Mat}(R)$ is a tuple (m, μ, η) where $m \in \mathbb{N}$, $\mu: m + m \rightarrow m$, and $\eta: 0 \rightarrow m$
fong-spivak-seven-sketches F03610	312	■ 1.000	$\mu(\eta, x) = x = \mu(x, \eta)$	$\mu(\eta, x) = x = \mu(x, \eta)$	satisfy the properties $\mu(\eta, x) = x = \mu(x, \eta)$ and $\mu(x, \mu(y, z)) = \mu(\mu(x, y), z)$. Note that there is
fong-spivak-seven-sketches F03611	312	■ 1.000	$\mu(x, \mu(y, z)) = \mu(\mu(x, y), z)$	$\mu(x, \mu(y, z)) = \mu(\mu(x, y), z)$	satisfy the properties $\mu(\eta, x) = x = \mu(x, \eta)$ and $\mu(x, \mu(y, z)) = \mu(\mu(x, y), z)$. Note that there is
fong-spivak-seven-sketches F03612	312	■ 0.970	$0 \rightarrow m$	$0 \rightarrow m$	only one morphism $0 \rightarrow m$ in $\operatorname{Mat}(R)$ for any m . It is not hard to show that for any $m \in \mathbb{N}$ there
fong-spivak-seven-sketches F03613	312	■ 0.970	$m \in \mathbb{N}$	$m \in \mathbb{N}$	only one morphism $0 \rightarrow m$ in $\operatorname{Mat}(R)$ for any m . It is not hard to show that for any $m \in \mathbb{N}$ there

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03614	312	■ 1.000	$m=2, \ \mu$	$m = 2, \mu$	is only one monoid structure. For example, when $m = 2$, μ must be the following matrix
fong-spivak-seven-sketches F03615	313	■ 0.999	$U(\eta): R^{\{0\}} \rightarrow R^{\{m\}}$	$U(\eta): R^0 \rightarrow R^m$	Anyway, for any monoid (m, μ, η) , the morphism $U(\eta): R^0 \rightarrow R^m$ is given by $U(\eta)(1) :=$
fong-spivak-seven-sketches F03616	313	■ 0.999	$U(\eta)(1) :=$	$U(\eta)(1) :=$	Anyway, for any monoid (m, μ, η) , the morphism $U(\eta): R^0 \rightarrow R^m$ is given by $U(\eta)(1) :=$
fong-spivak-seven-sketches F03617	313	■ 0.985	$(\emptyset, \ \ldots, \ \emptyset)$	$(0, \dots, 0)$	$(0, \dots, 0)$, and the morphism $U(\mu): R^m \times R^m \rightarrow R^m$ is given by
fong-spivak-seven-sketches F03618	313	■ 0.985	$U(\mu): R^{\{m\}} \times R^{\{m\}} \rightarrow R^{\{m\}}$	$U(\mu): R^m \times R^m \rightarrow R^m$	$(0, \dots, 0)$, and the morphism $U(\mu): R^m \times R^m \rightarrow R^m$ is given by
fong-spivak-seven-sketches F03619	313	■ 1.000	$R^{\{m\}}$	R^m	These give R^m the structure of a monoid.
fong-spivak-seven-sketches F03620	313	■ 0.860	$(1, \ \supset -, \ \sim)$	$(1, \supset -, \sim)$	3. The triple $(1, \supset, -)$ corresponds to the additive monoid structure on $\mathbb{R}$, e.g. with $(5, 3) \mapsto 8$.
fong-spivak-seven-sketches F03621	313	■ 0.860	$(5, 3) \mapsto 8$	$(5, 3) \mapsto 8$	3. The triple $(1, \supset, -)$ corresponds to the additive monoid structure on $\mathbb{R}$, e.g. with $(5, 3) \mapsto 8$.
fong-spivak-seven-sketches F03622	313	■ 0.989	$\left\{ (x, y, z) \in R^{\{3\}} \mid x = \right.$	$\{(x, y, z) \in R^3 \mid x =$	1. The behavior $B(\neg\hookleftarrow)$ of the reversed addition icon $\neg\hookleftarrow: 1 \rightarrow 2$ is the relation $\{(x, y, z) \in R^3 \mid x =$
fong-spivak-seven-sketches F03623	313	■ 1.000	$y+z\}$	$y + z\}$	$y + z$.
fong-spivak-seven-sketches F03624	313	■ 0.395	$\left\{ (x, y, z) \in R^{\{3\}} \mid x=y=\right.$	$\{(x, y, z) \in R^3 \mid x = y =$	2. The behavior $B(\neg\hookrightarrow)$ of the reversed copy icon, $\neg\hookrightarrow: 2 \rightarrow 1$ is the relation $\{(x, y, z) \in R^3 \mid x = y =$
fong-spivak-seven-sketches F03625	313	■ 1.000	$z\}$	$z\}$	—
fong-spivak-seven-sketches F03626	313	■ 0.926	$B+C \subseteq R^{\{m+p\}} \times R^{\{n+q\}}$	$B + C \subseteq R^{m+p} \times R^{n+q}$	If $B \subseteq R^m \times R^n$ and $C \subseteq R^p \times R^q$ are morphisms in $\mathbf{Rel}_R$, then take $B + C \subseteq R^{m+p} \times R^{n+q}$ to be the set
fong-spivak-seven-sketches F03627	313	■ 0.624	$\mathrm{B}\left(g\left\{ \right\}^{\circ}\right)\left(h^{\mathrm{op}}\right) ; \mathrm{B}\left(h^{\mathrm{op}}\right) ; \mathrm{B}\left(g\right) ; \mathrm{B}\left(h^{\mathrm{op}}\right) ; \mathrm{B}\left(h\right)$	$B(g^{\circ}(h^{\mathrm{op}})) = B(g); B(h^{\mathrm{op}})$	and by Eq. (5.78), the composite $B(g \circ (h^{\mathrm{op}})) = B(g) \circ B(h^{\mathrm{op}})$ is:
fong-spivak-seven-sketches F03628	313	■ 1.000	$S(h)$	$S(h)$	Since $S(g)$ and $S(h)$ are functions, the above immediately reduces to the desired formula:
fong-spivak-seven-sketches F03629	313	■ 0.829	$g^{\mathrm{op}}: n \rightarrow m$	$g^{\mathrm{op}}: n \rightarrow m$	The behavior of $g^{\mathrm{op}}: n \rightarrow m$ and $h: m \rightarrow p$ are respectively
fong-spivak-seven-sketches F03630	313	■ 0.829	$h: m \rightarrow p$	$h: m \rightarrow p$	The behavior of $g^{\mathrm{op}}: n \rightarrow m$ and $h: m \rightarrow p$ are respectively
fong-spivak-seven-sketches F03631	313	■ 0.882	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right) ; h\right) ; \mathrm{B}\left(h^{\mathrm{op}}\right) ; \mathrm{B}\left(h\right)$	$B((g^{\mathrm{op}}); h) = B(g^{\mathrm{op}}); B(h)$	and by Eq. (5.78), the composite $B((g^{\mathrm{op}}); h) = B(g^{\mathrm{op}}) \circ B(h)$ is:
fong-spivak-seven-sketches F03632	313	■ 0.597	$\{y \in R \mid y=0\}$	$\{y \in R \mid y = 0\}$	1. The behavior of the 0-reverse $\dashv$ is the subset $\{y \in R \mid y = 0\}$, and its n -fold tensor is similarly
fong-spivak-seven-sketches F03633	313	■ 1.000	$\left\{ y \in R^{\{n\}} \mid y=0\right.\}$	$\{y \in R^n \mid y = 0\}$	$\{y \in R^n \mid y = 0\}$. Composing this relation with $S(g) \subseteq R^m \times R^n$ gives $\{x \in R^m \mid S(g) = 0\}$,
fong-spivak-seven-sketches F03634	313	■ 1.000	$S(g) \subseteq R^{\{m\}} \times R^{\{n\}}$	$S(g) \subseteq R^m \times R^n$	$\{y \in R^n \mid y = 0\}$. Composing this relation with $S(g) \subseteq R^m \times R^n$ gives $\{x \in R^m \mid S(g) = 0\}$,
fong-spivak-seven-sketches F03635	313	■ 1.000	$\left\{ x \in R^{\{m\}} \mid S(g)=0\right.\}$	$\{x \in R^m \mid S(g) = 0\}$	$\{y \in R^n \mid y = 0\}$. Composing this relation with $S(g) \subseteq R^m \times R^n$ gives $\{x \in R^m \mid S(g) = 0\}$,
fong-spivak-seven-sketches F03636	313	■ 0.494	$\{x \in R\}$	$\{x \in R\}$	2. The behavior of the discard-inverse $\dashv$ is the subset $\{x \in R\}$, i.e. the largest subset of R , and
fong-spivak-seven-sketches F03637	313	■ 1.000	$R^{\{n\}} \subseteq R^{\{n\}}$	$R^n \subseteq R^n$	similarly its m -fold tensor is $R^n \subseteq R^n$. Composing this relation with $S(g) \subseteq R^m \times R^n$ gives
fong-spivak-seven-sketches F03638	313	■ 0.999	$\left\{ y \in R^{\{n\}} \mid \right.$	$\{y \in R^n \mid$	$\{y \in R^n \mid$ there exists $x \in R^m$ such that $S(g)(x) = y\}$, which is exactly the image of $S(g)$.
fong-spivak-seven-sketches F03639	313	■ 0.999	$\left. S(g)(x)=y\right.\}$	$S(g)(x) = y\}$	$\{y \in R^n \mid$ there exists $x \in R^m$ such that $S(g)(x) = y\}$, which is exactly the image of $S(g)$.
fong-spivak-seven-sketches F03640	314	■ 0.994	$\mathrm{B}(g)=\{(x, y) \mid S(g)(x)=y\}$	$B(g) = \{(x, y) \mid S(g)(x) = y\}$	3. For any $g: m \rightarrow n$, we first claim that the behavior $B(g) = \{(x, y) \mid S(g)(x) = y\}$ is linear, i.e. it
fong-spivak-seven-sketches F03641	314	■ 1.000	$S(g)(x)=y$	$S(g)(x) = y$	so if $S(g)(x) = y$ then $S(g)(rx) = ry$ and $S(g)(x_1 + x_2) = S(g)(x_1) + S(g)(x_2)$. Thus we con-

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03642	314	■ 1.000	$S(g)(r\ x)=r\ y$	$S(g)(rx) = ry$	so if $S(g)(x) = y$ then $S(g)(rx) = ry$ and $S(g)(x_1 + x_2) = S(g)(x_1) + S(g)(x_2)$. Thus we con-
fong-spivak-seven-sketches F03643	314	■ 1.000	$S(g)\left(x_{\{1\}}+x_{\{2\}}\right)=S(g)\left(x_{\{1\}}\right)+S(g)\left(x_{\{2\}}\right)$	$S(g)\left(x_1+x_2\right)=S(g)\left(x_1\right)+S(g)\left(x_2\right)$	so if $S(g)(x) = y$ then $S(g)(rx) = ry$ and $S(g)(x_1 + x_2) = S(g)(x_1) + S(g)(x_2)$. Thus we con-
fong-spivak-seven-sketches F03644	314	■ 1.000	$(x,\ y)\in\mathrm{B}(g)$	$(x,y)\in B(g)$	clude that $(x,y)\in B(g)$ implies $(rx,ry)\in B(g)$, so it's closed under scalar multiplication, and
fong-spivak-seven-sketches F03645	314	■ 1.000	$(r\ x,\ r\ y)\in\mathrm{B}(g)$	$(rx,ry)\in B(g)$	clude that $(x,y)\in B(g)$ implies $(rx,ry)\in B(g)$, so it's closed under scalar multiplication, and
fong-spivak-seven-sketches F03646	314	■ 0.999	$\left(x_{\{1\}},\ y_{\{1\}}\right),\left(x_{\{2\}},\ y_{\{2\}}\right)\in\mathrm{B}(g)$	$(x_1,y_1),(x_2,y_2)\in B(g)$	$(x_1,y_1),(x_2,y_2)\in B(g)$ implies $(x_1+x_2,y_1+y_2)\in B(g)$ so it's closed under addition. Similarly,
fong-spivak-seven-sketches F03647	314	■ 0.999	$\left(x_{\{1\}}+x_{\{2\}},\ y_{\{1\}}+y_{\{2\}}\right)\in\mathrm{B}(g)$	$(x_1+x_2,y_1+y_2)\in B(g)$	$(x_1,y_1),(x_2,y_2)\in B(g)$ implies $(x_1+x_2,y_1+y_2)\in B(g)$ so it's closed under addition. Similarly,
fong-spivak-seven-sketches F03648	314	■ 1.000	$\mathrm{B}\left(g^{\mathrm{op}}\right)$	$B(g^{\mathrm{op}})$	the behavior $B(g^{\mathrm{op}})$ is also linear; the proof is similar.
fong-spivak-seven-sketches F03649	314	■ 1.000	$C\subseteq R^{\{n\}}\times R^{\{p\}}$	$C\subseteq R^n\times R^p$	$B\subseteq R^m\times R^n$ and $C\subseteq R^n\times R^p$ are linear. Take $(x_1,z_1),(x_2,z_2)\in B\circ C$ and take $r\in R$.
fong-spivak-seven-sketches F03650	314	■ 1.000	$\left(x_{\{1\}},\ z_{\{1\}}\right),\left(x_{\{2\}},\ z_{\{2\}}\right)\in B\ ;\ C$	$(x_1,z_1),(x_2,z_2)\in B;C$	$B\subseteq R^m\times R^n$ and $C\subseteq R^n\times R^p$ are linear. Take $(x_1,z_1),(x_2,z_2)\in B\circ C$ and take $r\in R$.
fong-spivak-seven-sketches F03651	314	■ 1.000	$y_{\{1\}},\ y_{\{2\}}\in R^{\{n\}}$	$y_1,y_2\in R^n$	By definition, there exist $y_1,y_2\in R^n$ such that $(x_1,y_1),(x_2,y_2)\in B$ and $(y_1,z_1),(y_2,z_2)\in C$.
fong-spivak-seven-sketches F03652	314	■ 1.000	$\left(x_{\{1\}},\ y_{\{1\}}\right),\left(x_{\{2\}},\ y_{\{2\}}\right)\in B$	$(x_1,y_1),(x_2,y_2)\in B$	By definition, there exist $y_1,y_2\in R^n$ such that $(x_1,y_1),(x_2,y_2)\in B$ and $(y_1,z_1),(y_2,z_2)\in C$.
fong-spivak-seven-sketches F03653	314	■ 1.000	$\left(y_{\{1\}},\ z_{\{1\}}\right),\left(y_{\{2\}},\ z_{\{2\}}\right)\in C$	$(y_1,z_1),(y_2,z_2)\in C$	By definition, there exist $y_1,y_2\in R^n$ such that $(x_1,y_1),(x_2,y_2)\in B$ and $(y_1,z_1),(y_2,z_2)\in C$.
fong-spivak-seven-sketches F03654	314	■ 1.000	$\left(r\ x_{\{1\}},\ r\ y_{\{1\}}\right)\in B$	$(rx_1,ry_1)\in B$	Since B and C are linear, $(rx_1,ry_1)\in B$ and $(ry_1,rz_1)\in C$, and also $(x_1+x_2,y_1+y_2)\in B$ and
fong-spivak-seven-sketches F03655	314	■ 1.000	$\left(r\ y_{\{1\}},\ r\ z_{\{1\}}\right)\in C$	$(ry_1,rz_1)\in C$	Since B and C are linear, $(rx_1,ry_1)\in B$ and $(ry_1,rz_1)\in C$, and also $(x_1+x_2,y_1+y_2)\in B$ and
fong-spivak-seven-sketches F03656	314	■ 1.000	$\left(x_{\{1\}}+x_{\{2\}},\ y_{\{1\}}+y_{\{2\}}\right)\in B$	$(x_1+x_2,y_1+y_2)\in B$	Since B and C are linear, $(rx_1,ry_1)\in B$ and $(ry_1,rz_1)\in C$, and also $x_1+x_2,y_1+y_2\in B$ and
fong-spivak-seven-sketches F03657	314	■ 0.996	$\left(y_{\{1\}}+y_{\{2\}},\ z_{\{1\}}+z_{\{2\}}\right)\in C$	$(y_1+y_2,z_1+z_2)\in C$	$(y_1+y_2,z_1+z_2)\in C$. Hence $(rx_1,rz_1)\in (B\circ C)$ and $(x_1+x_2,z_1+z_2)\in (B\circ C)$, as desired.
fong-spivak-seven-sketches F03658	314	■ 0.996	$\left(r\ x_{\{1\}},\ r\ z_{\{1\}}\right)\in (B\ ;\ C)$	$(rx_1,rz_1)\in (B;C)$	$(y_1+y_2,z_1+z_2)\in C$. Hence $(rx_1,rz_1)\in (B\circ C)$ and $(x_1+x_2,z_1+z_2)\in (B\circ C)$, as desired.
fong-spivak-seven-sketches F03659	314	■ 0.996	$\left(x_{\{1\}}+x_{\{2\}},\ z_{\{1\}}+z_{\{2\}}\right)\in (B\ ;\ C)$	$(x_1+x_2,z_1+z_2)\in (B;C)$	$(y_1+y_2,z_1+z_2)\in C$. Hence $(rx_1,rz_1)\in (B\circ C)$ and $(x_1+x_2,z_1+z_2)\in (B\circ C)$, as desired.
fong-spivak-seven-sketches F03660	314	■ 0.996	$(B\ ;\ C)\subseteq R^{\{m\}}\times R^{\{p\}}$	$(B;C)\subseteq R^m\times R^p$	Suppose that $B\subseteq R^m\times R^n$ and $C\subseteq R^n\times R^p$ are linear. Their composite is the relation $(B\circ C)\subseteq R^m\times R^p$
fong-spivak-seven-sketches F03661	314	■ 0.999	$(x,\ z)$	(x,z)	consisting of all (x,z) for which there exists $y\in R^n$ with $(x,y)\in B$ and $(y,z)\in C$. We want to show
fong-spivak-seven-sketches F03662	314	■ 0.999	$(y,\ z)\in C$	$(y,z)\in C$	consisting of all (x,z) for which there exists $y\in R^n$ with $(x,y)\in B$ and $(y,z)\in C$. We want to show
fong-spivak-seven-sketches F03663	314	■ 1.000	$(B\ ;\ C)$	$(B;C)$	that the set $(B\circ C)$ is linear, i.e. closed under scalar multiplication and addition.
fong-spivak-seven-sketches F03664	314	■ 0.976	$(x,\ z)\in (B\ ;\ C)$	$(x,z)\in (B;C)$	For scalar multiplication, take an $(x,z)\in (B\circ C)$ and any $r\in R$. Since B is linear, we have $(r*x,r*y)\in B$
fong-spivak-seven-sketches F03665	314	■ 0.976	$(r\ * \ x,\ r\ * \ y)\in B$	$(r*x,r*y)\in B$	For scalar multiplication, take an $(x,z)\in (B\circ C)$ and any $r\in R$. Since B is linear, we have $(r*x,r*y)\in B$
fong-spivak-seven-sketches F03666	314	■ 0.997	$(r\ * \ y,\ r\ * \ z)\in C$	$(r*y,r*z)\in C$	and since C is linear we have $(r*y,r*z)\in C$, so then $(r*x,r*z)\in (B\circ C)$. For addition, if we also
fong-spivak-seven-sketches F03667	314	■ 0.997	$(r\ * \ x,\ r\ * \ z)\in (B\ ;\ C)$	$(r*x,r*z)\in (B;C)$	and since C is linear we have $(r*y,r*z)\in C$, so then $(r*x,r*z)\in (B\circ C)$. For addition, if we also
fong-spivak-seven-sketches F03668	314	■ 1.000	$\left(x^{\{\prime\}},\ z^{\{\prime\}}\right)\in (B\ ;\ C)$	$(x',z')\in (B;C)$	have $(x',z')\in (B\circ C)$ then there is some $y'\in R^n$ with $(x',y')\in B$ and $(y',z')\in C$, so since B and C are
fong-spivak-seven-sketches F03669	314	■ 1.000	$y^{\{\prime\}}\in R^{\{n\}}$	$y'\in R^n$	have $(x',z')\in (B\circ C)$ then there is some $y'\in R^n$ with $(x',y')\in B$ and $(y',z')\in C$, so since B and C are

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-F03670	314	1.000	$\left(y^{\{\prime\}}, z^{\{\prime\}}\right) \in C$	$(y', z') \in C$	have $(x', z') \in (B \circ C)$ then there is some $y' \in R^n$ with $(x', y') \in B$ and $(y', z') \in C$, so since B and C are
fong-spivak-seven-sketches-F03671	314	0.998	$\left(x+x^{\{\prime\}}, y+y^{\{\prime\}}\right) \in B$	$(x+x', y+y') \in B$	linear we have $(x+x', y+y') \in B$ and $(y+y', z+z') \in C$, hence $(x+x', z+z') \in (B \circ C)$.
fong-spivak-seven-sketches-F03672	314	0.998	$\left(y+y^{\{\prime\}}, z+z^{\{\prime\}}\right) \in C$	$(y+y', z+z') \in C$	linear we have $(x+x', y+y') \in B$ and $(y+y', z+z') \in C$, hence $(x+x', z+z') \in (B \circ C)$.
fong-spivak-seven-sketches-F03673	314	0.998	$\left(x+x^{\{\prime\}}, z+z^{\{\prime\}}\right) \in (B ; C)$	$(x+x', z+z') \in (B; C)$	linear we have $(x+x', y+y') \in B$ and $(y+y', z+z') \in C$, hence $(x+x', z+z') \in (B \circ C)$.
fong-spivak-seven-sketches-F03674	315	0.361	$\stackrel{\bullet}{a} \bullet \stackrel{\bullet}{b}, \quad \stackrel{a}{\bullet} \rightarrow \stackrel{b}{\bullet}, \quad \stackrel{a}{\bullet} \xrightarrow{\quad} \stackrel{b}{\bullet}$ $\stackrel{\bullet}{a} \bullet \stackrel{\bullet}{b} \rightarrow \stackrel{\bullet}{a} \bullet \stackrel{\bullet}{b}, \quad \stackrel{\bullet}{a} \bullet \stackrel{\bullet}{b} \xrightarrow{\quad} \stackrel{\bullet}{a} \bullet \stackrel{\bullet}{b}$	$\begin{array}{ccc} a & & b \\ \bullet & \bullet & \bullet \\ \bullet & \rightarrow & \bullet \\ \bullet & \xrightarrow{\quad} & \bullet \end{array}$	Let $A = \{a, b\}$, and consider the preorders shown here: $\begin{array}{ c c } \hline a & b \\ \hline \bullet & \bullet \\ \hline \end{array}, \quad \begin{array}{ c c } \hline a & b \\ \hline \bullet & \bullet \rightarrow \bullet \\ \hline \end{array}, \quad \begin{array}{ c c } \hline a & b \\ \hline \bullet & \bullet \xrightarrow{\quad} \bullet \\ \hline \end{array}$.
fong-spivak-seven-sketches-F03675	315	0.998	$a \not\leq b$	$a \not\leq b$	1. The left-most (the discrete preorder on A) has no initial object, because $\overline{a \not\leq b}$ and $b \not\leq a$.
fong-spivak-seven-sketches-F03676	315	0.998	$b \not\leq a$	$b \not\leq a$	1. The left-most (the discrete preorder on A) has no initial object, because $\overline{a \not\leq b}$ and $b \not\leq a$.
fong-spivak-seven-sketches-F03677	315	1.000	$f\left(1_{\{R\}}\right)=1_{\{S\}}$	$f(1_R) = 1_S$	1. The remaining conditions are that $f(1_R) = 1_S$, and that $f(r_1 * r_2) = f(r_1) * f(r_2)$.
fong-spivak-seven-sketches-F03678	315	1.000	$f\left(r_{\{1\}} *_R r_{\{2\}}\right)=f\left(r_{\{1\}}\right) *_S f\left(r_{\{2\}}\right)$	$f(r_1 * r_2) = f(r_1) * f(r_2)$	1. The remaining conditions are that $f(1_R) = 1_S$, and that $f(r_1 * r_2) = f(r_1) * f(r_2)$.
fong-spivak-seven-sketches-F03679	315	0.966	$R=\left(R, 0_{\{R\}},+_R, 1_{\{R\}},\{ \}^{\{*\}} R\right)$	$R = (R, 0_R, +_R, 1_R, *R)$	initial means that for any other rig $R = (R, 0_R, +_R, 1_R, *R)$, there is a unique rig homomorphism
fong-spivak-seven-sketches-F03680	315	1.000	$f: \mathbb{N} \rightarrow R$	$f: \mathbb{N} \rightarrow R$	$f: \mathbb{N} \rightarrow R$.
fong-spivak-seven-sketches-F03681	315	1.000	$0_{\{R\}}, 1$	$0_R, 1$	What is this homomorphism? Well, to be a rig homomorphism, f must send 0 to 0_R , 1 to 1_R .
fong-spivak-seven-sketches-F03682	315	1.000	$1_{\{R\}}$	1_R	What is this homomorphism? Well, to be a rig homomorphism, f must send 0 to 0_R , 1 to 1_R .
fong-spivak-seven-sketches-F03683	315	1.000	$f(m+n)=f(m)+_R f(n)$	$f(m+n) = f(m) +_R f(n)$	Furthermore, we must also have $f(m+n) = f(m) +_R f(n)$, and hence
fong-spivak-seven-sketches-F03684	315	0.610	$f(m * n)=f(m) *_R f(n)$	$f(m * n) = f(m) *_R f(n)$	It remains to check $f(m * n) = f(m) *_R f(n)$, and this will follow from distributivity. Noting that
fong-spivak-seven-sketches-F03685	315	1.000	$f(m * n)$	$f(m * n)$	$f(m * n)$ is equal to the sum of mn copies of 1_R , we have
fong-spivak-seven-sketches-F03686	315	1.000	$m n$	mn	$f(m * n)$ is equal to the sum of mn copies of 1_R , we have
fong-spivak-seven-sketches-F03687	315	1.000	$\varnothing \rightarrow c$	$\emptyset \rightarrow c$	unique map $\emptyset \rightarrow c$ coming from the initial object.
fong-spivak-seven-sketches-F03688	315	1.000	$c_{\{1\}} \rightarrow c$	$c_1 \rightarrow c$	If c_1 is initial then by the universal property, for any c there is a unique morphism $c_1 \rightarrow c$; in particular,
fong-spivak-seven-sketches-F03689	315	1.000	$c_{\{2\}} \rightarrow c_{\{1\}}$	$c_2 \rightarrow c_1$	$c_2 \rightarrow c_1$, call it g . But how do we know that f and g are mutually inverse? Well since c_1 is initial
fong-spivak-seven-sketches-F03690	316	0.705	$c_{\{1\}} \rightarrow c_{\{1\}}$	$c_1 \rightarrow c_1$	there is a unique morphism $c_1 \rightarrow c_1$. But we can think of two: id_{c_1} and $f \circ g$. Thus they must be
fong-spivak-seven-sketches-F03691	316	0.705	$\text{id}_{c_{\{1\}}}$	id_{c_1}	there is a unique morphism $c_1 \rightarrow c_1$. But we can think of two: id_{c_1} and $f \circ g$. Thus they must be
fong-spivak-seven-sketches-F03692	316	0.705	$f ; g$	$f; g$	there is a unique morphism $c_1 \rightarrow c_1$. But we can think of two: id_{c_1} and $f \circ g$. Thus they must be
fong-spivak-seven-sketches-F03693	316	0.735	$f ; g=\text{id}_{c_{\{1\}}}$	$f; g = \text{id}_{c_1}$	equal. Similarly for c_2 , so we have $f \circ g = \text{id}_{c_1}$ and $g \circ f = \text{id}_{c_2}$, which is the definition of f and g being
fong-spivak-seven-sketches-F03694	316	0.735	$g ; f=\text{id}_{c_{\{2\}}}$	$g; f = \text{id}_{c_2}$	equal. Similarly for c_2 , so we have $f \circ g = \text{id}_{c_1}$ and $g \circ f = \text{id}_{c_2}$, which is the definition of f and g being
fong-spivak-seven-sketches-F03695	316	1.000	$p+q$	$p + q$	Translating Definition 6.11 to this case, a coproduct $p+q$ is P is an element of P such that $p \leq p+q$ and
fong-spivak-seven-sketches-F03696	316	1.000	$p \leq p+q$	$p \leq p + q$	Translating Definition 6.11 to this case, a coproduct $p+q$ is P is an element of P such that $p \leq p+q$ and

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03725	317	■ 0.990	$\iota_A: A \rightarrow B+A$	$\iota_A: A \rightarrow B + A$	confusing: there are two maps named ι_A , (i) $\iota_A: A \rightarrow A + B$, and (ii) $\iota_A: A \rightarrow B + A$, and
fong-spivak-seven-sketches F03726	317	■ 0.991	$\left[\iota_A, \iota_B\right]: B+A \rightarrow A+B$	$[\iota_A, \iota_B]: B + A \rightarrow A + B$	similarly for ι_B . In the above we mean the map (ii). It has inverse $[\iota_A, \iota_B]: B + A \rightarrow A + B$,
fong-spivak-seven-sketches F03727	317	■ 0.924	Disc_S	Disc_S	has a pushout. The only morphisms in Disc_S are identities, so in particular $A = B = C$, and the
fong-spivak-seven-sketches F03728	317	■ 0.924	$A=B=C$	$A = B = C$	has a pushout. The only morphisms in Disc_S are identities, so in particular $A = B = C$, and the
fong-spivak-seven-sketches F03729	317	■ 1.000	$s^{\prime}$	s'	than one object, because if s' is another object then there is a morphism $s \rightarrow s'$, but the only
fong-spivak-seven-sketches F03730	317	■ 1.000	$s \rightarrow s^{\prime}$	$s \rightarrow s'$	than one object, because if s' is another object then there is a morphism $s \rightarrow s'$, but the only
fong-spivak-seven-sketches F03731	317	■ 1.000	$s=s^{\prime}$	$s = s'$	morphisms in S are identities so $s = s'$. Hence the set S must consist of exactly one element.
fong-spivak-seven-sketches F03732	318	■ 0.995	$\underline{5} \sqcup \underline{3}$	$\underline{5} \sqcup \underline{3}$	equivalence classes in $\underline{5} \sqcup \underline{3}$ generated by the relation $\{f(a) \sim g(a) \mid a \in \underline{4}\}$. If we denote elements of $\underline{5}$
fong-spivak-seven-sketches F03733	318	■ 0.995	$\{f(a) \sim g(a) \mid a \in \underline{4}\}$	$\{f(a) \sim g(a) \mid a \in \underline{4}\}$	equivalence classes in $\underline{5} \sqcup \underline{3}$ generated by the relation $\{f(a) \sim g(a) \mid a \in \underline{4}\}$. If we denote elements of $\underline{5}$
fong-spivak-seven-sketches F03734	318	■ 0.995	$\underline{5}$	$\underline{5}$	equivalence classes in $\underline{5} \sqcup \underline{3}$ generated by the relation $\{f(a) \sim g(a) \mid a \in \underline{4}\}$. If we denote elements of $\underline{5}$
fong-spivak-seven-sketches F03735	318	■ 1.000	$\{1, \ldots, 5\}$	$\{1, \dots, 5\}$	as $\boxed{1, \dots, 5}$ and those of $\underline{3}$ as $\{1', 2', 3'\}$, we can redraw the functions f, g :
fong-spivak-seven-sketches F03736	318	■ 1.000	$\underline{3}$	$\underline{3}$	as $\{1, \dots, 5\}$ and those of $\underline{3}$ as $\{1', 2', 3'\}$, we can redraw the functions f, g :
fong-spivak-seven-sketches F03737	318	■ 1.000	$\left\{1^{\prime}, 2^{\prime}, 3^{\prime}\right\}$	$\{1', 2', 3'\}$	as $\{1, \dots, 5\}$ and those of $\underline{3}$ as $\boxed{1', 2', 3'}$, we can redraw the functions f, g :
fong-spivak-seven-sketches F03738	318	■ 1.000	$\left\{1, 1^{\prime}, 3\right\},\{2\},\{4\}$	$\{1, 1', 3\}, \{2\}, \{4\}$	The equivalence classes are $\{1, 1', 3\}, \{2\}, \{4\}$, and $\{5, 2', 3'\}$. These four are exactly the four elements
fong-spivak-seven-sketches F03739	318	■ 1.000	$\left\{5, 2^{\prime}, 3^{\prime}\right\}$	$\{5, 2', 3'\}$	The equivalence classes are $\{1, 1', 3\}, \{2\}, \{4\}$, and $\{5, 2', 3'\}$. These four are exactly the four elements
fong-spivak-seven-sketches F03740	318	■ 0.962	$\varnothing \rightarrow X+\Upsilon$	$\emptyset \rightarrow X + \Upsilon$	1. The diagram to the left commutes because $\emptyset$ is initial, and so has a unique map $\emptyset \rightarrow X + Y$. This
fong-spivak-seven-sketches F03741	318	■ 0.947	$f ; \iota_X = g^{\circ} \iota_Y$	$f; \iota_X = g^{\circ} \iota_Y$	implies we must have $f \circ \iota_X = g \circ \iota_Y$.
fong-spivak-seven-sketches F03742	318	■ 0.454	$t: X+\varnothing Y \rightarrow T$	$t: X + \emptyset Y \rightarrow T$	map $t: X +_{\emptyset} Y \rightarrow T$ such that $\iota_X \circ t = x$ and $\iota_Y \circ t = y$. Thus $X +_{\emptyset} Y$ is the coproduct $X + Y$.
fong-spivak-seven-sketches F03743	318	■ 0.454	$\iota_X: t=x$	$\iota_X: t = x$	map $t: X +_{\emptyset} Y \rightarrow T$ such that $\iota_X \circ t = x$ and $\iota_Y \circ t = y$. Thus $X +_{\emptyset} Y$ is the coproduct $X + Y$.
fong-spivak-seven-sketches F03744	318	■ 0.454	$\iota_Y: t=y$	$\iota_Y: t = y$	map $t: X +_{\emptyset} Y \rightarrow T$ such that $\iota_X \circ t = x$ and $\iota_Y \circ t = y$. Thus $X +_{\emptyset} Y$ is the coproduct $X + Y$.
fong-spivak-seven-sketches F03745	319	■ 1.000	A, B, X, Y	A, B, X, Y	That is, we need to show that S , together with the maps from A, B, X, Y , and Z , has the required
fong-spivak-seven-sketches F03746	319	■ 1.000	$S \rightarrow T$	$S \rightarrow T$	We need to show there is a unique map $S \rightarrow T$ making everything commute. Since Q is a pushout of
fong-spivak-seven-sketches F03747	319	■ 1.000	$X \leftarrow A \rightarrow Y$	$X \leftarrow A \rightarrow Y$	$X \leftarrow A \rightarrow Y$, there is a unique map $Q \rightarrow T$ making a commutative triangle with Y , and since R is the
fong-spivak-seven-sketches F03748	319	■ 1.000	$Q \rightarrow T$	$Q \rightarrow T$	$X \leftarrow A \rightarrow Y$, there is a unique map $Q \rightarrow T$ making a commutative triangle with Y , and since R is the
fong-spivak-seven-sketches F03749	319	■ 1.000	$Y \leftarrow B \rightarrow Z$	$Y \leftarrow B \rightarrow Z$	pushout of $Y \leftarrow B \rightarrow Z$, there is a unique map $R \rightarrow T$ making a commutative triangle with Y . This
fong-spivak-seven-sketches F03750	319	■ 1.000	$R \rightarrow T$	$R \rightarrow T$	pushout of $Y \leftarrow B \rightarrow Z$, there is a unique map $R \rightarrow T$ making a commutative triangle with Y . This
fong-spivak-seven-sketches F03751	319	■ 0.964	(Y, Q, R, T)	(Y, Q, R, T)	implies that there is a commuting (Y, Q, R, T) square, and hence a unique map $S \rightarrow T$ from its pushout
fong-spivak-seven-sketches F03752	319	■ 0.998	$X+_N Y$	$X +_N Y$	The formula in Theorem 6.37 says that the pushout $X +_N Y$ is given by the set of equivalence classes of
fong-spivak-seven-sketches F03753	319	■ 1.000	$X \sqcup N \sqcup Y$	$X \sqcup N \sqcup Y$	$X \sqcup N \sqcup Y$ under the equivalence relation generated by $x \sim n$ if $x = f(n)$, and $y \sim n$ if $y = g(n)$, where

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03754	319	1.000	$x \sim n$	$x \sim n$	$X \sqcup N \sqcup Y$ under the equivalence relation generated by $x \sim n$ if $x = f(n)$, and $y \sim n$ if $y = g(n)$, where
fong-spivak-seven-sketches F03755	319	1.000	$x=f(n)$	$x = f(n)$	$X \sqcup N \sqcup Y$ under the equivalence relation generated by $x \sim n$ if $x = f(n)$, and $y \sim n$ if $y = g(n)$, where
fong-spivak-seven-sketches F03756	319	1.000	$y \sim n$	$y \sim n$	$X \sqcup N \sqcup Y$ under the equivalence relation generated by $x \sim n$ if $x = f(n)$, and $y \sim n$ if $y = g(n)$, where
fong-spivak-seven-sketches F03757	319	1.000	$y=g(n)$	$y = g(n)$	$X \sqcup N \sqcup Y$ under the equivalence relation generated by $x \sim n$ if $x = f(n)$, and $y \sim n$ if $y = g(n)$, where
fong-spivak-seven-sketches F03758	319	1.000	$x \in X, y \in Y, n \in N$	$x \in X, y \in Y, n \in N$	$x \in X, y \in Y, n \in N$. Since for every $n \in N$ there exists an $x \in X$ such that $x = f(n)$, this set is equal to
fong-spivak-seven-sketches F03759	319	1.000	$n \in N$	$n \in N$	$x \in X, y \in Y, n \in N$. Since for every $n \in N$ there exists an $x \in X$ such that $x = f(n)$, this set is equal to
fong-spivak-seven-sketches F03760	320	1.000	$B \rightarrowtail D \otimes D$	$B \rightarrow D \otimes D$	morphism $B \rightarrowtail D \otimes D$.
fong-spivak-seven-sketches F03761	320	1.000	$g: X \rightarrowtail Y$	$g: X \rightarrow Y$	$f: X \rightarrow Y$ and $g: X \rightarrow Y$ such that
fong-spivak-seven-sketches F03762	320	1.000	$\iota_1: X \rightarrowtail X+X$	$\iota_1: X \rightarrow X + X$	Write $\iota_1: X \rightarrow X + X$ for the map into the first copy of X in $X + X$, given by the definition of coproduct.
fong-spivak-seven-sketches F03763	320	1.000	$X+X$	$X + X$	Write $\iota_1: X \rightarrow X + X$ for the map into the first copy of X in $X + X$, given by the definition of coproduct.
fong-spivak-seven-sketches F03764	320	0.748	ι_1	ι_1	Then, using the fact that $\iota_1 \circ [\text{id}, \text{id}] = \text{id}$ from Exercise 6.17 1, and the commutativity of the above
fong-spivak-seven-sketches F03765	320	0.748	$[\text{id}, \text{id}] = \text{id}$	$[\text{id}, \text{id}] = \text{id}$	Then, using the fact that $\iota_1 \circ [\text{id}, \text{id}] = \text{id}$ from Exercise 6.17 1, and the commutativity of the above
fong-spivak-seven-sketches F03766	320	0.732	$f=\iota_1; [\text{id}, \text{id}]; f=\iota_1; [\text{id}, \text{id}]; g=g$	$f = \iota_1; [\text{id}, \text{id}]; f = \iota_1; [\text{id}, \text{id}]; g = g$	square, we have $f = \iota_1 \circ [\text{id}, \text{id}] \circ f = \iota_1 \circ [\text{id}, \text{id}] \circ g = g$. This means that $f: X \rightarrow T$ is the unique map
fong-spivak-seven-sketches F03767	320	0.732	$f: X \rightarrowtail T$	$f: X \rightarrow T$	square, we have $f = \iota_1 \circ [\text{id}, \text{id}] \circ f = \iota_1 \circ [\text{id}, \text{id}] \circ g = g$. This means that $f: X \rightarrow T$ is the unique map
fong-spivak-seven-sketches F03768	321	0.384	$\mathbb{C}$	$\mathbb{C}$	They mean that every category $\mathbf{Cospan}_{\mathbb{C}}$ is equal to a category $\mathbf{Cospan}_F$, for some well-chosen F . They
fong-spivak-seven-sketches F03769	321	0.809	$X \leftarrow N \rightarrowtail Y$	$X \leftarrow N \rightarrow Y$	$\mathbb{C}$. Next, observe that a morphism in $\mathbf{Cospan}_F$ is a cospan $X \leftarrow N \rightarrow Y$ in $\mathbb{C}$ together with an element
fong-spivak-seven-sketches F03770	321	0.974	$FN = \{*\}$	$FN = \{*\}$	of $FN = \{*\}$. But FN also has a unique element, $*$! So there's no choice here, and we can consider
fong-spivak-seven-sketches F03771	321	0.974	FN	FN	of $FN = \{*\}$. But FN also has a unique element, $*$! So there's no choice here, and we can consider
fong-spivak-seven-sketches F03772	321	0.758	$\operatorname{Cospan}_{\mathcal{C}} = \operatorname{Cospan}_{\mathcal{C}}$	$\operatorname{Cospan}_F = \operatorname{Cospan}_{\mathcal{C}}$	simply the usual composition of cospans via pushout, so $\mathbf{Cospan}_F = \mathbf{Cospan}_{\mathbb{C}}$.
fong-spivak-seven-sketches F03773	321	0.484	$\operatorname{Cospan}_{\mathcal{C}} \rightarrowtail \operatorname{Cospan}_F$	$\operatorname{Cospan}_{\mathcal{C}} \rightarrow \operatorname{Cospan}_F$	is the identity-on-objects functor $\mathbf{Cospan}_{\mathbb{C}} \rightarrow \mathbf{Cospan}_F$ that simply decorates each cospan with $*$, and
fong-spivak-seven-sketches F03774	321	0.983	$V = \{\text{ul}, \text{ur}, \text{dl}, \text{dr}\}, A =$	$V = \{\text{ul}, \text{ur}, \text{dl}, \text{dr}\}, A =$	We can represent the circuit in Eq. (6.71) by the tuple (V, A, s, t, ℓ) where $V = \{\text{ul}, \text{ur}, \text{dl}, \text{dr}\}$, $A =$
fong-spivak-seven-sketches F03781	321	1.000	$\psi_{\underline{2}, \underline{2}}(b, s)$	$\psi_{\underline{2}, \underline{2}}(b, s)$	The circuit $\psi_{\underline{2}, \underline{2}}(b, s)$ is the disjoint union of the two labelled graphs b and s :
fong-spivak-seven-sketches F03782	321	0.220	$1 \xrightarrow{f} \underline{2} \xleftarrow{g} \underline{1}$	$1 \xrightarrow{f} \underline{2} \xleftarrow{g} \underline{1}$	The cospan is the cospan $\underline{1} \xrightarrow{f} \underline{2} \xleftarrow{g} \underline{1}$, where $f(1) = 1$ and $g(1) = 2$. The decoration is the C-circuit
fong-spivak-seven-sketches F03783	321	0.220	$f(1)=1$	$f(1) = 1$	The cospan is the cospan $\underline{1} \xrightarrow{f} \underline{2} \xleftarrow{g} \underline{1}$, where $f(1) = 1$ and $g(1) = 2$. The decoration is the C-circuit
fong-spivak-seven-sketches F03784	321	0.220	$g(1)=2$	$g(1) = 2$	The cospan is the cospan $\underline{1} \xrightarrow{f} \underline{2} \xleftarrow{g} \underline{1}$, where $f(1) = 1$ and $g(1) = 2$. The decoration is the C-circuit
fong-spivak-seven-sketches F03785	321	0.954	$(\underline{2}, \{a\}, s, t, \ell)$	$(\underline{2}, \{a\}, s, t, \ell)$	$(\underline{2}, \{a\}, s, t, \ell)$, where $s(a) = 1, t(a) = 2$ and $\ell(a) = \text{battery}$.
fong-spivak-seven-sketches F03786	321	0.954	$s(a)=1, t(a)=2$	$s(a) = 1, t(a) = 2$	$(\underline{2}, \{a\}, s, t, \ell)$, where $s(a) = 1, t(a) = 2$ and $\ell(a) = \text{battery}$.
fong-spivak-seven-sketches F03787	321	0.954	$\ell(a)=$	$\ell(a) =$	$(\underline{2}, \{a\}, s, t, \ell)$, where $s(a) = 1, t(a) = 2$ and $\ell(a) = \text{battery}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03788	322	■ 1.000	$C := (V, A, s, t, \ell)$	$C := (V, A, s, t, \ell)$	Recall the circuit $C := (V, A, s, t, \ell)$ from the solution to Exercise 6.79. Then the first decorated cospan
fong-spivak-seven-sketches F03789	322	■ 0.462	$\underline{1} \xrightarrow{\text{f}} V \stackrel{\text{g}}{\leftarrow} \underline{2}, f(1) = \text{ul}, g(1) = \text{ur}$ $\underline{2}, f(1) = \mathrm{ul}, g(1) = \mathrm{ur}$	$\underline{1} \xrightarrow{f} V \xleftarrow{g} \underline{2}, f(1) = \text{ul}, g(1) = \text{ur}$	is given by the cospan $\underline{1} \xrightarrow{f} V \xleftarrow{g} \underline{2}, f(1) = \text{ul}, g(1) = \text{ur}$, and $g(2) = \text{ur}$, decorated by circuit C . The
fong-spivak-seven-sketches F03790	322	■ 0.462	$g(2) =$	$g(2) =$	is given by the cospan $\underline{1} \xrightarrow{f} V \xleftarrow{g} \underline{2}, f(1) = \text{ul}, g(1) = \text{ur}$, and $g(2) = \text{ur}$, decorated by circuit C . The
fong-spivak-seven-sketches F03791	322	■ 0.953	$\underline{2} \xrightarrow{\text{f}^{\prime}} V^{\prime} \stackrel{\text{g}^{\prime}}{\leftarrow} \underline{2}$ $\stackrel{\text{g}^{\prime}}{\leftarrow} \underline{2}$	$\underline{2} \xrightarrow{f'} V' \xleftarrow{g'} \underline{2}$	second decorated cospan is given by the cospan $\underline{2} \xrightarrow{f'} V' \xleftarrow{g'} \underline{2}$ and the circuit $C' := (V', A', s', t', \ell')$,
fong-spivak-seven-sketches F03792	322	■ 0.953	$C^{\prime} := \left(V^{\prime}, A^{\prime}, s^{\prime}, t^{\prime}, \ell^{\prime} \right)$ $C^{\prime} := (V', A', s', t', \ell')$	$C' := (V', A', s', t', \ell')$	second decorated cospan is given by the cospan $\underline{2} \xrightarrow{f'} V' \xleftarrow{g'} \underline{2}$ and the circuit $C' := (V', A', s', t', \ell')$,
fong-spivak-seven-sketches F03793	322	■ 1.000	$V^{\prime} = \{ l, r, d \}, A^{\prime} = \{ r1^{\prime}, r2^{\prime} \}$ $V^{\prime} = \{l, r, d\}, A^{\prime} = \{\mathrm{r1}', \mathrm{r2}'\}$	$V' = \{l, r, d\}, A' = \{\text{r1}', \text{r2}'\}$	where $V' = \{l, r, d\}, A' = \{\text{r1}', \text{r2}'\}$, and the functions are given by the tables
fong-spivak-seven-sketches F03798	322	■ 0.971	$V \stackrel{\text{g}}{\leftarrow} \underline{2} \xrightarrow{\text{f}^{\prime}} V^{\prime}$ $V \stackrel{\text{g}}{\leftarrow} \underline{2} \xrightarrow{f'} V'$	$V \xleftarrow{g} \underline{2} \xrightarrow{f'} V'$	To compose these, we first take the pushout of $V \xleftarrow{g} \underline{2} \xrightarrow{f'} V'$. This gives the a new apex $V'' =$
fong-spivak-seven-sketches F03799	322	■ 0.971	$V^{\prime} \stackrel{\text{g}^{\prime}}{\leftarrow} \underline{2}$ $V^{\prime} \stackrel{\text{g}^{\prime}}{\leftarrow} \underline{2}$	$V'' =$	To compose these, we first take the pushout of $V \xleftarrow{g} \underline{2} \xrightarrow{f'} V'$. This gives the a new apex $V'' =$
fong-spivak-seven-sketches F03800	322	■ 0.823	$\underline{1} \xrightarrow{\text{h}} V^{\prime} \stackrel{\text{k}}{\leftarrow} \underline{2}$ $\stackrel{\text{k}}{\leftarrow} \underline{2}$	$\underline{1} \xrightarrow{h} V'' \xleftarrow{k} \underline{2}$	$\{\text{ul}, \text{dl}, \text{dr}, \text{m}, \text{r}\}$ with five elements, and composite cospan $\underline{1} \xrightarrow{h} V'' \xleftarrow{k} \underline{2}$ given by $h(1) = \text{ul}, k(1) = \text{r}$
fong-spivak-seven-sketches F03801	322	■ 0.823	$h(1) = \mathrm{ul}, k(1) = \mathrm{r}$	$h(1) = \text{ul}, k(1) = \text{r}$	$\{\text{ul}, \text{dl}, \text{dr}, \text{m}, \text{r}\}$ with five elements, and composite cospan $\underline{1} \xrightarrow{h} V'' \xleftarrow{k} \underline{2}$ given by $h(1) = \text{ul}, k(1) = \text{r}$
fong-spivak-seven-sketches F03802	322	■ 0.536	$k(2) = \mathrm{m}$	$k(2) = \text{m}$	and $k(2) = \text{m}$. The new circuit is given by $(V'', A + A', s'', t'', \ell'')$ where the functions are given by
fong-spivak-seven-sketches F03803	322	■ 0.536	$\left(V^{\prime}, A^{\prime}, s^{\prime}, t^{\prime}, \ell^{\prime} \right)$ $(V'', A + A', s'', t'', \ell'')$	$(V'' A + A', s'', t'', \ell'')$	and $k(2) = \text{m}$. The new circuit is given by $(V'', A + A', s'', t'', \ell'')$ where the functions are given by
fong-spivak-seven-sketches F03807	322	■ 1.000	η	η	Composing η and x we have
fong-spivak-seven-sketches F03808	324	■ 0.846	(X, p, q)	(X, p, q)	Suppose first that (A, B, A', B') is a pullback, and take any (X, p, q) as in the following diagram:
fong-spivak-seven-sketches F03809	324	■ 0.444	$q : f^{\prime} ; g^{\prime} = p ; h_3$	$q : f'; g' = p; h_3$	where $q \circ f' \circ g' = p \circ h_3$. Then by the universal property of the (B, C, B', C') pullback, we get a unique
fong-spivak-seven-sketches F03810	324	■ 0.180	$r ; h_2 = g$	$r; h_2 = g$	In other words $r \circ h_2 = g \circ f'$ and $r \circ g = p$. Then by the universal property of the (A, B, A', B') pullback,
fong-spivak-seven-sketches F03811	324	■ 0.180	$r ; g = p$	$r; g = p$	In other words $r \circ h_2 = g \circ f'$ and $r \circ g = p$. Then by the universal property of the (A, B, A', B') pullback,
fong-spivak-seven-sketches F03812	324	■ 0.991	$r^{\prime} : X \rightarrow A$	$r' : X \rightarrow A$	we get a unique dotted arrow $r' : X \rightarrow A$ making the right-hand diagram commute, i.e. $r' \circ f = r$ and
fong-spivak-seven-sketches F03813	324	■ 0.991	$r^{\prime} ; f = r$	$r'; f = r$	we get a unique dotted arrow $r' : X \rightarrow A$ making the right-hand diagram commute, i.e. $r' \circ f = r$ and
fong-spivak-seven-sketches F03814	324	■ 1.000	$r^{\prime} ; h_1 = q$	$r'; h_1 = q$	$r' \circ h_1 = q$. This gives the existence of an r with the required property, $r' \circ f = r$ and $r' \circ f \circ g = r \circ g = p$.
fong-spivak-seven-sketches F03815	324	■ 1.000	$r^{\prime} ; f ; g = r ; g = p$	$r'; f; g = r; g = p$	$r' \circ h_1 = q$. This gives the existence of an r with the required property, $r' \circ f = r$ and $r' \circ f \circ g = r \circ g = p$.
fong-spivak-seven-sketches F03816	324	■ 0.626	$r_{\{0\}}$	r_0	To see uniqueness, suppose given another morphisms r_0 such that $r_0 \circ f \circ g = p$ and $r_0 \circ h_1 = q$:
fong-spivak-seven-sketches F03817	324	■ 0.626	$f ; g = p$	$f; g = p$	To see uniqueness, suppose given another morphisms r_0 such that $r_0 \circ f \circ g = p$ and $r_0 \circ h_1 = q$:
fong-spivak-seven-sketches F03818	324	■ 0.626	$r_{\{0\}} ; h_{\{1\}} = q$	$r_0; h_1 = q$	To see uniqueness, suppose given another morphisms r_0 such that $r_0 \circ f \circ g = p$ and $r_0 \circ h_1 = q$:
fong-spivak-seven-sketches F03819	324	■ 0.741	$r_{\{0\}} : f = r$	$r_0 : f = r$	Then by the uniqueness of r , we must have $r_0 \circ f = r$, and then by the uniqueness of r' , we must have
fong-spivak-seven-sketches F03820	324	■ 0.741	$r^{\prime}$	r'	Then by the uniqueness of r , we must have $r_0 \circ f = r$, and then by the uniqueness of r' , we must have
fong-spivak-seven-sketches F03821	324	■ 1.000	$r_{\{0\}} = r^{\prime}$	$r_0 = r'$	$r_0 = r'$. This proves the first result.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03822	324	■ 0.994	$r ; h_{\{2\}}=q ; f^{\{\backslash\prime\}}$	$r;h_2 = q;f'$	i.e. where $r \circ h_2 = q \circ f'$. Then letting $p := r \circ g$, we have
fong-spivak-seven-sketches F03823	324	■ 0.994	$p:=r ; g$	$p := r;g$	i.e. where $r \circ h_2 = q \circ f'$. Then letting $p := r \circ g$, we have
fong-spivak-seven-sketches F03824	325	■ 0.999	$r^{\{\backslash\prime\}} ; f ; g=p$	$r';f;g = p$	that $r' \circ f \circ g = p$ and $r_0 \circ h_1 = q$, as shown:
fong-spivak-seven-sketches F03825	325	■ 0.422	$r_{\{0\}}:=r^{\{\backslash\prime\}} ; f$	$r_0 := r';f$	But now let $r_0 := r' \circ f$. It satisfies $r_0 \circ g = p$ and $r_0 \circ h_2 = q \circ f'$, and r satisfies the same equations:
fong-spivak-seven-sketches F03826	325	■ 0.422	$r_{\{0\}} ; g=p$	$r_0;g = p$	But now let $r_0 := r' \circ f$. It satisfies $r_0 \circ g = p$ and $r_0 \circ h_2 = q \circ f'$, and r satisfies the same equations:
fong-spivak-seven-sketches F03827	325	■ 0.422	$r_{\{0\}} ; h_{\{2\}}=q ; f^{\{\backslash\prime\}}$	$r_0;h_2 = q;f'$	But now let $r_0 := r' \circ f$. It satisfies $r_0 \circ g = p$ and $r_0 \circ h_2 = q \circ f'$, and r satisfies the same equations:
fong-spivak-seven-sketches F03828	325	■ 0.554	$A, B, A^{\{\backslash\prime\}}, B^{\{\backslash\prime\}}$	A, B, A', B'	follows that r' is a pullback of the (A, B, A', B') square, as desired.
fong-spivak-seven-sketches F03829	325	■ 1.000	$a_{\{1\}}, a_{\{2\}} \in A$	$a_1, a_2 \in A$	A function $f: A \rightarrow B$ is injective iff for all $a_1, a_2 \in A$, if $f(a_1) = f(a_2)$ then $a_1 = a_2$. It is a monomorphism
fong-spivak-seven-sketches F03830	325	■ 1.000	$f\left(a_{\{1\}}\right)=f\left(a_{\{2\}}\right)$	$f\left(a_1\right)=f\left(a_2\right)$	A function $f: A \rightarrow B$ is injective iff for all $a_1, a_2 \in A$, if $f(a_1) = f(a_2)$ then $a_1 = a_2$. It is a monomorphism
fong-spivak-seven-sketches F03831	325	■ 1.000	$a_{\{1\}}=a_{\{2\}}$	$a_1 = a_2$	A function $f: A \rightarrow B$ is injective iff for all $a_1, a_2 \in A$, if $f(a_1) = f(a_2)$ then $a_1 = a_2$. It is a monomorphism
fong-spivak-seven-sketches F03832	325	■ 0.402	$g_{\{1\}}, g_{\{2\}}: X \rightarrowtail A$	$g_1, g_2: X \rightarrow A$	iff for all sets X and functions $g_1, g_2: X \rightarrow A$, if $g_1 \circ f = g_2 \circ f$ then $g_1 = g_2$. Indeed, this comes directly
fong-spivak-seven-sketches F03833	325	■ 0.402	$f=g_{\{2\}}$	$f = g_2$	iff for all sets X and functions $g_1, g_2: X \rightarrow A$, if $g_1 \circ f = g_2 \circ f$ then $g_1 = g_2$. Indeed, this comes directly
fong-spivak-seven-sketches F03834	325	■ 0.402	$g_{\{1\}}=g_{\{2\}}$	$g_1 = g_2$	iff for all sets X and functions $g_1, g_2: X \rightarrow A$, if $g_1 \circ f = g_2 \circ f$ then $g_1 = g_2$. Indeed, this comes directly
fong-spivak-seven-sketches F03835	325	■ 1.000	$X=\{*\}$	$X = \{*\}$	$X = \{*\}$ be a one element set, and let $g_1, g_2: X \rightarrow A$ be given by $g_1(*) := a_1$ and $g_2(*) := a_2$. Then
fong-spivak-seven-sketches F03836	325	■ 1.000	$g_{\{1\}}(*):=a_{\{1\}}$	$g_1(*) := a_1$	$X = \{*\}$ be a one element set, and let $g_1, g_2: X \rightarrow A$ be given by $g_1(*) := a_1$ and $g_2(*) := a_2$. Then
fong-spivak-seven-sketches F03837	325	■ 1.000	$g_{\{2\}}(*):=a_{\{2\}}$	$g_2(*) := a_2$	$X = \{*\}$ be a one element set, and let $g_1, g_2: X \rightarrow A$ be given by $g_1(*) := a_1$ and $g_2(*) := a_2$. Then
fong-spivak-seven-sketches F03838	325	■ 0.626	$=g_{\{2\}}$	$= g_2$	$g_1 \circ f = g_2 \circ f$, so $g_1 = g_2$, so $a_1 = a_2$.
fong-spivak-seven-sketches F03839	325	■ 1.000	$g_{\{1\}}(x)=g_{\{2\}}(x)$	$g_1(x) = g_2(x)$	We will have $g_1 = g_2$ if we can show that $g_1(x) = g_2(x)$ for every $x \in X$. So take any $x \in X$; since
fong-spivak-seven-sketches F03840	325	■ 1.000	$f\left(g_{\{1\}}(x)\right)=f\left(g_{\{2\}}(x)\right)$	$f\left(g_1(x)\right)=f\left(g_2(x)\right)$	$f(g_1(x)) = f(g_2(x))$ and f is injective, we have $g_1(x) = g_2(x)$ as desired.
fong-spivak-seven-sketches F03841	325	■ 0.974	$j:=i^{\{-1\}}$	$j := i^{-1}$	Let $j := i^{-1}$ be the inverse of i , and consider $g := (f \circ j): A \rightarrow B'$. Then $g \circ i = f$, so by the
fong-spivak-seven-sketches F03842	325	■ 0.974	$g:=(f: j): A \rightarrowtail B^{\{\backslash\prime\}}$	$g := (f : j): A \rightarrow B'$	Let $j := i^{-1}$ be the inverse of i , and consider $g := (f \circ j): A \rightarrow B'$. Then $g \circ i = f$, so by the
fong-spivak-seven-sketches F03843	325	■ 0.974	$g \circ i=f$	$g \circ i = f$	Let $j := i^{-1}$ be the inverse of i , and consider $g := (f \circ j): A \rightarrow B'$. Then $g \circ i = f$, so by the
fong-spivak-seven-sketches F03844	325	■ 0.553	$j^{\{\backslash\prime\}}: A \rightarrowtail A^{\{\backslash\prime\}}$	$j': A \rightarrow A'$	existence part of the universal property, there is a map $j': A \rightarrow A'$ such that $j' \circ i' = \text{id}_A$ and
fong-spivak-seven-sketches F03845	325	■ 0.553	$j^{\{\backslash\prime\}} ; i^{\{\backslash\prime\}}=\operatorname{id}_{\{A\}}$	$j';i' = \text{id}_A$	existence part of the universal property, there is a map $j': A \rightarrow A'$ such that $j' \circ i' = \text{id}_A$ and
fong-spivak-seven-sketches F03846	325	■ 0.392	$j^{\{\backslash\prime\}} ; f^{\{\backslash\prime\}}=f ; j$	$j';f' = f;j$	$j' \circ f' = f \circ j$. We will be done if we can show $i' \circ j' = \text{id}_{A'}$. One checks that $(i' \circ j') \circ i' = i'$ and
fong-spivak-seven-sketches F03847	325	■ 0.392	$i^{\{\backslash\prime\}} ; j^{\{\backslash\prime\}}=\operatorname{id}_{\{A^{\{\backslash\prime\}}\}}$	$i';j' = \text{id}_{A'}$	$j' \circ f' = f \circ j$. We will be done if we can show $i' \circ j' = \text{id}_{A'}$. One checks that $(i' \circ j') \circ i' = i'$ and
fong-spivak-seven-sketches F03848	325	■ 0.392	$\left(i^{\{\backslash\prime\}} ; j^{\{\backslash\prime\}}\right) ; i^{\{\backslash\prime\}}=i^{\{\backslash\prime\}}$	$(i';j');i' = i'$	$j' \circ f' = f \circ j$. We will be done if we can show $i' \circ j' = \text{id}_{A'}$. One checks that $(i' \circ j') \circ i' = i'$ and
fong-spivak-seven-sketches F03849 [conf 0.024]	325	■ 0.024	$\left(i^{\{\backslash\prime\}}\{ \}_{\{9\}} j^{\{\backslash\prime\}}\right), f^{\{\backslash\prime\}}=i^{\{\backslash\prime\}} ; f ; j=f^{\{\backslash\prime\}}: i, j=f^{\{\backslash\prime\}}$	$(i'_9j'), f' = i';f;j = f': i, j = f'$	that $(i' \circ j') \circ f' = i' \circ f \circ j = f' \circ i \circ j = f'$. But $\text{id}_{A'}$ also satisfies those properties: $\text{id}_{A'} \circ i' = i'$ and

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches-325 F03850 [conf 0.024]	325	■ 0.024	$\mathrm{id}_{A^{\{\prime\}}$	$\mathrm{id}_{A'}$	that $(i' \circ j') \circ f' = i' \circ f \circ j = f' \circ i \circ j = f'$. But $\mathrm{id}_{A'}$ also satisfies those properties: $\mathrm{id}_{A'} \circ i' = i'$ and
fong-spivak-seven-sketches-325 F03851 [conf 0.024]	325	■ 0.024	$\mathrm{id}_{A^{\{\prime\}}}\{ \}^{\{\circ\}} i^{\{\prime\}} = i^{\{\prime\}}$	$\mathrm{id}_{A'} \circ i' = i'$	that $(i' \circ j') \circ f' = i' \circ f \circ j = f' \circ i \circ j = f'$. But $\mathrm{id}_{A'}$ also satisfies those properties: $\mathrm{id}_{A'} \circ i' = i'$ and
fong-spivak-seven-sketches-325 F03852	325	■ 0.511	$f^{\{\prime\}} = f^{\{\prime\}}$	$f' = f'$	$\mathrm{id}_{A'} \circ f' = f'$, so by the uniqueness part of the universal property, $(i' \circ j') = \mathrm{id}_{A'}$.
fong-spivak-seven-sketches-325 F03853	325	■ 0.511	$\left(i^{\{\prime\}} ; j^{\{\prime\}} \right) = \mathrm{id}_{A^{\{\prime\}}}$	$(i'; j') = \mathrm{id}_{A'}$	$\mathrm{id}_{A'} \circ f' = f'$, so by the uniqueness part of the universal property, $(i' \circ j') = \mathrm{id}_{A'}$.
fong-spivak-seven-sketches-325 F03854	325	■ 0.549	$g: X \rightarrowtail A$	$g: X \rightarrow A$	So take any object X and morphisms $g: X \rightarrow A$ and $h: X \rightarrow B$ such that $g \circ f = h \circ \mathrm{id}_B$. We need
fong-spivak-seven-sketches-325 F03855	325	■ 0.549	$h: X \rightarrowtail B$	$h: X \rightarrow B$	So take any object X and morphisms $g: X \rightarrow A$ and $h: X \rightarrow B$ such that $g \circ f = h \circ \mathrm{id}_B$. We need
fong-spivak-seven-sketches-325 F03856	325	■ 0.549	$g \circ f = h \circ \mathrm{id}_B$	$g \circ f = h \circ \mathrm{id}_B$	So take any object X and morphisms $g: X \rightarrow A$ and $h: X \rightarrow B$ such that $g \circ f = h \circ \mathrm{id}_B$. We need
fong-spivak-seven-sketches-325 F03857	325	■ 0.554	$r: X \rightarrowtail A$	$r: X \rightarrow A$	to show there is a unique morphism $r: X \rightarrow A$ such that $r \circ \mathrm{id}_A = g$ and $r \circ f = h$. That's easy:
fong-spivak-seven-sketches-325 F03858	325	■ 0.554	$r; \operatorname{id}_A = g$	$r; \mathrm{id}_A = g$	to show there is a unique morphism $r: X \rightarrow A$ such that $r \circ \mathrm{id}_A = g$ and $r \circ f = h$. That's easy:
fong-spivak-seven-sketches-325 F03859	325	■ 0.554	$r; f = h$	$r; f = h$	to show there is a unique morphism $r: X \rightarrow A$ such that $r \circ \mathrm{id}_A = g$ and $r \circ f = h$. That's easy:
fong-spivak-seven-sketches-325 F03860	325	■ 1.000	$r = g$	$r = g$	the first requirement forces $r = g$ and the second requirement is then fulfilled.
fong-spivak-seven-sketches-326 F03861	326	■ 0.666	$v \otimes w$	$v \otimes w$	• $v \otimes w$ serves as a meet operation, i.e. it satisfies the same universal property as $\wedge$, namely
fong-spivak-seven-sketches-326 F03862	326	■ 0.988	$v \leq (w \multimap x)$	$v \leq (w \multimap x)$	namely $v \leq (w \multimap x)$ iff $v \otimes W \leq x$. So $\mathcal{V}$ is a cartesian closed category, and of course it is a
fong-spivak-seven-sketches-326 F03863	326	■ 0.988	$v \otimes W \leq x$	$v \otimes W \leq x$	namely $v \leq (w \multimap x)$ iff $v \otimes W \leq x$. So $\mathcal{V}$ is a cartesian closed category, and of course it is a
fong-spivak-seven-sketches-326 F03864	326	■ 0.998	$\mathbb{N}^{\mathrm{op}} \times \mathbb{N}^{\mathrm{op}}$	$\mathbb{N}^{\mathrm{op}} \times \mathbb{N}^{\mathrm{op}}$	This is the product preorder $\mathbb{N}^{\mathrm{op}} \times \mathbb{N}^{\mathrm{op}}$: its objects are pairs $(a, b) \in \mathbb{N} \times \mathbb{N}$ with $(a, b) \leq (a', b')$
fong-spivak-seven-sketches-326 F03865	326	■ 0.998	$(a, b) \in \mathbb{N} \times \mathbb{N}$	$(a, b) \in \mathbb{N} \times \mathbb{N}$	This is the product preorder $\mathbb{N}^{\mathrm{op}} \times \mathbb{N}^{\mathrm{op}}$: its objects are pairs $(a, b) \in \mathbb{N} \times \mathbb{N}$ with $(a, b) \leq (a', b')$
fong-spivak-seven-sketches-326 F03866	326	■ 0.998	$(a, b) \leq (a^{\{\prime\}}, b^{\{\prime\}})$	$(a, b) \leq (a', b')$	This is the product preorder $\mathbb{N}^{\mathrm{op}} \times \mathbb{N}^{\mathrm{op}}$: its objects are pairs $(a, b) \in \mathbb{N} \times \mathbb{N}$ with $(a, b) \leq (a', b')$
fong-spivak-seven-sketches-326 F03867	326	■ 1.000	$a^{\{\prime\}} \leq a$	$a' \leq a$	iff, in the usual ordering on $\mathbb{N}$ we have $a' \leq a$ and $b' \leq b$. But you can just look at the diagram.
fong-spivak-seven-sketches-326 F03868	326	■ 1.000	$b^{\{\prime\}} \leq b$	$b' \leq b$	iff, in the usual ordering on $\mathbb{N}$ we have $a' \leq a$ and $b' \leq b$. But you can just look at the diagram.
fong-spivak-seven-sketches-326 F03869	326	■ 0.658	$(0, 0)$	$(0, 0)$	It has a top element, $(0, 0)$, and it has binary meets, $(a, b) \wedge (a', b') = (\max(a, a'), \max(b, b'))$. But
fong-spivak-seven-sketches-326 F03870	326	■ 0.658	$(a, b) \wedge (a^{\{\prime\}}, b^{\{\prime\}}) = (\max(a, a'), \max(b, b^{\{\prime\}}))$	$(a, b) \wedge (a', b') = (\max(a, a'), \max(b, b'))$	It has a top element, $(0, 0)$, and it has binary meets, $(a, b) \wedge (a', b') = (\max(a, a'), \max(b, b'))$. But
fong-spivak-seven-sketches-327 F03871	327	■ 0.958	$x \multimap y$	$x \multimap y$	each x, y , the hom-object $x \multimap y$ exists. The formula for it is $x \multimap y = \bigvee \{w \mid w \wedge x \leq y\}$, i.e. we
fong-spivak-seven-sketches-327 F03872	327	■ 0.958	$x \multimap y = \bigvee \{w \mid w \wedge x \leq y\}$	$x \multimap y = \bigvee \{w \mid w \wedge x \leq y\}$	each x, y , the hom-object $x \multimap y$ exists. The formula for it is $x \multimap y = \bigvee \{w \mid w \wedge x \leq y\}$, i.e. we
fong-spivak-seven-sketches-327 F03873	327	■ 0.997	$y \wedge x \leq y$	$y \wedge x \leq y$	need these particular joins to exist. Since $y \wedge x \leq y$, we have $y \leq x \multimap y$. So we can replace the
fong-spivak-seven-sketches-327 F03874	327	■ 0.997	$y \leq x \multimap y$	$y \leq x \multimap y$	need these particular joins to exist. Since $y \wedge x \leq y$, we have $y \leq x \multimap y$. So we can replace the
fong-spivak-seven-sketches-327 F03875	327	■ 0.431	$x \multimap y = \bigvee \{w \mid y \leq w$	$x \multimap y = \bigvee \{w \mid y \leq w$	formula with $x \multimap y = \bigvee \{w \mid y \leq w$ and $w \wedge x \leq y\}$. But the set of elements in $\mathbb{N}^{\mathrm{op}} \times \mathbb{N}^{\mathrm{op}}$ that
fong-spivak-seven-sketches-327 F03876	327	■ 0.431	$w \wedge x \leq y\}$	$w \wedge x \leq y\}$	formula with $x \multimap y = \bigvee \{w \mid y \leq w$ and $w \wedge x \leq y\}$. But the set of elements in $\mathbb{N}^{\mathrm{op}} \times \mathbb{N}^{\mathrm{op}}$ that
fong-spivak-seven-sketches-327 F03877	327	■ 0.431	$\mathbb{N}^{\mathrm{op}} \times \mathbb{N}^{\mathrm{op}}$	$\mathbb{N}^{\mathrm{op}} \times \mathbb{N}^{\mathrm{op}}$	formula with $x \multimap y = \bigvee \{w \mid y \leq w$ and $w \wedge x \leq y\}$. But the set of elements in $\mathbb{N}^{\mathrm{op}} \times \mathbb{N}^{\mathrm{op}}$ that

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fong-spivak-seven-sketches F03878	327	1.000	$m: \mathbb{Z} \rightarrow \mathbb{B}$	$m: \mathbb{Z} \rightarrow \mathbb{B}$	Let $m: \mathbb{Z} \rightarrow \mathbb{B}$ be the characteristic function of the inclusion $\mathbb{N} \subseteq \mathbb{Z}$.
fong-spivak-seven-sketches F03879	327	1.000	$\mathbb{N} \subseteq \mathbb{Z}$	$\mathbb{N} \subseteq \mathbb{Z}$	Let $m: \mathbb{Z} \rightarrow \mathbb{B}$ be the characteristic function of the inclusion $\mathbb{N} \subseteq \mathbb{Z}$.
fong-spivak-seven-sketches F03880	327	1.000	$\ulcorner m \urcorner(-5) =$	$\ulcorner m \urcorner(-5) =$	1. $\ulcorner m \urcorner(-5) = \text{false}$.
fong-spivak-seven-sketches F03881	327	1.000	$\ulcorner m \urcorner(0) =$	$\ulcorner m \urcorner(0) =$	2. $\ulcorner m \urcorner(0) = \text{true}$.
fong-spivak-seven-sketches F03882	327	1.000	$A \subseteq \mathbb{B}$	$A \subseteq \mathbb{B}$	1. The sort of thing (**) we're looking for is a subobject of $\mathbb{B}$, say $A \subseteq \mathbb{B}$. This would have a
fong-spivak-seven-sketches F03883	327	0.993	$\ulcorner \Rightarrow \urcorner: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	$\ulcorner \Rightarrow \urcorner: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	3. The characteristic function for $P \Rightarrow Q$ is the function $\ulcorner \Rightarrow \urcorner: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$ given by the first, second,
fong-spivak-seven-sketches F03884	327	0.991	$\subseteq \mathbb{B} \times \mathbb{B}$	$\subseteq \mathbb{B} \times \mathbb{B}$	4. It classifies the subset $\{(\text{true}, \text{true}), (\text{false}, \text{true}), (\text{false}, \text{false})\}$ $\{(\text{true}, \text{true}), (\text{false}, \text{true}), (\text{false}, \text{false})\}$.
fong-spivak-seven-sketches F03885	327	1.000	$\ulcorner E \urcorner, \ulcorner P \urcorner, \ulcorner T \urcorner: \mathbb{N} \rightarrow \mathbb{B}$	$\ulcorner E \urcorner, \ulcorner P \urcorner, \ulcorner T \urcorner: \mathbb{N} \rightarrow \mathbb{B}$	Say that $\ulcorner E \urcorner, \ulcorner P \urcorner, \ulcorner T \urcorner: \mathbb{N} \rightarrow \mathbb{B}$ classify respectively the subsets $E := \{n \in \mathbb{N} \mid n \text{ is even}\}$, $P := \{n \in \mathbb{N} \mid$
fong-spivak-seven-sketches F03886	327	1.000	$\}, P := \{n \in \mathbb{N} \mid$	$\}, P := \{n \in \mathbb{N} \mid$	Say that $\ulcorner E \urcorner, \ulcorner P \urcorner, \ulcorner T \urcorner: \mathbb{N} \rightarrow \mathbb{B}$ classify respectively the subsets $E := \{n \in \mathbb{N} \mid n \text{ is even}\}$, $P := \{n \in \mathbb{N} \mid$
fong-spivak-seven-sketches F03887	327	0.994	$\ulcorner E \urcorner(17) =$	$\ulcorner E \urcorner(17) =$	1. $\ulcorner E \urcorner(17) = \text{false}$ because 17 is not even.
fong-spivak-seven-sketches F03888	327	1.000	$\ulcorner P \urcorner(17) =$	$\ulcorner P \urcorner(17) =$	2. $\ulcorner P \urcorner(17) = \text{true}$ because 17 is prime.
fong-spivak-seven-sketches F03889	327	0.857	$\ulcorner T \urcorner(17) =$	$\ulcorner T \urcorner(17) =$	3. $\ulcorner T \urcorner(17) = \text{true}$ because $17 \geq 10$.
fong-spivak-seven-sketches F03890	327	0.857	$17 \geq 10$	$17 \geq 10$	3. $\ulcorner T \urcorner(17) = \text{true}$ because $17 \geq 10$.
fong-spivak-seven-sketches F03891	327	0.679	$B(x, \epsilon) := \{x' \in \mathbb{R} \mid x - x' < \epsilon\}$	$B(x, \epsilon) := \{x' \in \mathbb{R} \mid x - x' < \epsilon\}$	1. The 1-dimensional analogue of an ϵ -ball around a point $x \in \mathbb{R}$ is $B(x, \epsilon) := \{x' \in \mathbb{R} \mid x - x' < \epsilon\}$,
fong-spivak-seven-sketches F03892	327	1.000	$x \in U$	$x \in U$	2. A subset $U \subseteq \mathbb{R}$ is open if, for every $x \in U$ there is some $\epsilon > 0$ such that $B(x, \epsilon) \subseteq U$.
fong-spivak-seven-sketches F03893	327	1.000	$B(x, \epsilon) \subseteq U$	$B(x, \epsilon) \subseteq U$	2. A subset $U \subseteq \mathbb{R}$ is open if, for every $x \in U$ there is some $\epsilon > 0$ such that $B(x, \epsilon) \subseteq U$.
fong-spivak-seven-sketches F03894	327	1.000	$U_1 := \{x \in \mathbb{R} \mid 0 < x < 2\}$	$U_1 := \{x \in \mathbb{R} \mid 0 < x < 2\}$	3. Let $U_1 := \{x \in \mathbb{R} \mid 0 < x < 2\}$ and $U_2 := \{x \in \mathbb{R} \mid 1 < x < 3\}$. Then $U := U_1 \cup U_2 = \{x \in \mathbb{R} \mid 0 <$
fong-spivak-seven-sketches F03895	327	1.000	$U_2 := \{x \in \mathbb{R} \mid 1 < x < 3\}$	$U_2 := \{x \in \mathbb{R} \mid 1 < x < 3\}$	3. Let $U_1 := \{x \in \mathbb{R} \mid 0 < x < 2\}$ and $U_2 := \{x \in \mathbb{R} \mid 1 < x < 3\}$. Then $U := U_1 \cup U_2 = \{x \in \mathbb{R} \mid 0 <$
fong-spivak-seven-sketches F03896	327	1.000	$U := U_1 \cup U_2 = \{x \in \mathbb{R} \mid 0 <$	$U := U_1 \cup U_2 = \{x \in \mathbb{R} \mid 0 <$	3. Let $U_1 := \{x \in \mathbb{R} \mid 0 < x < 2\}$ and $U_2 := \{x \in \mathbb{R} \mid 1 < x < 3\}$. Then $U := U_1 \cup U_2 = \{x \in \mathbb{R} \mid 0 <$
fong-spivak-seven-sketches F03897	327	1.000	$x < 3\}$	$x < 3\}$	$x < 3\}$.
fong-spivak-seven-sketches F03898	327	1.000	$y = (a, b)$	$y = (a, b)$	² If $y = (a, b)$ then there are exactly $(a + 1) * (b + 1)$ elements y' for which $y \leq y'$.
fong-spivak-seven-sketches F03899	327	1.000	$(a + 1) * (b + 1)$	$(a + 1) * (b + 1)$	² If $y = (a, b)$ then there are exactly $(a + 1) * (b + 1)$ elements y' for which $y \leq y'$.
fong-spivak-seven-sketches F03900	328	1.000	$I = \{1, 2, 3, 4, \dots\}$	$I = \{1, 2, 3, 4, \dots\}$	4. Let $I = \{1, 2, 3, 4, \dots\}$ and for each $i \in I$ let $U_i := \{x \in \mathbb{R} \mid \frac{1}{i} < x < 1\}$, so we have $U_1 \subseteq U_2 \subseteq$
fong-spivak-seven-sketches F03901	328	1.000	$U_i := \left\{x \in \mathbb{R} \mid \frac{1}{i} < x < 1\right\}$	$U_i := \left\{x \in \mathbb{R} \mid \frac{1}{i} < x < 1\right\}$	4. Let $I = \{1, 2, 3, 4, \dots\}$ and for each $i \in I$ let $U_i := \{x \in \mathbb{R} \mid \frac{1}{i} < x < 1\}$ so we have $U_1 \subseteq U_2 \subseteq$
fong-spivak-seven-sketches F03902	328	1.000	$U_1 \subseteq U_2 \subseteq$	$U_1 \subseteq U_2 \subseteq$	4. Let $I = \{1, 2, 3, 4, \dots\}$ and for each $i \in I$ let $U_i := \{x \in \mathbb{R} \mid \frac{1}{i} < x < 1\}$, so we have $U_1 \subseteq U_2 \subseteq$
fong-spivak-seven-sketches F03903	328	1.000	$U_3 \subseteq \dots$	$U_3 \subseteq \dots$	$U_3 \subseteq \dots$. Their union is $U := \bigcup_{i \in I} U_i = \{x \in \mathbb{R} \mid 0 < x < 1\}$.
fong-spivak-seven-sketches F03904	328	1.000	$U := \bigcup_{i \in I} U_i = \{x \in \mathbb{R} \mid 0 < x < 1\}$	$U := \bigcup_{i \in I} U_i = \{x \in \mathbb{R} \mid 0 < x < 1\}$	$U_3 \subseteq \dots$. Their union is $U := \bigcup_{i \in I} U_i = \{x \in \mathbb{R} \mid 0 < x < 1\}$.
fong-spivak-seven-sketches F03905	328	1.000	$\varnothing \subseteq X$	$\varnothing \subseteq X$	1. The coarse topology on X is the one whose only open sets are $X \subseteq X$ and $\varnothing \subseteq X$. This is a

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fong-spivak-seven-sketches F03906	328	■ 0.999	$\bigcup_{i \in I} A_i$	$\bigcup_{i \in I} A_i$	$\bigcup_{i \in I} A_i$ is X iff $A_i = X$ for some $i \in I$.
fong-spivak-seven-sketches F03907	328	■ 0.999	$A_{\{i\}}=X$	$A_i = X$	$\bigcup_{i \in I} A_i$ is X iff $A_i = X$ for some $i \in I$.
fong-spivak-seven-sketches F03908	328	■ 1.000	$A \subseteq X$	$A \subseteq X$	2. The fine topology on X is the one where every subset $A \subseteq X$ is considered open. All the
fong-spivak-seven-sketches F03909	328	■ 0.405	$(X, \mathbf{P}(X))$	$(X, \mathbf{P}(X))$	3. If $(X, \mathbf{P}(X))$ is discrete, $(Y, \mathbf{Op}_Y)$ is any topological space, and $f: X \rightarrow Y$ is any function then it
fong-spivak-seven-sketches F03910	328	■ 1.000	$f^{-1}(U) \subseteq X$	$f^{-1}(U) \subseteq X$	is continuous. Indeed, this just means that for any open set $U \subseteq Y$ the preimage $f^{-1}(U) \subseteq X$ is
fong-spivak-seven-sketches F03911	328	■ 1.000	$\varnothing \rightarrow \{1\} \rightarrow \{1, 2\}$	$\emptyset \rightarrow \{1\} \rightarrow \{1, 2\}$	1. The Hasse diagram for the Sierpinsky topology is  .
fong-spivak-seven-sketches F03912	328	■ 0.994	$U_{\{i\}}=U$	$U_i = U$	• $U_i = U$ for some $i \in I$.
fong-spivak-seven-sketches F03913	328	■ 1.000	$Y=B \cap Y$	$Y = B \cap Y$	1. We want to show that $Y \in \mathbf{Op}_{\cap Y}$. We need to find $B \in \mathbf{Op}$ such that $Y = B \cap Y$; this is easy, you
fong-spivak-seven-sketches F03914	328	■ 1.000	$B=Y$	$B = Y$	could take $B = Y$ or $B = X$, or anything in between.
fong-spivak-seven-sketches F03915	328	■ 1.000	$B=X$	$B = X$	could take $B = Y$ or $B = X$, or anything in between.
fong-spivak-seven-sketches F03916	328	■ 1.000	$\varnothing=\varnothing \cap Y$	$\emptyset = \emptyset \cap Y$	union. $\emptyset = \emptyset \cap Y$, so according to the formula, $\emptyset \in \mathbf{Op}_{\cap Y}$. Suppose that $A_1, A_2 \in \mathbf{Op}_{\cap Y}$. Then
fong-spivak-seven-sketches F03917	328	■ 1.000	$\varnothing \in \mathbf{Op}_{\cap Y}$	$\emptyset \in \mathbf{Op}_{\cap Y}$	union. $\emptyset = \emptyset \cap Y$, so according to the formula, $\emptyset \in \mathbf{Op}_{\cap Y}$. Suppose that $A_1, A_2 \in \mathbf{Op}_{\cap Y}$. Then
fong-spivak-seven-sketches F03918	328	■ 1.000	$A_{\{1\}}, A_{\{2\}} \in \mathbf{Op}_{\cap Y}$	$A_1, A_2 \in \mathbf{Op}_{\cap Y}$	union. $\emptyset = \emptyset \cap Y$, so according to the formula, $\emptyset \in \mathbf{Op}_{\cap Y}$. Suppose that $A_1, A_2 \in \mathbf{Op}_{\cap Y}$. Then
fong-spivak-seven-sketches F03919	328	■ 0.999	$B_{\{1\}}, B_{\{2\}} \in \mathbf{Op}$	$B_1, B_2 \in \mathbf{Op}$	there exist $B_1, B_2 \in \mathbf{Op}$ with $A_1 = B_1 \cap Y$ and $A_2 = B_2 \cap Y$. But then $A_1 \cap A_2 = (B_1 \cap Y) \cap (B_2 \cap Y) =$
fong-spivak-seven-sketches F03920	328	■ 0.999	$A_{\{1\}}=B_{\{1\}} \cap Y$	$A_1 = B_1 \cap Y$	there exist $B_1, B_2 \in \mathbf{Op}$ with $A_1 = B_1 \cap Y$ and $A_2 = B_2 \cap Y$. But then $A_1 \cap A_2 = (B_1 \cap Y) \cap (B_2 \cap Y) =$
fong-spivak-seven-sketches F03921	328	■ 0.999	$A_{\{2\}}=B_{\{2\}} \cap Y$	$A_2 = B_2 \cap Y$	there exist $B_1, B_2 \in \mathbf{Op}$ with $A_1 = B_1 \cap Y$ and $A_2 = B_2 \cap Y$. But then $A_1 \cap A_2 = (B_1 \cap Y) \cap (B_2 \cap Y) =$
fong-spivak-seven-sketches F03922	328	■ 0.999	$A_{\{1\}} \cap A_{\{2\}}=\left(B_{\{1\}} \cap Y\right) \cap \left(B_{\{2\}} \cap Y\right)=$	$A_1 \cap A_2 = (B_1 \cap Y) \cap (B_2 \cap Y) =$	there exist $B_1, B_2 \in \mathbf{Op}$ with $A_1 = B_1 \cap Y$ and $A_2 = B_2 \cap Y$. But then $A_1 \cap A_2 = (B_1 \cap Y) \cap (B_2 \cap Y) =$
fong-spivak-seven-sketches F03923	328	■ 0.995	$\left(B_{\{1\}} \cap B_{\{2\}}\right) \cap Y$	$(B_1 \cap B_2) \cap Y$	$(B_1 \cap B_2) \cap Y$, so it is in $\mathbf{Op}_{\cap Y}$ since $B_1 \cap B_2 \in \mathbf{Op}$. The same idea works for arbitrary unions:
fong-spivak-seven-sketches F03924	328	■ 0.995	$B_{\{1\}} \cap B_{\{2\}} \in \mathbf{Op}$	$B_1 \cap B_2 \in \mathbf{Op}$	$(B_1 \cap B_2) \cap Y$, so it is in $\mathbf{Op}_{\cap Y}$ since $B_1 \cap B_2 \in \mathbf{Op}$. The same idea works for arbitrary unions:
fong-spivak-seven-sketches F03925	328	■ 1.000	$A_{\{i\}}$	A_i	given a set I and A_i for each $i \in I$, we have $A_i = B_i \cap Y$ for some $B_i \in \mathbf{Op}$, and
fong-spivak-seven-sketches F03926	328	■ 1.000	$A_{\{i\}}=B_{\{i\}} \cap Y$	$A_i = B_i \cap Y$	given a set I and A_i for each $i \in I$, we have $A_i = B_i \cap Y$ for some $B_i \in \mathbf{Op}$, and
fong-spivak-seven-sketches F03927	328	■ 1.000	$B_{\{i\}} \in \mathbf{Op}$	$B_i \in \mathbf{Op}$	given a set I and A_i for each $i \in I$, we have $A_i = B_i \cap Y$ for some $B_i \in \mathbf{Op}$, and
fong-spivak-seven-sketches F03928	328	■ 1.000	$U \subseteq \mathbf{Op}$	$U \subseteq \mathbf{Op}$	by an open set $U \subseteq \mathbf{Op}$. It might look something like this:
fong-spivak-seven-sketches F03929	328	■ 1.000	$\mathcal{C}(B, C)=$	$\mathcal{C}(B, C) =$	paths between them, of the monoidal product of distances along that path. For example, $\mathcal{C}(B, C) =$
fong-spivak-seven-sketches F03930	329	■ 1.000	$\left(U_{\{3\}} \vee U_{\{1\}}\right) \vee \left(U_{\{4\}} \vee U_{\{2\}}\right)$	$(U_3 \wedge U_1) \vee (U_4 \wedge U_2)$	 , because $\wedge$ is the monoidal product in $\mathcal{V}$.
fong-spivak-seven-sketches F03931	329	■ 1.000	$\mathcal{C}(a, b)$	$\mathcal{C}(a, b)$	In general, we can thus imagine the open set $\mathcal{C}(a, b)$ as a kind of ‘size restriction’ for getting from a to b ,
fong-spivak-seven-sketches F03932	329	■ 1.000	$\left\{a_{\{1\}}, a_{\{2\}}\right\}$	$\{a_1, a_2\}$	1. The fiber of f over a is $\{a_1, a_2\}$.
fong-spivak-seven-sketches F03933	329	■ 1.000	$\left\{c_{\{1\}}\right\}$	$\{c_1\}$	2. The fiber of f over c is $\{c_1\}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03934	329	■ 1.000	$\operatorname{Sec}_f(V_2)=\varnothing$	$\operatorname{Sec}_f(V_2) = \emptyset$	2. When $V_2 = \{a, b, c, d\}$, there are no sections: $\operatorname{Sec}_f(V_2) = \emptyset$.
fong-spivak-seven-sketches F03935	329	■ 0.949	$\left.\operatorname{Sec}_f(V_3)\right _{V_2}$	$\operatorname{Sec}_f(V_3) _{V_2}$	3. When $V_3 = \{a, b, d, e\}$, the set $\operatorname{Sec}_f(V_3)$ has $2 * 3 * 1 * 2 = 12$ elements.
fong-spivak-seven-sketches F03936	329	■ 0.949	$2 * 3 * 1 * 2=12$	$2 * 3 * 1 * 2 = 12$	3. When $V_3 = \{a, b, d, e\}$, the set $\operatorname{Sec}_f(V_3)$ has $2 * 3 * 1 * 2 = 12$ elements.
fong-spivak-seven-sketches F03937	330	■ 1.000	$g_1:=\left(a_1, b_1\right)$	$g_1 := (a_1, b_1)$	1. Let $g_1 := (a_1, b_1)$ and $g_2 := (b_2, e_1)$; these do not agree on the overlap.
fong-spivak-seven-sketches F03938	330	■ 1.000	$g_2:=\left(b_2, e_1\right)$	$g_2 := (b_2, e_1)$	1. Let $g_1 := (a_1, b_1)$ and $g_2 := (b_2, e_1)$; these do not agree on the overlap.
fong-spivak-seven-sketches F03939	330	■ 1.000	$g \in \operatorname{Sec}_f\left(U_1 \cup U_2\right)$	$g \in \operatorname{Sec}_f\left(U_1 \cup U_2\right)$	2. No, there's no section $g \in \operatorname{Sec}_f\left(U_1 \cup U_2\right)$ for which $g _{U_1}=g_1$ and $g _{U_2}=g_2$.
fong-spivak-seven-sketches F03940	330	■ 1.000	$\left.g\right _{U_1}=g_1$	$g _{U_1}=g_1$	2. No, there's no section $g \in \operatorname{Sec}_f\left(U_1 \cup U_2\right)$ for which $g _{U_1}=g_1$ and $g _{U_2}=g_2$.
fong-spivak-seven-sketches F03941	330	■ 1.000	$\left.g\right _{U_2}=g_2$	$g _{U_2}=g_2$	2. No, there's no section $g \in \operatorname{Sec}_f\left(U_1 \cup U_2\right)$ for which $g _{U_1}=g_1$ and $g _{U_2}=g_2$.
fong-spivak-seven-sketches F03942	330	■ 0.997	$F(\{1,2\}) \rightarrow F(\{1\}) \rightarrow F(\varnothing)$	$F(\{1,2\}) \rightarrow F(\{1\}) \rightarrow F(\emptyset)$	2. A presheaf F on $\mathbf{Op}$ consists of any three sets and any two functions $F(\{1,2\}) \rightarrow F(\{1\}) \rightarrow F(\emptyset)$
fong-spivak-seven-sketches F03943	330	■ 1.000	$F(\varnothing) \cong\{1\}$	$F(\emptyset) \cong \{1\}$	4. As explained in Example 7.36, F will be a sheaf iff $F(\emptyset) \cong \{1\}$. Thus we the category of sheaves
fong-spivak-seven-sketches F03944	330	■ 1.000	$F(\{1,2\}) \rightarrow F(\{1\})$	$F(\{1,2\}) \rightarrow F(\{1\})$	is equivalent to that of just two sets and one function $F(\{1,2\}) \rightarrow F(\{1\})$.
fong-spivak-seven-sketches F03945	330	■ 1.000	$S(\varnothing)=\{()\}$	$S(\emptyset) = \{()\}$	$S(\emptyset) = \{()\}$ by the sheaf condition (see Example 7.36). So the only data in a sheaf $S \in \mathbf{Shv}(X)$ is the set
fong-spivak-seven-sketches F03946	330	■ 1.000	$S(\{1\})$	$S(\{1\})$	$S(\{1\})$. This is how we get the correspondence between sets and sheaves on the one point space.
fong-spivak-seven-sketches F03947	330	■ 0.751	$\Omega: \mathbf{Op}(X)^{\mathrm{op}} \rightarrow \mathbf{Set}$	$\Omega: \mathbf{Op}(X)^{\mathrm{op}} \rightarrow \mathbf{Set}$	According to Eq. (7.50), the subobject classifier $\Omega: \mathbf{Op}(X)^{\mathrm{op}} \rightarrow \mathbf{Set}$ in $\mathbf{Shv}(X)$ should be the functor
fong-spivak-seven-sketches F03948	330	■ 0.947	$\Omega(\{1\})$	$\Omega(\{1\})$	where $\Omega(\{1\})$ is the set of open sets of $\{1\}$. So we're hoping to see that there is a one-to-one correspon-
fong-spivak-seven-sketches F03949	330	■ 1.000	$\mathbf{Op}(\{1\})$	$\mathbf{Op}(\{1\})$	dence between the set $\mathbf{Op}(\{1\})$ and the set $\mathbb{B} = \{\text{true}, \text{false}\}$ of booleans. Indeed there is: there are
fong-spivak-seven-sketches F03950	330	■ 0.600	$\Omega(U):=\left\{U^{\prime} \in \mathbf{Op} \mid U^{\prime} \subseteq U\right\}$	$\Omega(U) := \{U' \in \mathbf{Op} \mid U' \subseteq U\}$	By Eqs. (7.50) and (7.51) the definition of $\Omega(U)$ is $\Omega(U) := \{U' \in \mathbf{Op} \mid U' \subseteq U\}$, and the definition of
fong-spivak-seven-sketches F03951	330	■ 1.000	$U^{\prime} \mapsto U^{\prime} \cap V$	$U' \mapsto U' \cap V$	the restriction map for $V \subseteq U$ is $U' \mapsto U' \cap V$.
fong-spivak-seven-sketches F03952	331	■ 0.998	$U^{\prime} \subseteq U$	$U' \subseteq U$	1. It is functorial: given $W \subseteq V \subseteq U$ and $U' \subseteq U$, we indeed have $(U' \cap V) \cap W = U' \cap W$, since
fong-spivak-seven-sketches F03953	331	■ 0.998	$\left(U^{\prime} \cap V\right) \cap W=U^{\prime} \cap W$	$(U' \cap V) \cap W = U' \cap W$	1. It is functorial: given $W \subseteq V \subseteq U$ and $U' \subseteq U$, we indeed have $(U' \cap V) \cap W = U' \cap W$, since
fong-spivak-seven-sketches F03954	331	■ 1.000	$W \subseteq V$	$W \subseteq V$	$W \subseteq V$. For functoriality, we also need preservation of identities, and this amounts to $U' \cap U = U'$
fong-spivak-seven-sketches F03955	331	■ 1.000	$U^{\prime} \cap U=U^{\prime}$	$U' \cap U = U'$	$W \subseteq V$. For functoriality, we also need preservation of identities, and this amounts to $U' \cap U = U'$
fong-spivak-seven-sketches F03956	331	■ 0.947	$G^{\prime}$	G'	There is only one that classifies G' , and here it is. Let's write $\gamma := \ulcorner G' \urcorner$.
fong-spivak-seven-sketches F03957	331	■ 0.947	$\gamma:=\ulcorner G^{\prime} \urcorner$	$\gamma := \ulcorner G' \urcorner$	There is only one that classifies G' , and here it is. Let's write $\gamma := \ulcorner G' \urcorner$.
fong-spivak-seven-sketches F03958	331	■ 1.000	$\gamma(D)=0$	$\gamma(D) = 0$	• Since D is missing from G' , we have $\gamma(D) = 0$ (vertex: missing).
fong-spivak-seven-sketches F03959	331	■ 1.000	A, B, C	A, B, C	• Since vertices A, B, C are present in G' we have $\gamma(A) = \gamma(B) = \gamma(C) = V$ (vertex: present).
fong-spivak-seven-sketches F03960	331	■ 1.000	$\gamma(A)=\gamma(B)=\gamma(C)=V$	$\gamma(A) = \gamma(B) = \gamma(C) = V$	• Since vertices A, B, C are present in G' we have $\gamma(A) = \gamma(B) = \gamma(C) = V$ (vertex: present).
fong-spivak-seven-sketches F03961	331	■ 1.000	$\gamma(i)=(V, 0 ; 0)$	$\gamma(i) = (V, 0; 0)$	• The above forces $\gamma(i) = (V, 0; 0)$ (arrow from present vertex to missing vertex: missing).

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F03962	331	1.000	$\gamma(f)=(V, V ; A)$	$\gamma(f) = (V, V; A)$	Since the arrow f is in G', we have $\gamma(f) = (V, V; A)$ (arrow from present vertex to present vertex:
fong-spivak-seven-sketches F03963	331	1.000	$\gamma(g)=\gamma(h)=(V, V ; \emptyset)$	$\gamma(g) = \gamma(h) = (V, V; \emptyset)$	Since the arrows g and h are missing in G', we have $\gamma(g) = \gamma(h) = (V, V; \emptyset)$ (arrow from present
fong-spivak-seven-sketches F03964	331	1.000	$U=\mathbb{R}-\{0\} \subseteq \mathbb{R}$	$U = \mathbb{R} - \{0\} \subseteq \mathbb{R}$	With $U = \mathbb{R} - \{0\} \subseteq \mathbb{R}$, we have:
fong-spivak-seven-sketches F03965	331	1.000	$\mathbb{R}-U=\{0\}$	$\mathbb{R} - U = \{0\}$	1. The complement of U is $\mathbb{R} - U = \{0\}$ and $\neg U$ is its interior, which is $\neg U = \emptyset$.
fong-spivak-seven-sketches F03966	331	1.000	$\neg U=\varnothing$	$\neg U = \emptyset$	1. The complement of U is $\mathbb{R} - U = \{0\}$ and $\neg U$ is its interior, which is $\neg U = \emptyset$.
fong-spivak-seven-sketches F03967	331	1.000	$\mathbb{R}-\varnothing=\mathbb{R}$	$\mathbb{R} - \emptyset = \mathbb{R}$	2. The complement of $\neg U$ is $\mathbb{R} - \emptyset = \mathbb{R}$, and this is open, so $\neg \neg U = \mathbb{R}$.
fong-spivak-seven-sketches F03968	331	1.000	$\neg \neg U=\mathbb{R}$	$\neg \neg U = \mathbb{R}$	2. The complement of $\neg U$ is $\mathbb{R} - \emptyset = \mathbb{R}$, and this is open, so $\neg \neg U = \mathbb{R}$.
fong-spivak-seven-sketches F03969	331	0.943	$\neg \neg U \subseteq$	$\neg \neg U \subseteq$	4. It is false that $\neg \neg U \subseteq U$.
fong-spivak-seven-sketches F03970	331	0.995	$V \in \mathbf{Op}$	$V \in \mathbf{Op}$	1. If for any $V \in \mathbf{Op}$ we have $\top \wedge V = V$ then when $V = X$ we have $\top \wedge X := \top \cap X = X$, but
fong-spivak-seven-sketches F03971	331	0.995	$\top \wedge V=V$	$\top \wedge V = V$	1. If for any $V \in \mathbf{Op}$ we have $\top \wedge V = V$ then when $V = X$ we have $\top \wedge X := \top \cap X = X$, but
fong-spivak-seven-sketches F03972	331	0.995	$V=X$	$V = X$	1. If for any $V \in \mathbf{Op}$ we have $\top \wedge V = V$ then when $V = X$ we have $\top \wedge X := \top \cap X = X$, but
fong-spivak-seven-sketches F03973	331	0.995	$\top \wedge X:=\top \cap X=X$	$\top \wedge X := \top \cap X = X$	1. If for any $V \in \mathbf{Op}$ we have $\top \wedge V = V$ then when $V = X$ we have $\top \wedge X := \top \cap X = X$, but
fong-spivak-seven-sketches F03974	331	0.953	$\top=\top \cap X=X$	$\top = \top \cap X = X$	anything intersected with X is itself, so $\top = \top \cap X = X$.
fong-spivak-seven-sketches F03975	331	0.477	$(\top \vee V):=(X \cup V)=X$	$(\top \vee V) := (X \cup V) = X$	2. $(\top \vee V) := (X \cup V) = X$ holds and $(V \Rightarrow X) = \bigcup_{\{R \in \mathbf{Op} \mid R \cap V \subseteq X\}} R = X$ holds because $(X \cap V) \subseteq X$.
fong-spivak-seven-sketches F03976	331	0.477	$(V \Rightarrow X)=\bigcup_{\{R \in \mathbf{Op} \mid R \cap V \subseteq X\}} R=X$	$(V \Rightarrow X) = \bigcup_{\{R \in \mathbf{Op} \mid R \cap V \subseteq X\}} R = X$	2. $(\top \vee V) := (X \cup V) = X$ holds and $(V \Rightarrow X) = \bigcup_{\{R \in \mathbf{Op} \mid R \cap V \subseteq X\}} R = X$ holds because $(X \cap V) \subseteq X$.
fong-spivak-seven-sketches F03977	331	0.477	$(X \cap V) \subseteq X$	$(X \cap V) \subseteq X$	2. $(\top \vee V) := (X \cup V) = X$ holds and $(V \Rightarrow X) = \bigcup_{\{R \in \mathbf{Op} \mid R \cap V \subseteq X\}} R = X$ holds because $(X \cap V) \subseteq X$.
fong-spivak-seven-sketches F03978	331	0.885	$(\perp \vee V)=V$	$(\perp \vee V) = V$	3. If for any set $V \in \mathbf{Op}$ we have $(\perp \vee V) = V$, then when $V = \emptyset$ we have $(\perp \vee \emptyset) = (\perp \cup \emptyset) = \emptyset$, but
fong-spivak-seven-sketches F03979	331	0.885	$V=\emptyset$	$V = \emptyset$	3. If for any set $V \in \mathbf{Op}$ we have $(\perp \vee V) = V$, then when $V = \emptyset$ we have $(\perp \vee \emptyset) = (\perp \cup \emptyset) = \emptyset$, but
fong-spivak-seven-sketches F03980	331	0.885	$(\perp \vee \varnothing)=(\perp \cup \varnothing)=\varnothing$	$(\perp \vee \emptyset) = (\perp \cup \emptyset) = \emptyset$	3. If for any set $V \in \mathbf{Op}$ we have $(\perp \vee V) = V$, then when $V = \emptyset$ we have $(\perp \vee \emptyset) = (\perp \cup \emptyset) = \emptyset$, but
fong-spivak-seven-sketches F03981	331	0.807	$(\perp \wedge V)=(\varnothing \cap V)=\varnothing$	$(\perp \wedge V) = (\emptyset \cap V) = \emptyset$	4. $(\perp \wedge V) = (\emptyset \cap V) = \emptyset$ holds, and $(\perp \Rightarrow V) = \bigcup_{\{R \in \mathbf{Op} \mid R \cap \emptyset \subseteq V\}} R = X$ holds because $(X \cap \emptyset) \subseteq V$.
fong-spivak-seven-sketches F03982	331	0.807	$(\perp \Rightarrow V)=\bigcup_{\{R \in \mathbf{Op} \mid R \cap \emptyset \subseteq V\}} R=X$	$(\perp \Rightarrow V) = \bigcup_{\{R \in \mathbf{Op} \mid R \cap \emptyset \subseteq V\}} R = X$	4. $(\perp \wedge V) = (\emptyset \cap V) = \emptyset$ holds, and $(\perp \Rightarrow V) = \bigcup_{\{R \in \mathbf{Op} \mid R \cap \emptyset \subseteq V\}} R = X$ holds because $(X \cap \emptyset) \subseteq V$.
fong-spivak-seven-sketches F03983	331	0.807	$(X \cap \varnothing) \subseteq V$	$(X \cap \emptyset) \subseteq V$	4. $(\perp \wedge V) = (\emptyset \cap V) = \emptyset$ holds, and $(\perp \Rightarrow V) = \bigcup_{\{R \in \mathbf{Op} \mid R \cap \emptyset \subseteq V\}} R = X$ holds because $(X \cap \emptyset) \subseteq V$.
fong-spivak-seven-sketches F03984	331	1.000	$\{S \mid p\}$	$\{S \mid p\}$	a person who is alive throughout that interval. A section in the subobject $\{S \mid p\}$ over any interval of
fong-spivak-seven-sketches F03985	331	0.983	$S=\mathbb{N}$	$S = \mathbb{N}$	$S = \mathbb{N}$, let $p(s)$ be the predicate “ $24 \leq s \leq 28$,” and let $q(s)$ be the predicate “ s is not prime.” Then
fong-spivak-seven-sketches F03986	331	0.983	$24 \leq s \leq 28$	$24 \leq s \leq 28$	$S = \mathbb{N}$, let $p(s)$ be the predicate “ $24 \leq s \leq 28$,” and let $q(s)$ be the predicate “ s is not prime.” Then
fong-spivak-seven-sketches F03987	331	0.983	$q(s)$	$q(s)$	$S = \mathbb{N}$, let $p(s)$ be the predicate “ $24 \leq s \leq 28$,” and let $q(s)$ be the predicate “ s is not prime.” Then
fong-spivak-seven-sketches F03988	332	1.000	$p(n, z)$	$p(n, z)$	We have the predicate $p: \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{B}$ given by $p(n, z)$ iff $n \leq z $.
fong-spivak-seven-sketches F03989	332	1.000	$n \leq z $	$n \leq z $	We have the predicate $p: \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{B}$ given by $p(n, z)$ iff $n \leq z $.

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fong-spivak-seven-sketches F03990	332	1.000	$\{\emptyset\} \subseteq \mathbb{N}$	$\{0\} \subseteq \mathbb{N}$	1. The predicate $\forall(z : \mathbb{Z}). p(n, z)$ holds for $\{0\} \subseteq \mathbb{N}$.
fong-spivak-seven-sketches F03991	332	1.000	$\mathbb{N} \subseteq \mathbb{N}$	$\mathbb{N} \subseteq \mathbb{N}$	2. The predicate $\exists(z : \mathbb{Z}). p(n, z)$ holds for $\mathbb{N} \subseteq \mathbb{N}$.
fong-spivak-seven-sketches F03992	332	0.988	$\varnothing \subseteq \mathbb{Z}$	$\emptyset \subseteq \mathbb{Z}$	3. The predicate $\forall(n : \mathbb{N}). p(n, z)$ holds for $\emptyset \subseteq \mathbb{Z}$.
fong-spivak-seven-sketches F03993	332	1.000	$\mathbb{Z} \subseteq \mathbb{Z}$	$\mathbb{Z} \subseteq \mathbb{Z}$	4. The predicate $\exists(n : \mathbb{N}). p(n, z)$ holds for $\mathbb{Z} \subseteq \mathbb{Z}$.
fong-spivak-seven-sketches F03994	332	0.786	$\forall(t : T)$	$\forall(t : T)$	1. The formula says that $\forall(t : T). p(s, t)$ “returns the largest open set $V \subseteq U$ for which $p(s _V, t) = V$
fong-spivak-seven-sketches F03995	332	1.000	$t_{\{i\}}$	t_i	union of all time intervals V_i for which there is some item t_i in the news about which s is worried
fong-spivak-seven-sketches F03996	332	0.990	$V =$	$V =$	something!,” she is saying that it’s like $V =$ “this past year.”
fong-spivak-seven-sketches F03997	332	0.695	$s^{\backslash\prime \prime}$	s''	2. This seems like a good thing for “there exists a piece of news that worries s ” to mean: the news
fong-spivak-seven-sketches F03998	332	1.000	$p:=j\left(q\right)$	$p := j(q)$	hypothesis, that $p \leq j(p)$ for all $U \subseteq X$ and $p \in \Omega(U)$. If $j(j(q)) \leq j(q)$, then letting $p := j(q)$ we have
fong-spivak-seven-sketches F03999	332	0.999	$j(p) \leq p$	$j(p) \leq p$	both $j(p) \leq p$ and $p \leq j(p)$. This means $p \cong j(p)$, but Ω is a poset (not just a preorder) so $p = j(p)$, i.e.
fong-spivak-seven-sketches F04000	332	0.999	$p \cong j(p)$	$p \cong j(p)$	both $j(p) \leq p$ and $p \leq j(p)$. This means $p \cong j(p)$, but Ω is a poset (not just a preorder) so $p = j(p)$, i.e.
fong-spivak-seven-sketches F04001	332	0.999	$p=j\left(p\right)$	$p = j(p)$	both $j(p) \leq p$ and $p \leq j(p)$. This means $p \cong j(p)$, but Ω is a poset (not just a preorder) so $p = j(p)$, i.e.
fong-spivak-seven-sketches F04002	333	1.000	$s \in S(U), p(s)$	$s \in S(U), p(s)$	2. Let U be the interval 2019/01/01 – 2019/02/01. For an arbitrary person $s \in S(U)$, $p(s)$ is a subset
fong-spivak-seven-sketches F04003	333	1.000	$j\left(j\left(p\left(s\right)\right)\right)=j\left(p\left(s\right)\right)$	$j(j(p(s)) = j(p(s))$	5. It is true that $j(j(p(s)) = j(p(s))$, because suppose given a time during which “if Bob is in San
fong-spivak-seven-sketches F04004	333	1.000	$0 \leq 2 \leq 6 \leq 8$	$0 \leq 2 \leq 6 \leq 8$	1. Since $0 \leq 2 \leq 6 \leq 8$, we have $[2, 6] \in o_{[0,8]}$ by the above formula.
fong-spivak-seven-sketches F04005	333	0.977	$[2,6] \in^{?} o_{[0,5]} \cup o_{[4,8]}$	$[2,6] \in^{?} o_{[0,5]} \cup o_{[4,8]}$	2. In order to have $[2,6] \in^{?} o_{[0,5] \cup o_{[4,8]}}$, we would need to have either $[2,6] \in^{?} o_{[0,5]}$ or $[2,6] \in^{?} o_{[4,8]}$.
fong-spivak-seven-sketches F04006	333	0.977	$[2,6] \in^{?} o_{[0,5]}$	$[2,6] \in^{?} o_{[0,5]}$	2. In order to have $[2,6] \in^{?} o_{[0,5] \cup o_{[4,8]}}$, we would need to have either $[2,6] \in^{?} o_{[0,5]}$ or $[2,6] \in^{?} o_{[4,8]}$.
fong-spivak-seven-sketches F04007	333	0.977	$[2,6] \in^{?} o_{[4,8]}$	$[2,6] \in^{?} o_{[4,8]}$	2. In order to have $[2,6] \in^{?} o_{[0,5] \cup o_{[4,8]}}$, we would need to have either $[2,6] \in^{?} o_{[0,5]}$ or $[2,6] \in^{?} o_{[4,8]}$.
fong-spivak-seven-sketches F04008	333	0.510	$U^{\backslash\prime} \subseteq \mathbb{R}$	$U' \subseteq \mathbb{R}$	A subset $U \subseteq \mathbb{R}$ is open in the subspace topology of $\mathbb{R} \subseteq \mathbb{R}$ iff there is an open set $U' \subseteq \mathbb{R}$ with
fong-spivak-seven-sketches F04009	333	1.000	$U=U^{\backslash\prime} \cap \mathbb{R}$	$U = U' \cap \mathbb{R}$	$U = U' \cap \mathbb{R}$. We want to show that this is the case iff U is open in the usual topology.
fong-spivak-seven-sketches F04010	333	0.999	$U^{\backslash\prime}=\bigcup_{i \in I} o_{\left[\left[a_{\{i\}}, b_{\{i\}}\right]\right]}$	$U' = \bigcup_{i \in I} o_{[a_i, b_i]}$	basic opens, $U' = \bigcup_{i \in I} o_{[a_i, b_i]}$, where $o_{[a_i, b_i]} = \{[d, u] \in \mathbb{R} \mid a_i < d < u < b_i\}$. Since $\mathbb{R} = \{[x, x] \in \mathbb{R}\}$,
fong-spivak-seven-sketches F04011	333	0.999	$o_{\left[\left[a_{\{i\}}, b_{\{i\}}\right]\right]}=\left\{\left[d, u\right] \in \mathbb{R} \mid a_{\{i\}}<d<u<b_{\{i\}}\right\}$	$o_{[a_i, b_i]} = \{[d, u] \in \mathbb{R} \mid a_i < d < u < b_i\}$	basic opens, $U' = \bigcup_{i \in I} o_{[a_i, b_i]}$, where $o_{[a_i, b_i]} = \{[d, u] \in \mathbb{R} \mid a_i < d < u < b_i\}$. Since $\mathbb{R} = \{[x, x] \in \mathbb{R}\}$,
fong-spivak-seven-sketches F04012	333	0.999	$\mathbb{R}=\left\{\left[x, x\right] \in \mathbb{R}\right\}$	$\mathbb{R} = \{[x, x] \in \mathbb{R}\}$	basic opens, $U' = \bigcup_{i \in I} o_{[a_i, b_i]}$, where $o_{[a_i, b_i]} = \{[d, u] \in \mathbb{R} \mid a_i < d < u < b_i\}$. Since $\mathbb{R} = \{[x, x] \in \mathbb{R}\}$,
fong-spivak-seven-sketches F04013	333	1.000	$U^{\backslash\prime} \cap \mathbb{R}$	$U' \cap \mathbb{R}$	the intersection $U' \cap \mathbb{R}$ will then be
fong-spivak-seven-sketches F04014	333	1.000	$B\left(m_{\{i\}}, r_{\{i\}}\right)$	$B\left(m_i, r_i\right)$	and this is just the union of open balls $B(m_i, r_i)$ where $m_i := \frac{a_i+b_i}{2}$ is the midpoint and $r_i := \frac{b_i-a_i}{2}$ is
fong-spivak-seven-sketches F04015	333	1.000	$m_{\{i\}}:=\frac{a_{\{i\}}+b_{\{i\}}}{2}$	$m_i := \frac{a_i + b_i}{2}$	and this is just the union of open balls $B(m_i, r_i)$ where $m_i := \frac{a_i+b_i}{2}$ is the midpoint and $r_i := \frac{b_i-a_i}{2}$ is
fong-spivak-seven-sketches F04016	333	1.000	$r_{\{i\}}:=\frac{b_{\{i\}}-a_{\{i\}}}{2}$	$r_i := \frac{b_i - a_i}{2}$	and this is just the union of open balls $B(m_i, r_i)$ where $m_i := \frac{a_i+b_i}{2}$ is the midpoint and $r_i := \frac{b_i-a_i}{2}$ is
fong-spivak-seven-sketches F04017	333	0.877	$\left(a_{\{i\}}, b_{\{i\}}\right)$	(a_i, b_i)	the radius of the interval (a_i, b_i) . The open balls $B(m_i, r_i)$ are open in the usual topology on $\mathbb{R}$ and the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches F04018	333	■ 1.000	$U=\bigcup_{j\in J} B(m_j, \epsilon_j)$	$U = \bigcup_{j \in J} B(m_j, \epsilon_j)$	Suppose that U is open in the usual topology. Then $U = \bigcup_{j \in J} B(m_j, \epsilon_j)$ for some set J . Let $a_j := m_j - \epsilon_j$
fong-spivak-seven-sketches F04019	333	■ 1.000	$a_j:=m_j-\epsilon_j$	$a_j := m_j - \epsilon_j$	Suppose that U is open in the usual topology. Then $U = \bigcup_{j \in J} B(m_j, \epsilon_j)$ for some set J . Let $a_j := m_j - \epsilon_j$
fong-spivak-seven-sketches F04020	333	■ 1.000	$b_j:=m_j+\epsilon_j$	$b_j := m_j + \epsilon_j$	and $b_j := m_j + \epsilon_j$. Then
fong-spivak-seven-sketches F04021	333	■ 0.950	$H_X(U):=$	$H_X(U) :=$	Fix any topological space $(X, \mathbf{Op}_X)$ and any subset $R \subseteq \mathbb{IR}$ of the interval domain. Define $H_X(U) :=$
fong-spivak-seven-sketches F04022	333	■ 1.000	$\{f: U \cap R \rightarrowtail X \mid f$	$\{f: U \cap R \rightarrow X \mid f$	$\{f: U \cap R \rightarrow X \mid f$ is continuous}.
fong-spivak-seven-sketches F04023	333	■ 1.000	$f: U \cap R \rightarrowtail X$	$f: U \cap R \rightarrow X$	1. H_X is a presheaf: given $V \subseteq U$ the restriction map sends the continuous function $f: U \cap R \rightarrow X$
fong-spivak-seven-sketches F04024	333	■ 1.000	$V \cap R \subseteq U \cap R$	$V \cap R \subseteq U \cap R$	to its restriction along the subset $V \cap R \subseteq U \cap R$.
fong-spivak-seven-sketches F04025	333	■ 1.000	$U=\bigcup_i U_i$	$U = \bigcup_i U_i$	2. It is a sheaf: given any family U_i of open sets with $U = \bigcup_i U_i$ and a continuous function
fong-spivak-seven-sketches F04026	333	■ 1.000	$f_i: U_i \cap R \rightarrowtail X$	$f_i: U_i \cap R \rightarrow X$	$f_i: U_i \cap R \rightarrow X$ for each i , agreeing on overlaps, they can be glued together to give a continuous
fong-spivak-seven-sketches F04027	333	■ 0.913	$U \cap R$	$U \cap R$	function on all of $U \cap R$, since $U \cap R = (\bigcup_i U_i) \cap R = \bigcup_i (U_i \cap R)$.
fong-spivak-seven-sketches F04028	333	■ 0.913	$U \cap R=\left(\bigcup_i U_i\right) \cap R=\bigcup_i\left(U_i \cap R\right)$	$U \cap R = \left(\bigcup_i U_i\right) \cap R = \bigcup_i (U_i \cap R)$	function on all of $U \cap R$, since $U \cap R = (\bigcup_i U_i) \cap R = \bigcup_i (U_i \cap R)$.